Global Proof Transfers and Structural Classifications

Unified Research Manuscript
Sixteen Constituent Papers, Consolidated Results, and an Editorial Top Ten

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Abstract

This manuscript consolidates the sixteen distinct papers developed in the present conversation. Its central themes are exact analytic connection formulas, minimum differential and recurrence complexity, algebraic and differential-Galois classifications, optimal recovery by mixture likelihood, and structural obstructions across mathematical fields. The analytic parts treat physical Jacobi trace densities at resonances and collided knots. The period and recurrence parts treat hypercubic and banana integrals, weighted Franel and multinomial sequences, rational diagonals, and arithmetic classifications. Statistical results include full-parameter binomial, Poisson, and known-interval Gaussian recovery, contamination, unequal depths, and precise limitations of likelihood adaptation. Further parts address spectral network design, periodic operators, entropy under heat flow, homogeneous information algorithms, quantum information, finite groups, preorder polytopes, semifinite operator inequalities, diffusion means, and categorical products of generalized cellular automata.

The mathematical bodies are retained with their proofs and qualifications. Duplicate constituent files are included only once, local identifiers are made unique, references are consolidated, and an editorial guide records which later results strengthen earlier ones. A ranked selection of ten achievement families is provided as an assessment of breadth, exactness, proof development, and reuse, rather than a claim of certified priority or publication-level significance.

Research status. This is a research draft. Source reconciliation, separate in-session mathematical readings, and typesetting checks do not constitute external peer review or formal proof verification. The rankings assess the arguments and scopes recorded here. Historical novelty is subject to the literature qualifications in each part; this consolidation makes no new novelty claim.

Editorial guide: coverage, dependencies, and revisions

Seventeen retrieved TEX files reduce to sixteen distinct constituent manuscripts: an earlier collection already contains the complete mathematical bodies of its eight separately delivered source papers. Those eight parts are included once, followed by the eight later papers. The expanded entropy–transport draft agrees substantively with Part VIII of that earlier collection. No distinct result is omitted merely because it falls outside the editorial top ten.

What is consolidated. There is one title, one global abstract, one table of contents, one numbering framework, and one bibliography. Each part retains a short local overview and its substantive arguments. Repeated supporting lemmas remain when they record a separate proof or explain a local dependency; a repeated theorem is not counted twice in the ranking. Parts and appendices have qualified numbers, and links point to the corresponding statements. The TEX file contains all mathematical text and diagrams and requires no external input, image, manuscript, or bibliography database to compile.

Imported mathematical prerequisites. Being a standalone TEX file does not mean that every cited theorem is reproved. Classical ingredients are cited where used. In particular, the full banana-period Galois classifications in Parts VI–VII use the all-mass rank, irreducibility, pairing, and infinity-monodromy results of the previously supplied manuscript [032], cited as [49]. Its full development is outside these sixteen papers. Those claims retain that dependency; they are not presented as independently established from this volume alone. The separate diagonal-variable theorem in Part XI has its own lower- and upper-bound argument.

How later results change earlier scope statements. Statements such as "not proved here" inside a constituent paper describe that paper. The final status must be read with the following developments.

1. The nonresonant ordinary Jacobi connection of Part II is extended to all admissible real parameters in Part IV. Part IX adds local analytic amplitudes, resonant regularity, and exact polynomial collision tests. Part X gives a global formula under entire interactions. The graphical noninteger-repulsion result in Part IX is local; it is not silently replaced by a global arbitrary-repulsion assertion.

2. The weighted-Franel phase and cubic cases are followed by the all-exponent, all-complex-weight width theorem in Part VII and the arithmetic-subsequence development in Part VIII. Width means largest shift minus smallest shift, not the number of nonzero terms. Earlier sparse-term theorems therefore retain independent content.

3. Hypercubic rank and algebraicity in Part V are complemented by the symplectic and orthogonal group arguments in Parts VI–VII. These are functional statements about differential modules and jets, not automatic transcendence assertions for numerical special values.

4. Part VIII includes the repaired local-Fourier formulation of the Laplace–Jordan argument, Theorem VIII.8.3. Its stated regularity hypotheses are essential. The odd-exponent invariant-summand correction in Part VII is retained.

5. The full Poisson rate theorem of Part IV is reproduced and extended to the two small sample sizes in Part VIII. These are one result family. The support-free growing-support Gaussian theorem in Part V and the known-interval all-scale theorem in Part VI use different optimization domains. The latter does not imply unrestricted support adaptation.

6. Robust binomial likelihood, the exact two-depth problem, and the heteroscedastic Gaussian counterexample concern different observation models. None supplies a universal theorem for every heterogeneous or contaminated mixture experiment.

7. Graph design reaches the correct $\sqrt{n}$ order over all connected graphs and an exact leading constant on block graphs. The leading constant over all connected graphs remains unresolved in this volume.

8. Later factorial-reduction and irrationality theorems strengthen earlier sufficient families. The necessary direction of the original factorial-reduction conjecture and its higher-degree irrationality target are not thereby settled.

Established work that is not a new achievement here. The original Shannon-entropy counterexample is credited in Part XIII; that part's increment is a Tsallis/Rényi parameter classification. The bipartite palindromicity criterion in Part XVI overlaps Hatasa and Ohsugi–Hibi, and the delayed-star proof is an elementary consequence of a known formula. Earlier lemniscate-length results and subsequently located automata and general-tree counterexamples are credited or excluded as specified in the constituent texts. No conclusion about the irrationality of Euler's constant or another unresolved numerical constant is inferred from this assembly.

Editorial top ten

The ranking is an editorial judgment using four criteria: breadth of the exact statement, completion of exceptional and boundary cases, the supplied proof mechanism, and usefulness for further problems. A common mechanism and its dependent corollaries occupy one position. These are ten *achievement families*, not a certified count of ten independently new major conjecture resolutions. All share the research status stated on the title page.

1. **Global connections for physical Jacobi trace laws.** The ordinary trace formula covers every $N \geq 1$, $a, b > -1$, and $\beta \geq 0$, including resonance. It identifies the actual density across its intervals. Weighted local cone gluing, grouped collisions, and exact regularity extend the mechanism, while entire interactions admit a global continuation theorem. This ranks first for combining exact normalization, global continuation, and exceptional-parameter control. The local graphical result is not a full arbitrary-basis connection matrix. See Theorems IV.3.1, IX.2.5, IX.5.1, and X.5.1; Parts IV, IX, X.

2. **Hypercubic periods: exact rank, algebraicity, and full groups.** For arbitrary nonzero complex masses, including collided thresholds, the rank is given by an explicit multiplicity formula. The algebraicity classification and the orthogonal/symplectic parity classification then describe the whole differential module. This is a broad structural result with a substantial chain of local-Fourier and representation-theoretic inputs. Functional independence must be distinguished from numerical special-value transcendence. See Theorems V.3.1, V.3.2, VI.3.1, and VII.2.1; Parts V–VII.

3. **The exact rational-diagonal variable count for weighted banana series.** With $n \geq 2$ nonzero complex weights, the minimum number of variables of a rational function realizing the stated multinomial-square series as a complete diagonal is exactly n. Positivity, separated thresholds, and genericity are unnecessary. The value of the result lies in a matching impossibility theorem and construction, rather than a convenient upper representation alone. See Theorem XI.1.1, Part XI.

4. **Mixture likelihood at complete statistical parameter ranges.** The binomial, Poisson, and known-interval Gaussian NPMLE theorems provide matching uniform recovery rates over their stated full parameter ranges. Binomial extensions cover unknown whole-report contamination and an arbitrarily imbalanced depth-1/depth-T experiment. These are distinct models sharing a likelihood-contact and two-scale inversion mechanism; known support/noise and the precise contamination model remain essential. See Theorems VIII.6.1, VIII.5.1, VI.7.1, III.2.1, and III.3.1; Parts III, VI, VIII.

5. **Weighted Franel sequences: complete recurrence and algebraicity classifications.** For every exponent $r \geq 2$ and nonzero complex weight, the centered sequence has width $\lceil r/2 \rceil$ at 1, $\lfloor r/2 \rfloor$ at -1, and r otherwise. The companion generating-function algebraicity classification, arithmetic subsequences, and multinomial extensions show the reach of the rank argument. Width and sparse term count are kept distinct, and the special invariant-summand case is corrected. See Theorems VII.4.1, V.2.1, VIII.7.2, and X.1.2; Parts V, VII, VIII, X.

6. **Complete heat-flow parameter classification for Tsallis and Rényi entropy.** For smooth positive line densities with Gaussian tails, the universal alternating-derivative property holds among Tsallis orders $\alpha > 0$ exactly at $\alpha = 2$; no Rényi order survives. The proof isolates a negative exponential coefficient and transfers a periodic construction to a Gaussian window on the line. It determines all orders, but does not give a uniform lowest derivative order for failure. The earlier Shannon counterexample is explicitly credited. See Theorem XIII.2.1, Part XIII.

7. **Spectral-supremum submajorization in every semifinite algebra.** For a finite family of positive measurable affiliated operators, $\bigvee_{j\,\mathrm{sp}} A_j \prec\prec \sum_j A_j$. The proof includes unbounded operators and infinite traces, yields the sharp norm constant one, and retains an overlap remainder for integrable pairs. This addresses the full stated semifinite conjecture; type III algebras and infinite families are not included. See Theorem XVI.1.1 and Proposition XVI.1.3, Part XVI.

8. **A unified preorder-polytope reciprocity and gamma-positivity framework.** Transfer maps, lattice counting, and one multivariate binomial identity give polar and matrix dualities, a nonnegative gamma expansion, and full weighted q-zeta reciprocity for arbitrary finite preorders. Several numbered source conjectures follow from this shared mechanism and are counted here once. Real-rootedness and polytope realization are separate questions. See Theorems XV.4.1, XV.4.2, and XV.4.3, Part XV.

9. **Spectral design: the correct worst-case order and exact block solutions.** The determinant-optimal versus inverse-trace-optimal design gap is $\Theta(\sqrt{n})$ over connected graphs. Every connected block graph has explicit optimal conductances, and that subclass has limiting constant $1/(4\sqrt{2})$. Matching order and exact optimization make this more than a construction of bad examples. The unrestricted leading constant remains open. See Theorems III.5.5, IV.4.2, and IV.4.4; Parts III–IV.

10. **Finite multivariate logistic mixtures with all component parameters identifiable.** Finite linear independence holds in every dimension with arbitrary locations, positive diagonal scales, and positive shape parameters. Consequently the component count, mixing weights, and component parameters are identifiable up to permutation. The result concerns finite mixtures; unrestricted infinite mixing measures are a different question. See Theorems XII.3.1 and XII.3.4, Part XII.

Strong results outside this selection. The ranking does not diminish the GCA product-nonexistence theorem, the all-time unique diffusion-mean construction, periodic-potential rigidity and singularity counterexamples, uniform quadratic irrationality, the sharp quantum depolarizing coefficient, or the homogeneous-algorithm separations. They remain in full in Parts XI–XVI. The all-mass banana Galois group family is retained with its explicit [032] dependency rather than counted as an additional internally self-contained top-ten proof.

Constituent map

Each of the sixteen parts below is included exactly once. The first eight are taken from the previously reconciled collection, whose mathematical bodies were compared against their individual source files. The remaining eight are the subsequent standalone papers.

Part	Constituent manuscript
1	Exact Complexity and Arithmetic Transfers
2	Sharp Bounds and Connection Formulas Beyond Existing Restrictions
3	Entropy, Likelihood Geometry, and Sharp Proof Transfers
4	Global Parameter Theorems from Entropy, Transport, and Spectral Structure
5	Global Algebraicity and Exact Complexity
6	Global Galois Groups and Complete Parameter Classifications
7	Beyond Generic Parameters: Global Classifications
8	Entropy–Transport, Critical Values, and Laplace Monodromy
9	Exact Cone Gluing and Collision Tests for Weighted Trace Laws
10	Global Transfers from Recurrences, Periods and Statistical Recovery
11	Diagonal Grade and Homogeneous Algorithms
12	Floquet Rigidity, Logistic Mixtures, and Factorial Reduction
13	Further Open Problems: Proofs and Counterexamples
14	Open Problems Across Further Domains
15	Structural Results on Published Open Problems
16	Spectral Suprema, Heat-Kernel Selection, and Cellular-Automaton Products

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Part I
Exact Complexity and Arithmetic Transfers

Manuscript overview. We transfer proof mechanisms from recurrence minimality, period arithmetic, weighted Laplacian spectra, and mixture identification to six concrete problems. Products of separated central-trinomial moments have exact recurrence width $\prod_i (m_i + 1)$. Jacobi transforms have exact differential order $N + 1$, increasing to 2^N under arbitrarily small separated weights at even nonnegative Dyson index. For every inverse-power spectral loss, we determine the unique optimal conductances when a matching of links is forbidden. Holomorphic time differentials imply simultaneous transcendence of Hamiltonian coordinates at nonzero algebraic-logarithmic clock times and a harmonic exception for action-quantized energies. For a mean-reverting square-root diffusion, we verify power-call stopping for every positive real power, classify algebraic exercise boundaries by an exact discount resonance, and prove transcendence of zero-state values at positive integer discount ratios. Finally, a known emission family with finite-dimensional mixing defect admits hidden-population recovery from three visits despite unknown persistence; an explicit positive analytic family shows both the failure of two visits and the sharp risk $\Theta(\min\{1, (r^2\sqrt{N})^{-1}\})$ on a specified submodel. The proofs identify the assumptions and classical inputs of each transfer.

Integration note. This opening manuscript supplies six distinct transfers. Its methodological sources are the older numbered manuscripts cited in its introduction. The nonlinear clock, stopping, spectral-design, and hidden-state results are retained here in full.

I.1 What is transferred, and what is proved

The starting collection contains several distinct kinds of proof: positive integrals make singularities visible; a finite-dimensional representation gives an upper bound that can be matched by those singularities; an exact spectral region converts inverse existence into optimization; and arithmetic separation of Abelian integrals from logarithms constrains dynamical motion. A separate observation-theoretic principle distinguishes local differential relations from the kernel of a global mixing operator. We use these mechanisms as starting points, and reconstruct the needed arguments for the target problems below.

There are three different senses of an exact conclusion in this paper. A *complexity theorem* supplies matching upper and lower bounds. A *classification theorem* states the exceptional parameter set as well as the generic alternative. An *observation theorem* proves both a sufficient experiment and an obstruction to a smaller or noisier one. These claims must not be conflated with historical priority.

The target problems are deliberately concrete. They ask for a minimal recurrence for products of central-trinomial moments, the change in Jacobi-trace differential complexity when weights are separated, an optimal weighted network when several links are forbidden, arithmetic constraints on polynomial Hamiltonian motion, the optimal exercise boundary of a mean-reverting diffusion, and recovery of a hidden population with unknown temporal persistence. The problem definitions are part of the statements; none of the conclusions is obtained merely by relabelling a constant from the source collection.

Classical inputs include Selberg and Kummer identities, analytic subgroup and E-function theorems, finite-dimensional linear algebra, and elementary statistical testing. Whenever a result relies on such an input, it is stated at the point of use. In the mechanical application, action quantization defines semiclassical levels; the exact quantum spectrum is a separate problem.

The source methods were selected from the supplied manuscripts [8, 9, 50, 10, 11, 30, 12, 49]. Their role is methodological provenance; the target results below include their own arguments and identify any external theorem they use.

Mechanism	Target field	Conclusion proved
Visible critical values and tensor bounds	Lattice return probabilities	Exact shift width, including repeated parameters
Distinct endpoint phases	Random-matrix linear statistics	Exact differential order and a symmetry-breaking jump
Spectral feasibility and strict convexity	Network design	Unique optimum for every inverse-power loss
Holomorphic time forms	Classical and semiclassical mechanics	Arithmetic clocks and exceptional quantization levels
E-function specialization	Mean-reverting optimal stopping	Discount-resonance classification of a genuine exercise boundary
Mixture-kernel repair and temporal moments	Hidden-state statistics	Minimal observation count and persistence-dependent minimax rate

I.2 Exact recurrence widths for anisotropic return probabilities

I.2.1 A transfer from critical values to Hadamard products

The singularity-counting argument for Franel sequences has an application to products of classical moment sequences and to return probabilities of finite-range random walks. It determines an exact recurrence width in every dimension, including all positive integer multiplicities of the parameters. The result is not presented as the resolution of a separately named published conjecture: the literature checked below does not supply such a conjecture for this family. Its contribution here is a uniform, explicit lower bound matching the standard symmetric-power upper bound.

For real numbers $p > 1$ define

$$L_p(x) = (p-1) + \frac{p+1}{2}(x + x^{-1}), \qquad u_p(n) = \mathrm{CT}_x\, L_p(x)^n \quad (n \geq 0). \tag{I.2.1}$$

The generalized central trinomial coefficient is

$$T_n(b,c) = \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j}\binom{2j}{j} b^{n-2j} c^j.$$

Thus $u_p(n) = T_n(p-1, (p+1)^2/4)$. In particular $u_p(n)$ is a positive integer when p is an odd positive integer. The generating function and three-term recurrence are classical; see [1, 2]. For completeness the special case needed here will be derived below.

Fix $s \geq 1$, distinct odd primes $p_1, \ldots, p_s$, and integers $m_1, \ldots, m_s \geq 1$. Put

$$D = \sum_{i=1}^{s} m_i, \qquad N = \prod_{i=1}^{s} (m_i + 1), \qquad A_n = \prod_{i=1}^{s} u_{p_i}(n)^{m_i}. \tag{I.2.2}$$

A recurrence of shift width r means an identity

$$\sum_{j=0}^{r} q_j(n) A_{n+j} = 0, \qquad q_j \in \mathbb{C}(n), \quad q_0 q_r \neq 0,$$

valid for all sufficiently large integers n. No degree bound is imposed on its rational coefficients.

Theorem I.2.1 (Exact product width)**.** *For the sequence in* (I.2.2)*, the least recurrence width over* $\mathbb{C}(n)$ *is exactly*

$$\boxed{\prod_{i=1}^{s} (m_i + 1).}$$

A recurrence of this width exists over $\mathbb{Q}(n)$*. The same conclusion holds for multiplicatively independent rational parameters* $p_i > 1$*, without the assertion that every* A_n *is an integer.*

The conclusion measures the width in the discrete variable n. It is not a claim about the minimum differential order of the generating function. Repeated parameters contribute $m_i + 1$, whereas distinct independent parameters contribute multiplicatively. In particular the width can be 2^D even though the underlying Laurent polynomial has degree at most one in each individual variable.

I.2.2 Visibility and the recurrence budget

We recall the precise analytic principle, including the condition that prevents cancellation on an interior critical fiber.

Lemma I.2.2 (Definite-fiber visibility)**.** *Let M be a compact real-analytic manifold of positive dimension d with a positive analytic density $d\mu$, and let $\Phi : M \to \mathbb{R}$ be analytic. Suppose $v \neq 0$ is a critical value whose critical fiber contains finitely many critical points, all nondegenerate and all local maxima, or all nondegenerate and all local minima. Regular points on the same fiber are allowed. If*

$$a_n = \int_M \Phi^n \, d\mu, \qquad F(t) = \sum_{n \geq 0} a_n t^n,$$

then a branch continued from the germ of F at 0 is singular at $t = 1/v$. Consequently every rational-coefficient recurrence for (a_n) has width at least the number of distinct nonzero values with this property.

Proof. The analytic Morse lemma and polar coordinates show that the pushforward $\Phi_* \mu$ has, near v, density

$$\rho(v + x) = r(x) + \mathbf{1}_{\{\epsilon x > 0\}} |x|^{d/2-1} h(|x|), \qquad h(0) > 0,$$

where r is smooth and h is analytic. Here $\epsilon = 1$ at minima and $\epsilon = -1$ at maxima. To justify this with regular points on the fiber, take disjoint Morse neighborhoods and cutoffs equal to one near each critical point. The complement is a compact submersion region over a sufficiently small interval about v, whose pushforward is smooth. Inside a Morse chart, the density contribution at distance $w > 0$ from v is

$$\frac{1}{2} w^{d/2-1} \int_{\mathbb{S}^{d-1}} g(\sqrt{w}\, \omega) \, d\omega.$$

The spherical average is analytic in w, with positive constant term. Every critical point contributes on the same side, so the leading coefficients add. The resulting one-sided term is not smooth, for either integer or half-integer exponent, and cannot be canceled by r.

The Cauchy transform

$$G(z) = \int_M \frac{d\mu(x)}{z - \Phi(x)}$$

is holomorphic in the upper half-plane and satisfies $G(z) = z^{-1} F(z^{-1})$ near infinity. Its boundary distribution obeys $-\pi^{-1} \operatorname{Im} G(x + i0) = \Phi_* \mu$ by the Poisson kernel. If this branch extended holomorphically through v, its imaginary part on the real axis would be analytic, contradicting the nonsmooth density. Inversion gives the asserted singularity of the continued germ of F.

For the last assertion clear the denominators of a width-r recurrence, writing its coefficients as $q_j(n) \in \mathbb{C}[n]$. With $\vartheta = t\, d/dt$, summation gives

$$\left(\sum_{j=0}^{r} t^{r-j} q_j(\vartheta - j) \right) F(t) = R(t),$$

where R is a polynomial of initial terms. If $e = \max_j \deg q_j$ and $c_j = [n^e] q_j(n)$, the coefficient of the highest ordinary derivative is $t^e H(t)$, with

$$H(t) = \sum_{j=0}^{r} c_j t^{r-j} \neq 0, \qquad \deg H \leq r.$$

An ordinary point of this inhomogeneous linear differential equation cannot be a singularity of any continued solution branch. Thus every nonzero finite singularity just obtained is a zero of H, proving the bound. □

I.2.3 All nonzero critical fibers are definite

Use one angle $\theta_{i,j}$ for each pair $1 \le i \le s$, $1 \le j \le m_i$, and normalized Haar measure on $\mathbb{T}^D$. Constant-term extraction gives

$$A_n = \int_{\mathbb{T}^D} \Phi(\theta)^n \, d\theta, \qquad \Phi(\theta) = \prod_{i=1}^{s} \prod_{j=1}^{m_i} ((p_i - 1) + (p_i + 1)\cos\theta_{i,j}). \tag{I.2.3}$$

The convention in the integral includes normalization by $(2\pi)^{-D}$.

Lemma I.2.3 (The complete critical-value list)**.** *All critical points with $\Phi \neq 0$ have every angle equal to 0 or π. For k_i angles equal to 0 in block i, their value is*

$$v_{\boldsymbol{k}} = 2^D (-1)^{D - \sum_i k_i} \prod_{i=1}^{s} p_i^{k_i}, \qquad 0 \le k_i \le m_i. \tag{I.2.4}$$

These N values are distinct. Every critical point over a positive $v_{\boldsymbol{k}}$ is a nondegenerate maximum, and every one over a negative $v_{\boldsymbol{k}}$ is a nondegenerate minimum.

Proof. At a nonzero value no factor in (I.2.3) vanishes. The logarithmic derivative with respect to $\theta_{i,j}$ is

$$-\frac{(p_i + 1)\sin\theta_{i,j}}{(p_i - 1) + (p_i + 1)\cos\theta_{i,j}}.$$

It vanishes exactly when $\theta_{i,j} \in \{0, \pi\}$. The corresponding factor values are $2p_i$ and -2, respectively, giving (I.2.4). Unique factorization shows that equality of two absolute values forces equality of every k_i. Multiplicative independence gives the same argument for rational p_i.

At a corner, every off-diagonal Hessian entry is zero. The diagonal entry for angle $\theta_{i,j}$ is

$$-v_{\boldsymbol{k}} \frac{(p_i + 1)\cos\theta_{i,j}}{(p_i - 1) + (p_i + 1)\cos\theta_{i,j}}.$$

The quotient is $(p_i + 1)/(2p_i) > 0$ at 0, and $(p_i + 1)/2 > 0$ at π. Thus all Hessian eigenvalues are nonzero and have sign $-\operatorname{sgn}(v_{\boldsymbol{k}})$. This also proves that all critical points on any one nonzero fiber use the same side; their contributions cannot cancel in Lemma I.2.2. □

The lemma and the visibility principle give a lower bound of N over $\mathbb{C}(n)$. This argument does not require irreducibility of a telescoper, numerical guessing, or noncancellation of complex saddle-point expansions.

I.2.4 A matching upper bound from symmetric powers

The one-coordinate generating function is

$$\sum_{n \ge 0} u_p(n) t^n = \frac{1}{\sqrt{1 - 2(p-1)t - 4pt^2}}. \tag{I.2.5}$$

For example, integrate the geometric series for $1/(1 - t((p-1) + (p+1)\cos\theta))$; the constant-term binomial expansion gives the same identity formally. Differentiating (I.2.5) and comparing coefficients proves

$$(n+1)u_p(n+1) = (2n+1)(p-1)u_p(n) + 4pn\, u_p(n-1) \quad (n \ge 1), \tag{I.2.6}$$

with $u_p(0) = 1$ and $u_p(1) = p - 1$.

For a fixed exponent m, the $m+1$ monomials

$$u_p(n)^{m-j}u_p(n+1)^j, \qquad 0 \le j \le m,$$

form a vector whose shift is a rational matrix times the original vector: replace $u_p(n+2)$ using (I.2.6) and expand binomially. For rational p all matrix entries lie in $\mathbb{Q}(n)$. Taking the tensor product of these symmetric-power systems over the s blocks gives an N-dimensional system containing A_n as a coordinate. The $N+1$ coordinate rows expressing $A_n, \dots, A_{n+N}$ in terms of the state at n are linearly dependent over $\mathbb{Q}(n)$. Consequently A_n satisfies a recurrence of width at most N, allowing finitely many initial exceptions when rational denominators vanish. Combined with the lower bound already proved, this completes the proof of Theorem I.2.1.

Corollary I.2.4 (Powers of the usual central trinomial coefficients)**.** *Let $C_n = \mathrm{CT}_x(1+x+x^{-1})^n$. For every integer $m \ge 1$, the least recurrence width of C_n^m over $\mathbb{C}(n)$ is $m+1$.*

Proof. For $p=3$, $L_3(x) = 2(1+x+x^{-1})$ and $u_3(n) = 2^n C_n$. Multiplication of a sequence by a nonzero geometric sequence preserves its minimum recurrence width: absorb the geometric factors into the individual coefficients. Apply the main theorem with one block. □

I.2.5 Rigidity of the leading recurrence symbol

Theorem I.2.5 (Exact singularity set and leading symbol)**.** *Let $F(t) = \sum_{n\ge 0} A_n t^n$ for (I.2.2). The set of nonzero finite points at which some branch continued from the germ of F is singular is exactly*

$$\{v_{\boldsymbol{k}}^{-1} : 0 \le k_i \le m_i\}.$$

For any width-N polynomial-coefficient recurrence $\sum_{j=0}^{N} q_j(n) A_{n+j} = 0$, put $e = \max_j \deg q_j$ and $c_j = [n^e]q_j(n)$. There is a nonzero constant c such that

$$\sum_{j=0}^{N} c_j t^{N-j} = c \prod_{0\le k_i \le m_i} (1 - v_{\boldsymbol{k}} t). \tag{I.2.7}$$

Proof. The budget polynomial on the left is nonzero and has degree at most N. The N distinct reciprocal critical values are all zeros of it, by the visibility lemma. Its degree must therefore be exactly N, and these are all its roots, proving (I.2.7). The associated inhomogeneous differential equation is ordinary away from these points and 0, and therefore has no other nonzero finite singularities for any branch of its solution F. Visibility already proves the reverse inclusion. □

This is an exact constraint on every recurrence of minimum width, including ones with larger polynomial degrees than necessary. It can be used to reject an incorrectly guessed recurrence without first factoring its Ore operator. It also records every secondary exponential scale, although a single dominant asymptotic ratio sees only $2^D \prod_i p_i^{m_i}$.

I.2.6 A completely explicit two-parameter certificate

Take $p_1 = 3$, $p_2 = 5$, $m_1 = m_2 = 1$. Then

$$A_n = T_n(2,4)T_n(4,9) = 1, 8, 408, 15680, 750880, 36334848, 1844830464, \dots.$$

The exact recurrence is

$$\begin{aligned} 0 = {} & 57600(n+1)^2(2n+7)A_n \\ & - 1920(2n+3)^2(2n+7)A_{n+1} \\ & - 32(2n+5)(49n^2+245n+261)A_{n+2} \\ & - 8(2n+3)(2n+7)^2A_{n+3} + (n+4)^2(2n+3)A_{n+4}, \qquad n \ge 0. \end{aligned} \tag{I.2.8}$$

It has minimum width four, since the visible critical values are $4, -12, -20, 60$. The leading-symbol identity is

$$115200t^4 - 15360t^3 - 3136t^2 - 64t + 2 = 2(1-4t)(1+12t)(1+20t)(1-60t).$$

Here is an exact algebraic certificate of the recurrence, independent of checking initial values. Set

$$M_p(n) = \begin{pmatrix} 0 & 1 \\ 4p(n+1)/(n+2) & (p-1)(2n+3)/(n+2) \end{pmatrix}, \qquad K(n) = M_3(n) \otimes M_5(n),$$

and let $e_1 = (1,0,0,0)$. If $q_0, \dots, q_4$ are the coefficients displayed in (I.2.8), direct rational arithmetic gives the row identity

$$q_0(n)e_1 + \sum_{j=1}^{4} q_j(n)e_1K(n+j-1)\cdots K(n) = 0.$$

Multiplication by $(u_3(n), u_3(n+1))^{\mathsf{T}} \otimes (u_5(n), u_5(n+1))^{\mathsf{T}}$ proves the recurrence. The accompanying exact-arithmetic certificate verifies this rational matrix identity; initial-term checks serve only as additional diagnostics.

I.2.7 A Markov-chain interpretation

Consider a discrete-time walk on $\mathbb{Z}^D$. At each step all coordinate increments are independent. A coordinate in block i has distribution

$$\mathbb{P}(\Delta = 0) = \frac{p_i - 1}{2p_i}, \qquad \mathbb{P}(\Delta = 1) = \mathbb{P}(\Delta = -1) = \frac{p_i + 1}{4p_i}. \tag{I.2.9}$$

The step distribution has finite support, positive holding probability, and positive probabilities for both signs in every coordinate. Let R_n be its probability of returning to the origin after n steps.

Corollary I.2.6 (Exact width of anisotropic return probabilities)**.** *For the walk* (I.2.9)*,*

$$R_n = \frac{A_n}{(2^D \prod_i p_i^{m_i})^n}, \qquad \operatorname{width}_{\mathbb{C}(n)}(R) = \prod_i (m_i + 1).$$

Consequently anisotropic walks with one coordinate for each of D distinct odd primes have exact recurrence width 2^D. When all coordinates have the same parameter, their exact width is $D+1$.

Proof. The one-coordinate probability generating Laurent polynomial is $L_{p_i}(x)/(2p_i)$. Independence, followed by constant-term extraction for the origin, proves the formula for R_n. A nonzero geometric normalization preserves recurrence width, as in Corollary I.2.4. □

This return-probability statement is an application of the exact-width theorem, not a second independent solution of an open problem. The sharp change from $D+1$ to 2^D quantifies how distinct coordinate parameters remove the symmetric-power reduction.

I.2.8 Literature boundary and remaining work

Generalized central trinomial coefficients, their Legendre-polynomial representations, moment properties, and three-term recurrence are classical; the studies [1, 2] concern this established family. Closure under Hadamard products and the symmetric-power construction are also

standard. The substantive assertion proved here is the matching lower bound for the explicit independent-parameter family, together with the complete nonzero finite singularity set and forced leading symbol. The proof is self-contained once elementary analytic Morse theory and the ordinary-point theorem for linear differential equations are available.

The wider methodological context is recurrence minimality. Straub and Zudilin [74] explicitly distinguish minimum telescoping order from minimum order among all rational-coefficient recurrences for Franel sums. The present lower bound addresses the latter notion for a different family. We do not claim that their Franel problem is newly settled by this section, nor that any existing central-trinomial congruence or real-rootedness question has been resolved here. A more exhaustive priority search for the exact product-width formula remains appropriate before journal submission. Multiplicatively dependent distinct parameters can cause critical-value collisions; the proof still supplies a lower bound equal to the number of distinct values, but does not assert a complete minimum-width formula on those collision loci.

I.3 Jacobi ensembles: exact differential complexity under symmetry breaking

The transfer in this section replaces the special Bessel equations of the tableau problem by arbitrary beta-integral factors. The main lower bound comes from the Fourier asymptotics of the *particular integral*; it does not assume that a differential system is irreducible. The existence of the unweighted Jacobi differential system is due to Forrester and Kumar. What is proved here is exact minimality, and an exact increase from $N+1$ to 2^N after a small perturbation of the weights in the linear statistic. These are mathematical deductions; historical priority for the resulting formulations is not asserted.

For $N \geq 1$, real $a_i, b_i > -1$, $c_i > 0$, and $\beta \geq 0$, put

$$Z_{\mathbf{a},\mathbf{b},\beta,\mathbf{c}}(t) = \int_{[0,1]^N} e^{t\sum_i c_i u_i} |\Delta(u)|^\beta \prod_{i=1}^N u_i^{a_i}(1-u_i)^{b_i}\, du, \qquad \Delta(u) = \prod_{i<j}(u_j - u_i). \tag{I.3.1}$$

The convention at $\beta = 0$ is $|\Delta|^0 = 1$. The integral is an entire function, since its weight is integrable and the integration domain is compact. Dividing it by $Z(0) > 0$ gives the moment generating function of the corresponding weighted linear statistic. Such a constant normalization does not change its minimal linear differential order over $\mathbb{C}(t)$.

I.3.1 An endpoint principle for a single integral

For $A \subseteq \{1, \dots, N\}$ and exponents $q_i > -1$, define

$$d_A(\mathbf{q}) = \sum_{i\in A}(q_i+1) + \frac{\beta}{2}|A|(|A|-1), \qquad \mathcal{L}_A(\mathbf{q},\mathbf{c}) = \int_{(0,\infty)^A} e^{-\sum_{i\in A} c_i x_i} |\Delta(x_A)|^\beta \prod_{i\in A} x_i^{q_i}\, dx_A.$$

For the empty set, set $d_\varnothing = 0$ and $\mathcal{L}_\varnothing = 1$. Every nonempty integral on the right is finite and strictly positive: the endpoint powers are integrable, the Vandermonde has no singularity for $\beta \geq 0$, and the exponential controls infinity.

Lemma I.3.1 (All endpoint frequencies occur)**.** *For a subset $S \subseteq \{1, \dots, N\}$, write*

$$\lambda_S = \sum_{i\in S} c_i, \qquad d_{S,-} = d_{S^c}(\mathbf{a}), \qquad d_{S,+} = d_S(\mathbf{b}), \qquad \gamma_S = d_{S,-} + d_{S,+}.$$

As $R \to +\infty$, the function in (I.3.1) has a complete oscillatory expansion

$$Z(iR) \sim \sum_{S\subseteq[N]} e^{i\lambda_S R} R^{-\gamma_S} \sum_{k\geq 0} A_{S,k} R^{-k}, \tag{I.3.2}$$

where

$$A_{S,0} = e^{\frac{\pi i}{2}(d_{S,-}-d_{S,+})}\mathcal{L}_{S^c}(\mathbf{a},\mathbf{c})\mathcal{L}_S(\mathbf{b},\mathbf{c}) \neq 0. \tag{I.3.3}$$

Complete means that truncating each series after $k = K-1$ leaves $\sum_S O(R^{-\gamma_S-K})$; expansions to any larger prescribed order are available. For every fixed $j \geq 0$ the same assertion holds for $Z^{(j)}(iR)$, with the leading coefficient at the S-th frequency equal to $\lambda_S^j A_{S,0}$. All the expansions can therefore be used in any fixed polynomial-coefficient differential expression in Z.

Proof. The corner indexed by S has $u_i = 1$ for $i \in S$ and $u_i = 0$ otherwise. Near it put $x_i = u_i$ for $i \notin S$ and $y_i = 1-u_i$ for $i \in S$. The phase is

$$\lambda_S + \sum_{i\notin S} c_i x_i - \sum_{i\in S} c_i y_i.$$

The weight is the product of the two homogeneous functions

$$H_-(x) = |\Delta(x_{S^c})|^\beta \prod_{i\notin S} x_i^{a_i}, \qquad H_+(y) = |\Delta(y_S)|^\beta \prod_{i\in S} y_i^{b_i},$$

and a real analytic function $B(x,y)$ with $B(0,0) = 1$. Indeed, the cross-cluster differences are $1-x_i-y_j$, which remain positive, and the remaining endpoint factors are powers of $1-x_i$ or $1-y_j$. The degrees of H_- and H_+, including their measures, are $d_{S,-}$ and $d_{S,+}$.

Here is an elementary conical version of Watson's endpoint argument. For a homogeneous, locally integrable function H on a positive orthant, of degree $d-r$ in r variables with $d > 0$, use the radial coordinate $v = \sum c_i x_i$. Integration over the compact simplex $\sum c_i x_i = 1$ gives

$$\int H(x) f\Big(\sum c_i x_i\Big)\, dx = C_H \int_0^\infty v^{d-1} f(v)\, dv, \qquad C_H = \frac{\int e^{-\sum c_i x_i} H(x)\, dx}{\Gamma(d)}.$$

With a smooth cutoff equal to one at zero, the one-dimensional endpoint expansion of the last integral, for $f(v) = e^{\pm iRv}$, starts with $\Gamma(d)e^{\pm\pi id/2}R^{-d}$. This follows, for example, by first inserting $e^{-\epsilon v}$, evaluating $\int_0^\infty e^{-(\epsilon\mp iR)v} v^{d-1} dv$, and integrating by parts outside zero. Taylor expansion of a smooth amplitude gives the expansion to every order with a remainder of the stated order. For two nonempty clusters, choose the corner cutoff as $\chi_-(v)\chi_+(w)$, where $v = \sum_{i\notin S} c_i x_i$ and $w = \sum_{i\in S} c_i y_i$. After integration over the two compact simplices, the remaining amplitude is jointly smooth in v, w; all its derivatives are dominated by the integrable angular weights. The same endpoint argument therefore applies to the two radial variables independently. Each monomial of $B(x,y)$ increases the total exponent by its nonnegative integer total degree. This proves the corner expansion and (I.3.3). An empty cluster simply contributes the factor one.

It remains to exclude contributions away from the corners, including points on diagonals where the weight need not be smooth. At any such point at least one cluster of equal coordinates lies strictly between zero and one. In a sufficiently small neighbourhood, translate all coordinates in that cluster together. The associated constant vector field leaves all within-cluster differences unchanged. Differences between distinct clusters and endpoint factors of the moving cluster are smooth and nonzero there. All remaining singular factors are independent of this translation. Its derivative of the phase is $\sum_{i \text{ in the cluster}} c_i > 0$. A smooth partition of unity and repeated integration by parts in this direction consequently give $O(R^{-M})$ for every M. There are no boundary terms in the moving direction: the cutoff is compactly supported in that neighbourhood, and the direction is tangent to the collision faces. All differentiated amplitudes are bounded by integrable endpoint and collision weights. Compactness gives a finite cover.

Finally, differentiation in t inserts the polynomial $(\sum c_i u_i)^j$. Its value at the corner is λ_S^j, and the same Taylor and integration-by-parts arguments apply to it. This establishes the derivative assertion, including the case $\lambda_S = 0$, in which a displayed leading coefficient can vanish. □

Lemma I.3.2 (Frequency lower bound)**.** *Suppose an entire function and all its fixed-order derivatives have complete expansions of the form above with M distinct real frequencies $\lambda_1, \dots, \lambda_M$, real power exponents, and nonzero leading amplitudes. Suppose the derivative leading amplitudes are multiplied by λ_j^k. Every nonzero rational-function-coefficient differential operator annihilating the function has order at least M. More precisely, if*

$$L = \sum_{k=0}^{r} p_k(t)\partial_t^k, \qquad D = \max_k \deg p_k, \qquad \chi_L(X) = \sum_k [t^D]p_k(t)X^k,$$

then $\prod_{j=1}^{M}(X - \lambda_j)$ divides $\chi_L(X)$.

Proof. Clear denominators. The top coefficient at frequency λ_j in $(Lf)(iR)$ is $i^D A_{j,0}\chi_L(\lambda_j)$ times $R^{D-\gamma_j}e^{i\lambda_j R}$. A complete finite-frequency power expansion has unique coefficients. For clarity, take its largest remaining real power, divide by that power, and average against $e^{-i\lambda_j R}$ on $[T, 2T]$. Terms of smaller power tend to zero, and distinct frequencies average to zero. The coefficient at the chosen frequency must therefore vanish. Repeating in decreasing powers is legitimate because only finitely many exponents occur above any fixed bound. Thus every formal coefficient of an identically zero expression vanishes. In particular $\chi_L(\lambda_j) = 0$ for all j. The polynomial χ_L is nonzero by the definition of D and has degree at most r, proving both conclusions. □

I.3.2 Exact order for the ordinary Jacobi trace

Write $Z_{N,a,b,\beta}$ for (I.3.1) when all $a_i = a$, all $b_i = b$, and all $c_i = 1$.

Theorem I.3.3 (Exact Jacobi trace order)**.** *For every $N \geq 1$, $a, b > -1$ and $\beta \geq 0$, the minimal linear differential order of $Z_{N,a,b,\beta}$ over $\mathbb{C}(t)$ is exactly $N + 1$. For any primitive polynomial-coefficient minimal operator $L = \sum_{k=0}^{N+1} p_k(t)\partial_t^k$, if $D = \max_k \deg p_k$, then*

$$\sum_{k=0}^{N+1} [t^D]p_k(t)X^k = C\prod_{j=0}^{N}(X - j), \qquad C \neq 0.$$

In particular $\deg p_{N+1} = D$.

Proof. For the upper bound, we recall the published Jacobi differential system with the parameter conventions made explicit. Set

$$H_p(x) = \binom{N}{p}^{-1} \int_{[0,1]^N} e^{-x\sum u_i} \prod u_i^a (1 - u_i)^b |\Delta(u)|^\beta e_p(1 - u_1, \dots, 1 - u_N)\, du.$$

Forrester–Kumar, Corollary 2.5, equations (2.19)–(2.20) of `arXiv:2011.00787v2`, gives

$$(N - p)xH_{p+1} = ((N - p)x + B_p)H_p + xH_p' - D_pH_{p-1},$$

where

$$B_p = p\left(a + b + 2 + \frac{\beta}{2}(2N - p - 1)\right), \qquad D_p = p\left(b + 1 + \frac{\beta}{2}(N - p)\right).$$

Its parameters there are $a = a$ and the parameter denoted by b in that system is $b + 1$ here. The same identity at $\beta = 0$ follows directly by integration by parts, or by continuity in β. These identities hold for all real endpoint parameters in the stated integrability range.

Let $\mathscr{D} = \partial_x$, $Q_0 = 1$, $Q_1 = 1 + \mathscr{D}/N$, and recursively, for $1 \leq p < N$, set

$$(N - p)Q_{p+1} = (\mathscr{D} + N - p + B_p/x)\, Q_p - (D_p/x)Q_{p-1}.$$

Then $H_p = Q_p H_0$, and Q_p has order p and leading coefficient $(N-p)!/N!$. The last equation of the system is

$$((\mathscr{D} + B_N/x)Q_N - (D_N/x)Q_{N-1})\, H_0 = 0.$$

Its order is $N+1$, with leading coefficient $1/N!$. Since $Z(t) = H_0(-t)$, this proves the upper bound.

For the lower bound, the corners with $|S| = j$ all have the same frequency j, the same exponent

$$\gamma_j = (N-j)(a+1) + j(b+1) + \frac{\beta}{2}((N-j)(N-j-1) + j(j-1)),$$

and the same nonzero leading amplitude in Lemma I.3.1. Their amplitudes add with the positive integer multiplier $\binom{N}{j}$; no cancellation occurs. Thus all $N+1$ distinct frequencies $0, \dots, N$ occur. Apply Lemma I.3.2. For a minimal operator the degree of χ_L is at most $N+1$, so its $N+1$ prescribed roots give the displayed exact polynomial. □

I.3.3 Weighted statistics and an exponential increase in order

Theorem I.3.4 (Exact anisotropic order at even Dyson index)**.** *Let $N \geq 1$, $m \in \mathbb{Z}_{\geq 0}$, $a_i, b_i > -1$ and $c_i > 0$. Assume that the 2^N subset sums $\lambda_S = \sum_{i \in S} c_i$ are pairwise distinct. Then $Z_{\mathbf{a},\mathbf{b},2m,\mathbf{c}}$ has minimal linear differential order exactly 2^N over $\mathbb{C}(t)$. For every polynomial-coefficient minimal operator, its top-degree coefficient polynomial satisfies*

$$\chi_L(X) = C \prod_{S \subseteq [N]} (X - \lambda_S), \qquad C \neq 0.$$

The assertion includes $m = 0$, that is, independent beta variables.

Proof. When $\beta = 2m$, the Vandermonde factor is the polynomial $\Delta(u)^{2m}$. Expand it as a finite linear combination of monomials $u_1^{\nu_1} \cdots u_N^{\nu_N}$. Put

$$f_i(t) = \int_0^1 e^{c_i t u} u^{a_i} (1-u)^{b_i}\, du.$$

Integration by parts of $e^{c_i t u} u^{a_i+1}(1-u)^{b_i+1}$, whose endpoint values vanish, gives the Kummer equation

$$t f_i'' + (a_i + b_i + 2 - c_i t) f_i' - c_i(a_i+1) f_i = 0.$$

Consequently every derivative of f_i is in the $\mathbb{C}(t)$-span of f_i, f_i'. The vector space spanned by the 2^N products

$$\prod_{i=1}^{N} f_i^{(\epsilon_i)}, \qquad \epsilon_i \in \{0,1\},$$

is closed under differentiation. The monomial expansion of Δ^{2m} expresses the integral as a linear combination of products $\prod_i c_i^{-\nu_i} f_i^{(\nu_i)}$, so the integral lies in this space. Its derivative span therefore has dimension at most 2^N. This proves an annihilator of order at most 2^N, without any assumption about the Galois groups of the factors.

By Lemma I.3.1, each of the 2^N corners has a nonzero leading amplitude. The subset-sum hypothesis keeps their frequencies distinct, so Lemma I.3.2 gives the matching lower bound. It also proves the top-degree polynomial identity, as in Theorem I.3.3. □

Corollary I.3.5 (An arbitrarily small perturbation changes exact order)**.** *Fix $N \geq 2$, $a, b > -1$ and $m \geq 0$. At equal weights $c_i = 1$ the order is $N+1$. For every $0 < \epsilon < 1$, set*

$$c_i(\epsilon) = 1 + \epsilon 2^{-i}, \qquad 1 \leq i \leq N.$$

The order at these perturbed weights is exactly 2^N. Thus weights arbitrarily close to equal weights change the exact differential order by $2^N - N - 1$.

Proof. For two subsets of different cardinality, the difference of their subset sums has a nonzero integer part and a perturbation of absolute value less than $\epsilon < 1$, so it is nonzero. For subsets of the same cardinality, uniqueness of finite binary sums shows that their perturbations are different. Apply the two theorems. Clearly $\max_i |c_i(\epsilon) - 1| \to 0$ as $\epsilon \to 0^+$. □

An explicit order-three to order-four example. For $N = 2$, $a = b = 0$ and $\beta = 2$, direct integration gives

$$Z_{(1,1)}(t) = \frac{2e^{2t} - (2t^2 + 4)e^t + 2}{t^4},$$

whereas weights $(1, 2)$ give

$$Z_{(1,2)}(t) = \frac{3e^{3t} - (2t^2 + 2t + 3)e^{2t} - (2t^2 - 2t + 3)e^t + 3}{4t^4}.$$

Both expressions have removable singularities at zero, with value $1/6$. The exact minimal orders are respectively three and four. These formulas follow by writing $(u - v)^2 = u^2 - 2uv + v^2$ and differentiating $(e^x - 1)/x$; thus they are exact elementary checks of both the tensor upper bound and the endpoint lower bound.

Remark I.3.6 (Scope and what has not been inferred). The upper bound for the unweighted trace is a consequence of the Forrester–Kumar system, not a new discovery of that system. The lower bounds above are for the specific transforms, not merely for ambient differential systems. No irreducibility, full differential Galois group, absence of apparent singularities, or arithmetic nature of special values follows from the present argument. In particular integer endpoint parameters can give elementary exponential expressions and reducible minimal operators; the exact-order conclusions remain valid. For general non-even β and unequal weights the endpoint argument still gives the same 2^N lower bound under the subset-sum hypothesis, but no matching upper bound is claimed here. At colliding subset sums, the unequal-parameter amplitudes can combine, so the anisotropic theorem is not extended silently across that locus.

Primary source. P. J. Forrester and S. Kumar, *Differential recurrences for the distribution of the trace of the β-Jacobi ensemble*, Physica D **434** (2022), 133220; `arXiv:2011.00787v2`, Corollary 2.5. `https://doi.org/10.1016/j.physd.2022.133220`.

I.4 Exact statistical network design with missing interactions

The inverse spectral results of the earlier manuscript have consequences for experimental design, electrical networks, and stochastic consensus. The general interpretation is classical: total effective resistance is $n \operatorname{tr} L^+$, and its optimization over conductances is a convex problem; see Ghosh–Boyd–Saberi, *SIAM Review* **50** (2008), 37–66, doi:10.1137/050645452. Symmetry reduction is also standard. The results below are explicit solutions for all inverse spectral powers on a missing-edge graph and on an arbitrary matching-defect graph. They are new deductions in this manuscript, without a historical-priority claim for these particular graph families.

Let L be a nonnegatively weighted graph Laplacian on n labeled vertices, with connected positive support. Write its positive eigenvalues as $\lambda_2, \dots, \lambda_n$, fix

$$T = \operatorname{tr} L = 2 \sum_e w_e > 0, \qquad J_p(L) = \sum_{i=2}^{n} \lambda_i^{-p} \quad (p > 0).$$

Disconnected positive supports have $J_p = +\infty$ by convention; this is the extended objective for estimation of all vertex contrasts. All optimizations below permit zero weights on allowed edges, but their optimizers will have strictly positive weights on every allowed edge.

Lemma I.4.1 (Strict convexity and symmetry reduction)**.** *On the connected Laplacian cone with fixed vertex set, J_p is strictly convex for every $p > 0$. So is $-\log\det'(L)$, where $\det'(L)$ is the product of the positive eigenvalues. A minimizer under a fixed trace and fixed allowed graph is unique and invariant under every automorphism of that graph.*

Proof. Restrict L to $\mathbf{1}^\perp$, obtaining a positive definite matrix A. For a strictly convex twice differentiable scalar function f, the second derivative of $\operatorname{tr} f(A)$ in a nonzero symmetric direction H, in an eigenbasis of A, is

$$\sum_i f''(\lambda_i)H_{ii}^2 + 2\sum_{i<j}\frac{f'(\lambda_i)-f'(\lambda_j)}{\lambda_i-\lambda_j}H_{ij}^2,$$

with the continuous interpretation at repeated eigenvalues. This is strictly positive for $f(x) = x^{-p}$ and $f(x) = -\log x$. The passage from weights to a Laplacian is injective, since its off-diagonal entries recover all the weights. The fixed-trace weight simplex is compact, and the objectives diverge on disconnected limits; thus a minimizer exists and is unique whenever a connected weighting exists. Averaging over the finite automorphism group preserves feasibility and cannot increase the objective. Uniqueness gives invariance. □

I.4.1 One missing edge: every inverse power and the exact design penalty

Theorem I.4.2 (All-power optimality with one forbidden interaction)**.** *Let $n \geq 3$, $p > 0$, and $T > 0$. Put*

$$c = \frac{n}{n-2}, \qquad k = n-3, \qquad \rho_p = \left(\frac{1+c}{1+c^{-p}}\right)^{1/(p+1)}, \qquad x = \frac{T}{1+c+k\rho_p}.$$

Among all connected noncomplete weighted graphs on n vertices with trace T, the minimum of J_p is

$$\boxed{T^{-p}\left[k + \left((1+c)^p(1+c^{-p})\right)^{1/(p+1)}\right]^{p+1}.} \tag{I.4.1}$$

Every minimizing graph is $K_n - e$. With the missing edge labeled 12, the unique minimizing weighting is

$$w_{1j} = w_{2j} = \alpha := \frac{x}{n-2} \quad (j \geq 3), \qquad w_{ij} = \beta := \frac{x}{n-2}\left(\rho_p - \frac{2}{n-2}\right) \quad (3 \leq i < j \leq n). \tag{I.4.2}$$

The second family is absent when $n = 3$. Its spectrum is

$$\{0, x, (\rho_p x)^{(n-3)}, cx\}.$$

For a fixed forbidden edge, the optimizer is unique even when all additional allowed edges may be assigned zero weight.

Proof. We reproduce the spectral obstruction needed for the lower bound. If $L_{12} = 0$, set $x_0 = \lambda_2$, $z_0 = \lambda_n$, $a = (x_0 + z_0)/2$, $d = (z_0 - x_0)/2$, and $H = I - \mathbf{1}\mathbf{1}^\mathsf{T}/n$. On $\mathbf{1}^\perp$, $\|L - aH\| \leq d$. For $u = He_1$ and $v = He_2$, the missing edge gives

$$a/n = u^\mathsf{T}(L - aH)v \leq d\|u\|\|v\| = d(n-1)/n.$$

Consequently $z_0 \geq cx_0$.

Minimize $\sum \lambda_i^{-p}$ over the larger convex spectral set

$$0 < \lambda_2 \leq \cdots \leq \lambda_n, \qquad \sum \lambda_i = T, \qquad c\lambda_2 - \lambda_n \leq 0.$$

The candidate in the statement has $1 < \rho_p < c$, so its intermediate eigenvalues lie strictly between its extreme eigenvalues. For the trace multiplier choose $\nu = p(\rho_p x)^{-p-1}$ and for the last inequality choose $\mu = p((\rho_p x)^{-p-1} - (cx)^{-p-1}) > 0$. The stationarity equations for the two endpoints reduce to

$$(1+c)(\rho_p x)^{-p-1} = x^{-p-1} + c(cx)^{-p-1},$$

which is exactly the defining formula for ρ_p. All middle-coordinate stationarity equations hold as well. The order constraints need no multiplier. Convexity proves optimality and strict convexity proves uniqueness of this spectral list. For $n = 3$ the two-coordinate problem gives $x = T/(1+c)$ directly; the same stated value follows.

To realize the list, put $m = n - 2$ and use the weighting (I.4.2). The endpoint difference has eigenvalue $m\alpha = x$, the vector constant on the middle vertices and opposite on the endpoints has eigenvalue $n\alpha = cx$, and middle zero-sum vectors have eigenvalue $2\alpha + m\beta = \rho_p x$. All allowed weights are positive: for $n \geq 4$, $2/(n-2) \leq 1 < \rho_p$. The trace and the desired objective therefore agree with the lower bound. Algebraic simplification gives (I.4.1).

For each fixed missing pair, Lemma I.4.1 gives uniqueness on the entire weight simplex supported on $K_n - e$. Since its unique optimizer makes every other edge positive, any global optimizer with an additional absent edge is impossible. This proves the support classification as well as uniqueness. □

Corollary I.4.3 (Exact resistance and uncertainty-volume costs)**.** *Under the same trace constraint, every connected noncomplete graph obeys*

$$\operatorname{tr} L^{+} \geq \frac{1}{T}\left(n - 3 + \sqrt{\frac{n}{n-2}} + \sqrt{\frac{n-2}{n}}\right)^{2}. \tag{I.4.3}$$

Equality has the unique weighting of Theorem I.4.2 with $\rho_1 = \sqrt{c}$. The maximum spanning-tree partition function is

$$\tau(L) := \frac{\det'(L)}{n} \leq \frac{T^{n-1}(n-2)}{(n-1)^{n+1}}. \tag{I.4.4}$$

Here equality holds uniquely, up to the missing-edge label, at

$$\alpha_D = \frac{T}{(n-1)^2}, \qquad \beta_D = \frac{T(n-3)}{(n-2)(n-1)^2},$$

with the core-edge formula omitted for $n = 3$. Relative to the complete-graph optimum, the loss in the maximum determinant is exactly the factor $1 - (n-1)^{-2}$.

Proof. The first assertion is the case $p = 1$. For the determinant, maximize $\log x + \log z + (n-3)\log t$ under the same spectral constraints. The strictly concave maximization has $z = cx$ and $t = (1+c)x/2$; the multiplier check is the logarithmic version of the preceding proof. Thus

$$x = \frac{T(n-2)}{(n-1)^2}, \quad t = \frac{T}{n-1}, \quad z = \frac{Tn}{(n-1)^2}.$$

The same explicit realization and strict convexity prove uniqueness. Multiply these eigenvalues and use the weighted matrix-tree theorem. The complete-graph maximum is $(T/(n-1))^{n-1}$ for $\det'(L)$, which gives the claimed exact ratio. □

For example, take $n = 4$, $T = 1$, and forbid 12. The resistance-optimal design has

$$x = \frac{1}{3+\sqrt{2}}, \qquad \alpha = \frac{1}{2(3+\sqrt{2})}, \qquad \beta = \frac{\sqrt{2}-1}{2(3+\sqrt{2})},$$

and $\operatorname{tr} L^{+} = (1+\sqrt{2}+1/\sqrt{2})^2$. The determinant-optimal design instead has $\alpha_D = 1/9$, $\beta_D = 1/18$, and $\tau = 2/243$. Thus two standard statistical criteria already prescribe different conductance allocations on this smallest nontrivial example.

I.4.2 Any number of disjoint missing interactions

Write M_r for a matching of size r and let $G = K_n - M_r$. The $2r$ vertices meeting the matching are called paired, and the other $s = n - 2r$ vertices are called unpaired. This graph is connected for the parameters used below.

Theorem I.4.4 (All-power optimizer for arbitrary matching defects)**.** *Let $r \geq 2$, $s \geq 1$, $n = 2r+s$, $p > 0$, and $q = p + 1$. There is a unique $t \in (1, r/(r-1))$ satisfying*

$$\frac{n\{r-(r-1)t\}}{s} = \left(\frac{n}{r+(n-r)t^{-q}}\right)^{1/q}. \tag{I.4.5}$$

Define

$$h = \frac{n\{r-(r-1)t\}}{s}, \qquad v = \left(\frac{2}{1+t^{-q}}\right)^{1/q}, \qquad B = r+(r-1)t+(s-1)v+h, \qquad d = T/B.$$

The unique minimizer of J_p over all weights on $K_n - M_r$ of trace T is the following three-orbit weighting:

$$\begin{aligned} a &= \tfrac{d}{2}(t-1) && \text{between distinct paired pairs,} \\ b &= dh/n && \text{between paired and unpaired vertices,} \\ c_0 &= \frac{d}{s}\left(v - \frac{2r}{n}h\right) && \text{between distinct unpaired vertices.} \end{aligned} \tag{I.4.6}$$

The last family is absent for $s = 1$. All present weights are positive. Its positive spectrum, with multiplicities, is

$$\{d^{(r)}, (td)^{(r-1)}, (vd)^{(s-1)}, hd\}, \tag{I.4.7}$$

so the minimum is explicitly

$$(B/T)^p\{r+(r-1)t^{-p}+(s-1)v^{-p}+h^{-p}\}. \tag{I.4.8}$$

The case $r = 1$ is Theorem I.4.2. If $s = 0$ and $n = 2r \geq 4$, the unique minimizer has the uniform edge weight $T/[n(n-2)]$ and positive spectrum

$$\{(T/n)^{(r)}, (T/(n-2))^{(r-1)}\}.$$

Proof. The left side of (I.4.5) is strictly decreasing, from $n/s > 1$ to 0, and the right side is strictly increasing and positive, with value 1 at $t = 1$. This proves existence and uniqueness of the root. In particular $a, b > 0$. Put $\theta = 2r/n \in (0, 1)$. The root equation gives

$$\left(\frac{v}{h}\right)^q = \frac{2\{r+(n-r)t^{-q}\}}{n(1+t^{-q})} > \theta.$$

Since $q > 1$, this implies $v/h > \theta^{1/q} > \theta$, and hence $c_0 > 0$ whenever that edge family is present.

By Lemma I.4.1, it suffices to optimize among the three automorphism orbits just described. For orbit weights (a, b, c_0), decompose the vertex space into pair differences, pairwise constant vectors whose paired-coordinate sum vanishes, unpaired zero-sum vectors, the paired/unpaired quotient mode, and the global constant mode. The positive eigenvalues are respectively

$$D = 2(r-1)a+sb\ (r \text{ times}), \quad U = 2ra+sb\ (r-1 \text{ times}), \quad V = 2rb+sc_0\ (s-1 \text{ times}), \quad W = nb.$$

These subspaces are orthogonal and their dimensions sum to n after including the constant mode. The trace constraint is

$$4r(r-1)a + 4rsb + s(s-1)c_0 = T. \tag{I.4.9}$$

Substitution of (I.4.6) gives $(D, U, V, W) = d(1, t, v, h)$ and the trace $dB = T$.

Divide the stationarity multiplier for J_p by p and call it μ. The a-equation is

$$\mu = \tfrac{1}{2}(D^{-q} + U^{-q}).$$

When $s \geq 2$, the c_0-equation is $\mu = V^{-q}$. After these substitutions, the b-equation is equivalent to

$$nW^{-q} = rD^{-q} + (n - r)U^{-q}. \tag{I.4.10}$$

For $s = 1$ there is no c_0-equation and its coefficient in the b-equation is zero; the same displayed relation follows directly. The definitions of v and h give exactly these stationarity relations. Thus the stated positive point is a constrained stationary point of a strictly convex objective on the orbit-weight simplex. It is its unique minimum, and symmetry reduction makes it the unique minimum on the full weight simplex. The spectral value formula follows immediately.

For $s = 0$, the graph is edge-transitive, so uniqueness and symmetry force all weights to agree. There are $n(n-2)/2$ edges. The eigenvalues follow from $L = w(nH - \sum_{j=1}^{r}(e_{2j-1} - e_{2j})(e_{2j-1} - e_{2j})^{\mathsf{T}})$. □

Corollary I.4.5 (Quartic resistance equation and quadratic determinant equation)**.** *For $r \geq 2$ and $s = n - 2r \geq 1$, the resistance-optimal parameter is the unique root in $(1, r/(r-1))$ of*

$$s^2t^2 = n(rt^2 + n - r)\{r - (r-1)t\}^2. \tag{I.4.11}$$

For maximizing $\det'(L)$*, the same weighting formulas hold with*

$$q = 1, \qquad t = \frac{1 - s + \sqrt{(n-1)^2 - 4s}}{2(r-1)}, \qquad v = \frac{2t}{1+t}, \qquad h = \frac{nt}{rt + n - r}.$$

The determinant maximum equals $d^{n-1}t^{r-1}v^{s-1}h$, with $d = T/B$ as above.

Proof. For $p = 1$, square (I.4.5) and multiply by the positive denominators. Both sides of the unsquared equation are positive on the specified interval, so the quartic introduces no extra root there. For the determinant, apply the same convexity and stationarity proof to $-\log\det'(L)$. Its derivative replaces the power $q = p + 1$ by $q = 1$. The root equation simplifies to

$$(r-1)t^2 + (s-1)t - (n-r) = 0.$$

Its unique positive root is the formula above. Positivity follows from the preceding proof: at $q = 1$ the strict inequality $v/h > \theta$ remains valid. The spectrum gives the determinant. □

For an explicit non-edge-transitive example take $K_6 - M_2$, $T = 1$. For resistance minimization, t is the unique root in $(1, 2)$ of

$$t^2 = 3(t^2 + 2)(2 - t)^2,$$

and the weights on the paired–paired, paired–unpaired, and unpaired–unpaired edge orbits are approximately

$$(a, b, c_0) = (0.04717362924,\ 0.03545488293,\ 0.02766641956).$$

The exact value is given by the unique-root formulas, not by these rounded numbers. For determinant maximization on the same graph, $t = (\sqrt{17} - 1)/2$. This supplies exact formulas for a family whose full inverse-eigenvalue region need not be known.

I.4.3 Statistical and stochastic meanings of the optimized quantities

Consider noisy pairwise measurements of a vertex signal modulo its additive constant, with independent Gaussian errors and edge precision w_{ij}. The Fisher information on $\mathbf{1}^{\perp}$ is L; the efficient estimator's covariance is L^{+}. Consequently J_1 is total contrast-estimation variance, J_p is the sum of the pth powers of the covariance eigenvalues, and maximizing $\det'(L)$ minimizes the volume of the confidence ellipsoid. The preceding theorems give the exact designs under a total precision budget and either one forbidden comparison or an arbitrary set of disjoint forbidden comparisons.

For the projected Ornstein–Uhlenbeck consensus dynamics

$$dX_t = -LX_t\,dt + \sigma H\,dB_t, \qquad X_t \in \mathbf{1}^{\perp}, \qquad H = I - \mathbf{1}\mathbf{1}^{\mathsf{T}}/n,$$

the stationary covariance is $\Sigma = (\sigma^2/2)L^{+}$, since $L\Sigma + \Sigma L = \sigma^2 H$. Thus the exact minimum stationary mean-square disagreement is $(\sigma^2/2)\min J_1$, and every higher covariance power sum is obtained by multiplying (I.4.1) or (I.4.8) by $(\sigma^2/2)^p$. Finally, summing $(e_i - e_j)^{\mathsf{T}}L^{+}(e_i - e_j)$ over $i < j$ gives nJ_1, the total effective resistance. These interpretations use standard identities; the explicit constrained optimizers and their strict-support classification are the content of the preceding results.

I.5 Arithmetic clocks and action quantization in algebraic Hamiltonian dynamics

The previous manuscripts already established transcendence of periods of regular compact algebraic trajectories. The additional ingredient here is *global holomorphy of the time differential*, including at complex points at infinity. It upgrades transcendence of a period to separation from all algebraic linear combinations of logarithms, then gives a simultaneous arithmetic statement for position and momentum at prescribed times. A superelliptic extension treats nonquadratic kinetic energies and yields an exact algebraic/transcendental dichotomy for action-quantized energy levels. These are transfers of the classical analytic subgroup theorem, not claims of a new transcendence theorem or of a solution of an open quantum spectral problem.

Throughout, arithmetic statements refer to the displayed dimensionless coordinates and coefficients. Rescalings by nonzero algebraic constants preserve them; arbitrary real changes of units need not do so. Put

$$\mathcal{L}_{\mathrm{alg}} = \overline{\mathbb{Q}} + \mathrm{span}_{\overline{\mathbb{Q}}}\{\lambda \in \mathbb{C} : e^{\lambda} \in \overline{\mathbb{Q}}^{\times}\}, \qquad \mathcal{C}_{\mathrm{alg}} = \mathcal{L}_{\mathrm{alg}} \cap \mathbb{R}.$$

All logarithm branches are allowed. This is a vector space, not an assertion about closure under multiplication. It contains every real algebraic number, π, and the real logarithm of every positive algebraic number. We call its real elements the prescribed arithmetic clock times.

I.5.1 The separation input and a general clock principle

Lemma I.5.1 (First-kind integrals and logarithmic separation)**.** *Let C_j be smooth projective curves over $\overline{\mathbb{Q}}$, η_j holomorphic differentials defined over $\overline{\mathbb{Q}}$, and δ_j paths with algebraic endpoints; closed paths are allowed without a restriction on their basepoint. For algebraic coefficients c_j, set*

$$S = \sum_{j=1}^{k} c_j \int_{\delta_j} \eta_j.$$

Then $S \in \mathcal{L}_{\mathrm{alg}}$ implies $S = 0$. In particular, if $S \neq 0$, both S and S/π are transcendental.

Proof. Use Abel–Jacobi maps, based at algebraic points, to place the integrals on the product A of the Jacobians. There are an invariant algebraic differential η on A and a logarithm $v \in \mathrm{Lie}(A)_{\mathbb{C}}$ such that $\exp_A(v) \in A(\overline{\mathbb{Q}})$ and $S = \eta(v)$. The chosen paths determine the logarithm, including its periods. For a closed path based at a nonalgebraic point, prepend a connector from any algebraic point and append its inverse. The added integrals cancel, and the resulting closed path has algebraic basepoint. If $S \in \mathcal{L}_{\mathrm{alg}}$, write

$$\eta(v) - c_0 - \sum_{j=1}^{s} c_j \lambda_j = 0, \quad c_j \in \overline{\mathbb{Q}}, \quad e^{\lambda_j} \in \overline{\mathbb{Q}}^{\times}.$$

Apply the refined analytic subgroup theorem in [3, Theorem 6.2] to

$$G = A \times \mathbb{G}_a \times \mathbb{G}_m^s, \qquad u = (v, 1, \lambda_1, \dots, \lambda_s).$$

Its exponential is algebraic. The theorem supplies a connected algebraic subgroup H with $u \in \mathrm{Lie}(H)_{\mathbb{C}}$ on which the displayed linear form vanishes identically.

A connected subgroup of a product of an abelian variety and a connected commutative affine group splits between these two types. Indeed, the algebraic-group Goursat argument reduces any mixing to a common connected quotient of an abelian variety and an affine group; such a quotient is both proper and affine and hence trivial. Thus $H = B \times H_0$ for an abelian subvariety $B \subseteq A$ and an affine subgroup H_0. Restriction of the linear form to $\mathrm{Lie}(B)$ gives $\eta|_{\mathrm{Lie}(B)} = 0$. Since v lies in this Lie algebra, $S = 0$. Finally, $\overline{\mathbb{Q}}$ and $\overline{\mathbb{Q}}\pi$ are contained in $\mathcal{L}_{\mathrm{alg}}$. □

Proposition I.5.2 (Arithmetic clock exclusion)**.** *Let C be a smooth projective curve over the real algebraic numbers, and let η be an algebraic holomorphic differential. Suppose a real trajectory $P(t)$, defined for t in an interval, satisfies*

$$\eta(\dot{P}(t)) = 1.$$

If $P(t_1)$ and $P(t_2)$ are algebraic and $t_1 \neq t_2$, then

$$t_2 - t_1 \notin \mathcal{C}_{\mathrm{alg}}.$$

Consequently the times at which the trajectory is algebraic meet every coset of $\mathcal{C}_{\mathrm{alg}}$ in at most one point. Every positive return time T satisfies $T \notin \mathcal{C}_{\mathrm{alg}}$ and T/π is transcendental.

Proof. Integrate the time form along the actual oriented trajectory segment:

$$\int_{P|_{[t_1,t_2]}} \eta = t_2 - t_1.$$

This is a nonzero first-kind integral with algebraic endpoints when the endpoints are algebraic, so Lemma I.5.1 applies. For a return time the segment is a closed path; a closed period is covered by the same lemma, without any arithmetic restriction on the chosen starting point. Its integral is the nonzero real number T. □

I.5.2 Polynomial potentials: a global condition and its exact role

Consider the classical Hamiltonian system

$$H(x,p) = \tfrac{1}{2}p^2 + V(x), \qquad \dot{x} = p, \qquad \dot{p} = -V'(x), \tag{I.5.1}$$

with $V \in (\overline{\mathbb{Q}} \cap \mathbb{R})[x]$. An energy E is called algebraically regular if $E - V(x)$ is squarefree over $\mathbb{C}$. This is stronger than absence of real critical points on one selected orbit.

Theorem I.5.3 (Position and momentum at arithmetic clock times)**.** *Let* $\deg V = d \geq 3$ *and let* $E \in \overline{\mathbb{Q}} \cap \mathbb{R}$ *be algebraically regular. Suppose* (I.5.1) *has a nonconstant bounded real orbit of energy* E *and an algebraic initial phase point* $(x(0), p(0))$*. Then for every* $t \in \mathcal{C}_{\mathrm{alg}} \setminus \{0\}$*,*

$$x(t) \notin \overline{\mathbb{Q}}, \qquad p(t) \notin \overline{\mathbb{Q}}.$$

The period T *and angular frequency* $\Omega = 2\pi/T$ *satisfy*

$$T \notin \mathcal{C}_{\mathrm{alg}}, \qquad T/\pi \notin \overline{\mathbb{Q}}, \qquad \Omega \notin \overline{\mathbb{Q}}.$$

No such orbit is rationally resonant with a harmonic oscillator of positive algebraic angular frequency.

Proof. Let C_E be the smooth projective model of $p^2 = 2(E - V(x))$. Squarefreeness makes the affine curve smooth. The time differential is $\eta = dx/p$. It is regular at a finite point where $p \neq 0$. At a simple root x_0 of $E - V$, use p as local parameter: $x - x_0$ has order two, so dx/p is again regular.

We check infinity, since finite regularity is insufficient. If $d = 2g + 1 \geq 3$, the unique point above infinity has $\mathrm{ord}(x) = -2$ and $\mathrm{ord}(p) = -(2g + 1)$. Thus

$$\mathrm{ord}(dx/p) = 2g - 2 \geq 0.$$

If $d = 2g + 2 \geq 4$, the two points above infinity have $\mathrm{ord}(x) = -1$ and $\mathrm{ord}(p) = -(g + 1)$, so $\mathrm{ord}(dx/p) = g - 1 \geq 0$ at both. Therefore η is holomorphic on all of C_E.

The real orbit is compact, and its vector field never vanishes: a zero would give $p = 0$ and $V'(x) = 0$ at energy E, contrary to squarefreeness. Hence it is periodic and exists for all real times. The energy identity extends $\eta(\dot{P}) = 1$ through turning points. Proposition I.5.2 rules out an algebraic phase point at any nonzero prescribed clock time.

On this energy curve either coordinate is algebraic if and only if both are algebraic. In one direction this follows by taking the square root in $p^2 = 2(E - V(x))$; in the other, an algebraic p makes x a root of the nonconstant polynomial $V(X) - E + p^2/2$ over $\overline{\mathbb{Q}}$. Thus both displayed coordinates are transcendental. The period assertion follows from the same proposition. If Ω were algebraic, $T/\pi = 2/\Omega$ would be algebraic. Finally, a rational relation between T and the harmonic period $2\pi/\omega$, for algebraic $\omega > 0$, would have the same forbidden consequence. □

For example, consider $H = p^2/2 + x^4$, with $x(0) = 0$, $p(0) = \sqrt{2}$. The energy is 1 and $1 - x^4$ is squarefree. Position and momentum are therefore both transcendental at each of the exact times 1, π, and $\log 2$. This is an arithmetic assertion about those exact values; numerical orbit sampling is not its proof.

Corollary I.5.4 (The harmonic exception)**.** *Among algebraic polynomial potentials admitting a bounded nonconstant orbit at an algebraically regular algebraic energy, the period is an algebraic multiple of* π *if and only if the potential has degree two.*

Proof. Degrees zero and one admit no bounded nonconstant orbit. A quadratic potential admitting one has the form $V(x) = a(x - b)^2 + c$ with $a > 0$ and algebraic a, b, c; its period is $2\pi/\sqrt{2a}$. Degrees at least three are excluded by Theorem I.5.3. □

Example I.5.5 (A real regular oval with singular complex energy curve)**.** The squarefreeness assumption cannot be replaced by regularity of the chosen real oval. Take

$$V(x) = -\tfrac{1}{2}(1 - x^2)(1 + x^2)^2, \qquad E = 0.$$

The orbit above $-1 \le x \le 1$ has $p = \pm(1+x^2)\sqrt{1-x^2}$ and is a smooth compact real oval with a nonvanishing vector field. Nevertheless

$$T = 2\int_{-1}^{1} \frac{dx}{(1+x^2)\sqrt{1-x^2}} = \sqrt{2}\,\pi.$$

For the last equality, put $x = \sin\theta$ and then $u = \tan\theta$; the resulting integral is $2\int_{-\infty}^{\infty} du/(1+2u^2)$. The polynomial $2(E-V)$ has repeated roots at $x = \pm i$. On the normalization $u^2 = 1 - x^2$, its time differential is $dx/((1+x^2)u)$, which has complex poles. The loss of global holomorphy explains the algebraic multiple of π without contradicting the theorem. In particular, starting from $(0,1)$ returns to the algebraic initial point at an arithmetic clock time.

I.5.3 Superelliptic kinetic energies and an action-level dichotomy

Let $a, b > 0$ be real algebraic numbers, and let r, m be positive integers. Consider

$$H_{r,m}(x,p) = ap^{2r} + bx^{2m}, \qquad E > 0. \tag{I.5.2}$$

The real energy curve is a smooth compact oval. Its Hamiltonian vector field is

$$(2rap^{2r-1}, -2mbx^{2m-1}),$$

and never vanishes on the oval. Write $T_{r,m}(E)$ for its period and

$$\mathcal{A}_{r,m}(E) = \oint p\,dx > 0$$

for the action, with the orientation of the Hamiltonian flow.

Lemma I.5.6 (Global time form for the superelliptic family)**.** *For algebraic $E > 0$, the time differential on the smooth projective energy curve is*

$$\eta = \frac{dx}{2rap^{2r-1}}.$$

It is holomorphic everywhere if $(r,m) \neq (1,1)$. For $(r,m) = (1,1)$ it has simple poles at infinity.

Proof. Set $N = 2r$, $D = 2m$, and $g = \gcd(N, D)$. The curve $p^N = (E - bx^D)/a$ is irreducible because its right side has simple zeros and hence is not a proper power in $\overline{\mathbb{Q}}(x)$. At a finite zero of p, p is a local parameter and $x - x_0$ has order N. Accordingly dx/p^{N-1} has order zero. There are no other finite poles.

At each of the g points above infinity on the normalization,

$$\operatorname{ord}(x) = -N/g, \qquad \operatorname{ord}(p) = -D/g.$$

These orders follow by balancing p^N and x^D, or from the local normalization of $p^N x^{-D} = -b/a + O(x^{-D})$. Thus

$$\operatorname{ord}(\eta) = \frac{D(N-1) - N}{g} - 1 = \frac{2m(2r-1) - 2r}{\gcd(2r, 2m)} - 1. \tag{I.5.3}$$

If $r = 1$ and $m \ge 2$, the quotient before subtraction is $m - 1 \ge 1$; the case $m = 1, r \ge 2$ is symmetric and gives $r - 1 \ge 1$. If $r, m \ge 2$, then $2m(2r-1) - 2r \ge 2\min(r,m) \ge g$. The order is therefore nonnegative in every nonharmonic case. For $r = m = 1$ the numerator is zero, giving order -1. □

Theorem I.5.7 (Arithmetic action quantization)**.** *Fix algebraic $a, b, \hbar, \nu > 0$ and positive integers r, m. There is a unique positive solution E_ν of*

$$\mathcal{A}_{r,m}(E_\nu) = 2\pi\hbar\nu. \tag{I.5.4}$$

Its arithmetic type is completely determined by

$$E_\nu \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad (r, m) = (1, 1).$$

In the harmonic case $E_\nu = 2\hbar\sqrt{ab}\,\nu$. In every other case E_ν is transcendental. Moreover, at every algebraic $E > 0$ in a nonharmonic case,

$$T_{r,m}(E) \notin \mathcal{C}_{\mathrm{alg}}, \qquad \mathcal{A}_{r,m}(E)/\pi \notin \overline{\mathbb{Q}}.$$

An algebraic initial point on that energy oval has transcendental position and momentum at every nonzero time in $\mathcal{C}_{\mathrm{alg}}$.

Proof. The clock statements follow from Lemma I.5.6 and Proposition I.5.2. Algebraicity of either coordinate implies that of the other by the energy equation.

The positive action is

$$\mathcal{A}_{r,m}(E) = 4a^{-1/(2r)} \int_0^{(E/b)^{1/(2m)}} (E - bx^{2m})^{1/(2r)}\, dx.$$

Rescaling $x = (E/b)^{1/(2m)}u$ gives

$$\mathcal{A}_{r,m}(E) = C_{r,m,a,b}E^\sigma, \quad \sigma = \frac{1}{2r} + \frac{1}{2m}, \quad C_{r,m,a,b} = 4a^{-1/(2r)}b^{-1/(2m)} \int_0^1 (1 - u^{2m})^{1/(2r)}\, du > 0. \tag{I.5.5}$$

Consequently the action is continuous and strictly increasing from zero to infinity, proving existence and uniqueness in (I.5.4). Differentiation under the integral gives $\mathcal{A}'_{r,m}(E) = T_{r,m}(E)$. This can be justified after the same fixed-interval substitution; the differentiated turning-point singularity has exponent $1/(2r) - 1 > -1$ and is integrable. It follows that

$$\mathcal{A}_{r,m}(E) = \frac{E}{\sigma} T_{r,m}(E). \tag{I.5.6}$$

For nonharmonic (r, m) and algebraic $E > 0$, Lemma I.5.1 makes $T_{r,m}(E)/\pi$ transcendental. Multiplication by the nonzero algebraic number E/σ proves the asserted action statement. If E_ν in (I.5.4) were algebraic, the same statement would contradict $\mathcal{A}_{r,m}(E_\nu)/\pi = 2\hbar\nu$.

For $r = m = 1$, the energy oval is an ellipse with semiaxes $\sqrt{E/b}$ and $\sqrt{E/a}$. Its action is its enclosed phase-space area, $\pi E/\sqrt{ab}$, giving the stated algebraic solution. □

Remark I.5.8 (What is, and is not, quantized here). For the usual quadratic kinetic energy $r = 1$, the choice $\nu = n + 1/2$, $n \geq 0$, is the standard leading Bohr–Sommerfeld rule for a smooth two-turning-point well. Thus all these levels for even power potentials of degree at least four are transcendental when $a, b, \hbar$ are algebraic. Leading Bohr–Sommerfeld levels and exact Schrödinger eigenvalues are different objects; this distinction is explicit in [4]. The theorem makes no arithmetic claim about the exact eigenvalues. For $r > 1$, it classifies the stated action equation and does not assign a universal Maslov correction to every higher-order quantum operator.

Remark I.5.9 (No independence claim for the resulting levels). For two positive algebraic labels μ, ν, equation (I.5.5) gives

$$\frac{E_\nu}{E_\mu} = \left(\frac{\nu}{\mu}\right)^{2rm/(r+m)} \in \overline{\mathbb{Q}}.$$

The infinitely many transcendental levels therefore belong to one algebraic proportionality class. Their individual transcendence is not algebraic independence. No analogous action conclusion for an arbitrary mixed polynomial potential is inferred here: its action differential need not be holomorphic, and the homogeneous identity (I.5.6) is essential to this argument.

I.6 Discount resonance and arithmetic of CIR stopping boundaries

The transfer in this section changes both the stochastic model and its parameter geometry. The dimension and payoff exponent of a finite-horizon Bessel bridge no longer determine the same Kummer parameter. Instead an independent discount rate determines the upper parameter, while a fixed cost translates the payoff. This produces an arithmetic resonance in the discount-to-mean-reversion ratio. The case of linear payoff is the published CIR liquidation problem of Leung, Li, and Wang [5, Theorem 3.1]; their optimality theorem is credited, not claimed anew. We prove the verification for all positive power payoffs and then establish arithmetic conclusions. We do not claim that the arithmetic question was posed as an open problem in that source, or that historical priority has been established.

Let Y be the nonnegative CIR diffusion

$$dY_t = \kappa(\theta - Y_t)\,dt + \sigma\sqrt{Y_t}\,dW_t, \qquad \kappa, \theta, \sigma > 0. \tag{I.6.1}$$

We use its usual nonexplosive solution, including the instantaneously reflecting boundary when $2\kappa\theta < \sigma^2$ and the entrance interpretation when started at zero in the other cases. Fix $r, m > 0$ and $K \geq 0$. Define

$$V(y) = \sup_{\tau} \mathbb{E}_y\big[e^{-r\tau}(Y_\tau - K)_+^m\big], \qquad y \geq 0, \tag{I.6.2}$$

where infinite stopping times are permitted and have payoff zero. Put

$$s = \frac{2\kappa}{\sigma^2}, \qquad a = \frac{r}{\kappa}, \qquad b = \frac{2\kappa\theta}{\sigma^2}, \qquad k = sK, \qquad f(z) = M(a; b; z). \tag{I.6.3}$$

The generator of $Z = sY$, divided by κ, is $z\partial_{zz} + (b - z)\partial_z$. Thus

$$zf'' + (b - z)f' - af = 0. \tag{I.6.4}$$

All Taylor coefficients of f, and all its derivatives on the nonnegative real axis, are strictly positive.

Theorem I.6.1 (The global stopping rule, including concave powers)**.** *For every set of real parameters specified above, there is a unique $z_* > k$ satisfying*

$$(z_* - k)f'(z_*) = mf(z_*). \tag{I.6.5}$$

With $y_* = z_*/s$, the value and an optimal stopping time are

$$V(y) = \begin{cases} s^{-m}(z_* - k)^m \dfrac{f(sy)}{f(z_*)}, & 0 \leq y < y_*, \\ (y - K)^m, & y \geq y_*, \end{cases} \qquad \tau_* = \inf\{t \geq 0 : Y_t \geq y_*\}. \tag{I.6.6}$$

The continuation inequality is strict at every $y < y_*$. No Feller inaccessibility condition on zero is needed.

Proof. For $w > 0$ write $f(k + w) = \sum_{j \geq 0} c_j w^j$, where $c_j = f^{(j)}(k)/j! > 0$. Its tilted coefficients $c_j w^j / f(k + w)$ form a probability distribution. Their mean is

$$H(w) = \frac{wf'(k + w)}{f(k + w)}, \qquad wH'(w) = \mathrm{Var}_w(j) > 0.$$

The mean tends to zero at zero. It tends to infinity at infinity: for any fixed J, the mass on $0 \leq j \leq J$ tends to zero by comparison with the positive term of degree $J + 1$. Hence there

is exactly one $w_* > 0$ with $H(w_*) = m$, giving $z_* = k + w_*$. Moreover $w^m/f(k+w)$ increases strictly up to w_* and decreases strictly thereafter. Consequently the proposed value majorizes the payoff strictly below the boundary. Value matching and continuous first derivative follow from the contact equation.

It remains to verify superharmonicity in the whole stopping region; this is needed in addition to locating a maximum of a ratio. For $z = k + w > k$ and $g(z) = w^m$,

$$\big[z\partial_{zz} + (b-z)\partial_z - a\big]g(z) = w^{m-2}R(w), \tag{I.6.7}$$

where

$$R(w) = m(m-1)k + m(m+b-1-k)w - (m+a)w^2.$$

At the contact equation, direct use of (I.6.4) gives

$$R(w_*) = -z_* w_* H'(w_*) < 0. \tag{I.6.8}$$

If $m \geq 1$, the concave quadratic R has nonnegative constant term, so it is negative after its possible positive root; (I.6.8) then gives $R(w) < 0$ for $w \geq w_*$. If $k = 0$, the same conclusion follows immediately by factoring out w. Suppose $0 < m < 1$ and $k > 0$. If R is never positive for $w > 0$, nothing remains. Otherwise it has two positive roots $w_- < w_+$, and

$$0 < w_- < \frac{m(m+b-1-k)}{2(m+a)} < \frac{m(b-k)}{m+a} < b - k.$$

At $z = k + w_- < b$, positivity of f'' and (I.6.4) imply $f'/f < a/(b-z)$. The preceding bound on w_- therefore gives

$$H(w_-) < \frac{aw_-}{b-k-w_-} < m.$$

Thus $w_* > w_-$. In view of (I.6.8), the contact cannot lie in $[w_-, w_+]$, and hence $w_* > w_+$. This proves the required sign everywhere above the boundary, also for concave powers.

The proposed function is nonnegative, continuously differentiable, piecewise twice continuously differentiable, and analytic near zero. The generalized Itô formula has no local-time contribution at the boundary because of smooth fit. Applied after localization, it shows that $e^{-rt}V(Y_t)$ is a nonnegative local supermartingale, hence a supermartingale. Optional sampling and Fatou's lemma give $V(y) \geq \mathbb{E}_y[e^{-r\tau}(Y_\tau - K)_+^m]$ for every stopping time, with the stipulated zero payoff at infinity. This argument also proves finiteness of the supremum.

For $y < y_*$ the stopped process $e^{-r(t\wedge\tau_*)}V(Y_{t\wedge\tau_*})$ is a martingale: its drift is zero and its state remains in the compact interval $[0, y_*]$, where the function and its derivative are bounded. As $t \to \infty$, the term on $\{\tau_* > t\}$ is bounded by $e^{-rt}\max_{[0,y_*]} V$ and tends to zero. The term on $\{\tau_* \leq t\}$ tends, boundedly, to the discounted boundary payoff. Equality in the value follows. For $y \geq y_*$ stop at zero. Analyticity at zero makes the same Itô calculation valid for the usual CIR solution whether zero is attainable or not. □

Theorem I.6.2 (A discount-resonance arithmetic phase diagram)**.** *Assume $a, b, m \in \mathbb{Q}_{>0}$, $s \in \overline{\mathbb{Q}}_{>0}$, and $K \in \overline{\mathbb{Q}}_{\geq 0}$. Then*

$$y_* \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad a - b \in \mathbb{Z}_{\geq 0}. \tag{I.6.9}$$

Outside this set the stopping boundary is transcendental, for every algebraic fixed cost $K \geq 0$ and every positive rational payoff power. On the resonance $a - b = N \in \mathbb{Z}_{\geq 0}$, put $P_N(z) = M(-N; b; -z)$. The boundary is the unique root $z_* > k$ of

$$(z-k)\{P_N'(z) + P_N(z)\} - mP_N(z) = 0, \qquad y_* = z_*/s. \tag{I.6.10}$$

In this resonant case, for every algebraic starting state $y \geq 0$,

$$V(y) \text{ is transcendental} \quad \Longleftrightarrow \quad y < y_*; \qquad V(y) \text{ is algebraic} \quad \Longleftrightarrow \quad y \geq y_*. \tag{I.6.11}$$

Proof. For rational positive a, b, f and f' are E-functions. The usual rational Pochhammer-ratio denominator estimate gives the arithmetic coefficient bounds, and (I.6.4) supplies their two-dimensional differential system, whose only finite singularity is zero. We recall precisely why its reducible special solutions are the only exceptions. If the logarithmic derivative of the entire exponential-type function f is rational, removing its finitely many zero factors gives $f = e^{\lambda z}P(z)$ with P a polynomial. Substitution into (I.6.4), at the two highest degrees, forces either $\lambda = 0$, $a = -\deg P$, or $\lambda = 1$, $a - b = \deg P$. The first is impossible because $a > 0$. Conversely Kummer's transformation gives $f = e^z P_N(z)$ when $a - b = N \geq 0$.

If $a - b$ is not a nonnegative integer, f, f' are linearly independent over $\overline{\mathbb{Q}}(z)$. Beukers' ordinary-point specialization theorem [68, Corollary 1.4] makes $f(\xi), f'(\xi)$ linearly independent over $\overline{\mathbb{Q}}$ at every nonzero algebraic ξ. An algebraic $z_* > k$ would give the nontrivial algebraic linear relation (I.6.5), a contradiction. Since s is nonzero algebraic, y_* is then transcendental.

On resonance the contact equation is (I.6.10), a nonzero polynomial of degree $N + 1$ with algebraic coefficients. Its specified root is algebraic. For an algebraic $y < y_*$, formula (I.6.6) becomes

$$V(y) = s^{-m}(z_* - k)^m \frac{P_N(sy)}{P_N(z_*)} e^{sy - z_*}.$$

The prefactor is nonzero algebraic: P_N has positive coefficients and m is rational. The exponent is a nonzero algebraic number. Hermite–Lindemann proves transcendence. If $y \geq y_*$, the value $(y - K)^m$ is algebraic, including the boundary case. □

Remark I.6.3 (The OU interpretation). If X is a d-dimensional Ornstein–Uhlenbeck process satisfying $dX_t = -\lambda X_t\, dt + \eta\, dW_t$, then $Y_t = \|X_t\|^2$ has (I.6.1) with $\kappa = 2\lambda$, $\theta = d\eta^2/(2\lambda)$, and $\sigma = 2\eta$. Consequently $a = r/(2\lambda)$, $b = d/2$, and $s = \lambda/\eta^2$. For rational $r/\lambda > 0$, integer $d \geq 1$, algebraic s, K, and rational $m > 0$, the optimal radial boundary for the payoff $(\|X\|^2 - K)_+^m$ is algebraic exactly when

$$\frac{r}{\lambda} - d \in 2\mathbb{Z}_{\geq 0}.$$

For $K = 0$, this covers every positive rational power of $\|X\|$. The resonance depends on discount and dimension, rather than on the reward parity appearing in the bridge classification.

I.6.1 Transcendence of zero-state values at every integer discount ratio

Boundary transcendence alone does not imply transcendence of the optimized value. The following elimination supplies that extra step for an infinite family including nonresonant parameters and nonzero transaction costs.

Theorem I.6.4 (Integer discount ratios give transcendental optimized values)**.** *Suppose $a = n \in \mathbb{Z}_{>0}$, $b, m \in \mathbb{Q}_{>0}$, $s \in \overline{\mathbb{Q}}_{>0}$, and $K \in \overline{\mathbb{Q}}_{\geq 0}$. Then $V(0)$ is transcendental. This holds for every positive rational payoff power, not only integer powers.*

Proof. The resonant cases $b \in \{1, \dots, n\}$ follow from Theorem I.6.2. Suppose otherwise, so that z_* is transcendental. Put $Y(z) = M(1; b; z)$. There are polynomials $A_n, B_n \in \mathbb{Q}[z]$ such that

$$F_n(z) := M(n; b; z) = A_n(z)Y(z) + B_n(z).$$

They are constructed from $A_1 = 1$, $B_1 = 0$ by

$$A_{j+1} = \frac{zA_j' + (z + j + 1 - b)A_j}{j}, \qquad B_{j+1} = \frac{(b-1)A_j + zB_j' + jB_j}{j}.$$

These follow by coefficient comparison from $zY' = (z + 1 - b)Y + (b - 1)$ and $M(j + 1; b; z) = (z\partial_z + j)M(j; b; z)/j$. In particular $\deg A_n = n - 1$ and its leading coefficient is $1/(n - 1)!$.

For completeness the determinant that prevents hidden division by zero is

$$z(B_nA_n' - A_nB_n') + (z+1-b)A_nB_n - (b-1)A_n^2 = D_n(b) := \frac{\prod_{j=1}^{n}(j-b)}{(n-1)!} \neq 0. \tag{I.6.12}$$

Indeed F_n and $A_n e^z z^{1-b}$ solve the same Kummer equation with upper parameter n, by the preceding contiguous operator. Their Wronskian is a constant times $e^z z^{-b}$; direct expansion identifies that constant with the polynomial on the left. At zero, $A_n(0) = \prod_{j=2}^{n}(j-b)/(n-1)!$ and $A_n(0) + B_n(0) = 1$, giving exactly $D_n(b)$.

Write $w = z - k$ and define rational functions

$$U = w\left[A_n' + \left(1 + \frac{1-b}{z}\right)A_n\right] - mA_n, \qquad T = w\left[B_n' + \frac{b-1}{z}A_n\right] - mB_n.$$

Then $(w\partial_z - m)F_n = UY + T$ and $B_nU - A_nT = wD_n(b)/z$. At the boundary, therefore,

$$F_n(z_*)U(z_*) = \frac{(z_* - k)D_n(b)}{z_*} \neq 0.$$

Let

$$Q(z) = zU(z) = (z-k)\{zA_n'(z) + (z+1-b)A_n(z)\} - mzA_n(z). \tag{I.6.13}$$

This is a polynomial of degree $n+1$ with leading coefficient $1/(n-1)!$. Since $f(0) = 1$, the value formula reduces to

$$V(0) = \frac{s^{-m}}{D_n(b)}(z_* - k)^{m-1}Q(z_*). \tag{I.6.14}$$

Write $m = u/v$ in lowest positive terms. If $V(0)$ were algebraic, raising (I.6.14) to the vth power would make

$$(z_* - k)^{u-v}Q(z_*)^v$$

algebraic. The rational function $(z-k)^{u-v}Q(z)^v$ has algebraic coefficients and has pole order $vn + u > 0$ at infinity, so it is nonconstant. Equating it to an algebraic number and clearing denominators gives a nonzero polynomial over $\overline{\mathbb{Q}}$ vanishing at z_*. This contradicts the transcendence of z_*. All denominators used above are nonzero by $z_* > k \geq 0$ and (I.6.12). □

Example I.6.5 (A concrete fixed-cost liquidation value)**.** Take $dY_t = (2 - Y_t)\,dt + \sqrt{2Y_t}\,dW_t$, discount rate $r = 1$, sale cost $K = 1$, and linear payoff. Then $s = a = m = 1$, $b = 2$, and $f(z) = (e^z - 1)/z$. The optimal stopping level $z_* > 1$ is the unique such solution of

$$(z^2 - 3z + 1)e^z + 2z - 1 = 0.$$

Both the boundary and the optimized zero-state value are transcendental, with the exact algebraic relation

$$V(0) = -z_*^2 + 3z_* - 1.$$

Numerically $z_* \approx 2.45628765889609$ and $V(0) \approx 0.335513913443030$. The decimals illustrate the theorem and are not used in its proof. In contrast, if the mean level is changed from 2 to 1, then $a = b = 1$ is resonant, $y_* = 2$ is algebraic, and $V(y) = e^{y-2}$ for $0 \leq y < 2$. Thus an algebraic stopping boundary can coexist with a transcendental value at every algebraic continuation state.

Corollary I.6.6 (No algebraic knife edge in the published CIR entry test)**.** *For the linear-payoff CIR liquidation and subsequent entry model of Leung–Li–Wang, assume $a = r/\kappa \in \mathbb{Z}_{>0}$, $b \in \mathbb{Q}_{>0}$, and algebraic $s > 0$ and sale cost $K \geq 0$. For every algebraic entry cost $c \geq 0$, the equality $V(0) = c$ is impossible. Consequently the nontrivial-entry condition $V(0) > c$ of [5, Equation (3.7)] has no algebraic equality case, and it can be decided by a terminating certified interval-refinement calculation from exact algebraic input.*

Proof. Take $m = 1$ in Theorem I.6.4. The linear liquidation value with payoff $Y_\tau - K$ equals our positive-part version: the verified optimizer stops only above K, and allowing never to stop cannot improve a negative exercise. The primary source gives the stated entry criterion. The unique boundary can be enclosed by bisection of the strictly increasing function H; the entire hypergeometric series and its derivative are effectively evaluated with certified tails on compact intervals. Thus $V(0)$ has arbitrarily precise computable intervals. Since $V(0) \neq c$, refinement eventually separates it from the exact algebraic number c, deciding the strict inequality. No complexity bound or new entry-boundary formula is asserted. □

Scope. The first arithmetic theorem classifies every stopping boundary in the stated rational-parameter family. The second classifies a substantial family of optimized values, but does not classify all nonresonant values at arbitrary starting states. This is a transfer to established diffusion and real-option models, using the classical Siegel–Shidlovskii theorem in its refined specialization form.

I.7 Unknown temporal persistence turns a two-view inverse into a three-visit experiment

The transfer in this section is from finite-dimensional defects of mixture operators to a hidden Markov experiment. Repeated observations need not share one permanent latent state: the state is retained with an unknown probability and is otherwise refreshed from the unknown population. The conclusions are an exact three-visit identification theorem, a sharp persistence-dependent inverse bound, and an exact minimax rate in a continuous-population example.

Model and conventions. Let Θ and $\mathcal{X}$ be compact metric spaces, and let the known map $\theta \mapsto P_\theta \in \mathcal{P}(\mathcal{X})$ be weakly continuous. For a finite signed measure ν on Θ, put

$$A_m \nu = \int P_\theta^{\otimes m}\, d\nu(\theta), \qquad K_m = \ker A_m.$$

The hidden process is stationary, with population $G \in \mathcal{P}(\Theta)$ and transition kernel

$$M_{G,\rho}(\theta, d\eta) = \rho \delta_\theta(d\eta) + (1-\rho) G(d\eta), \qquad 0 < \rho \leq 1. \tag{I.7.1}$$

Conditionally on its latent path, the emissions X_j are independent with law P_{θ_j}. The population G and persistence ρ are unknown. Write $T_{G,\rho}$ for the law of (X_0, X_1, X_2) and $\mu_G = A_1 G$. The total-variation norm of a signed measure is its full variation, denoted $\|\cdot\|_{\mathrm{var}}$; thus the distance of two probabilities is at most two. Constants below do not depend on the sample size or on the persistence lower bound.

Lemma I.7.1 (Finite defects supply a two-view inverse)**.** *Suppose* $\dim K_1 = d < \infty$ *and* K_1 *contains no nonzero finitely supported signed measure. Then* A_2 *is injective on finite signed measures. In particular, the component map is injective.*

Proof. For $f \in C(\mathcal{X}; \mathbb{R})$ put $Tf(\theta) = \int f\, dP_\theta$, and let V be the uniform closure of the range of T in $C(\Theta; \mathbb{R})$. It contains 1 and has annihilator K_1, so its codimension is d. Products of functions of the two coordinates are dense in $C(\mathcal{X}^2)$ by Stone–Weierstrass. Hence $\nu \in K_2$ implies $v\nu \in K_1$ for every $v \in V$. A vector-space complement of V of dimension d shows that the multiplication map $g \mapsto g\nu$ has rank at most $2d$. If $|\nu|$ had more than $2d$ support points, continuous functions on disjoint neighborhoods of these points would give more than $2d$ linearly independent measures $g\nu$. Thus ν is finitely supported. Also $K_2 \subset K_1$, by marginalization, so the hypothesis forces $\nu = 0$. Finally, equality of two component laws would put $\delta_\theta - \delta_\eta$ in K_1. □

Theorem I.7.2 (Blind three-visit recovery)**.** *Assume that A_2 is injective on finite signed measures. The law $T_{G,\rho}$ determines G for every $0 < \rho \le 1$. If G is not a point mass, it also determines ρ. If $G = \delta_\theta$, the emissions are independent with law P_θ, so ρ is unidentifiable and G remains identifiable. The assumptions of Lemma I.7.1 are sufficient.*

Proof. Let Q_h denote the law of (X_0, X_h). The transition powers are $M_{G,\rho}^h = \rho^h I + (1-\rho^h)G$. Therefore

$$Q_h = \mu_G^{\otimes 2} + \rho^h C_G, \qquad C_G = A_2 G - \mu_G^{\otimes 2}. \tag{I.7.2}$$

The triple law supplies μ_G, Q_1, Q_2. If $C_G \neq 0$, set $D_h = Q_h - \mu_G^{\otimes 2}$. Then

$$\rho = \frac{\|D_2\|_{\mathrm{var}}}{\|D_1\|_{\mathrm{var}}}, \qquad A_2 G = \mu_G^{\otimes 2} + \rho^{-1} D_1.$$

Injectivity of A_2 recovers G.

It remains to characterize the zero-signal case. For every real continuous f,

$$\int f(x) f(y)\, dC_G(x,y) = \mathrm{Var}_G\Big(\int f\, dP_\theta\Big).$$

If $C_G = 0$, choose a countable uniform-dense subset of $C(\mathcal{X})$ and intersect the resulting full-G-measure sets. All component probabilities on that set coincide, by density and uniqueness of measures. Injectivity of A_2 implies injectivity of $\theta \mapsto P_\theta$, so the set contains at most one point and G is a point mass. Its observed marginal identifies the point. Conversely a point mass plainly gives $C_G = 0$ and independent emissions for every ρ. □

Theorem I.7.3 (Uniform inverse bound without a covariance floor)**.** *For any two populations G, H and any $\rho, \sigma \in [r, 1]$, where $0 < r \le 1$, let $\epsilon = \|T_{G,\rho} - T_{H,\sigma}\|_{\mathrm{var}}$. Then*

$$\|A_2 G - A_2 H\|_{\mathrm{var}} \le (2 + 9r^{-2})\epsilon. \tag{I.7.3}$$

This bound does not assume a positive lower bound on $\|C_G\|$ or $\|C_H\|$. If A_2 is injective, define

$$\omega_2(u) = \sup\{W_1(G,H) : \|A_2 G - A_2 H\|_{\mathrm{var}} \le u\}.$$

Here W_1 uses the given compact metric on Θ. Then $\omega_2(u) \to 0$ as $u \downarrow 0$, and

$$W_1(G,H) \le \omega_2((2 + 9r^{-2})\epsilon).$$

Proof. Marginalization is a contraction in variation norm, and $\|\mu^{\otimes 2} - \nu^{\otimes 2}\|_{\mathrm{var}} \le 2\|\mu - \nu\|_{\mathrm{var}}$. Put $D = \rho C_G$ and $E = \sigma C_H$. Equation (I.7.2) gives

$$\|D - E\|_{\mathrm{var}} \le 3\epsilon, \qquad \|\rho D - \sigma E\|_{\mathrm{var}} \le 3\epsilon.$$

It follows that $|\rho - \sigma|\|D\|_{\mathrm{var}} \le 6\epsilon$. Since $\rho\sigma(C_G - C_H) = \sigma D - \rho E$, the triangle inequality gives $\|C_G - C_H\|_{\mathrm{var}} \le 9\epsilon/(\rho\sigma)$. Adding the product-marginal difference proves the bound.

To establish the modulus assertion, suppose it failed and take populations G_n, H_n with a fixed positive W_1 separation and $\|A_2 G_n - A_2 H_n\|_{\mathrm{var}} \to 0$. Compactness gives weakly convergent subsequences $G_n \to G$, $H_n \to H$. Weak continuity of the component map implies weak continuity of A_2, so $A_2 G = A_2 H$. Injectivity gives $G = H$. On a compact metric space weak convergence implies W_1 convergence, a contradiction. □

I.7.1 A positive entire kernel for which two visits fail

Let U be the uniform probability on $[0,1]$, let $L_n(x) = P_n(2x-1)$ be the shifted Legendre polynomials, and put

$$v_n(\theta) = \theta^n - \frac{2\theta}{n+1}, \qquad k(x \mid \theta) = 1 + \sum_{n=2}^{\infty} \frac{L_n(x) v_n(\theta)}{8n!}, \quad x, \theta \in [0,1]. \tag{I.7.4}$$

Proposition I.7.4 (Exact two-versus-three threshold)**.** *The kernel in* (I.7.4) *is jointly entire, is a probability density in x, and satisfies $3/4 < k(x \mid \theta) < 5/4$ on the real unit square. Its one-view nullspace is exactly*

$$K_1 = \mathbb{R}(U - \delta_0). \tag{I.7.5}$$

Thus A_2 is injective and its components are distinct. Nevertheless two consecutive visits with unknown persistence do not identify all populations. Three visits do, so the universal threshold for this kernel is exactly three.

Proof. The elementary bounds $|L_n(x)| \le 1$, $|v_n(\theta)| \le 2$ show $|k-1| \le (e-2)/4 < 1/4$. All L_n, $n \ge 1$, have integral zero, so the density integrates to one. On a compact complex set the two polynomial factors grow at most exponentially in n; the factorial denominator gives locally uniform convergence and hence joint entire extension.

If $A_1\nu = 0$, integration against 1 gives $\nu([0,1]) = 0$ and integration against each $L_n(x)$ gives $\int v_n \, d\nu = 0$, $n \ge 2$. Set $b = 2\int \theta \, d\nu$. Then $\int \theta^n \, d\nu = b/(n+1)$ for every $n \ge 1$. These are precisely the moments of $b(U - \delta_0)$, including the zeroth moment. Polynomial density proves (I.7.5); the converse follows from $v_n(0) = \int v_n \, dU = 0$. The nonzero elements of this nullspace are not finitely supported, so Lemma I.7.1 applies.

For $0 < s \le 1$ define $G_s = (1-s)\delta_0 + sU$. Since $P_0 = U$ and $A_1 U = U$, all these populations have $\mu_{G_s} = U$. Set $C = A_2 U - U^{\otimes 2} \ne 0$. The inequality follows from the zero-signal argument in Theorem I.7.2, or from $\int v_2^2 \, dU > 0$. Their consecutive-pair laws are

$$Q_1(G_s, \rho) = U^{\otimes 2} + s\rho C. \tag{I.7.6}$$

Distinct choices with the same product $s\rho$ are indistinguishable from two visits. Three-visit identification follows from the theorem. □

For later use let S be the signed measure on $[0,1]^3$ which is the sum of C on coordinates $(0,1)$ tensored with U on coordinate 2, and U on coordinate 0 tensored with C on coordinates $(1,2)$. Set $D = A_3 U - U^{\otimes 3}$ and $K = D - S$. Conditioning on the two refresh decisions gives the exact identity

$$T_{G_s,\rho} = U^{\otimes 3} + s\rho S + s\rho^2 K. \tag{I.7.7}$$

The $(0,2)$ marginal of K is C, so $K \ne 0$. All these measures have bounded densities, and the probability densities of the triple laws are bounded below by $(3/4)^3$.

Corollary I.7.5 (The inverse persistence exponent is sharp)**.** *The order r^{-2} in* (I.7.3) *cannot be replaced by $o(r^{-2})$, even for the fixed kernel* (I.7.4)*.*

Proof. Compare $(s,\rho) = (1,r)$ and $(s,\rho) = (1/2, 2r)$, $0 < r \le 1/2$. Their A_2 laws differ by $C/2$, while (I.7.7) shows that their triple laws differ by $r^2 K$. Both C and K are fixed nonzero signed measures. □

I.7.2 An exact statistical price for unknown persistence

The next experiment consists of N independent stationary three-visit records, for example N independent subjects. It is not one long dependent record. This distinction is part of the theorem's assumptions.

Theorem I.7.6 (Sharp population risk near independent refresh)**.** *Use the fixed kernel* (I.7.4)*. Let $0 < r \le 1/2$, let $s \in [1/2, 1]$ and $\rho \in [r, 2r]$, and observe $N \ge 1$ independent records with law $T_{G_s,\rho}$. The estimator knows the kernel, r, and the stated parameter ranges, but does not know s or ρ. There are fixed constants $0 < c < C < \infty$ such that*

$$c \min\{1, (r^2\sqrt{N})^{-1}\} \le \inf_{\widehat{G}} \sup_{s,\rho} \mathbb{E}_{s,\rho} W_1(\widehat{G}, G_s) \le C \min\{1, (r^2\sqrt{N})^{-1}\}. \tag{I.7.8}$$

Consequently, for a sequence $r = r_N$, uniform consistent population recovery in this experiment is possible exactly when $Nr_N^4 \to \infty$.

Proof. Let $\phi(x) = 80L_2(x)$. Legendre orthogonality and the coefficient of L_2 in (I.7.4) give $\mathbb{E}_\theta \phi(X) = v_2(\theta)$. Moreover

$$a := \int_0^1 v_2(\theta)^2 \, d\theta = \frac{2}{135}, \qquad \int v_2 \, dG_s = 0, \qquad \int v_2^2 \, dG_s = sa.$$

Thus the two bounded statistics $Z_1 = \phi(X_0)\phi(X_1)/a$ and $Z_2 = \phi(X_0)\phi(X_2)/a$ have expectations

$$v = s\rho, \qquad w = s\rho^2. \tag{I.7.9}$$

Let $(\bar{Z}_1, \bar{Z}_2)$ be their empirical averages. Choose a measurable least-squares minimizer $(\widehat{s}, \widehat{\rho})$ of its distance from $(s\rho, s\rho^2)$ over the compact rectangle $[1/2, 1] \times [r, 2r]$. A deterministic lexicographic tie rule, or any measurable minimum selector, may be used. Because the true parameter is admissible, the fitted moment vector differs from the true vector by at most twice the empirical error. The boundedness of Z_1, Z_2 gives expected Euclidean error $O(N^{-1/2})$.

On this rectangle, $w \ge r^2/2$, $0 \le v \le 2r$, and $s = v^2/w$. For two points in its moment image, direct subtraction gives

$$|s - s'| \le \frac{8}{r}|v - v'| + \frac{2}{r^2}|w - w'|.$$

For example, use $|v^2 - v'^2|/w \le 8|v - v'|/r$ and $(v'^2/w')|w - w'|/w \le 2|w - w'|/r^2$. Consequently $\mathbb{E}|\widehat{s} - s| \le C/(r^2\sqrt{N})$, clipped at the diameter of the parameter interval. Since

$$W_1(G_s, G_{s'}) = \frac{1}{2}|s - s'|, \tag{I.7.10}$$

the estimator $\widehat{G} = G_{\widehat{s}}$ proves the upper bound.

For the lower bound, set $s_0 = 3/4$, $\rho_0 = 3r/2$, and, for $0 < h \le 1/8$, put $s_1 = s_0 + h$ and $\rho_1 = \rho_0 s_0/s_1$. Both points lie in the parameter rectangle and $s_0\rho_0 = s_1\rho_1$. Writing $v = s_0\rho_0$, (I.7.7) gives

$$T_{G_{s_1},\rho_1} - T_{G_{s_0},\rho_0} = v^2(s_1^{-1} - s_0^{-1})K.$$

Its density is $O(hr^2)$ uniformly. The common positive lower bound on the probability densities therefore implies

$$\chi^2(T_{G_{s_1},\rho_1} \| T_{G_{s_0},\rho_0}) \le C_0 h^2 r^4.$$

For N independent records the chi-squared divergence is at most $\exp(C_0 N h^2 r^4) - 1$. Choose $h = c_0 \min\{1, (r^2\sqrt{N})^{-1}\}$, with a sufficiently small fixed c_0. The two product laws then have total-variation distance bounded away from one in the half-variation convention. The elementary two-point testing lower bound and (I.7.10) give the asserted risk. The necessity and sufficiency of $Nr_N^4 \to \infty$ follow directly. □

Relation to prior theory and limits of the transfer. Three-observation identification is already a major theme of hidden Markov theory. In particular, Gassiat, Cleynen, and Robin's finite-state nonparametric HMM theorem treats full-rank transitions and linearly independent emissions. General state-space translation models are treated by Gassiat, Le Corff, and Lehéricy, arXiv:1902.01070. Those results are not being claimed here. The precise contribution proved in this section is a known, possibly one-view-nonidentifiable emission family with an arbitrary Borel population, a restricted unknown refresh transition, an explicit two-visit counterexample with positive entire densities, and its sharp Nr^4 statistical boundary. The proof does not identify an arbitrary unknown Markov transition or an unknown emission family, and it does not establish a rate from one long trajectory. Historical priority for this particular combination remains to be assessed through a dedicated literature review.

I.8 Transfer boundaries and reproducibility

The results answer the target problems defined in this manuscript. The literature comparison identifies their established inputs, but does not turn the absence of a matching search result into a priority claim. In particular, the CIR linear exercise rule and the unweighted Jacobi differential system are credited to their original sources. The additional statements proved here concern arithmetic, exact minimality, explicit optimization, and recovery under an unknown persistence parameter.

Necessary restrictions. Several tempting extensions are deliberately excluded because a proof would require an additional ingredient.

1. The recurrence theorem requires separation of multiplicative critical values. The anisotropic Jacobi theorem requires distinct subset sums, and its matching upper bound is proved for even nonnegative Dyson index.

2. The Hamiltonian time form is required to be holomorphic on the *complex* compactified curve. A smooth real orbit alone is insufficient; the explicit singular-complex-curve example above demonstrates the failure.

3. Action quantization defines semiclassical levels. It gives no theorem about the arithmetic of exact quantum eigenvalues.

4. The CIR boundary classification does not automatically classify every optimized value at every nonresonant starting point. The integer-discount zero-state theorem supplies an additional elimination argument for its stated family.

5. Three-visit recovery uses a known emission family and a restricted refresh transition. The sharp risk theorem concerns the explicit population submodel and independent three-observation records, not an arbitrary hidden Markov model or observations from one long chain.

6. A direct extension of the binomial-mixture likelihood-transport proof to arbitrary heterogeneous replication counts is not obtained here. The likelihood score normalizes the aggregate posterior, rather than each observation stratum separately; replacing all replication counts by their minimum does not by itself prove the required inequality.

Exact checks and their role. The fourth-order recurrence displayed in Section I.2 was checked as an exact rational matrix identity and against directly computed initial coefficients. The contiguous determinant identity (I.6.12) was expanded symbolically for integer upper parameters through six, retaining the lower parameter as an indeterminate. These checks detect normalization

and sign errors; the general proofs do not rely on a finite sample of parameters. The $K_6 - M_2$ example was also checked by direct construction of its weighted Laplacian and comparison with (I.4.7). The hidden-state moment constant $2/135$ was checked by exact integration of $(\theta^2 - 2\theta/3)^2$ on $[0, 1]$.

The arguments were also read separately for their vulnerable steps: noncancellation at critical fibers, diagonal collision strata in Fourier asymptotics, positivity of the optimized conductances, complex points at infinity of the time differential, concave-power stopping verification, and the two-point statistical lower bound. These checks are useful evidence, not a substitute for external expert review or formal verification.

Part II

Sharp Bounds and Connection Formulas Beyond Existing Restrictions

Manuscript overview. We give four independent results that remove a stated restriction or determine a previously unmatched boundary in a specified problem. First, the worst ratio between the total effective resistance obtained by normalized infinity Lewis weights and its optimum on an n-vertex tree is $\sqrt{n}/8 + 1/2 + O(n^{-1/2})$. Balanced double-brooms attain this bound. They disprove the logarithmic approximation conjecture in Suliman–Weinstein's 2023 preprint, and a rational certificate contradicts its stated constant tree bound. Second, McIntosh's alternating binomial sums have minimum recurrence width $\lfloor r/2 \rfloor$ for every integer $r \geq 2$, removing the prime-or-power-of-two restriction in his minimality theorem. Even a recurrence with arbitrary gaps requires at least $\lfloor r/2 \rfloor + 1$ nonzero summands. Third, we determine the physical-density Frobenius connection coefficients for Jacobi traces in a nonresonant real-parameter range, extending the integer-parameter formula of Forrester–Kumar. Finally, for a specified known-emission refresh model observed along one stationary path, the population minimax risk on $\rho \in [r, 1-q]$ is $\Theta(\min\{1, n^{-1/2}(r^{-2} + q^{-1/2})\})$. The low-persistence lower bound uses a uniform full-history filtering argument; the nearly frozen bound counts fresh latent draws. All constructions and proof assumptions are stated explicitly.

Integration note. The alternating-sum theorem is extended in Parts VII and VIII. The tree comparison is followed by broader graph families in Parts III–V, and the nonresonant Jacobi connection is extended in Part IV. The hidden-trajectory argument remains a distinct contribution.

II.1 Targets, bottlenecks, and proof transfers

The four results have different relationships to the literature. The spectral-design result is a negative answer to an explicit conjecture in a preprint. The binomial-sum result extends a published minimality theorem using that paper's published upper bound. The Jacobi result extends a published connection formula within a stated nonresonant range. The statistical result closes a limitation of the preceding research draft in a fixed observation model; it is not attributed to a general hidden-Markov-model conjecture.

1. **Spectral design.** Suliman and Weinstein [13] conjecture an $O(\log n)$ approximation ratio for normalized infinity Lewis weights relative to optimal total effective resistance. Their Theorem 1.3 states a constant bound for trees. The tree objective reduces to a ratio of two moments of square roots of cut sizes. A sharp interval moment inequality identifies the extremal distribution, and a double-broom realizes it asymptotically. This gives both a counterexample and a sharp replacement theorem for trees.

2. **Alternating binomial sums.** McIntosh [29] constructs a recurrence of width $\lfloor r/2 \rfloor$ for every $r \geq 2$, but proves minimality only for prime r or powers of two. His asymptotic lower bound sees the algebraic degree of one dominant growth rate. A positive torus-moment representation exposes all positive critical values, including those in other cyclotomic orbits. Genuine singularities force the missing lower bound, and Descartes' rule also excludes sparse recurrences with fewer nonzero terms.

3. **Jacobi trace connections.** Forrester and Kumar [84] solve the physical-density connection problem for integer b, β. Their Fuchsian system supplies the local solution spaces; complete corner Fourier expansions determine the connection coefficients for nonresonant real parameters. Both sides of an interior knot must be included. The continuation convention and the distinction from a full arbitrary-basis connection matrix are essential.

4. **Dependent hidden populations.** The preceding draft [15] treats independent three-observation records. We replace those records with one complete dependent path. A posterior filter contraction proves that the whole history cannot overcome the low-persistence information loss. A fresh-draw oracle gives the other boundary, and overlapping moment statistics attain both costs with one estimator.

These transfers use, respectively, exact moment optimization, positivity of singular fibers, complete endpoint asymptotics, and matching statistical upper and lower bounds. Each following section supplies its own proof and identifies the results it imports. The source-specific conclusions do not rely on interpreting a literature search as proof of historical novelty.

II.2 A sharp correction to the Lewis-weight approximation problem on trees

Suliman and Weinstein formulate a comparison between the D-optimal (infinity-Lewis) design and the A-optimal effective-resistance design [13]. Their Conjecture 1.5 asserts an $O(\log n)$ approximation ratio on every n-vertex graph, while their Theorem 1.3 asserts the constant bound 3.12 on trees. We show that the tree ratio instead has worst-case growth $\sqrt{n}/8$. More precisely, its maximum over n-vertex trees equals

$$\frac{\sqrt{n}}{8} + \frac{1}{2} + O(n^{-1/2}).$$

An explicit 441-vertex tree already contradicts the stated constant bound, with a certificate involving only rational numbers. The target statements here are those of arXiv:2302.05966v1.

II.2.1 The exact tree design ratio

Let $T = (V, E)$ be a tree with $|V| = n \geq 2$. Assign a positive conductance w_e to each edge, with $\sum_{e \in E} w_e = 1$. Deleting e produces two components of sizes s_e and $n - s_e$; write

$$c_e = s_e(n - s_e).$$

Let $R_{uv}(w)$ denote effective resistance and set

$$K_T(w) = \sum_{\{u,v\} \subset V} R_{uv}(w).$$

The unordered-pair convention is immaterial for approximation ratios: ordered pairs multiply the objective by two, and the standard identity $K_T(w) = n \operatorname{tr} L(w)^\dagger$ gives the same A-optimal design. The effective-resistance design problem and its convex formulation are classical; see [48].

Lemma II.2.1 (Exact comparison on a tree)**.** *The unique A-optimal and D-optimal designs on T are respectively*

$$w_e^A = \frac{\sqrt{c_e}}{\sum_{f \in E} \sqrt{c_f}}, \qquad w_e^D = \frac{1}{n-1}.$$

Consequently, if $\alpha(T) = K_T(w^D)/K_T(w^A)$, then

$$\alpha(T) = \frac{(n-1) \sum_{e \in E} c_e}{\left(\sum_{e \in E} \sqrt{c_e}\right)^2}. \tag{II.2.1}$$

Proof. The effective resistance between two vertices of a tree is the sum of $1/w_e$ along their unique connecting path. Exactly c_e unordered vertex pairs have paths traversing e, so

$$K_T(w) = \sum_{e \in E} \frac{c_e}{w_e}.$$

Cauchy–Schwarz and the budget constraint give

$$\left(\sum_e \sqrt{c_e}\right)^2 \leq \left(\sum_e \frac{c_e}{w_e}\right)\left(\sum_e w_e\right) = K_T(w),$$

with equality exactly at the displayed w^A. For the D-optimal design, the product of the $n-1$ nonzero Laplacian eigenvalues equals $n \prod_e w_e$ by the matrix-tree identity. The arithmetic–geometric mean inequality therefore gives the unique maximizer $w_e^D = 1/(n-1)$. Substitution proves (II.2.1). □

The normalized infinity-Lewis weights in [13] coincide with this D-optimal design. In particular, (II.2.1) is exactly the ratio in their Claim 4.1 and equation (4.7), rather than a different normalization or another loss function.

II.2.2 A finite bound and its sharp asymptotic

Define

$$M_n = \max\{\alpha(T) : T \text{ is a tree on } n \text{ vertices}\}, \quad a_n = \sqrt{n-1}, \quad b_n = \sqrt{\lfloor n^2/4 \rfloor}.$$

The maximum exists because there are finitely many tree topologies on n vertices.

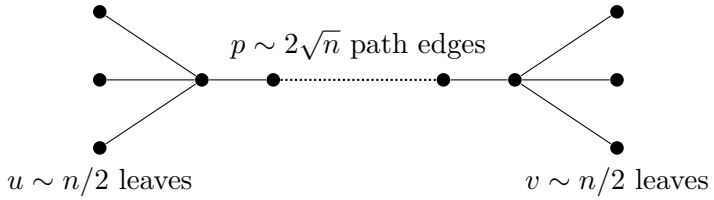


Figure 1: The balanced double-broom. Each fan denotes a larger family of leaves. Leaf-edge cut counts are $n-1$, while central-path cut counts are $n^2/4 + O(n)$. Their proportions realize the extremal moment ratio.

Theorem II.2.2 (Sharp tree approximation asymptotic)**.** *For every $n \geq 2$ and every n-vertex tree T,*

$$\alpha(T) \leq \frac{(a_n + b_n)^2}{4a_n b_n}. \tag{II.2.2}$$

Moreover,

$$M_n = \frac{\sqrt{n}}{8} + \frac{1}{2} + O(n^{-1/2}) \qquad (n \longrightarrow \infty). \tag{II.2.3}$$

Balanced double-brooms whose central paths have $\lfloor 2\sqrt{n} \rfloor$ edges attain the asymptotic in (II.2.3).

Proof. We first prove the upper bound. Every tree edge satisfies

$$n - 1 \leq c_e \leq \lfloor n^2/4 \rfloor.$$

Put $x_e = \sqrt{c_e}$ and $\mu = (n-1)^{-1} \sum_e x_e$. The pointwise inequality $(x_e - a_n)(x_e - b_n) \leq 0$ yields

$$\frac{1}{n-1} \sum_e x_e^2 \leq (a_n + b_n)\mu - a_n b_n.$$

By (II.2.1),

$$\alpha(T) = \frac{(n-1)^{-1} \sum_e x_e^2}{\mu^2} \leq \frac{a_n + b_n}{\mu} - \frac{a_n b_n}{\mu^2} \leq \frac{(a_n + b_n)^2}{4a_n b_n}.$$

For the last inequality one may maximize the scalar expression, whose maximum occurs at $\mu = 2a_n b_n/(a_n + b_n)$, or complete the square. In particular,

$$\frac{(a_n + b_n)^2}{4a_n b_n} = \frac{b_n}{4a_n} + \frac{1}{2} + \frac{a_n}{4b_n} = \frac{\sqrt{n}}{8} + \frac{1}{2} + O(n^{-1/2}). \tag{II.2.4}$$

For the lower bound, fix $n \geq 9$ and put

$$p = \lfloor 2\sqrt{n} \rfloor, \qquad L = n - p - 1, \qquad u = \lfloor L/2 \rfloor, \qquad v = L - u.$$

Take a path with p edges, attach u leaves to its left endpoint and v leaves to its right endpoint, and denote the resulting tree by T_n. It has exactly n vertices and $n - 1 = L + p$ edges. The L leaf edges have cut count $n - 1$. For the jth path edge, $1 \leq j \leq p$, the cut count is

$$c_j = (u + j)(n - u - j) = \frac{n^2}{4} - d_j^2, \qquad d_j = u + j - \frac{n}{2}. \tag{II.2.5}$$

The choice of balanced leaf counts gives $|d_j| \leq p/2$.

Write $r = \sqrt{n}$ and $p = 2r + t$, where $-1 < t \leq 0$. Using $\sum_j d_j^2 = O(p^3)$, we obtain

$$\begin{aligned} S_n := \sum_{e \in E(T_n)} c_e &= L(n-1) + \frac{pn^2}{4} - \sum_{j=1}^{p} d_j^2 \\ &= \frac{r^5}{2} + \left(\frac{t}{4} + 1\right) r^4 + O(r^3). \end{aligned} \tag{II.2.6}$$

For the square-root sum, the exact identity

$$0 \le \frac{n}{2} - \sqrt{\frac{n^2}{4} - d_j^2} = \frac{d_j^2}{n/2 + \sqrt{n^2/4 - d_j^2}} \le \frac{2d_j^2}{n}$$

shows that the total error on the path is $O(p^3/n) = O(r)$. Since $\sqrt{n-1} = r - 1/(2r) + O(r^{-3})$, it follows that

$$\begin{aligned} Q_n := \sum_{e \in E(T_n)} \sqrt{c_e} &= L\sqrt{n-1} + \sum_{j=1}^{p} \sqrt{n^2/4 - d_j^2} \\ &= 2r^3 + \left(\frac{t}{2} - 2\right) r^2 + O(r). \end{aligned} \tag{II.2.7}$$

All remainder constants above are uniform in $t \in [-1, 0]$ and in the parity of L. Substituting (II.2.6) and (II.2.7) into (II.2.1) gives

$$\begin{aligned} \alpha(T_n) &= \frac{(r^2-1)\{r^5/2 + (t/4+1)r^4 + O(r^3)\}}{4r^6 + (2t-8)r^5 + O(r^4)} \\ &= \frac{r}{8} + \frac{1}{2} + O(r^{-1}). \end{aligned}$$

The terms depending on t cancel from the constant coefficient. Together with (II.2.4), this proves (II.2.3). □

Remark II.2.3 (Why the extremal scale is a square root). The ratio is the empirical second moment of the numbers $\sqrt{c_e}$ divided by their squared empirical mean. Leaf edges have $\sqrt{c_e} \simeq \sqrt{n}$, whereas the central edges of a balanced double-broom have $\sqrt{c_e} \simeq n/2$. About $2\sqrt{n}$ central edges balance these two contributions to the first moment and asymptotically attain the interval moment bound. Thus both the exponent and the leading constant arise from a genuine extremal design calculation.

II.2.3 A rational certificate and the stated conjecture

Proposition II.2.4 (An explicit finite counterexample)**.** *Let T consist of a path with 40 edges, with 200 new leaves attached to each endpoint. Then T has 441 vertices and*

$$\alpha(T) > \frac{775676}{247107} > \frac{78}{25} = 3.12.$$

Proof. More generally, for an integer $k \ge 1$, take a path with $4k$ edges and attach $2k^2$ leaves to each endpoint. Then $n = (2k+1)^2$ and

$$c_{\text{leaf}} = n - 1, \qquad c_j = (2k^2 + j)(2k^2 + 4k + 1 - j), \quad 1 \le j \le 4k.$$

Summation gives

$$S_k := \sum_e c_e = \frac{4k}{3}(12k^4 + 36k^3 + 26k^2 + 6k + 1).$$

Since $\sqrt{n-1} < 2k+1$ and $\sqrt{c_j} \le n/2$,

$$Q_k := \sum_e \sqrt{c_e} < 4k^2(2k+1) + 2kn = 2k(2k+1)(4k+1).$$

Therefore

$$\alpha(T_k) > \frac{4(k+1)(12k^4 + 36k^3 + 26k^2 + 6k + 1)}{3(2k+1)^2(4k+1)^2}. \tag{II.2.8}$$

At $k = 10$, the right-hand side is $775676/247107$, and

$$\frac{775676}{247107} - \frac{78}{25} = \frac{117554}{6177675} > 0.$$

Thus the certificate requires no floating-point approximation. □

Corollary II.2.5 (Negative resolution of the logarithmic comparison)**.** *The D-optimal/infinity-Lewis effective-resistance approximation ratio is not $O(\log n)$ on general graphs, or even on trees. In particular, Conjecture 1.5 of [13], as stated in arXiv v1, is false. The stronger Conjecture 4.6 in the same version is false as well.*

Proof. Theorem II.2.2 supplies trees with $\alpha(T_n) \sim \sqrt{n}/8$, and $\sqrt{n}/\log n \to \infty$. The source's Theorem 1.4 bounds the actual approximation ratio above by the quantity in its Conjecture 4.6, so that stronger proposed logarithmic bound cannot hold either. □

II.2.4 The subtree reduction does not preserve the ratio

The discrepancy with the asserted constant tree bound can be traced to a specific monotonicity step. Claim 4.3 of [13] asserts that $T \subset T'$ implies $\alpha(T) \le \alpha(T')$ for trees. Under ordinary subtree containment this assertion is false, as the following elementary argument shows independently of the preceding asymptotic theorem.

Proposition II.2.6 (Failure of subtree monotonicity)**.** *There are trees $T \subset T'$ with $\alpha(T) > \alpha(T')$.*

Proof. Take $T = P_4$, the path on four vertices. Its cut counts are $3, 4, 3$; hence

$$\alpha(P_4) = \frac{30}{(2\sqrt{3}+2)^2} > 1.$$

For $N \ge 1$, attach N new leaves to an endpoint of P_4, forming $T_N \supset P_4$. The new leaf edges have cut count $N+3$ and the three original edges have cut counts $3(N+1)$, $2(N+2)$, and $N+3$. Thus

$$\sum_e c_e = N^2 + 9N + 10,$$

$$\sum_e \sqrt{c_e} = (N+1)\sqrt{N+3} + \sqrt{3(N+1)} + \sqrt{2(N+2)} = N^{3/2} + O(N^{1/2}).$$

Since $|E(T_N)| = N + 3$, formula (II.2.1) gives $\alpha(T_N) \to 1$. For all sufficiently large N, $\alpha(T_N) < \alpha(P_4)$, proving the assertion. □

In the proposed monotonicity proof, a design on T is treated as feasible for the resistance problem on T'. When vertices are added, extending the old design by zero conductances disconnects those new vertices and yields infinite total effective resistance; it does not give the asserted comparison of optimal values. The subsequent reduction from arbitrary bowties to a family with three proportional size parameters therefore does not apply. The extremal double-brooms above have a path of order $\sqrt{n}$ and two leaf populations of order n, which is exactly the scale missed by that reduction.

II.3 Exact recurrence minimality for alternating binomial power sums

For an integer $r \ge 2$, consider McIntosh's alternating binomial sum

$$A_r(n) = \sum_{k=-n}^{n} (-1)^k \binom{2n}{n+k}^r, \qquad n \ge 0. \tag{II.3.1}$$

McIntosh proved that this sequence has a polynomial-coefficient recurrence with consecutive shift width at most $\lfloor r/2 \rfloor$, and proved that this width is minimal when r is prime or a power of two [29, Theorems 1 and 2]. His lower-bound argument uses the algebraic degree of the dominant exponential growth rate. That degree is $\varphi(2r)/2$, which can be strictly smaller than $\lfloor r/2 \rfloor$. Here all the positive critical values of a real moment representation provide the missing lower bounds.

By a recurrence we mean an eventual identity

$$\sum_{j \in J} q_j(n) A_r(n+j) = 0, \qquad q_j \in \mathbb{C}(n) \setminus \{0\}, \tag{II.3.2}$$

where $J \subset \mathbb{Z}$ is finite and nonempty. Its *shift width* is $\max J - \min J$, and its *number of terms* is $|J|$. These are different quantities when the shifts have gaps.

Theorem II.3.1. *For every integer $r \geq 2$, put $m = \lfloor r/2 \rfloor$. Every recurrence (II.3.2) has shift width at least m and at least $m+1$ nonzero terms. Both bounds are attained by a recurrence with coefficients in $\mathbb{Q}[n]$. Consequently the minimum shift width is $\lfloor r/2 \rfloor$, and the minimum number of terms, even allowing arbitrary gaps and coefficients in $\mathbb{C}(n)$, is $\lfloor r/2 \rfloor + 1$.*

The new assertion is the lower bound for every r, including the stronger bound for sparse recurrences over $\mathbb{C}(n)$. Attainment uses the general upper bound in McIntosh's Theorem 1. Thus the theorem removes the prime-or-power-of-two restriction from his minimality conclusion; no claim about an exhaustive historical priority search is needed for this precise comparison.

II.3.1 A positive moment representation

Set $d = r - 1$, $X = x_1 \cdots x_d$, and

$$P_r(x) = \prod_{i=1}^{d} (1 + x_i)(1 - X^{-1}). \tag{II.3.3}$$

Here CT denotes the constant term in all d variables. For every integer $\ell \geq 0$, direct expansion gives

$$\mathrm{CT}\, P_r^{\ell} = \sum_{j=0}^{\ell} (-1)^j \binom{\ell}{j}^r. \tag{II.3.4}$$

Indeed, the term indexed by j in the last factor forces the same exponent j in each of the other d factors.

On $\mathbb{T}^d = (\mathbb{R}/2\pi\mathbb{Z})^d$, write $x_i = e^{i\theta_i}$ and $S = \sum_i \theta_i$. The elementary half-angle identities yield

$$P_r(e^{i\theta}) = i\Psi_r(\theta), \qquad \Psi_r(\theta) = 2^r \prod_{i=1}^{d} \cos(\theta_i/2) \sin(S/2). \tag{II.3.5}$$

The half-angle expression is globally periodic: increasing one θ_i by 2π changes both its cosine and the final sine by a minus sign. Thus $\Psi_r = -iP_r|_{\mathbb{T}^d}$ is a globally real-analytic real-valued function. Define

$$\Phi_r = \Psi_r^2 = -P_r^2|_{\mathbb{T}^d} \geq 0. \tag{II.3.6}$$

For normalized Haar probability μ on the torus,

$$A_r(n) = (-1)^n \,\mathrm{CT}\, P_r^{2n} = \int_{\mathbb{T}^d} \Phi_r(\theta)^n \, d\mu(\theta). \tag{II.3.7}$$

The factor $(-1)^n$ comes from replacing k by $j - n$ in (II.3.1). It is also exactly the factor contributed by $P_r^{2n} = (-1)^n \Psi_r^{2n}$.

II.3.2 The complete positive critical set

Lemma II.3.2. *The distinct positive critical values of Φ_r are*

$$v_j = \left(2\cos\frac{(2j+1)\pi}{2r}\right)^{2r}, \qquad 0 \le j < m, \qquad m = \lfloor r/2 \rfloor. \tag{II.3.8}$$

Every critical point on each of these fibers is a nondegenerate local maximum. There are no other critical points with $\Phi_r > 0$.

Proof. At a point where $P_r \neq 0$, all its factors are nonzero and

$$x_i \frac{\partial}{\partial x_i} \log P_r = \frac{x_i}{1+x_i} + \frac{1}{X-1}. \tag{II.3.9}$$

Its vanishing is equivalent to $x_i X = -1$. On the torus, $\partial_{\theta_i} P_r = i x_i \partial_{x_i} P_r$; also $P_r = i\Psi_r$. It follows that these equations are precisely the critical equations for Ψ_r away from its zero set. They force $x_1 = \cdots = x_d = \omega$ with $\omega^r = -1$. Conversely each such root with $\omega \neq -1$ gives a critical point. The root -1 occurs only for odd r and has $P_r = 0$, so it is excluded. Since $d\Phi_r = 2\Psi_r\, d\Psi_r$, this also exhausts the positive critical set of Φ_r.

At such a point $X = -1/\omega$, hence $P_r = (1+\omega)^r$. With $\omega = e^{i(2j+1)\pi/r}$ this is

$$P_r = i(-1)^j \left(2\cos\frac{(2j+1)\pi}{2r}\right)^r.$$

Conjugate roots produce equal values of Φ_r. After pairing them, the resulting values are exactly (II.3.8). Their angles lie strictly between 0 and $\pi/2$, so they are positive and strictly decreasing. There are m of them.

For the Hessian, let I_d be the identity and J_d the all-ones matrix. Differentiating (II.3.9) gives, at the critical point,

$$\mathrm{Hess}_{\log x} \log P_r = \frac{\omega}{(1+\omega)^2}(I_d + J_d).$$

The diagonal contribution is $\omega/(1+\omega)^2$. Every entry also receives $-X/(X-1)^2$, which has the same value when $X = -1/\omega$. If $\omega = e^{i\alpha}$, then

$$\frac{\omega}{(1+\omega)^2} = \frac{1}{4\cos^2(\alpha/2)} > 0.$$

Angular differentiation contributes a minus sign. On the real torus one may use the real logarithm $\log|\Psi_r|$, whose Hessian agrees with that of a local complex logarithm of P_r. Since the gradient vanishes, this gives

$$\mathrm{Hess}_\theta\, \Phi_r = -\frac{\Phi_r}{2\cos^2(\alpha/2)}(I_d + J_d). \tag{II.3.10}$$

The eigenvalues of $I_d + J_d$ are 1, with multiplicity $d-1$, and $d+1 = r$. The Hessian is therefore negative definite at every positive critical point. □

II.3.3 Visibility of every positive critical value

We record the real-variable argument that prevents the smaller critical values from disappearing under analytic continuation. In particular, the argument includes the integer exponents that occur in even dimensions.

Lemma II.3.3. *Let M be a compact real-analytic manifold of dimension $d \geq 1$, let μ have a positive real-analytic density, and let $\Phi : M \to \mathbb{R}$ be real analytic. Suppose that the critical points in the fiber over $v \neq 0$ form a finite nonempty set and are all nondegenerate local maxima. Regular points in that fiber are allowed. Then the generating germ*

$$F(t) = \int_M \frac{d\mu}{1 - t\Phi}$$

has a branch continued from $t = 0$ that is singular at $t = 1/v$.

Proof. The analytic Morse lemma gives local coordinates near each critical point in which $\Phi = v - |y|^2$ and $d\mu = g(y)\, dy$, with g analytic and $g(0) > 0$. For $w > 0$ sufficiently small, the local density of $\Phi_*\mu$ at $v - w$ is

$$\frac{1}{2} w^{d/2-1} \int_{S^{d-1}} g(\sqrt{w}\,\omega)\, d\omega.$$

The spherical average is analytic in w: odd Taylor terms integrate to zero. Its constant term is positive. Choose smooth cutoffs equal to one near the critical points. After restricting to a sufficiently small interval about v, the remaining compact region contains only regular points, so its pushforward has a smooth density. Summing the critical contributions therefore gives

$$\frac{d\Phi_*\mu}{dx}(v + x) = b(x) + \mathbf{1}_{\{x<0\}}(-x)^{d/2-1} h(-x), \tag{II.3.11}$$

where b is smooth, h is analytic near zero, and $h(0) > 0$. There is no cancellation because all critical points contribute from the same side with positive leading coefficients.

The density in (II.3.11) is not smooth at zero. If d is even, a finite derivative has a jump. If d is odd and $d \geq 3$, a noninteger power prevents smoothness; for $d = 1$, the density is locally integrable but unbounded.

Consider the Cauchy transform

$$G(z) = \int_M \frac{d\mu}{z - \Phi}.$$

It is holomorphic in the upper half-plane and satisfies $G(z) = z^{-1} F(z^{-1})$ near infinity. Its boundary distribution is

$$-\frac{1}{\pi} \operatorname{Im} G(x + i0) = \Phi_*\mu,$$

as follows from the Poisson kernel approximation to the identity. If this upper-half-plane branch extended holomorphically through v, its boundary imaginary part would be real analytic there. This contradicts (II.3.11). Thus the branch is singular at v. The change of variables $t = 1/z$, which is locally biholomorphic at $v \neq 0$, gives the claimed singularity at $1/v$. The continuation from infinity in the upper half-plane corresponds to a continuation of the generating germ from zero, including when v is an interior point of the real range of Φ. □

Applying Lemmas II.3.2 and II.3.3 to (II.3.7) proves that

$$F_r(t) = \sum_{n \geq 0} A_r(n) t^n \tag{II.3.12}$$

has genuine continued-branch singularities at the m distinct positive points $t_j = 1/v_j$. This assertion concerns all these continued singularities, not merely the first singularity on the circle of convergence.

II.3.4 Singularities force width and support bounds

Lemma II.3.4. *Let a_n have a generating function analytic near zero. Suppose its continued germ has singularities at m distinct positive finite points. Then every eventual rational-coefficient recurrence for a_n has shift width at least m and at least $m+1$ nonzero terms, even if its coefficients lie in $\mathbb{C}(n)$ and its shifts have gaps.*

Proof. After translating the index, write the recurrence as

$$\sum_{j\in J} q_j(n)a_{n+j} = 0, \qquad \min J = 0, \qquad \max J = L.$$

Clear denominators so that $q_j \in \mathbb{C}[n]$. Set $e = \max_j \deg q_j$ and $c_j = [n^e]q_j$. Not all c_j vanish. Let $F(t) = \sum_{n\geq 0} a_n t^n$ and $\vartheta = t\,d/dt$. Multiplying by t^{n+L} and summing over all sufficiently large n gives

$$\left(\sum_{j\in J} t^{L-j} q_j(\vartheta - j)\right) F(t) = R(t), \tag{II.3.13}$$

where R is a polynomial accounting for finitely many initial terms. The coefficient of the highest ordinary derivative is $t^e H(t)$, where

$$H(t) = \sum_{j\in J} c_j t^{L-j} \neq 0. \tag{II.3.14}$$

Distinct shifts give distinct monomials, so this leading coefficient cannot vanish identically. Every nonzero finite singularity of a continued solution of (II.3.13) must be a zero of H: otherwise the equation is an inhomogeneous linear differential equation with an ordinary point and analytic right-hand side. For $e = 0$, the same conclusion follows directly from $H(t)F(t) = R(t)$.

Consequently H has at least m distinct positive roots. Since $\deg H \leq L$, this implies $L \geq m$. For the term bound, choose a nonzero real or imaginary part of H. It is a real polynomial with the same m positive roots and no more nonzero monomials than H. Descartes' rule of signs implies that a real polynomial with k nonzero monomials has at most $k-1$ positive roots. Thus H has at least $m+1$ nonzero monomials. Formula (II.3.14) now gives $|J| \geq m+1$. □

Proof of Theorem II.3.1. The moment representation and the complete critical-point calculation give m distinct positive singularities of F_r by Lemma II.3.3. Lemma II.3.4 proves both lower bounds over $\mathbb{C}(n)$. McIntosh's general existence theorem supplies a recurrence over $\mathbb{Q}[n]$ with shifts in an interval of length m [29, Theorem 1]. It has at most $m+1$ positions; the lower bounds force it to attain both claimed minima. □

Corollary II.3.5. *For any minimum-width recurrence, translate the shifts to $0, \ldots, m$ and define its highest-degree symbol H by (II.3.14). Then*

$$H(t) = c\prod_{j=0}^{m-1}(1 - v_j t), \qquad c \neq 0. \tag{II.3.15}$$

The set of nonzero finite points at which some branch of F_r continued from zero is singular is exactly $\{1/v_j : 0 \leq j < m\}$.

Proof. The polynomial H has degree at most m and already has the m distinct roots $1/v_j$, so (II.3.15) follows. The differential equation (II.3.13) has no other nonzero finite nonordinary points, hence no continued solution branch can acquire a singularity elsewhere. Visibility supplies a singular branch at every displayed point. □

II.3.5 The first missing composite case

For $r = 6$, the three critical scales are

$$v_0 = 1351 + 780\sqrt{3}, \qquad v_1 = 64, \qquad v_2 = 1351 - 780\sqrt{3}.$$

The dominant scale has algebraic degree two, so the dominant-growth argument alone cannot prove the correct width three. The rational interior scale 64 supplies the third visible singularity. Theorem II.3.1 gives width three and at least four nonzero terms for every sparse recurrence. Moreover,

$$\prod_{j=0}^{2}(X - v_j) = (X - 64)(X^2 - 2702X + 1) = X^3 - 2766X^2 + 172929X - 64.$$

Thus a minimum-width recurrence has highest-degree generating symbol proportional to

$$1 - 2766t + 172929t^2 - 64t^3.$$

For comparison, $r = 2$ and $r = 3$ have the sole critical scales 4 and 27, respectively, in agreement with the familiar hypergeometric formulas $A_2(n) = \binom{2n}{n}$ and $A_3(n) = (3n)!/(n!)^3$.

II.4 Nonresonant connection coefficients for Jacobi trace densities

II.4.1 The published connection problem and the new scope

Forrester and Kumar explicitly identify the determination of the scalars in the Frobenius representation of the Jacobi trace density as the connection problem left by Davis. Their Proposition 1.3 and equation (3.55) solve that problem when the endpoint parameter b and the Dyson index β are nonnegative integers. Their discussion after equation (3.54) states that the general problem remains; see *Differential recurrences for the distribution of the trace of the β-Jacobi ensemble*, Physica D **434** (2022), 133220, `arXiv:2011.00787v2`, Section 3.3 [84].

Here the corner Fourier amplitudes determine those physical-density connection scalars for real, noninteger parameters, provided the local exponents are nonintegral. The answer uses an explicitly specified upper half-plane continuation. It generalizes the sign in the integer-parameter formula to a complex phase. The theorem concerns the particular physical density, not the full connection matrix between arbitrary bases. Resonant exponents and numerical stability of long continuation chains are not settled by it.

Fix

$$N \geq 1, \qquad a > -1, \qquad b > 0, \qquad \beta \geq 0. \tag{II.4.1}$$

The stronger restriction $b > 0$ makes every component of the published auxiliary integral system absolutely integrable. Define

$$S_N(a, b, \beta) = \int_{[0,1]^N} |\Delta(u)|^\beta \prod_{i=1}^{N} u_i^a (1 - u_i)^b \, du$$

and let $\rho(t)$ be the density of $\sum_{i=1}^{N} u_i$ under this normalized measure, extended by zero outside $[0, N]$. For $r \geq 0$ put

$$\begin{aligned} W_r(q, \beta) &= \int_{(0,\infty)^r} e^{-\sum x_i} |\Delta(x)|^\beta \prod x_i^q \, dx, \qquad W_0 = 1, \\ &= \prod_{j=0}^{r-1} \frac{\Gamma(q + 1 + j\beta/2)\Gamma(1 + (j+1)\beta/2)}{\Gamma(1 + \beta/2)}. \end{aligned}$$

Only $q = a, b > -1$ will occur. The Laguerre–Selberg and Selberg evaluations give this product and

$$S_N(a,b,\beta) = \prod_{j=0}^{N-1} \frac{\Gamma(a+1+j\beta/2)\Gamma(b+1+j\beta/2)\Gamma(1+(j+1)\beta/2)}{\Gamma(a+b+2+(N+j-1)\beta/2)\Gamma(1+\beta/2)}.$$

For $0 \le p \le N$ define

$$\begin{aligned}
\alpha_p &= (N-p)(a+1) + \frac{\beta}{2}(N-p)(N-p-1), \\
\eta_p &= p(b+1) + \frac{\beta}{2}p(p-1), \\
D_p &= \alpha_p + \eta_p, \qquad \gamma_p = D_p - 1, \qquad \text{(II.4.2)} \\
K_p &= \binom{N}{p} \frac{W_{N-p}(a,\beta)W_p(b,\beta)}{S_N(a,b,\beta)}. \qquad \text{(II.4.3)}
\end{aligned}$$

Here $D_p > 0$ and $K_p > 0$. Our nonresonance assumption is

$$\gamma_p \notin \mathbb{Z} \qquad (0 \le p < N). \qquad \text{(II.4.4)}$$

It removes a countable collection of parameter hyperplanes; it does not require a, b or β to be rational. The exponent at the final endpoint $p = N$ need not be nonintegral, since no continuation across that endpoint is needed to represent the density on $(0, N)$.

II.4.2 The normalized Frobenius branches

We specify the branches through the Forrester–Kumar system, so that the normalization of the coefficients below is unambiguous. With indices $j = 0, \dots, N$, set

$$\Lambda = -\operatorname{diag}(N, N-1, \dots, 0), \qquad h = a + b + 1 + \tfrac{\beta}{2}(2N-1), \qquad k = b + \tfrac{\beta}{2}N,$$

and let X be tridiagonal, with entries

$$\begin{aligned}
X_{j,j-1} &= u_j = -jk + \tfrac{\beta}{2}j^2, \\
X_{j,j} &= v_j = hj - \tfrac{\beta}{2}(N-j)(2j+1) + k(N-2j) - 1, \\
X_{j,j+1} &= w_j = h(j-N) + \tfrac{\beta}{2}(N-j)(N-j-1) + k(N-j).
\end{aligned}$$

Entries outside the index range are omitted. The published system is

$$(tI + \Lambda)G'(t) = XG(t). \qquad \text{(II.4.5)}$$

To relate it to the physical density, form

$$H_j(t) = \binom{N}{j}^{-1} \int_{[0,1]^N} \delta\Big(t - \sum u_i\Big) |\Delta(u)|^\beta \prod u_i^a (1-u_i)^{b-1} e_j(1-u_1, \dots, 1-u_N)\, du.$$

If $P_{jk} = \binom{N-j}{N-k}$, then $G = P^{-1}H$ satisfies (II.4.5). The last row of P is the last coordinate row, so

$$G_N(t) = H_N(t) = S_N(a,b,\beta)\rho(t). \qquad \text{(II.4.6)}$$

These are Proposition 3.9 and equation (3.48) of the cited paper. They also hold at $\beta = 0$, by the same integration identities or by continuity. On every interval between the integers the system is ordinary, so its distributional solution is real analytic there.

Lemma II.4.1 (A uniquely normalized singular branch)**.** *Fix $p \in \{0, \dots, N\}$ with $\gamma_p \notin \mathbb{Z}$. At $t = p$, the system (II.4.5) has N holomorphic local solutions and one nonintegral Frobenius solution. The last component of the latter, after a unique constant normalization, has the form*

$$F_p(t) = (t-p)^{\gamma_p} U_p(t-p), \qquad U_p(0) = 1, \tag{II.4.7}$$

where U_p is holomorphic with real Taylor coefficients. Initially $t - p > 0$ and the power is positive real. Any last component of a local solution is the sum of a holomorphic function and a constant multiple of this Frobenius branch, with the branch of the power specified.

Proof. Put $j_0 = N - p$ and $s = t - p$. The residue matrix is the matrix with only row j_0 nonzero, that row being row j_0 of X. It has rank one, with its nonzero eigenvalue

$$v_{j_0} = \gamma_p - p =: v.$$

This identity follows by substitution into the displayed definition of v_j. The hypothesis $\gamma_p \notin \mathbb{Z}$ implies that v is nonintegral, so the residue is diagonalizable with eigenvalues 0 and v, and no difference between them is a nonzero integer. The ordinary nonresonant Frobenius construction gives N holomorphic solutions and one solution

$$s^v \sum_{\ell \geq 0} g_\ell s^\ell, \qquad g_0 = e_{j_0}.$$

For completeness, substitution gives the recurrence

$$(v + \ell)(pI + \Lambda) g_\ell = (X - (v + \ell - 1)I) g_{\ell - 1}.$$

Tridiagonality shows that its last component vanishes up to $\ell = p - 1$, while

$$(g_p)_N = \frac{\prod_{\ell=1}^{p} u_{N-p+\ell}}{p! \prod_{\ell=1}^{p} (v + \ell)} \neq 0. \tag{II.4.8}$$

Indeed, the first contribution to row $N - p + \ell$ travels through its subdiagonal entry $u_{N-p+\ell}$, and that row of $pI + \Lambda$ has diagonal entry ℓ. Each numerator is $-j(b + \beta(N - j)/2)$ with $j \geq 1$, hence is nonzero by $b > 0$; the denominators are nonzero by nonresonance. For $p = 0$, the empty product is one. Division by (II.4.8) gives (II.4.7), since $v + p = \gamma_p$. The nonintegral Frobenius line is unique, and all coefficients are real. □

Write $F_p^{\uparrow}(t)$ for its continuation from just to the right of p to a real $t > p$, moving to the right and passing every intervening integer by a small semicircle in the *upper* half-plane. This fixes a homotopy class and hence a branch. In particular, when a germ on the left of a knot is continued to its right in this convention, $(-s)^\gamma$ becomes $e^{-\pi i \gamma} s^\gamma$.

II.4.3 The connection formula

Theorem II.4.2 (Physical-density connection coefficients)**.** *Assume (II.4.1) and (II.4.4). For $0 \leq p < N$ set*

$$\boxed{C_p = e^{-\pi i \eta_p} \frac{K_p}{\Gamma(D_p)} = e^{-\pi i \eta_p} \binom{N}{p} \frac{W_{N-p}(a, \beta) W_p(b, \beta)}{S_N(a, b, \beta) \Gamma(D_p)}.} \tag{II.4.9}$$

For each $p \in \{0, \dots, N-1\}$, let $\rho_{p,-}^{\uparrow}$ be the continuation to the right of p of the density germ on its left, with the upper convention. Let $\rho_{p,+}$ be its physical germ on the right. At $p = 0$ use the zero density on the exterior left interval. Then

$$\rho_{p,+}(t) - \rho_{p,-}^{\uparrow}(t) = C_p F_p(t) \qquad (t > p \text{ sufficiently close to } p). \tag{II.4.10}$$

Consequently, for $k = 0, \dots, N-1$ and $k < t < k+1$,

$$\boxed{\rho(t) = \sum_{p=0}^{k} C_p F_p^{\uparrow}(t).} \tag{II.4.11}$$

Although the summands can be complex, their sum is the real, positive physical density. The lower-half-plane formula is its complex conjugate. If in addition $\gamma_N \notin \mathbb{Z}$, the local jump (II.4.10) also holds at $p = N$, with the same formula (II.4.9) and zero physical density on the right.

The distinction between (II.4.10) and a one-sided real coefficient is essential. At noninteger parameters the density generally has nonanalytic terms on *both* sides of an interior knot. The coefficient C_p is the discrepancy after a specified analytic continuation; it is not, in general, the coefficient on the right alone.

Lemma II.4.3 (The complete local Fourier symbol)**.** *Fix $p \in \{0, \dots, N\}$. Let χ be smooth, supported in a sufficiently small interval about p, and identically one near p. Then as $R \to +\infty$,*

$$\int e^{iRt} \chi(t) \rho(t)\, dt \sim e^{ipR} R^{-D_p} \sum_{\ell \geq 0} A_{p,\ell} R^{-\ell}, \qquad A_{p,0} = K_p e^{\frac{\pi i}{2}(\alpha_p - \eta_p)}. \tag{II.4.12}$$

The expansion is complete to every prescribed finite order. If D_p is nonintegral, there are no additional integer-power terms at this frequency.

Proof. Insert the defining pushforward integral for ρ. The relevant corners of the cube have p coordinates equal to one and $N - p$ equal to zero. Near one of these corners use $x_i = u_i$ in the zero cluster and $y_i = 1 - u_i$ in the one cluster. The phase becomes $p + \sum x_i - \sum y_i$. The weight is

$$|\Delta(x)|^{\beta} \prod x_i^{a} \; |\Delta(y)|^{\beta} \prod y_i^{b} \; B(x, y), \qquad B(0, 0) = 1,$$

where B is analytic near the origin. The two homogeneous weights, including their measures, have degrees α_p and η_p. Radial integration with $r = \sum x_i$ and $q = \sum y_i$ reduces their leading Fourier integrals to the one-dimensional endpoint formula

$$\int_0^{\infty} e^{\pm iRr} r^{d-1} \chi_0(r)\, dr \sim \Gamma(d) e^{\pm \pi i d/2} R^{-d}, \quad d > 0,$$

where χ_0 is smooth and equals one near zero. For nonempty clusters the angular constants are $W_{N-p}(a, \beta)/\Gamma(\alpha_p)$ and $W_p(b, \beta)/\Gamma(\eta_p)$, respectively; an empty cluster contributes one without radial integration. The angular integrals are over compact simplexes with integrable weights. Consequently, after choosing smooth cutoffs in r and q, angular integration of B produces a smooth amplitude in these two radial variables, with Taylor remainders of every order. Applying the endpoint formula successively in r and q gives a complete expansion. Taylor expansion of B gives only integer increments of D_p. Thus one corner contributes the claimed leading coefficient without the binomial factor, and all $\binom{N}{p}$ corners contribute identically.

To justify the complete expansion and exclude other contributions, cover the complement of the corner neighbourhoods by small sets. At every point of this complement, some cluster of coincident coordinates lies strictly between zero and one. Translating all coordinates of that cluster together leaves its internal differences unchanged, is tangent to all relevant collision faces, and has nonzero phase derivative equal to the cluster size. Other factors in the translated coordinates are smooth and nonzero. A smooth partition of unity and repeated integration by parts in this direction give an error $O(R^{-M})$ for every M, even when β is not integral. Differentiated factors are bounded by integrable endpoint and collision weights. The radial endpoint expansions of the corner pieces have remainders to arbitrary order. This proves the complete expansion and the absence of another power scale in (II.4.12). □

Proof of Theorem II.4.2. Fix a knot p with $\gamma_p \notin \mathbb{Z}$, and abbreviate $D = D_p$, $\gamma = \gamma_p$, $\alpha = \alpha_p$, $\eta = \eta_p$, and $K = K_p$. By Lemma II.4.1, on the two real sides of $s = t - p = 0$ the physical density has the form

$$\rho(p+s) = \begin{cases} a_+(s) + c_+ s^\gamma U_p(s), & s > 0, \\ a_-(s) + c_-(-s)^\gamma U_p(s), & s < 0, \end{cases} \tag{II.4.13}$$

where $a_\pm$ are holomorphic near zero and $c_\pm$ are real. At an endpoint the exterior expression is zero. The reality follows from the real equation and the real normalization of U_p.

We first prove $a_+ = a_-$. After multiplication by a cutoff, a nonzero first Taylor coefficient of $a_+ - a_-$ of degree q would give a nonzero term of order R^{-q-1} in the Fourier transform at frequency p. In contrast, the nonanalytic terms in (II.4.13) give only powers $R^{-D-\ell}$, $\ell \geq 0$. Since D is nonintegral, the two power scales are disjoint. Lemma II.4.3 supplies a complete expansion containing only the second scale. Uniqueness of an asymptotic power expansion forces every Taylor coefficient of $a_+ - a_-$ to vanish. The two holomorphic functions are therefore identical. This step is why leading asymptotics alone would not suffice for the proof.

Comparing the leading Fourier coefficient in (II.4.13) with (II.4.12) gives

$$\Gamma(D)\left(c_+ e^{\pi i D/2} + c_- e^{-\pi i D/2}\right) = K e^{\pi i(\alpha-\eta)/2}.$$

The real and imaginary parts uniquely determine the two real coefficients because $\sin(\pi D) \neq 0$. They are

$$c_+ = \frac{K \sin(\pi\alpha)}{\Gamma(D)\sin(\pi D)}, \qquad c_- = \frac{K \sin(\pi\eta)}{\Gamma(D)\sin(\pi D)}. \tag{II.4.14}$$

These formulas remain meaningful when an individual sine in the numerator vanishes. They also cover the empty-cluster endpoints.

Under upper-half-plane continuation of the left germ, $(-s)^\gamma$ becomes $e^{-\pi i\gamma} s^\gamma$. As the analytic backgrounds agree, the discrepancy is

$$(c_+ - e^{-\pi i\gamma} c_-) F_p(t).$$

Using $D = \alpha + \eta$ and $\gamma = D - 1$,

$$\sin(\pi\alpha) - e^{-\pi i\gamma}\sin(\pi\eta) = e^{-\pi i\eta}\sin(\pi D).$$

Substitution into (II.4.14) gives exactly $C_p = K e^{-\pi i\eta}/\Gamma(D)$ and proves (II.4.10).

At $p = 0$ the density on the left is zero, so the first interval satisfies $\rho = C_0 F_0$. Continue this identity to the next knot through the specified upper path and add the local discrepancy $C_1 F_1$. The identity extends throughout the next interval by analytic continuation of the same differential system. Inductively, this proves (II.4.11). Reversing upper and lower paths conjugates the coefficients and branches. □

II.4.4 Comparison, checks, and limitations

When b is a positive integer and β is a nonnegative integer,

$$\eta_p = p(b+1) + \frac{\beta}{2} p(p-1) \in \mathbb{Z}, \qquad e^{-\pi i\eta_p} = (-1)^{p(b+1)+\beta p(p-1)/2}.$$

If a is irrational, all exponents γ_p with $p < N$ are nonintegral. Thus the theorem has a nonempty intersection with the integer-parameter result, and (II.4.9) there is exactly the sign and Laguerre–Selberg constant of Forrester–Kumar (1.12), Proposition 3.4, and (3.55). Their $b = 0$ boundary case is not included in the present auxiliary-integral hypothesis. The new range is the nonresonant

real-parameter range outside those integer restrictions. In that range the coefficient on the left of an interior knot generally does not vanish, and the continuation convention is indispensable.

As an exact consistency test of the additional final-endpoint statement, take $N = 1$ and $a, b \notin \mathbb{Z}$. The density is $t^a(1-t)^b/B(a+1, b+1)$ on $(0, 1)$. At $t = 1$ its upper continuation is $e^{-\pi i b}(t-1)^b t^a/B(a+1, b+1)$. Formula (II.4.9) gives $C_1 = e^{-\pi i(b+1)}/B(a+1, b+1)$, precisely the coefficient that cancels this branch to produce the zero exterior density.

A second, noninteger numerical diagnostic uses the exact $N = 2$ hypergeometric density in Forrester–Kumar (3.36)–(3.37). For $a = 0.2$, $b = 0.35$, $\beta = 0.7$, the predicted coefficient at $p = 1$ is

$$C_1 = -3.7279987849354161471 + 7.3166095798246842339\, i.$$

Evaluating the difference between the physical right density and the upper continuation of the left density, then dividing by $(t-1)^{a+b+1}$, converges to this value as $t \downarrow 1$. This diagnostic does not replace the proof.

Remark II.4.4 (Remaining questions). This result does not assert a full connection matrix for arbitrary Frobenius bases, a uniform error bound for numerical continuation, or a treatment of resonant $\gamma_p \in \mathbb{Z}$. Resonance can introduce logarithms and requires a separately normalized limiting basis; simply substituting resonant parameters into (II.4.14) would be invalid. The condition $b > 0$ has also been retained instead of asserting an unproved extension through divergent auxiliary integrals. A broader priority audit of general Selberg connection formulas remains necessary before any claim of historical first solution.

II.5 One hidden-refresh trajectory: sharp costs at both persistence boundaries

A population that is invisible from one observation can become identifiable through temporal dependence. For a fixed known emission family, the preceding three-visit construction gives a persistence-dependent rate from independent triples. This section removes that independence assumption: all observations now come from one stationary trajectory. Two distinct obstructions must be controlled simultaneously. As persistence tends to zero, the one-view defect reappears. As persistence tends to one, the trajectory contains too few fresh population draws. We obtain matching upper and lower bounds over an interval containing both boundaries, using an estimator that knows neither boundary margin. The proof of the small-persistence lower bound compares full-history filters; a calculation for isolated triples alone would not suffice.

II.5.1 A fixed analytic emission family and the statistical experiment

Let U be uniform probability on $[0, 1]$, and write $L_j(x) = P_j(2x-1)$ for the shifted Legendre polynomial of degree j. Set

$$v_j(\theta) = \theta^j - \frac{2\theta}{j+1}, \qquad k(x \mid \theta) = 1 + \sum_{j=2}^{\infty} \frac{L_j(x) v_j(\theta)}{8 j!}, \quad x, \theta \in [0, 1]. \tag{II.5.1}$$

The family k is known. Factorial decay makes the series normally convergent on every compact subset of $\mathbb{C}^2$, so it defines an entire function. The identities $\int L_j\, dU = 0$ and $\int v_j\, dU = 0$ show that

$$\int k(x \mid \theta)\, dU(x) = 1, \qquad k(x \mid 0) = 1, \qquad \int k(x \mid \theta)\, dU(\theta) = 1. \tag{II.5.2}$$

Since $|L_j| \le 1$ and $|v_j| \le 2$ on the real unit interval,

$$|k - 1| \le \frac{e-2}{4} < \frac{1}{4}, \qquad \frac{3}{4} \le k \le \frac{5}{4}. \tag{II.5.3}$$

In particular $k(\cdot \mid \theta)$ is a strictly positive probability density.

For $s \in [1/2, 1]$ let

$$G_s = (1 - s)\delta_0 + sU. \tag{II.5.4}$$

The latent chain is stationary with population G_s and transition kernel

$$M_{s,\rho}(\theta, d\eta) = \rho\delta_\theta(d\eta) + (1 - \rho)G_s(d\eta), \qquad 0 < \rho < 1. \tag{II.5.5}$$

Conditionally on the latent path, observations are independent with density $k(\cdot \mid \theta_i)$. Both s and ρ are unknown. The data consist of *one complete trajectory* $(X_1, \ldots, X_n)$; denote its law by $\mathsf{P}_{s,\rho}^{(n)}$. By (II.5.2), every individual observation has law U, independently of s and ρ.

We estimate the population in Wasserstein distance W_1 for the usual metric on $[0, 1]$. The one-dimensional distribution-function formula gives

$$W_1(G_s, G_t) = \frac{|s - t|}{2}. \tag{II.5.6}$$

The infima below allow arbitrary measurable probability-valued estimators, not just estimates within the family (II.5.4).

Theorem II.5.1 (Two-boundary minimax rate from one path)**.** *For $0 < r \le 1/8$, $0 < q \le 1/4$ and $n \ge 1$, define*

$$R_n(r, q) = \inf_{\widehat{G}} \sup_{\substack{s\in[1/2,1]\\ \rho\in[r,1-q]}} \mathbb{E}_{s,\rho} W_1(\widehat{G}, G_s).$$

There are constants $0 < c < C < \infty$, depending only on the fixed emission family, such that

$$c \min\left\{1, \frac{r^{-2} + q^{-1/2}}{\sqrt{n}}\right\} \le R_n(r, q) \le C \min\left\{1, \frac{r^{-2} + q^{-1/2}}{\sqrt{n}}\right\}. \tag{II.5.7}$$

A single estimator $\widehat{G}_n$, using neither r nor q, satisfies the stronger pointwise upper bound

$$\mathbb{E}_{s,\rho} W_1(\widehat{G}_n, G_s) \le C \min\left\{1, \frac{1}{\sqrt{n}\,\rho^2\sqrt{1 - \rho}}\right\}, \qquad s \in [1/2, 1], \quad 0 < \rho < 1. \tag{II.5.8}$$

The rate in (II.5.7) contains the sum, equivalently the maximum, of two boundary costs. A product $r^{-2}q^{-1/2}$ would lose this information: one value of ρ cannot lie close to both zero and one. The restrictions on r, q guarantee that both lower-bound experiments lie in the parameter interval; the upper bound applies throughout that interval, including its interior.

Corollary II.5.2 (Exact uniform-consistency criterion)**.** *For sequences $r_n \in (0, 1/8]$ and $q_n \in (0, 1/4]$, uniform consistent population estimation over the corresponding classes is possible if and only if*

$$nr_n^4 \longrightarrow \infty \qquad \text{and} \qquad nq_n \longrightarrow \infty.$$

Proof. Apply both sides of (II.5.7). A sum of two nonnegative sequences tends to zero exactly when both summands do. □

II.5.2 An estimator that needs neither persistence margin

Let $\phi = 80L_2$ and $a = 2/135$. Legendre orthogonality gives

$$\int \phi(x)k(x \mid \theta)\, dU(x) = v_2(\theta), \qquad \int v_2(\theta)^2\, dU(\theta) = \frac{1}{5} - \frac{1}{3} + \frac{4}{27} = a.$$

Also $v_2(0) = 0$ and $\int v_2\, dU = 0$. The refresh transition has powers $M_{s,\rho}^h = \rho^h I + (1-\rho^h)G_s$. Conditional independence of emissions therefore yields, for every integer $h \geq 1$,

$$\mathbb{E}_{s,\rho}\frac{\phi(X_i)\phi(X_{i+h})}{a} = s\rho^h. \tag{II.5.9}$$

For $n \geq 4$, put $m = n-2$ and define the overlapping-window averages

$$\widehat{V} = \frac{1}{m}\sum_{i=1}^{m}\frac{\phi(X_i)\phi(X_{i+1})}{a}, \qquad \widehat{W} = \frac{1}{m}\sum_{i=1}^{m}\frac{\phi(X_i)\phi(X_{i+2})}{a}.$$

Their mean vector is $(v, w) = (s\rho, s\rho^2)$.

Lemma II.5.3 (Variance of overlapping windows)**.** *There is a fixed constant C such that*

$$\mathbb{E}_{s,\rho}[(\widehat{V}-v)^2 + (\widehat{W}-w)^2] \leq \frac{C}{n(1-\rho)}. \tag{II.5.10}$$

Proof. Each summand is a bounded function of the block (X_i, X_{i+1}, X_{i+2}). For two such blocks with starting positions separated by $\ell \geq 3$, condition on the latent state at the right endpoint of the first block and the state at the left endpoint of the second. Across the intervening gap, the transition is

$$M_{s,\rho}^{\ell-2} = \rho^{\ell-2}I + (1-\rho^{\ell-2})G_s.$$

The second term produces the product of the two expectations. It follows that the absolute covariance is at most $C\rho^{\ell-2}$. Overlaps with $\ell \leq 2$ have covariance at most C. Summing the covariance series, and using $m \geq n/2$, proves (II.5.10). □

Choose a measurable minimizer $(\widehat{s}, \widehat{\rho})$ of

$$(\widehat{V} - s'\rho')^2 + (\widehat{W} - s'\rho'^2)^2 \quad \text{over} \quad (s', \rho') \in [1/2, 1] \times [0, 1]. \tag{II.5.11}$$

Existence follows from compactness and continuity; a fixed measurable tie-breaking rule supplies measurability. Set $\widehat{G}_n = G_{\widehat{s}}$. For $n < 4$, use a fixed member of the population family.

Lemma II.5.4 (A global inverse with the true persistence in its denominator)**.** *Let $s, s' \in [1/2, 1]$, $0 < \rho \leq 1$, $0 \leq \rho' \leq 1$, and put $(v, w) = (s\rho, s\rho^2)$ and $(v', w') = (s'\rho', s'\rho'^2)$. Then*

$$|s - s'| \leq \frac{4}{\rho^2}(|v - v'| + |w - w'|). \tag{II.5.12}$$

The candidate persistence ρ' may be zero.

Proof. The polynomial identities $v^2 = sw$ and $v'^2 = s'w'$ hold even at $\rho' = 0$. Consequently

$$w(s - s') = v^2 - v'^2 + s'(w' - w).$$

Since $w \geq \rho^2/2$, $v, v' \in [0, 1]$ and $s' \leq 1$, the claimed bound follows. □

Proof of the upper bounds in Theorem II.5.1. Least-squares projection makes the Euclidean error of the fitted moment vector at most twice the empirical error from the true moment vector. Lemma II.5.4, Cauchy–Schwarz and Lemma II.5.3 give

$$\mathbb{E}_{s,\rho}|\widehat{s} - s| \leq \frac{C}{\sqrt{n}\,\rho^2\sqrt{1-\rho}}.$$

The error is also bounded by $1/2$. Equation (II.5.6) proves (II.5.8), after increasing the constant to include $n < 4$. Crucially, the persistence in this bound is the true parameter, not a worst-case lower bound substituted before the variance calculation.

For $\rho \leq 1/2$, the right-hand side before clipping is at most $C/(\sqrt{n}\,r^2)$. For $\rho \geq 1/2$, it is at most $C/\sqrt{nq}$. Taking the supremum over $[r, 1-q]$, then using the bounded loss, proves the upper half of (II.5.7). □

II.5.3 Small persistence: a full-path information bound

The moment map has first coordinate $s\rho$. We compare two parameter points along a curve on which this coordinate is constant. The next lemma shows that the entire observed trajectory, including all its long-range dependence, has only fourth-order information in small persistence along that curve. Write $D(P\|Q)$ for relative entropy.

Lemma II.5.5 (Uniform filter comparison along $s\rho$ = constant)**.** *Suppose $s, s' \in [1/2, 1]$, $0 < \rho, \rho' \le 1/4$ and $\alpha = s\rho = s'\rho'$. Then, for every $n \ge 1$,*

$$D(\mathsf{P}_{s,\rho}^{(n)} \| \mathsf{P}_{s',\rho'}^{(n)}) \le Cn\alpha^2(\rho - \rho')^2. \tag{II.5.13}$$

The constant is independent of the time horizon and both parameters.

Proof. Put $q_x(\theta) = k(x \mid \theta) - 1$. For a fixed observed history $(x_1, \dots, x_t)$, let $f_t^{s,\rho}$ be the posterior density of θ_t relative to G_s. Use the recursive formula below to extend this density to a function on all of $[0, 1]$. With $f_0 = 1$, Bayes' formula gives

$$f_t^{s,\rho}(\theta) = \frac{k(x_t \mid \theta)[1 - \rho + \rho f_{t-1}^{s,\rho}(\theta)]}{p_t^{s,\rho}(x_t)}, \tag{II.5.14}$$

$$p_t^{s,\rho}(x) = 1 + s\rho \int q_x(\theta) f_{t-1}^{s,\rho}(\theta)\, dU(\theta). \tag{II.5.15}$$

Here $p_t^{s,\rho}$ is the conditional density of X_t given the preceding history. To check the second formula, integrate the numerator of the first before observing x_t. The constant part integrates to one because $\int f_{t-1}^{s,\rho}\, dG_s = 1$; the remaining part uses $\int q_x\, dG_s = 0$ and $q_x(0) = 0$. For $t = 1$, the convention $f_0 = 1$ gives the stationary initial law and $p_1 = 1$.

Every predictive distribution is a mixture of emission densities, so

$$\frac{3}{4} \le p_t^{s,\rho} \le \frac{5}{4} \tag{II.5.16}$$

for every history, including histories of probability zero. Indeed all formulas are pointwise defined with positive denominators. An induction in (II.5.14) gives

$$\|f_t^{s,\rho}\|_\infty \le 3. \tag{II.5.17}$$

If the preceding norm is at most three, the next one is at most $(5/3)(1 - \rho + 3\rho) \le 5/2 < 3$; the induction starts at $f_0 = 1$.

Compare both filters on the same arbitrary history, and write $\Delta_t = \|f_t^{s,\rho} - f_t^{s',\rho'}\|_\infty$. Equation (II.5.15), the fixed value of α, and $\|q_x\|_\infty \le 1/4$ imply

$$\sup_x |p_t^{s,\rho}(x) - p_t^{s',\rho'}(x)| \le \frac{\alpha}{4}\Delta_{t-1}. \tag{II.5.18}$$

If N, N' denote the numerators in (II.5.14), then

$$\|N - N'\|_\infty \le \frac{5}{4}(\rho\Delta_{t-1} + 4|\rho - \rho'|).$$

This follows by adding and subtracting $\rho f_{t-1}^{s',\rho'}$ and using (II.5.17). Using $N'/p' = f_t^{s',\rho'}$, whose norm is at most three, and (II.5.16), we obtain

$$\Delta_t \le \left(\frac{5}{3}\rho + \alpha\right)\Delta_{t-1} + \frac{20}{3}|\rho - \rho'| \le \frac{2}{3}\Delta_{t-1} + \frac{20}{3}|\rho - \rho'|.$$

The last inequality uses $\alpha \le \rho \le 1/4$. Since $\Delta_0 = 0$, geometric summation proves $\Delta_t \le 20|\rho - \rho'|$ uniformly in time and history. Therefore

$$\sup_x |p_t^{s,\rho}(x) - p_t^{s',\rho'}(x)| \le 5\alpha|\rho - \rho'|. \tag{II.5.19}$$

For two densities p, p' relative to the probability measure U, $D(p\|p') \le \int (p - p')^2/p'\, dU$. Equations (II.5.16) and (II.5.19) thus bound every conditional relative entropy by $C\alpha^2(\rho - \rho')^2$, uniformly in the preceding history. Summing by the chain rule for relative entropy gives (II.5.13). □

Proposition II.5.6 (The small-persistence obstruction)**.** *On the parameter class of Theorem II.5.1,*

$$R_n(r, q) \ge c \min\{1, (\sqrt{n}\, r^2)^{-1}\}. \tag{II.5.20}$$

Proof. Take

$$s_0 = \frac{3}{4}, \quad \rho_0 = \frac{3r}{2}, \qquad s_1 = s_0 + h, \quad \rho_1 = \rho_0 \frac{s_0}{s_1}, \qquad 0 < h \le \frac{1}{8}.$$

Both persistence values belong to $[r, 1 - q]$ and are at most $1/4$. Their products $s_i\rho_i$ agree, and $|\rho_0 - \rho_1| \le Crh$. Lemma II.5.5 gives

$$D(\mathsf{P}^{(n)}_{s_0,\rho_0} \| \mathsf{P}^{(n)}_{s_1,\rho_1}) \le Cnh^2r^4.$$

Choose $h = c_0 \min\{1, (\sqrt{n}\, r^2)^{-1}\}$ with a fixed sufficiently small $c_0 > 0$. Then the total variation distance of the two observed-path laws is at most $1/2$, by Pinsker's inequality. Their population separation is $h/2$ by (II.5.6). For any probability-valued estimator, the nearest-population test and the triangle inequality give a maximal expected loss at least one quarter of this separation times one minus the total variation distance. This proves (II.5.20). □

II.5.4 Nearly frozen paths: only fresh latent draws carry population information

Proposition II.5.7 (The nearly frozen obstruction)**.** *On the parameter class of Theorem II.5.1,*

$$R_n(r, q) \ge \frac{c}{\sqrt{1 + nq}} \ge c' \min\{1, (nq)^{-1/2}\}. \tag{II.5.21}$$

Proof. Fix $\rho = 1 - q$. Consider an oracle that reveals all refresh indicators, the initial latent value, and every newly drawn latent value. The oracle can simulate the whole observed path using a parameter-independent Markov kernel, so data processing bounds observed-path relative entropy by oracle relative entropy. The number of independent population draws revealed is

$$M = 1 + \mathrm{Binomial}(n - 1, q), \qquad \mathbb{E}M = 1 + (n - 1)q.$$

Refresh indicators have the same law at all values of s. Because δ_0 and U are mutually singular,

$$D(G_s\|G_t) = s \log \frac{s}{t} + (1 - s) \log \frac{1 - s}{1 - t}.$$

For $s_0 = 3/4$ and $s_1 = s_0 + h$, $0 < h \le 1/8$, this is at most Ch^2. Conditioning on the indicators and averaging the product-law relative entropies therefore gives

$$D(\mathsf{P}^{(n)}_{s_0,1-q} \| \mathsf{P}^{(n)}_{s_1,1-q}) \le C(1 + nq)h^2.$$

Choose $h = c_0/\sqrt{1 + nq}$ and apply the same two-point argument as before. The resulting bound is $c/\sqrt{1 + nq}$. This is comparable, with absolute constants, to $\min\{1, (nq)^{-1/2}\}$. □

Completion of Theorem II.5.1. The upper bound has already been proved. The maximum of the lower bounds in Propositions II.5.6 and II.5.7 is, within an absolute constant, their clipped sum. Explicitly, for $A, B \geq 0$,

$$\max\{\min(1, A), \min(1, B)\} \geq \tfrac{1}{2} \min(1, A + B).$$

Taking $A = (\sqrt{n}\, r^2)^{-1}$ and $B = (nq)^{-1/2}$ proves the lower half of (II.5.7). □

Remark II.5.8 (The two endpoints are different impossibility mechanisms). At $\rho = 0$, all observed paths are independent uniform samples for every s, so the population is unidentifiable. At $\rho = 1$, the oracle above reveals just one population draw, regardless of the length of the observed path. The same two-point argument then gives a positive constant minimax risk for every n. Thus the obstruction at one is lack of fresh population draws, even though the emission law of the retained latent state may be learned accurately from its long run.

II.5.5 Transfer conditions and precise relation to earlier work

Proposition II.5.9 (The same rate for other fixed emission families)**.** *Let Θ be a compact metric space, let $\theta_0 \in \Theta$, and let H be a probability measure with $H(\{\theta_0\}) = 0$. Let μ be a probability measure on the observation space. Suppose the known emission densities satisfy, pointwise,*

$$|k(x \mid \theta) - 1| \leq \frac{1}{4}, \qquad k(x \mid \theta_0) = 1, \qquad \int k(x \mid \theta)\, dH(\theta) = 1,$$
$$\int k(x \mid \theta)\, d\mu(x) = 1.$$

Assume there is a bounded measurable real function ϕ with $\int \phi\, d\mu = 0$ such that

$$b(\theta) = \int \phi(x) k(x \mid \theta)\, d\mu(x), \qquad a = \int b(\theta)^2\, dH(\theta) > 0.$$

Replace U by H and 0 by θ_0 in the population (II.5.4), and use the same refresh transition. Then Theorem II.5.1 and Corollary II.5.2 hold, with constants depending on this fixed experiment and its metric.

Proof. The assumptions give $b(\theta_0) = 0$ and $\int b\, dH = 0$. For $h = 1, 2$, the bounded statistic $\phi(X_i)\phi(X_{i+h})/a$ again has expectation $s\rho^h$. The covariance bound and moment inverse are unchanged. Also

$$W_1(G_s, G_t) = |s - t| \int d(\theta_0, \theta)\, dH(\theta).$$

The upper bound follows by coupling the excess mass between δ_{θ_0} and H, and the reverse inequality follows from the 1-Lipschitz function $d(\theta_0, \cdot)$. The displayed multiplier is strictly positive.

For the small-persistence lower bound, the filter proof is identical with H replacing U in latent integrals and μ replacing U in predictive relative entropies. Every numerical bound in that proof used only the stated bound $|k - 1| \leq 1/4$ and the two cancellation identities. Finally, $H(\{\theta_0\}) = 0$ gives the same Bernoulli relative entropy for population draws, proving the nearly frozen lower bound. This verifies all assertions. □

The direct advance over the independent-three-visit result is the matching rate for a single dependent path, including both persistence boundaries and an estimator that does not require their margins. The small-persistence lower bound required a separate full-path argument; it cannot be obtained by declaring overlapping triples independent or by substituting an informal effective sample size.

For comparison, Abraham, Gassiat and Naulet study minimax frontiers for nonparametric hidden Markov models with unknown emission densities [14]. Their Section 1.2 distinguishes the independence boundary from an absolute spectral-gap condition, and their informal Theorem 1 allows constants to depend on that gap. Section 4 states that exact dependence on the gap was not its focus, and poses a different adaptation question concerning emission densities of different smoothness. The present theorem does not resolve that question: the emission family is fixed and known, the population submodel is one-dimensional, and transitions have the restricted refresh form (II.5.5). Within this specified experiment, both boundary dependences in (II.5.7) are proved sharply.

Although the population has only one scalar unknown coefficient, the latent state is continuous. A value drawn from the continuous part retains its own emission law throughout a persistence run. Collapsing all such values into one of two state labels and assigning that label a fixed emission law would remove the within-run dependence and would change the experiment.

II.6 What has and has not been resolved

The double-broom construction is an explicit obstruction to the logarithmic Lewis-weight claim under the source's own normalization. The result determines the worst tree asymptotic, including its leading coefficient and constant term. It does not determine the worst asymptotic over all connected graphs.

For alternating binomial sums, the lower bound holds for every integer exponent and for coefficients in $\mathbb{C}(n)$, with no coefficient-degree restriction. The matching upper bound is credited to McIntosh. We remove the restriction in his theorem; we do not assert that an exhaustive survey has certified the absence of every subsequent proof.

For Jacobi traces, the conclusion gives the scalars for the physical density in the stated nonresonant range. Resonance, the remaining endpoint parameter range, and stable numerical continuation remain separate tasks. For the hidden-population model, both boundary rates concern the explicitly defined known emission family and population class. Unknown emissions and general transition kernels are outside the theorem.

The heterogeneous binomial-mixture NPMLE was also examined during target selection. A posterior-coupling bound works for comparable observation depths, but a matching all-profile, all-range theorem was not obtained. It is therefore not counted among the four completed results here.

Part III

Entropy, Likelihood Geometry, and Sharp Proof Transfers

Manuscript overview. We transfer entropy–transport and singularity methods to five specified problems. For binomial population recovery, every ordinary unrestricted nonparametric maximum-likelihood estimator attains the sharp Wasserstein risk under an unknown fraction of adversarially replaced reports; the estimator requires no modification and no contamination parameter. A second theorem proves uniform minimax optimality for a joint likelihood with arbitrarily many depth-one observations and any number of observations at a common deeper level. Its key identity is an affine posterior tilt, which repairs the failure of stratumwise normalization. For exponential families, likelihood contact gives a conditional moment-generating-function bound and an entropy bound for posterior barycenter movement. Gaussian and Poisson applications give martingale couplings and high-information minimax regimes, with explicit boundary and approximation limitations. Beyond statistics, the worst D-optimal/A-optimal design ratio over all connected graphs is of order $\sqrt{n}$: an electrical maximum-principle argument yields a universal upper bound $\sqrt{n/8}+1/2$, and complete-graph barbells improve the preceding tree lower constant. Finally, the minimum recurrence width of phase-weighted central binomial power sums is classified on the entire unit circle: it is r at every nonreal phase and drops at the two real phases. A polynomial moment reduction supplies the matching upper bound. All five developments distinguish previously proved ingredients from new arguments and state the unresolved extensions explicitly.

Integration note. The robust binomial, unequal-depth, and posterior-contact results retain their own hypotheses. The phase-weight recurrence result is extended in Parts VII and VIII. The all-connected-graph result complements the more specialized extremal theorems in Parts II, IV, and V.

III.1 Questions closed and methods transferred

The statistical starting point is the entropy–transport argument in the supplied observation compilation. Likelihood scores define a posterior coupling; a relative-entropy decomposition then converts componentwise concentration into control of the fitted distribution. This transfer does not assert that an arbitrary mixture inherits its components' concentration. The new arguments address three obstructions to a direct application: adversarial selection of observations, failure of normalization within an observation stratum, and infinite relative entropy between an empirical measure and a continuous density.

1. **Ordinary likelihood under report corruption.** The supplied compilation already establishes a sharp robust rate using different estimators, but expressly leaves ordinary NPMLE robustness unproved. We establish that property, including strong, data-dependent replacements and adaptation to their unknown budget. A bounded retained-sample score gives a linear posterior repair cost. Uniform likelihood events over all retained subsets handle the adversary without an invalid iid assumption.

2. **Arbitrary imbalance between two observation depths.** For depths 1 and T, the shallow likelihood forces the deep posterior update to be an affine tilt of the fitted mixing law. Its transport cost is exactly its change of mean. This restores a uniform predictive bound and matches lower bounds whose alternatives have identical means. The number of shallow observations may be arbitrarily large.

3. **Likelihood contact in continuous and count families.** A natural-parameter perturbation turns the global likelihood score condition into a conditional moment-generating-function inequality. Entropy is compared between two weightings of the same observed sample, so no empirical-to-density KL divergence is used. Boundary-constrained fits require a separate clipping argument.

4. **All connected graph topologies.** The previous paper settled the tree asymptotic and refuted the logarithmic conjecture stated by Suliman and Weinstein. The present extension determines the correct worst-case growth order over all connected graphs. The exact leading constant in that larger class is still undetermined.

5. **The complete unit-circle phase family.** The Franel singularity method and the preceding alternating-sum calculation extend to all complex phases of modulus one. A new finite-dimensional moment reduction supplies the generic upper recurrence. This is a self-posed parameter classification, not an asserted solution of a named external open problem.

These are five related developments, not five independent claims to have settled famous conjectures. Their precise statements and proofs follow. The binomial approximation and likelihood-cover tools are reproduced in Section III.A for independent checking; those supporting lemmas are credited to the preceding compilation and are not new claims.

III.2 The ordinary mixture MLE withstands unknown adversarial report replacement

The sharp robust population-recovery rate was previously attained by minimum-distance estimators. The preceding compilation explicitly left open whether the ordinary, unmodified nonparametric maximum likelihood estimator attains that rate [20, Parts VII–VIII]. We answer that question affirmatively, including replacement locations chosen after inspecting the entire clean sample. The contamination budget is not supplied to the estimator. The rate itself and its

lower bound are prior results; the new conclusion concerns the ordinary MLE and a certificate sufficient for its numerical approximation.

Write

$$K_\theta(j) = \binom{t}{j}\theta^j(1-\theta)^{t-j}, \qquad p_G(j) = \int_0^1 K_\theta(j)\,dG(\theta), \quad 0 \le j \le t.$$

On the count alphabet, $W_{1,t}$ uses cost $|i-j|/t$; population W_1 uses the ordinary distance on $[0,1]$. Total variation is one half of the ℓ_1 distance, and $h(p,q)^2 = \sum_j(\sqrt{p_j} - \sqrt{q_j})^2$. Set $\log(e/0) = +\infty$ and

$$r_{N,t}(\varepsilon) = \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{t\log(eN)}} + \frac{1}{\sqrt{N}} + \varepsilon + \frac{1}{\sqrt{t\log(e/\varepsilon)}}\right\}. \tag{III.2.1}$$

Theorem III.2.1 (Sharp robustness of the unmodified NPMLE)**.** *Let $J_1, \dots, J_N$ be independent with common law p_G, where G is an arbitrary probability law on $[0,1]$. A possibly randomized adversary sees the entire clean sample and replaces at most $\lfloor \varepsilon N \rfloor$ reports by arbitrary values in $\{0,\dots,t\}$, where $0 \le \varepsilon \le 1/4$. Every measurable choice $\widehat{G}$ of an unrestricted mixing MLE of the corrupted reports satisfies*

$$\sup_G \sup_A \mathbb{E} W_1(\widehat{G}, G) \le C r_{N,t}(\varepsilon). \tag{III.2.2}$$

The corresponding minimax risk over all estimators is at least $c r_{N,t}(\varepsilon)$. The constants are absolute, uniformly in $N, t \ge 1$. Thus the ordinary MLE adapts to the unknown replacement budget and attains the sharp robust rate.

The same upper bound holds for any fitted law H whose corrupted-data score satisfies

$$\sup_{\theta\in[0,1]} \sum_{j:q(j)>0} \frac{q(j)K_\theta(j)}{p_H(j)} \le 1 + \frac{1}{N}, \tag{III.2.3}$$

with $p_H > 0$ on the observed support. Here q is the empirical law of the corrupted reports.

This result concerns replacement of whole reports, not a budget of individual Bernoulli-bit changes. Unrestricted mixing is essential to the score condition; no claim is made for an MLE with a prescribed number of mixture components. The clean population problem was studied by Vinayak, Kong, Valiant and Kakade [70], with subsequent Chebyshev-moment improvements by Musco, Musco, Rosenblatt and Singh [16]. General contamination lower bounds based on total variation moduli are classical [17]. We do not attribute the estimator-specific question here to a named conjecture in those papers.

III.2.1 A retained-sample score and entropy–transport repair

For a probability vector a on the count alphabet, put $\ell_a(H) = \sum_j a(j)\log p_H(j)$, with zero terms omitted. The following finite-space lemma isolates the mechanism. Its likelihood geometry and entropy–transport duality are classical [18, 19]; the use of a retained subsample and a linear repair cost is what is needed below.

Lemma III.2.2 (Score-controlled posterior repair)**.** *Let $(\mathcal{X}, d)$ be a finite metric space of diameter one, and suppose the component kernels satisfy*

$$W_{1,d}(R, K_\theta)^2 \le 2cD(R\|K_\theta).$$

Let H be a mixing law with $p_H > 0$ on the support of a, and suppose

$$S_a(\theta) = \sum_{x:a(x)>0} \frac{a(x)K_\theta(x)}{p_H(x)} \le \sigma, \qquad \sigma \ge 1.$$

Then

$$\sup_F \ell_a(F) - \ell_a(H) \le \log \sigma, \tag{III.2.4}$$

$$W_{1,d}(a, p_H) \le \sqrt{2cD(a\|p_H)} + 1 - \sigma^{-1}. \tag{III.2.5}$$

Proof. Since $\int S_a \, dH = 1$, the measure $dH^+ = S_a \, dH$ is a probability law. For $S_a(\theta) > 0$, define

$$R_\theta(x) = \frac{a(x)K_\theta(x)}{p_H(x)S_a(\theta)}.$$

It is a probability vector and $\int R_\theta \, dH^+ = a$. Direct expansion of the logarithms gives

$$D(a\|p_H) = D(H^+\|H) + \int D(R_\theta\|K_\theta) \, dH^+(\theta). \tag{III.2.6}$$

Zero-density terms are omitted. Convexity of transportation cost under mixtures, the component inequality, and Cauchy–Schwarz give $W_{1,d}(a, p_{H^+}) \le \sqrt{2cD(a\|p_H)}$. Also, $(u-1)_+ \le (1-\sigma^{-1})u$ for $0 \le u \le \sigma$, so

$$\|H^+ - H\|_{\mathrm{TV}} = \int (S_a - 1)_+ \, dH \le 1 - \sigma^{-1}.$$

Coupling their common mixing mass costs at most this amount because the observation diameter is one. This proves (III.2.5). Finally, Jensen's inequality gives, for every F,

$$\ell_a(F) - \ell_a(H) \le \log \sum_x a(x) \frac{p_F(x)}{p_H(x)} = \log \int S_a \, dF \le \log \sigma.$$

□

For the normalized binomial count metric the component constant is $c = 1/(4t)$. Indeed, a function with count-grid Lipschitz constant one changes by at most $1/t$ when one of the t underlying Bernoulli bits is changed. Conditional Hoeffding inequalities for the revealing martingale imply

$$\log \mathbb{E}_{K_\theta} e^{\lambda(f - \mathbb{E}_{K_\theta} f)} \le \frac{\lambda^2}{8t}.$$

The entropy variational inequality, optimized over $\lambda > 0$, and Wasserstein duality therefore give

$$W_{1,t}(R, K_\theta)^2 \le \frac{D(R\|K_\theta)}{2t}. \tag{III.2.7}$$

This argument also covers the degenerate endpoint kernels whenever the relative entropy is finite.

Now put $k = \lfloor \varepsilon N \rfloor$, $\delta = k/N$, and $n = N - k$. Pad the actual changed set to exactly k indices. Let I be its complement, q_I the empirical law of the retained clean observations, q_J that of the k removed observations, and q^0 the full clean empirical law. When $k = 0$, terms involving q_J are absent. Then

$$q \ge (1-\delta)q_I \quad \text{coordinatewise}, \qquad q^0 = (1-\delta)q_I + \delta q_J. \tag{III.2.8}$$

No independence of the data-selected retained sample is assumed.

The MLE exists: the binomial moment body is compact and contains a vector positive at every cell, so the likelihood maximum is finite and attained. Its directional score satisfies $S_q(\theta) \le 1$. More generally, suppose $S_q \le 1 + \zeta$. Equation (III.2.8) implies

$$S_{q_I} \le \sigma_\delta := \frac{1+\zeta}{1-\delta}, \qquad e_\delta := \log \sigma_\delta, \qquad a_\delta := 1 - \sigma_\delta^{-1} = \frac{\delta + \zeta}{1+\zeta}. \tag{III.2.9}$$

Consequently the fit is within e_δ of the optimal retained normalized likelihood, and its transportation repair costs at most a_δ, not a square root of the contamination fraction.

Lemma III.2.3 (Uniform predictive transportation under deletions)**.** *Let $0 \le \delta \le 1/4$ and $0 \le \zeta \le 1/N$. Every fit satisfying $S_q \le 1 + \zeta$ obeys, uniformly over the replacement adversary,*

$$\mathbb{E}W_{1,t}(p_H, p_G) \le C \left\{ N^{-1/2} + \delta + \sqrt{\frac{b(\delta)}{t}} \right\}, \qquad b(u) = -u \log u - (1-u)\log(1-u). \tag{III.2.10}$$

Here $b(0) = 0$.

Proof. Introduce the binary label specifying which part of (III.2.8) generated a count. Its mutual information with the count is at most $b(\delta)$. The entropy chain rule gives

$$(1-\delta)D(q_I \| p_G) + \delta D(q_J \| p_G) \le D(q^0 \| p_G) + b(\delta). \tag{III.2.11}$$

All clean observations lie in the support of p_G almost surely, so this identity requires no lower bound on its nonzero cell probabilities. By (III.2.4),

$$D(q_I \| p_H) \le D(q_I \| p_G) + e_\delta.$$

Apply (III.2.5) with (III.2.7). Since $W_{1,t}(q_I, q^0) \le \delta$, it follows samplewise that

$$\begin{aligned} W_{1,t}(p_H, p_G) \le{}& W_{1,t}(q^0, p_G) + \delta + a_\delta \\ &+ \sqrt{\frac{1}{2t}\left\{ \frac{D(q^0 \| p_G) + b(\delta)}{1-\delta} + e_\delta \right\}}. \end{aligned} \tag{III.2.12}$$

The full clean empirical law satisfies

$$\mathbb{E}D(q^0 \| p_G) \le \mathbb{E}\chi^2(q^0, p_G) \le t/N, \qquad \mathbb{E}W_{1,t}(q^0, p_G) \le 1/(2\sqrt{N}).$$

The second bound follows by integrating empirical-CDF standard deviations on the normalized interval. Jensen's inequality, $e_\delta \le C\delta + \zeta$, $a_\delta \le \delta + \zeta$, and $\delta \le Cb(\delta)$ prove (III.2.10). Crucially, (III.2.12) holds for every retained subset on the same clean sample; it is not an expectation conditional on an adaptively chosen subset. □

III.2.2 Likelihood envelopes uniform over the deletion set

We record the approximation inputs from the earlier clean likelihood analysis [20, Parts VI–VIII]. Their full proofs are reproduced in Section III.A; they concern only the binomial family, not the corruption mechanism, and are not new claims.

Lemma III.2.4 (Binomial cover and two-scale inversion)**.** *For $t \le N$, $u = (eN)^{-2}$, and $d = t+1$, the binomial mixture moment body has an ℓ_1 cover of radius u/d, with centers in that body and cardinality satisfying*

$$\log M \le C\sqrt{t}\,\log(eN)^{3/2}. \tag{III.2.13}$$

For arbitrary mixing laws F, G and integers $1 \le S \le K \le t/2$,

$$W_1(F,G) \le \frac{C}{K} + Ce^{4S^2/t} W_{1,t}(p_F, p_G) + \frac{Ce^{2K^2/t}}{S} h(p_F, p_G). \tag{III.2.14}$$

Proof. For (III.2.13), take $\beta = u/(t+1)$ in Lemma III.A.1; its degree cutoff is $O(\sqrt{t\log(eN)})$ when $t \le N$. Equation (III.2.14) is Proposition III.A.4, whose proof uses the derivative-stable approximation and coefficient estimates in Lemma III.A.3. The weighted Chebyshev-moment approach is also developed in [16]. □

Lemma III.2.5 (Hellinger control for all retained samples)**.** *Under the hypotheses of Lemma III.2.3, if $t \le N$, then*

$$\mathbb{E}h(p_G, p_H) \le C\left\{\frac{t^{1/4}\log(eN)^{3/4}}{\sqrt{N}} + \sqrt{b(\delta)}\right\}. \tag{III.2.15}$$

The bound is uniform over every data-dependent choice of the changed indices and every corresponding score-certified fit.

Proof. Let $p_1, \dots, p_M$ be the cover from Lemma III.2.4, and set $u_a(j) = p_a(j) + u/d$. Every model law is dominated coordinatewise by at least one u_a, and $\sum_j u_a(j) = 1 + u$. If $p_H \le u_a$, then $h(p_H, u_a)^2 \le u$.

Fix a deterministic index set I of size $n = N - k$. For every fitted law satisfying $\ell_{q_I}(H) \ge \ell_{q_I}(G) - e_\delta$, domination implies

$$\prod_{i\in I} \frac{u_a(J_i)}{p_G(J_i)} \ge e^{-ne_\delta}.$$

Under the original clean experiment these n observations are iid. Markov's inequality with exponent $1/2$ gives

$$\begin{aligned}\mathbb{P}\left\{\prod_{i\in I} \frac{u_a(J_i)}{p_G(J_i)} \ge e^{-ne_\delta}\right\} &\le e^{ne_\delta/2}\left(\sum_j \sqrt{p_G(j)u_a(j)}\right)^n \\ &\le \exp\left\{\frac{n}{2}(e_\delta + u - h(p_G, u_a)^2)\right\}.\end{aligned}$$

Ratios need be defined only on the support of p_G. Take a union over the M envelopes and all $\binom{N}{k}$ index sets. Except on an event of probability e^{-x}, simultaneously for all these subsets and all such fits,

$$h(p_G, p_H) \le \sqrt{u} + \sqrt{e_\delta + u + \frac{2\{\log M + \log\binom{N}{k} + x\}}{N-k}}. \tag{III.2.16}$$

This is the step allowing adversarial selection of the retained set. There is no assertion that it remains iid after that selection.

Since $\log\binom{N}{k} \le Nb(\delta)$, $N - k \ge 3N/4$, and $e_\delta \le Cb(\delta) + \zeta$, integrating (III.2.16) and using (III.2.13) proves (III.2.15). The additional terms u, $1/N$, and $\zeta \le 1/N$ are absorbed by its sampling term. □

III.2.3 Proof of the sharp upper bound and matching lower bound

Upper bound in Theorem III.2.1. First let $t \le N$. Bounded t is covered by increasing the constant in (III.2.2). For larger t put

$$L_N = \log(eN), \quad L_\delta = \log(e/\delta), \quad L = \min(L_N, L_\delta), \quad a = 1/64,$$

and choose

$$K = \left\lfloor \min\{t/2, \sqrt{atL}\} \right\rfloor, \qquad S = \lfloor \sqrt{t}/16 \rfloor.$$

For sufficiently large absolute t these satisfy $1 \le S \le K \le t/2$, $S \asymp \sqrt{t}$, and $e^{4S^2/t} \le C$. Also

$$\frac{1}{K} \le C\left\{\frac{1}{t} + \frac{1}{\sqrt{tL_N}} + \frac{1}{\sqrt{tL_\delta}}\right\}. \tag{III.2.17}$$

Apply (III.2.14), then Lemmas III.2.3 and III.2.5. The high-frequency sampling term is at most

$$\frac{C(eN)^{2a}L_N^{3/4}}{\sqrt{N}\,t^{1/4}} \le \frac{C}{\sqrt{tL_N}}.$$

Indeed, its ratio to the last expression is bounded by $C(eN)^{2a}N^{-1/4}L_N^{5/4}$ because $t \le N$, and this is uniformly bounded as $2a < 1/4$.

For $0 < \delta \le 1/4$ we have $b(\delta) \le C\delta L_\delta$. Consequently

$$\sqrt{\frac{b(\delta)}{t}} \le \frac{C}{\sqrt{tL_\delta}}, \qquad \frac{e^{2aL_\delta}\sqrt{b(\delta)}}{\sqrt{t}} \le \frac{C}{\sqrt{tL_\delta}}. \tag{III.2.18}$$

After multiplying by $\sqrt{tL_\delta}$, the two left sides are bounded respectively by $C\sqrt{\delta}L_\delta$ and $C\delta^{1/2-2a}L_\delta$, uniformly in this interval. At $\delta = 0$ the entropy contributions vanish. Thus (III.2.17)–(III.2.18) prove the upper bound with δ in place of ε.

If $t > N$, the elementary component coupling gives

$$W_1(H, G) \le \frac{1}{\sqrt{t}} + W_{1,t}(p_H, p_G).$$

Here the normalized count law of each mixture is coupled to its latent parameter at expected cost at most $1/(2\sqrt{t})$. The boundedness of $b(\delta)$ and (III.2.10) now give $\mathbb{E}W_1(H, G) \le C(N^{-1/2} + \delta)$. Finally, $\delta \le \varepsilon$ and all terms in (III.2.1) are nondecreasing in the contamination argument. The loss is at most one, so clipping gives the asserted bound. The proof allowed $\zeta \le 1/N$ throughout, establishing the certificate assertion as well. □

Matching lower bound. We recall the existing lower-bound construction to make the matching claim explicit. The clean experiment already gives

$$c \min\left\{1, t^{-1} + [t\log(eN)]^{-1/2} + N^{-1/2}\right\}.$$

The first term follows from mixing laws matching moments through degree t. For the second, let $\pi(dx) = dx/(\pi\sqrt{x(1-x)})$ and $G_m^{\pm} = (1 \pm T_m(2x-1)/2)\pi$. Their population distance is comparable to $1/m$. The beta–binomial conditional-expectation operator has squared singular value

$$\lambda_{t,m} = \prod_{j=0}^{m-1} \frac{t-j}{t+1+j} \le e^{-m^2/(2t)}$$

on this Chebyshev mode, and zero when $m > t$. Choosing m a sufficiently large multiple of $\sqrt{t\log(eN)}$, or using exact moment matching if it exceeds t, makes the product experiments close and proves the inverse sampling term. Endpoint mixing laws with masses $1/2 \pm c/\sqrt{N}$ prove the parametric term.

The same witnesses, together with an endpoint mass change of size η, prove the sharp total-variation inverse lower bound

$$\sup_{\|p_F - p_G\|_{\mathrm{TV}} \le \eta} W_1(F, G) \ge c \min\left\{1, t^{-1} + [t\log(e/\eta)]^{-1/2} + \eta\right\}. \tag{III.2.19}$$

Indeed the observed TV of the Chebyshev witness is at most $C\sqrt{\lambda_{t,m}}$, so $m \asymp \sqrt{t\log(e/\eta)}$ suffices. These are the lower witnesses in [20, Parts VII–VIII]; no estimator-specific assumption enters them.

If $k \ge 1$, use (III.2.19) with $\eta = k/(16N)$. Independently maximally couple N observations from the resulting p_F, p_G. The expected number of unequal reports is at most $k/16$, and its probability of exceeding k is at most $1/16$. In the F experiment an adversary draws the conditionally coupled G sample, makes all replacements if their number is at most k, and otherwise leaves the sample unchanged. Its output law is within TV $1/16$ of the clean G experiment. This adversary always respects the hard budget. Le Cam's two-point inequality proves the contamination lower bound, since $k/N \ge \varepsilon/2$ whenever $k = \lfloor \varepsilon N \rfloor \ge 1$. If $k = 0$, then $\varepsilon < 1/N$ and both contamination terms in (III.2.1) are bounded by clean terms. Combining the lower bounds and taking a clipped maximum, equivalent to the clipped sum, completes the proof. □

III.2.4 Independent Huber contamination and the meaning of the result

Corollary III.2.6 (The same MLE adapts to Huber contamination)**.** *Suppose the reports are independent with common law* $(1-\varepsilon)p_G + \varepsilon Q$*, where* Q *is any law on the count alphabet and* $0 \le \varepsilon \le 1/4$*. Every ordinary NPMLE, and every fit satisfying* (III.2.3)*, has worst-case expected population loss at most* $Cr_{N,t}(\varepsilon)$*. The minimax risk is at least* $cr_{N,t}(\varepsilon)$*.*

Proof. Condition on the contamination indicators. Their count K is binomial with parameters N, ε. Fill the corrupt positions with independent unobserved clean reports; conditional on the indicator set, the experiment is a replacement experiment with budget K. Theorem III.2.1 applies when $K \le N/4$. Put $\Delta = K/N$, $\Delta' = \min(\Delta, 1/4)$ and $g(x) = [t\log(e/x)]^{-1/2}$, with $g(0) = 0$. This function is increasing and concave on $[0, 1/4]$: its second derivative has the sign of $3/2 - \log(e/x)$, which is negative there. Therefore

$$\mathbb{E}g(\Delta') \le g(\mathbb{E}\Delta') \le g(\varepsilon), \qquad \mathbb{E}\Delta = \varepsilon, \qquad \mathbb{P}\{\Delta > 1/4\} \le 4\varepsilon.$$

Bound the loss by one on the last event and average the theorem on its complement. This proves the upper bound without a logarithmic loss from random-budget control.

For the lower bound, the clean experiment is a subcase. Huber neighborhoods of two laws intersect if their TV distance is at most $\varepsilon/(1-\varepsilon)$: the vector $(1-\varepsilon)\max(p_F, p_G)$ has mass at most one and can be completed to a common probability law. Apply (III.2.19) and the triangle inequality for population loss. □

The proof does not make arbitrary mixture MLEs robust in arbitrary models. It uses unrestricted mixing, bounded count geometry, component concentration at scale $t^{-1/2}$, and a two-scale inverse bound specific to the binomial operator. The entropy–transport step is substantive: a likelihood-deficit bound alone repairs the mixing law at order $\sqrt{\varepsilon}$, whereas the retained-score bound repairs it at order ε. The remaining binary-entropy cost is carried by the high-frequency inverse term and is absorbed in the sharp contamination modulus.

Thus the advance is an estimator theorem closing the explicitly recorded unmodified-MLE question. It is not a rederivation of the already known robust minimax curve, and it is not claimed as a resolution of a general external robust-mixture conjecture. A bounded check of the primary sources above did not identify this precise strong-replacement theorem; that check is not a proof of historical priority.

III.3 A sharp two-depth binomial NPMLE theorem from affine posterior repair

The homogeneous binomial-mixture likelihood problem asks whether an unrestricted mixing-measure maximum likelihood estimator attains the optimal population-recovery rate at every replication depth. The preceding integrated manuscript proves that assertion using likelihood transportation and a two-scale inverse inequality. Heterogeneous replication introduces a separate obstruction: likelihood optimality normalizes an aggregate posterior score, but generally does not normalize the score in any one replication stratum.

We resolve a nontrivial heterogeneous case with arbitrary imbalance. There may be any number of depth-one observations, together with n observations of a common, unrestricted depth T. Every maximizer of their joint likelihood attains the same worst-case order as the optimal estimator based on the n depth-T observations alone. The new argument identifies an affine posterior tilt and controls its Wasserstein cost exactly by its change in mean.

III.3.1 Experiment and result

For a mixing law $F \in \mathcal{P}([0,1])$, define

$$\mu_F = \int \theta \, dF(\theta), \qquad p_F(j) = \int B_j^T(\theta) \, dF(\theta), \qquad B_j^T(\theta) = \binom{T}{j} \theta^j (1-\theta)^{T-j}.$$

Observe $m \geq 0$ depth-one variables $Y_1, \ldots, Y_m$ and $n \geq 1$ depth-T counts $X_1, \ldots, X_n$, where $T \geq 1$. Every observation has its own independent latent parameter drawn from the same unknown law G. Thus the Y_i are independent Bernoulli variables of parameter μ_G, the X_i are independent with law p_G, and the two samples are independent. The joint log likelihood is

$$L(F) = \sum_{i=1}^{n} \log p_F(X_i) + \sum_{i=1}^{m} \{Y_i \log \mu_F + (1 - Y_i) \log(1 - \mu_F)\}. \tag{III.3.1}$$

Let H be any measurable choice of an unrestricted maximizer. Existence follows by maximizing the continuous likelihood on the compact space of mixing laws; the competitor $\delta_{1/2}$ gives positive likelihood, so all fitted probabilities at observed cells are positive.

Write $W_{1,T}$ for Wasserstein distance between count laws with cost $|j - l|/T$, and use W_1 for the usual cost on $[0,1]$. Our Hellinger convention is $h(p,v)^2 = \sum_j (\sqrt{p(j)} - \sqrt{v(j)})^2$, including nonnormalized nonnegative v. Relative entropy is denoted by D; D_B denotes its Bernoulli specialization.

Theorem III.3.1 (All-depth optimality under arbitrary two-stratum imbalance)**.** *Put*

$$r_{n,T} = \min \left\{ 1, \frac{1}{T} + \frac{1}{\sqrt{T \log(en)}} + \frac{1}{\sqrt{n}} \right\}.$$

There are absolute constants $0 < c < C < \infty$ such that, for every $m \geq 0$, $n, T \geq 1$,

$$c r_{n,T} \leq \inf_{\widetilde{G}} \sup_{G} \mathbb{E}_G W_1(\widetilde{G}, G) \leq \sup_{G} \mathbb{E}_G W_1(H, G) \leq C r_{n,T}. \tag{III.3.2}$$

The infimum allows all estimators using both samples. The upper bound holds for every measurable joint NPMLE, with constants independent of the number m of depth-one observations. In particular, it covers $m \gg n$, $T \gg n$, and the transition $T \asymp n/\log(en)$.

The number of depth-one observations can substantially improve mean estimation. The theorem says that this improvement does not change the worst-case order for recovering the whole mixing law; all lower-bound pairs below have exactly the same mean. The upper bound concerns the actual joint maximizer, rather than an estimator that discards the shallow sample.

III.3.2 The likelihood advantage of the shallow sample

Let q be the empirical law of the depth-T sample, set $\mu_0 = \mu_G$, and, for $m > 0$, write $z = m^{-1} \sum_i Y_i$. Define

$$A = m D_B(z \| \mu_0),$$

with $A = 0$ if $m = 0$ and all expressions involving z omitted in that case. This is precisely the maximal possible shallow-sample log-likelihood gain over its value at the truth. Joint optimality therefore implies

$$n \sum_j q_j \log p_H(j) \geq n \sum_j q_j \log p_G(j) - A, \qquad D(q \| p_H) \leq D(q \| p_G) + A/n. \tag{III.3.3}$$

The elementary inequality $D \le \chi^2$ and the multinomial variance identity give

$$\mathbb{E}_G A \le 1, \qquad \mathbb{E}_G D(q\|p_G) \le T/n, \qquad \mathbb{E}_G D(q\|p_H) \le (T+1)/n. \tag{III.3.4}$$

For example, if $0 < \mu_0 < 1$ and $m > 0$, $A \le m(z-\mu_0)^2/[\mu_0(1-\mu_0)]$, whose expectation is one. True zero-probability cells are omitted from the multinomial identity. At $\mu_0 \in \{0,1\}$ the sample is deterministic and the fitted law is the same endpoint point mass, so the risk is zero. Hence this boundary case also causes no exception.

III.3.3 The affine posterior tilt

Define the depth-T score and its posterior update by

$$S_T(\theta) = \sum_{j:q_j>0} \frac{q_j B_j^T(\theta)}{p_H(j)}, \qquad dH^+(\theta) = S_T(\theta)\, dH(\theta). \tag{III.3.5}$$

The measure H^+ is always a probability measure, since $\int S_T\, dH = 1$. It need not equal H.

Lemma III.3.2 (Exact cost of repairing the deep stratum)**.** *Let $\mu = \mu_H$ and $\mu^+ = \mu_{H^+}$. If $0 < \mu < 1$ and $m > 0$, then*

$$dH^+(\theta) = \{1 - b(\theta-\mu)\}\, dH(\theta), \qquad b = \frac{m}{n}\frac{z-\mu}{\mu(1-\mu)}. \tag{III.3.6}$$

Consequently

$$W_1(H, H^+) = |\mu - \mu^+| \le |\mu^+ - \mu_0| + w|z-\mu_0|, \qquad w = \min\{1, m/n\}. \tag{III.3.7}$$

When $m = 0$, take $w = 0$ and omit z; then $H^+ = H$. If the fitted mean is an endpoint, $H^+ = H$ also holds directly.

Proof. The depth-one score is

$$S_1(\theta) = \frac{z\theta}{\mu} + \frac{(1-z)(1-\theta)}{1-\mu} = 1 + \frac{z-\mu}{\mu(1-\mu)}(\theta-\mu).$$

The likelihood directional derivative in the direction $\delta_\theta - H$ gives

$$nS_T(\theta) + mS_1(\theta) \le n + m.$$

Its integral against H is $n+m$, so equality holds H-almost surely. Solving for S_T on this full-measure set proves (III.3.6). No individual-stratum score normalization is assumed.

For every $x \in [0,1]$,

$$F_{H^+}(x) - F_H(x) = -b \int_{[0,x]} (\theta - \mu)\, dH(\theta).$$

The integral on the right is nonpositive: it decreases up to μ and then increases to its full value zero. Hence the distribution-function difference has a constant sign. The CDF formula for W_1 proves $W_1(H, H^+) = |\mu - \mu^+|$.

Integrating (III.3.6) against θ gives

$$\mu^+ = \mu - a(z-\mu), \qquad a = \frac{m}{n}\frac{\mathrm{Var}_H(\theta)}{\mu(1-\mu)} \in [0, m/n].$$

Thus $\mu = (\mu^+ + az)/(1+a)$, and

$$|\mu - \mu^+| = \frac{a}{1+a}|z - \mu^+| \le w|z-\mu_0| + |\mu^+ - \mu_0|.$$

For $m = 0$, the joint score condition is $S_T = 1$ H-almost surely. If μ is an endpoint, H is the corresponding point mass; finite likelihood forces all depth-T observations to be that endpoint count, and $S_T = 1$ on the support of H. These cases establish the remaining assertions without division by zero. □

III.3.4 Entropy–transport after the repair

Lemma III.3.3 (Posterior transportation for an arbitrary fitted law)**.** *For the update in* (III.3.5)*, whether or not H maximizes a likelihood,*

$$W_{1,T}(q, p_{H^+}) \le \sqrt{\frac{D(q\|p_H)}{2T}}. \tag{III.3.8}$$

Proof. For $S_T(\theta) > 0$, set

$$R_\theta(j) = \frac{q_j B_j^T(\theta)}{p_H(j) S_T(\theta)}.$$

This is a probability vector, and $\int R_\theta \, dH^+(\theta) = q$. Direct expansion gives the entropy identity

$$\int D(R_\theta \| B_\theta^T) \, dH^+(\theta) = D(q\|p_H) - D(H^+\|H) \le D(q\|p_H). \tag{III.3.9}$$

Terms with $S_T = 0$ have zero H^+ mass.

For completeness, every binomial kernel satisfies

$$W_{1,T}(r, B_\theta^T)^2 \le \frac{D(r\|B_\theta^T)}{2T}.$$

Indeed a 1-Lipschitz function of the normalized sum of T independent Bernoulli variables has centered logarithmic moment generating function at most $\lambda^2/(8T)$ by the bounded-differences martingale argument. The entropy variational inequality bounds its expectation difference by $D/\lambda + \lambda/(8T)$; optimizing $\lambda > 0$ and using Wasserstein duality proves the displayed inequality. Mix the corresponding couplings over H^+ and use Cauchy–Schwarz and (III.3.9) to obtain (III.3.8). □

Proposition III.3.4 (Predictive transport uniform in the shallow sample size)**.** *Every joint maximizer satisfies*

$$\mathbb{E}_G W_{1,T}(p_H, p_G) \le C n^{-1/2}, \tag{III.3.10}$$

with an absolute constant independent of m, T, G.

Proof. Coupling binomial sums with the same uniforms at two parameters shows

$$W_{1,T}(p_F, p_J) \le W_1(F, J)$$

for any two mixing laws F, J. Also the normalized predictive mean is the mixing mean. Combining Lemmas III.3.2 and III.3.3 therefore gives the deterministic bound

$$\begin{aligned} W_{1,T}(p_H, p_G) &\le 2W_{1,T}(p_{H^+}, p_G) + w|z - \mu_0| \\ &\le 2\sqrt{\frac{D(q\|p_H)}{2T}} + 2W_{1,T}(q, p_G) + w|z - \mu_0|. \end{aligned} \tag{III.3.11}$$

At a fitted endpoint the same bound holds since $H^+ = H$. The empirical CDF identity gives $\mathbb{E}W_{1,T}(q, p_G) \le 1/(2\sqrt{n})$. If $m > 0$,

$$w\mathbb{E}|z - \mu_0| \le \frac{\min(1, m/n)}{2\sqrt{m}} \le \frac{1}{2\sqrt{n}}.$$

Finally (III.3.4) and Jensen's inequality bound the expected square-root term by $1/\sqrt{n}$. This proves (III.3.10). □

This is the step absent from the homogeneous argument. Directly asserting $S_T = 1$ would be wrong. Bounding an arbitrary posterior correction only by total variation would also lose the uniformity in m/n supplied by the affine structure.

III.3.5 Predictive Hellinger control with a conditional likelihood deficit

We record the finite-cover and likelihood-envelope ingredients to make their application to the joint estimator explicit. These two ingredients, and the polynomial inverse used below, are inherited from Section 6.3 of the preceding integrated manuscript; the conditional application here is new.

Lemma III.3.5 (A cover of the binomial moment body)**.** *For $0 < \beta < 1$, the set of laws $\{p_F : F \in \mathcal{P}([0,1])\}$ has an ℓ_1 cover of radius β, with centers in that set and cardinality M satisfying*

$$\log M \le d \log \frac{C(d+1)}{\beta}, \qquad d = \min\left\{T, \left\lceil \sqrt{2T \log(16T/\beta)} \right\rceil\right\}. \tag{III.3.12}$$

Proof. This is Lemma III.A.1, whose Fourier contour and moment-cell proof is included in the supporting appendix. □

Lemma III.3.6 (Conditional envelopes)**.** *For $T \le n$, every joint maximizer satisfies*

$$\mathbb{E}_G h(p_H, p_G)^2 \le \frac{C\{\sqrt{T}\log(en)^{3/2} + 1\}}{n}, \qquad \mathbb{E}_G h(p_H, p_G) \le \frac{CT^{1/4}\log(en)^{3/4}}{\sqrt{n}}. \tag{III.3.13}$$

Proof. Condition on the shallow sample. Then A in (III.3.3) is fixed, while the deep observations retain their independent product law. Let $\eta = (en)^{-2}$ and take the cover in Lemma III.3.5 with $\beta = \eta/(T+1)$. For each center p_a, put $u_a(j) = p_a(j) + \eta/(T+1)$. Every model law is bounded coordinatewise by some u_a, each u_a has mass $1+\eta$, and

$$\widetilde{p} \le u_a \quad \Longrightarrow \quad h(u_a, \widetilde{p})^2 \le \eta.$$

For a fixed envelope, Markov's inequality at exponent $1/2$ gives

$$\begin{aligned}\Pr\left\{\prod_{i=1}^n \frac{u_a(X_i)}{p_G(X_i)} \ge e^{-A} \,\middle|\, Y_1, \dots, Y_m\right\} &\le e^{A/2}\left(\sum_j \sqrt{p_G(j)u_a(j)}\right)^n \\ &= e^{A/2}\left(1 + \eta/2 - h(p_G, u_a)^2/2\right)^n.\end{aligned}$$

Ratios need be defined only on the true support, where the sample lies almost surely. The triangle inequality and a union over M envelopes now show that, conditionally with probability at least $1 - e^{-x}$, simultaneously for every $\widetilde{p}$ obeying the likelihood comparison in (III.3.3),

$$h(p_G, \widetilde{p}) \le \sqrt{\eta} + \sqrt{A/n + \eta + 2(\log M + x)/n}.$$

Thus the conclusion applies to the joint maximizer even though it uses both samples. Integrating this conditional tail and then averaging over the shallow sample gives

$$\mathbb{E}h(p_G, p_H)^2 \le C\{\eta + (\mathbb{E}A + \log M + 1)/n\}.$$

For $T \le n$, (III.3.12) gives $\log M \le C\sqrt{T}\log(en)^{3/2}$. Use $\mathbb{E}A \le 1$ and Jensen's inequality to finish the proof. □

III.3.6 The inherited inverse inequality and the complete upper bound

To state the required analytic input explicitly, for every 1-Lipschitz $f : [0,1] \to \mathbb{R}$ and integer $k \geq 1$ there is a polynomial P_k of degree at most k such that

$$\|f - P_k\|_\infty \leq C/k, \qquad \|P_k'\|_\infty \leq 2. \tag{III.3.14}$$

Also, if $P = \sum_{j=0}^{T} b_j B_j^T$ has degree $k \leq T/2$, then

$$\max_j |b_j| \leq Ce^{2k^2/T}\|P\|_\infty, \qquad \max_{j<T} T|b_{j+1} - b_j| \leq Ce^{4k^2/T} \quad \text{when } P = P_k. \tag{III.3.15}$$

The proofs of (III.3.14) and (III.3.15) are given in Lemma III.A.3 of the supporting appendix. They are inherited ingredients. The following form of Proposition III.A.4 states the exact inverse used here.

Lemma III.3.7 (Two-scale inverse inequality)**.** *For integers $1 \leq s \leq k \leq T/2$ and any two mixing laws F, J,*

$$W_1(F, J) \leq \frac{C}{k} + Ce^{4s^2/T} W_{1,T}(p_F, p_J) + \frac{Ce^{2k^2/T}}{s} h(p_F, p_J). \tag{III.3.16}$$

Proof. For a 1-Lipschitz test f, approximate it by P_k and P_s as in (III.3.14). Replacing f by P_k costs C/k. The Bernstein coefficients of P_s have normalized-grid Lipschitz constant at most $Ce^{4s^2/T}$, so their integral against $p_F - p_J$ is bounded by the middle term of (III.3.16). Since $\|P_k - P_s\|_\infty \leq C/s$, the remaining coefficient vector has supremum norm at most $Ce^{2k^2/T}/s$. Use $\|p_F - p_J\|_1 \leq 2h(p_F, p_J)$ to bound its integral. Take the Wasserstein dual supremum over f. □

Upper bound in Theorem III.3.1. First suppose $T \leq n$. Bounded T or n are absorbed by the loss bound one. Write $L_n = \log(en)$, fix $a = 1/64$, and take

$$k = \left\lfloor \min\{T/2, \sqrt{aTL_n}\} \right\rfloor, \qquad s = \lfloor \sqrt{T}/4 \rfloor.$$

In the remaining range these satisfy $1 \leq s \leq k \leq T/2$, $s \asymp \sqrt{T}$, $e^{4s^2/T} \leq C$, and $k^{-1} \leq C\{T^{-1} + (TL_n)^{-1/2}\}$. Proposition III.3.4, Lemma III.3.6, and (III.3.16) give

$$\mathbb{E}W_1(H, G) \leq \frac{C}{k} + \frac{C}{\sqrt{n}} + \frac{Ce^{2k^2/T} L_n^{3/4}}{\sqrt{n}\, T^{1/4}}.$$

Multiplying the last term by $\sqrt{TL_n}$ bounds it by

$$C(en)^{2a} n^{-1/2} T^{1/4} L_n^{5/4} \leq C(en)^{2a} n^{-1/4} L_n^{5/4} \leq C,$$

because $2a < 1/4$. This proves the required rate for $T \leq n$.

For $T \geq n$, coupling a mixing parameter with its own normalized binomial count gives $W_1(F, p_F) \leq 1/(2\sqrt{T})$, after viewing p_F as a law on the normalized grid. Therefore

$$W_1(H, G) \leq T^{-1/2} + W_{1,T}(p_H, p_G).$$

Proposition III.3.4 yields the root-n bound. Clipping at the diameter one completes the upper bound in every range. □

III.3.7 Matching lower bounds with identical shallow-data laws

Lower bound in Theorem III.3.1. Every pair below has common mean $1/2$. Thus the entire depth-one sample has exactly the same product Bernoulli law at the two alternatives, regardless of m.

For the root-n term, consider

$$G_u = \frac{u}{2}\delta_0 + \frac{u}{2}\delta_1 + (1-u)\delta_{1/2}, \qquad u_0 = \frac{1}{2}, \quad u_1 = \frac{1}{2} + h, \quad h = \frac{1}{16\sqrt{n}}.$$

Their Wasserstein distance is $h/2$. An oracle revealing the n deep latent parameters has relative entropy $nD_B(u_0\|u_1) \le 5nh^2 = 5/256$ between these alternatives. The observed deep counts are generated by a parameter-independent kernel, so data processing, Pinsker's inequality and the two-point metric-risk bound give a universal lower bound $c/\sqrt{n}$.

For the other terms, let

$$\pi(d\theta) = \frac{d\theta}{\pi\sqrt{\theta(1-\theta)}}, \qquad g_k(\theta) = \mathcal{T}_k(2\theta - 1), \qquad G_k^{\pm} = (1 \pm g_k/2)\pi, \quad k \ge 2.$$

Orthogonality makes these probability measures of common mean $1/2$. Direct integration of their signed cumulative function gives

$$W_1(G_k^+, G_k^-) = \frac{1}{2\pi k}\int_0^\pi |\sin(ku)| \sin u \, du \asymp k^{-1}. \tag{III.3.17}$$

The lower constant is uniform: restrict the integral to $[\pi/4, 3\pi/4]$ and use the period of $|\sin(ku)|$. If $k = T+1$, orthogonality to all degree-T polynomials makes the two deep predictive laws identical. This gives the lower bound c/T.

For $k \le T$, draw $\theta \sim \pi$ and $J \mid \theta \sim \mathrm{Binomial}(T, \theta)$, and define $Qf(j) = \mathbb{E}[f(\theta) \mid J = j]$. The posterior is $\mathrm{Beta}(j + 1/2, T - j + 1/2)$. The self-adjoint operator Q^*Q preserves degree-k polynomial spaces and has leading coefficient

$$\lambda_{T,k} = \prod_{i=0}^{k-1} \frac{T-i}{T+1+i}.$$

Indeed the degree-k coefficient in the posterior moment comes from $(J+1/2)\cdots(J+k-1/2)/[(T+1)\cdots(T+k)]$; taking its expectation under $\mathrm{Binomial}(T,\theta)$ contributes the falling factorial $(T)_k$. Self-adjointness and orthogonality therefore imply $Q^*Qg_k = \lambda_{T,k} g_k$. Since $\int g_k^2 \, d\pi = 1/2$, $\mathbb{E}(Qg_k(J))^2 = \lambda_{T,k}/2$.

The predictive laws are $p_\pi(1 \pm Qg_k/2)$, with $|Qg_k| \le 1$. Consequently

$$\chi^2(p_{G_k^+}\|p_{G_k^-}) \le 2\mathbb{E}(Qg_k(J))^2 = \lambda_{T,k} \le e^{-k^2/(2T)}. \tag{III.3.18}$$

For the last inequality, apply $\log(1-x) \le -x$ to each factor and use $T + 1 + i \le 2T$.

Take $k = \lceil\sqrt{8T\log(8n)}\rceil$ whenever this is at most T. The deep product chi-squared divergence is at most $\exp\{n(8n)^{-4}\} - 1 < 10^{-3}$, by tensorization. Two-point testing and (III.3.17) give $c/\sqrt{T\log(en)}$. If this choice exceeds T, then $T < 8\log(8n) \le C\log(en)$, so $1/T$ already dominates the desired logarithmic term up to an absolute constant. All degrees used are at least two, preserving equality of means. Finally the maximum of the three lower bounds is comparable to their sum clipped at one, proving the result. □

Corollary III.3.8 (Equal total budgets can have different recovery orders)**.** *For $N \ge 2$, consider two profiles with exactly N independent individuals and exactly N^2 Bernoulli trials. If all depths equal N, the minimax population risk is $\Theta(N^{-1/2})$. If $N-1$ depths equal one and the last equals $N^2 - N + 1$, the minimax risk is $\Theta(1)$. Every joint NPMLE has the corresponding upper order.*

Proof. Apply Theorem III.3.1 with $(m, n, T) = (0, N, N)$ and $(N-1, 1, N^2 - N + 1)$ respectively. □

III.3.8 Scope, provenance, and the unresolved general profile

The homogeneous all-depth theorem, finite moment-body covers, posterior entropy decomposition, and two-scale polynomial inverse are existing ingredients in the supplied integrated manuscript. We have reproduced the needed statements; proofs are supplied here and in the supporting appendix to specify all hypotheses. The new ingredient is the affine score identity (III.3.6), which repairs the deep stratum and restores the predictive root-n transportation bound uniformly over all shallow sample sizes. Conditional envelopes then preserve the full replication transition.

Vinayak, Kong, Valiant and Kakade study homogeneous mixture likelihood [70]; the Chebyshev-moment framework of Musco, Musco, Rosenblatt and Singh is also homogeneous in the relevant binomial application [16]. Recent heterogeneous-trial papers address different losses and parameter spaces. Lee, Baćak and Kennedy analyze smooth-density estimators [21]. Lee and Kennedy's October 2026 manuscript treats pointwise estimation of a Hölder-smooth mixing density, with profile-dependent effective sample sizes and a regularized series estimator [22]. The present theorem concerns arbitrary mixing laws, Wasserstein loss, and every unrestricted joint likelihood maximizer. It is not asserted to resolve a question posed in those smooth-density papers, nor is this comparison a comprehensive historical-priority assessment.

An arbitrary replication profile remains unresolved here. At shallow depth $d \geq 2$, the corresponding score is generally a polynomial of degree d. The posterior correction can have several sign changes, and its Wasserstein cost is no longer determined by its mean shift. Neither aggregate posterior normalization nor the present affine argument supplies the missing control for that general case.

III.4 Likelihood contact and empirical-posterior transportation

An empirical measure on Euclidean space is singular with respect to a Gaussian mixture density. Consequently its relative entropy to that density is infinite, and a finite-space empirical-entropy argument cannot be transferred directly. We use a different consequence of mixture likelihood optimality: translating the parameter at a likelihood-score contact point controls the entire moment generating function of an *empirical posterior*. This gives a sharp entropy bound for one EM update after changing sample weights, exact Gaussian and Poisson transport bounds, and minimax bounds in a specified high-precision regime.

The score contact condition is classical; see [47, Proposition 1.1] and its attribution to Lindsay. Rigollet–Weed's identification of maximum-likelihood deconvolution with entropic transport [23, Theorem 4.2] is closely related prior work. Posterior reweighting also underlies the information inequalities in [24, Theorems 3.1 and 3.3]. The results below concern finite empirical-posterior kernels and sample-weight entropy. They do not assert a new proof of those variational or functional-inequality theorems, nor historical priority for every consequence of likelihood stationarity.

III.4.1 Moment domination at a likelihood contact point

Let $x_1, \dots, x_N$ be fixed observations and let $q_i > 0$ satisfy $\sum_i q_i = 1$. Repeated observation values are allowed: i is an observation label. Consider a regular natural exponential family

$$k_\eta(x) = h(x) \exp\{\langle \eta, T(x) \rangle - A(\eta)\}, \qquad \eta \in U \subset \mathbb{R}^d,$$

where U is open and convex, A is twice continuously differentiable, and $0 < h(x_i) < \infty$, $T(x_i) \in \mathbb{R}^d$ for every observed label. For a mixing law H on U, put

$$f_H(x_i) = \int k_\eta(x_i)\, dH(\eta), \qquad \ell_q(H) = \sum_i q_i \log f_H(x_i).$$

The following theorem assumes that a maximizer exists and that its fitted values at the observations are finite and positive. These conditions will be verified in the applications.

Theorem III.4.1 (Empirical-posterior moment domination)**.** *Let H be any maximizer of ℓ_q. For H-almost every η,*

$$R_\eta(i) = \frac{q_i k_\eta(x_i)}{f_H(x_i)} \tag{III.4.1}$$

is a probability vector. The joint law $\Gamma(d\eta, i) = H(d\eta)R_\eta(i)$ has marginals H and q. For every u such that $\eta + u \in U$,

$$\mathbb{E}_{R_\eta} e^{\langle u, T(x_i)\rangle} \le e^{A(\eta+u)-A(\eta)}. \tag{III.4.2}$$

In particular,

$$\mathbb{E}_{R_\eta} T(x_i) = \nabla A(\eta), \qquad \mathrm{Cov}_{R_\eta}(T(x_i)) \preceq \nabla^2 A(\eta). \tag{III.4.3}$$

Thus the law $(\nabla A)_{\#}H$ is dominated in convex order by $\sum_i q_i \delta_{T(x_i)}$.

Proof. Taking the directional derivative toward a point mass gives

$$S(\zeta) := \sum_i \frac{q_i k_\zeta(x_i)}{f_H(x_i)} \le 1 \quad (\zeta \in U).$$

Since $\int S\, dH = 1$, equality holds H-almost surely. This proves normalization in (III.4.1); summing over the parameter gives the label marginal q_i. At a point with $S(\eta) = 1$, direct substitution shows

$$S(\eta + u) = e^{A(\eta)-A(\eta+u)} \mathbb{E}_{R_\eta} e^{\langle u, T(x_i)\rangle}.$$

This proves (III.4.2). The difference between $A(\eta+u) - A(\eta)$ and the logarithm of the displayed moment generating function has a minimum of zero at $u = 0$. Its gradient vanishes and its Hessian is positive semidefinite, giving (III.4.3). Finally Jensen's inequality, conditional on η, gives

$$\int \varphi(\nabla A(\eta))\, dH(\eta) \le \sum_i q_i \varphi(T(x_i))$$

for every convex function φ for which the expression is defined. □

The theorem concerns an unrestricted natural-parameter domain. If the optimization is restricted to a closed set, only increments staying inside that set are available. Interior support points retain (III.4.3); boundary support points need a separate argument.

III.4.2 A sharp entropy bound for reweighted EM updates

For a contact point η, define

$$\psi_\eta(u) = A(\eta + u) - A(\eta) - \langle u, \nabla A(\eta)\rangle, \qquad \eta + u \in U,$$

and let

$$\psi_\eta^*(z) = \sup_{u:\eta+u\in U} \{\langle u, z\rangle - \psi_\eta(u)\}.$$

Now change the observation weights from q to an arbitrary probability vector q'. Keep the old posterior responsibilities, and set

$$S'(\eta) = \sum_i \frac{q_i' k_\eta(x_i)}{f_H(x_i)}, \qquad dH'(\eta) = S'(\eta)\, dH(\eta), \tag{III.4.4}$$

$$R_\eta'(i) = \frac{q_i' k_\eta(x_i)}{f_H(x_i) S'(\eta)}, \qquad m_\eta = \sum_i R_\eta'(i) T(x_i).$$

The last line is needed only where $S' > 0$.

Theorem III.4.2 (Entropy budget for the location update)**.** *Under the assumptions of Theorem III.4.1, H' is a probability measure and*

$$\int \psi_\eta^*(m_\eta - \nabla A(\eta))\, dH'(\eta) \le D(q'\|q) - D(H'\|H). \tag{III.4.5}$$

All relative entropies on the right are finite. In particular, for a Gaussian location family with common positive definite covariance Σ, using the mean θ as the parameter,

$$W_{2,\Sigma^{-1}}(H', B)^2 \le \int \|m_\theta - \theta\|_{\Sigma^{-1}}^2\, dH'(\theta) \le 2\{D(q'\|q) - D(H'\|H)\}, \qquad B = (m_\cdot)_\# H'. \tag{III.4.6}$$

Here B is one EM update of both weights and component locations using the old responsibilities. The bound does not assert stability of an arbitrary fully refitted NPMLE. The factor 2 is best possible. The notation $W_{2,\Sigma^{-1}}$ denotes quadratic Wasserstein distance with cost $(x-y)^\mathsf{T}\Sigma^{-1}(x-y)$.

Proof. The joint measure $\Gamma'(d\eta, i) = H'(d\eta)R'_\eta(i)$ is obtained from Γ by multiplying its density on label i by q'_i/q_i. Consequently its label marginal is q', its parameter marginal is H', and the relative-entropy chain rule gives

$$D(q'\|q) = D(\Gamma'\|\Gamma) = D(H'\|H) + \int D(R'_\eta\|R_\eta)\, dH'(\eta). \tag{III.4.7}$$

Finiteness follows, for example, from $S' \le \max_i(q'_i/q_i) < \infty$ and the finite label space. The entropy variational inequality gives, for every admissible u,

$$\langle u, m_\eta - \nabla A(\eta)\rangle \le D(R'_\eta\|R_\eta) + \log \mathbb{E}_{R_\eta} e^{\langle u, T(x_i) - \nabla A(\eta)\rangle} \le D(R'_\eta\|R_\eta) + \psi_\eta(u).$$

Take the supremum over u and integrate, then use (III.4.7). For the Gaussian family, $\psi_\theta^*(z) = \frac{1}{2} z^\mathsf{T}\Sigma^{-1} z$. The coupling $\theta \mapsto m_\theta$ proves the first inequality in (III.4.6).

For sharpness it suffices to take one dimension, observations $-\sigma, \sigma$, and $q = (1/2, 1/2)$. The unique NPMLE is $H = \delta_0$: its score is $e^{-\theta^2/(2\sigma^2)}\cosh(\theta/\sigma) \le 1$, with equality only at zero. For $q' = (1/2 + \varepsilon, 1/2 - \varepsilon)$, $H' = \delta_0$ and the updated mean has magnitude $2\sigma\varepsilon$. The squared normalized transport is $4\varepsilon^2$, whereas $D(q'\|q) = 2\varepsilon^2 + O(\varepsilon^4)$. No factor smaller than 2 can therefore hold uniformly. □

For Poisson observations of mean $\lambda = t\theta$, the conjugate in (III.4.5) is the Poisson mean divergence

$$\psi_\lambda^*(m - \lambda) = m\log(m/\lambda) - m + \lambda. \tag{III.4.8}$$

Thus the same theorem controls the component-mean update in a nonquadratic cost adapted to counts. The multiplicative constant 1 in (III.4.5) is sharp even in this family. Indeed, an empirical Bernoulli law with mass p at one has Poisson-mixture NPMLE δ_p in the mean parameter: its score is $e^{p-\lambda}(1 - p + \lambda) \le 1$. Reweighting p to $p + h$ gives

$$\frac{(p+h)\log((p+h)/p) - h}{D(\mathrm{Bernoulli}(p+h)\,\|\,\mathrm{Bernoulli}(p))} \longrightarrow 1 - p \quad (h \longrightarrow 0).$$

Letting p tend to zero proves sharpness. At $\lambda = 0$ the old posterior is concentrated on zero counts, and the corresponding term is zero whenever its reweighted mass is nonzero.

III.4.3 Exact Gaussian and Poisson coupling bounds

Theorem III.4.3 (A sharp Gaussian martingale coupling)**.** *For arbitrary $x_i \in \mathbb{R}^d$ and positive q_i, let H be any NPMLE for Gaussian locations with common covariance $\Sigma \succ 0$, optimized over all probability measures on $\mathbb{R}^d$. There is a coupling (Θ, X) of H and $Q = \sum_i q_i \delta_{x_i}$ such that*

$$\mathbb{E}[X \mid \Theta] = \Theta, \qquad \mathbb{E}[e^{\langle v, X - \Theta\rangle} \mid \Theta] \le e^{v^\mathsf{T}\Sigma v/2} \quad (v \in \mathbb{R}^d). \tag{III.4.9}$$

Consequently

$$W_2(H,Q)^2 \le \operatorname{tr}\Sigma, \qquad \operatorname{Cov}(Q)-\Sigma \preceq \operatorname{Cov}(H) \preceq \operatorname{Cov}(Q). \tag{III.4.10}$$

The constant 1 *multiplying* $\operatorname{tr}\Sigma$ *is sharp.*

Proof. Projection onto the convex hull of the observations in the Σ^{-1} metric decreases every squared distance to an observation, strictly for a point outside the hull. Therefore the maximization may be restricted to that compact hull, where a maximizer exists; every maximizer is supported there. All fitted Gaussian likelihoods are positive. Apply Theorem III.4.1 in natural coordinates $\eta = \Sigma^{-1}\theta$ to obtain (III.4.9) and $\operatorname{Cov}(X \mid \Theta) \preceq \Sigma$. The coupling cost is at most $\operatorname{tr}\Sigma$. The total-covariance identity gives the sandwich in (III.4.10).

For equality, take the 2^d points $\Sigma^{1/2}s$, uniformly weighted, where $s \in \{-1,1\}^d$. The score of δ_0 is

$$e^{-\|z\|^2/2}\prod_{j=1}^{d}\cosh z_j \le 1, \qquad z=\Sigma^{-1/2}\theta,$$

with equality only at $z=0$. The contact criterion implies that δ_0 is the unique NPMLE. Its squared transport distance to the empirical law is exactly $\operatorname{tr}\Sigma$. □

Theorem III.4.4 (A sharp Poisson martingale coupling)**.** *Fix $t>0$ and nonnegative integer observations x_i. Let H be any NPMLE for* Poisson$(t\theta)$ *mixtures optimized over $\theta \in [0,\infty)$, with sample weights $q_i>0$. Put $\bar{x}=\sum_i q_i x_i$ and $Q_t=\sum_i q_i\delta_{x_i/t}$. There is a coupling of $\Theta \sim H$ and the empirical count X such that*

$$\mathbb{E}[X \mid \Theta] = t\Theta, \qquad \mathbb{E}[e^{s(X-t\Theta)} \mid \Theta] \le \exp\{t\Theta(e^s-1-s)\} \quad (s\in\mathbb{R}). \tag{III.4.11}$$

In particular,

$$W_2(H,Q_t)^2 \le \frac{\bar{x}}{t^2}. \tag{III.4.12}$$

The constant 1 *is best possible uniformly over finite datasets.*

Proof. If $x_{\max}>0$, replacing any parameter greater than $x_{\max}/t$ by $x_{\max}/t$ strictly increases every observed kernel likelihood. Thus maximizers exist on a compact interval and are supported there. If all counts are zero, the unique maximizer is δ_0. For positive support parameters, Theorem III.4.1 with natural parameter $\eta=\log(t\theta)$ proves (III.4.11), including $\operatorname{Var}(X \mid \Theta) \le t\Theta$. At $\theta=0$ the empirical posterior can put mass only on zero counts, so the same formulas hold with both sides degenerate. Integrating the conditional mean gives $t\mathbb{E}_H\Theta=\bar{x}$. The squared cost of the coupling $(\Theta, X/t)$ is therefore at most $\mathbb{E}_H\Theta/t=\bar{x}/t^2$.

For an empirical Bernoulli law with mass $p\in(0,1)$ at one, the unique maximizing intensity is p, by the score calculation following (III.4.8). Hence $H=\delta_{p/t}$ and the actual squared transport distance is $p(1-p)/t^2$. Taking rational p tending to zero realizes finite equally weighted datasets and proves sharpness. □

III.4.4 Boundary repair and the high-precision statistical regime

The preceding martingale conclusions cannot simply be asserted for a constrained parameter space. For example, a Poisson fit over $[0,M]$ to a single count $x>tM$ is δ_M, and $(x/t-M)^2$ can exceed x/t^2 arbitrarily. The correct comparison for a bounded fit uses clipped observations.

Theorem III.4.5 (Clipped-data transport for bounded NPMLEs)**.** *Let* $C_{a,b}(y) = \min(b,\max(a,y))$.

(i) For every NPMLE H of $\mathrm{Poisson}(t\theta)$ *mixtures over* $[0, M]$*, where* $t, M > 0$*,*

$$W_2\left(H, \sum_i q_i \delta_{C_{0,M}(x_i/t)}\right)^2 \le \frac{2M}{t}. \tag{III.4.13}$$

(ii) For every NPMLE H of one-dimensional $\mathcal{N}(\theta, \sigma^2)$ *mixtures over* $[a, b]$*, where* $a < b$*,*

$$W_2\left(H, \sum_i q_i \delta_{C_{a,b}(x_i)}\right)^2 \le 2\sigma^2. \tag{III.4.14}$$

Proof. Compactness gives existence, and the score contact condition again normalizes the empirical posterior kernels. At an interior Poisson parameter, the conditional mean and variance identities give

$$\mathbb{E}[(C_{0,M}(X/t) - \theta)^2 \mid \theta] \le \mathbb{E}[(X/t - \theta)^2 \mid \theta] \le M/t.$$

At zero the conditional count is zero. At $\theta = M$, natural increments $s \le 0$ are admissible, giving

$$\mathbb{E}[e^{s(X-tM)} \mid \theta = M] \le e^{tM(e^s - 1 - s)} \le e^{tMs^2/2}.$$

Chernoff's bound with $s = -r/(tM)$ yields $\mathbb{P}(tM - X > r \mid \theta = M) \le e^{-r^2/(2tM)}$. Integrating this bound gives

$$\mathbb{E}[(tM - X)_+^2 \mid \theta = M] = \int_0^\infty 2r\mathbb{P}(tM - X > r \mid \theta = M)\, dr \le 2tM.$$

This is exactly the required conditional cost after clipping at M. Integration over H proves (III.4.13).

For Gaussian locations, interior parameters have conditional variance at most σ^2. At $\theta = a$, a shift $v \in [0, b - a]$ gives

$$\mathbb{E}[e^{v(X-a)/\sigma^2} \mid \theta = a] \le e^{v^2/(2\sigma^2)}.$$

For a tail level $0 < r < b - a$ take the admissible shift $v = r$. Then

$$\mathbb{P}(C_{a,b}(X) - a > r \mid \theta = a) \le e^{-r^2/(2\sigma^2)}.$$

Integration up to $b - a$ bounds the conditional squared cost by $2\sigma^2$. The endpoint b is identical with negative shifts. Integrating over H proves (III.4.14). □

Corollary III.4.6 (Minimax optimality at high exposure or small noise)**.** *Suppose* $\Theta_1, \dots, \Theta_N$ *are independent with common law* G*.*

(i) If G is supported on $[0, M]$ *and* $X_i \mid \Theta_i \sim \mathrm{Poisson}(t\Theta_i)$ *independently, every measurable NPMLE H over* $[0, M]$ *satisfies*

$$\sup_G \mathbb{E}_G W_1(H, G) \le (1 + \sqrt{2})\sqrt{M/t} + \frac{M}{2\sqrt{N}}. \tag{III.4.15}$$

The unrestricted NPMLE over $[0, \infty)$ *satisfies the same bound with* $1 + \sqrt{2}$ *replaced by* 2*. There are absolute constants* $c, C > 0$ *such that, whenever* $tM \ge N$*,*

$$c\frac{M}{\sqrt{N}} \le \inf_{\widetilde{G}} \sup_{G \in \mathcal{P}([0,M])} \mathbb{E}_G W_1(\widetilde{G}, G) \le \sup_G \mathbb{E}_G W_1(H, G) \le C\frac{M}{\sqrt{N}}.$$

(ii) If G is supported on $[a, b]$ *and* $X_i = \Theta_i + \sigma Z_i$ *for independent standard Gaussian* Z_i*, every NPMLE over* $[a, b]$ *satisfies*

$$\sup_G \mathbb{E}_G W_1(H, G) \le (\sqrt{2} + \sqrt{2/\pi})\sigma + \frac{b - a}{2\sqrt{N}}. \tag{III.4.16}$$

The minimax risk is of order $(b - a)/\sqrt{N}$ *uniformly when* $\sigma \le (b - a)/\sqrt{N}$*.*

Proof. Write $Q_\Theta = N^{-1}\sum_i \delta_{\Theta_i}$. For a law supported on an interval of length L, the one-dimensional CDF formula gives

$$\mathbb{E}W_1(Q_\Theta, G) \le \int \sqrt{F_G(x)(1-F_G(x))/N}\,dx \le \frac{L}{2\sqrt{N}}.$$

In the Poisson model, clipping cannot increase the distance to $\Theta_i \in [0, M]$, and

$$\mathbb{E}|C_{0,M}(X_i/t) - \Theta_i| \le \mathbb{E}|X_i/t - \Theta_i| \le \mathbb{E}\sqrt{\Theta_i/t} \le \sqrt{M/t}.$$

The triangle inequality, (III.4.13), and $W_1 \le W_2$ prove (III.4.15). For the unrestricted estimator use (III.4.12) and $\mathbb{E}\bar{X} = t\mathbb{E}\Theta_i \le tM$ instead. In the Gaussian case the corresponding noise term is $\sigma\mathbb{E}|Z_i| = \sigma\sqrt{2/\pi}$, and (III.4.14) proves (III.4.16).

For the lower bounds, use mixing laws on the two endpoints of the parameter interval, with upper-endpoint probabilities $1/2 \pm h$, where $h = 1/(16\sqrt{N})$. Their W_1 separation is $2Lh$. The relative entropy between their latent N-sample experiments is bounded by an absolute constant smaller than one, using $D(\text{Bernoulli}(p)\|\,\text{Bernoulli}(q)) \le (p-q)^2/[q(1-q)]$. Passing the latent variables through either observation kernel cannot increase total variation. The two-point testing bound therefore gives $cL/\sqrt{N}$ for every estimator. Substitute $L = M$ or $L = b - a$ and impose the respective high-precision condition to match the upper bound. □

In the rate-one Poisson parameterization, with support $[0, B]$, the high-exposure condition is $B \ge N$ and the risk is $\Theta(B/\sqrt{N})$. This supplies a complete statement for that parameter regime. Theorem 6.1 of the version of [78] inspected here covers smaller support scales, up to $N^{1/3-\varepsilon}$. We do not claim that Corollary III.4.6 closes the intervening range, determines all noise-dependent minimax rates, or resolves a named conjecture from that paper.

III.4.5 Two limits of the transfer

Proposition III.4.7 (Heteroscedastic contact is precision weighted)**.** *For Gaussian observations with known positive definite covariances Σ_i, let H maximize $\sum_i q_i \log \int \phi_{\Sigma_i}(x_i - \theta)\,dH(\theta)$ over unrestricted locations. At a contact point, the empirical posterior satisfies*

$$\mathbb{E}_{R_\theta} \exp\{v^\mathsf{T}\Sigma_i^{-1}(x_i - \theta) - \tfrac{1}{2}v^\mathsf{T}\Sigma_i^{-1}v\} \le 1. \tag{III.4.17}$$

In particular,

$$\mathbb{E}_{R_\theta}[\Sigma_i^{-1}(x_i - \theta)] = 0.$$

The ordinary martingale identity $\mathbb{E}_{R_\theta} x_i = \theta$ can fail.

Proof. The likelihood score at $\theta + v$ is exactly the left side of (III.4.17), which proves the inequality. Its derivative at zero vanishes. For a counterexample to the unweighted identity, take equal weights, observations $x_1 = 0, x_2 = 1$, and variances $1, 4$. For $H = \delta_{1/5}$ the score is

$$S(\theta) = \tfrac{1}{2}e^{((1/5)^2-\theta^2)/2} + \tfrac{1}{2}e^{((4/5)^2-(1-\theta)^2)/8}.$$

It is strictly concave on $[0, 1]$, has derivative zero at $1/5$, and has positive derivative for $\theta < 0$ and negative derivative for $\theta > 1$. Its unique maximum is $S(1/5) = 1$. The score criterion therefore makes $\delta_{1/5}$ the unique NPMLE, while its conditional empirical mean is $1/2$. □

Thus the common-covariance convex-order and transport conclusions cannot be transferred to arbitrary heteroscedastic models by merely replacing Σ with an average covariance. No sharp heteroscedastic minimax conclusion is asserted here.

Proposition III.4.8 (Likelihood accuracy alone does not control unbounded latent transport)**.** *In the unrestricted Gaussian location model, no uniform bound on latent W_2 error follows from a fixed positive likelihood tolerance, even if the fit also has a uniformly bounded likelihood-score certificate.*

Proof. For the single observation zero, the exact NPMLE is δ_0. Fix $0 < \varepsilon < 1$ and set $H_R = (1-\varepsilon)\delta_0 + \varepsilon\delta_R$. Its normalized likelihood deficit is

$$-\log\{1 - \varepsilon + \varepsilon e^{-R^2/(2\sigma^2)}\} \le -\log(1-\varepsilon).$$

Its score supremum is at most $1/(1-\varepsilon)$, independently of R, whereas $W_2(H_R, \delta_0) = \sqrt{\varepsilon}R \to \infty$. For any specified positive deficit or score tolerance, choose a sufficiently small fixed ε and then let R increase. □

The exact posterior results and the reweighted EM bound therefore address a different question from unrestricted stability of all approximate likelihood fits. This distinction is necessary, not an artifact of using empirical entropy.

III.5 The square-root order of the design gap on all connected graphs

The comparison between D-optimal and A-optimal graph designs asks how much total effective resistance is lost by optimizing determinant instead of inverse trace. Suliman and Weinstein give graph-dependent bounds and propose an $O(\log n)$ bound in Conjecture 1.5 of [13]. The preceding tree counterexample rules out that conjecture. Here we determine the correct worst-case order on *all* connected graphs: it is $\Theta(\sqrt{n})$. We also give an exact solution on complete-graph barbells, whose asymptotic ratio exceeds the worst-tree ratio by a factor of $\sqrt{2}$.

III.5.1 A universal bound from electrical potentials

Let G be a connected undirected graph on $n \ge 2$ vertices. For each edge choose an orientation and write $b_e = e_u - e_v$. A design assigns nonnegative conductances w_e with $\sum_e w_e = 1$; its Laplacian is

$$L(w) = \sum_e w_e b_e b_e^{\mathsf{T}}.$$

Only connected supports have finite objective. On $H = \mathbf{1}^{\perp}$, of dimension $d = n - 1$, let

$$A = L(w^D)|_H, \qquad B = L(w^A)|_H$$

be respectively a determinant-maximizing and an inverse-trace-minimizing design. Define

$$t = \operatorname{tr} A^{-1}, \qquad \tau = \operatorname{tr} B^{-1}, \qquad T = \frac{t}{d^2}, \qquad \alpha(G) = \frac{t}{\tau}.$$

Both matrices have trace two. The unordered-pair total resistance is n times the inverse trace, so this is precisely the infinity-Lewis/ERMP approximation ratio of [13].

Lemma III.5.1 (Edge potentials at a determinant-optimal design)**.** *For every candidate edge e, including an edge assigned zero weight,*

$$R_e = b_e^{\mathsf{T}} A^{-1} b_e \le d, \qquad \|L(w^D)^{\dagger} b_e\|^2 \le \frac{n R_e^2}{4}.$$

Consequently,

$$\frac{1}{2} \le T \le \frac{n}{4}. \tag{III.5.1}$$

Proof. The derivative of $\log\det A$ in the edge coordinate is R_e. The simplex optimality conditions give $R_e \le d$, with equality on positive-weight edges: the common multiplier equals $\sum_e w_e^D R_e = \operatorname{tr}(A^{-1}A) = d$.

Write $v = L(w^D)^{\dagger} b_e$, considered as a zero-mean vector on the n vertices. It is harmonic away from the source and sink of b_e. The maximum principle on the connected support puts its maximum at the source and its minimum at the sink. Thus its range is $v_u - v_v = R_e$, even when e itself has weight zero. A zero-mean vector of range R_e has squared norm at most $nR_e^2/4$: if its minimum and maximum are $a \le 0 \le b$, then averaging $(v_i - a)(v_i - b) \le 0$ gives $n^{-1}\sum_i v_i^2 \le -ab \le (b-a)^2/4$. This proves the potential bound. Finally,

$$t = \sum_e w_e^D \|A^{-1} b_e\|^2 \le \frac{nd^2}{4}.$$

The opposite bound $t \ge d^2/2$ follows from $\operatorname{tr} A = 2$ and Cauchy–Schwarz on its eigenvalues. □

Theorem III.5.2 (Universal square-root upper bound)**.** *For $n > 2$, every connected n-vertex graph satisfies*

$$\alpha(G) \le \frac{2T(n-2)}{8T^2 - 8T + n} \le \sqrt{\frac{n}{8}} + \frac{1}{2}. \tag{III.5.2}$$

For $n = 2$, the ratio is one and the last inequality remains valid.

Proof. For every symmetric operator Y on H, matrix Cauchy–Schwarz gives

$$(\operatorname{tr} Y)^2 \le \operatorname{tr}(B^{-1}) \operatorname{tr}(Y^2 B).$$

Choose $Y = A^{-1} + \eta I$ with $\eta \ge 0$. Because $B = \sum_e w_e^A b_e b_e^{\mathsf{T}}$, the preceding lemma gives

$$\begin{aligned} \operatorname{tr}(Y^2 B) &= \sum_e w_e^A (\|A^{-1} b_e\|^2 + 2\eta R_e + 2\eta^2) \\ &\le \frac{nd^2}{4} + 2\eta d + 2\eta^2. \end{aligned}$$

Put $\eta = dz$. It follows that, for every $z \ge 0$,

$$\alpha(G) \le \frac{T(n/4 + 2z + 2z^2)}{(T+z)^2}. \tag{III.5.3}$$

If $T = 1/2$, equality in the eigenvalue Cauchy–Schwarz bound forces $A = (2/d)I$; this already minimizes inverse trace, so $\alpha(G) = 1$. If $T > 1/2$, choose

$$z = \frac{n/4 - T}{2T - 1} \ge 0,$$

where nonnegativity follows from (III.5.1). Substitution into (III.5.3) yields the first inequality of (III.5.2). The derivative of that rational function of T has the sign of $n - 8T^2$. Its maximum on $[1/2, n/4]$ therefore occurs at $T = \sqrt{n/8}$ and equals $\sqrt{n/8} + 1/2$. For $n = 2$, the normalized Laplacian on H is the scalar two, so both designs coincide. □

Remark III.5.3 (An elementary first bound). Taking $z = 0$ above gives $\alpha(G) \le n/(4T)$. The bound $\tau \ge d^2/2$ also gives $\alpha(G) \le 2T$. Thus already

$$\alpha(G) \le \min\{2T, n/(4T)\} \le \sqrt{n/2}.$$

Optimizing the nonnegative shift improves its leading constant by a factor of two. The graph-specific input is the maximum principle for electrical potentials; a general positive semidefinite design need not satisfy that input.

III.5.2 Exact designs on complete-graph barbells

Let $G_{m,p}$ consist of two disjoint copies of K_m, $m \geq 2$, with a distinguished root in each, joined by a path of $p \geq 1$ edges. Its $p-1$ interior vertices are new. Set

$$n = 2m + p - 1, \qquad d = n - 1, \qquad h = n - m + 1,$$

$$c_j = (m + j - 1)(m + p - j) \quad (1 \leq j \leq p), \qquad q = \sqrt{\frac{mh}{2}} + (m-2)\sqrt{\frac{n}{2}},$$

$$Q = 2q + \sum_{j=1}^{p} \sqrt{c_j}.$$

Here c_j is the cut count of the jth bridge on the joining path.

Theorem III.5.4 (Exact barbell objectives and optimal conductances)**.** *For the unordered-pair resistance objective K, the optimal values and the D-design's resistance are*

$$K(w^A) = Q^2, \qquad K(w^D) = d\left((m-1)(2n-m) + \sum_{j=1}^{p} c_j\right). \tag{III.5.4}$$

The D-optimal design assigns $1/d$ to each joining-path edge and $2/(md)$ to every clique edge. The A-optimal design assigns $\sqrt{c_j}/Q$ to the jth joining-path edge and total conductance $s = q/Q$ to each clique. Within either clique, root-to-other conductances are

$$a = \frac{s\sqrt{2h/m}}{q},$$

and, for $m \geq 3$, other-to-other conductances are

$$b = \frac{1}{m-1}\left(\frac{s\sqrt{2n}}{q} - a\right).$$

Proof. The displayed D-weights sum to one. Each bridge has effective resistance d, and the uniform clique weights likewise give resistance $2/(m \cdot 2/(md)) = d$ between every clique pair. Attachments at a single root do not change resistances inside a clique. Thus every candidate edge has resistance d, which verifies the determinant optimality conditions.

For the A-optimal objective, resistances add across the graph's blocks. Each clique contributes a weighted resistance sum with mass h at its root and mass one at each other vertex: all vertices outside the clique project to the root. For a fixed clique budget s, symmetry and convexity allow equal weights a on root-to-other edges and b on other-to-other edges. For $m \geq 3$ put $\lambda = a + (m-1)b$. The root contrast eigenvalue is ma, while the peripheral zero-sum eigenspace has eigenvalue λ and dimension $m-2$. Consequently

$$R_{\mathrm{root},i} = \frac{1}{(m-1)a} + \frac{m-2}{(m-1)\lambda}, \qquad R_{i,j} = \frac{2}{\lambda}$$

for distinct peripheral vertices i, j. The weighted contribution and budget are therefore

$$H(a,\lambda) = \frac{h}{a} + \frac{n(m-2)}{\lambda}, \qquad s = \frac{m}{2}a + \frac{m-2}{2}\lambda.$$

Cauchy–Schwarz gives $H \geq q^2/s$, with equality at

$$a = \frac{s\sqrt{2h/m}}{q}, \qquad \lambda = \frac{s\sqrt{2n}}{q}.$$

Since $nm \geq h$, these weights satisfy $\lambda \geq a$ and hence $b \geq 0$. When $m = 2$, the clique is a single edge with contribution h/a and budget $a = s$, and the same formula follows directly from $q = \sqrt{h}$.

The joining-path contribution is $\sum_j c_j/w_j$. Allocating the unit total budget among the two clique blocks and the p bridges yields

$$\min\left\{\frac{q^2}{s_1} + \frac{q^2}{s_2} + \sum_j \frac{c_j}{w_j} : s_1 + s_2 + \sum_j w_j = 1\right\} = Q^2.$$

Equality gives exactly the asserted conductances. This proves global optimality, not just stationarity.

At the D-weights, every clique pair has resistance d. The weighted-pair coefficient of one clique is

$$h(m-1) + \binom{m-1}{2} = (m-1)(n - m/2).$$

Summing over the two cliques and adding the bridge contributions $d\sum_j c_j$ proves the second formula in (III.5.4). □

III.5.3 The general-graph order and the remaining constant gap

Theorem III.5.5 (Square-root worst-case order and a denser obstruction)**.** *Let $\mathcal{R}_n$ be the maximum of $\alpha(G)$ over connected simple graphs on n vertices. Then*

$$\left(\frac{1}{4\sqrt{2}} + o(1)\right)\sqrt{n} \leq \mathcal{R}_n \leq \frac{\sqrt{n}}{2\sqrt{2}} + \frac{1}{2}. \tag{III.5.5}$$

In particular, $\mathcal{R}_n = \Theta(\sqrt{n})$. The lower-bound graphs have an asymptotic leading constant $\sqrt{2}$ times the worst-tree leading constant $1/8$.

Proof. The upper bound is Theorem III.5.2. For each sufficiently large n, choose an integer $p = \sqrt{2n} + O(1)$ with $p \equiv n + 1 \pmod 2$, and set $m = (n - p + 1)/2$. Then $G_{m,p}$ has exactly n vertices, $m = n/2 + O(\sqrt{n})$, and uniformly along the joining path

$$c_j = n^2/4 + O(n).$$

The quantities in Theorem III.5.4 satisfy

$$2q = \frac{n^{3/2}}{\sqrt{2}} + O(n), \qquad \sum_j \sqrt{c_j} = \frac{n^{3/2}}{\sqrt{2}} + O(n),$$

$$\sum_j c_j = \frac{n^{5/2}}{2\sqrt{2}} + O(n^2).$$

The clique term $(m-1)(2n-m)$ is $O(n^2)$. It follows that

$$K(w^D) = \frac{n^{7/2}}{2\sqrt{2}} + O(n^3), \qquad K(w^A) = 2n^3 + O(n^{5/2}),$$

and hence

$$\alpha(G_{m,p}) = \frac{\sqrt{n}}{4\sqrt{2}} + O(1).$$

This proves the lower bound and the claimed order. Comparison with the previously established tree leading constant $1/8$ gives the final assertion. □

The exact asymptotic leading constant for all connected graphs is not determined by these bounds. It lies between $1/(4\sqrt{2})$ and $1/(2\sqrt{2})$ if a limiting constant exists; without assuming such existence, the same numbers bound the liminf and limsup. Neither endpoint is asserted sharp here.

III.6 A phase classification for weighted binomial power sums

For $r \geq 2$ and $\phi \in \mathbb{R}$, define

$$B_{r,\phi}(n) = \sum_{k=-n}^{n} \binom{2n}{n+k}^{r} e^{ik\phi}, \qquad n \geq 0. \tag{III.6.1}$$

The sum is real by pairing k with $-k$. Weighted binomial powers belong to the hypergeometric polynomial families considered, for example, by Matala-Aho for powers three and four [26] and by Simsek for general powers [27]. The theorem below is a classification of this phase family; it is not presented as the resolution of a named external conjecture. Its new content relative to the preceding alternating-sum result is the complete nonreal phase range. The real phases are included to identify exactly where the recurrence width drops.

A recurrence is an eventual identity $\sum_{j\in J} q_j(n) B_{r,\phi}(n+j) = 0$, with finite nonempty $J \subset \mathbb{Z}$ and nonzero $q_j \in \mathbb{C}(n)$. Its width is $\max J - \min J$ and its number of terms is $|J|$.

Theorem III.6.1. *Put*

$$m(r,\phi) = \begin{cases} \lceil r/2 \rceil, & \phi \equiv 0 \pmod{2\pi}, \\ \lfloor r/2 \rfloor, & \phi \equiv \pi \pmod{2\pi}, \\ r, & \phi \notin \pi\mathbb{Z}. \end{cases} \tag{III.6.2}$$

The minimum recurrence width of $B_{r,\phi}$ over $\mathbb{C}(n)$ is $m(r,\phi)$. The minimum number of nonzero terms, even allowing arbitrary gaps between shifts, is $m(r,\phi)+1$. A recurrence attaining both minima can be chosen with coefficients in $\mathbb{Q}(2\cos\phi)[n]$.

In particular, every nonreal root of unity gives width r, independently of its order. The widths at 1 and -1 are roughly half as large. This distinction will follow from collisions of positive critical values, not from an asymptotic comparison involving only the largest one. The theorem makes no assertion for weights off the unit circle or about the minimum differential order of a generating function.

III.6.1 Positive moments and critical-value collisions

Write $\lambda = e^{i\phi}$, $d = r-1$, $X = x_1 \cdots x_d$, and

$$P_{r,\lambda}(x) = \prod_{i=1}^{d} (1 + x_i)(1 + \lambda/X).$$

Direct expansion gives

$$\mathrm{CT}\, P_{r,\lambda}^{\ell} = \sum_{j=0}^{\ell} \binom{\ell}{j}^{r} \lambda^{j}. \tag{III.6.3}$$

On $x_i = e^{i\theta_i}$, with $S = \sum_i \theta_i$, we have

$$P_{r,\lambda}(e^{i\theta}) = e^{i\phi/2} \Psi_{r,\phi}(\theta), \qquad \Psi_{r,\phi}(\theta) = 2^{r} \prod_{i=1}^{d} \cos(\theta_i/2) \cos((\phi - S)/2). \tag{III.6.4}$$

The real function $\Psi_{r,\phi}$ is globally periodic and analytic: shifting a single angle by 2π changes two factors by a minus sign. Thus $\Phi_{r,\phi} = \Psi_{r,\phi}^2 \geq 0$ is a real-analytic function on $\mathbb{T}^d$. For normalized Haar measure μ,

$$B_{r,\phi}(n) = \lambda^{-n}\,\mathrm{CT}\,P_{r,\lambda}^{2n} = \int_{\mathbb{T}^d} \Phi_{r,\phi}^n \, d\mu. \tag{III.6.5}$$

The first equality is the change of index $j = n + k$ in (III.6.1). The phase in the second equality cancels because $(e^{i\phi/2})^{2n} = \lambda^n$. In particular these real sums are strictly positive.

Lemma III.6.2. *The function $\Phi_{r,\phi}$ has exactly $m(r,\phi)$ distinct positive critical values. Every critical point on every positive critical fiber is a nondegenerate local maximum. For $0 \leq \phi < 2\pi$, the values are the distinct nonzero members of*

$$v_j = \left(2\cos\frac{\phi + 2\pi j}{2r}\right)^{2r}, \qquad j = 0, \ldots, r-1. \tag{III.6.6}$$

Proof. At $P_{r,\lambda} \neq 0$, the logarithmic critical equations are

$$\frac{x_i}{1+x_i} - \frac{\lambda}{X+\lambda} = 0 \quad \Longleftrightarrow \quad x_i X = \lambda.$$

They force $x_1 = \cdots = x_d = \omega$, where $\omega^r = \lambda$. Every such root is on the unit circle, and it gives a critical point unless $\omega = -1$, when $P = 0$. These are all positive critical points of Φ: angular differentiation of $P = e^{i\phi/2}\Psi$ is equivalent to differentiation of the real Ψ, and away from zero $d\Phi = 2\Psi\, d\Psi$. At a critical point $P = (1+\omega)^r$, giving (III.6.6).

For completeness, differentiation once more gives

$$\mathrm{Hess}_{\log x} \log P = \frac{\omega}{(1+\omega)^2}(I_d + J_d),$$

where J_d is the all-ones matrix. Indeed the derivative of $-\lambda/(X+\lambda)$ in each logarithmic direction is $\lambda X/(X+\lambda)^2 = \omega/(1+\omega)^2$. For $\omega = e^{i\alpha} \neq -1$, the common factor is $1/(4\cos^2(\alpha/2)) > 0$. Angular differentiation introduces a minus sign; using the local real logarithm $\log|\Psi|$ yields

$$\mathrm{Hess}_\theta\, \Phi = -\frac{\Phi}{2\cos^2(\alpha/2)}(I_d + J_d). \tag{III.6.7}$$

The eigenvalues of $I_d + J_d$ are 1 and $d+1 = r$, so this Hessian is negative definite.

It remains to count distinct values. The half-angles $\theta_j = (\phi + 2\pi j)/(2r)$ lie in $[0, \pi)$. For distinct indices, equality of the nonzero values in (III.6.6) is equivalent to $\theta_j + \theta_k = \pi$, hence to

$$\phi/\pi = r - j - k.$$

If $\phi \notin \pi\mathbb{Z}$, no collision is possible and no θ_j equals $\pi/2$, giving r positive values. At $\phi = 0$, pairing j with $r-j$ leaves exactly $\lceil r/2 \rceil$ positive values, represented by $0 \leq j < \lceil r/2 \rceil$; the root -1, if present, contributes zero and is excluded. At $\phi = \pi$, pairing j with $r-1-j$ leaves $\lfloor r/2 \rfloor$ positive values, represented by $0 \leq j < \lfloor r/2 \rfloor$. □

III.6.2 Visibility and the lower bounds

We recall the precise real-variable mechanism, including even-dimensional integer exponents. If a real-analytic function on a compact manifold has finitely many critical points on a positive fiber v, all nondegenerate maxima, the analytic Morse lemma gives local coordinates $\Phi = v - |y|^2$. A positive analytic measure has the form $g(y)\,dy$, $g(0) > 0$. Its local pushforward density at $v - w$, $w > 0$, is

$$\frac{1}{2} w^{d/2-1} \int_{S^{d-1}} g(\sqrt{w}\,\omega)\, d\omega.$$

The spherical average is analytic in w and has positive constant term. Smooth cutoffs leave a compact submersion region, whose pushforward is smooth. Consequently the complete density near v has the form

$$b(x) + \mathbf{1}_{\{x<0\}}(-x)^{d/2-1}h(-x), \qquad x = t - v, \tag{III.6.8}$$

with b smooth, h analytic, and $h(0) > 0$. This is not smooth: in even dimension a finite derivative jumps, whereas in odd dimension there is a noninteger power (an integrable unbounded density when $d = 1$). All critical contributions have the same side and a positive leading coefficient.

The Cauchy transform $G(z) = \int (z - \Phi)^{-1}\, d\mu$ is holomorphic in the upper half-plane, and its boundary distribution satisfies $-\operatorname{Im} G(x + i0)/\pi = \Phi_* \mu$. A holomorphic extension through v would give a real-analytic boundary density, contradicting (III.6.8). Since $G(z) = z^{-1}F(z^{-1})$ near infinity for $F(t) = \sum_n (\int \Phi^n\, d\mu) t^n$, the generating germ has a genuine continued-branch singularity at $1/v$. This argument applies also to critical values inside the range of Φ.

By Lemma III.6.2, the generating function of $B_{r,\phi}$ therefore has $m = m(r, \phi)$ distinct positive finite singularities. To see their implications for an arbitrary recurrence, translate its shifts to $\min J = 0$, $\max J = L$ and clear denominators in $q_j(n)$. Put $e = \max_j \deg q_j$ and $c_j = [n^e]q_j$. With $\vartheta = t\, d/dt$, summation gives

$$\left(\sum_{j \in J} t^{L-j} q_j(\vartheta - j) \right) F(t) = R(t),$$

where R is a polynomial of initial terms. Its highest ordinary derivative coefficient is $t^e H(t)$, where

$$H(t) = \sum_{j \in J} c_j t^{L-j} \neq 0.$$

Every nonzero finite singularity of a continued solution must be a zero of H, by ordinary-point theory for linear differential equations; for $e = 0$ this follows directly from $HF = R$. Thus H has at least m positive roots. Its degree is at most L, giving $L \geq m$. Moreover, a nonzero real or imaginary part of H has the same positive roots and no more monomials. Descartes' rule of signs gives at least $m + 1$ nonzero monomials, hence $|J| \geq m + 1$. Both lower bounds hold over $\mathbb{C}(n)$ and allow gaps in J.

III.6.3 An independent upper bound for every nonreal phase

The matching upper bound uses an explicit r-dimensional state. It requires neither a general weighted-recurrence theorem nor an invertibility assumption about a moment matrix.

For an integer $N \geq 0$, define a polynomial functional

$$L_N[p] = \sum_{k=0}^{N} p(k) \binom{N}{k}^r \lambda^k.$$

For every $h \geq 0$, shifting the summation index gives

$$L_N[R_h] = 0, \qquad R_h(k) = k^{r+h} - \lambda (N - k)^r (k + 1)^h. \tag{III.6.9}$$

Indeed $k^r \binom{N}{k}^r \lambda^k = \lambda (N - k + 1)^r \binom{N}{k-1}^r \lambda^{k-1}$, and the boundary terms vanish. The leading coefficient of every R_h is

$$1 - (-1)^r \lambda. \tag{III.6.10}$$

It is nonzero if λ is nonreal. Successive subtraction of multiples of R_h reduces any polynomial in k to degree less than r, without changing L_N. Denote this triangular reduction by $\mathcal{R}_N$. All reduction coefficients belong to $\mathbb{Q}(\lambda, N)$.

Form the column vector

$$M_N = (L_N[1], L_N[k], \dots, L_N[k^{r-1}])^{\mathsf{T}}.$$

For $N \geq 1$ and $0 \leq j < r$, the binomial identity implies

$$(M_{N-1})_j = N^{-r} L_N[k^j (N-k)^r].$$

Define the rational matrix $T_r(N, \lambda)$ by taking its jth row to be the coefficients of $N^{-r}\mathcal{R}_N[k^j(N-k)^r]$ in the basis $1, k, \dots, k^{r-1}$. Then

$$M_{N-1} = T_r(N, \lambda) M_N. \tag{III.6.11}$$

The index N here is the upper binomial index and is distinct from the index n of (III.6.1).

Set $V_n = \lambda^{-n} M_{2n}$. Equations (III.6.3) and (III.6.11) give

$$\begin{aligned} V_{n-1} &= \lambda T_r(2n-1, \lambda) T_r(2n, \lambda) V_n, \\ B_{r,\phi}(n) &= (V_n)_0. \end{aligned} \tag{III.6.12}$$

Thus $B_{r,\phi}(n), \dots, B_{r,\phi}(n-r)$ are $r+1$ row functionals of the same r-dimensional vector V_n. Their rows have entries in $\mathbb{Q}(\lambda)(n)$, so they are linearly dependent over that field. This is a nonzero eventual recurrence of width at most r. No matrix inverse is involved, and possible special initial indices are immaterial to an eventual recurrence. The lower bounds already proved force its width to be r and all $r+1$ positions to be nonzero.

III.6.4 Exceptional phases and the coefficient field

At $\phi = \pi$, the sequence is exactly McIntosh's alternating sum. His general existence theorem gives width at most $\lfloor r/2 \rfloor$ [29, Theorem 1]. This matches the critical-value lower bound above. This boundary assertion agrees with the previously established alternating minimality result and is not counted as a second new application.

At $\phi = 0$, write $f_r(N) = \sum_{k=0}^{N} \binom{N}{k}^r$, so $B_{r,0}(n) = f_r(2n)$. Stoll's existence theorem gives f_r a recurrence of width at most $\lceil r/2 \rceil$ [28]. To pass to the even subsequence, solve such a recurrence for its highest shift over $\mathbb{Q}(N)$ and use its usual companion state. Composing two consecutive transitions produces a state for the even subsequence of the same dimension. Linear dependence of one more consecutive row functional therefore gives width at most $\lceil r/2 \rceil$, not twice that quantity. The lower bound proves equality.

Finally, in the nonreal case the construction provides a recurrence over $K(n)$, where $K = \mathbb{Q}(\lambda)$. Translate its shifts to $0, \dots, r$ and normalize the coefficient of shift r to one. Such a monic recurrence on this fixed shift interval is unique: the difference of two would be a nonzero recurrence of smaller width. The sequence is real, so complex conjugation gives the same normalized recurrence. Consequently its coefficients lie in the conjugation-fixed field

$$K^{\mathrm{conj}}(n) = \mathbb{Q}(\lambda + \lambda^{-1})(n) = \mathbb{Q}(2\cos\phi)(n).$$

For clarity, the fixed-field equality also holds for transcendental λ: conjugation sends λ to λ^{-1}, and λ satisfies a quadratic equation over $\mathbb{Q}(\lambda + \lambda^{-1})$; as λ is nonreal, the extension has degree two. Clearing denominators gives coefficients in $\mathbb{Q}(2\cos\phi)[n]$. At the real phases the cited upper recurrences already have rational coefficients. This completes the proof of Theorem III.6.1.

III.6.5 A concrete nonreal phase

For $r = 3$ and $\phi = \pi/2$, the sequence is integer-valued:

$$B(n) = \sum_{\substack{-n \le k \le n \\ k \text{ even}}} (-1)^{k/2} \binom{2n}{n+k}^3.$$

Its critical scales are $26 + 15\sqrt{3}$, 8, and $26 - 15\sqrt{3}$, with scale polynomial

$$(X - 8)(X^2 - 52X + 1) = X^3 - 60X^2 + 417X - 8.$$

Therefore the minimum width is three and the minimum number of terms is four, over $\mathbb{C}(n)$ as well as $\mathbb{Q}(n)$. Every minimum-width recurrence has highest-degree generating symbol proportional to

$$1 - 60t + 417t^2 - 8t^3.$$

Indeed the symbol has degree at most three and must vanish at the reciprocals of all three visible scales. In contrast, the same power $r = 3$ has width two at phase zero and width one at phase π.

The phase classification consequently describes an exact change in recurrence complexity under the weight, while keeping the power fixed. It supplies no uniform bound on coefficient sizes near the exceptional phases, where the critical values collide.

III.7 What is closed, and what remains open

The main statistical closure is estimator-specific. The robust minimax curve was already established in the supplied compilation; Theorem III.2.1 proves that the ordinary NPMLE itself attains it without knowing the contamination fraction. Theorem III.3.1 closes the full parameter range for depths 1 and T, but not for arbitrary replication profiles. The affine identity behind that theorem is special to a depth-one stratum.

For continuous families, Theorem III.4.2 gives an exact entropy budget for one reweighted posterior update, with optimal Gaussian constant. Corollary III.4.6 supplies matching statistical rates in the stated high-precision regimes. It does not settle all intermediate Poisson exposure or Gaussian noise regimes. The counterexamples in Section III.4 explain why unrestricted approximate fits and heteroscedastic observations require additional ideas.

For graph design, Theorem III.5.5 determines the worst growth order over all connected graphs. The leading constants in its universal upper and constructed lower bounds still differ by a factor of two. Determining the exact constant and the extremal graph families remains open here. For weighted binomial sums, Theorem III.6.1 classifies the full unit circle; it does not classify arbitrary complex weights or identify recurrence width with differential-equation order.

The supporting analytic tools and the preceding real-phase and tree results are credited to the earlier manuscripts [20, 25]. The proofs here establish the specified assertions subject to their assumptions. Internal proof checks are not a substitute for external expert review or a comprehensive assessment of historical priority.

III.A Supporting binomial approximation and likelihood lemmas

The following tools were established in the supplied observation compilation. They are reproduced with proofs so that the new statistical arguments do not require an external supplement. They are not counted as new results. Write $B_j^t(x) = \binom{t}{j} x^j (1-x)^{t-j}$ and $p_G(j) = \int B_j^t \, dG$. The distance $W_{1,t}$ uses ground cost $|j - k|/t$, and $h(p, q)^2 = \sum_j (\sqrt{p_j} - \sqrt{q_j})^2$. The same formula defines h for nonnegative vectors of unequal mass. All constants below are absolute.

Lemma III.A.1 (Small covers of the binomial moment body)**.** *For $0 < \beta < 1$ the set $\{p_G : G \in \mathcal{P}([0,1])\}$ has a finite ℓ_1 cover of radius β, with centers in that same set, whose cardinality M satisfies*

$$\log M \le m \log \frac{C(m+1)}{\beta}, \qquad m = \min\left\{t, \left\lceil \sqrt{2t \log(16t/\beta)} \right\rceil \right\}. \tag{III.A.1}$$

Proof. View $K_x = (B_0^t(x), \dots, B_t^t(x))$ as a vector-valued polynomial. Its shifted Chebyshev expansion is

$$K_x = \sum_{k=0}^{t} v_k T_k(2x-1).$$

We claim $\|v_k\|_1 \le 2\exp\{-k^2/(2t)\}$ for $k \ge 1$. Indeed, the even periodic function $F(z) = K_{(1+\cos z)/2}$ has Fourier coefficient $v_k/2$. For $z = \theta + iu$, the ellipse identity gives

$$\left|\frac{1+\cos z}{2}\right| + \left|\frac{1-\cos z}{2}\right| = \cosh u.$$

The binomial formula therefore yields $\|F(\theta+iu)\|_1 = (\cosh u)^t \le e^{tu^2/2}$. Shift the Fourier contour in the appropriate direction and take $u = k/t$. This proves the claim. In particular, truncating after degree $m < t$ has vector ℓ_1 error at most $2te^{-(m+1)^2/(2t)} \le \beta/8$, uniformly in x. For $m = t$ the error is zero.

Each mixing law contributes its first m shifted Chebyshev moments in $[-1,1]^m$. Partition that cube into cells of coordinate diameter at most $\beta/[8(m+1)]$, and choose one mixing-law representative from every nonempty cell. If G, H lie in one cell, their truncated transforms differ in ℓ_1 norm by at most $2m\beta/[8(m+1)] \le \beta/4$. The two tails contribute at most $\beta/4$. Thus their predictive laws have distance at most β. The number of cells is at most $[C(m+1)/\beta]^m$, as asserted. □

Lemma III.A.2 (Likelihood envelopes, including zero cells)**.** *Let $\varepsilon \ge 0$ and $\eta > 0$. Let p be the true law on a finite alphabet with d cells, and suppose a model admits an ℓ_1 cover of radius η/d, with M probability centers in the model. Let $\widetilde{p}$ be any fitted model law satisfying*

$$\sum_{i=1}^{N} \log \widetilde{p}(X_i) \ge \sum_{i=1}^{N} \log p(X_i) - N\varepsilon.$$

For every $x \ge 0$, simultaneously for all such fitted laws, except on an event of probability at most e^{-x},

$$h(p, \widetilde{p}) \le \sqrt{\eta} + \sqrt{\varepsilon + \eta + 2(\log M + x)/N}. \tag{III.A.2}$$

Consequently

$$\mathbb{E}h(p, \widetilde{p})^2 \le C\left\{\varepsilon + \eta + \frac{\log M + 1}{N}\right\}. \tag{III.A.3}$$

Proof. For each center p_a, form the nonnegative vector $u_a(j) = p_a(j) + \eta/d$. Every model law is bounded coordinatewise by at least one u_a, and $\sum_j u_a(j) = 1 + \eta$. If $\widetilde{p} \le u_a$, then

$$h(u_a, \widetilde{p})^2 \le \sum_j (u_a(j) - \widetilde{p}(j)) = \eta.$$

Thus $h(p, \widetilde{p}) > \sqrt{\eta} + r$ implies $h(p, u_a) > r$. Moreover the likelihood comparison implies $\prod_i u_a(X_i)/p(X_i) \ge e^{-N\varepsilon}$. The ratios need only be defined on the support of p, where all sampled

observations lie almost surely. Markov's inequality with exponent $1/2$ gives

$$\begin{aligned}\Pr\left\{\prod_i \frac{u_a(X_i)}{p(X_i)} \ge e^{-N\varepsilon}\right\} &\le e^{N\varepsilon/2}\left(\sum_j \sqrt{p(j)u_a(j)}\right)^N \\ &= e^{N\varepsilon/2}\left(1+\frac{\eta}{2}-\frac{h(p,u_a)^2}{2}\right)^N \\ &\le \exp\{N(\varepsilon+\eta-r^2)/2\}\end{aligned}$$

when $h(p,u_a) > r$. Taking the union over the M envelopes proves (III.A.2). Integrating that tail, and using $(a+b)^2 \le 2a^2+2b^2$, proves (III.A.3). □

Lemma III.A.3 (Stable approximation and bounded-polynomial inversion)**.** *There is an absolute C with the following properties. For every 1-Lipschitz $f:[0,1]\to\mathbb{R}$ and integer $k\ge 1$, there is a polynomial P_k of degree at most k such that*

$$\|P_k - f\|_\infty \le C/k, \qquad \|P_k'\|_\infty \le 2.$$

If $P=\sum_{j=0}^t b_j B_j^t$ has degree $k\le t/2$, then

$$\max_j |b_j| \le Ce^{2k^2/t}\|P\|_\infty. \tag{III.A.4}$$

For the derivative-stable P_k, its coefficient vector also satisfies

$$\max_{j<t} t|b_{j+1}-b_j| \le Ce^{4k^2/t}. \tag{III.A.5}$$

Proof. Subtract $f(0)$, reflect f on $[0,1]$ to a period-two 1-Lipschitz function, and convolve with the normalized positive Jackson kernel $c_s[\sin(s\pi u/2)/\sin(\pi u/2)]^4$. Its first absolute moment is $O(1/s)$, so the resulting trigonometric polynomial Q_s has error $O(1/s)$ and derivative norm at most one. Its frequencies have magnitude at most $2s$, and its Fourier coefficients have modulus at most one. Replacing each exponential by its Taylor polynomial at $1/2$ of degree $100s$ changes the function and derivative by at most e^{-100s}. Indeed $|\pi l(x-1/2)| \le \pi s$ and the factorial inequality $M! \ge (M/e)^M$ bounds both remainders, including the $4s+1$ terms and the extra derivative factor. Take $s=\lfloor k/100\rfloor$ for $k\ge 100$; use a constant polynomial and enlarge C for smaller k.

For (III.A.4), put $M=\|P\|_\infty$ and $F(z)=(1+z)^tP(z/(1+z))$. The maximum principle in the inverse Joukowski variable gives $|P((2+w+w^{-1})/4)| \le M|w|^k$ for $|w|\ge 1$. For $0<j<t$, set $r=j/(t-j)$, $v=r/(1+r)^2$, and $u=2\sqrt{v}\,|\sin(\phi/2)|$. The larger inverse Joukowski modulus for $re^{i\phi}/(1+re^{i\phi})$ is $(1+u)/\sqrt{1-u^2}$. Hence

$$\frac{|F(re^{i\phi})|}{M(1+r)^t} \le (1-u^2)^{(t-k)/2}(1+u)^k \le e^{2k^2/t}e^{-tu^2/8}.$$

The last inequality follows from $k\le t/2$ and completing the square. The average of the final Gaussian factor is at most $C/\sqrt{1+tv}$. Cauchy's coefficient formula, together with $\binom{t}{j}(j/t)^j(1-j/t)^{t-j} \ge c/\sqrt{1+tv}$, proves (III.A.4). The last lower bound follows directly from Stirling's factorial bounds; endpoint coefficients are $P(0)$ and $P(1)$. Finally differentiate the Bernstein expansion: $P_k' = t\sum_{j<t}(b_{j+1}-b_j)B_j^{t-1}$. Apply (III.A.4) at degree $k-1\le (t-1)/2$ and use $2(k-1)^2/(t-1)\le 4k^2/t$. □

Proposition III.A.4 (Two-scale deterministic inversion)**.** *For integers $1\le s\le k\le t/2$ and any two mixing laws G,H,*

$$W_1(G,H) \le \frac{C}{k} + Ce^{4s^2/t}W_{1,t}(p_G,p_H) + \frac{Ce^{2k^2/t}}{s}h(p_G,p_H). \tag{III.A.6}$$

Proof. For any 1-Lipschitz test f, take the polynomials P_k, P_s from Lemma III.A.3. The approximation error in integrating P_k instead of f against $G - H$ is at most C/k. The Bernstein coefficients of P_s have normalized-grid Lipschitz constant at most $Ce^{4s^2/t}$. On the other hand,

$$\|P_k - P_s\|_\infty \le C/k + C/s \le C/s.$$

Thus the Bernstein coefficients of $P_k - P_s$ have supremum norm at most $Ce^{2k^2/t}/s$. Test P_s by count transportation and $P_k - P_s$ by $\|p_G - p_H\|_1 \le 2h(p_G, p_H)$. All constants are uniform over f, so taking the Wasserstein dual supremum proves the result. □

The small amplitude $1/s$ is essential. Applying only the Hellinger bound to the entire polynomial loses the low-frequency empirical $N^{-1/2}$ scale; applying only transportation loses the critical transition. Proposition III.A.4 retains both.

Part IV
Global Parameter Theorems from Entropy, Transport, and Spectral Structure

Manuscript overview. We transfer likelihood entropy, two-scale inversion, canonical singular germs, and spectral optimization to four precisely specified problems. The main statistical theorem determines the Wasserstein minimax risk for Poisson population recovery for every sample size $N \geq 3$ and every known support radius $B > 0$, and proves that every ordinary unrestricted Poisson mixture maximum-likelihood estimator attains it. An exact occupancy channel converts Poisson kernels into binomial kernels while preserving the posterior entropy argument for the original estimator. A second global theorem determines the physical-density connection coefficients for the Jacobi trace for every $N \geq 1$, $a, b > -1$, and $\beta \geq 0$, including all resonances. Convergent corner integrals and a shifted auxiliary differential system give parameter-continuous continuation without singular normalization factors. For every connected block graph, we compute both optimal designs and prove that the worst D-optimal/A-optimal resistance ratio has leading constant $1/(4\sqrt{2})$ times $\sqrt{n}$. Finally, in a known two-Gaussian dictionary, we prove that ordinary likelihood incurs contamination error of order $\sqrt{\varepsilon}$, whereas the minimax error is of order ε, with matching finite-sample bounds and an exact population bias equation. The supporting binomial approximation arguments are reproduced, and the unresolved extensions are stated explicitly.

Integration note. The all-support Poisson theorem is incorporated again in Part VIII with a complete supporting argument. The all-real-parameter Jacobi connection, block-graph extremal constant, and Gaussian corruption obstruction are retained as separate results.

IV.1 The global questions and their precise scope

The first two theorems remove parameter exclusions from specific likelihood and connection problems. The third solves an exact extremal problem throughout a graph class. The fourth identifies a sharp obstruction to extending the same statistical method without additional assumptions. Their conclusions are summarized below; each has a complete proof in its indicated section.

Problem	Scope of the theorem	What is determined
Poisson recovery	Every $N \geq 3$, every known $B > 0$, all mixing laws on $[0, B]$	The minimax W_1 rate and optimality, up to absolute factors, of the original NPMLE; Theorem IV.2.1.
Jacobi trace	Every $N \geq 1$, $a, b > -1$, $\beta \geq 0$, including resonances	All connection discrepancies of the physical density in a canonical normalization; Theorem IV.3.1.
Graph design	Every connected block graph	Exact optimal weights and the sharp asymptotic extremal constant; Theorems IV.4.2 and IV.4.4.
Likelihood robustness	Two fixed Gaussian components, every $N \geq 1$, $0 \leq \varepsilon \leq 1/4$	Matching MLE and minimax risks under both strong replacement and Huber contamination; Theorem IV.5.1.

Provenance of the methods. The entropy–transport and two-scale binomial tools come from the supplied manuscript *Observation and Recovery: Geometry, Mixtures, and Spectral Inversion*, especially its complete binomial-mixture likelihood transition [30]. The preceding proof-transfer drafts developed their robustness consequences and the nonresonant Jacobi and graph comparisons [25, 31]. The new Poisson argument requires an exact channel identity, an infinite-alphabet entropy estimate, and a Poisson analytic cover. The new Jacobi argument requires a full vector of convergent corner integrals; its scalar leading term alone would not justify continuation through resonance. The graph argument requires feasibility of a matrix square-root design on every complete block. The Gaussian result uses a bounded likelihood contrast and an explicit finite outlier attack to determine the limitation of the transfer.

Relation to external questions. The Poisson comparison is with the support-growth restriction in Miao et al.'s 2021 author version, Theorem 6.1(b) [78]. The Jacobi comparison is with the integer-parameter physical connection formula of Forrester–Kumar [84]. These are concrete theorem extensions. The block-graph extremal question and the precise Gaussian separation are formulated and answered here; they are not presented as named conjectures from those sources. These comparisons do not establish priority over all subsequent or related work.

All constants denoted c, C are positive and may change between displays. Unless a dependency is stated, constants in statistical rate comparisons are numerical. W_1 denotes transportation with the usual absolute-distance cost on the underlying real parameter space.

IV.2 Poisson mixture likelihood across all support regimes

Consider the experiment

$$p_G(j) = \int_0^B e^{-\lambda} \frac{\lambda^j}{j!}\, dG(\lambda), \qquad X_1, \dots, X_N \overset{\text{iid}}{\sim} p_G,$$

where the support radius $B > 0$ is known. Let $\widehat{G}$ be any measurable maximizer of the original Poisson mixture likelihood over all probability laws on $[0, B]$. There is no restriction on the number of mixing atoms. Set $L = \log(eN)$ and, for $N \geq 3$, define

$$r(N, B) = \begin{cases} B \min\{1, L^{-1} \log(e + L/B)\}, & 0 < B \leq L, \\ \max\{\sqrt{B/L}, B/\sqrt{N}\}, & B > L. \end{cases} \tag{IV.2.1}$$

The two expressions at the breakpoint agree up to absolute constants.

Theorem IV.2.1 (Global Poisson population recovery)**.** *There exist absolute constants $0 < c < C < \infty$ such that, for every $N \geq 3$ and every $B > 0$,*

$$c\, r(N, B) \leq \inf_{\widetilde{G}} \sup_{G \in \mathcal{P}([0,B])} \mathbb{E}_G W_1(\widetilde{G}, G) \leq \sup_{G \in \mathcal{P}([0,B])} \mathbb{E}_G W_1(\widehat{G}, G) \leq C\, r(N, B).$$

In particular, the ordinary NPMLE is minimax up to absolute factors throughout the entire support/sample-size plane. For observations $X_i \mid \Theta_i \sim \mathrm{Poi}(t\Theta_i)$ with $G \in \mathcal{P}([0, M])$, the corresponding rate is $r(N, tM)/t$.

The small-support rate was established for the NPMLE by Miao et al. [78, Theorem 6.1(a)]. Their Theorem 6.1(b) gives the growing-support rate $\sqrt{B/\log N}$ under $B \leq C_0 N^{1/3-\epsilon_0}$. The point addressed here is the unrestricted growing-support range for the same estimator. We also include the necessary diameter clipping as $B \downarrow 0$, and give direct lower bounds rather than extending a displayed lower-rate formula beyond its meaningful range. The proof uses the two-scale binomial inverse inequality proved in Section IV.A; all additional ingredients are proved below. No claim about the historical priority of every ingredient is intended.

IV.2.1 A parameter-free occupancy channel

For an integer $m \geq 1$, define a Markov kernel A_m as follows. Given a count x, independently assign x balls uniformly to m boxes, and output the number of occupied boxes. Poisson splitting proves

$$A_m \,\mathrm{Poi}(\lambda) = \mathrm{Bin}(m, \phi_m(\lambda)), \qquad \phi_m(\lambda) = 1 - e^{-\lambda/m}. \tag{IV.2.2}$$

Indeed the box counts are independent Poisson random variables of mean λ/m, so their nonempty indicators are independent Bernoulli variables. In particular, if $F = \phi_m \# G$, then $A_m p_G$ is exactly the binomial mixture with mixing law F. On $[0, \phi_m(B)]$ the inverse map has derivative at most $me^{B/m}$, and consequently

$$W_1(G, H) \leq me^{B/m} W_1(\phi_m \# G, \phi_m \# H). \tag{IV.2.3}$$

This channel is a proof device. We do not transform the data before fitting, or assert that a transformed MLE remains an MLE.

We use $W_{1,m}$ for transportation on $\{0, \dots, m\}$ with cost $|i - j|/m$, and $h(p, q)^2 = \sum_j (\sqrt{p_j} - \sqrt{q_j})^2$.

Lemma IV.2.2 (Empirical entropy on the infinite alphabet)**.** *Let q be the empirical law of N observations from p_G. Uniformly over $G \in \mathcal{P}([0, B])$,*

$$\mathbb{E} D(q \| p_G) \leq C(B + L)/N. \tag{IV.2.4}$$

Proof. For every coordinate with probability $p_j > 0$, convexity and $\log u \le u - 1$ give

$$0 \le \mathbb{E}\{q_j \log(q_j/p_j) - q_j + p_j\} \le \min\{(1-p_j)/N,\ p_j \log(1/p_j)\}.$$

The second bound also uses $q_j \log q_j \le 0$. The expression inside braces is nonnegative, so its infinite sum can be averaged by Tonelli's theorem. Coordinates with $p_j = 0$ contribute zero almost surely. For $j \ge B$, the Poisson mass is increasing in its mean throughout $[0, B]$, and hence

$$p_G(j) \le e^{-B} B^j / j! \le (eB/j)^j \le e^{-j} \quad (j \ge e^2 B).$$

Take $J = \lceil e^2 B + 3L \rceil$. The first $J+1$ coordinates contribute at most $(J+1)/N$. Since $u \log(1/u)$ is increasing on $(0, e^{-1}]$, the remaining contribution is bounded by $\sum_{j>J} j e^{-j} \le C(J+1)e^{-J} \le C/N$. This proves (IV.2.4). Notice that the expected empirical chi-square divergence on the full infinite alphabet would not give this conclusion. □

Lemma IV.2.3 (Root-N prediction after occupancy)**.** *For $m = \lceil \max(B, L) \rceil$ and every Poisson NPMLE $\widehat{G}$,*

$$\mathbb{E} W_{1,m}(A_m p_{\widehat{G}}, A_m p_G) \le C/\sqrt{N}. \tag{IV.2.5}$$

Proof. Existence follows from compactness of the mixing parameter space and upper semicontinuity of the likelihood; the admissible fit δ_B is strictly positive at every observed count. Write $H = \widehat{G}$. The likelihood directional derivative gives

$$S(\lambda) = \sum_{j: q_j > 0} \frac{q_j K_\lambda(j)}{p_H(j)} \le 1 \quad (0 \le \lambda \le B).$$

Since $\int S\, dH = 1$, equality holds H-almost everywhere. Thus $R_\lambda(j) = q_j K_\lambda(j)/p_H(j)$ is a probability kernel H-almost everywhere and $\int R_\lambda\, dH = q$. Its entropy chain rule and the likelihood comparison yield

$$\int D(R_\lambda \| K_\lambda)\, dH(\lambda) = D(q\|p_H) \le D(q\|p_G). \tag{IV.2.6}$$

The binomial transport-entropy inequality is

$$W_{1,m}(R, \mathrm{Bin}(m,p))^2 \le D(R\|\,\mathrm{Bin}(m,p))/(2m).$$

For completeness, a normalized-count 1-Lipschitz function changes by at most $1/m$ on changing one of the m Bernoulli bits. The conditional Hoeffding bound gives centered log-mgf at most $u^2/(8m)$; entropy duality and Wasserstein duality give the displayed inequality, including degenerate endpoints whenever the entropy is finite. Apply it to $A_m R_\lambda$, use entropy data processing, and integrate (IV.2.6). Convexity of transportation under mixtures and Cauchy–Schwarz give

$$W_{1,m}(A_m q, A_m p_H) \le \sqrt{D(q\|p_G)/(2m)}.$$

For each count threshold, the CDF of $A_m q$ is an average of independent $[0,1]$-valued functions of the X_i. Its expected absolute error is at most $1/(2\sqrt{N})$. The CDF formula for $W_{1,m}$ therefore gives $\mathbb{E} W_{1,m}(A_m q, A_m p_G) \le 1/(2\sqrt{N})$. Now use Lemma IV.2.2 and $(B+L)/m \le 2$. □

IV.2.2 A growing-support Hellinger bound

Lemma IV.2.4 (Poisson mixture entropy and likelihood control)**.** *For every fixed absolute $b_0 > 0$, uniformly over $b_0 L \le B \le N$,*

$$\mathbb{E} h(p_{\widehat{G}}, p_G) \le C_{b_0} B^{1/4} L^{3/4} / \sqrt{N}. \tag{IV.2.7}$$

Proof. For complex z the vector $K_z = (e^{-z} z^j/j!)_{j\ge 0}$ belongs to ℓ_1 and satisfies $\|K_z\|_1 = e^{|z|-\operatorname{Re} z}$. On the ellipse $z = B(1+\cos(\theta+iu))/2$, direct calculation gives

$$|z| - \operatorname{Re} z = \tfrac{B}{2}(\cosh u - 1)(1-\cos\theta) \le B(\cosh u - 1).$$

The kth vector Chebyshev coefficient v_k of $x \mapsto K_{Bx}$ therefore satisfies, by contour shifting,

$$\|v_k\|_1 \le 2\exp\{B(\cosh u - 1) - ku\} \le 2\exp\{-c\min(k^2/B, k)\}.$$

For the second inequality take $u = k/(2B)$ if $k \le 2B$, using $\cosh u - 1 \le u^2$ for $0 \le u \le 1$, and take $u = 1$ otherwise. It follows by summing the Gaussian and exponential tails that truncating the expansion at

$$d = \left\lceil C\left\{\sqrt{B\log(C(B+1)/\beta)} + \log(C(B+1)/\beta)\right\}\right\rceil$$

has uniform ℓ_1 error at most $\beta/8$. Every Chebyshev moment of a probability law on $[0,B]$ belongs to $[-1,1]$, and $\|v_k\|_1 \le 2$ also follows directly from its real integral formula. Grid the first d moments with mesh $\beta/(16(d+1))$. Choose one actual mixture in each occupied grid box. This gives an ℓ_1 cover of radius β, with centers in the mixture class and cardinality M satisfying

$$\log M \le Cd\log(C(d+1)/\beta). \tag{IV.2.8}$$

Take $J = \lceil e^2 B + 6L\rceil$ and $\beta = N^{-3}/(J+1)$. In the indicated range, $d \le C_{b_0}\sqrt{BL}$ and $\log M \le C_{b_0}\sqrt{B}L^{3/2}$. For every cover center p_a, define the pointwise envelope

$$u_a(j) = \begin{cases} p_a(j) + \beta, & j \le J, \\ e^{-j}, & j > J. \end{cases}$$

Every mixture is bounded coordinatewise by its associated envelope, whose total mass U_a lies in $[1, 1+\eta]$ for $\eta = CN^{-3}$, whenever that envelope is used for at least one mixture. Unused envelopes can be discarded. Normalize $r_a = u_a/U_a$. An associated mixture p satisfies $h(p, r_a) \le \sqrt{2\eta}$.

Here is the elementary likelihood argument, to specify the treatment of the infinite alphabet. If $\widehat{p}$ is a likelihood maximizer and $u_a \ge \widehat{p}$, then

$$\prod_{i=1}^{N} \frac{r_a(X_i)}{p_G(X_i)} \ge e^{-N\eta}.$$

Under p_G, the expectation of the square root of this product is the Nth power of the affinity, at most $\exp\{-Nh(p_G, r_a)^2/2\}$. Markov's inequality followed by a union bound shows, with probability at least $1 - e^{-x}$,

$$h(p_G, \widehat{p}) \le \sqrt{2\eta} + \sqrt{\eta + 2(\log M + x)/N}.$$

Integration gives $\mathbb{E}h(p_G, \widehat{p})^2 \le C\eta + C(\log M + 1)/N$, which proves (IV.2.7). □

IV.2.3 The two-scale argument closes every growing-support regime

Recall the already proved binomial inverse inequality: if F, H are mixing laws on $[0,1]$, b_F, b_H their m-trial binomial mixtures, and $1 \le S \le K \le m/2$, then

$$W_1(F,H) \le C/K + Ce^{4S^2/m} W_{1,m}(b_F, b_H) + Ce^{2K^2/m} h(b_F, b_H)/S. \tag{IV.2.9}$$

Its small high-frequency amplitude $1/S$ is indispensable. The proof approximates a Lipschitz test function at two polynomial degrees, preserves a bounded derivative at the lower degree, and uses

the degree-elevated Bernstein coefficient bounds. This is Proposition IV.A.2, not an additional unproved input.

Suppose $b_0 L \le B \le N$, with $b_0 > 0$ an absolute constant to be fixed below. Set $m = \lceil \max(B, L) \rceil$, so $m \asymp_{b_0} B$. For sufficiently large absolute N, take

$$S = \lfloor \sqrt{m}/16 \rfloor, \qquad K = \left\lfloor \min\{m/2, \sqrt{amL}\} \right\rfloor, \qquad a = 1/64.$$

These are admissible, $S \asymp \sqrt{m}$, $e^{4S^2/m} \le C$, and $m/K \le C\sqrt{m/L}$ because $m \ge L$. Apply (IV.2.9) to the pushforwards under ϕ_m, then use (IV.2.3), Lemmas IV.2.3 and IV.2.4, and Hellinger data processing. This gives

$$\mathbb{E}W_1(\widehat{G}, G) \le C_{b_0} \left\{ \sqrt{B/L} + \frac{B}{\sqrt{N}} + \frac{e^{2aL} B^{3/4} L^{3/4}}{\sqrt{N}} \right\}.$$

The ratio of the last term to $\sqrt{B/L}$ is at most $Ce^{2aL} N^{-1/4} L^{5/4}$, which is uniformly bounded since $B \le N$ and $2a < 1/4$. We have proved

$$\mathbb{E}W_1(\widehat{G}, G) \le C_{b_0} \{ \sqrt{B/L} + B/\sqrt{N} \}. \tag{IV.2.10}$$

Bounded N is absorbed in the absolute constant by the diameter bound; this is uniform in B because $B \ge b_0 L$ in this step.

If $B > N$, no Hellinger estimate or approximation is necessary. For any binomial mixing law, coupling its parameter p with its normalized count gives expected cost at most $1/(2\sqrt{m})$. Thus

$$W_1(F, H) \le m^{-1/2} + W_{1,m}(b_F, b_H).$$

Together with (IV.2.3) and Lemma IV.2.3, this gives $\mathbb{E}W_1(\widehat{G}, G) \le C(\sqrt{B} + B/\sqrt{N}) \le CB/\sqrt{N}$. Every $B \ge L$ is now covered.

IV.2.4 The small-support regime, including arbitrarily small radii

We include a direct proof to ensure uniformity as $B \downarrow 0$. Let f be 1-Lipschitz on $[0, B]$ and normalize $f(0) = 0$. Jackson approximation gives a degree-k polynomial $P(\lambda) = \sum_{l=0}^{k} a_l \lambda^l$ such that

$$\|f - P\|_{[0,B]} \le CB/k, \qquad \|P\|_{[0,B]} \le CB.$$

Its monomial coefficients satisfy

$$|a_l| \le BC_1^k B^{-l}. \tag{IV.2.11}$$

Indeed expand $P(Bx)$ in shifted Chebyshev polynomials: each Chebyshev coefficient has magnitude at most $2\|P\|$, and the sum of absolute monomial coefficients of $T_j(2x-1)$ is at most 7^j by its three-term recurrence. Summing the finite geometric series proves (IV.2.11) after enlarging C_1.

Take $k \ge 2eB$ and put $Q_k(\lambda) = \sum_{r=0}^{k} \lambda^r / r!$. The exponential remainder implies

$$\|e^{-\lambda} P(\lambda) Q_k(\lambda) - f(\lambda)\|_{[0,B]} \le CB/k.$$

For example, $0 \le 1 - e^{-\lambda} Q_k(\lambda) \le C(eB/(k+1))^{k+1} \le C2^{-k}$, which suffices. Write this weighted polynomial as $\sum_{j=0}^{2k} b_j K_\lambda(j)$. Since

$$b_j = \sum_{\substack{0 \le l \le k \\ 0 \le j-l \le k}} a_l \frac{j!}{(j-l)!},$$

equation (IV.2.11) gives

$$\max_{0\le j\le 2k}|b_j| \le B(C_2k/B)^k. \tag{IV.2.12}$$

The factors $k+1$ and C_1^k have been absorbed into C_2^k; $2k/B \ge 1$ by hypothesis. Wasserstein duality consequently gives the deterministic inequality

$$W_1(F,H) \le CB/k + B(C_2k/B)^k\|p_F - p_H\|_1. \tag{IV.2.13}$$

For the MLE, likelihood comparison, $h^2 \le D$, the Hellinger triangle inequality and $\|p-q\|_1 \le 2h(p,q)$ imply

$$\mathbb{E}\|p_{\widehat{G}} - p_G\|_1 \le 4\sqrt{\mathbb{E}D(q\|p_G)} \le C\sqrt{L/N} \qquad (B \le L).$$

Put $d_0 = \log(e + L/B)$ and $k = \lfloor \gamma L/d_0 \rfloor$, where $\gamma > 0$ is a sufficiently small absolute constant. If $\gamma L/d_0 < 2$, the diameter bound $W_1 \le B$ is already at most a constant times $B\min(1, d_0/L)$. Otherwise $k \ge \gamma L/(2d_0)$. Choose a fixed $b_0 > 0$ sufficiently small that $B \le b_0 L$ implies $\gamma(L/B)/(2\log(e+L/B)) \ge 2e$; then $k \ge 2eB$. Furthermore

$$k\log(C_2k/B) \le C_3\gamma L \le L/8$$

after decreasing γ once, since $\log(C_2\gamma(L/B)/d_0) \le C_3d_0$. The stochastic term in (IV.2.13) is therefore at most $CBe^{L/8}\sqrt{L/N} \le CB/k$: here $k \le \gamma L$ and $e^{-3L/8}L^{3/2}$ is bounded. This proves the required rate for $B \le b_0L$. On $b_0L \le B \le L$, equation (IV.2.10) is bounded by an absolute constant, whereas $B\min(1, d_0/L)$ is bounded below by a positive absolute constant. This proves the upper bound in Theorem IV.2.1 for every B.

IV.2.5 Matching lower bounds without support-range restrictions

We first record a convenient moment-matching pair. On an interval $[a-M, a+M]$, push forward the probability measures

$$(1 \pm \tfrac{1}{2}T_k(x))\frac{dx}{\pi\sqrt{1-x^2}}, \qquad -1 < x < 1,$$

under $x \mapsto a + Mx$. Denote them by F_k^+, F_k^-. They match all moments of degree less than k. Their CDF difference is, up to orientation, $\sin(k\theta)/(\pi k)$ for $x = \cos\theta$. Consequently

$$W_1(F_k^+, F_k^-) = \frac{M}{\pi k}\int_0^\pi |\sin(k\theta)|\sin\theta\, d\theta \ge \frac{M}{\pi k}. \tag{IV.2.14}$$

The last inequality follows from $|\sin u| \ge \sin^2 u$ and $\int_0^\pi \sin^2(k\theta)\sin\theta\, d\theta = 1 + (4k^2-1)^{-1}$.

For $B \le L$, use this pair on $[0, B]$. Expanding $e^{-\lambda}$ and using the matched moments gives the elementary bound

$$\|p_{F_k^+} - p_{F_k^-}\|_1 \le 2\sum_{r\ge k}\frac{(2B)^r}{r!}. \tag{IV.2.15}$$

Indeed the term with observed count j and exponential-expansion index l vanishes if $j + l < k$; summing the absolute values of all remaining terms gives (IV.2.15). Set $d_0 = \log(e + L/B)$ and $k = \lceil 100L/d_0 \rceil$. Writing $t = L/B \ge 1$, we have $t = e^{d_0} - e \ge e^{d_0}/(e+1)$ and $e^{d_0/2}/d_0 \ge e/2$. These inequalities show

$$\log\frac{k}{2eB} \ge \frac{d_0}{2}, \qquad k\log\frac{k}{2eB} \ge 50L.$$

They also ensure $k \ge 4B$. Thus the tail in (IV.2.15) is at most $4(2eB/k)^k \le 4e^{-50L}$. The two N-sample laws have TV at most $2Ne^{-50L}$, whereas (IV.2.14) is at least $cB\min\{1, d_0/L\}$. Le Cam's two-point inequality proves the small-support lower bound, including arbitrarily tiny B.

For $B \ge L$, set $a = B/2$, $M = \sqrt{BL}/8$, and $k = \lceil 20L \rceil$. The moment-matching interval lies inside $[0, B]$. The exact Poisson identity

$$\sum_{x=0}^{\infty} \frac{K_\lambda(x) K_\mu(x)}{K_a(x)} = \exp\{(\lambda - a)(\mu - a)/a\}$$

shows, upon integrating against the signed difference $F_k^+ - F_k^-$, that

$$\sum_x \frac{(p_{F_k^+}(x) - p_{F_k^-}(x))^2}{K_a(x)} \le 4 \sum_{r \ge k} \frac{(M^2/a)^r}{r!} = 4 \sum_{r \ge k} \frac{(L/32)^r}{r!}.$$

All exchanges are justified by boundedness of λ, μ on the integration interval and the absolutely convergent exponential series. The last tail is at most Ce^{-40L}, for instance by $r! \ge (r/e)^r$ and a geometric-tail bound. Cauchy–Schwarz bounds the predictive TV by half the square root of this quantity. Le Cam's inequality and (IV.2.14) therefore give the lower rate $c\sqrt{B/L}$.

Finally consider mixing laws on $\{0, B\}$ with respective weights $1/2 \pm c_0/\sqrt{N}$ at B, for a sufficiently small absolute $c_0 > 0$. Their latent N-sample laws have TV bounded away from one, by the Bernoulli KL formula and Pinsker's inequality. The Poisson sample is a Markov image of that latent sample, so its TV is no larger. The mixing-law distance is $2c_0 B/\sqrt{N}$. Le Cam's inequality yields the remaining lower rate $cB/\sqrt{N}$. Taking the maximum completes the proof of Theorem IV.2.1.

The proof uses a common known exposure and a known compact support. It makes no assertion for heterogeneous exposures, unknown support radii, or an MLE constrained to a prescribed number of components. The gain over a direct one-scale inverse estimate is structural: the occupancy channel imports a binomial concentration inequality, likelihood contact supplies posterior entropy, and the two-scale inverse places predictive transportation and high-frequency Hellinger error in different terms.

IV.3 Jacobi trace connections for all real parameters

Forrester and Kumar formulate the physical-density connection problem for the trace of the β-Jacobi ensemble and give its coefficients when b and β are nonnegative integers [84, Proposition 1.3 and Section 3.3]. The preceding research draft extended the formula to nonintegral local exponents, retaining $b > 0$. Here we remove both restrictions. The normalization at a resonant exponent is specified by a convergent simplex integral, and continuation uses an auxiliary system whose coefficients have no parameter poles. The result concerns the physical density in this canonical normalization, not a connection matrix between arbitrary choices of local bases.

IV.3.1 Canonical corner functions

Fix $N \ge 1$, $a, b > -1$, and $\beta \ge 0$. Write

$$S_N = \int_{[0,1]^N} |\Delta(x)|^\beta \prod_i x_i^a (1-x_i)^b \, dx, \qquad \rho(t) = \frac{1}{S_N} \int_{[0,1]^N} \delta\Big(t - \sum_i x_i\Big) |\Delta(x)|^\beta \prod_i x_i^a (1-x_i)^b \, dx.$$

The density is zero outside $[0, N]$. At an integer it can diverge; statements about values there are understood by one-sided limits or as identities of locally integrable distributions. Define the Laguerre integrals

$$W_r(q) = \int_{(0,\infty)^r} e^{-\sum x_i} |\Delta(x)|^\beta \prod_i x_i^q \, dx = \prod_{j=0}^{r-1} \frac{\Gamma(q + 1 + j\beta/2)\Gamma(1 + (j+1)\beta/2)}{\Gamma(1 + \beta/2)}, \qquad W_0 = 1.$$

The normalizing integral is

$$S_N = \prod_{j=0}^{N-1} \frac{\Gamma(a+1+j\beta/2)\Gamma(b+1+j\beta/2)\Gamma(1+(j+1)\beta/2)}{\Gamma(a+b+2+(N+j-1)\beta/2)\Gamma(1+\beta/2)}.$$

For $0 \le p \le N$, put $m = N - p$ and

$$\alpha_p = m(a+1) + \tfrac{\beta}{2} m(m-1), \qquad \eta_p = p(b+1) + \tfrac{\beta}{2} p(p-1),$$
$$D_p = \alpha_p + \eta_p, \qquad \gamma_p = D_p - 1,$$
$$Z_p = \frac{W_m(a) W_p(b)}{\Gamma(D_p)}.$$

In particular $D_p > 0$ and $Z_p > 0$.

On the simplex

$$\mathcal{D}_p = \{(u,v) \in [0,\infty)^m \times [0,\infty)^p : \sum_i u_i + \sum_j v_j = 1\},$$

use the measure $d\sigma$ obtained by eliminating the last coordinate with Jacobian one, equivalently the Dirac-delta simplex measure. For $N = 1$ it is the unit mass on the single point. Set

$$w_p(u,v) = |\Delta(u)|^\beta |\Delta(v)|^\beta \prod_i u_i^a \prod_j v_j^b,$$
$$A_p(s;u,v) = \prod_i (1-su_i)^b \prod_j (1+sv_j)^a \prod_{i,j} (1+s(v_j-u_i))^\beta,$$
$$U_p(s) = Z_p^{-1} \int_{\mathcal{D}_p} w_p(u,v) A_p(s;u,v)\, d\sigma, \qquad |s| < 1,$$
$$F_p(p+s) = s^{\gamma_p} U_p(s), \qquad 0 < s < 1.$$

All analytic factors in A_p use the branches equal to one at $s = 0$. Empty products are one. Radial integration of the two Laguerre clusters gives $\int_{\mathcal{D}_p} w_p\, d\sigma = Z_p$. Consequently $U_p(0) = 1$. The integral converges for every stated parameter and is holomorphic in s on $|s| < 1$. On compact subsets of the parameter domain it and all its s derivatives depend continuously on the parameters. No division by $\sin(\pi D_p)$ occurs.

Theorem IV.3.1 (All-parameter physical connection formula)**.** *For every $N \ge 1$, $a, b > -1$, and $\beta \ge 0$, each canonical F_p has analytic continuation along paths avoiding $0, 1, \dots, N$. Let $F_p^{\uparrow}(t)$ denote continuation to the right with every intervening integer passed in the upper half-plane. Define*

$$C_p = e^{-\pi i \eta_p} \binom{N}{p} \frac{Z_p}{S_N}. \tag{IV.3.1}$$

Then, for $k = 0, \dots, N-1$ and $k < t < k+1$,

$$\boxed{\rho(t) = \sum_{p=0}^{k} C_p F_p^{\uparrow}(t).} \tag{IV.3.2}$$

Equivalently, at every knot p, the physical right germ minus the upper continuation of the physical left germ is $C_p F_p$. This includes $p = N$, with zero physical density on its right. The lower-half-plane formula is the complex conjugate. All assertions include resonances $\gamma_p \in \mathbb{Z}$ and the entire range $-1 < b \le 0$.

IV.3.2 An auxiliary system that remains regular in the parameters

The following construction supplies both continuation and the normalization needed at resonance. Let $\widetilde{b} = b + 1 > 0$ and

$$\Lambda = -\operatorname{diag}(N, N-1, \dots, 0), \qquad h = a + \widetilde{b} + 1 + \tfrac{\beta}{2}(2N-1), \qquad k = \widetilde{b} + \tfrac{\beta}{2}N.$$

For indices $j = 0, \dots, N$, define the tridiagonal matrix X by

$$\begin{aligned} X_{j,j-1} &= -jk + \tfrac{\beta}{2}j^2, \\ X_{j,j} &= hj - \tfrac{\beta}{2}(N-j)(2j+1) + k(N-2j) - 1, \\ X_{j,j+1} &= h(j-N) + \tfrac{\beta}{2}(N-j)(N-j-1) + k(N-j). \end{aligned}$$

Missing entries are zero. Write $P_{j\ell} = \binom{N-j}{N-\ell}$, with a binomial coefficient zero outside its usual range. The Forrester–Kumar system is

$$(tI + \Lambda)G'(t) = XG(t), \qquad H = PG. \tag{IV.3.3}$$

With the shifted parameter, its physical auxiliary vector is

$$H_j(t) = \binom{N}{j}^{-1} \int_{[0,1]^N} \delta\Big(t - \sum_i x_i\Big) |\Delta(x)|^\beta \prod_i x_i^a (1-x_i)^b e_j(1-x_1, \dots, 1-x_N)\, dx.$$

Thus $H_0 = S_N \rho$, and every defining integral is absolutely convergent throughout the parameter range of the theorem.

Lemma IV.3.2 (Explicit corner vectors and parameter-continuous transport)**.** *For a fixed p, define the vector germ*

$$\begin{aligned} H_{p,j}(p+s) = \frac{s^{\gamma_p}}{\binom{N}{j}} \int_{\mathcal{D}_p} w_p A_p(s; u, v) \\ \cdot e_j(1 - su_1, \dots, 1 - su_m, -sv_1, \dots, -sv_p)\, d\sigma. \end{aligned} \tag{IV.3.4}$$

Then $G_p = P^{-1}H_p$ solves (IV.3.3), and $H_{p,0} = Z_p F_p$. Its continuation along any fixed path avoiding the integers is continuous in all real parameters $a, b > -1$, $\beta \geq 0$.

Proof. First take a, b, β sufficiently large and positive. The vector integral is the slice integral on the cone having m coordinates $x_i \geq 0$ and p coordinates $x_j \geq 1$, with $x_i = su_i$ and $x_j = 1 + sv_j$. Restrict to $0 < s < 1$. The slice is compact, the lower coordinates stay below one, and the upper coordinates exceed one. Cross-cluster differences are $1 + s(v_j - u_i) > 0$. The upper factors $(x_j - 1)^b$ differ from the corresponding analytic Jacobi powers by a constant phase, so their logarithmic derivatives are the same. Sum over the choices of the upper coordinates to make the domain symmetric; this multiplies every vector component by the same factor $\binom{N}{p}$.

The integration-by-parts identities giving (IV.3.3) therefore apply on this cone as well. Endpoint and within-cluster collision boundary terms vanish for the chosen large parameters; the compact slice has no boundary at infinity. Equivalently these identities may be tested against a smooth compactly supported function of $s \in (0,1)$ before integration by parts. The change of variables gives exactly (IV.3.4), including its factor $s^{D_p - 1}$ and its elementary symmetric polynomials.

For fixed $0 < s < 1$, the vector integral and its derivative are holomorphic for $\operatorname{Re} a, \operatorname{Re} b > -1$ and $\operatorname{Re} \beta > 0$. On compact subsets the endpoint factors have a common integrable majorant; the collision powers only improve integrability. The identity theorem continues the differential

identity from the large-parameter set to this connected domain. Dominated convergence includes $\beta = 0$. The zeroth coordinate is $Z_p F_p$ by definition. More explicitly, the leading coefficients are

$$H_{p,j}(p+s) = Z_p s^{\gamma_p} \left(\frac{\binom{N-p}{j}}{\binom{N}{j}} + O(s) \right), \qquad G_p(p+s) = \frac{Z_p}{\binom{N}{p}} s^{\gamma_p} \left(e_{N-p} + O(s) \right),$$

because $\binom{N-p}{j}/\binom{N}{j} = P_{j,N-p}/\binom{N}{p}$.

Finally, away from the integers, the coefficient matrix $(tI + \Lambda)^{-1}X$ is holomorphic in t and polynomial in the parameters. Its fundamental matrix along a fixed compact path depends continuously, indeed locally holomorphically, on these parameters. Start at any fixed $s_0 \in (0,1)$ with the explicit vector integral. This proves the claimed continuation and parameter dependence without dividing by resonant exponent differences. □

IV.3.3 Determination of the coefficients off resonance

We give the local argument in order to distinguish the complex connection coefficient from a one-sided real singular coefficient. At $t = p$, the residue of the shifted system has only row $N - p$ nonzero, and its nonzero eigenvalue is γ_p. If γ_p is nonintegral, the residue is diagonalizable with eigenvalues 0 and γ_p. The regular-singular recurrence gives N holomorphic solutions and one nonintegral solution. The zeroth H coordinate of its leading vector is nonzero because $P_{0,N-p} = \binom{N}{p} \neq 0$. Lemma IV.3.2 therefore identifies its normalized zeroth coordinate with F_p.

For a smooth cutoff χ equal to one near p, the complete local Fourier expansion of the physical density is

$$\int e^{iRt} \chi(t) \rho(t)\, dt \sim e^{ipR} R^{-D_p} \sum_{\ell \geq 0} A_{p,\ell} R^{-\ell}, \quad A_{p,0} = K_p e^{\pi i(\alpha_p - \eta_p)/2}, \quad K_p = \binom{N}{p} \frac{W_m(a) W_p(b)}{S_N}. \tag{IV.3.5}$$

Here and below completeness means a remainder to any prescribed finite order. To verify it, near a corner write the lower coordinates as x and the upper coordinates as $1 - y$. The phase is $p + \sum x - \sum y$, and the weight is

$$|\Delta(x)|^{\beta} |\Delta(y)|^{\beta} \prod x_i^a \prod y_j^b \prod_i (1 - x_i)^b \prod_j (1 - y_j)^a \prod_{i,j} (1 - x_i - y_j)^{\beta}.$$

Its final three factors are analytic near the corner. Radial variables $r = \sum x_i$ and $q = \sum y_j$ have powers $r^{\alpha_p - 1}$ and $q^{\eta_p - 1}$. Taylor expansion of the analytic factor and successive one-dimensional endpoint Fourier integrals give only the powers $R^{-D_p - \ell}$. The angular constants are the two Laguerre integrals divided by $\Gamma(\alpha_p)$ and $\Gamma(\eta_p)$; empty clusters are omitted. This proves the leading coefficient in (IV.3.5) and the entire power scale.

Away from these corners, a cluster of coincident coordinates lies strictly inside $(0,1)$. Translating that whole cluster leaves its internal differences fixed and gives a nonzero phase derivative. It also stays tangent to any other fixed boundary faces. A local partition of unity and repeated integration by parts in that direction give arbitrarily rapid decay. All differentiated noncluster factors are smooth on the selected neighborhood, while the remaining endpoint and collision powers are integrable. Thus there are no other power contributions at frequency p.

Suppose now that D_p is nonintegral. On the two sides of $s = 0$, regular-singular theory gives

$$\rho(p+s) = \begin{cases} a_+(s) + c_+ s^{\gamma_p} U_p(s), & s > 0, \\ a_-(s) + c_- (-s)^{\gamma_p} U_p(s), & s < 0, \end{cases}$$

where $a_\pm$ are holomorphic and $c_\pm$ are real. If the analytic backgrounds first differ at Taylor degree q, their cutoff Fourier transform has a nonzero R^{-q-1} term. This contradicts (IV.3.5),

whose powers are all nonintegral. Hence $a_+ = a_-$. Comparing the leading coefficient yields

$$c_+ = \frac{K_p \sin(\pi\alpha_p)}{\Gamma(D_p)\sin(\pi D_p)}, \qquad c_- = \frac{K_p \sin(\pi\eta_p)}{\Gamma(D_p)\sin(\pi D_p)}.$$

Upper continuation changes $(-s)^{\gamma_p}$ into $e^{-\pi i\gamma_p} s^{\gamma_p}$. Therefore the discrepancy is

$$(c_+ - e^{-\pi i\gamma_p} c_-)F_p = e^{-\pi i\eta_p} \frac{K_p}{\Gamma(D_p)} F_p = C_p F_p.$$

The sine identity uses $D_p = \alpha_p + \eta_p$. It includes both endpoints with the exterior density zero. Starting from the first interval and successively adding these discrepancies proves (IV.3.2) whenever all relevant γ_p are nonintegral.

IV.3.4 Crossing every resonant hyperplane

Completion of Theorem IV.3.1. Fix any admissible parameters and a noninteger t. Approximate the parameters by admissible real triples with $\beta > 0$ and all γ_p nonintegral; the excluded affine hyperplanes are locally finite. Each C_p has a finite continuous limit. By Lemma IV.3.2, so does each continued $F_p^{\uparrow}(t)$, using the same paths and starting points throughout.

The physical slice integral is continuous in the parameters at this fixed t. To see a common integrable bound, note that, since t is not an integer, at least one coordinate of every point of the slice is bounded away from both zero and one. A finite cover selects such a coordinate and eliminates it with Jacobian one. Its endpoint factors are bounded, the other endpoint factors have a uniform integrable majorant with powers greater than -1, and $|\Delta|^\beta \le 1$ for real $\beta \ge 0$ on the cube. Dominated convergence now applies on each patch. The normalizer is positive and continuous as well.

Pass to the limit in the nonresonant identity to obtain (IV.3.2) at the chosen parameters. The same argument applies to the local discrepancy, using ODE continuation for the left germ and zero exterior density at the endpoints. It also includes $\beta = 0$. No one-sided formula containing $1/\sin(\pi D_p)$ has been evaluated at a pole: the continued canonical functions and their connection coefficients are separately finite before the limit is taken. □

Example IV.3.3 (Complete resonance and logarithmic resonance)**.** If $a = b = \beta = 0$, then $U_p = 1$ and $Z_p = 1/(N-1)!$. The theorem gives the familiar identity

$$\rho(t) = \frac{1}{(N-1)!} \sum_{p=0}^{\lfloor t \rfloor} (-1)^p \binom{N}{p} (t-p)^{N-1}, \qquad 0 < t < N,$$

with the sum taken away from integer t. Every exponent is resonant. For $N = 1$ it gives the beta density directly. For $N = 2$, $a = b = -1/2$, $\beta = 0$, the physical density has a logarithmic singularity at $t = 1$, while $\gamma_1 = 0$ and U_1 is the convergent analytic simplex integral above. The logarithm appears in the continued earlier branch; it does not force a pole in C_1 or in the canonical U_1.

This removes the nonresonance and auxiliary-integrability restrictions of the preceding connection theorem for every integrable real Jacobi parameter with $\beta \ge 0$. The system itself and its classical Selberg constants are inherited from the cited literature. No claim is made about stable floating-point evaluation for large N, arbitrary-basis connection matrices, or a complete historical-priority assessment of related Selberg connection formulas.

IV.4 Exact designs and a sharp extremal constant on block graphs

The universal connected-graph comparison leaves a factor of two between its upper and lower asymptotic constants. We close that gap for the entire class of connected *block graphs*, namely graphs whose blocks are complete graphs, with every bridge regarded as a block K_2. This class contains all trees, all complete graphs, and arbitrary configurations of cliques joined at articulation vertices. In addition to its sharp leading constant, it admits an exact finite-graph formula for every optimal conductance. We do not claim that the remaining constant gap on arbitrary connected graphs is closed.

For clarity, a design consists of conductances $w_e \geq 0$ with $\sum_{e\in E} w_e = 1$ and connected positive-weight support. Choose an orientation of each edge and let $b_e = e_u - e_v$; its Laplacian is $L(w) = \sum_e w_e b_e b_e^{\mathsf{T}}$. The D-optimal design w^D maximizes $\det(L(w)|_{\mathbf{1}^\perp})$, while the A-optimal design w^A minimizes the unordered-pair total effective resistance

$$K_G(w) = n \operatorname{tr} L(w)^{\dagger}.$$

Throughout this section, $\alpha(G) = K_G(w^D)/K_G(w^A)$.

IV.4.1 The demand matrix of a block

Let G be a connected block graph with $n \geq 2$ vertices and $d = n - 1$. Write $\mathcal{B}$ for its blocks. If $B \in \mathcal{B}$ has vertex set $\{1, \dots, m_B\}$, delete all edges of B from G. The resulting components contain exactly one vertex of B each. Let their sizes be

$$a_{B,1}, \dots, a_{B,m_B}, \qquad a_{B,i} \geq 1, \qquad \sum_i a_{B,i} = n.$$

Define the block demand Laplacian

$$C_B = n \operatorname{diag}(a_B) - a_B a_B^{\mathsf{T}}. \tag{IV.4.1}$$

It is the complete-graph Laplacian with conductance $a_{B,i}a_{B,j}$ on the pair $\{i, j\}$. Put

$$q_B = \operatorname{tr} C_B^{1/2}, \qquad Q = \sum_{B\in\mathcal{B}} q_B.$$

All square roots here are the positive semidefinite matrix square roots.

The unordered-pair total effective resistance decomposes by blocks:

$$K_G(w) = \sum_{B\in\mathcal{B}} \sum_{i<j} a_{B,i}a_{B,j} R_{ij}(L_B) = \sum_{B\in\mathcal{B}} \operatorname{tr}(C_B L_B^{\dagger}). \tag{IV.4.2}$$

Indeed, the block-cut tree specifies the blocks traversed by each pair of vertices. Resistances add along that chain, and precisely $a_{B,i}a_{B,j}$ pairs project to distinct vertices i, j of B.

Lemma IV.4.1 (Feasibility of the square-root block design)**.** *For each block B, the matrix $C_B^{1/2}$ has zero row sums and strictly negative off-diagonal entries. It is therefore a feasible positive-conductance Laplacian on the complete block B.*

Proof. Zero row sums follow from $C_B\mathbf{1} = 0$ and spectral calculus. For a positive semidefinite matrix C, the scalar integral for the square root gives

$$C^{1/2} = \frac{1}{\pi} \int_0^{\infty} t^{-1/2} C (tI + C)^{-1} \, dt.$$

For the special matrix in (IV.4.1), put $D_t = tI + n\,\mathrm{diag}(a_B)$. Sherman–Morrison gives

$$(tI + C_B)^{-1} = D_t^{-1} + \frac{D_t^{-1} a_B a_B^{\mathsf{T}} D_t^{-1}}{1 - a_B^{\mathsf{T}} D_t^{-1} a_B}.$$

The denominator is strictly positive for $t > 0$, since

$$\sum_i \frac{a_{B,i}^2}{t + n a_{B,i}} < \sum_i \frac{a_{B,i}}{n} = 1.$$

Every off-diagonal resolvent entry is therefore positive. For $i \neq j$,

$$\big[C_B(tI + C_B)^{-1}\big]_{ij} = -t\big[(tI + C_B)^{-1}\big]_{ij} < 0.$$

Integration proves the claimed strict sign. □

IV.4.2 Exact optimal weights and a spectral moment formula

Theorem IV.4.2 (Exact designs on every connected block graph)**.** *The A-optimal Laplacian on each block is*

$$L_B^A = \frac{2}{Q} C_B^{1/2}, \qquad w_{ij}^A = -\frac{2}{Q}(C_B^{1/2})_{ij} \quad (i \neq j). \tag{IV.4.3}$$

The total conductance assigned to B is q_B/Q, and

$$K_G(w^A) = \frac{Q^2}{2}.$$

The D-optimal conductance on every edge of a block of size m_B is

$$w_e^D = \frac{2}{m_B d}, \qquad K_G(w^D) = \frac{d}{2} \sum_B \operatorname{tr} C_B.$$

If $\lambda_{B,1}, \dots, \lambda_{B,m_B-1}$ are the nonzero eigenvalues of C_B, then the approximation ratio is exactly

$$\alpha(G) = \frac{d \sum_{B,j} \lambda_{B,j}}{\left(\sum_{B,j} \sqrt{\lambda_{B,j}}\right)^2}. \tag{IV.4.4}$$

There are exactly d eigenvalues in these sums.

Proof. Fix a positive budget s_B for block B, so $\operatorname{tr} L_B = 2s_B$. On its subspace $\mathbf{1}^\perp$, matrix Cauchy–Schwarz gives

$$(\operatorname{tr} C_B^{1/2})^2 \le \operatorname{tr}(C_B L_B^{-1}) \operatorname{tr} L_B.$$

Equality holds precisely when L_B is proportional to $C_B^{1/2}$. By Lemma IV.4.1, that choice is feasible; hence the block minimum is $q_B^2/(2s_B)$, attained at $L_B = (2s_B/q_B) C_B^{1/2}$. Allocating the total unit budget among the blocks yields

$$\min_{s_B>0,\ \sum_B s_B = 1} \sum_B \frac{q_B^2}{2s_B} = \frac{1}{2}\left(\sum_B q_B\right)^2, \qquad s_B = \frac{q_B}{Q}.$$

This proves (IV.4.3) and its global optimality.

The block-cut tree identity is $\sum_B (m_B - 1) = n - 1 = d$. Consequently the stated D-weights sum to one. Within a uniformly weighted K_{m_B}, every pair has resistance $2/(m_B w_e^D) = d$; attachments at a single articulation vertex do not alter that resistance. Thus every candidate edge has effective resistance d, verifying the determinant-optimality conditions. Using (IV.4.2) now gives

$$K_G(w^D) = d \sum_B \sum_{i<j} a_{B,i} a_{B,j} = \frac{d}{2} \sum_B \operatorname{tr} C_B.$$

Dividing by the A-optimal value proves (IV.4.4). The same block-cut identity counts the nonzero eigenvalues. □

IV.4.3 A common spectral interval and the sharp constant

Lemma IV.4.3. *Every nonzero eigenvalue of every C_B lies in*

$$n \le \lambda \le n^2/2.$$

Proof. Suppress the block subscript and put $D = \operatorname{diag}(a)$, $u = (\sqrt{a_i})_i$, and $P = I - uu^{\mathsf{T}}/n$. Since $\|u\|^2 = n$, P is an orthogonal projection and

$$C = nD^{1/2}PD^{1/2}.$$

The nonzero eigenvalues of $D^{1/2}PD^{1/2}$ equal those of PDP on range P. There the quadratic form is at least $\min_i a_i \ge 1$, which proves the lower bound. For the upper bound, the diagonal entry in row i of C is $a_i(n - a_i)$, and the sum of the absolute off-diagonal entries is the same. Gershgorin's bound therefore gives

$$\lambda \le 2 \max_i a_i(n - a_i) \le n^2/2.$$

□

Theorem IV.4.4 (Sharp leading constant for all block graphs)**.** *Every connected n-vertex block graph satisfies*

$$\alpha(G) \le \frac{\sqrt{n}}{4\sqrt{2}} + \frac{1}{2} + \frac{1}{2\sqrt{2n}}. \tag{IV.4.5}$$

If $\mathcal{B}_n$ is the largest ratio among all such graphs, then

$$\lim_{n\to\infty} \frac{\mathcal{B}_n}{\sqrt{n}} = \frac{1}{4\sqrt{2}}. \tag{IV.4.6}$$

Proof. Set $x_{B,j} = \sqrt{\lambda_{B,j}}$, $a = \sqrt{n}$, and $b = n/\sqrt{2}$. There are d such numbers and each lies in $[a, b]$. By (IV.4.4), the ratio is their empirical second moment divided by the squared empirical mean. The pointwise inequality $(x - a)(x - b) \le 0$ gives

$$\alpha(G) \le \frac{(a+b)^2}{4ab} = \frac{\sqrt{n}}{4\sqrt{2}} + \frac{1}{2} + \frac{1}{2\sqrt{2n}},$$

proving the upper bound.

For the lower bound, take two K_m blocks joined at distinguished roots by a path with p edges, with

$$n = 2m + p - 1, \qquad p = \sqrt{2n} + O(1).$$

An appropriate parity choice of p gives such a graph for every sufficiently large n. Put $h = n-m+1$,

$$q = \sqrt{mh/2} + (m-2)\sqrt{n/2}, \qquad c_j = (m + j - 1)(m + p - j).$$

The exact design formulas reduce to

$$K_G(w^A) = \left(2q + \sum_{j=1}^{p} \sqrt{c_j}\right)^2, \qquad K_G(w^D) = d\left((m-1)(2n-m) + \sum_{j=1}^{p} c_j\right).$$

For example, the nonzero eigenvalues of an end-clique demand matrix are n with multiplicity $m - 2$ and mh once, which gives q; a bridge contributes $\sqrt{c_j}$ to the square-root budget. Since $m = n/2 + O(\sqrt{n})$,

$$2q = \frac{n^{3/2}}{\sqrt{2}} + O(n), \qquad \sum_j \sqrt{c_j} = \frac{n^{3/2}}{\sqrt{2}} + O(n),$$

$$K_G(w^D) = \frac{n^{7/2}}{2\sqrt{2}} + O(n^3), \qquad K_G(w^A) = 2n^3 + O(n^{5/2}).$$

Thus $\alpha(G) = \sqrt{n}/(4\sqrt{2}) + O(1)$. These graphs belong to the block-graph class, so the upper and lower bounds prove (IV.4.6). □

Proposition IV.4.5 (Exact spectral deficit identity)**.** *Let x range uniformly over the d numbers in the preceding proof, and define*

$$\mu = \mathbb{E}x, \qquad \delta = \mathbb{E}[(x-a)(b-x)].$$

Then

$$\frac{(a+b)^2}{4ab} - \alpha(G) = \frac{((a+b)\mu - 2ab)^2}{4ab\mu^2} + \frac{\delta}{\mu^2}. \tag{IV.4.7}$$

In particular, equality in the finite bound requires and is implied by all $x \in \{a, b\}$ and $\mu = 2ab/(a+b)$. If a sequence of block graphs attains the leading constant in (IV.4.6), then

$$\frac{\mu}{2\sqrt{n}} \longrightarrow 1, \qquad \delta = o(n^{3/2}).$$

Proof. The identity $\mathbb{E}x^2 = (a+b)\mu - ab - \delta$ gives (IV.4.7) by completing the square. Both terms on its right are nonnegative. Dividing the first term by $(a+b)^2/(4ab)$ shows that a vanishing relative deficit forces $(a+b)\mu/(2ab) \to 1$. Since $a/b \to 0$, this gives $\mu/(2\sqrt{n}) \to 1$. The second term then gives $\delta = o(n^{3/2})$. □

The stability statement concerns the block demand spectra; it does not assert that all nearly extremal graphs are barbells. For a noncomplete block, the matrix $C_B^{1/2}$ may require edges absent from the graph, so its trace-optimal design need not be feasible. Completing such blocks is not a valid monotonicity argument for the ratio. Accordingly, the exact leading constant on all connected graphs, and a full combinatorial characterization of their extremizers, remain open in this manuscript.

IV.5 A sharp limit to universal robustness of mixture likelihood

The component concentration needed for entropy–transport does not, by itself, make an ordinary mixture maximum-likelihood estimator optimal under contamination. We prove this in a fixed, identifiable, one-dimensional model. Both Gaussian components are known, their variances equal one, and only their mixing weight is estimated. The comparison is uniform in the sample size and contamination level. An exact population bias equation identifies the mechanism behind the finite-sample separation.

IV.5.1 Model and the two risk laws

Let f_j denote the density of $N(j, 1)$, $j = 0, 1$, and put

$$p_w = (1-w)f_0 + wf_1, \qquad G_w = (1-w)\delta_0 + w\delta_1, \qquad 0 \le w \le 1.$$

Write P_w for the observed distribution with density p_w. Then $W_1(G_w, G_v) = |w - v|$. Throughout this section, $\mathrm{TV}(P, Q) = \frac{1}{2}\int |p - q|$.

For the strong contamination experiment, start with independent $X_1, \dots, X_N \sim P_w$. An adversary, possibly randomized and depending on the whole clean sample, produces $Y_1, \dots, Y_N$ by replacing at most $k = \lfloor \varepsilon N \rfloor$ coordinates. Denote the class of these transformations by $\mathcal{A}_{N,k}$.

In the Huber experiment the Y_i are independent with common law $(1-\varepsilon)P_w + \varepsilon Q$, where Q is any probability measure on $\mathbb{R}$.

An ordinary MLE is any measurable choice

$$\widehat{w} \in \operatorname*{argmax}_{0\le v\le 1} \sum_{i=1}^{N} \log p_v(Y_i).$$

Such choices exist: the objective is finite and continuous on a compact interval for every finite real sample. Bounds below hold for every measurable choice, including possible nonunique maximizers.

Theorem IV.5.1 (Sharp finite-sample separation)**.** *There are numerical constants $0 < c < C < \infty$ such that, for every $N \ge 1$ and $0 \le \varepsilon \le 1/4$,*

$$c\min\{1, N^{-1/2} + \sqrt{\varepsilon}\} \le \sup_{w\in[0,1]} \sup_{A\in\mathcal{A}_{N,\lfloor \varepsilon N\rfloor}} \mathbb{E}_w|\widehat{w}(A(X)) - w| \le C\min\{1, N^{-1/2} + \sqrt{\varepsilon}\}, \quad \text{(IV.5.1)}$$

$$c\min\{1, N^{-1/2} + \varepsilon\} \le \inf_{\widetilde{w}} \sup_{w\in[0,1]} \sup_{A\in\mathcal{A}_{N,\lfloor \varepsilon N\rfloor}} \mathbb{E}_w|\widetilde{w}(A(X)) - w| \le C\min\{1, N^{-1/2} + \varepsilon\}. \quad \text{(IV.5.2)}$$

The infimum is over measurable estimators with values in $[0,1]$. The same two inequalities hold when the adversarial supremum is replaced by the supremum over Q in the Huber experiment. An estimator attaining the minimax upper bound does not need to know ε.

Thus, along $\varepsilon \to 0$ and $N\varepsilon \to \infty$, the ratio between the worst MLE risk and the minimax risk tends to infinity. This is a statement about the two-component dictionary just specified. It does not by itself prove a lower bound for the unrestricted Gaussian location-mixture NPMLE with a continuum of allowed components.

IV.5.2 Uniform upper bound for every likelihood maximizer

Set

$$r(x) = \frac{f_1(x)}{f_0(x)} = e^{x-1/2}, \qquad U(x) = r(x) - 1, \qquad a = \int_{\mathbb{R}} \frac{(f_1 - f_0)^2}{f_0 + f_1}\, dx > 0.$$

Let P_N be the clean empirical measure, let q be the corrupted empirical measure, and suppose the true parameter is w_0. The likelihood directional condition towards p_{w_0} gives

$$q\left(\frac{p_{w_0}}{p_{\widehat{w}}}\right) \le 1. \quad \text{(IV.5.3)}$$

This condition remains valid at an endpoint maximizer because the direction towards w_0 is feasible. The function $g(z) = (1-z)/(1+z)$ is convex and decreasing for $z \ge 0$. Jensen's inequality applied to (IV.5.3) therefore implies

$$q\phi \ge 0, \qquad \phi = \frac{p_{\widehat{w}} - p_{w_0}}{p_{\widehat{w}} + p_{w_0}}.$$

Put $\Delta = \widehat{w} - w_0$ and $v = (\widehat{w} + w_0)/2$. Then

$$\phi = \Delta\psi_v, \qquad \psi_v(x) = \frac{U(x)}{2(1 + vU(x))}.$$

The equality of the total masses of $p_{\widehat{w}}$ and p_{w_0} gives the exact identity

$$P_{w_0}\phi = -\frac{1}{2}\int_{\mathbb{R}} \frac{(p_{\widehat{w}} - p_{w_0})^2}{p_{\widehat{w}} + p_{w_0}}\, dx \le -\frac{a}{4}\Delta^2,$$

where the inequality uses $p_{\widehat{w}} + p_{w_0} \le 2(f_0 + f_1)$. Since $|\phi| \le 1$, replacing at most k observations changes its empirical integral by at most $2k/N$, even for this data-dependent function. Consequently, with $\delta = k/N$,

$$\frac{a}{4}\Delta^2 \le |\Delta| Z_N + 2\delta, \qquad Z_N = \sup_{0 \le v \le 1} |(P_N - P_{w_0})\psi_v|. \tag{IV.5.4}$$

It remains to control Z_N without a logarithmic loss. Direct differentiation yields

$$0 \le \psi_v'(x) = \frac{r(x)}{2(1 + v(r(x) - 1))^2} \le \frac{1}{2} e^{|x - 1/2|}.$$

Let F_N, F_{w_0} be the corresponding empirical and true distribution functions. Integration by parts gives

$$Z_N \le \frac{1}{2} \int_{\mathbb{R}} e^{|x-1/2|} |F_N(x) - F_{w_0}(x)| \, dx.$$

The boundary terms vanish also for $v = 0, 1$: ψ_v grows at most exponentially, the true tails are Gaussian, and an empirical sample has finite extrema. The variance of the empirical distribution function gives

$$\mathbb{E} Z_N \le \frac{1}{2\sqrt{N}} \int_{\mathbb{R}} e^{|x-1/2|} \sqrt{F_{w_0}(x)(1 - F_{w_0}(x))} \, dx \le \frac{C_0}{\sqrt{N}}. \tag{IV.5.5}$$

The last constant is uniform over $w_0 \in [0, 1]$. For example, on the left tail $F_{w_0} \le \Phi(x)$, and on the right tail $1 - F_{w_0} \le 1 - \Phi(x - 1)$; these bounds have Gaussian decay. Solving (IV.5.4) gives

$$|\Delta| \le \frac{4}{a} Z_N + \sqrt{\frac{8\delta}{a}}. \tag{IV.5.6}$$

Together with $|\Delta| \le 1$, this proves the strong-contamination upper bound in (IV.5.1). In particular, the argument requires no moment condition on the replacement values.

IV.5.3 A finite outlier attack attaining the square-root term

We first establish a clean-sample event whose probability is uniformly positive. Under P_0, let $U = e^{X - 1/2} - 1$, so

$$\mathbb{E}_0 U = 0, \qquad A := \mathbb{E}_0 U^2 = e - 1, \qquad B := \mathbb{E}_0 U^4 = e^6 - 4e^3 + 6e - 3.$$

For $n \ge 1$ independent copies, put $S = n^{-1} \sum_{i=1}^n U_i$ and $V = n^{-1} \sum_{i=1}^n U_i^2$. Then

$$\mathbb{E} S^2 = A/n, \qquad \mathbb{E} S^4 \le (B + 3A^2)/n^2.$$

Interpolation between the first, second, and fourth moments gives

$$\mathbb{E}|S| \ge \frac{(\mathbb{E} S^2)^{3/2}}{(\mathbb{E} S^4)^{1/2}} \ge \frac{A^{3/2}}{\sqrt{B + 3A^2}\sqrt{n}}.$$

Since $\mathbb{E} S_+ = \mathbb{E}|S|/2$, Cauchy–Schwarz implies

$$\mathbb{P}(S > 0) \ge p_* := \frac{A^2}{4(B + 3A^2)} > 0.$$

Choose $M = 2A/p_*$. Markov's inequality, using $\mathbb{E} V = A$, gives

$$\mathbb{P}(S \ge 0, \ V \le M) \ge p_*/2. \tag{IV.5.7}$$

Suppose $1 \le k \le N/4$, set $\delta = k/N$, and take true parameter $w_0 = 0$. Replace a predetermined set of k observations by a common value z, leaving $n = N - k$ independent P_0 observations. Choose a numerical $c_0 > 0$ such that $c_0 \le 1/2$ and $c_0^2 \le 1/(16M)$, and put

$$w_* = c_0\sqrt{\delta}, \qquad z = \frac{1}{2} + \log(1 + 1/w_*). \tag{IV.5.8}$$

In particular, this is a finite replacement value. On the event in (IV.5.7), the average clean likelihood score at w_* satisfies

$$\frac{1}{n}\sum_{i=1}^{n} \frac{U_i}{1 + w_* U_i} = S - \frac{w_*}{n}\sum_{i=1}^{n} \frac{U_i^2}{1 + w_* U_i} \ge -2M w_*,$$

because $U_i \ge -1$ and $w_* \le 1/2$. The choice of z gives $U(z) = 1/w_*$, so its score equals $1/(2w_*)$. The normalized corrupted log likelihood has derivative at least

$$-2M w_* + \frac{\delta}{2w_*} > 0.$$

By concavity every maximizer therefore exceeds w_*. It follows that every measurable MLE has expected error at least $(p_* c_0/2)\sqrt{k/N}$. If $k = \lfloor \varepsilon N \rfloor \ge 1$, then $k/N \ge \varepsilon/2$. If $k = 0$, then $\sqrt{\varepsilon} \le N^{-1/2}$; the clean lower bound proved next already covers this case.

IV.5.4 The minimax rate and the clean lower bound

Let

$$b = \Phi(-1/2), \qquad d = 2\Phi(1/2) - 1 > 0.$$

The threshold probability is affine in the parameter: $P_w(X > 1/2) = b + dw$. Hence the estimator

$$\widetilde{w} = \Pi_{[0,1]}\left\{ \frac{N^{-1}\sum_{i=1}^{N} \mathbf{1}_{\{Y_i > 1/2\}} - b}{d} \right\} \tag{IV.5.9}$$

satisfies

$$\mathbb{E}|\widetilde{w} - w| \le \frac{1}{d}\left(\frac{1}{2\sqrt{N}} + \frac{k}{N} \right)$$

under strong contamination. Here $\Pi_{[0,1]}$ denotes projection onto the interval, which cannot increase the distance to w. This proves the minimax upper bound without knowing the corruption budget.

For clarity, a clean lower bound of order $N^{-1/2}$ holds uniformly for every estimator. Choose $w_\pm = 1/2 \pm h/\sqrt{N}$, with a fixed sufficiently small $h > 0$. Both parameters lie in $[1/4, 3/4]$, and

$$\chi^2(P_{w_+}, P_{w_-}) = \frac{4h^2}{N}\int \frac{(f_1 - f_0)^2}{p_{w_-}} \le \frac{16 a h^2}{N}.$$

The product chi-squared divergence is at most $\exp(16ah^2) - 1$. Taking h so this is at most $1/4$ bounds the product total variation by $1/4$. The two-point testing inequality then yields expected absolute error at least $c/\sqrt{N}$ for one of the two parameters. This applies in both contamination experiments: the strong adversary may leave all reports unchanged, and in the Huber experiment one may choose $Q = P_w$.

To obtain the strong-contamination lower bound, note the exact identity

$$\mathrm{TV}(P_w, P_v) = d|w - v|. \tag{IV.5.10}$$

For $k \ge 1$, choose $w = 1/4$ and $v = w + k/(16Nd)$; these parameters lie in $[0, 1]$. Couple independent pairs (X_i, Z_i) maximally with marginals P_w, P_v. The expected number J of mismatches is

$k/16$. Conditional on X, an adversary can sample this coupling and output Z when $J \le k$, and X otherwise. It makes at most k replacements in every realization, and its output law is within total variation $1/16$ of $P_v^{\otimes N}$ because $\mathbb{P}(J > k) \le 1/16$. The experiment at parameter v may use the identity adversary. Two-point testing therefore gives a lower bound ck/N. Randomization is harmless: for any estimator a randomized attack's risk is an average of risks of deterministic attacks. For $k \ge 1$ this is at least $c'\varepsilon$; for $k = 0$ the term ε is absorbed by $N^{-1/2}$. Combining the clean and contamination lower bounds proves (IV.5.2); combining the clean bound with (IV.5.8) proves the lower bound in (IV.5.1). The sum and maximum of the two terms differ by at most a factor two.

IV.5.5 Huber contamination

Represent the observations by independent contamination indicators $B_i \sim \text{Bernoulli}(\varepsilon)$, clean $X_i \sim P_w$, and outliers $Z_i \sim Q$, all mutually independent, with $Y_i = X_i$ when $B_i = 0$ and $Y_i = Z_i$ otherwise. Set $K = \sum B_i$. On $K \le N/4$, the pathwise inequality (IV.5.6) applies with $\delta = K/N$ to the complete latent clean sample. Therefore

$$\mathbb{E}|\widehat{w} - w| \le C\mathbb{E}Z_N + C\mathbb{E}\sqrt{K/N} + \mathbb{P}(K > N/4) \le C'(N^{-1/2} + \sqrt{\varepsilon}).$$

We used Jensen's inequality and $\mathbb{P}(K > N/4) \le 4\varepsilon$. For (IV.5.9), replacement changes the threshold average by at most K/N, giving the minimax upper bound $d^{-1}(1/(2\sqrt{N}) + \varepsilon)$.

For the minimax Huber lower bound, take $w = 1/4$ and $v - w = \varepsilon/(2d)$. These parameters remain in $[0, 1]$. By (IV.5.10), $\mathrm{TV}(P_w, P_v) = \varepsilon/2 \le \varepsilon/(1 - \varepsilon)$. Their Huber neighborhoods intersect. Indeed, the measure with density $(1 - \varepsilon)\max(p_w, p_v)$ has total mass at most one; adding arbitrary nonnegative mass to make it a probability gives a law of the form $(1-\varepsilon)P_w+\varepsilon Q_w$ and also $(1 - \varepsilon)P_v + \varepsilon Q_v$. Identical observed laws force absolute-error risk at least $|v - w|/2$ for one parameter. The clean lower bound handles $\varepsilon = 0$ and supplies the $N^{-1/2}$ term.

Finally, take $w_0 = 0$ to prove the MLE lower bound. If $N\varepsilon < 16$, the clean lower bound absorbs $\sqrt{\varepsilon} \le 4N^{-1/2}$. Otherwise choose $w_* = c_0\sqrt{\varepsilon}$ and a single point-mass contamination at $z = 1/2 + \log(1 + 1/w_*)$, with c_0 as above. Chebyshev's inequality gives

$$\mathbb{P}\{N\varepsilon/2 \le K \le 2N\varepsilon\} \ge 1 - \frac{5}{N\varepsilon} \ge \frac{11}{16}.$$

On this event $K \le N/2$. Conditional on the contamination indices, the retained $N - K \ge 1$ clean observations satisfy (IV.5.7) with probability at least $p_*/2$. At w_* the normalized score is at least

$$-2Mw_* + \frac{\varepsilon}{4w_*} > 0.$$

Thus every MLE exceeds w_* on an event of fixed positive probability, proving the lower bound $c\sqrt{\varepsilon}$. This completes the proof of Theorem IV.5.1.

IV.5.6 Exact population bias

The preceding attack is not merely a finite-sample artifact. For $0 < \varepsilon \le 1/4$ and finite $z \in \mathbb{R}$, let $w_z(\varepsilon)$ maximize

$$L_z(w) = (1 - \varepsilon)\mathbb{E}_0 \log(1 + wU) + \varepsilon \log(1 + wU(z)), \qquad 0 \le w \le 1.$$

Subtracting the fixed f_0 log likelihood gives this finite contrast. It is strictly concave, since the clean component assigns positive probability to $U \ne 0$, so the maximizer is unique.

Theorem IV.5.2 (Exact worst positive point-contamination bias)**.** *Define*

$$A(w) = \mathbb{E}_0 \frac{U^2}{1+wU}, \qquad 0 \le w \le 1.$$

For each $0 < \varepsilon \le 1/4$ *there is a unique* $w_\varepsilon \in (0,1)$ *solving*

$$w_\varepsilon^2 A(w_\varepsilon) = \frac{\varepsilon}{1-\varepsilon}. \tag{IV.5.11}$$

It satisfies

$$\sup_{z\in\mathbb{R}} w_z(\varepsilon) = \lim_{z\to+\infty} w_z(\varepsilon) = w_\varepsilon, \qquad w_\varepsilon \sim \sqrt{\frac{\varepsilon}{e-1}} \quad (\varepsilon \downarrow 0). \tag{IV.5.12}$$

Set $w_0 = 0$ *when* $\varepsilon = 0$.

Proof. All positive and negative moments of $r = e^{X-1/2}$ are finite. For $w \in [0,1]$, the integrand in $A(w)$ is dominated by $U^2 + U^2/r$, an integrable function. Thus A is finite and continuous on this interval. Since $\mathbb{E}_0 U = 0$,

$$\mathbb{E}_0 \frac{U}{1+wU} = -wA(w).$$

For every nonzero $u > -1$, the function $w^2u^2/(1+wu)$ is strictly increasing on $[0,1]$; for $w > 0$ its derivative is $wu^2(2+wu)/(1+wu)^2 > 0$. Consequently $w^2A(w)$ is continuous and strictly increasing, starts at zero, and ends at

$$A(1) = \mathbb{E}_0(r - 2 + r^{-1}) = e - 1.$$

Since $\varepsilon/(1-\varepsilon) \le 1/3 < e-1$, (IV.5.11) has exactly one interior solution.

For $w > 0$ the outlier score is $U(z)/(1+wU(z)) \le 1/w$. Hence

$$L_z'(w) \le -(1-\varepsilon)wA(w) + \varepsilon/w.$$

The right side is negative for $w > w_\varepsilon$, so $w_z(\varepsilon) \le w_\varepsilon$. As $z \to +\infty$, the outlier score converges locally uniformly for $w > 0$ to $1/w$. At any fixed $0 < u < w_\varepsilon$ the limiting derivative is positive, so eventually $L_z'(u) > 0$ and $w_z(\varepsilon) > u$. Letting $u \uparrow w_\varepsilon$ establishes the limit and supremum. Finally, strict monotonicity shows $w_\varepsilon \to 0$, and $A(0) = e - 1$ in (IV.5.11) proves the asymptotic formula. □

IV.5.7 What entropy–transport does and does not imply

The following posterior argument works even when the observation space is unbounded. It only needs a bound on the distances between the component distributions.

Lemma IV.5.3 (Posterior repair with finite component diameter)**.** *Let* $K_1, \dots, K_m$ *be probability measures with finite first moments on a metric space, with densities* k_j *relative to a common measure. Suppose*

$$W_1(R, K_j) \le \sqrt{2c\, D(R\|K_j)} \quad \textit{for every } R, \qquad \max_{i,j} W_1(K_i, K_j) \le D_K.$$

Let $H = (H_j)_{j=1}^m$ *be a probability vector and* $p_H = \sum_j H_j k_j$. *Let* a *be a probability density with* $D(a\|p_H) < \infty$. *Suppose, for all* j *with* $H_j > 0$,

$$S_j := \int \frac{ak_j}{p_H} \le \sigma, \qquad \sigma \ge 1.$$

Then

$$W_1(a, p_H) \le \sqrt{2cD(a\|p_H)} + D_K(1 - \sigma^{-1}). \tag{IV.5.13}$$

Proof. Set $H_j^+ = H_j S_j$; these nonnegative numbers sum to one. For $H_j^+ > 0$, let R_j have density $ak_j/(S_j p_H)$; zero-mass terms may use an arbitrary conditional distribution. Then $\sum_j H_j^+ R_j = a$. Apply the chain rule for relative entropy to the joint law with (j, x) density $H_j a(x) k_j(x)/p_H(x)$ and reference law $H_j k_j(x)$. It gives

$$D(a\|p_H) = D(H^+\|H) + \sum_j H_j^+ D(R_j\|K_j).$$

Convexity of transport under mixing and Cauchy–Schwarz therefore give

$$W_1(a, p_{H^+}) \le \sum_j H_j^+ W_1(R_j, K_j) \le \sqrt{2cD(a\|p_H)}.$$

The domination $H^+ \le \sigma H$ yields $\sum_j \min(H_j, H_j^+) \ge 1/\sigma$, hence $\mathrm{TV}(H^+, H) \le 1 - 1/\sigma$. Couple the common mixing mass identically, and couple the residual component laws at cost at most D_K per unit mass. This gives $W_1(p_{H^+}, p_H) \le D_K(1 - 1/\sigma)$. The triangle inequality proves (IV.5.13). □

For the Gaussian pair, each component satisfies the classical $T_1(1)$ inequality, and $D_K = 1$. Consider a population MLE p_w under $(1-\varepsilon)P_0 + \varepsilon\delta_z$. The directional likelihood conditions towards either component give

$$(1-\varepsilon)\int f_0 \frac{f_j}{p_w} + \varepsilon \frac{f_j(z)}{p_w(z)} \le 1, \qquad j = 0, 1.$$

In particular $S_j \le \sigma = (1-\varepsilon)^{-1}$. Jensen's inequality also gives the clean likelihood deficit bound

$$D(f_0\|p_w) \le \log \int f_0 \frac{f_0}{p_w} \le \log \frac{1}{1-\varepsilon}.$$

Applying Lemma IV.5.3 yields

$$W_1(P_0, P_w) \le \sqrt{2 \log \frac{1}{1-\varepsilon}} + \varepsilon. \tag{IV.5.14}$$

Finally, $W_1(P_0, P_w) = w$: coupling $X \sim N(0,1)$ with $X + B$, where $B \sim \mathrm{Bernoulli}(w)$, gives the upper bound w, and the difference of means gives the lower bound. Theorem IV.5.2 shows that the square-root order in (IV.5.14) cannot generally be improved to a linear contamination term for the ordinary likelihood maximizer. Thus Gaussian component concentration, finite component diameter, convexity, and a well-conditioned scalar mixing parameter together do not imply contamination-minimax ordinary MLE risk.

IV.5.8 Relation to existing robustness results

The nonrobustness of ordinary likelihood in normal mixture problems is established motivation for robust alternatives; see, for example, Fujisawa–Eguchi [33]. Baraud–Birgé [32, Section 6] show that MLEs on suitable convex separable density models are ρ-estimators. Lecestre [34] develops nonasymptotic Hellinger robustness for finite mixtures, including contamination and outlier guarantees. These facts are consistent with the present result: a squared Hellinger error of order ε permits parameter error of order $\sqrt{\varepsilon}$ in this model, while a bounded threshold statistic attains parameter error of order ε.

The role of this section is a precise limitation theorem for a proposed universal entropy–transport extension, with matching finite-sample bounds for two contamination experiments and an exact population worst-bias identity. It makes no historical-priority claim and does not assert resolution of a named external conjecture.

IV.6 Consequences and remaining boundaries

Theorem IV.2.1 supplies a single risk curve for the entire known-support Poisson experiment. With $L = \log(eN)$, its growing-support part changes from $\sqrt{B/L}$ to $B/\sqrt{N}$ at $B \asymp N/L$. The first term expresses the difficulty of inversion; the second is already forced by uncertainty in the weights of two endpoint atoms. For fixed positive support radius, the small-support expression has the familiar order $(\log\log N)/\log N$. The new proof covers the intermediate and unbounded growing-support ranges for the same original likelihood estimator. It does not settle unknown-support adaptation, heterogeneous exposures, or an imposed finite-component constraint.

Theorem IV.3.1 gives a single canonical connection formula throughout the real integrable Jacobi parameter region considered here. Its separate summands may be complex, and transported branches may contain logarithms at resonance; their prescribed sum is the physical density. The theorem resolves that physical combination, not every connection matrix for arbitrary local bases or a stable large-N numerical algorithm.

For block graphs, the exact ratio is a spectral second moment divided by a squared spectral first moment. A common spectral interval then determines the sharp leading extremal constant. The feasibility argument depends on each block being complete. The exact constant for all connected graphs remains unresolved here, as does a combinatorial classification of all nearly extremal graphs.

The Gaussian separation closes a universal-transfer question negatively: Gaussian component concentration and a finite component transportation diameter do not suffice for contamination-minimax performance of ordinary mixture likelihood. The square-root contamination cost predicted by the entropy bound is attained in the specified two-component model. This lower bound has not been extended here to an unrestricted continuum of Gaussian locations.

The global statements in this manuscript are therefore global in their explicitly stated model and parameter classes. Arbitrary heterogeneous observation profiles and arbitrary complex weights in the binomial-power recurrence problem are also outside the completed results. Independent internal checks found no substantive gaps in the four arguments; external expert review, broader novelty assessment, and publication decisions remain separate steps.

IV.A Supporting two-scale binomial inversion

These supporting tools are reproduced from the complete binomial-mixture likelihood argument in the supplied manuscript [30], also used in [31]. They are included for independent verification and are not counted as new results here. Write $B_j^t(x) = \binom{t}{j}x^j(1-x)^{t-j}$ and $p_G(j) = \int B_j^t\,dG$. The distance $W_{1,t}$ uses ground cost $|j-k|/t$, and $h(p,q)^2 = \sum_j(\sqrt{p_j}-\sqrt{q_j})^2$. All constants below are absolute.

Lemma IV.A.1 (Stable approximation and bounded-polynomial inversion)**.** *There is an absolute C with the following properties. For every 1-Lipschitz $f:[0,1]\to\mathbb{R}$ and integer $k\ge 1$, there is a polynomial P_k of degree at most k such that*

$$\|P_k - f\|_\infty \le C/k, \qquad \|P_k'\|_\infty \le 2.$$

If $P = \sum_{j=0}^t b_j B_j^t$ has degree $k \le t/2$, then

$$\max_j |b_j| \le Ce^{2k^2/t}\|P\|_\infty. \tag{IV.A.1}$$

For the derivative-stable P_k, its coefficient vector also satisfies

$$\max_{j<t} t|b_{j+1}-b_j| \le Ce^{4k^2/t}. \tag{IV.A.2}$$

Proof. Subtract $f(0)$, reflect f on $[0,1]$ to a period-two 1-Lipschitz function, and convolve with the normalized positive Jackson kernel $c_s[\sin(s\pi u/2)/\sin(\pi u/2)]^4$. Its first absolute moment is $O(1/s)$, so the resulting trigonometric polynomial Q_s has error $O(1/s)$ and derivative norm at most one. Its frequencies have magnitude at most $2s$, and its Fourier coefficients have modulus at most one. Replacing each exponential by its Taylor polynomial at $1/2$ of degree $100s$ changes the function and derivative by at most e^{-100s}. Indeed $|\pi l(x-1/2)| \le \pi s$ and the factorial inequality $M! \ge (M/e)^M$ bounds both remainders, including the $4s+1$ terms and the extra derivative factor. Take $s = \lfloor k/100\rfloor$ for $k \ge 100$; use a constant polynomial and enlarge C for smaller k.

For (IV.A.1), put $M = \|P\|_\infty$ and $F(z) = (1+z)^t P(z/(1+z))$. The maximum principle in the inverse Joukowski variable gives $|P((2+w+w^{-1})/4)| \le M|w|^k$ for $|w| \ge 1$. For $0 < j < t$, set $r = j/(t-j)$, $v = r/(1+r)^2$, and $u = 2\sqrt{v}|\sin(\phi/2)|$. The larger inverse Joukowski modulus for $re^{i\phi}/(1+re^{i\phi})$ is $(1+u)/\sqrt{1-u^2}$. Hence

$$\frac{|F(re^{i\phi})|}{M(1+r)^t} \le (1-u^2)^{(t-k)/2}(1+u)^k \le e^{2k^2/t}e^{-tu^2/8}.$$

The last inequality follows from $k \le t/2$ and completing the square. The average of the final Gaussian factor is at most $C/\sqrt{1+tv}$. Cauchy's coefficient formula, together with $\binom{t}{j}(j/t)^j(1-j/t)^{t-j} \ge c/\sqrt{1+tv}$, proves (IV.A.1). The last lower bound follows directly from Stirling's factorial bounds; endpoint coefficients are $P(0)$ and $P(1)$. Finally differentiate the Bernstein expansion: $P_k' = t\sum_{j<t}(b_{j+1}-b_j)B_j^{t-1}$. Apply (IV.A.1) at degree $k-1 \le (t-1)/2$ and use $2(k-1)^2/(t-1) \le 4k^2/t$. □

Proposition IV.A.2 (Two-scale deterministic inversion)**.** *For integers $1 \le s \le k \le t/2$ and any two mixing laws G, H,*

$$W_1(G,H) \le \frac{C}{k} + Ce^{4s^2/t}W_{1,t}(p_G,p_H) + \frac{Ce^{2k^2/t}}{s}h(p_G,p_H). \tag{IV.A.3}$$

Proof. For any 1-Lipschitz test f, take the polynomials P_k, P_s from Lemma IV.A.1. The approximation error in integrating P_k instead of f against $G-H$ is at most C/k. The Bernstein coefficients of P_s have normalized-grid Lipschitz constant at most $Ce^{4s^2/t}$. On the other hand,

$$\|P_k - P_s\|_\infty \le C/k + C/s \le C/s.$$

Thus the Bernstein coefficients of $P_k - P_s$ have supremum norm at most $Ce^{2k^2/t}/s$. Test P_s by count transportation and $P_k - P_s$ by $\|p_G - p_H\|_1 \le 2h(p_G, p_H)$. All constants are uniform over f, so taking the Wasserstein dual supremum proves the result. □

The small amplitude $1/s$ is essential. Applying only the Hellinger bound to the entire polynomial loses the low-frequency empirical $N^{-1/2}$ scale; applying only transportation loses the critical transition. Proposition IV.A.2 retains both.

Part V
Global Algebraicity and Exact Complexity

Manuscript overview. We establish three complete classifications using methods from coefficient asymptotics, differential modules, and arithmetic inverse periods. First, for every integer $r \geq 1$ and every complex weight λ, the weighted Franel generating function $\sum_{n\geq 0} \sum_{k=0}^{n} \binom{n}{k}^{r} \lambda^{k} z^{n}$, and its even section, are algebraic precisely when $r \leq 2$ or $\lambda = 0$. A global torus maximum inequality resolves complex cancellation, and an arithmetic Puiseux obstruction finishes the classification. Second, the exact minimal differential order of every anisotropic hypercubic lattice Green function is determined solely by the multiplicities of its nonzero squared weights. The formula holds in every dimension and at every complex specialization, including colliding thresholds. Irreducibility and local monodromy imply that the function is algebraic exactly when at most one weight is nonzero. Third, all algebraic quantiles at algebraic levels for rational polynomial tilts of the arcsine law are classified by cyclotomic factors. Every allowable finite exceptional set is realized, with minimum polynomial degree one plus its cardinality. Two further transfers prove sharp support-free Gaussian likelihood recovery throughout a specified growing-support regime and an exact design-comparison constant for dense inflations of every fixed graph skeleton. The arguments are given in full, with classical inputs, previously known cases, and remaining limitations stated explicitly.

Integration note. The rank and algebraicity statements are the foundation for the Galois-group refinements in Parts VI and VII. Its support-free Gaussian theorem differs from the known-support all-scale theorem of Part VI; both estimation statements are retained. The graph-inflation comparison and quantile constructions remain included.

V.1 What is global, and where the methods come from

The central questions concern entire families, rather than finitely many examples. The first three results classify every parameter in their stated domains. Their conclusions have different meanings: nonalgebraicity of a generating function, minimum order of a differential equation, and arithmetic of inverse distribution values must not be conflated.

Family	Complete domain	Conclusion
Weighted Franel series	Every integer $r \geq 1$ and every $\lambda \in \mathbb{C}$	Algebraic exactly for $r \leq 2$ or $\lambda = 0$; also for the even section.
Hypercubic lattice periods	Every dimension and every complex weight tuple	Exact minimal differential order, irreducibility, and an algebraicity dichotomy.
Polynomial arcsine tilts	Every positive rational polynomial weight	Exact exceptional quantile sets, inverse realization, and minimum degree.
Gaussian likelihood	Every $N \geq 3$ and $A \geq \sigma\sqrt{\log(eN)}$	Uniform minimax W_1 risk without supplying the support bound.
Graph inflation	Every fixed connected simple skeleton, positive masses and lengths	Exact asymptotic ratio and the sharp optimized constant within this class.

The method transfers are as follows.

1. **From Franel singularities to all complex weights.** The coefficient-asymptotic obstruction is already present in the unweighted literature. A pair-averaging inequality identifies every dominant saddle for arbitrary complex weights. The two competing saddles on the negative real axis are treated explicitly, including their collision after taking the even section. The result is an algebraicity classification, not a new claim about every minimum recurrence width.

2. **From Bessel periods to all anisotropic hypercubic lattices.** The supplied manuscript [49] uses Bessel differential modules, diagonal SL_2 decompositions, and Fourier transforms for exact order problems. Here a single Laplace transform in the Bessel variable produces a different period family. An upper bound in middle de Rham cohomology and a lower bound from a visible simple Fourier constituent agree. This also removes the possible extra rank from the unilateral integration endpoint.

3. **From arithmetic inverse periods to a complete design problem.** The supplied inverse-period work [50] motivates reducing a distribution function to a logarithmic period and an algebraic correction. For polynomial arcsine tilts the correction is an explicit rational polynomial. Its cyclotomic factors determine every exception, and the same operator constructs every possible exceptional set with optimal degree. Baker's and Gelfond–Schneider's theorems remain classical inputs.

4. **From likelihood contact to a continuous-data bound.** The posterior coupling method in [30, 31] gives an empirical-posterior martingale for an unrestricted Gaussian fit. A second-order Taylor cancellation and an exact Gaussian-to-binomial channel replace an inapplicable empirical-to-density entropy argument. The theorem is uniform on the stated growing-support range; the small-support problem is not claimed to be settled.

5. **From electrical maximum principles to all fixed skeletons.** A flow norm inequality has an exact equality classification. Combined with series reduction and clique budget allocation, it controls every fixed cyclic skeleton as well as trees. It does not provide the sharp upper constant for arbitrary graph sequences.

V.2 Algebraicity of weighted Franel series for every complex weight

For an integer $r \geq 1$ and $\lambda \in \mathbb{C}$, define

$$a_n^{(r)}(\lambda) = \sum_{k=0}^{n} \binom{n}{k}^r \lambda^k, \qquad A_{r,\lambda}(z) = \sum_{n\geq 0} a_n^{(r)}(\lambda) z^n, \qquad E_{r,\lambda}(z) = \sum_{n\geq 0} a_{2n}^{(r)}(\lambda) z^n.$$

All these series have positive radii of convergence. For nonzero λ, the normalized central sequence considered previously has generating function $E_{r,\lambda}(z/\lambda)$.

The unweighted nonalgebraicity result for $r \geq 3$ is already established by Banderier and Drmota [35, Proposition 7], using the exponent and the power of π in the coefficient asymptotics. Algebraic independence results for the unweighted Franel family also appear in Adamczewski, Bell, and Delaygue [36, Theorem 9.8]. The point here is the complete complex-weight classification, including algebraic weights off the unit circle and the cancellation boundary $\lambda < 0$. We do not identify that full classification as an explicitly stated open conjecture in those sources.

Theorem V.2.1 (Complete algebraicity classification)**.** *For every integer $r \geq 1$ and every $\lambda \in \mathbb{C}$, each of $A_{r,\lambda}$ and $E_{r,\lambda}$ is algebraic over $\mathbb{C}(z)$ if and only if $r \leq 2$ or $\lambda = 0$. Consequently, for every $r \geq 3$ and every nonzero complex λ, the normalized central generating function $E_{r,\lambda}(z/\lambda)$ is transcendental over $\mathbb{C}(z)$.*

The proof does not require the visibility of every complex critical value, nor a classification of minimum recurrence widths. It finds all dominant saddles and proves that their relevant amplitudes cannot cancel.

V.2.1 A global maximum inequality on the coefficient torus

Lemma V.2.2 (Pair averaging)**.** *Fix $\rho > 0$. Among r unit complex numbers with prescribed product, every maximizer of $\prod_{j=1}^{r} |1 + \rho e^{it_j}|$ has all e^{it_j} equal. The maximizing common angles are precisely those rth roots of the prescribed product having the largest real part.*

Proof. For two angles choose a midpoint a modulo π with $\cos a \geq 0$, and choose b so that $e^{it} = e^{i(a+b)}$ and $e^{iu} = e^{i(a-b)}$. Replacing both angles by a preserves their product. Direct expansion gives

$$|1 + \rho e^{ia}|^4 - |(1 + \rho e^{it})(1 + \rho e^{iu})|^2 = 4\rho(1 - \cos b)\big((1 + \rho^2)\cos a + \rho(1 + \cos b)\big) \geq 0.$$

Equality requires either $\cos b = 1$, or $\cos a = 0$ and $\cos b = -1$; both cases imply $e^{it} = e^{iu}$. The maximum of the full product is positive. Thus every pair at a maximizer must have equal phases. The prescribed product then forces the common phase to be an rth root, and comparison of $|1 + \rho e^{it}|^2 = 1 + \rho^2 + 2\rho\cos t$ proves the last assertion. □

For $\lambda \neq 0$, let $\rho = |\lambda|^{1/r}$ and let Ω_λ consist of the roots of $\omega^r = \lambda$ with largest real part. If $\lambda \notin (-\infty, 0)$, this is the single root with $\arg\omega = (\operatorname{Arg}\lambda)/r$, where $-\pi < \operatorname{Arg}\lambda < \pi$. If $\lambda < 0$, it consists of the two roots $\rho e^{\pm\pi i/r}$. Put

$$d = r - 1, \qquad \nu = d/2, \qquad \sigma_\omega = (1 + \omega)^r, \qquad V_\omega = \frac{\omega}{(1 + \omega)^2}, \qquad \kappa_\omega = \frac{(2\pi)^{-\nu}}{\sqrt{r}\, V_\omega^\nu}.$$

For $r \geq 3$ every root in Ω_λ has positive real part, and

$$\operatorname{Re} V_\omega = \frac{(1 + |\omega|^2)\operatorname{Re}\omega + 2|\omega|^2}{|1 + \omega|^4} > 0.$$

The power of V_ω in κ_ω uses the logarithm on the right half-plane.

Lemma V.2.3 (Dominant asymptotics at every complex weight)**.** *Fix $r \geq 3$ and $\lambda \neq 0$, and write $M = |\sigma_\omega|$ for $\omega \in \Omega_\lambda$. Then*

$$a_n^{(r)}(\lambda) = n^{-\nu} \sum_{\omega \in \Omega_\lambda} \kappa_\omega \sigma_\omega^n + O(M^n n^{-\nu-1}). \tag{V.2.1}$$

For the ordinary coefficients the displayed exponential bases are distinct. For the even coefficients their squares are distinct except when $\lambda = -1$. In that exceptional case the two amplitudes are equal positive real numbers and therefore add without cancellation.

Proof. With $X = x_1 \cdots x_d$, define the Laurent polynomial

$$P_\lambda(x) = \prod_{j=1}^{d} (1 + x_j)(1 + \lambda/X).$$

Coefficient extraction gives $a_n^{(r)}(\lambda) = \mathrm{CT}\, P_\lambda^n$. Evaluate this constant term on the torus $|x_j| = \rho$. Its r factors have arguments $t_1, \ldots, t_r$ satisfying $\sum_j t_j = \operatorname{Arg} \lambda$ modulo 2π. Lemma V.2.2 identifies the complete set of maxima of $|P_\lambda|$: they are $x_1 = \cdots = x_d = \omega$ with $\omega \in \Omega_\lambda$.

In local angular coordinates $x_j = \omega e^{i\theta_j}$, the last factor is $1 + \omega e^{-i\sum_j \theta_j}$. The first derivatives of $\log P_\lambda$ vanish and its Hessian is

$$-V_\omega(I_d + J_d), \qquad \det(I_d + J_d) = r,$$

where J_d has all entries one. Its real part is negative definite. Outside fixed small neighborhoods of these finitely many points, $|P_\lambda| \leq M - \epsilon$. Inside a neighborhood, Taylor expansion, the scaling $\theta = n^{-1/2} y$, and dominated Gaussian integration give the contribution

$$\sigma_\omega^n (2\pi)^{-d} \frac{(2\pi/n)^{d/2}}{\sqrt{r}\, V_\omega^{d/2}} (1 + O(n^{-1})).$$

The Gaussian branch is determined by $\operatorname{Re} V_\omega > 0$; odd Taylor terms integrate to zero on a symmetric local neighborhood. This proves (V.2.1).

Only the case $\lambda < 0$ has two contributions. Write $\omega = \rho e^{i\pi/r}$ and $\alpha = \arg(1 + \omega)$, so $0 < \alpha < \pi/r$. Their ordinary bases have arguments $\pm r\alpha \in (-\pi, \pi)$ and are distinct. Their squares coincide if and only if $2r\alpha = \pi$, equivalently $\alpha = \pi/(2r)$. Since the argument of $1 + \rho e^{i\pi/r}$ is strictly increasing in ρ, this occurs exactly at $\rho = 1$, that is, $\lambda = -1$. At that value

$$V_\omega = \frac{1}{4\cos^2(\pi/(2r))} > 0$$

for both roots, so their κ_ω are the same positive number. □

V.2.2 Algebraic-series asymptotics with several dominant phases

Lemma V.2.4 (Arithmetic amplitude obstruction)**.** *Let $f(z) = \sum_{n \geq 0} c_n z^n$ have algebraic coefficients. Suppose that, for distinct nonzero algebraic numbers $\tau_1, \ldots, \tau_s$ of the same modulus $T > 0$, nonzero constants b_j, and $\nu > 0$,*

$$c_n = n^{-\nu} \sum_{j=1}^{s} b_j \tau_j^n + o(T^n n^{-\nu}).$$

If f is algebraic over $\mathbb{C}(z)$, then ν is not a positive integer. If ν is a positive half-integer that is not an integer, every b_j belongs to $\overline{\mathbb{Q}}/\sqrt{\pi}$.

Proof. First the algebraic relation descends to $\overline{\mathbb{Q}}(z)$. Indeed, fix the finitely many monomials $z^i f^j$ in any polynomial relation. Their coefficients form an infinite linear system with finitely many columns over $\overline{\mathbb{Q}}$. Its rank is attained by finitely many rows, so a nonzero complex kernel implies a nonzero kernel over $\overline{\mathbb{Q}}$.

The given expansion implies a radius of convergence T^{-1}. For the lower bound on coefficient growth, use the elementary fact that a nonzero finite sum $\sum_j b_j \zeta_j^n$ with distinct $|\zeta_j| = 1$ cannot tend to zero: its Cesàro squared mean tends to $\sum_j |b_j|^2 > 0$.

An algebraic function has finitely many singularities on this circle and convergent Puiseux expansions there, with rational exponents and algebraic coefficients. A nonintegral local term $a(1 - z/\zeta)^q$ contributes

$$\frac{a}{\Gamma(-q)} \zeta^{-n} n^{-q-1} (1 + O(n^{-1})).$$

This follows from the exact binomial coefficient formula and Stirling's formula. Nonnegative integral powers are locally analytic and make no such contribution; poles give nonnegative powers of n. Truncating the finitely many Puiseux expansions and deforming the Cauchy contour gives their coefficient expansions to any prescribed order.

Distinct phases in these expansions cannot cancel each other. More explicitly, if $\sum_j d_j \zeta_j^n = o(1)$, multiply by $\overline{\zeta_k}^n$ and take Cesàro means to obtain $d_k = 0$. Apply this first to any powers of n larger than $n^{-\nu}$, then at $n^{-\nu}$. Thus each specified phase τ_j^n must have a nonzero leading Puiseux contribution with exponent $q = \nu - 1$. When ν is a positive integer, this exponent is a nonnegative integer, which is impossible. When ν is a nonintegral half-integer, $\Gamma(1 - \nu)$ is a nonzero rational multiple of $\sqrt{\pi}$, so $b_j \in \overline{\mathbb{Q}}/\sqrt{\pi}$. □

V.2.3 Proof of the classification

Proof of Theorem V.2.1. First suppose $r \geq 3$ and $\lambda \neq 0$ is algebraic. Every dominant ω, σ_ω, and V_ω is algebraic, and

$$\kappa_\omega \in \overline{\mathbb{Q}}^\times \pi^{-\nu}.$$

For $A_{r,\lambda}$, Lemma V.2.3 supplies distinct dominant bases and nonzero amplitudes. For $E_{r,\lambda}$ use the same formula with n replaced by $2n$. The only collision of squared bases occurs at $\lambda = -1$ and adds two equal nonzero amplitudes; the extra factor $2^{-\nu}$ is algebraic. Thus in both cases every combined amplitude is still in $\overline{\mathbb{Q}}^\times \pi^{-\nu}$.

If r is odd, $\nu = (r-1)/2$ is a positive integer, contradicting Lemma V.2.4. If r is even, then $r \geq 4$ and ν is a nonintegral half-integer. Algebraicity would force a nonzero amplitude to belong simultaneously to $\overline{\mathbb{Q}}^\times \pi^{-\nu}$ and $\overline{\mathbb{Q}}^\times \pi^{-1/2}$. This would make $\pi^{\nu - 1/2} = \pi^{(r-2)/2}$ algebraic, contradicting the transcendence of π.

Now suppose λ is transcendental and one of the two series has a polynomial relation over $\mathbb{C}(z)$. Its coefficients lie in $K = \mathbb{Q}(\lambda)$. The finite-column linear-system argument in Lemma V.2.4 descends a nonzero polynomial relation to $K[z, Y]$. Replace the transcendental λ by an indeterminate w. Each coefficient identity becomes an identity in $\mathbb{Q}(w)$, so the corresponding symbolic weighted series is algebraic over $\mathbb{Q}(w, z)$. After clearing coefficient denominators, specialize w to a positive rational number for which the relation does not become the zero polynomial. There are only finitely many excluded values. This contradicts the algebraic-weight case already proved. Therefore both series are nonalgebraic at every nonzero complex weight.

Finally, if $\lambda = 0$, then $a_n^{(r)}(0) = 1$ and both series are $(1 - z)^{-1}$. For $r = 1$,

$$A_{1,\lambda}(z) = \frac{1}{1 - (1 + \lambda)z}, \qquad E_{1,\lambda}(z) = \frac{1}{1 - (1 + \lambda)^2 z}.$$

For $r = 2$, constant-term extraction of a quadratic Laurent polynomial gives

$$A_{2,\lambda}(z) = \left(1 - 2(1 + \lambda)z + (1 - \lambda)^2 z^2\right)^{-1/2},$$

with value one at zero. Its even section

$$E_{2,\lambda}(z) = \tfrac{1}{2}\big(A_{2,\lambda}(\sqrt{z}) + A_{2,\lambda}(-\sqrt{z})\big)$$

is algebraic as well. These cases exhaust the classification. □

Remark V.2.5 (What has and has not been classified). The theorem includes all algebraic collision parameters, such as $\lambda = 8$ and $1/8$ for the cubic central family. It does not determine the minimum recurrence width at those parameters. The mechanism is a global maximum inequality followed by arithmetic asymptotics, rather than an assertion that all complex critical values are visible in the physical period. The unweighted case and the general Puiseux obstruction are inherited ideas; the complex-weight maximum analysis and its cancellation check supply the extension.

V.3 Exact order and algebraicity for every anisotropic hypercubic period

For $n \geq 1$ and nonzero complex numbers $m_1, \dots, m_n$, define

$$c_k(m) = \sum_{a_1+\cdots+a_n=k} \binom{k}{a_1, \dots, a_n}^2 \prod_i m_i^{2a_i}, \tag{V.3.1}$$

$$F_m(z) = \sum_{k\geq 0} \binom{2k}{k} c_k(m) z^k. \tag{V.3.2}$$

The coefficient is the weighted number of length-$2k$ closed walks on the n-dimensional hypercubic lattice when each step in either coordinate direction i has weight m_i. Complex weights are allowed in the analytic statement. The series converges near zero.

Partition the masses by equality of their squares. Write their distinct nonzero squared values as $\mu_1^2, \dots, \mu_q^2$, with multiplicities $n_1, \dots, n_q$. Thus $\sum_j n_j = n$. For the standard representation $V = \mathbb{C}^2$ of SL_2, put

$$W = \bigotimes_{j=1}^{q} \mathrm{Sym}^{n_j} V = \bigoplus_{d\geq 0} (\mathrm{Sym}^d V)^{\oplus a_d}, \qquad Q = \prod_j (n_j + 1), \qquad b = \sum_d a_d. \tag{V.3.3}$$

The decomposition in the middle uses the diagonal SL_2; the tensor product itself is irreducible for the product group SL_2^q.

Theorem V.3.1 (All weights and all dimensions)**.** *The minimal order of a nonzero operator in $\mathbb{C}(z)\langle \partial_z \rangle$ annihilating F_m is*

$$r(m) = Q - b = \sum_{d\geq 1} d\, a_d = \prod_j (n_j + 1) - [u^{\lfloor n/2 \rfloor}] \prod_j (1 + u + \cdots + u^{n_j}). \tag{V.3.4}$$

The minimal operator is irreducible over $\mathbb{C}(z)$. No hypothesis on distinct signed sums, thresholds, or generic choices of the nonzero masses is required.

More precisely, define

$$H_m(s) = s^{-1} F_m(s^{-2}).$$

Its minimal operator over $\mathbb{C}(s)$ is irreducible of the same order $r(m)$. The local monodromy of this equation at $s = \infty$ is unipotent, with one Jordan block of size d for each of the a_d summands with $d \geq 1$ in (V.3.3).

Corollary V.3.2 (Algebraicity classification)**.** *Allow zero weights and delete them before counting the dimension. Then $F_m(z)$ is algebraic over $\mathbb{C}(z)$ if and only if at most one weight is nonzero. With no nonzero weights it is 1; with one nonzero weight m, it is $(1-4m^2z)^{-1/2}$. For two or more nonzero weights it is transcendental, even when signed thresholds collide or vanish.*

For equal squared masses, (V.3.4) is $r = n$. For pairwise distinct squared masses it is

$$r = 2^n - \binom{n}{\lfloor n/2 \rfloor},$$

whose first values are $1, 2, 5, 10, 22, 44, 93$. For example the three-dimensional multiplicity patterns $[3]$, $[2,1]$, $[1,1,1]$ give orders $3, 4, 5$. The first two cases agree with classical isotropic and tetragonal formulae. Generic three-dimensional order five has also been announced in recent work of Elkies, Grimm and Schwartz; it is not presented here as a new generic-three-dimensional discovery. The scope of Theorem V.3.1 is its uniform formula in all dimensions and all squared-weight multiplicities, including every nonzero complex specialization.

V.3.1 Provenance and standard differential-module inputs

The Bessel representation and the central-binomial relation to hypercubic walks are classical; see Guttmann [37]. Delves and Joyce's anisotropic three-dimensional order-four equation concerns the tetragonal family with two equal coordinate weights [38]. The recent public announcement [39] describes a generic order-five result and a genus-two period formula in dimension three; its stated exceptional-set restriction and the absence of a linked full paper are kept separate from the theorem proved here. A literature search does not certify historical priority for the all-dimensional statement.

We use the following standard facts about algebraic connections on a curve. They are stated explicitly to fix the extension convention. The Fourier–Laplace transform is the Weyl-algebra equivalence $x \mapsto -\partial_s$, $\partial_x \mapsto s$, and preserves simple holonomic modules. For an irreducible connection M on $\mathbb{G}_m$, its intermediate extension $j_{!*}M$ to $\mathbb{A}^1$ is simple. At a regular singular point with unipotent monodromy T, its vanishing cycles are $\operatorname{im}(T-1)$, with the restricted monodromy. A Jordan block of size $d+1$ therefore contributes a block of size d, and a size-one block contributes nothing. The last assertion follows directly from the nearby/vanishing-cycle maps: absence of a point-supported quotient makes the canonical map surjective, while absence of a point-supported submodule makes the variation map injective, with composite $T-1$.

The stationary-phase theorem gives, if M has only the finite singularity zero and all slopes at infinity are at most one,

$$\operatorname{rank} \mathcal{F}(j_{!*}M) = \dim \operatorname{im}(T_0 - 1). \tag{V.3.5}$$

The formal germ of the Fourier transform at infinity is regular singular with this vanishing-cycle monodromy. Slope-one exponential factors at the original infinity contribute at finite transformed points, not to this rank formula through a slope-greater-than-one term. These are the formal stationary-phase statements of García López and Sabbah [40, 41]; in particular one must use the intermediate extension, rather than retain the entire nearby cycle space at zero.

We also use the curve de Rham index formula and the local-to-global compact-support sequence: for a connection N on $\mathbb{G}_m$,

$$\chi_{\mathrm{dR}}(\mathbb{G}_m, N) = -\operatorname{irr}_0(N) - \operatorname{irr}_\infty(N),$$

and

$$H^1_c \longrightarrow H^1 \longrightarrow H^1_{\mathrm{loc},0} \oplus H^1_{\mathrm{loc},\infty} \longrightarrow H^2_c \longrightarrow 0, \qquad H^2_c \cong H^0(N^\vee)^\vee.$$

All applications below have zero H^0 for the connection and its dual, and no formal slope-zero part at infinity. These standard index and duality inputs [45], as well as the Bessel differential Galois fact used next, are not asserted as new results.

V.3.2 The Bessel connection and its complete local partition

Set $\theta_x = x\partial_x$. Let B_μ be the rank-two connection associated with

$$(\theta_x^2 - 4\mu^2 x^2)y = 0. \tag{V.3.6}$$

It has solutions $I_0(2\mu x), K_0(2\mu x)$. Its differential Galois group over $\mathbb{C}(x)$ is SL_2; this is the classical non-half-integral Bessel case [42, 43]. At zero, I_0 is even and analytic, while

$$K_0(2\mu x) = -(\log x) I_0(2\mu x) + \text{an even analytic germ.}$$

The local monodromy is therefore a nontrivial unipotent block. At infinity the formal exponential factors are $e^{2\mu x}, e^{-2\mu x}$, with a common regular prefactor $x^{-1/2}$.

Lemma V.3.3. *If the μ_j^2 are pairwise distinct and nonzero, the direct sum of the B_{μ_j} has differential Galois group $\mathrm{SL}_2^{\,q}$. Consequently*

$$M = \bigotimes_j \mathrm{Sym}^{n_j} B_{\mu_j}$$

is irreducible of rank Q, and the function $g(x) = \prod_i I_0(2m_i x)$ generates this connection. At zero its monodromy has blocks of sizes $d+1$, with multiplicities a_d from (V.3.3); its slopes at infinity are at most one.

Proof. The projections of the Galois group, and of its identity component, onto each SL_2 factor are surjective. The Goursat lemma for products of simple groups [44, Proposition 1.8.2] says that failure of the full product would identify two standard rank-two representations, after a finite algebraic base change and a rank-one twist. At a point over infinity their exponential-factor pairs would be

$$\{2\mu_i x, -2\mu_i x\} = \{2\mu_j x + q(x), -2\mu_j x + q(x)\}.$$

Comparing the sums gives $q(x) = 0$ for their irregular parts, and comparing their differences gives $\mu_i = \pm\mu_j$, a contradiction. The same argument remains valid after ramification at infinity. Thus the full product acts, and its external tensor product of symmetric powers is irreducible.

Evaluation at the product of the I_0 solutions is a nonzero morphism from this simple differential module to a differential field. Its kernel is zero, so its image is precisely the cyclic module generated by g. At zero all factors have the same nonzero logarithmic-monodromy constant. The tensor monodromy is $\exp(\kappa e)$ for the diagonal raising operator e on W. On $\mathrm{Sym}^d V$ this has one block of size $d+1$. At infinity tensor products of the Bessel formal factors have exponentials $\exp(2x\sum_j \lambda_j\mu_j)$, where $\lambda_j = n_j, n_j - 2, \dots, -n_j$. Some sums may coincide or vanish; this only produces slope-zero or slope-one components, never slopes greater than one. □

Every summand in (V.3.3) has the parity of n and contains exactly one weight-zero vector if n is even, or exactly one weight-one vector if n is odd. Thus b is the corresponding central weight multiplicity. The character

$$\prod_j (u^{n_j} + u^{n_j-2} + \cdots + u^{-n_j})$$

proves the coefficient formula in (V.3.4). In particular $\mathrm{Sym}^n V$ occurs once, so $r \geq n$.

V.3.3 An upper bound that eliminates the endpoint ambiguity

For $\operatorname{Re} s > 2\sum_i |m_i|$, termwise integration gives

$$H_m(s) = \int_0^\infty e^{-sx} g(x)\,dx = \sum_{k\ge 0} \frac{(2k)!}{s^{2k+1}} \sum_{|a|=k} \prod_i \frac{m_i^{2a_i}}{a_i!^2} = s^{-1} F_m(s^{-2}). \tag{V.3.7}$$

To prove its order bound one cannot simply identify a unilateral Laplace transform with a Fourier transform: derivatives produce endpoint terms. We instead place this particular integral in a middle de Rham space and construct its closed cycle explicitly.

Work over $K = \mathbb{C}(s)$ and twist M by e^{-sx}; call the resulting connection N. On the cyclic vector η evaluating to g, its x-connection is $\nabla_x^M - s$ and its commuting s-connection is $\nabla_s = \partial_s - x$. The Bessel bases $b_\mu, \theta_x b_\mu$ and their symmetric tensor products give a $K[x]$-lattice N_0 stable under θ_x. Its residue at zero is nilpotent with the same b Jordan blocks as M.

The connection N is irreducible of rank $Q \ge 2$, also after extending the constant field algebraically, so it and its dual have no global horizontal vector. Since s is transcendental over the mass constants, every formal exponential factor at infinity is nonzero and linear in x after the twist. Thus $\operatorname{irr}_\infty(N) = Q$, while zero is regular singular. The index formula gives $\dim_K H^1_{\mathrm{dR}}(\mathbb{G}_m, N) = Q$. Local cohomology at infinity vanishes, while at zero it is the cokernel of the nilpotent residue and has dimension b. The compact-support sequence therefore gives

$$H^1_{\mathrm{mid}}(N) := \ker\{H^1_{\mathrm{dR}}(N) \longrightarrow H^1_{\mathrm{loc},0}(N)\}, \qquad \dim_K H^1_{\mathrm{mid}}(N) = Q - b = r. \tag{V.3.8}$$

Here $\omega = [\eta\,dx]$ belongs to this kernel. To check this without conflating dx and dx/x, complete the lattice to $\Lambda = N_0 \otimes K[[x]]$. The operator θ_x acts on $x^k\Lambda/x^{k+1}\Lambda$ as $k + \mathrm{Res}$, which is invertible for $k \ge 1$. It is therefore bijective on $x\Lambda$. There is $v \in x\Lambda$ with $\theta_x v = x\eta$, hence $\nabla_x v = \eta$, so $\eta\,dx$ is formally exact at zero. The commuting connection ∇_s preserves N_0, acts on both global and local cohomology, and hence preserves the kernel in (V.3.8).

It remains to verify that (V.3.7) is a period of this space. Choose branches along the positive ray for

$$h(x) = K_0(2m_1 x) \prod_{i>1} I_0(2m_i x).$$

This is a solution of M, including when some squared masses coincide. Continuing once around zero changes it by $-2\pi i\, g(x)$. Start at infinity on this ray, travel to ε, circle zero once, and return to infinity on the continued branch. With the factor e^{-sx} this is a closed rapid-decay twisted cycle: both ends decay at infinity for the stated half-plane, and there is no endpoint at zero. The integral of ω along its small circle is $O(\varepsilon(1 + |\log\varepsilon|))$ and tends to zero. Its two rays therefore give, up to a nonzero constant and orientation, exactly $H_m(s)$. The construction is valid for complex masses: fixed-ray continuations of K_0 have at most exponential growth, which is dominated by e^{-sx} in a sufficiently far right half-plane. Integrals of exact forms on this closed twisted cycle vanish.

Since $\omega, \nabla_s\omega, \dots, \nabla_s^r\omega$ are linearly dependent over K in (V.3.8), period integration proves

$$\operatorname{ord}_{\mathbb{C}(s)} H_m \le r. \tag{V.3.9}$$

This argument accounts for the unilateral endpoint instead of silently dropping its possible rank-one contribution.

V.3.4 Visibility of the simple Fourier constituent

Let $\mathcal{K} = j_{!*}M$ on the affine x-line. Lemma V.3.3 and (V.3.5) imply that $\mathcal{F}\mathcal{K}$ is simple of generic rank r. Its germ at $s = \infty$ has exactly the Jordan blocks of sizes d stated in the theorem.

We show that this constituent occurs in the module generated by the actual integral, not merely in a formal transform. Put $A_x = \mathbb{C}[x]\langle\partial_x\rangle$ and $\mathcal{E} = A_x g$. All its elements are entire functions of exponential type. Its restriction to $\mathbb{G}_m$ is M; therefore its composition factors are $\mathcal{K}$, once, and possibly copies of the point module δ_0. For an entire function f in this module, write $\widehat{f}(s) = \int_0^\infty e^{-sx} f(x)\,dx$ and set

$$\mathcal{L} = \{\widehat{f} : f \in \mathcal{E}\} + \mathbb{C}[s].$$

All integrals converge in a common sufficiently far right half-plane for each finite set of functions involved. Integration by parts gives

$$\widehat{xf} = -\partial_s \widehat{f}, \qquad \widehat{f'} = s\widehat{f} - f(0).$$

Consequently $\mathcal{L}$ is an A_s-module, and

$$\mathcal{L}/\mathbb{C}[s] \cong \mathcal{F}\mathcal{E}. \tag{V.3.10}$$

The map is injective: $\widehat{f}(s) \to 0$ along the positive real axis, so if it is a polynomial it is zero, and uniqueness of the Laplace transform then gives $f = 0$. Thus (V.3.10) is an actual isomorphism. Since Fourier sends δ_0 to the trivial module $\mathbb{C}[s]$, every composition factor of $A_s H_m \subseteq \mathcal{L}$ is either $\mathcal{F}\mathcal{K}$ (at most once) or the trivial affine-line connection.

If the former were absent, H_m would lie in an iterated extension of trivial affine-line connections. Every function in such a module is a polynomial: choose successive lifts of horizontal generators; the derivative of a lift is a polynomial combination of earlier lifts, so induction and integration prove the assertion. This contradicts $H_m(s) = s^{-1} + O(s^{-3})$. Hence $\mathcal{F}\mathcal{K}$ occurs and $\operatorname{ord} H_m \geq r$. Combining with (V.3.9) proves equality. Since this constituent already uses the full generic rank, the generic differential module generated by H_m is isomorphic to the irreducible generic module of $\mathcal{F}\mathcal{K}$. Its monodromy at infinity is therefore exactly the one obtained by stationary phase.

V.3.5 Descent and the algebraicity dichotomy

Multiplication by s does not change differential order or irreducibility over $\mathbb{C}(s)$. To descend from $sH_m(s) = F_m(s^{-2})$ to $z = s^{-2}$, use the derivation $\partial_z = -(s^3/2)\partial_s$, which preserves even functions. A relation of order r over $\mathbb{C}(s)$ can be decomposed using

$$\mathbb{C}(s) = \mathbb{C}(s^2) \oplus s\mathbb{C}(s^2).$$

Its even and odd parts are separately zero, because all derivatives of $F_m(s^{-2})$ with respect to z are even. At least one part is a nonzero relation over $\mathbb{C}(s^2) = \mathbb{C}(z)$ of no larger order. Conversely every relation over $\mathbb{C}(z)$ pulls back, so the minimal orders are equal. A nontrivial factorization over $\mathbb{C}(z)$ would pull back to one over $\mathbb{C}(s)$, contrary to the irreducibility already proved. This completes the proof of Theorem V.3.1.

For $n \geq 2$, the summand $\operatorname{Sym}^n V$ in W supplies a unipotent monodromy block of size $n \geq 2$ at $s = \infty$. This monodromy has infinite order. The minimal differential module of an algebraic function has finite monodromy: after taking a finite normal algebraic extension containing all its conjugates, their constant linear span supplies a finite-branch solution space and a differential annihilator over the base field. Its minimal operator is a factor of this finite-monodromy operator. Thus H_m, and consequently F_m, cannot be algebraic for $n \geq 2$. The explicit formulas when zero or one weights remain prove the converse and Corollary V.3.2.

The argument proves a statement about a function of the variable z. It does not assert transcendence of every special value at algebraic z, nor does it prove a general formula for the differential Galois group at all weight specializations. Those are distinct arithmetic and differential-algebraic questions.

V.4 Cyclotomic classification and design of algebraic quantiles

We classify all algebraic quantiles at algebraic probability levels for every rational polynomial weighting of the arcsine distribution. The exceptional levels form finite sets determined exactly by cyclotomic factors of an explicit polynomial. Conversely, every allowable finite set can be realized, and its minimum polynomial degree is determined. A real-variable root count reduces the entire two-parameter family of strict half-integer beta shapes to three possible levels. The classical transcendence inputs are Baker's theorem on linear forms in logarithms and the Gelfond–Schneider theorem; the classification and construction below are proved from those inputs.

V.4.1 The correction polynomial

Let $w \in \mathbb{Q}[X]$ satisfy $w(x) > 0$ for $0 < x < 1$. Zeros at the two endpoints are permitted. Define

$$c_w = \frac{1}{\pi}\int_0^1 \frac{w(t)}{\sqrt{t(1-t)}}\,dt > 0, \qquad F_w(x) = \frac{1}{\pi c_w}\int_0^x \frac{w(t)}{\sqrt{t(1-t)}}\,dt. \tag{V.4.1}$$

This is a continuous strictly increasing distribution function. Write

$$\mathcal{E}_w = \{r \in \overline{\mathbb{Q}} \cap (0,1) : F_w^{-1}(r) \in \overline{\mathbb{Q}}\}$$

for its set of algebraic probability levels with algebraic quantiles. Only the untruncated distribution on $[0,1]$ is considered here.

Lemma V.4.1 (Exact polynomial reduction)**.** *There is a unique polynomial $R_w \in \mathbb{Q}[X]$ such that*

$$w(X) = c_w + X(1-X)R_w'(X) + (1/2 - X)R_w(X). \tag{V.4.2}$$

If $w(X) = \sum_j w_j X^j$, then

$$c_w = \sum_j w_j\, 4^{-j}\binom{2j}{j} \in \mathbb{Q}_{>0}.$$

For $0 \le x \le 1$, with the positive square root and $\arcsin\sqrt{x} \in [0, \pi/2]$,

$$F_w(x) = \frac{2}{\pi}\arcsin\sqrt{x} + \frac{\sqrt{x(1-x)}}{\pi c_w}R_w(x). \tag{V.4.3}$$

If w is nonconstant of degree d, then R_w is nonzero of degree $d-1$. If w is constant, then $R_w = 0$.

Proof. For the linear operator $\mathcal{L}R = X(1-X)R' + (1/2 - X)R$,

$$\mathcal{L}(X^k) = (k + 1/2)X^k - (k+1)X^{k+1}.$$

Starting with the highest degree, this triangular identity gives unique rational coefficients for a polynomial R such that $w - \mathcal{L}R$ is constant. A nonzero polynomial R of degree k has $\mathcal{L}R$ of degree $k+1$, proving uniqueness and the degree assertion.

Since

$$\frac{d}{dx}\{\sqrt{x(1-x)}R(x)\} = \frac{\mathcal{L}R(x)}{\sqrt{x(1-x)}},$$

the integral of $\mathcal{L}R/\sqrt{x(1-x)}$ over $[0,1]$ vanishes. Hence the constant just obtained is c_w. The beta integral for each monomial gives $\int_0^1 x^{j-1/2}(1-x)^{-1/2}\,dx = \pi 4^{-j}\binom{2j}{j}$; alternatively this follows by $x = \sin^2\theta$ and the elementary reduction of even powers of sine. Integrating (V.4.2) from zero to x proves (V.4.3). □

Theorem V.4.2 (Arithmetic reduction)**.** *For every algebraic $x \in (0,1)$,*

$$F_w(x) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad R_w(x) = 0 \ \text{ and } \ \frac{2}{\pi}\arcsin\sqrt{x} \in \mathbb{Q}. \tag{V.4.4}$$

Whenever these conditions hold, $F_w(x) = 2\arcsin\sqrt{x}/\pi$ is rational. In particular, every algebraic irrational level has a transcendental quantile for every w in (V.4.1)*.*

Proof. Suppose x and $r = F_w(x)$ are algebraic, and put $\theta = \arcsin\sqrt{x} \in (0, \pi/2)$. Equation (V.4.3) shows that

$$2\theta - \pi r = -\frac{\sqrt{x(1-x)}}{c_w} R_w(x) \in \overline{\mathbb{Q}}.$$

The number $\zeta = e^{2i\theta} = 1 - 2x + 2i\sqrt{x(1-x)}$ is algebraic. Choose $\log\zeta = 2i\theta$ and $\log(-1) = i\pi$. Then

$$\log\zeta - r\log(-1) = i(2\theta - \pi r)$$

is algebraic. Baker's theorem says that an algebraic-coefficient linear form in logarithms of algebraic numbers is either zero or transcendental [46]. It must therefore vanish. Consequently $R_w(x) = 0$ and $\zeta = e^{i\pi r}$. If the algebraic number r were irrational, the Gelfond–Schneider theorem, applied to this determination of $(-1)^r$, would make ζ transcendental. Thus $r \in \mathbb{Q}$. The converse follows immediately from (V.4.3). □

V.4.2 All exceptional sets and a finite exact algorithm

For each integer $d \geq 3$, put

$$\mathcal{Q}_d = \left\{ \frac{2j}{d} : 1 \leq j < d/2,\ \gcd(j,d) = 1 \right\}, \tag{V.4.5}$$

$$P_d(X) = \prod_{\substack{1 \leq j < d/2 \\ (j,d)=1}} \left(X - \sin^2\frac{\pi j}{d} \right). \tag{V.4.6}$$

We call $\mathcal{Q}_d$ a cyclotomic packet of probability levels. The sets $\mathcal{Q}_d$ are pairwise disjoint: d is the exact order of $e^{i\pi r}$ for every $r \in \mathcal{Q}_d$.

Lemma V.4.3. *The polynomial P_d is monic, rational, and irreducible, of degree $\varphi(d)/2$. Its roots are distinct and belong to $(0,1)$. The polynomials P_d, $d \geq 3$, are pairwise distinct.*

Proof. For a primitive dth root ζ_d, write

$$\sin^2\frac{\pi j}{d} = \frac{2 - \zeta_d^j - \zeta_d^{-j}}{4}.$$

The Galois group of $\mathbb{Q}(\zeta_d)$ acts transitively on these numbers. Two such numbers agree exactly when their indices agree up to sign modulo d: the sum $z + z^{-1}$ determines the unordered pair $\{z, z^{-1}\}$. Thus the orbit has $\varphi(d)/2$ distinct elements, proving rationality and irreducibility. None of these roots is an endpoint because $d \geq 3$ and the root of unity is primitive. The same unordered-pair observation shows that a root cannot belong to both P_d and P_e unless $d = e$. □

Theorem V.4.4 (Complete packet classification)**.** *If w is nonconstant, its entire exceptional set is*

$$\mathcal{E}_w = \coprod_{\substack{d \geq 3 \\ P_d \mid R_w}} \mathcal{Q}_d. \tag{V.4.7}$$

For r in this set the quantile is exactly $\sin^2(\pi r/2)$. If ν_w is the number of distinct roots of R_w in $(0,1)$, every packet appearing in (V.4.7) satisfies

$$\frac{\varphi(d)}{2} \le \nu_w, \qquad d \le 8\nu_w^2 \le 8(\deg w - 1)^2. \tag{V.4.8}$$

Moreover $|\mathcal{E}_w| \le \nu_w$. Thus the whole exceptional set is determined by finitely many exact divisions in $\mathbb{Q}[X]$. For a positive constant w, instead, $\mathcal{E}_w = \mathbb{Q} \cap (0,1)$.

Proof. Theorem V.4.2 makes every exceptional level rational. If r is such a level, $e^{i\pi r}$ has an exact order $d \ge 3$, so $r = 2j/d$ with the index conditions in (V.4.5). Its quantile $x = \sin^2(\pi j/d)$ is a root of R_w. Lemma V.4.3 shows that $P_d \mid R_w$. Conversely this divisibility makes every root of P_d a root of R_w, and (V.4.3) gives precisely the quantiles and levels in $\mathcal{Q}_d$.

All $\varphi(d)/2$ roots of a dividing P_d belong to $(0,1)$, and different packets give different roots. This proves the root count and cardinality bounds. For completeness, the elementary inequality $\varphi(d) \ge \sqrt{d/2}$ follows from

$$\frac{\varphi(d)^2}{d} = \prod_{p^a \| d} p^{a-2}(p-1)^2 \ge \frac{1}{2}.$$

Every odd-prime factor on the right is at least one; the only possible factor below one is $p = 2, a = 1$, equal to $1/2$. This proves the finite bound in (V.4.8). For constant w, use $R_w = 0$ in Theorem V.4.2; rational angles have algebraic sine. □

An exact algorithm is now explicit. Compute c_w, R_w by the triangular identity in Lemma V.4.1; for all $3 \le d \le 8(\deg w - 1)^2$, form P_d and test $P_d \mid R_w$ over $\mathbb{Q}$. Output the packets for the successful tests. A Sturm count for R_w can replace $\deg w - 1$ by ν_w and shorten the list. The polynomial P_d itself can be obtained from the usual cyclotomic polynomial by the substitution $4X = 2 - Z - Z^{-1}$ or by its minimal-polynomial description. No test of numerical equality between transcendental quantities is required. When $\deg w = 1$, R_w is a nonzero constant and the output is empty.

V.4.3 Converse design and the minimum degree

Theorem V.4.5 (Realization of every possible finite exceptional set)**.** *A finite set $\mathcal{E} \subset \overline{\mathbb{Q}} \cap (0,1)$ equals $\mathcal{E}_w$ for some rational polynomial w strictly positive on $[0,1]$ if and only if*

$$\mathcal{E} = \coprod_{d \in D} \mathcal{Q}_d$$

for a finite set $D \subset \{3, 4, \dots\}$. For this prescribed set, the minimum possible degree of w is

$$1 + \sum_{d \in D} \frac{\varphi(d)}{2} = 1 + |\mathcal{E}|. \tag{V.4.9}$$

This includes the empty set, whose minimum degree is one.

Proof. Necessity follows from Theorem V.4.4; a constant positive weight has infinitely many exceptions. For sufficiency put $R = \prod_{d \in D} P_d$, with $R = 1$ if D is empty. Choose a positive rational number $c > \max_{x \in [0,1]} |\mathcal{L}R(x)|$ and set

$$w = c + \mathcal{L}R.$$

Then $w > 0$ on $[0,1]$. The integral of $\mathcal{L}R/\sqrt{x(1-x)}$ is zero, so $c_w = c$ and $R_w = R$. Pairwise distinct irreducible factors in Lemma V.4.3 and (V.4.7) show that the exceptional set is exactly the prescribed one. The construction has degree $1 + \deg R = 1 + \sum_{d \in D} \varphi(d)/2$.

Any polynomial weight realizing this finite set is nonconstant, and its correction polynomial must be divisible by every P_d, $d \in D$. Those polynomials are pairwise coprime, so its degree is at least the displayed sum. Lemma V.4.1 then proves the lower bound in (V.4.9). When D is empty, the same construction is a positive linear weight and constants are excluded, proving that case too. □

Example V.4.6 (A single exceptional probability)**.** Choose $R = X - 1/4$ and $c = 1$. Then

$$w(X) = \frac{7}{8} + \frac{7}{4}X - 2X^2$$

is positive on $[0, 1]$: it is concave and its endpoint values are $7/8$ and $5/8$. Its entire exceptional set is $\mathcal{Q}_6 = \{1/3\}$ and $F_w^{-1}(1/3) = 1/4$. Every other algebraic internal level has a transcendental quantile. Degree two is minimal.

Example V.4.7 (An algebraic quantile forces its conjugate packet)**.** The packet $\mathcal{Q}_{10}$ is $\{1/5, 3/5\}$ and

$$P_{10}(X) = X^2 - \frac{3}{4}X + \frac{1}{16}.$$

Therefore a rational polynomial weighting cannot have $1/5$ as its only algebraic level with an algebraic quantile: $3/5$ must occur too. The two quantiles are $(3 - \sqrt{5})/8$ and $(3 + \sqrt{5})/8$. Theorem V.4.5 realizes exactly this pair with a positive polynomial of degree three, and this degree is minimal.

V.4.4 Real-variable shape bounds

Put $g(\theta) = w(\sin^2 \theta)$ for $0 \le \theta \le \pi/2$. Its average is c_w. Define

$$h(\theta) = \int_0^\theta (g(u) - c_w)\, du.$$

Formula (V.4.3) gives, with $x = \sin^2 \theta$,

$$h(\theta) = \tfrac{1}{2}\sqrt{x(1-x)}R_w(x), \qquad h(0) = h(\pi/2) = 0. \tag{V.4.10}$$

Proposition V.4.8 (Mean crossings bound arithmetic complexity)**.** *Suppose w is nonconstant and $g - c_w$ has s distinct zeros in $(0, \pi/2)$. Then $\nu_w \le s - 1$. In particular:*

1. *If g is strictly monotone, then $\nu_w = 0$ and $\mathcal{E}_w$ is empty.*
2. *Suppose g is strictly increasing up to one interior maximum and strictly decreasing thereafter, and $g(0), g(\pi/2) < c_w$. Then $\nu_w = 1$. Consequently $\mathcal{E}_w$ has at most one member, and*

$$\mathcal{E}_w \subset \{1/3, 1/2, 2/3\}.$$

Proof. If h has ν_w distinct interior zeros, its two endpoint zeros and Rolle's theorem force at least $\nu_w + 1$ distinct interior zeros of $h' = g - c_w$. This gives the first bound. Strict monotonicity gives exactly one mean crossing.

Under the single-maximum assumptions there are exactly two mean crossings, say $\theta_- < \theta_+$. The function h strictly decreases before θ_-, strictly increases between the crossings, and strictly decreases after θ_+. It is negative near zero and positive just before $\pi/2$, because its derivative is negative there and its final value is zero. It therefore has exactly one interior zero, located between the two mean crossings. Every possible packet now has $\varphi(d) \le 2$. The integers $d \ge 3$ with this property are exactly $3, 4, 6$, whose packets are respectively $\{2/3\}, \{1/2\}, \{1/3\}$. □

V.4.5 The full half-integer beta lattice

The normalized incomplete beta function is $I_x(\alpha,\beta) = B_x(\alpha,\beta)/B(\alpha,\beta)$; we use its usual strictly increasing inverse on $(0,1)$ [57]. First take strict half-integer shapes

$$\alpha = m + \tfrac{1}{2}, \qquad \beta = n + \tfrac{1}{2}, \qquad m,n \in \mathbb{Z}_{\geq 0}.$$

This is (V.4.1) with $w_{m,n}(X) = X^m(1-X)^n$. Let $R_{m,n}$ be its correction polynomial and let $\mathcal{E}_{m,n}$ denote its exceptional set.

Theorem V.4.9 (Three-level theorem and exact decision for every pair)**.** *For all nonnegative integers m,n the following statements hold.*

1. *If $m = n = 0$, then $\mathcal{E}_{0,0} = \mathbb{Q} \cap (0,1)$.*
2. *If exactly one of m,n is zero, then $\mathcal{E}_{m,n}$ is empty.*
3. *If $m,n \geq 1$, then $\mathcal{E}_{m,n}$ has at most one member, and is exactly the subset of the following three candidates passing the indicated rational test:*

level	*quantile*	*necessary and sufficient test*
1/3	1/4	$R_{m,n}(1/4) = 0$
1/2	1/2	$m = n$
2/3	3/4	$R_{m,n}(3/4) = 0.$

In particular, $\mathcal{E}_{m,m} = \{1/2\}$ for $m \geq 1$.

In every case, every algebraic irrational level has a transcendental quantile. The tests are finite rational arithmetic, for example by the recurrences

$$\begin{aligned} R_{0,0}(X) &= 0, \\ R_{0,n}(X) &= \frac{2n-1}{2n} R_{0,n-1}(X) + \frac{(1-X)^{n-1}}{n}, \qquad n \geq 1, \\ R_{m,n}(X) &= \frac{2m-1}{2(m+n)} R_{m-1,n}(X) - \frac{X^{m-1}(1-X)^n}{m+n}, \qquad m \geq 1. \end{aligned} \tag{V.4.11}$$

Proof. The arcsine case follows from Theorem V.4.4. With exactly one positive exponent, $w_{m,n}$ is strictly monotone. With both exponents positive, it vanishes at the endpoints and has exactly one strict maximum, because its logarithmic derivative is $m/X - n/(1-X)$. Proposition V.4.8 gives all the candidate restrictions. For each candidate the corresponding P_d is linear, so Theorem V.4.4 gives precisely the stated evaluation test.

It remains to simplify the middle test. If $m = n$, symmetry gives $I_{1/2} = 1/2$, so that test passes. If $m > n$, then for $0 < x < 1/2$,

$$\frac{x^{m-1/2}(1-x)^{n-1/2}}{(1-x)^{m-1/2}x^{n-1/2}} = \left(\frac{x}{1-x}\right)^{m-n} < 1.$$

The probability below $1/2$ is strictly less than $1/2$. For $m < n$ it is strictly greater. Hence the middle test passes if and only if $m = n$.

For completeness, put

$$c_{m,n} = \frac{(2m)!(2n)!}{4^{m+n} m!\, n!\, (m+n)!}.$$

The primitive is $2c_{m,n} \arcsin\sqrt{x} + \sqrt{x(1-x)} R_{m,n}(x)$. Integration by parts, or differentiation of $x^{m-1/2}(1-x)^{n+1/2}$, gives for $m \geq 1$

$$B_x(m+\tfrac{1}{2}, n+\tfrac{1}{2}) = \frac{2m-1}{2(m+n)} B_x(m-\tfrac{1}{2}, n+\tfrac{1}{2}) - \frac{x^{m-1/2}(1-x)^{n+1/2}}{m+n}.$$

The analogous reduction in n, starting with $m = 0$, has positive boundary term $x^{1/2}(1-x)^{n-1/2}/n$. Substitution of the primitive proves (V.4.11). □

Example V.4.10 (A genuine asymmetric exception)**.** For $(m, n) = (1, 2)$,

$$c_{1,2} = \frac{1}{16}, \qquad R_{1,2}(X) = -\frac{(2X-3)(4X-1)}{24}.$$

The unique interior root is $1/4$. Therefore

$$I_{1/4}(3/2, 5/2) = 1/3, \qquad \mathcal{E}_{1,2} = \{1/3\}.$$

By reflection $\mathcal{E}_{2,1} = \{2/3\}$, with quantile $3/4$. Every other algebraic internal probability level has a transcendental quantile for these two distributions.

Corollary V.4.11 (All positive half-integer shapes)**.** *Let $\alpha, \beta \in \frac{1}{2}\mathbb{Z}_{>0}$. If at least one of α, β is an integer, every algebraic level in $(0, 1)$ has an algebraic beta quantile. If neither is an integer, Theorem V.4.9 gives the exact finite tests and all the possible exceptional levels.*

Proof. When both shapes are integers, the incomplete beta integral is a polynomial with rational coefficients. Suppose one shape is integral and the other is a strict half-integer. If α is integral, the substitution $t = 1 - y^2$ turns the integral into a polynomial primitive in $y = \sqrt{1-x}$ with rational coefficients; if β is integral, use $t = y^2$. The complete integral is then a nonzero algebraic number. Equating the incomplete integral to an algebraic multiple of the complete one is a nonconstant polynomial equation for y with algebraic coefficients. Thus y, and hence the unique quantile x, is algebraic. The remaining shapes are exactly those in Theorem V.4.9. □

Corollary V.4.12 (Quantiles of the binomial Jeffreys posterior)**.** *After m successes and n failures, the prior with density proportional to $[x(1-x)]^{-1/2}$ gives the posterior* Beta$(m + 1/2, n + 1/2)$. *If $m + n \geq 1$, every algebraic level outside $\{1/3, 1/2, 2/3\}$ has a transcendental posterior quantile. In particular, both endpoints of the equal-tailed 90%, 95%, and 99% posterior intervals are transcendental for every nonempty data set.*

Proof. Multiply the prior density by the binomial likelihood $x^m(1-x)^n$ and apply Theorem V.4.9. The lower-tail probabilities of these intervals are respectively $1/20$, $1/40$, and $1/200$, and the upper-tail probabilities are their complements. None is one of the three candidates. □

Exact scope of the classification. Theorem V.4.4 classifies all exceptional sets for rational polynomial arcsine weightings, and Theorem V.4.5 proves both converse realization and optimal degree. Theorem V.4.9 supplies an exact finite decision for every half-integer beta pair. It does not assert that the only asymmetric pairs passing a test are $(1, 2)$ and $(2, 1)$: a closed list of all integer pairs satisfying $R_{m,n}(1/4) = 0$ is not proved here. The restriction to full-interval normalization is also essential to the proof; no statement for arbitrary algebraic truncation is inferred.

Relation to the literature. Baker's logarithmic theorem is an established input, not a new transcendence principle. The beta identities are classical [57]. Dhillon–Saha study arithmetic values of incomplete beta functions [55]. Their publisher abstract and bibliographic record were available in preparing this draft, but their full article was not obtained; consequently no claim is made that all potential overlaps with that paper have been excluded. The independently proved claims here are the explicit packet criterion, the converse design with its sharp degree, and the resulting uniform restrictions. No historical-priority or named-open-problem claim is made.

V.5 Support-free Gaussian likelihood in the complete growing-support regime

Let $X_i = \Theta_i + \sigma Z_i$, where the Θ_i are independent with law G, the Z_i are independent standard normal variables, and $\sigma > 0$ is known. In this section $\widehat{G}$ maximizes the Gaussian mixture likelihood over *all* probability measures on $\mathbb{R}$. Neither a support center nor a support radius is supplied to the estimator.

Theorem V.5.1 (Uniform growing-support minimaxity)**.** *There exist absolute constants $c, C > 0$ such that the following holds. For $N \geq 3$, put $L = \log(eN)$. If $A \geq \sigma\sqrt{L}$, then*

$$\inf_{\widetilde{G}} \sup_{G \in \mathcal{P}([\mu-A,\mu+A])} \mathbb{E}_G W_1(\widetilde{G}, G) \geq c\left\{\frac{\sigma}{\sqrt{L}} + \frac{A}{\sqrt{N}}\right\},$$

$$\sup_{G \in \mathcal{P}([\mu-A,\mu+A])} \mathbb{E}_G W_1(\widehat{G}, G) \leq C\left\{\frac{\sigma}{\sqrt{L}} + \frac{A}{\sqrt{N}}\right\}.$$

The infimum can even be taken over estimators that know A and μ. Every measurable choice of the unrestricted NPMLE satisfies the upper bound, uniformly in A, μ, σ, N in this range.

The result includes every intermediate support/noise scale between $A \asymp \sigma\sqrt{\log N}$ and $A \asymp \sigma\sqrt{N}$, as well as the larger-support range. It does not assert the entire known-support minimax curve when $A < \sigma\sqrt{\log N}$. Soloff, Guntuboyina and Sen [64, Theorem 11] prove a deconvolution bound for a fixed compact support once the sample size exceeds a support-dependent threshold. The issue here is a bound with absolute constants throughout the stated growing-radius range. The classical likelihood contact condition is used explicitly [47, Proposition 1.1]; that condition itself is not new.

Translation and scaling reduce the proof to $\mu = 0$, $\sigma = 1$. Write φ and Φ for the standard normal density and CDF, $p_H = H * \varphi$, and $q = N^{-1}\sum_i \delta_{X_i}$. For probability densities p, r, use the Hellinger convention $h(p,r)^2 = \int_{\mathbb{R}} (\sqrt{p} - \sqrt{r})^2\, dx$.

V.5.1 An exact Gaussian-to-binomial experiment

For an integer $m \geq 1$, let ε be independent of X, with centered Gaussian law on $\mathbb{R}^m$ and covariance $mI_m - \mathbf{1}\mathbf{1}^{\mathsf{T}}$. This matrix is positive semidefinite; its zero eigenvalue causes no difficulty. Given X, set $Y_j = X + \varepsilon_j$ and retain $Z = \sum_{j=1}^m \mathbf{1}_{\{Y_j \geq 0\}}$. This defines a parameter-free Markov channel A_m. If $X \sim N(\theta, 1)$, then the joint Gaussian vector Y has mean $\theta\mathbf{1}$ and covariance mI_m. Thus its coordinates are independent and

$$A_m N(\theta, 1) = \mathrm{Bin}(m, \phi_m(\theta)), \qquad \phi_m(\theta) = \Phi(\theta/\sqrt{m}). \tag{V.5.1}$$

For $m = \lceil D^2 \rceil$, $D \geq 1$, the inverse of ϕ_m on $[-D, D]$ has Lipschitz constant at most CD. In particular, Hellinger data processing transfers predictive Hellinger accuracy to the binomial experiment, while latent distances can be recovered with a factor CD.

This exact channel does not by itself justify a Poisson-style entropy transfer: $D(q\|p_H)$ is infinite for continuous Gaussian observations. We instead use a second-order cancellation.

V.5.2 Likelihood contact supplies the second-order cancellation

Every unrestricted Gaussian likelihood maximizer is supported in $[\min_i X_i, \max_i X_i]$: moving a parameter outside that interval to its nearest endpoint strictly increases every component likelihood. Compactness then proves existence. Its score satisfies

$$S_H(\theta) = \frac{1}{N}\sum_i \frac{\varphi(X_i - \theta)}{p_H(X_i)} \leq 1 \quad (\theta \in \mathbb{R}), \qquad S_H = 1 \quad H\text{-almost everywhere.}$$

At each contact parameter the empirical-posterior weights $R_\theta(i) = N^{-1}\varphi(X_i-\theta)/p_H(X_i)$ sum to one. Evaluating the score at $\theta+u$ gives

$$\mathbb{E}_{R_\theta} e^{u(X-\theta)} = e^{u^2/2} S_H(\theta+u) \le e^{u^2/2}.$$

Consequently $\mathbb{E}_{R_\theta} X = \theta$ and $\mathbb{E}_{R_\theta}(X-\theta)^2 \le 1$, and the joint law $H(d\theta)R_\theta(i)$ has observation marginal q. For every globally twice continuously differentiable F with bounded second derivative, Taylor's theorem therefore yields

$$\left| \int F\, d(H-q) \right| \le \tfrac{1}{2}\|F''\|_\infty. \tag{V.5.2}$$

The true Gaussian coupling similarly gives $|\int F\, d(G-p_G)| \le \|F''\|_\infty/2$. Hence

$$\left| \int F\, d(H-G) \right| \le \mathrm{Lip}(F) W_1(q,p_G) + \|F''\|_\infty. \tag{V.5.3}$$

Unlike a transport bound applied to an arbitrary Lipschitz test, this estimate is small when the retained test has small curvature.

Lemma V.5.2 (Polynomial approximation with two controlled derivatives)**.** *For every 1-Lipschitz $g : [0,1] \to \mathbb{R}$ and every integer $k \ge 1$, there is a real polynomial P_k of degree at most k such that*

$$\|g - P_k\|_\infty \le C/k, \qquad \|P_k'\|_\infty \le C, \qquad \|P_k''\|_\infty \le Ck.$$

All norms in this display are on $[0,1]$.

Proof. Subtract $g(0)$ and reflect g to a 2-periodic 1-Lipschitz function. Convolve with a positive normalized Jackson trigonometric kernel of degree $2s$. The resulting trigonometric polynomial Q_s approximates g within C/s, and $\|Q_s'\|_\infty \le 1$, because convolution preserves the Lipschitz constant. The trigonometric Bernstein inequality applied to Q_s' gives $\|Q_s''\|_\infty \le Cs$.

To replace Q_s by an ordinary polynomial, expand its finitely many exponentials $e^{i\pi j(x-1/2)}$, $|j| \le 2s$, through Taylor degree $100s$. Each Fourier coefficient has magnitude at most the bounded supremum norm of the reflected function. For $|x-1/2| \le 1/2$, the exponential-series remainder and its first two derivatives are at most a polynomial in s times e^{-cs}, since $|\pi j(x-1/2)| \le \pi s$ while the truncation degree is $100s$. Their sum over $O(s)$ modes is still exponentially small. Take the real part and add back $g(0)$. With $s = \lfloor k/100 \rfloor$, this proves the assertions for large k; bounded k is handled by a constant polynomial and an enlarged absolute constant. □

Lemma V.5.3 (A two-scale likelihood inverse)**.** *Suppose the true mixing law G and an unrestricted Gaussian MLE H are supported in $[-D,D]$, where $D \ge 1$, and put $m = \lceil D^2 \rceil$. For $1 \le S \le K \le m/2$,*

$$W_1(H,G) \le C\left\{ \frac{D}{K} + \frac{S}{D} + W_1(q,p_G) + \frac{De^{2K^2/m}}{S} h(p_H,p_G) \right\}. \tag{V.5.4}$$

Proof. For a 1-Lipschitz test f on $[-D,D]$, define $g(x) = f(\phi_m^{-1}(x))/D$ on $\phi_m([-D,D])$. It has an absolute Lipschitz constant, so it extends to $[0,1]$ with the same constant. Apply Lemma V.5.2, with constants rescaled if necessary, at degrees S and K.

The global function $F_S(\theta) = DP_S(\phi_m(\theta))$ satisfies $\mathrm{Lip}(F_S) \le C$. Writing $z = \theta/\sqrt{m}$,

$$F_S''(\theta) = \frac{D}{m}\{P_S''(\Phi(z))\varphi(z)^2 - P_S'(\Phi(z))z\varphi(z)\},$$

so $\|F_S''\|_\infty \le CS/D$. Equation (V.5.3) controls this part by $C\{W_1(q,p_G) + S/D\}$.

The residual polynomial $P_K - P_S$ has norm at most C/S. The binomial Bernstein coefficient bound proved in Lemma V.A.1 expands it at degree m with coefficients of magnitude at most $Ce^{2K^2/m}/S$. Its expectation difference under the pushforward mixing laws is therefore bounded by this quantity times $\|A_m p_H - A_m p_G\|_1 \le 2h(p_H,p_G)$. Finally the approximation of f by $DP_K \circ \phi_m$ costs CD/K. Wasserstein duality proves (V.5.4). □

V.5.3 Uniform likelihood accuracy on the random support range

Let $T = \sqrt{40L}$, $D = A + T$, and $E = \{\max_i |Z_i| \le T\}$. Every unrestricted MLE is supported in $[-D, D]$ on E. Also $\mathbb{P}(E^c) \le 2Ne^{-20L}$ and

$$\mathbb{E}[\mathbf{1}_{E^c} W_1(\widehat{G}, G)] \le \mathbb{E}[\mathbf{1}_{E^c}(2A + \max_i |Z_i|)] \le C(A+T)N^{-19}. \tag{V.5.5}$$

Since $A \ge \sqrt{L}$, $D \asymp A$ with absolute constants.

We give the density bound used on this event explicitly. If $A \le \sqrt{N}$, then

$$\mathbb{E}[\mathbf{1}_E h(p_{\widehat{G}}, p_G)] \le C\sqrt{D}\, L^{3/4}/\sqrt{N}. \tag{V.5.6}$$

To verify it, consider Gaussian mixtures with means in $[-D, D]$. For complex z, $\sup_x |\varphi(x-z)| \le (2\pi)^{-1/2} e^{(\operatorname{Im} z)^2/2}$. On $z = D\cos(\theta + iu)$ this is at most $C\exp(D^2 \sinh^2 u/2)$. Contour shifting, with u a small multiple of k/D^2 when $k \le D^2$ and a fixed small constant otherwise, bounds the kth Chebyshev coefficient in supremum norm by

$$C\exp\{-c\min(k^2/D^2, k)\}.$$

Truncating at degree $d = C\{D\sqrt{\log(C(D+1)/\beta)} + \log(C(D+1)/\beta)\}$ and gridding the Chebyshev moments gives a uniform density cover of radius β, with centers that are actual mixtures and $\log M \le Cd\log(C(d+1)/\beta)$.

Take $U = D + \sqrt{12L}$ and $\beta = N^{-3}/(2U)$. As $\sqrt{L} \le D \le C\sqrt{N}$, this gives $\log M \le CDL^{3/2}$. Replace each cover center p_a by the envelope

$$u_a(x) = \begin{cases} p_a(x) + \beta, & |x| \le U, \\ \varphi(|x| - D), & |x| > U. \end{cases}$$

It dominates every associated mixture and has total mass in $[1, 1 + CN^{-3}]$ whenever used. Normalizing the envelope changes an associated density by Hellinger distance at most $CN^{-3/2}$. The square-root likelihood-ratio union bound in Lemma V.A.2 now gives

$$\mathbb{E}[\mathbf{1}_E h(p_{\widehat{G}}, p_G)^2] \le C(\log M + 1)/N + CN^{-3}.$$

The argument applies to every fit in the covered class with likelihood at least that of G. Consequently it applies on E without conditioning on E or pretending that the conditional observations remain independent. Cauchy–Schwarz proves (V.5.6).

The ordinary empirical distribution also satisfies

$$\mathbb{E}W_1(q, p_G) \le C(A+1)/\sqrt{N}. \tag{V.5.7}$$

Integrate empirical-CDF standard deviations: on $[-A, A]$ use $\operatorname{Var}(\mathbf{1}_{X \le x}) \le 1/4$, and on the two tails use $\mathbb{P}(|X| > A + u) \le 2\mathbb{P}(Z > u)$ and the finiteness of $\int_0^\infty \sqrt{\mathbb{P}(Z > u)}\, du$.

V.5.4 Proof of the uniform upper and lower bounds

Assume first $\sqrt{L} \le A \le \sqrt{N}$. For sufficiently large absolute N, put $m = \lceil D^2 \rceil$ and choose

$$S = \left\lfloor \frac{D}{16\sqrt{L}} \right\rfloor \vee 1, \qquad K = \left\lfloor \min\{m/2, \sqrt{amL}\} \right\rfloor, \qquad a = 1/64.$$

These choices satisfy $1 \le S \le K \le m/2$, $D/K \le C/\sqrt{L}$, $S/D \le C/\sqrt{L}$, and $D/S \le C\sqrt{L}$. Apply Lemma V.5.3 on E, then equations (V.5.5)–(V.5.7). The only term needing absorption is

$$C\sqrt{L}\, e^{2aL} \frac{\sqrt{D} L^{3/4}}{\sqrt{N}}.$$

Its ratio to $L^{-1/2}$ is bounded by $Ce^{2aL}N^{-1/4}L^{7/4}$, because $D \le C\sqrt{N}$. This is uniformly bounded as $2a < 1/4$. Thus $\mathbb{E}W_1(\widehat{G}, G) \le C\{L^{-1/2} + A/\sqrt{N}\}$. Bounded N is absorbed by the elementary bound in the next paragraph.

If $A > \sqrt{N}$, the empirical-posterior coupling gives $W_1(\widehat{G}, q) \le 1$. Also coupling the true mixing law to its Gaussian observation law costs at most $\mathbb{E}|Z|$. Equation (V.5.7) therefore yields

$$\mathbb{E}W_1(\widehat{G}, G) \le 1 + \mathbb{E}|Z| + C(A+1)/\sqrt{N} \le CA/\sqrt{N}.$$

This proves the upper bound throughout the claimed range.

For the nonparametric lower bound, use the Chebyshev-arcsine pair from Lemma V.A.3 on $[-M, M]$, with $M = \sqrt{L}/8$ and $k = \lceil 20L \rceil$. Its support lies in $[-A, A]$, its moments through degree $k-1$ agree, and its W_1 separation is at least $M/(\pi k) \ge c/\sqrt{L}$. The Gaussian identity

$$\int_{\mathbb{R}} \frac{\varphi(x-\lambda)\varphi(x-\mu)}{\varphi(x)}\, dx = e^{\lambda\mu}$$

gives

$$\int \frac{(p_{F_k^+} - p_{F_k^-})^2}{\varphi} \le 4\sum_{j\ge k} \frac{M^{2j}}{j!} = 4\sum_{j\ge k} \frac{(L/64)^j}{j!} \le Ce^{-40L}.$$

Predictive TV and hence the TV of the N-sample experiments are bounded away from one. Le Cam's inequality proves the $c/\sqrt{L}$ lower bound. Finally, endpoint mixing laws on $\{-A, A\}$ whose weights differ by $c/\sqrt{N}$ give the lower bound $cA/\sqrt{N}$, by the Bernoulli KL calculation and data processing from the latent experiment. Taking the maximum, equivalent to the displayed sum up to a factor two, completes the proof of Theorem V.5.1.

The distinction between constrained and unrestricted fits is essential to this proof. At a boundary atom of a constrained fit, posterior mean equality can fail, so equation (V.5.2) cannot simply be asserted. The result above genuinely concerns the unrestricted estimator and does not settle that separate small-support problem. Its mechanism is the transfer of the likelihood contact principle through a second-order Taylor cancellation, followed by a binomial inverse at a different scale.

V.6 A sharp comparison for fixed skeletons and their dense inflations

This section extends the sharp asymptotic comparison on block graphs to dense inflations of arbitrary fixed connected skeletons, including skeletons with cycles. It does not settle the sharp constant for all connected graphs. Both the finite skeleton inequality and the inflation limit are proved below without invoking the earlier block-graph classification. The external motivating problem is the comparison of determinant-optimal and total-resistance-optimal graph designs in [13]; the constant studied here is a continuation of that problem, not a constant explicitly conjectured in that paper. The convex effective-resistance design framework is developed in [48].

A design on a connected simple graph G with n vertices consists of conductances $w_e \ge 0$ with $\sum_e w_e = 1$ and connected positive support. For an arbitrary orientation, write $b_e = e_u - e_v$ and $L(w) = \sum_e w_e b_e b_e^{\mathsf{T}}$. The D-optimal design maximizes $\det(L(w)|_{\mathbf{1}^\perp})$, and the A-optimal design minimizes

$$K_G(w) = \sum_{u<v} R_{uv}(w) = n \operatorname{tr} L(w)^\dagger.$$

Write $\mathfrak{a}(G) = K_G(w^D)/K_G(w^A)$. Disconnected supports have infinite resistance objective.

V.6.1 A weighted flow inequality with an exact equality case

Fix a connected simple graph $H = (V, E)$ on $h \geq 2$ vertices, masses $\alpha_i > 0$ with $\sum_i \alpha_i = 1$, and lengths $\ell_e > 0$. For conductances c define

$$\mathcal{K}_\alpha(c) = \sum_{i<j} \alpha_i \alpha_j R_{ij}(c), \qquad K_\ell = \mathcal{K}_\alpha((1/\ell_e)_{e\in E}),$$

and set

$$A_\ell = \min_{\substack{c_e \geq 0 \\ \sum_e \ell_e^2 c_e = 1}} \mathcal{K}_\alpha(c). \tag{V.6.1}$$

Theorem V.6.1 (Sharp skeleton inequality)**.** *For every such H, α, and ℓ,*

$$K_\ell \leq \frac{1}{2}\sqrt{A_\ell}. \tag{V.6.2}$$

Equality holds if and only if $H = K_2$ and its two masses are both $1/2$. The length of the single edge is unrestricted.

Proof. Orient H and let B be its vertex-edge incidence matrix. Index commodities by unordered pairs $i < j$. A feasible simultaneous flow $F = (F_e^{ij})$ satisfies

$$BF^{ij} = \sqrt{\alpha_i \alpha_j}(e_i - e_j).$$

The row F_e is a vector in $\mathbb{R}^{\binom{h}{2}}$. Thomson's principle, summed over commodities, gives

$$\mathcal{K}_\alpha(c) = \min_{F \text{ feasible}} \sum_e \frac{\|F_e\|^2}{c_e}, \tag{V.6.3}$$

where a zero conductance requires a zero flow on that edge.

Let F^0 be the electrical flow for conductances $1/\ell_e$. For a fixed edge $e = (u, v)$ put $z = L(1/\ell)^\dagger(e_u - e_v)$. Reciprocity yields

$$F_e^{0,ij} = \frac{\sqrt{\alpha_i \alpha_j}}{\ell_e}(z_i - z_j).$$

The weighted variance identity therefore gives

$$\|F_e^0\|^2 = \frac{1}{\ell_e^2}\left(\sum_i \alpha_i z_i^2 - \left(\sum_i \alpha_i z_i\right)^2\right). \tag{V.6.4}$$

The maximum principle places every z_i between z_u and z_v; their difference is the effective resistance R_e. For a probability distribution on an interval of length R_e, the variance is at most $R_e^2/4$. For completeness, after translating the interval to $[0, R_e]$, the inequality $x^2 \leq R_e x$ bounds the variance by $\mu(R_e - \mu) \leq R_e^2/4$. The direct edge has resistance ℓ_e, so $R_e \leq \ell_e$. Consequently

$$\|F_e^0\| \leq \frac{R_e}{2\ell_e} \leq \frac{1}{2}. \tag{V.6.5}$$

Moreover $R_e < \ell_e$ when e is not a bridge, since the graph with that edge removed then supplies a positive parallel conductance.

For every feasible F, each column of $F - F^0$ is a circulation. Its scalar product with an electrical potential gradient is zero. Thus Thomson orthogonality, followed by (V.6.5), gives

$$K_\ell = \sum_e \ell_e \langle F_e^0, F_e\rangle \leq \frac{1}{2}\sum_e \ell_e \|F_e\|. \tag{V.6.6}$$

For a fixed feasible F, Cauchy–Schwarz gives

$$\left(\sum_e \ell_e \|F_e\|\right)^2 \le \left(\sum_e \frac{\|F_e\|^2}{c_e}\right)\left(\sum_e \ell_e^2 c_e\right).$$

Equality is attained by $c_e = \|F_e\|/(\ell_e \sum_f \ell_f \|F_f\|)$, allowing $c_e = 0$ when $F_e = 0$. Hence

$$\sqrt{A_\ell} = \min_{F \text{ feasible}} \sum_e \ell_e \|F_e\|. \tag{V.6.7}$$

This minimum is attained: the feasible set is a nonempty closed affine set, and the displayed positive weighted norm has compact sublevel sets. The support of any feasible flow is connected, since all pair demands are positive. In particular, the minimizing conductances above are an admissible connected design. Taking the minimum in (V.6.6) proves the inequality.

Suppose equality holds and let F^* minimize (V.6.7). Every edge with $F_e^* \neq 0$ must have $\|F_e^0\| = 1/2$; otherwise (V.6.6) is strict. Each used edge is therefore a bridge of H. Since these edges form a connected spanning support, H must itself be a tree. For a tree edge the potential z is constant on either side of its cut, so (V.6.4) says $\|F_e^0\|^2 = a_e(1-a_e)$, where a_e is the mass on one side. Every edge is used and thus every cut has mass $1/2$ on either side. If a tree has at least three vertices, two of its leaves would already have mass $1/2$ each, contradicting the strictly positive mass on a third vertex. Therefore $H = K_2$, with equal masses. Conversely, for one edge of length ℓ and equal masses, $K_\ell = \ell/4$ and $A_\ell = \ell^2/4$. □

V.6.2 Inflating a fixed skeleton

Fix H, α, and ℓ as above, and a scale $s > 0$. Construct G_n by replacing vertex i of H with a clique $K_{m_i(n)}$, designating one of its vertices as the root, and replacing each skeleton edge e with an internally disjoint path of $p_e(n)$ edges between the corresponding roots. Assume

$$n = \sum_i m_i(n) + \sum_e (p_e(n) - 1), \qquad m_i(n) = \alpha_i n + O(\sqrt{n}), \qquad p_e(n) = s\ell_e\sqrt{n} + O(1). \tag{V.6.8}$$

Such integer choices exist for all sufficiently large n, by rounding the path lengths and distributing their $O(\sqrt{n})$ interior vertices among the clique-size corrections.

Theorem V.6.2 (Exact inflation limit)**.** *For every family satisfying* (V.6.8)*,*

$$K_{G_n}(w^D) = sK_\ell n^{7/2} + o(n^{7/2}), \tag{V.6.9}$$

$$K_{G_n}(w^A) = n^3\left(\frac{1}{\sqrt{2}} + s\sqrt{A_\ell}\right)^2 + o(n^3). \tag{V.6.10}$$

In particular,

$$\lim_{n\to\infty} \frac{\mathfrak{a}(G_n)}{\sqrt{n}} = \frac{sK_\ell}{(1/\sqrt{2} + s\sqrt{A_\ell})^2}. \tag{V.6.11}$$

The maximum over $s > 0$ equals

$$\frac{K_\ell}{2\sqrt{2}\sqrt{A_\ell}} \le \frac{1}{4\sqrt{2}}, \tag{V.6.12}$$

and is attained uniquely at $s = (2A_\ell)^{-1/2}$. Equality in the final bound holds precisely for the balanced two-vertex skeleton. In that case the joining path has $p(n) \sim \sqrt{2n}$ edges.

Proof. Write $d = n - 1$, and suppress n from m_i and p_e. Call the graph of roots and subdivided paths the central graph.

The determinant-optimal design. The spanning-tree polynomial factors across the cliques attached at roots and the central graph. The determinant is n times this polynomial. Its central factor has the identity

$$\tau_{\mathrm{cen}}(w) = \left(\prod_{e\in E}\prod_{j=1}^{p_e} w_{ej}\right) \sum_{T\in\mathcal{T}(H)} \prod_{e\notin T} \left(\sum_{j=1}^{p_e} w_{ej}^{-1}\right). \tag{V.6.13}$$

Indeed, a spanning tree contains every segment of each path corresponding to an edge of some $T \in \mathcal{T}(H)$, and deletes exactly one segment from every other path. The identity is first written for positive weights and extends as a polynomial identity.

The logarithmic determinant is strictly concave as a function of $L|_{\mathbf{1}^\perp}$. For a simple graph the edge weights are determined by the off-diagonal entries of L, so the maximizing weights are unique. Formula (V.6.13) is invariant under arbitrary permutations of the segments of any one path. Therefore its D-optimal segments all have the same weight w_e^D. This common weight is positive, since $p_e \geq 2$ for large n and deleting the entire path would isolate its interior vertices. The same uniqueness and the complete-graph factor show that all edges of each attached clique have equal positive weight.

The derivative of the logarithmic determinant with respect to w_f is R_f. The budget multiplier is d by homogeneity; hence $R_f = d$ on every positive edge. A clique K_m with constant edge conductance a has pair resistance $2/(ma)$. Since it attaches at one vertex, other parts of the graph do not change this internal resistance. Thus its edge weights are exactly $2/(md)$ and its pair resistances are d.

In a weighted random spanning tree, the inclusion probability of an edge f is $w_f R_f = dw_f$. A tree omits at most one segment of any central path, and all its segments are interchangeable under (V.6.13). Consequently

$$\frac{1-1/p_e}{d} \leq w_e^D \leq \frac{1}{d}. \tag{V.6.14}$$

After series reduction, the root network has conductances $w_e^D/p_e = (1+o(1))/(sd\sqrt{n}\,\ell_e)$, uniformly over the fixed edge set. Its root resistances therefore satisfy

$$R_{ij}^{\mathrm{cen}}(w^D) = sd\sqrt{n}\,[R_{ij}(1/\ell) + o(1)].$$

Pairs in distinct cliques contribute a central-resistance term

$$\sum_{i<j} m_i m_j R_{ij}^{\mathrm{cen}}(w^D) = sK_\ell n^{7/2} + o(n^{7/2}).$$

All access resistances inside cliques are $O(n)$, so their total contribution is $O(n^3)$. There are $O(\sqrt{n})$ central interior vertices and hence $O(n^{3/2})$ pairs involving at least one such vertex. By (V.6.14), the resistance of a segment is at most $2d$ for large n. Connecting vertices along the fixed skeleton uses $O(\sqrt{n})$ segments, so every relevant pair resistance is $O(n^{3/2})$. These remaining contributions are also $O(n^3)$. This proves (V.6.9).

The resistance-optimal design. Resistance adds across a graph joined at a single vertex. In the objective of an attached clique K_m, its root has projected mass $b = n - m + 1$, and its other $m-1$ vertices each have mass one. Permutation averaging over the latter vertices does not increase the objective, by convexity of effective resistance in conductances. Thus write its root-to-other edge weights as a and its other edge weights as b'; put $\lambda = a + (m-1)b'$. The two nonzero Laplacian modes give the objective and budget

$$\frac{b}{a} + \frac{(m-2)n}{\lambda}, \qquad \frac{ma + (m-2)\lambda}{2} = t.$$

Cauchy–Schwarz shows that the minimum is q_m^2/t, where

$$q_m = \frac{\sqrt{m(n-m+1)} + (m-2)\sqrt{n}}{\sqrt{2}}. \tag{V.6.15}$$

The equality allocation has $\lambda/a = \sqrt{mn/(n-m+1)} \geq 1$, so $b' \geq 0$ and the minimum is feasible. All $m_i \geq 3$ for large n; the same formula also holds for $m = 2$ by the single-edge calculation. From (V.6.8),

$$\sum_i q_{m_i} = \frac{n^{3/2}}{\sqrt{2}} + O(n). \tag{V.6.16}$$

Let J_n be the minimum central resistance objective with unit central budget, assigning mass m_i to root i and mass one to every central interior vertex. The articulation decomposition and an optimal allocation of the total budget give the exact identity

$$K_{G_n}(w^A) = \left(\sum_i q_{m_i} + \sqrt{J_n}\right)^2. \tag{V.6.17}$$

Here one uses $\min_{t_j \geq 0, \sum t_j = 1} \sum_j a_j^2/t_j = (\sum_j a_j)^2$ for the clique components and the central component. It remains to prove

$$J_n/n^3 \longrightarrow s^2 A_\ell. \tag{V.6.18}$$

For a lower bound, discard pairs involving central interior vertices. If a path has segment weights w_{ej} and total budget t_e, its equivalent conductance is at most t_e/p_e^2, by the harmonic-arithmetic mean inequality, with equality for equal weights. Reducing all paths therefore leaves skeleton conductances $\widehat{c}_e$ with $\sum_e p_e^2 \widehat{c}_e \leq 1$. Put $c_e = n\widehat{c}_e$ and $\beta_i = m_i/n$. The root-only objective divided by n^3 is $\mathcal{K}_\beta(c)$, and

$$\sum_e (p_e/\sqrt{n})^2 c_e \leq 1.$$

The optimum of this finite skeleton problem converges to $s^2 A_\ell$. To see this without assuming an interior optimizer, observe that the pair coefficients $\beta_i\beta_j/(\alpha_i\alpha_j)$ and the cost coefficients $(p_e/\sqrt{n})^2/(s^2\ell_e^2)$ all converge to one uniformly over the fixed finite sets. Multiplicative comparison of objectives and budget sets, followed by homogeneity, squeezes their optima. This proves the required lower limit.

For the upper limit, fix any positive skeleton design c with $\sum_e \ell_e^2 c_e = 1$. Give path e budget $t_e = \ell_e^2 c_e$ and distribute it equally over its segments. Then its equivalent conductance is t_e/p_e^2 and the root-only objective divided by n^3 converges to $s^2\mathcal{K}_\alpha(c)$. Every central pair resistance is $O_c(n)$: a whole path has resistance $p_e^2/t_e = O_c(n)$ and the skeleton is fixed and connected. The weighted number of pairs involving an interior vertex is $O(n^{3/2})$, so their contribution is $O_c(n^{5/2}) = o(n^3)$. Finally, positive designs approach the optimum in (V.6.1): mix an optimal connected design with any fixed positive design and let the mixture parameter tend to zero. Effective resistances are continuous on connected supports. Taking first $n \to \infty$ and then this parameter to zero proves the upper limit in (V.6.18). Equations (V.6.16)–(V.6.18) prove (V.6.10).

The ratio limit follows. For $a, b > 0$, the function $s/(a+bs)^2$ has its unique maximum $1/(4ab)$ at $s = a/b$. Use $a = 1/\sqrt{2}$, $b = \sqrt{A_\ell}$, and Theorem V.6.1. In its equality case, $A_\ell = \ell^2/4$ and $s\ell = \sqrt{2}$. □

Remark V.6.3 (What is and is not closed). For every fixed simple skeleton with strictly positive fixed vertex masses, the optimized leading coefficient in (V.6.11) is at most $1/(4\sqrt{2})$. It is strictly smaller unless the skeleton is the balanced single edge. This includes cyclic skeletons whose dense inflations are not block graphs. The strictness is for each fixed choice; it is not a uniform gap when some masses tend to zero. This result does not cover a skeleton whose size grows with n,

arbitrary non-clique dense components, or general multiscale graph sequences. In particular it does not improve the previously obtained universal upper bound $\mathfrak{a}(G) \le \sqrt{n/8} + 1/2$, and the factor of two between that bound and the sharp constructed lower constant $1/(4\sqrt{2})$ remains open for arbitrary connected graphs. The quoted universal upper bound is background from the earlier general-graph comparison; it is not used in either proof here.

V.7 Boundaries of the results and next questions

The three algebraicity classifications are complete within their stated domains. They do not imply that every special value of a transcendental function is transcendental. In particular, the hypercubic theorem concerns the minimal differential equation in the energy variable, and does not classify all differential Galois groups or all arithmetic special values. The Franel theorem does not resolve the minimum recurrence width at every weight. For beta quantiles the finite rational tests are exact for every half-integer pair, but a closed list of all asymmetric pairs passing those tests is not proved.

The two additional transfers also have exact boundaries. The Gaussian theorem covers $A \ge \sigma\sqrt{\log(eN)}$ with absolute constants, not the whole small-support parameter plane. The graph theorem covers fixed skeletons with positive fixed masses and lengths; a growing skeleton or a different dense component can require new arguments. The universal all-graph leading constant remains undetermined.

Independent internal readers examined the principal arguments, including the Franel amplitude obstruction, the Fourier endpoint issue, the cyclotomic converse construction, the Gaussian inverse inequality, and the skeleton equality case. Exact symbolic checks and compilation checks were used where useful. These checks support the draft but do not replace external mathematical review. Related literature and known special cases are identified in each section; in particular, the full text of the recent incomplete-beta arithmetic paper was not available for a complete overlap assessment. The manuscript is therefore presented as a set of explicit proofs for scrutiny, without a claim of having resolved a verified famous open conjecture or attained a specified publication tier.

V.A Supporting estimates for the Gaussian theorem

The following estimates make the Gaussian proof independent of the preceding Poisson manuscript.

Lemma V.A.1 (Degree-elevated Bernstein coefficients)**.** *If P is a polynomial of degree $k \le m/2$ and*

$$P(x) = \sum_{j=0}^{m} b_j \binom{m}{j} x^j (1-x)^{m-j},$$

then

$$\max_j |b_j| \le C e^{2k^2/m} \|P\|_{[0,1]}.$$

Proof. Put $M = \|P\|_{[0,1]}$ and $F(z) = (1+z)^m P(z/(1+z))$; its jth coefficient is $\binom{m}{j} b_j$. The maximum principle in the inverse Joukowski variable gives $|P((2+w+w^{-1})/4)| \le M|w|^k$ for $|w| \ge 1$. For $0 < j < m$ take $r = j/(m-j)$, $v = r/(1+r)^2$, and $u = 2\sqrt{v}\,|\sin(\phi/2)|$. Direct inversion of the Joukowski map shows that the larger modulus associated with $re^{i\phi}/(1+re^{i\phi})$ is $(1+u)/\sqrt{1-u^2}$. Thus

$$\frac{|F(re^{i\phi})|}{M(1+r)^m} \le (1-u^2)^{(m-k)/2}(1+u)^k \le \exp(2k^2/m)\exp(-mu^2/8).$$

For the final inequality, use $k \le m/2$, $\log(1+u) \le u$, $\log(1-u^2) \le -u^2$, and complete the square in $ku - mu^2/8$. The limiting point $u = 1$ is interpreted by continuity. The average of $\exp(-mu^2/8)$ over ϕ is at most $C/\sqrt{1+mv}$, by comparison with a Gaussian integral near zero. Stirling's factorial bounds give

$$\binom{m}{j}\left(\frac{j}{m}\right)^j\left(1-\frac{j}{m}\right)^{m-j} \ge \frac{c}{\sqrt{1+mv}}.$$

Cauchy's coefficient formula cancels these two square-root factors and proves the claim. The endpoint coefficients are $P(0)$ and $P(1)$. □

Lemma V.A.2 (A finite likelihood-envelope bound)**.** *Let p be a probability density and let $u_1, \dots, u_M$ be nonnegative envelopes with total masses $U_a \in [1, 1+\eta]$. Suppose that, on an event E, a fitted density $\widehat{p}$ is dominated by some u_a and satisfies $\prod_i \widehat{p}(X_i) \ge \prod_i p(X_i)$, where the X_i are independent with density p. Then, for $x \ge 0$,*

$$\mathbb{P}\left(E \cap \left\{h(p,\widehat{p}) > \sqrt{2\eta} + \sqrt{\eta + 2(\log M + x)/N}\right\}\right) \le e^{-x},$$

and $\mathbb{E}[\mathbf{1}_E h(p,\widehat{p})^2] \le C\eta + C(\log M + 1)/N$.

Proof. Put $r_a = u_a/U_a$. If $u_a \ge \widehat{p}$, then $\|r_a - \widehat{p}\|_1 \le 2\eta$, so $h(r_a, \widehat{p}) \le \sqrt{2\eta}$, and the likelihood comparison implies $\prod_i r_a(X_i)/p(X_i) \ge e^{-N\eta}$. The expectation of the square root of this product is the Nth power of the affinity, at most $e^{-Nh(p,r_a)^2/2}$. Markov's inequality and a union bound over the M deterministic envelopes prove the tail inequality. Integrating it proves the second assertion. The event E is only intersected with the bad event; no conditioning or independence assertion on E is used. □

Lemma V.A.3 (A quantitative moment-matching pair)**.** *For every interval $[a-M, a+M]$ with $M > 0$ and integer $k \ge 1$, there are probability laws F_k^+, F_k^- on that interval which match moments of every degree less than k and satisfy*

$$W_1(F_k^+, F_k^-) \ge \frac{M}{\pi k}.$$

Proof. Push the measures $(1 \pm \frac{1}{2}T_k(x))\,dx/(\pi\sqrt{1-x^2})$ on $[-1,1]$ forward by $x \mapsto a + Mx$. Positivity, normalization and moment matching follow from $|T_k| \le 1$ and Chebyshev orthogonality. With $x = \cos\theta$, their CDF difference is $\pm\sin(k\theta)/(\pi k)$, with a sign depending only on the orientation. Therefore

$$W_1(F_k^+, F_k^-) = \frac{M}{\pi k}\int_0^\pi |\sin(k\theta)|\sin\theta\,d\theta \ge \frac{M}{\pi k},$$

because $|\sin u| \ge \sin^2 u$ and $\int_0^\pi \sin^2(k\theta)\sin\theta\,d\theta = 1 + (4k^2-1)^{-1}$. □

Part VI
Global Galois Groups and Complete Parameter Classifications

Manuscript overview. We develop six extensions of earlier proof methods, with explicit attention to exceptional parameters. For anisotropic hypercubic periods, the connected differential Galois group in dimension three is exactly SO_3, SO_4, or SO_5, according to the multiplicities of the squared weights. In every even dimension the full group is symplectic for every nonzero complex weight tuple. A unique extremal exponential factor supplies a transvection even when other thresholds collide; the same argument removes the threshold restriction from the odd-propagator banana Galois theorem, using the previously established all-mass order theorem. In arithmetic, we give a closed classification of algebraic quantiles at algebraic probability levels for every strict half-integer beta distribution. A prime-denominator obstruction eliminates all but finitely many shape rows. For cubic weighted binomial sums, local monodromy rank retains the multiplicity of colliding singularities and proves the minimum recurrence width for every nonzero complex weight. Finally, the ordinary constrained Gaussian mixture NPMLE is minimax, up to universal constants, simultaneously for all sample sizes, support radii, and noise scales. A boundary posterior drift estimate joins the small-support Hermite regime to the large-support transport regime. Proofs, supporting estimates, and the precise dependencies on preceding work are included.

Integration note. The remaining odd hypercubic and even-propagator Galois cases are treated in Part VII. The cubic-weight recurrence theorem is generalized there and then to all arithmetic subsequences in Part VIII. The all-scale Gaussian and exceptional half-integer quantile results remain distinct.

VI.1 The questions and the changes in scope

The organizing question is whether a method that succeeds for generic parameters can control its exceptional set. The extensions here address three different obstructions: loss of information when singularities collide, an arithmetic test that remains indexed by infinitely many integer pairs, and a likelihood argument that fails at a support boundary. They are related by a proof strategy, not by a common underlying model.

Question	Earlier scope or remaining restriction	Extension proved here
Three-dimensional lattice periods	A public announcement gives the connected SO_5 result for very general directions.	Exact connected groups for every nonzero complex triple; all distinct squared weights give SO_5.
Even-dimensional lattice periods	The preceding all-weight theorem determines order and algebraicity, but not the full group.	The full group is Sp_R for every even dimension and every nonzero complex tuple.
Odd-propagator banana periods	Manuscript [032] assumes distinct, nonzero squared signed thresholds for its group theorem.	The full symplectic group holds without either threshold condition.
Half-integer beta quantiles	Arithmetic reduction leaves a test depending on each integer shape pair.	A closed list of all exceptional pairs, with no unbounded search.
Cubic weighted sums	Distinct-singularity counting fails at $\lambda = 8, 1/8$.	Exact minimum width for every $\lambda \neq 0$, including both collisions.
Gaussian mixing recovery	Small support, growing support, and boundary-constrained fits require different arguments.	One uniform minimax curve for the actual constrained NPMLE over all $(N, A/\sigma)$.

The principal differential-algebraic advance. The Bessel–Fourier method from the supplied period manuscript [49] and the preceding hypercubic work is combined with reflected Poincaré duality. In even dimension, all relevant local monodromies are unipotent. An extremal signed sum is unique regardless of collisions elsewhere, and gives one transvection. Connectedness, irreducibility, and that transvection force the full symplectic group. In dimension three, the monodromy block $[3, 1, 1]$ rules out every proper connected orthogonal subgroup acting irreducibly in dimension five. The resulting obstruction to elliptic expressions is functional; it says nothing about reduction of an isolated numerical value.

The arithmetic and recurrence advances. The beta classification combines the earlier logarithmic-transcendence reduction with a new uniform exclusion argument. A single prime in a specified interval produces a unique denominator of negative p-adic valuation. Nagura's theorem and a one-sided variance bound cover all large first exponents, and exact rational signs handle the remaining rows. For cubic sums, the replacement for distinct-singularity counting is the rank of local monodromy: a collided rank-two action requires at least a double zero of every scalar leading coefficient.

The statistical advance. The likelihood-contact and transport method from the preceding mixture work [30, 31] is combined with Hermite inversion. At an endpoint of the known support interval, posterior centering can fail. Monotonicity of the endpoint posterior weight bounds its total drift, so the same likelihood estimator can be controlled across the whole support-to-noise range. Matching testing lower bounds are uniform in that range. The interval center and radius are known; the statement does not assert unrestricted adaptation to an unknown tiny interval.

Dependencies and unresolved questions. Appendix VI.A reproduces the preceding hypercubic rank proof; it is supporting material, not a second new claim. The banana extension uses the precise irreducibility, rank, pairing, and infinity-monodromy inputs of [032], listed in Section VI.4; it does not reprove that manuscript in full. The other new classifications and the Gaussian rate are proved here with their stated classical inputs. The public Watson summary is an announcement, not an available complete proof against which historical priority can be settled. The beta and cubic classifications close explicit parameter questions, not identified named conjectures from the external literature. In particular this draft does not claim to resolve the Schneider-type question about general incomplete beta integrals.

VI.2 The connected Galois group in dimension three, without exceptional directions

Let $m_1, m_2, m_3 \in \mathbb{C}^{\times}$. Put

$$F_m(z) = \sum_{k\geq 0} (2k)! \left(\sum_{a_1+a_2+a_3=k} \prod_{i=1}^{3} \frac{m_i^{2a_i}}{a_i!^2} \right) z^k, \qquad H_m(s) = s^{-1} F_m(s^{-2}).$$

For $\operatorname{Re} s > 2\sum_i |m_i|$, the latter also equals $\int_0^\infty e^{-sx} \prod_i I_0(2m_i x)\, dx$. The differential Galois group of a function means that of its minimal scalar differential equation, acting on its solution space.

Theorem VI.2.1 (Complete three-dimensional connected-group classification)**.** *Set $r = 3, 4, 5$ according as the number of distinct values among m_1^2, m_2^2, m_3^2 is $1, 2, 3$. The differential Galois groups of H_m over $\mathbb{C}(s)$ and F_m over $\mathbb{C}(z)$ have identity component $\mathrm{SO}_r(\mathbb{C})$ in its standard representation. In particular, for every pairwise-distinct triple of nonzero squared complex weights the connected group is SO_5. No condition on collisions or vanishing of signed mass sums is imposed. The minimal order remains r after every finite algebraic extension of the respective coefficient field.*

The public announcement of Elkies, Grimm and Schwartz [39] states the connected SO_5 result for very general directions, outside the equal-rate walls and a further countable union of algebraic curves. The theorem above removes this exceptional-direction restriction using the all-weight rank and local-monodromy theorem. It does not claim priority for the generic result, a new closed formula for the Watson integral, or information about isolated special values of the energy variable.

VI.2.1 The precise previously established input

We use the following specialization of the all-weight hypercubic rank theorem proved in the preceding work. Its hypotheses and its geometric realization are recorded because irreducibility of a fifth-order operator alone would not suffice here. Let B_μ be the rank-two Bessel connection

$$(\theta_x^2 - 4\mu^2 x^2) y = 0, \qquad \theta_x = x\partial_x.$$

If the distinct squared masses have multiplicities n_j, put $M = \bigotimes_j \operatorname{Sym}^{n_j} B_{\mu_j}$ and, over $K = \mathbb{C}(s)$, put $N_s = M \otimes e^{-sx}$. The connection in s on this last module is $\partial_s - x$. The cyclic differential module of H_m is the middle cohomology

$$\mathcal{P}_m = H^1_{\mathrm{mid}}(\mathbb{G}_{m,K}, N_s) = \operatorname{im}\{H^1_c(N_s) \longrightarrow H^1(N_s)\}. \tag{R}$$

It is irreducible over K. For three masses its rank and its unipotent local monodromy T_∞ at $s = \infty$ are

squared-mass multiplicities	$\bigotimes_j \operatorname{Sym}^{n_j} \mathbb{C}^2$	$\operatorname{rank} \mathcal{P}_m$	Jordan blocks of T_∞
$[3]$	Sym^3	3	$[3]$
$[2,1]$	$\operatorname{Sym}^3 \oplus \operatorname{Sym}^1$	4	$[3,1]$
$[1,1,1]$	$\operatorname{Sym}^3 \oplus 2\operatorname{Sym}^1$	5	$[3,1,1]$.

(R3)

The middle term is a decomposition for the diagonal SL_2. The rank theorem follows from the product Bessel Galois group, intermediate extension, and Fourier stationary phase; in particular a block of size $d+1$ at the original zero contributes a block of size d at the transformed infinity. The equality in (R), including visibility of this particular unilateral Laplace integral and removal of its endpoint terms, is part of that theorem. Thus it is not an assumption about a putative generic Picard–Fuchs equation.

VI.2.2 A reflected Poincaré pairing

The next construction works in every dimension and for every squared-mass multiplicity pattern. We give it directly rather than infer the sign of a form from a Fourier-transform convention.

Lemma VI.2.2 (A form for every choice of weights)**.** *For n nonzero complex weights, the middle connection $\mathcal{P}_m$ has a nondegenerate horizontal bilinear form of parity $(-1)^{n+1}$. Here parity $+1$ means symmetric. Thus its Galois group is contained in an orthogonal group for odd n and a symplectic group for even n.*

Proof. In the Bessel basis $b_\mu, \theta_x b_\mu$, the form $B_\mu(b_\mu, \theta_x b_\mu) = 1$ is alternating, nondegenerate and horizontal. Indeed, $\theta_x^2 b_\mu = 4\mu^2 x^2 b_\mu$ makes its connection matrix infinitesimally symplectic. Symmetric powers and tensor products give M a horizontal nondegenerate form B of parity $\epsilon = (-1)^n$.

Let $\iota : x \mapsto -x$. Its canonical action fixes each Bessel basis vector $b_\mu, \theta_x b_\mu$; the defining matrix is even in x. This identifies $\iota^* M$ with M and preserves B. Consequently

$$\iota^* N_s = M \otimes e^{sx} \simeq N_s^\vee,$$

compatibly both with the x-connection and with the s-connections $\partial_s + x$ on the left and the dual of $\partial_s - x$ on the right. Curve Poincaré duality therefore supplies a perfect pairing of $\mathcal{P}_m$ with itself. The use of the middle image makes this pairing nondegenerate: the annihilator of the image of $H^1_c \to H^1$ is the kernel of the dual map, so the perfect compact-support pairing descends to the two images. This is the same duality-and-involution construction used for exponential connections in [51, §2, especially (2.12) and Corollary 2.15].

For completeness, its sign can be checked by residues. Represent $a, b \in \mathcal{P}_m$ by N_s-valued one-forms α, β, with formal primitives f_a, f_b at 0 and ∞. The sum of local residues gives

$$P(a,b) = \sum_{v=0,\infty} \operatorname{Res}_v B(f_a, \iota^*\beta). \tag{P}$$

Local exactness at both points is precisely the middle condition; at infinity the twisted connection has no local cohomology. Formula (P) is independent of the choices by the compact-support duality just described. Pullback by ι preserves the sum of residues, and ι is an isometry for B. Hence

$$P(b,a) = \sum_v \operatorname{Res}_v B(\iota^* f_b, \alpha) = \epsilon \sum_v \operatorname{Res}_v B(\alpha, \iota^* f_b).$$

The one-form

$$dB(f_a, \iota^* f_b) = B(\alpha, \iota^* f_b) + B(f_a, \iota^*\beta)$$

has zero residues. It follows that $P(b,a) = -\epsilon P(a,b)$. Residues commute with ∂_s, and the coefficient pairing is horizontal for the two commuting parameter connections, so P is horizontal. This proves the assertion, including nondegeneracy and its sign. □

VI.2.3 Excluding decomposition on a finite cover

Lemma VI.2.3. *Let $G \subseteq \mathrm{GL}(V)$ be an algebraic group acting irreducibly, with $\dim V = r \in \{3,4,5\}$. Suppose G contains a unipotent whose Jordan partition is $[3, 1^{r-3}]$. Then G^0 acts irreducibly on V.*

Proof. The unipotent radical U of G^0 is characteristic in G^0 and therefore normal in G. A unipotent group has a nonzero fixed vector. Its fixed space V^U is G-invariant, hence equals V. Faithfulness gives $U = 1$, so G^0 is reductive and its representation on V is completely reducible.

The isotypic components for G^0 are permuted transitively by G: otherwise the sum over a proper orbit would contradict irreducibility. If their number is t, their common irreducible dimension is d, and their common multiplicity is e, then $r = tde$. Every unipotent element of G belongs to G^0, since the Zariski closure of its cyclic subgroup is the connected one-parameter unipotent group obtained by exponentiating its logarithm. On an isotypic component such an element acts as $A \otimes 1_e$, so its Jordan blocks all have sizes at most d. The block of size three forces $d \geq 3$. For $r = 3, 4, 5$, the equality $r = tde$ then forces $t = e = 1$ and $d = r$. Thus V is irreducible under G^0. □

This argument deals explicitly with both permutation of components and repeated copies of one component. In particular, it excludes imprimitive finite-cover explanations of the order-five equation.

VI.2.4 The small orthogonal representations

Lemma VI.2.4. *Let $G \subseteq \mathrm{O}_r(\mathbb{C})$ act irreducibly, where $r \in \{3, 4, 5\}$, and contain a unipotent of partition $[3, 1^{r-3}]$. Then $G^0 = \mathrm{SO}_r(\mathbb{C})$.*

Proof. By Lemma VI.2.3, G^0 is reductive and acts irreducibly. Its center acts by scalars by Schur's lemma; the only scalar isometries are ± 1. Its Lie algebra $\mathfrak{g}$ is therefore semisimple. A maximal torus embeds in one of SO_r, so $\operatorname{rank} \mathfrak{g} \leq 2$.

For $r = 3$, dimension at most $\dim \mathfrak{so}_3 = 3$ leaves only $\mathfrak{sl}_2 = \mathfrak{so}_3$. For $r = 4$, dimension at most six leaves $\mathfrak{sl}_2$ or $\mathfrak{sl}_2 \oplus \mathfrak{sl}_2$. The irreducible four-dimensional $\mathfrak{sl}_2$ representation is $\operatorname{Sym}^3 \mathbb{C}^2$; its unique invariant bilinear form is alternating, so it cannot preserve the symmetric form on V. The other possibility has dimension six and hence equals $\mathfrak{so}_4$.

For $r = 5$, the rank-at-most-two semisimple possibilities of dimension at most ten are

$$A_1, \quad A_1 \oplus A_1, \quad A_2, \quad B_2 = C_2.$$

The type G_2 has dimension fourteen and is too large. A faithful irreducible representation of a product of two simple factors is a tensor product of nontrivial representations, which cannot have prime dimension five. Type A_2 has irreducible dimensions

$$\frac{(a+1)(b+1)(a+b+2)}{2}, \qquad a, b \in \mathbb{Z}_{\geq 0},$$

none equal to five. This formula is the A_2 case of the Weyl dimension formula; the rank-two root systems and that formula are standard highest-weight theory, see [52]. For type A_1, the only five-dimensional irreducible is $\operatorname{Sym}^4 \mathbb{C}^2$. A nonidentity unipotent acts there as a single block of size five: its logarithm is the raising operator on binary quartics, whose nilpotence index is five. Our unipotent has partition $[3, 1, 1]$, a contradiction. Thus $\mathfrak{g}$ is B_2 of dimension ten, so it equals $\mathfrak{so}_5$.

In every case a connected closed subgroup of SO_r with its full Lie algebra is SO_r itself. □

Proof of Theorem VI.2.1. Input (R3) supplies irreducibility and the indicated unipotent for H_m. Lemma VI.2.2, with $n = 3$, supplies the orthogonal form. Lemma VI.2.4 gives the claimed connected group. The change $z = s^{-2}$ is a finite algebraic base change and $H_m = s^{-1} F_m(s^{-2})$ differs by a rational gauge. A finite algebraic base change preserves the identity component of a Picard–Vessiot group [43], so the conclusion holds for F_m as well. Finally, after any further finite algebraic base change the Galois group still contains the standard SO_r, which acts irreducibly. Thus the differential module remains irreducible of rank r and its nonzero solution cannot satisfy a smaller-order scalar equation. □

Corollary VI.2.5 (Exactly one algebraic relation among the first jets)**.** *Suppose m_1^2, m_2^2, m_3^2 are pairwise distinct and nonzero. Then*

$$\operatorname{trdeg}_{\mathbb{C}(s)} \mathbb{C}(s)(H_m, H_m', H_m'', H_m''', H_m^{(4)}) = 4.$$

Their polynomial relations over $\mathbb{C}(s)$ form the principal ideal generated by one nondegenerate quadratic polynomial. For F_m and its first four derivatives the same transcendence degree and single-quadric assertion hold over $\overline{\mathbb{C}(z)}$.

Proof. Choose the cyclic cohomology basis $\omega, \nabla_s\omega, \dots, \nabla_s^4\omega$ in (R), and let $C(s) \in \mathrm{GL}_5(\mathbb{C}(s))$ be the Gram matrix of Lemma VI.2.2. A scalar solution f is a horizontal linear functional in this basis. The dual form gives

$$(f, f', f'', f''', f^{(4)})\, C(s)^{-1} (f, f', f'', f''', f^{(4)})^T = c_f \in \mathbb{C}. \tag{Q}$$

For $f = H_m$ its solution vector is nonzero. The orbit of any nonzero vector under SO_5 has dimension four: it is a nonzero level of a nondegenerate quadratic form, or the punctured isotropic cone. Since the full group lies in O_5, it has the same orbit dimension. Picard–Vessiot Galois correspondence identifies the transcendence degree of the generated solution field with this orbit dimension. The polynomial relation ideal is therefore prime of height one and contains (Q). A nondegenerate quadratic polynomial in five variables, with or without a constant term, is absolutely irreducible: a factorization into affine linear forms would give quadratic rank at most two. Hence (Q) generates the ideal. Finite algebraic extension, the substitution $z = s^{-2}$, and the invertible triangular change between the two derivative vectors preserve these assertions over the algebraic closure. □

Corollary VI.2.6 (A precise obstruction to elliptic expressions)**.** *In the pairwise-distinct squared-mass case, let K' be any finite algebraic extension of $\mathbb{C}(s)$. Let E/K' be a compositum of finitely many Picard–Vessiot fields of rank-two elliptic Gauss–Manin connections, allowing algebraic pullbacks and finite monodromy factors. Then H_m is not algebraic over E. The analogous assertion holds for F_m after a finite algebraic extension of $\mathbb{C}(z)$.*

Proof. The connected Galois group of E is a subgroup of a product of elliptic SL_2 groups. Every simple factor of its semisimple Lie quotient is therefore of type A_1: a simple factor must project nontrivially to some $\mathfrak{sl}_2$ factor, where it injects and hence equals $\mathfrak{sl}_2$.

Suppose H_m were algebraic over E, and let P be the Picard–Vessiot field of its irreducible equation. The compositum EP is Picard–Vessiot over K'. All conjugates of H_m under $\mathrm{Gal}(EP/K')$ are algebraic over E, because E is stable under this group. Restriction to $\mathrm{Gal}(P/K')$ is surjective, so these conjugates span the whole solution space by irreducibility. A finite solution basis and its derivatives are therefore algebraic over E, so EP/E is finite. Restriction from the Galois group of EP to that of E has finite kernel and is surjective, and hence induces an isomorphism of Lie algebras. Restriction to the Galois group of P is surjective. It would follow that $\mathfrak{so}_5$, a simple Lie algebra of type B_2, is a quotient of a Lie algebra whose semisimple factors are all A_1, which is impossible. The finite base change and gauge relating F_m to H_m give the other assertion. □

This includes algebraic operations on finitely many elliptic period systems and their derivatives. It is not a claim about unrestricted compositions of special functions or about a single numerical value of the period.

VI.3 Full symplectic Galois groups in every even dimension

Keep the hypercubic series F_m and its Laplace form H_m, now for an arbitrary even number $n \geq 2$ of nonzero complex weights. Let the distinct squared weights have multiplicities $n_1, \dots, n_q$, and set

$$R = \prod_{j=1}^{q} (n_j + 1) - [u^{n/2}] \prod_{j=1}^{q} (1 + u + \cdots + u^{n_j}).$$

The all-weight order theorem identifies R as the rank of the irreducible middle connection (R), with no conditions on signed sums.

Theorem VI.3.1 (All even dimensions and all nonzero complex weights)**.** *For every even $n \geq 2$ and every $m \in (\mathbb{C}^\times)^n$, the full differential Galois groups of H_m over $\mathbb{C}(s)$ and F_m over $\mathbb{C}(z)$ are $\mathrm{Sp}_R(\mathbb{C})$ in their standard representations. Thus R is even, and for each function $f = H_m$ or F_m the R functions $f, f', \dots, f^{(R-1)}$ are algebraically independent over its coefficient field. These assertions hold even when signed mass sums collide or vanish.*

Orthogonal and symplectic structures for lattice Green differential operators have long been studied; see, for example, [53]. The statement here determines the full group uniformly over every nonzero complex weight tuple in every even dimension, including collided thresholds. No historical-priority claim is inferred from a literature search.

Here Sp_R denotes the group preserving a nondegenerate alternating form on an R-dimensional space. We first isolate the linear-algebra argument that will identify the group.

Lemma VI.3.2 (One transvection in a connected irreducible group)**.** *Let (V, ω) be a finite-dimensional complex symplectic space. If a connected algebraic subgroup $G \subseteq \mathrm{Sp}(V)$ acts irreducibly and contains a nonidentity transvection, then $G = \mathrm{Sp}(V)$.*

Proof. Write $N_u(v) = \omega(u, v)u$. A symplectic transvection is $1 + aN_u$ for some $u \neq 0$ and $a \neq 0$; this follows directly by writing its rank-one difference as $v \mapsto \ell(v)u$ and using the symplectic identity to conclude $\ell \in \mathbb{C}\omega(u, -)$. Rescaling u absorbs a. Its powers are $1 + kN_u$, so their Zariski closure is $\{1 + tN_u : t \in \mathbb{C}\}$. Consequently $N_u \in \mathfrak{g} = \mathrm{Lie}\, G$, as is N_v for every $v \in \mathcal{O} = G \cdot u$. The span of $\mathcal{O}$ is a nonzero invariant space, hence is V.

Join two orbit vectors when their symplectic pairing is nonzero. The spans W_i of different connected components of this graph are mutually orthogonal. Their sum is direct: if a finite sum $\sum_i w_i = 0$ with $w_i \in W_i$, then w_i is orthogonal to W_i by this equality and to all other W_j by construction; it is orthogonal to V, hence zero. Thus there are finitely many components. The group G permutes their spans. The stabilizer of each span is a closed subgroup of finite index, so connectedness makes every span G-invariant. Irreducibility forces a single component. Choose finitely many orbit vectors spanning V and add the finitely many vertices of paths connecting them. This produces a finite connected spanning set $S \subseteq \mathcal{O}$.

For $v, w \in V$ put

$$L_{v,w}(x) = \omega(v, x)w + \omega(w, x)v.$$

The identities

$$[N_v, N_w] = \omega(v, w)L_{v,w}, \qquad [L_{v,w}, N_y] = \omega(v, y)L_{w,y} + \omega(w, y)L_{v,y} \tag{T}$$

follow by direct multiplication. The first gives $L_{v,w} \in \mathfrak{g}$ for edges of the graph, and $L_{v,v} = 2N_v$ is also in $\mathfrak{g}$. Induct on the length of a path from v to y. If $\omega(v, y) \neq 0$, the first identity already applies. Otherwise, let w be the penultimate vertex. By induction $L_{v,w} \in \mathfrak{g}$, and the second identity gives $L_{v,y} \in \mathfrak{g}$ because $\omega(w, y) \neq 0$. Hence all $L_{v,w}$ with $v, w \in S$ belong to $\mathfrak{g}$. Since S spans V, these span $\mathrm{Sym}^2 V \simeq \mathfrak{sp}(V)$. Thus $\mathfrak{g} = \mathfrak{sp}(V)$, and connectedness proves the lemma. □

VI.3.1 Formal factors and collided thresholds

We state exactly the local Fourier input. If a holonomic module $\mathcal{K}$ has formal germ at infinity

$$\widehat{\mathcal{K}}_\infty = \bigoplus_{c\in\mathbb{C}} E^{cx}\otimes R_c, \qquad R_c \text{ regular},$$

then stationary phase at the finite transformed point $s = c$ gives

$$\phi_c(\mathcal{F}\mathcal{K}) \simeq \mathcal{F}^{(\infty,0)}(R_c). \tag{LF}$$

Here ϕ_c is the formal vanishing-cycle germ, including its regular unipotent part. For a regular germ the local transform on the right is regular of the same rank and preserves its monodromy (up to the harmless inversion convention). In particular a trivial rank-one germ becomes the one-dimensional vanishing cycles of δ_0, with identity monodromy. This statement includes $c = 0$. It follows either by applying the global Fourier transform to the trivial connection or from the inverse regular local-transform conventions. Precise references are Arinkin [77, Corollary 2.3(1), §3.3], García López [40, Proposition 2.1], and Sabbah [41, Remarks 5.2, 5.3(4), 5.7].

For an intermediate extension of positive rank, with nearby monodromy T, the canonical map from nearby to vanishing cycles is surjective and the variation map in the opposite direction is injective. Their composites are $T - 1$. Consequently

$$\phi_c \text{ has identity monodromy} \quad\Longrightarrow\quad T \text{ is unipotent and } (T-1)^2 = 0, \qquad \operatorname{rank}(T-1) = \dim \phi_c. \tag{VC}$$

Indeed the vanishing cycles contain every nonidentity generalized eigenspace of T and the image of $T - 1$ on its unipotent part. Thus (VC) does not discard a constant formal factor.

Lemma VI.3.3. *For even n, the minimal equation of H_m is regular singular everywhere. Each finite local monodromy is unipotent with square-zero difference from the identity. At some finite nonzero point this difference has rank one.*

Proof. Let μ_j^2 be the distinct squared weights and put $\Lambda = \prod_j \{n_j, n_j - 2, \dots, -n_j\}$. For $M = \bigotimes_j \operatorname{Sym}^{n_j} B_{\mu_j}$ the Bessel formal solutions at infinity give the direct-sum decomposition

$$\widehat{M}_\infty \simeq \bigoplus_{\lambda\in\Lambda} E^{c_\lambda x}\otimes R_n, \qquad c_\lambda = 2\sum_j \lambda_j\mu_j, \tag{B}$$

where R_n is rank one with a formal solution $x^{-n/2}$. The invertible formal power series in the Bessel expansions are absorbed by rank-one formal gauges. Since n is even, R_n is the trivial formal connection. Distinct indices with equal c_λ remain a direct sum of trivial regular factors; collision introduces no extension or logarithm in (B).

The all-weight theorem identifies the equation of H_m with the generic connection of the simple module $\mathcal{F}(j_{!*}M)$. Its positive rank ensures that this is an intermediate extension at every finite point. If $d_c = \#\{\lambda \in \Lambda : c_\lambda = c\}$, (LF) says its formal vanishing cycles at c are regular, of dimension d_c, and have identity monodromy. At other points the vanishing cycles are zero. The purely irregular part of a holonomic germ is unchanged by formal vanishing cycles, so regularity of these vanishing cycles excludes an irregular nearby part. By (VC), all these points are regular singular with the asserted unipotence. The infinity statement of the all-weight theorem already gives regularity and unipotence at infinity.

Choose $\xi \in \mathbb{C}^\times$ such that $\operatorname{Re}(\xi\mu_j) \neq 0$ for every j. Define $\lambda_j^* = n_j \operatorname{sign}\operatorname{Re}(\xi\mu_j)$. This is the unique maximizer of the real functional $\operatorname{Re}(\xi c_\lambda)$ on Λ, and its value is $2\sum_j n_j|\operatorname{Re}(\xi\mu_j)| > 0$. Thus $c_{\lambda^*} \neq 0$ and $d_{c_{\lambda^*}} = 1$. The corresponding local monodromy is a nonidentity transvection. □

Proof of Theorem VI.3.1 for H_m. The middle pairing in Lemma VI.2.2 is alternating, so the Galois group G lies in Sp_R. The all-weight rank theorem gives irreducibility. By Lemma VI.3.3 the equation is regular singular everywhere, so Schlesinger's density theorem identifies G with the Zariski closure of its monodromy group [43]. The fundamental group of the punctured sphere is generated by the local loops. All their matrices are unipotent, including the one at infinity, so all belong to G^0. Thus $G = G^0$. Lemma VI.3.3 supplies one transvection, and Lemma VI.3.2 gives $G = \mathrm{Sp}_R$. □

VI.3.2 Descent to the ordinary generating function

It remains to exclude a finite similitude multiplier on descending from s to $z = s^{-2}$. This can be checked on the explicit pairing. Let $\omega = [\eta\, dx]$ be the cyclic class evaluating to H_m in (R), where η is the even tensor product of the Bessel cyclic vectors. The simultaneous involution

$$J : (s, x) \longmapsto (-s, -x)$$

preserves N_s and induces a semilinear involution of $\mathcal{P}_m$ covering $s \mapsto -s$. It is an isometry for (P): applying J to both entries and then changing x to $-x$ leaves the sum of residues unchanged. On the cyclic class it has $J\omega = -\omega$ because dx changes sign. Therefore $\zeta = s\omega$ is fixed. The parameter derivative

$$\nabla_z = -\frac{s^3}{2}\nabla_s$$

commutes with J, so all $\nabla_z^i \zeta$ are fixed. They are a basis by the order theorem and the parity descent for $F_m(s^{-2}) = sH_m(s)$. In this basis the Gram matrix of (P) is fixed by $s \mapsto -s$, and hence belongs to $\mathbb{C}(s^2) = \mathbb{C}(z)$. It is nondegenerate, alternating and z-horizontal. Thus the full Galois group of F_m lies in Sp_R. Its identity component is unchanged by the quadratic extension to $\mathbb{C}(s)$, where the group just proved is Sp_R. The full group over $\mathbb{C}(z)$ must therefore be Sp_R as well.

Finally Sp_R acts transitively on the nonzero vectors of its standard representation. The orbit of the nonzero solution H_m has dimension R, and the same holds for F_m. Picard–Vessiot Galois correspondence gives transcendence degree R for the field generated by the first R jets. Since it has exactly R generators, they are algebraically independent. This completes the proof of Theorem VI.3.1.

VI.4 Removing all threshold restrictions in odd-propagator banana periods

The same transvection argument has a second application to the banana periods in the supplied manuscript [49]. Here the number of Bessel factors is odd, whereas it is even in the hypercubic application. The two integral transforms are different, and so are their parity laws. This section removes the nonzero, distinct-threshold hypothesis from the odd-factor Galois-group theorem in that manuscript. It does not prove the remaining leading-coefficient multiplicity conjecture.

For nonzero complex masses $m_1, \dots, m_n$, define

$$c_k(m) = \sum_{a_1+\cdots+a_n=k} \binom{k}{a_1, \dots, a_n}^2 \prod_i m_i^{2a_i}, \qquad \varpi_m(t) = \sum_{k\ge 0} \frac{c_k(m)}{t^{k+1}}.$$

The series converges near infinity. Group the masses by equality of their squares, with multiplicities $n_1, \dots, n_q$, and write

$$\bigotimes_{j=1}^{q} \operatorname{Sym}^{n_j} \mathbb{C}^2 = \bigoplus_{d\ge 0} (\operatorname{Sym}^d \mathbb{C}^2)^{\oplus a_d}$$

for restriction to the diagonal SL_2.

Precise prior input. The all-mass order theorem of [49] states that the minimal connection of ϖ_m is irreducible, has rank

$$o(m) = \sum_{d\ge 2} (d-1)a_d,$$

and has unipotent monodromy at infinity with blocks J_{d-1}, one for each summand with $d \ge 2$. In particular, the block J_{n-1} occurs once. Its middle-cohomology construction gives an invariant nondegenerate alternating form when n is odd. These statements do not require distinct or nonzero signed thresholds. They are used here as previously proved inputs, rather than counted again as new results. The cited manuscript contains their proofs; the new removal of threshold restrictions is proved below.

Theorem VI.4.1 (Every odd number of propagators, every nonzero mass tuple)**.** *Let $n \ge 3$ be odd and $m_i \in \mathbb{C}^\times$. The differential Galois group over $\mathbb{C}(t)$ of the minimal equation of ϖ_m is*

$$\mathrm{Sp}(o(m), \mathbb{C}).$$

This holds when nonzero thresholds coincide and when signed mass sums vanish. Consequently

$$\varpi_m, \varpi_m', \dots, \varpi_m^{(o(m)-1)}$$

are algebraically independent over $\mathbb{C}(t)$, and remain so over every finite algebraic extension of that differential field.

VI.4.1 Why collisions do not obstruct connectedness

Put $y = u^2$ and $g_m(y) = \prod_i I_0(m_i\sqrt{y})$. Its Bessel connection is

$$M = \bigotimes_j \operatorname{Sym}^{n_j} B_{\mu_j}, \qquad B_\mu : \quad (\theta_y^2 - \mu^2 y/4) f = 0, \quad \theta_y = y\partial_y.$$

On the formal u-cover at infinity, it splits as a direct sum of rank-one factors indexed by

$$\lambda_j \in \{n_j, n_j - 2, \dots, -n_j\}, \qquad c(\lambda) = \sum_j \lambda_j \mu_j.$$

Each factor has the form

$$e^{c(\lambda)u}u^{-n/2}(1+O(u^{-1})).$$

The splitting comes from the two formal Bessel solutions in each factor before taking symmetric and tensor powers. Equality of two exponential factors therefore gives a direct sum with multiplicity, not a new nontrivial formal extension.

For $c \neq 0$, the pair indexed by $\lambda, -\lambda$ descends to

$$A_c(n) = [u \mapsto y = u^2]_+(E^{cu} \otimes U_n), \qquad U_n : \ u^{-n/2}.$$

The standard local Fourier calculation sends this pair, after the first Fourier transform, to

$$E^{c^2/(4s)} \otimes R_c, \qquad R_c \text{ regular of rank one.}$$

The power in its Laplace asymptotic is $s^{(n-3)/2}$: under $y = u^2$, the measure contributes $2u\,du$, evaluation at the critical point $u = c/(2s)$ contributes $s^{n/2-1}$, and the Gaussian fluctuation contributes $s^{-1/2}$. Equivalently this is the formal stationary-phase formula [41, 40]. Thus the monodromy of R_c is $(-1)^{n-3}$, independent of c. Only its value 1 for odd n is needed here.

The construction of the banana connection first takes the Fourier transform of $j_{!*}M$, then pulls its restriction to $\mathbb{G}_m$ back by $s = 1/v$, and then takes the Fourier transform of the intermediate extension in v. The rational gauge v^{-1} in the scalar integral representation has no effect on monodromy. If τ is the second Fourier variable, the period normalization is $t = 4\tau$; the factor $E^{c^2v/4}$ is therefore located at $t = c^2$. The factor above produces vanishing cycles at $t = c^2$ with the same monodromy. If several pairs give the same nonzero c^2, their vanishing-cycle spaces are the direct sum of these lines. For odd n their monodromy is the identity.

The final Fourier module is simple of positive rank. Hence it is the intermediate extension at each finite point. If T_a denotes its local monodromy at a nonzero threshold a, the nearby/vanishing-cycle description gives

$$\phi_a \cong \operatorname{im}(T_a - I), \qquad (T_a - I)^2 = 0.$$

In particular T_a is unipotent, even at a collision. The regular part arising from $c = 0$ is still regular through the two transforms; it can affect the monodromy at zero, but creates no further nonzero singularity. All formal local factors of the final module are regular, so its global equation is regular singular.

The monodromy group is generated by the loops around infinity and the nonzero thresholds: in the fundamental group relation one can omit the loop around zero. Every one of these generators is unipotent. The Zariski closure of the cyclic group generated by a unipotent matrix in characteristic zero is connected. Schlesinger's theorem therefore shows that the differential Galois group G is connected. This argument needs no assertion that the monodromy at zero itself is unipotent.

VI.4.2 A simple threshold always survives

Choose $\xi \in \mathbb{C}$ such that $\operatorname{Re}(\xi\mu_j) \neq 0$ for every j, and maximize $\operatorname{Re}(\xi c(\lambda))$ over the finite index set. There is a unique maximizer, given by

$$\lambda_j^* = n_j \operatorname{sign}\operatorname{Re}(\xi\mu_j).$$

Its value has strictly positive real projection, so $c_* := c(\lambda^*) \neq 0$. The only tuple giving c_* is λ^*; the only one giving $-c_*$ is $-\lambda^*$. Over $\mathbb{C}$, equality $c(\lambda)^2 = c_*^2$ means $c(\lambda) = c_*$ or $-c_*$. Consequently the squared threshold c_*^2 has exactly one contributing pair, even if other thresholds collide or vanish.

Its vanishing-cycle space is one-dimensional with trivial monodromy. Since the final module is an intermediate extension, the corresponding T_* satisfies

$$\operatorname{rank}(T_* - I) = 1, \qquad (T_* - I)^2 = 0.$$

It is a nontrivial symplectic transvection.

Proof of Theorem VI.4.1. The prior input gives irreducibility and an invariant alternating form. The preceding arguments give connectedness and a nontrivial transvection. Lemma VI.3.2 therefore gives $G = \mathrm{Sp}(o(m))$.

Let E be a Picard–Vessiot extension and v the nonzero solution vector corresponding to ϖ_m. The field generated by its first $o(m)$ derivatives is the fixed field of its stabilizer. Its transcendence degree is consequently the dimension of the G-orbit of v. A symplectic group is transitive on nonzero vectors, so this dimension is $o(m)$. There are exactly $o(m)$ generators, and hence they are algebraically independent. Algebraic extension of the base field preserves their transcendence degree; equivalently, the connected Galois group is unchanged by finite algebraic base extension. □

For pairwise distinct squared masses, the rank in this theorem is the established formula

$$o(m) = 2^n - \binom{n+1}{\lfloor (n+1)/2 \rfloor}.$$

For all equal squared masses it is $n-1$. The theorem concerns functional algebraic independence, not arithmetic independence of values at a fixed energy. It also does not determine the full Galois group for every even n; the reflection case requires a separate argument.

VI.5 The complete exceptional-pair classification for half-integer beta quantiles

For the strict half-integer beta family, an arithmetic reduction leaves three candidate algebraic probability levels. The remaining issue is to classify the integer shape pairs that actually attain those levels at algebraic quantiles. We prove a closed list: apart from the arcsine family and symmetric medians, the only exceptions are

$$I_{1/4}(3/2, 5/2) = 1/3, \qquad I_{3/4}(5/2, 3/2) = 2/3.$$

The new ingredient is a prime-denominator obstruction for a finite Fourier sum. A central-coefficient product makes that obstruction explicit; Nagura's prime-interval theorem and an elementary probability bound then reduce the unbounded parameter lattice to eleven exact sign comparisons.

VI.5.1 Statement of the classification

For $a, b > 0$, write

$$B_x(a, b) = \int_0^x t^{a-1}(1-t)^{b-1}\, dt, \qquad I_x(a, b) = \frac{B_x(a, b)}{B_1(a, b)}.$$

The inverse $I^{-1}_{a,b}$ means the strictly increasing inverse in $x \in (0, 1)$; the subscript records the two fixed shapes. For integers $m, n \geq 0$, define

$$\mathcal{E}_{m,n} = \{r \in \overline{\mathbb{Q}} \cap (0, 1) : I^{-1}_{m+1/2,n+1/2}(r) \in \overline{\mathbb{Q}}\}.$$

Theorem VI.5.1 (Closed classification of all strict half-integer pairs)**.** *For all integers $m, n \geq 0$,*

$$\mathcal{E}_{m,n} = \begin{cases} \mathbb{Q} \cap (0, 1), & m = n = 0, \\ \{1/2\}, & m = n \geq 1, \\ \{1/3\}, & (m, n) = (1, 2), \\ \{2/3\}, & (m, n) = (2, 1), \\ \varnothing, & \textit{otherwise.} \end{cases} \tag{VI.5.1}$$

The corresponding quantiles are respectively $\sin^2(\pi r/2)$*,* $1/2$*,* $1/4$*, and* $3/4$*. Every algebraic probability level not listed in* (VI.5.1) *has a transcendental quantile.*

The distinction from the earlier finite arithmetic test is that (VI.5.1) contains no parameter-dependent polynomial evaluation or search. Its proof settles that test simultaneously for every integer pair. We give the arithmetic reduction as well, so that the proof is complete apart from its explicitly cited classical transcendence and prime-interval inputs.

VI.5.2 Reduction to three algebraic points

Let $w_{m,n}(X) = X^m(1-X)^n$, and put

$$c_{m,n} = \frac{(2m)!(2n)!}{4^{m+n} m!\, n!\, (m+n)!} > 0. \tag{VI.5.2}$$

There is a unique polynomial $R_{m,n} \in \mathbb{Q}[X]$ with

$$w_{m,n}(X) = c_{m,n} + X(1-X)R'_{m,n}(X) + (1/2 - X)R_{m,n}(X). \tag{VI.5.3}$$

Indeed, the operator on a monomial X^j is $(j+1/2)X^j-(j+1)X^{j+1}$, so descending elimination leaves a unique constant. Integration against $[x(1-x)]^{-1/2}$ shows that this constant is (VI.5.2). Consequently

$$I_x(m+1/2,n+1/2)=\frac{2\arcsin\sqrt{x}}{\pi}+\frac{\sqrt{x(1-x)}}{\pi c_{m,n}}R_{m,n}(x). \tag{VI.5.4}$$

In particular,

$$\operatorname{sgn} R_{m,n}(1/4)=\operatorname{sgn}\{I_{1/4}(m+1/2,n+1/2)-1/3\}. \tag{VI.5.5}$$

Lemma VI.5.2 (Arithmetic and shape reduction)**.** *If $x\in(0,1)$ is algebraic, then*

$$I_x(m+1/2,n+1/2)\in\overline{\mathbb{Q}}\quad\Longleftrightarrow\quad R_{m,n}(x)=0\quad\text{and}\quad\frac{2\arcsin\sqrt{x}}{\pi}\in\mathbb{Q}. \tag{VI.5.6}$$

If $m,n\geq 1$, $R_{m,n}$ has exactly one root in $(0,1)$, and any algebraic level with an algebraic quantile is one of

$$(r,x)=(1/3,1/4),\quad(1/2,1/2),\quad(2/3,3/4).$$

The middle pair occurs if and only if $m=n$. If exactly one of m,n is zero, there are no exceptions.

Proof. Suppose x and $r=I_x(m+1/2,n+1/2)$ are algebraic, and set $\theta=\arcsin\sqrt{x}\in(0,\pi/2)$. Then $\zeta=e^{2i\theta}=1-2x+2i\sqrt{x(1-x)}$ is algebraic. With $\log\zeta=2i\theta$ and $\log(-1)=i\pi$, (VI.5.4) makes

$$\log\zeta-r\log(-1)=-i\frac{\sqrt{x(1-x)}}{c_{m,n}}R_{m,n}(x)$$

algebraic. Baker's theorem on algebraic-coefficient linear forms in logarithms says that such a form is either zero or transcendental [46]. It must vanish. Thus $R_{m,n}(x)=0$ and $\zeta=e^{i\pi r}$. If r were algebraic irrational, the Gelfond–Schneider theorem applied to this value of $(-1)^r$ would contradict the algebraicity of ζ. Hence r is rational. The converse follows from (VI.5.4).

For the root count, put

$$g(\theta)=\sin^{2m}\theta\cos^{2n}\theta,\qquad h(\theta)=\int_0^\theta(g(u)-c_{m,n})\,du.$$

The constant $c_{m,n}$ is the average of g on $[0,\pi/2]$, and (VI.5.4) yields

$$h(\theta)=\tfrac{1}{2}\sqrt{x(1-x)}R_{m,n}(x),\quad x=\sin^2\theta,\qquad h(0)=h(\pi/2)=0. \tag{VI.5.7}$$

For $m,n\geq 1$, g vanishes at both endpoints and has one strict maximum. Its logarithmic derivative in the interior is $2m\cot\theta-2n\tan\theta$, strictly decreasing through zero. Thus $g-c_{m,n}$ has exactly two zeros. The function h first decreases, then increases, then decreases; it is negative near zero and positive just before $\pi/2$. It has exactly one interior zero.

For a rational candidate r, let d be the order of $e^{i\pi r}$, so $d\geq 3$. The Galois conjugates of $x=(2-\zeta-\zeta^{-1})/4$ are the distinct numbers $\sin^2(\pi j/d)$, with $(j,d)=1$, identified only under $j\mapsto -j$. All $\varphi(d)/2$ conjugates lie in $(0,1)$ and are roots of the rational polynomial $R_{m,n}$. The one-root count forces $\varphi(d)\leq 2$, hence $d\in\{3,4,6\}$. These give exactly the three displayed pairs.

For the middle pair, symmetry gives $I_{1/2}=1/2$ if $m=n$. If $m>n$, the density at $x\in(0,1/2)$ divided by its density at $1-x$ is $(x/(1-x))^{m-n}<1$, so $I_{1/2}<1/2$. The reverse inequality holds when $m<n$.

If precisely one exponent vanishes, g is strictly monotone, so $g-c_{m,n}$ has exactly one zero. Were h to have an interior zero, its two endpoint zeros would force at least two interior zeros of h' by Rolle's theorem. This is impossible. Equation (VI.5.6) then excludes every exception. □

VI.5.3 The prime obstruction

Lemma VI.5.3 (A central-coefficient product)**.** *For every integer $q \geq 0$ and every number A, as an identity of polynomial coefficients in A,*

$$[z^q](1-z)^A(1+z)^{2q-A} = \frac{(-2)^q}{q!}\prod_{j=1}^{q}(A-(2j-1)). \tag{VI.5.8}$$

The empty product for $q=0$ equals one. Generalized powers on the left are expanded as formal power series at zero.

Proof. The coefficient is a polynomial in A of degree at most q. In

$$(1+z)^{2q}\exp\!\left(A\log\frac{1-z}{1+z}\right), \qquad \log\frac{1-z}{1+z} = -2z + O(z^3),$$

the coefficient of $A^q z^q$ is $(-2)^q/q!$. For each odd integer $A = 1, 3, \ldots, 2q-1$, the expression $(1-z)^A(1+z)^{2q-A}$ is a polynomial of degree $2q$ satisfying $z^{2q}P(1/z) = -P(z)$. Its central coefficient therefore vanishes. These q roots and the leading coefficient prove (VI.5.8). □

Lemma VI.5.4 (Prime-denominator obstruction)**.** *Let $m, n \geq 1$. If there is a prime $p \geq 5$ such that*

$$\max\{2m, 2n/3\} < p \leq m+n, \tag{VI.5.9}$$

then $R_{m,n}(1/4) \neq 0$.

Proof. Set $L = m+n$ and write

$$(1-z)^{2m}(1+z)^{2n} = \sum_{j=0}^{2L} a_j z^j.$$

All a_j are integers, and reciprocity gives $a_{2L-j} = a_j$. On setting $z = e^{2i\theta}$, we obtain

$$\sin^{2m}\theta\cos^{2n}\theta = \frac{(-1)^m}{4^L}\left(a_L + 2\sum_{k=1}^{L} a_{L+k}\cos(2k\theta)\right).$$

Its average on $[0, \pi/2]$ is $(-1)^m 4^{-L} a_L = c_{m,n}$. Define the integer-valued function

$$\chi(k) = \frac{2}{\sqrt{3}}\sin\frac{k\pi}{3}; \qquad (\chi(0), \ldots, \chi(5)) = (0, 1, 1, 0, -1, -1),$$

extended periodically modulo six. Integrating the Fourier expansion to $\theta = \pi/6$, and using (VI.5.7), gives the exact identity

$$R_{m,n}(1/4) = (-1)^m 4^{1-L}\sum_{k=1}^{L}\frac{\chi(k)a_{L+k}}{k}. \tag{VI.5.10}$$

The inequality $\max\{2m, 2n/3\} \geq (m+n)/2$ follows by considering $n \leq 3m$ and $n \geq 3m$. Thus (VI.5.9) implies $p > L/2$. Set $q = L - p$, so $0 \leq q < p$. Among the denominators $1, \ldots, L$, exactly $k = p$ is divisible by p. Also $\chi(p) \in \{1, -1\}$, since $p \geq 5$. We show that a_{L+p} is not divisible by p.

By reciprocity, $a_{L+p} = a_q$. Since $2n = 2p + 2q - 2m$, the identity $(1+z)^p = 1 + z^p$ in $\mathbb{F}_p[[z]]$ shows that

$$a_q \equiv [z^q](1-z)^{2m}(1+z)^{2q-2m} \pmod{p}. \tag{VI.5.11}$$

This remains valid if $2q - 2m < 0$: the power series $1 + z$ is invertible, and $(1 + z)^{2p} = (1 + z^p)^2$ contributes only its constant term through degree $q < p$. By Lemma VI.5.3, the right side is

$$\frac{(-2)^q}{q!} \prod_{j=1}^{q} (2m - (2j - 1)) \pmod{p}.$$

The denominator $q!$ is a unit modulo p. Every factor in the product is a nonzero odd integer; its largest possible value is $2m - 1 < p$, and its smallest is

$$2m - 2q + 1 = 2p - 2n + 1 > -p,$$

where the last inequality uses $p > 2n/3$. No factor is divisible by p, proving that a_q is a unit. The summand with $k = p$ in (VI.5.10) now has p-adic valuation -1, while all other summands have nonnegative valuation. The sum is nonzero, as claimed. □

VI.5.4 Uniform exclusion of the unbounded shape ranges

We use Nagura's theorem in its original real-variable form: for every real $y \geq 25$ there is a prime

$$y < p < \frac{6}{5}y. \tag{VI.5.12}$$

This is the $n = 5$ case of the theorem on page 180 of [54].

Lemma VI.5.5 (All sufficiently large first exponents)**.** *For every integer $m \geq 13$ and every integer $n \geq 1$, $R_{m,n}(1/4) \neq 0$.*

Proof. For fixed $a > 0$ and $x \in (0, 1)$, the identity

$$I_x(a, b + 1) - I_x(a, b) = \frac{x^a(1 - x)^b}{bB_1(a, b)} > 0 \tag{VI.5.13}$$

follows by integration by parts and $B_1(a, b + 1) = bB_1(a, b)/(a + b)$. Thus the beta CDF at $1/4$ strictly increases with n for fixed m.

First suppose $n \leq n_0 = \lfloor 7m/5 \rfloor$. It suffices to bound the CDF with $n = n_0$. For $X \sim \mathrm{Beta}(m + 1/2, n_0 + 1/2)$ and $S = m + n_0 + 1$, its mean and variance satisfy

$$\mu = \frac{m + 1/2}{S} > \frac{5}{12}, \qquad \sigma^2 = \frac{(m + 1/2)(n_0 + 1/2)}{S^2(S + 1)} \leq \frac{1}{4(S + 1)} \leq \frac{1}{8m + 8}.$$

Here $S \leq 12m/5 + 1$ proves the strict mean inequality, and $n_0 \geq m$ proves the last variance inequality. The one-sided variance bound

$$\Pr(X - \mu \leq -d) \leq \frac{\sigma^2}{\sigma^2 + d^2} \qquad (d > 0)$$

is obtained by applying Markov's inequality to $(X - \mu - t)^2$ and minimizing $(\sigma^2 + t^2)/(d + t)^2$ over $t \geq 0$; the minimum is at $t = \sigma^2/d$. Taking $d = \mu - 1/4 > 1/6$ gives

$$I_{1/4}(m + 1/2, n_0 + 1/2) < \frac{36}{8m + 44} = \frac{9}{2m + 11} < \frac{1}{3}, \tag{VI.5.14}$$

because $m \geq 13$. Equation (VI.5.5) excludes equality for every $n \leq n_0$.

Next suppose $\lfloor 7m/5 \rfloor < n \leq 3m$. Then $m + n > 12m/5$. Apply (VI.5.12) with $y = 2m \geq 26$ to obtain

$$2m < p < 12m/5 < m + n.$$

Since $n \leq 3m$, this prime satisfies (VI.5.9); Lemma VI.5.4 applies.

Finally, if $n > 3m$, use $y = 2n/3 > 2m \geq 26$. Nagura supplies

$$2n/3 < p < 4n/5 < n < m + n.$$

Again (VI.5.9) holds, and the prime obstruction finishes this last range. □

VI.5.5 The finite exact certificate and completion

Theorem VI.5.6 (Unique quarter-point pair)**.** *For all integers $m, n \geq 1$,*

$$R_{m,n}(1/4) = 0 \quad \Longleftrightarrow \quad (m, n) = (1, 2). \tag{VI.5.15}$$

Equivalently, $I_{1/4}(m + 1/2, n + 1/2) = 1/3$ if and only if $(m, n) = (1, 2)$.

Proof. Lemma VI.5.5 handles $m \geq 13$. For the remaining finitely many first exponents define

$$T_{m,n} = 4^{m+n-1} 3^{-n} R_{m,n}(1/4). \tag{VI.5.16}$$

These numbers obey the exact rational recurrences

$$\begin{aligned} T_{0,0} &= 0, \\ T_{0,n} &= \frac{2(2n-1)T_{0,n-1} + 1}{3n} && (n \geq 1), \\ T_{m,n} &= \frac{2(2m-1)T_{m-1,n} - 1}{m+n} && (m \geq 1, n \geq 0). \end{aligned} \tag{VI.5.17}$$

To verify the recurrence, reduction of the incomplete beta integral gives

$$R_{0,n}(X) = \frac{2n-1}{2n} R_{0,n-1}(X) + \frac{(1-X)^{n-1}}{n},$$

and

$$R_{m,n}(X) = \frac{2m-1}{2(m+n)} R_{m-1,n}(X) - \frac{X^{m-1}(1-X)^n}{m+n}.$$

For example, the latter follows by differentiating $x^{m-1/2}(1-x)^{n+1/2}$ and integrating from zero to x; the former follows from the same calculation in the other parameter with $m = 0$. Substitution of $x = 1/4$ and (VI.5.16) proves (VI.5.17).

For $m = 1$, direct reduction gives

$$R_{1,2}(X) = -\frac{(2X-3)(4X-1)}{24}, \qquad R_{1,2}(1/4) = 0.$$

Strict monotonicity (VI.5.13) and (VI.5.5) exclude every other n in this row.

For $2 \leq m \leq 12$, Table 2 contains the exact values produced by (VI.5.17) at two consecutive integers. Each first value is negative and each second value positive. By (VI.5.5), the same signs hold for the CDF minus $1/3$. Its strict increase in n excludes every integer $n \geq 1$, including those outside the table. This completes all parameter ranges. □

Proof of Theorem VI.5.1. If $m = n = 0$, (VI.5.4) is the arcsine CDF and $R_{0,0} = 0$. Equation (VI.5.6) gives precisely the rational levels, all with quantile $\sin^2(\pi r/2)$. If exactly one exponent is zero, Lemma VI.5.2 excludes exceptions.

For positive exponents, that lemma leaves only the three candidate pairs. Theorem VI.5.6 settles the first candidate. Reflection of the beta density gives

$$I_x(m + 1/2, n + 1/2) = 1 - I_{1-x}(n + 1/2, m + 1/2),$$

so the $3/4$ candidate occurs exactly at $(m, n) = (2, 1)$. The median candidate occurs exactly when $m = n$. These cases are disjoint and exhaust the list. □

Corollary VI.5.7 (Exact algebraic exceptions for the Jeffreys posterior)**.** *After m successes and n failures, with $m + n \geq 1$, the binomial Jeffreys posterior has an algebraic quantile at an algebraic internal probability level only in the following cases: balanced counts $m = n$ at level $1/2$; counts $(1, 2)$ at level $1/3$; and counts $(2, 1)$ at level $2/3$. All other such posterior quantiles are transcendental.*

Proof. The posterior is $\mathrm{Beta}(m + 1/2, n + 1/2)$. Apply Theorem VI.5.1. □

m	n_0	T_{m,n_0}	T_{m,n_0+1}
2	4	$-1/45$	$1/35$
3	6	$-1/28$	$1/420$
4	9	$-2/143$	$67/6006$
5	12	$-71/19448$	$7949/525096$
6	14	$-2507/167960$	$179/83980$
7	17	$-1217/148580$	$2113/371450$
8	20	$-3839/1040060$	$120877/15080870$
9	23	$-231151/440006560$	$2112517/220003280$
10	25	$-11678941/1476337800$	$103517/57895600$
11	28	$-4411771/872155860$	$1079573/306019600$
12	31	$-1048804991/369448814700$	$627776153/129013871800$

Table 2: Exact rational sign certificate for the eleven remaining rows. Every entry follows from (VI.5.17); no floating-point comparison is used.

Scope and relation to earlier work. The new statement is the uniform classification (VI.5.15), which removes the unfinished parameter-dependent test in the preceding polynomial-arcsine analysis. The logarithmic transcendence input, the central-coefficient identity, and Nagura's theorem are not claimed as new. The related paper of Dhillon and Saha [55] concerns arithmetic properties of incomplete beta values. Saha's accompanying conference presentation [56] explicitly discusses unnormalized values with rational nonintegral a and $a + b \in \mathbb{N}$ using Baker's theory. Its stated open question concerning an analogue of Schneider's theorem is distinct from the normalized-CDF pair classification proved here. The paper's publisher abstract and bibliographic details were checked, but a full-text comparison has not been completed. Accordingly no historical-priority claim, nor a claim to resolve a named external conjecture, is made here. The theorem concerns the normalized full-interval beta CDF and every strict half-integer shape pair; it makes no assertion for general rational shapes or arbitrary truncations.

VI.6 Exact recurrence width for every complex cubic weight

For $\lambda \in \mathbb{C}$, define

$$a_n(\lambda) = \sum_{k=0}^{n} \binom{n}{k}^3 \lambda^k, \qquad E_\lambda(z) = \sum_{n\geq 0} a_{2n}(\lambda) z^n.$$

For $\lambda \neq 0$, its central normalization is

$$B_\lambda(n) = \lambda^{-n} a_{2n}(\lambda) = \sum_{k=-n}^{n} \binom{2n}{n+k}^3 \lambda^k.$$

The *width* of a nonzero eventual recurrence $\sum_{j\in J} q_j(n) f_{n+j} = 0$, with $q_j \in \mathbb{C}(n)$ and finite nonempty $J \subset \mathbb{Z}$, is $\max J - \min J$ after deleting zero coefficients. Geometric multiplication $f_n \mapsto c^n f_n$, $c \neq 0$, does not change the minimum width.

Theorem VI.6.1 (Complete cubic parameter classification)**.** *For every $\lambda \in \mathbb{C}^*$, the minimum recurrence width of $B_\lambda(n)$, equivalently of $a_{2n}(\lambda)$, over $\mathbb{C}(n)$ is*

$$\begin{cases} 1, & \lambda = -1, \\ 2, & \lambda = 1, \\ 3, & \lambda \notin \{-1, 1\}. \end{cases}$$

In particular, the collision parameters $\lambda = 8$ and $\lambda = 1/8$ have minimum width three. The conclusion concerns recurrence width, not the differential order of a generating function and not the minimum number of nonzero terms in a recurrence with arbitrary gaps.

Weighted binomial recurrences were studied by Matala-Aho [26]. The contribution here is the lower bound at every complex specialization, including the collision parameters. We do not identify this precise parameter classification with a named conjecture in the cited literature. McIntosh's alternating problem [29] concerns the different parameter family with fixed weight -1 and variable exponent.

VI.6.1 A hypergeometric period and its monodromy

Lemma VI.6.2 (Cubic generating function)**.** *With $c = 1 + \lambda$ and $H(u) = {}_2F_1(\frac{1}{3}, \frac{2}{3}; 1; u)$, one has the germ identity*

$$A_\lambda(t) := \sum_{n\geq 0} a_n(\lambda) t^n = \frac{1}{1-ct} H\left(\frac{27\lambda t^2}{(1-ct)^3}\right). \tag{VI.6.1}$$

Proof. The terminating Pfaff–Saalschutz identity [57, §16.4.3] gives

$$\sum_{j\geq 0} \frac{(n+j)!}{j!^3 (k-j)!(n-k-j)!} = \binom{n}{k} {}_3F_2\left(\begin{matrix} -k, k-n, n+1 \\ 1, 1 \end{matrix}; 1\right) = \binom{n}{k}^3,$$

where terms with negative factorial arguments are omitted. Taking coefficients of λ^k therefore proves

$$a_n(\lambda) = \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(n+j)!}{j!^3 (n-2j)!} \lambda^j (1+\lambda)^{n-2j}.$$

Summing first over $n - 2j \geq 0$ yields

$$A_\lambda(t) = \sum_{j\geq 0} \frac{(3j)!}{j!^3} \frac{\lambda^j t^{2j}}{(1-ct)^{3j+1}}.$$

Finally $(3j)!/j!^3 = 27^j (\frac{1}{3})_j (\frac{2}{3})_j / (j!)^2$. □

Lemma VI.6.3 (Irreducibility and genuine finite singularities)**.** *For $\lambda \neq 0$, the rank-two local system obtained by the pullback and rational gauge in (VI.6.1) is irreducible. Its genuine nonzero finite monodromy singularities are exactly the distinct roots of*

$$\Delta_\lambda(t) = (1-(1+\lambda)t)^3 - 27\lambda t^2. \tag{VI.6.2}$$

At each such root its local monodromy T is nontrivial unipotent and $\operatorname{rank}(T-I) = 1$.

Proof. The base hypergeometric equation is

$$u(1-u)y'' + (1-2u)y' - \frac{2}{9}y = 0.$$

Its local monodromies at 0 and 1 are nontrivial unipotents with different fixed lines. Indeed, $H(u)$ spans the line of holomorphic solutions at zero, while the logarithmic connection formula [57, §15.8.10] gives

$$H(u) = -\frac{\log(1-u)}{\Gamma(1/3)\Gamma(2/3)} + O(1) \quad (u \longrightarrow 1).$$

The equation is invariant under $u \mapsto 1-u$, which proves the corresponding assertion at zero. Thus the two fixed lines differ; in dimension two this proves irreducibility.

For completeness this irreducibility survives every nonconstant rational pullback. Remove the three base singularities and also the finitely many critical values of the rational map. After removing their preimages, the map is a finite unramified covering. Its fundamental group has finite-index image in the base fundamental group. Consequently its monodromy image has finite index in the original monodromy group. It contains positive powers of each of the two unipotents just described. These powers have the same two different fixed lines, so their common representation is irreducible. The rational map $27\lambda t^2/(1-ct)^3$ is nonconstant for every $\lambda \neq 0$. A nonzero rational scalar gauge does not change its monodromy.

At a root of Δ_λ, the rational map takes the value 1. A root of multiplicity $e \geq 1$ therefore gives a conjugate of the eth power of the base monodromy at 1, which is still nontrivial unipotent of rank one. There is no cancellation with $1-ct$ at a root of Δ_λ, since $\lambda \neq 0$. If $c \neq 0$, the point $t = 1/c$ maps to infinity with ramification degree three. The base exponents $1/3, 2/3$ have nonintegral difference; after this pullback and the scalar gauge their two local solutions are holomorphic, with leading powers 0 and 1. Their monodromy is therefore the identity. All other finite nonzero points map to ordinary base points, except for the roots already listed. This proves the assertion about genuine monodromy support. □

Lemma VI.6.4 (The even section generates both period spaces)**.** *For $\lambda \neq 0, -1$, the analytic continuations of*

$$E_\lambda(t^2) = \tfrac{1}{2}\big(A_\lambda(t) + A_\lambda(-t)\big)$$

span the direct sum of the two rank-two period spaces for $A_\lambda(t)$ and $A_\lambda(-t)$. At a nonzero point t_0, the local monodromy on this four-dimensional space consequently has

$$\operatorname{rank}(T-I) = \mathbf{1}_{\{\Delta_\lambda(t_0)=0\}} + \mathbf{1}_{\{\Delta_\lambda(-t_0)=0\}}.$$

Here multiplicities of roots of either individual polynomial do not alter their rank-one contributions.

Proof. Work on a common punctured sphere for both equations. Write S for the distinct roots of Δ_λ. Its discriminant is

$$\operatorname{disc}_t \Delta_\lambda = -19683\lambda^2(\lambda-1)^2.$$

If $\lambda \neq 0, \pm 1$, the set S has three nonzero elements, and hence $S \neq -S$. If $\lambda = 1$, then $\Delta_1(t) = (1-8t)(1+t)^2$, so again $S \neq -S$. The two irreducible representations have different genuine monodromy supports and cannot be isomorphic.

Regarded as spaces of local analytic functions at an ordinary base point, their intersection is zero: a nonzero intersection would be an invariant subspace of both irreducible spaces and would identify them. In their direct sum, the monodromy span of the displayed sum projects onto each summand, because both component functions are nonzero and each representation is irreducible. A proper invariant subspace with both projections onto two irreducible summands would be the graph of an isomorphism between them. This is impossible. The span is thus the full direct sum. The rank formula now follows from Lemma VI.6.3 by adding the ranks on the two summands. □

VI.6.2 Why a collision still consumes two units of width

Lemma VI.6.5 (A simple leading zero allows at most rank-one monodromy)**.** *Consider a scalar homogeneous differential equation of order $N \geq 1$ with holomorphic coefficients near $s = 0$. If its leading coefficient has a simple zero there, the local monodromy on its solution space satisfies* $\operatorname{rank}(T - I) \leq 1$.

Proof. After multiplication by a holomorphic unit, write the equation as

$$sy^{(N)} + \sum_{j=0}^{N-1} b_j(s) y^{(j)} = 0.$$

For a formal solution $y = \sum_{k \geq 0} c_k s^k$, the coefficient of s^m has coefficient

$$\frac{(m+N-1)!}{m!}(m + b_{N-1}(0))$$

on its highest-index unknown c_{m+N-1}. All remaining terms involve lower-index coefficients. Initially $c_0, \ldots, c_{N-2}$ are free. There is at most one $m \geq 0$ at which the displayed multiplier vanishes. If none exists, these $N-1$ coefficients determine $N-1$ independent formal holomorphic solutions. At the exceptional index, at most one linear compatibility condition is imposed, while c_{m+N-1} becomes a new free coefficient. Hence the dimension is still at least $N-1$. For $N = 1$ the claimed dimension lower bound is zero and is immediate. This is a regular singular equation: after division by s, each coefficient has at most a simple pole. The convergent Frobenius theorem therefore makes these formal holomorphic solutions convergent. At least $N-1$ independent local solutions are fixed by monodromy, proving the assertion, including resonant cases. □

Lemma VI.6.6 (A recurrence gives a finite leading-zero budget)**.** *If a convergent generating-function germ $F(z) = \sum_{n \geq 0} f_n z^n$ has an eventual recurrence of width L over $\mathbb{C}(n)$, it has a scalar homogeneous differential annihilator whose leading coefficient, away from zero, has at most L zeros counted with multiplicity.*

Proof. Translate the shifts and clear denominators to write $\sum_{j=0}^{L} p_j(n) f_{n+j} = 0$, with $p_j \in \mathbb{C}[n]$ not all zero. With $\vartheta = z\, d/dz$, coefficient extraction gives

$$\mathcal{D}F = P(z), \qquad \mathcal{D} = \sum_{j=0}^{L} z^{L-j} p_j(\vartheta - j),$$

where P is a polynomial accounting for the finitely many initial indices. If $D = \max_j \deg p_j$, the coefficient of $(d/dz)^D$ in $\mathcal{D}$ is $z^D h(z)$, where

$$h(z) = \sum_{\deg p_j = D} \operatorname{lc}(p_j) z^{L-j}$$

is nonzero and has degree at most L. Left-multiplying $\mathcal{D}$ by a sufficiently high power of d/dz kills P and does not change this leading coefficient. This gives the required annihilator. □

VI.6.3 Proof of the classification

Proof of Theorem VI.6.1. We first establish the upper bounds without requiring any recurrence matrix to be invertible. For $\lambda \neq -1$, put

$$L_N[p] = \sum_{k=0}^{N} p(k) \binom{N}{k}^3 \lambda^k.$$

Shifting the summation index gives, for every integer $h \geq 0$,

$$L_N\big[k^{3+h} - \lambda(N-k)^3(k+1)^h\big] = 0.$$

The polynomial inside brackets has degree $h+3$ and leading coefficient $1+\lambda \neq 0$. Successive elimination therefore reduces every polynomial moment to the three moments of $1, k, k^2$, with coefficients rational in N. Since

$$\binom{N-1}{k}^3 = N^{-3}(N-k)^3 \binom{N}{k}^3,$$

the column $M_N = (L_N[1], L_N[k], L_N[k^2])^{\mathsf{T}}$ satisfies $M_{N-1} = T(N) M_N$ for a rational 3×3 matrix. It follows that $V_n = M_{2n}$ satisfies $V_{n-1} = T(2n-1)T(2n)V_n$. The first-coordinate functionals expressing $a_{2n}, a_{2n-2}, a_{2n-4}, a_{2n-6}$ in terms of V_n are four rows in a three-dimensional vector space over $\mathbb{C}(n)$. Their linear dependence proves width at most three.

At $\lambda = 1$, the classical cubic Franel recurrence

$$(n+1)^2 a_{n+1}(1) = (7n^2 + 7n + 2)a_n(1) + 8n^2 a_{n-1}(1)$$

(also obtained by substituting (VI.6.1) into the Gauss equation) gives a two-dimensional rational companion system. Taking every second state gives width at most two. At $\lambda = -1$, (VI.6.1) gives

$$A_{-1}(t) = H(-27t^2), \qquad B_{-1}(n) = \frac{(3n)!}{n!^3},$$

which has width one and cannot have width zero.

We next prove all remaining lower bounds. Put $z = t^2$ and compute

$$\begin{aligned} Q_\lambda(z) &= \Delta_\lambda(t)\Delta_\lambda(-t) \\ &= 1 - 3(\lambda^2 + 20\lambda + 1)z \\ &\quad + 3(\lambda^4 - 50\lambda^3 + 141\lambda^2 - 50\lambda + 1)z^2 - (1+\lambda)^6 z^3. \end{aligned}$$

Its discriminant is

$$\operatorname{disc}_z Q_\lambda = -19683\lambda^2(\lambda-8)^2(\lambda-1)^2(\lambda+1)^2(8\lambda-1)^2.$$

For $\lambda \notin \{0, \pm 1, 8, 1/8\}$ it has three distinct nonzero roots. At each such root, the monodromy span of E_λ has a nontrivial local monodromy by Lemma VI.6.4. Here the change of variable $z = t^2$ is locally unramified, so its local monodromy is unchanged. Every scalar annihilator of E_λ must have a vanishing leading coefficient at each of these three points: at an ordinary point all solutions would be holomorphic and monodromy would be trivial. Lemma VI.6.6 therefore proves width at least three.

For $\lambda = 8$, the factorizations are

$$\Delta_8(t) = (1-27t)(1+27t^2), \qquad Q_8(z) = (1-729z)(1+27z)^2.$$

At $z = -1/27$, both t and $-t$ are roots of Δ_8, and Lemma VI.6.4 gives monodromy rank two on the four-dimensional space generated by the even period. Any scalar annihilator has a local solution space containing this invariant space, so its monodromy rank is at least two. Its leading coefficient can be neither nonzero nor simply zero there, by Lemma VI.6.5; its zero has multiplicity at least two. The point $z = 1/729$ requires a further zero. The budget is again at least three. Reciprocity $a_n(\lambda) = \lambda^n a_n(1/\lambda)$, or equivalently $B_\lambda = B_{1/\lambda}$, proves the same lower bound at $\lambda = 1/8$.

Finally, at $\lambda = 1$ the two distinct points $z = 1/64$ and $z = 1$ coming from $S = \{1/8, -1\}$ both carry nontrivial monodromy of the even period. They require two leading zeros and hence width at least two. Together with the upper bounds this completes the proof. □

Remark VI.6.7 (What replaces positivity). The generic distinct-singularity count alone gives no sharp conclusion at $\lambda = 8$. Two independent vanishing cycles become supported at one point of the even generating function. Irreducibility makes both cycles visible in the continuation space of the chosen period; the simple-leading-zero lemma then forces that collision to retain multiplicity two in the differential leading coefficient. Thus the relevant obstruction is local monodromy rank, rather than merely the number of distinct singular points. The argument is specific to the cubic hypergeometric period; it does not assert a classification for all exponents.

VI.7 Gaussian mixture likelihood across all support and noise scales

Observe $X_i = \Theta_i + \sigma Z_i$, where Θ_i are independent with law G, Z_i are independent standard normal variables, and $\sigma > 0$ is known. The interval $I = [\mu - A, \mu + A]$, $A > 0$, is known. Let $\widehat{G}_I$ be any measurable maximizer of the original Gaussian mixture likelihood over all probability measures on I. No bound is imposed on the number of mixing atoms. Set $L = \log(eN)$ and $a = A/\sigma$, and define

$$R(N, A, \sigma) = \sigma \begin{cases} a \min\{1, L^{-1} \log(e + L/a^2)\}, & 0 < a \le \sqrt{L}, \\ \max\{L^{-1/2}, a/\sqrt{N}\}, & a > \sqrt{L}. \end{cases} \tag{VI.7.1}$$

The formulas at the breakpoint agree up to absolute factors.

Theorem VI.7.1 (The complete Gaussian likelihood curve)**.** *There are absolute constants $c, C > 0$ such that, for every $N \ge 3$, $A, \sigma > 0$, and $\mu \in \mathbb{R}$,*

$$cR(N, A, \sigma) \le \inf_{\widetilde{G}} \sup_{G \in \mathcal{P}(I)} \mathbb{E}_G W_1(\widetilde{G}, G) \le \sup_{G \in \mathcal{P}(I)} \mathbb{E}_G W_1(\widehat{G}_I, G) \le CR(N, A, \sigma).$$

The infimum permits arbitrary estimators knowing I and σ. Thus the ordinary constrained NPMLE attains the minimax rate, up to absolute factors, throughout the entire parameter plane.

For a fixed nonzero A/σ, the rate is of order $A \log\log N / \log N$. The fixed-compact-support rate was already known for the denoised method of moments; see Wu and Yang [58, Theorem 5 and Proposition 8]. Soloff, Guntuboyina and Sen [64, Theorem 11 and Remark 12] prove a deconvolution guarantee for the unrestricted NPMLE above a support-dependent sample-size threshold, and discuss adaptation as an open direction. The precise claim here is the uniform curve (VI.7.1) for a constrained likelihood fit. It does not resolve every adaptation question for an unrestricted fit. In fact, Proposition VI.7.4 below rules out unrestricted adaptation to this curve for arbitrarily small known intervals.

Translation and scaling reduce the proof to $\mu = 0$, $\sigma = 1$. Write φ, Φ for the standard normal density and CDF, $p_H = H * \varphi$, $q = N^{-1} \sum_i \delta_{X_i}$, and $h(p, r)^2 = \int (\sqrt{p} - \sqrt{r})^2 \, dx$.

VI.7.1 Likelihood accuracy and a Hermite inverse

We first record a density bound, uniformly for $A \le \sqrt{N}$:

$$\mathbb{E} h(p_{\widehat{G}_I}, p_G) \le \frac{C}{\sqrt{N}} \begin{cases} L, & A \le \sqrt{L}, \\ \sqrt{A}\, L^{3/4}, & \sqrt{L} < A \le \sqrt{N}. \end{cases} \tag{VI.7.2}$$

Here and below C denotes an absolute constant whose value may change. We give the covering argument to make the uniformity clear. Put $D = A \vee 1$. For complex z, $\sup_x |\varphi(x - z)| \le C \exp((\operatorname{Im} z)^2/2)$. On the ellipse $z = D\cos(\theta + iu)$, contour shifting bounds the kth Chebyshev coefficient, in uniform norm over real x, by

$$C \exp\{-c \min(k^2/D^2, k)\}.$$

For $k \le D^2$ take u a sufficiently small multiple of k/D^2; for $k > D^2$ take a sufficiently small fixed u. Truncating at $d = C\{D\sqrt{\log(C(D+1)/\beta)} + \log(C(D+1)/\beta)\}$ and gridding the $d+1$ Chebyshev moments yields a uniform density cover of radius β with $\log M \le Cd \log(C(d+1)/\beta)$. One can take each center to be an actual mixture by choosing one representative from each nonempty grid cell.

Take $U = D + \sqrt{12L}$ and $\beta = N^{-3}/(2U)$. The resulting bound is $\log M \le C(A\sqrt{L} + L)L$. Replace each center p_j by $u_j = p_j + \beta$ on $[-U, U]$ and $u_j(x) = \varphi(|x| - D)$ outside. It dominates every associated mixture and has mass between 1 and $1 + CN^{-3}$. The finite-envelope likelihood lemma, Lemma VI.B.3, gives $\mathbb{E}h^2 \le C(\log M + 1)/N + CN^{-3}$, because the fitted likelihood is at least the true likelihood. This proves (VI.7.2). No empirical-to-density relative entropy is used.

Lemma VI.7.2 (Uniform small-support inverse)**.** *If F, H are supported on $[-A, A]$, then for every integer $k \ge 1$,*

$$W_1(F, H) \le \frac{CA}{k} + A\{C(1 + \sqrt{k}/A)\}^k h(p_F, p_H). \tag{VI.7.3}$$

Proof. For a 1-Lipschitz test f with $f(0) = 0$, approximate $f(Au)/A$ on $[-1, 1]$ by a degree-k polynomial $P(u) = \sum_{j=0}^{k} b_j u^j$ with error C/k and supremum norm at most C. Such a polynomial follows from Lemma VI.B.1 by affine rescaling. Expanding P in Chebyshev polynomials, whose coefficients are bounded by twice its supremum norm, gives $\sum_j |b_j| \le C^k$: the sum of absolute monomial coefficients of T_j is at most 3^j, by its three-term recurrence.

Let He_j be the probabilists' Hermite polynomial. Its Gaussian expectation equals θ^j, and its translation formula and Gaussian orthogonality give

$$\| \mathrm{He}_j(Z + \theta)\|_2 \le \sum_{r=0}^{j} \binom{j}{r} A^{j-r}\sqrt{r!} \le (A + \sqrt{j})^j \qquad (|\theta| \le A).$$

Writing $p_F - p_H = (\sqrt{p_F} - \sqrt{p_H})(\sqrt{p_F} + \sqrt{p_H})$ and applying Cauchy–Schwarz therefore bounds their jth mixing moment difference by $2(A+\sqrt{j})^j h(p_F, p_H)$. Insert these bounds into $A\sum b_j A^{-j}\theta^j$. The constant term cancels, and the approximation error costs CA/k. Wasserstein duality proves (VI.7.3). □

VI.7.2 A boundary drift estimate for the actual constrained fit

The following estimate is the additional ingredient needed to transfer the large-support argument to a constrained NPMLE. Existence of a maximizer follows from weak compactness on $[-A, A]$ and continuity and positivity of the Gaussian likelihood. Write $H = \widehat{G}_I$. The likelihood directional derivative gives

$$S(\theta) = \frac{1}{N}\sum_i \frac{\varphi(X_i - \theta)}{p_H(X_i)} \le 1 \quad (-A \le \theta \le A), \qquad S = 1 \quad H\text{-a.e.}$$

Thus $R_\theta(i) = N^{-1}\varphi(X_i - \theta)/p_H(X_i)$ is a probability law at each contact point, with observation marginal q. At an interior contact, differentiation gives $\mathbb{E}_{R_\theta}(X - \theta) = 0$ and $\mathbb{E}_{R_\theta}(X - \theta)^2 \le 1$. At the right endpoint the mean residual is nonnegative, and at the left endpoint it is nonpositive. Define

$$B = H\{A\}\mathbb{E}_{R_A}(X - A) - H\{-A\}\mathbb{E}_{R_{-A}}(X + A), \qquad V = \int \mathbb{E}_{R_\theta}(X - \theta)^2\, dH(\theta),$$

where a missing endpoint atom contributes zero.

Lemma VI.7.3 (Boundary drift and posterior second moments)**.** *For $A \ge 1$, $B \ge 0$ and*

$$\mathbb{E}B \le C\left\{\mathbb{E}h(p_H, p_G) + \frac{A+1}{\sqrt{N}}\right\}, \qquad \mathbb{E}V \le C. \tag{VI.7.4}$$

Proof. Let $w = H\{A\} > 0$ and $\tau(x) = w\varphi(x - A)/p_H(x)$. The function τ takes values in $[0, 1]$ and is nondecreasing: its derivative is $\tau(x)\{A - \mathbb{E}_H(\Theta \mid X = x)\} \ge 0$. With $f(x) = (x - A)\tau(x)$, the right endpoint contribution is

$$b_+ = wS'(A) = (q - p_H)f = (q - p_G)f + (p_G - p_H)f \ge 0. \tag{VI.7.5}$$

The equality $(p_H)f = 0$ uses the mean of $N(A, 1)$.

We need a weighted monotone empirical-process bound. For every square-integrable real random variable Y and its empirical law q_Y, and every constant a,

$$\mathbb{E} \sup_{\substack{0\le t\le 1\\ t \text{ nondecreasing}}} |(q_Y - P_Y)[(y-a)t(y)]| \le \frac{C\sqrt{\mathbb{E}(Y-a)^2}}{\sqrt{N}}. \tag{VI.7.6}$$

Indeed a monotone t is a subprobability mixture of threshold indicators $\mathbf{1}_{[s,\infty)}$, with a possible constant term, which is the threshold $s = -\infty$. Symmetrization reduces the supremum to weighted Rademacher sums $N^{-1}\sum_i \varepsilon_i(Y_i - a)\mathbf{1}_{Y_i\ge s}$. Conditional on the data, sort the Y_i in decreasing order. These sums form a subset of the partial sums of independent centered increments $\varepsilon_i(Y_i - a)$. Doob's L^2 inequality bounds their expected maximum by $2\{\sum_i (Y_i - a)^2\}^{1/2}$. Jensen's inequality proves (VI.7.6). Applying it with $Y = X$ and $a = A$ bounds the expected absolute empirical term in (VI.7.5) by $C(A + 1)/\sqrt{N}$.

Only an upper bound is needed for its population term. If $f_+ = \max(f, 0)$ and $f_- = \max(-f, 0)$, retaining only positive contributions and using the Hellinger factorization gives

$$(p_G - p_H)f \le 2h(p_G, p_H)\{\|f_-\|_{L^2(p_H)} + \|f_+\|_{L^2(p_G)}\}.$$

Both norms are bounded by absolute constants. For the first, $\tau^2 p_H \le w\varphi(\cdot - A)$ gives $\int f^2 p_H \le w \le 1$. For the second, $f_+ \le (x - A)_+$ and $\Theta \le A$ under the true law, so $\int f_+^2 p_G \le \mathbb{E}Z_+^2 = 1/2$. This proves the required bound for b_+; reflection gives the left endpoint bound and hence the first assertion of (VI.7.4).

For the second assertion the interior contribution to V is at most one. The feasible shift from A to $A - 1$ gives

$$\mathbb{E}_{R_A} e^{-(X-A)} = e^{1/2} S(A - 1) \le e^{1/2}.$$

Since $u^2 \le Ce^u$ for $u \ge 0$, the inward residual satisfies $\mathbb{E}_{R_A}(A - X)_+^2 \le C$. The weighted outward residual is at most $w\mathbb{E}_{R_A}(X - A)_+^2 = q[(X - A)_+^2\tau(X)] \le q[(X - A)_+^2]$, whose expectation is at most $1/2$. The reflected left endpoint gives the same bounds. Summing the three contributions proves $\mathbb{E}V \le C$. □

For a global C^2 function F with bounded first two derivatives, Taylor's theorem now gives the deterministic inequality

$$|(H - q)F| \le \|F'\|_\infty B + \tfrac{1}{2}\|F''\|_\infty V. \tag{VI.7.7}$$

This explicitly keeps the boundary term that would be lost by incorrectly treating a constrained posterior as a martingale. The true Gaussian coupling gives $|(G - p_G)F| \le \|F''\|_\infty/2$.

VI.7.3 A two-scale inverse with boundary terms

For $m \ge 1$, an exact Markov channel sends $N(\theta, 1)$ to $\mathrm{Bin}(m, \phi_m(\theta))$, where $\phi_m(\theta) = \Phi(\theta/\sqrt{m})$. To see this, add to one observation X an independent centered Gaussian vector with covariance $mI_m - \mathbf{1}\mathbf{1}^\mathsf{T}$, and count the nonnegative coordinates of $X\mathbf{1} + \varepsilon$. The resulting coordinates are independent $N(\theta, m)$ variables. The singular covariance of ε is positive semidefinite and causes no problem. This channel is used only in the proof. For $m = \lceil A^2 \rceil$, $A \ge 1$, the inverse of ϕ_m on $[-A, A]$ is CA-Lipschitz.

For integers $1 \le S \le K \le m/2$ we claim

$$W_1(H, G) \le C\left\{\frac{A}{K} + \frac{S}{A}(1 + V) + B + W_1(q, p_G) + \frac{A}{S}e^{2K^2/m}h(p_H, p_G)\right\}. \tag{VI.7.8}$$

For a 1-Lipschitz test f on $[-A,A]$, extend $g(u)=f(\phi_m^{-1}(u))/A$ from $\phi_m([-A,A])$ to $[0,1]$ with an absolute Lipschitz constant. Lemma VI.B.1 gives polynomials P_S,P_K approximating g with errors $C/S, C/K$, with first derivatives bounded by C and second derivatives by CS, CK, respectively. The function $F_S(\theta)=AP_S(\Phi(\theta/\sqrt{m}))$ is globally C-Lipschitz, with

$$F_S''(\theta)=\frac{A}{m}\{P_S''(\Phi(z))\varphi(z)^2-P_S'(\Phi(z))z\varphi(z)\},\qquad z=\theta/\sqrt{m},$$

so $\|F_S''\|_\infty\le CS/A$. Equation (VI.7.7), the true coupling, and empirical Wasserstein duality control this low-degree part. The residual P_K-P_S has norm at most C/S and Bernstein coefficients of degree m at most $Ce^{2K^2/m}/S$, by Lemma VI.B.2. Its expectation difference is thus bounded by that quantity times the predictive L^1 distance after the channel, which is at most $2h(p_H,p_G)$ by data processing. The remaining approximation error is CA/K. Taking the supremum over f proves (VI.7.8).

The empirical CDF also gives the elementary uniform bound

$$\mathbb{E}W_1(q,p_G)\le C(A+1)/\sqrt{N}. \tag{VI.7.9}$$

Integrate its pointwise standard deviations, using the bound $1/2$ on $[-A,A]$ and the Gaussian tail bound outside that interval. The integral of the square root of a Gaussian tail is finite.

VI.7.4 Completion of the upper bound

Suppose first $A\le\sqrt{L}$ and put $d=\log(e+L/A^2)$. Choose a sufficiently small absolute $c_0>0$. If $c_0L/d<2$, the deterministic diameter bound $W_1(H,G)\le 2A$ is at most $CA\min(1,d/L)$. Otherwise put $k=\lfloor c_0L/d\rfloor\ge 2$. Then $k\asymp L/d$, and

$$k\log\{C(1+\sqrt{k}/A)\}\le C_1kd\le C_1c_0L\le L/8.$$

Here $k/A^2\le L/A^2$ and $d\ge 1$ justify the first inequality. Equations (VI.7.2) and (VI.7.3) yield

$$\mathbb{E}W_1(H,G)\le CA/k+CAe^{L/8}L/\sqrt{N}\le CA/k\le CAd/L.$$

The absorption is uniform because $k\le L$ and $e^{L/8}L^2/\sqrt{N}$ is bounded for $N\ge 3$.

Next let $\sqrt{L}<A\le\sqrt{N}$ and $m=\lceil A^2\rceil$. For all sufficiently large absolute N, choose

$$S=\lfloor A/(16\sqrt{L})\rfloor\vee 1,\qquad K=\left\lfloor\min\{m/2,\sqrt{\alpha mL}\}\right\rfloor,\qquad \alpha=1/64.$$

Then $1\le S\le K\le m/2$, $A/K+S/A\le C/\sqrt{L}$, and $A/S\le C\sqrt{L}$. Apply (VI.7.8), Lemma VI.7.3, and (VI.7.9). The density bound (VI.7.2) controls the remaining terms. In particular the high-degree term is at most $Ce^{2\alpha L}\sqrt{A}L^{5/4}/\sqrt{N}$; its ratio to $L^{-1/2}$ is at most $Ce^{2\alpha L}N^{-1/4}L^{7/4}$, which is uniformly bounded. The term $\mathbb{E}h$ in the boundary bound is smaller still. This proves $\mathbb{E}W_1(H,G)\le C\{L^{-1/2}+A/\sqrt{N}\}$.

If $A>\sqrt{N}$, the empirical-posterior coupling and Lemma VI.7.3 give $\mathbb{E}W_1(H,q)\le\sqrt{\mathbb{E}V}\le C$. Coupling G to p_G costs at most $\mathbb{E}|Z|$, so (VI.7.9) proves the bound $CA/\sqrt{N}$. The same elementary argument handles the bounded set of smaller sample sizes excluded in the preceding choice of S,K. This completes the upper bound for all $A>0$ and $N\ge 3$.

VI.7.5 Matching lower bounds and the cost of unrestricted fitting

For $A^2\le L$ set $d=\log(e+L/A^2)$ and $k=\lceil 100L/d\rceil$. Use the Chebyshev-arcsine pair of Lemma VI.B.4 on $[-A,A]$. Its moments through degree $k-1$ agree, and its Wasserstein separation is at least $A/(\pi k)\ge cA\min(1,d/L)$. The Gaussian identity

$$\int\frac{\varphi(x-\lambda)\varphi(x-\eta)}{\varphi(x)}\,dx=e^{\lambda\eta}$$

therefore gives

$$\int \frac{(p_{F_k^+} - p_{F_k^-})^2}{\varphi} \le 4 \sum_{j \ge k} \frac{A^{2j}}{j!}. \tag{VI.7.10}$$

For completeness put $x = L/A^2 \ge 1$. Since $d \le \log(4x)$ and $\log d \le d/2$,

$$\log \frac{k}{eA^2} \ge \log \frac{100x}{ed} \ge d/2, \qquad k \ge 2eA^2.$$

The series in (VI.7.10) is consequently at most $2(eA^2/k)^k \le 2e^{-50L}$. Cauchy–Schwarz bounds the single-observation total variation by the square root of this expression, and tensorization bounds the N-sample total variation by N times that quantity. Le Cam's two-point inequality gives the required small-support lower bound, including the clipped regime $d \gg L$.

For $A > \sqrt{L}$, use instead radius $M = \sqrt{L}/8$ and $k = \lceil 20L \rceil$ in the same construction. Its separation is $c/\sqrt{L}$, while the series in (VI.7.10) is bounded by $4\sum_{j \ge k} (L/64)^j/j! \le Ce^{-40L}$. Finally the two endpoint laws on $\{-A, A\}$ with weights $1/2 \pm c_1/\sqrt{N}$ have Wasserstein separation $cA/\sqrt{N}$. For small absolute c_1, the N-sample KL divergence of their latent Bernoulli experiments is bounded, and observation can only decrease total variation. Le Cam's inequality gives the second lower bound. Their maximum proves Theorem VI.7.1.

Proposition VI.7.4 (Failure of unrestricted full-plane adaptation)**.** *Let $\widehat{G}_{\mathbb{R}}$ be any exact unrestricted Gaussian location NPMLE with known noise σ. Under $G = \delta_\mu$,*

$$\mathbb{E} W_1(\widehat{G}_{\mathbb{R}}, \delta_\mu) \ge \sigma \sqrt{\frac{2}{\pi N}}.$$

Consequently this estimator cannot attain the known-interval minimax rate uniformly over all positive support radii.

Proof. Every unrestricted maximizer is supported in the sample range: moving a parameter outside that range to its nearest endpoint increases its Gaussian likelihood at every observation. The unrestricted contact condition permits differentiation in both directions, so $\mathbb{E}_{R_\theta} X = \theta$ at every contact. Integrating shows that the fitted mixing mean is exactly $\bar{X}$. Wasserstein duality gives $W_1(\widehat{G}_{\mathbb{R}}, \delta_\mu) \ge |\bar{X} - \mu|$; under the specified truth its expectation is the displayed value. On a known interval $[\mu - A, \mu + A]$, the estimator δ_μ has worst-case risk at most A. Letting $A\sqrt{N}/\sigma \to 0$ proves the assertion. □

The benchmark in this proposition supplies the interval center. If the center is also unknown, the class itself contains an unknown Gaussian location and has an additional location-estimation cost; the proposition does not rule out adaptation for that different class.

VI.8 Boundaries of the present extensions

The proofs identify specific places where exceptional parameters can be controlled. Their limits are equally specific. The full hypercubic group in every odd dimension greater than three is not classified here. The banana theorem addresses odd numbers of propagators; the even case and the leading-coefficient multiplicity problem remain separate. Algebraic independence of functions does not imply algebraic independence of their values at a fixed argument.

The beta classification covers all strict half-integer shapes, not all rational shapes or truncated distributions. The cubic recurrence result concerns the exponent three and recurrence width, not all exponents or the number of nonzero terms in a recurrence with gaps. The Gaussian theorem assumes a known support interval and known noise variance and proves rates up to absolute constants, not exact minimax constants.

Two further transfers were explored without a complete theorem. The sharp leading constant for the worst design ratio over all connected graphs remains undetermined: a proposed local variance bound fails on a clique with pendant vertices. A general rational-lemniscate extension also remains open; arbitrary algebraic-curve length arguments are insufficient. Neither attempt is included among the results.

The strongest next verification step is a specialist review of the Fourier intermediate-extension conventions, the all-mass inputs from [032], and the relation to existing classifications. The proofs here are a concrete research draft for that review; a journal-level or historical-significance assessment requires more than internal checks.

VI.A Previously established hypercubic rank and algebraicity theorem

Reproduced supporting input from the preceding draft. This appendix is not counted as a new result of the present paper.

For $n \geq 1$ and nonzero complex numbers $m_1, \dots, m_n$, define

$$c_k(m) = \sum_{a_1+\cdots+a_n=k} \binom{k}{a_1, \dots, a_n}^2 \prod_i m_i^{2a_i}, \tag{VI.A.1}$$

$$F_m(z) = \sum_{k\geq 0} \binom{2k}{k} c_k(m) z^k. \tag{VI.A.2}$$

The coefficient is the weighted number of length-$2k$ closed walks on the n-dimensional hypercubic lattice when each step in either coordinate direction i has weight m_i. Complex weights are allowed in the analytic statement. The series converges near zero.

Partition the masses by equality of their squares. Write their distinct nonzero squared values as $\mu_1^2, \dots, \mu_q^2$, with multiplicities $n_1, \dots, n_q$. Thus $\sum_j n_j = n$. For the standard representation $V = \mathbb{C}^2$ of SL_2, put

$$W = \bigotimes_{j=1}^{q} \mathrm{Sym}^{n_j} V = \bigoplus_{d\geq 0} (\mathrm{Sym}^d V)^{\oplus a_d}, \qquad Q = \prod_j (n_j + 1), \qquad b = \sum_d a_d. \tag{VI.A.3}$$

The decomposition in the middle uses the diagonal SL_2; the tensor product itself is irreducible for the product group SL_2^q.

Theorem VI.A.1 (All weights and all dimensions)**.** *The minimal order of a nonzero operator in $\mathbb{C}(z)\langle\partial_z\rangle$ annihilating F_m is*

$$r(m) = Q - b = \sum_{d\geq 1} d\, a_d = \prod_j (n_j + 1) - [u^{\lfloor n/2 \rfloor}] \prod_j (1 + u + \cdots + u^{n_j}). \tag{VI.A.4}$$

The minimal operator is irreducible over $\mathbb{C}(z)$. No hypothesis on distinct signed sums, thresholds, or generic choices of the nonzero masses is required.

More precisely, define

$$H_m(s) = s^{-1} F_m(s^{-2}).$$

Its minimal operator over $\mathbb{C}(s)$ is irreducible of the same order $r(m)$. The local monodromy of this equation at $s = \infty$ is unipotent, with one Jordan block of size d for each of the a_d summands with $d \geq 1$ in (VI.A.3).

Corollary VI.A.2 (Algebraicity classification)**.** *Allow zero weights and delete them before counting the dimension. Then $F_m(z)$ is algebraic over $\mathbb{C}(z)$ if and only if at most one weight is nonzero. With no nonzero weights it is 1; with one nonzero weight m, it is $(1 - 4m^2 z)^{-1/2}$. For two or more nonzero weights it is transcendental, even when signed thresholds collide or vanish.*

For equal squared masses, (VI.A.4) is $r = n$. For pairwise distinct squared masses it is

$$r = 2^n - \binom{n}{\lfloor n/2 \rfloor},$$

whose first values are $1, 2, 5, 10, 22, 44, 93$. For example the three-dimensional multiplicity patterns $[3]$, $[2, 1]$, $[1, 1, 1]$ give orders $3, 4, 5$. The first two cases agree with classical isotropic and tetragonal formulae. Generic three-dimensional order five has also been announced in recent work of Elkies, Grimm and Schwartz; it is not presented here as a new generic-three-dimensional discovery. The scope of Theorem VI.A.1 is its uniform formula in all dimensions and all squared-weight multiplicities, including every nonzero complex specialization.

VI.A.1 Provenance and standard differential-module inputs

The Bessel representation and the central-binomial relation to hypercubic walks are classical; see Guttmann [37]. Delves and Joyce's anisotropic three-dimensional order-four equation concerns the tetragonal family with two equal coordinate weights [38]. The recent public announcement [39] describes a generic order-five result and a genus-two period formula in dimension three; its stated exceptional-set restriction and the absence of a linked full paper are kept separate from the theorem proved here. A literature search does not certify historical priority for the all-dimensional statement.

We use the following standard facts about algebraic connections on a curve. They are stated explicitly to fix the extension convention. The Fourier–Laplace transform is the Weyl-algebra equivalence $x \mapsto -\partial_s$, $\partial_x \mapsto s$, and preserves simple holonomic modules. For an irreducible connection M on $\mathbb{G}_m$, its intermediate extension $j_{!*}M$ to $\mathbb{A}^1$ is simple. At a regular singular point with unipotent monodromy T, its vanishing cycles are $\operatorname{im}(T-1)$, with the restricted monodromy. A Jordan block of size $d+1$ therefore contributes a block of size d, and a size-one block contributes nothing. The last assertion follows directly from the nearby/vanishing-cycle maps: absence of a point-supported quotient makes the canonical map surjective, while absence of a point-supported submodule makes the variation map injective, with composite $T-1$.

The stationary-phase theorem gives, if M has only the finite singularity zero and all slopes at infinity are at most one,

$$\operatorname{rank}\mathcal{F}(j_{!*}M) = \dim \operatorname{im}(T_0 - 1). \tag{VI.A.5}$$

The formal germ of the Fourier transform at infinity is regular singular with this vanishing-cycle monodromy. Slope-one exponential factors at the original infinity contribute at finite transformed points, not to this rank formula through a slope-greater-than-one term. These are the formal stationary-phase statements of García López and Sabbah [40, 41]; in particular one must use the intermediate extension, rather than retain the entire nearby cycle space at zero.

We also use the curve de Rham index formula and the local-to-global compact-support sequence: for a connection N on $\mathbb{G}_m$,

$$\chi_{\mathrm{dR}}(\mathbb{G}_m, N) = -\operatorname{irr}_0(N) - \operatorname{irr}_\infty(N),$$

and

$$H^1_c \longrightarrow H^1 \longrightarrow H^1_{\mathrm{loc},0} \oplus H^1_{\mathrm{loc},\infty} \longrightarrow H^2_c \longrightarrow 0, \qquad H^2_c \cong H^0(N^\vee)^\vee.$$

All applications below have zero H^0 for the connection and its dual, and no formal slope-zero part at infinity. These standard index and duality inputs [45], as well as the Bessel differential Galois fact used next, are not asserted as new results.

VI.A.2 The Bessel connection and its complete local partition

Set $\theta_x = x\partial_x$. Let B_μ be the rank-two connection associated with

$$(\theta_x^2 - 4\mu^2 x^2)y = 0. \tag{VI.A.6}$$

It has solutions $I_0(2\mu x), K_0(2\mu x)$. Its differential Galois group over $\mathbb{C}(x)$ is SL_2; this is the classical non-half-integral Bessel case [42, 43]. At zero, I_0 is even and analytic, while

$$K_0(2\mu x) = -(\log x) I_0(2\mu x) + \text{an even analytic germ.}$$

The local monodromy is therefore a nontrivial unipotent block. At infinity the formal exponential factors are $e^{2\mu x}, e^{-2\mu x}$, with a common regular prefactor $x^{-1/2}$.

Lemma VI.A.3. *If the μ_j^2 are pairwise distinct and nonzero, the direct sum of the B_{μ_j} has differential Galois group SL_2^q. Consequently*

$$M = \bigotimes_j \mathrm{Sym}^{n_j} B_{\mu_j}$$

is irreducible of rank Q, and the function $g(x) = \prod_i I_0(2m_i x)$ generates this connection. At zero its monodromy has blocks of sizes $d+1$, with multiplicities a_d from (VI.A.3); its slopes at infinity are at most one.

Proof. The projections of the Galois group, and of its identity component, onto each SL_2 factor are surjective. The Goursat lemma for products of simple groups [44, Proposition 1.8.2] says that failure of the full product would identify two standard rank-two representations, after a finite algebraic base change and a rank-one twist. At a point over infinity their exponential-factor pairs would be

$$\{2\mu_i x, -2\mu_i x\} = \{2\mu_j x + q(x), -2\mu_j x + q(x)\}.$$

Comparing the sums gives $q(x) = 0$ for their irregular parts, and comparing their differences gives $\mu_i = \pm\mu_j$, a contradiction. The same argument remains valid after ramification at infinity. Thus the full product acts, and its external tensor product of symmetric powers is irreducible.

Evaluation at the product of the I_0 solutions is a nonzero morphism from this simple differential module to a differential field. Its kernel is zero, so its image is precisely the cyclic module generated by g. At zero all factors have the same nonzero logarithmic-monodromy constant. The tensor monodromy is $\exp(\kappa e)$ for the diagonal raising operator e on W. On $\mathrm{Sym}^d V$ this has one block of size $d+1$. At infinity tensor products of the Bessel formal factors have exponentials $\exp(2x\sum_j \lambda_j\mu_j)$, where $\lambda_j = n_j, n_j-2, \dots, -n_j$. Some sums may coincide or vanish; this only produces slope-zero or slope-one components, never slopes greater than one. □

Every summand in (VI.A.3) has the parity of n and contains exactly one weight-zero vector if n is even, or exactly one weight-one vector if n is odd. Thus b is the corresponding central weight multiplicity. The character

$$\prod_j (u^{n_j} + u^{n_j-2} + \cdots + u^{-n_j})$$

proves the coefficient formula in (VI.A.4). In particular $\mathrm{Sym}^n V$ occurs once, so $r \geq n$.

VI.A.3 An upper bound that eliminates the endpoint ambiguity

For $\mathrm{Re}\, s > 2\sum_i |m_i|$, termwise integration gives

$$H_m(s) = \int_0^\infty e^{-sx} g(x)\, dx = \sum_{k\geq 0} \frac{(2k)!}{s^{2k+1}} \sum_{|a|=k} \prod_i \frac{m_i^{2a_i}}{a_i!^2} = s^{-1} F_m(s^{-2}). \tag{VI.A.7}$$

To prove its order bound one cannot simply identify a unilateral Laplace transform with a Fourier transform: derivatives produce endpoint terms. We instead place this particular integral in a middle de Rham space and construct its closed cycle explicitly.

Work over $K = \mathbb{C}(s)$ and twist M by e^{-sx}; call the resulting connection N. On the cyclic vector η evaluating to g, its x-connection is $\nabla_x^M - s$ and its commuting s-connection is $\nabla_s = \partial_s - x$. The Bessel bases $b_\mu, \theta_x b_\mu$ and their symmetric tensor products give a $K[x]$-lattice N_0 stable under θ_x. Its residue at zero is nilpotent with the same b Jordan blocks as M.

The connection N is irreducible of rank $Q \geq 2$, also after extending the constant field algebraically, so it and its dual have no global horizontal vector. Since s is transcendental

over the mass constants, every formal exponential factor at infinity is nonzero and linear in x after the twist. Thus $\mathrm{irr}_\infty(N) = Q$, while zero is regular singular. The index formula gives $\dim_K H^1_{\mathrm{dR}}(\mathbb{G}_m, N) = Q$. Local cohomology at infinity vanishes, while at zero it is the cokernel of the nilpotent residue and has dimension b. The compact-support sequence therefore gives

$$H^1_{\mathrm{mid}}(N) := \ker\{H^1_{\mathrm{dR}}(N) \longrightarrow H^1_{\mathrm{loc},0}(N)\}, \qquad \dim_K H^1_{\mathrm{mid}}(N) = Q - b = r. \tag{VI.A.8}$$

Here $\omega = [\eta\, dx]$ belongs to this kernel. To check this without conflating dx and dx/x, complete the lattice to $\Lambda = N_0 \otimes K[[x]]$. The operator θ_x acts on $x^k\Lambda/x^{k+1}\Lambda$ as $k + \mathrm{Res}$, which is invertible for $k \geq 1$. It is therefore bijective on $x\Lambda$. There is $v \in x\Lambda$ with $\theta_x v = x\eta$, hence $\nabla_x v = \eta$, so $\eta\, dx$ is formally exact at zero. The commuting connection ∇_s preserves N_0, acts on both global and local cohomology, and hence preserves the kernel in (VI.A.8).

It remains to verify that (VI.A.7) is a period of this space. Choose branches along the positive ray for

$$h(x) = K_0(2m_1 x) \prod_{i>1} I_0(2m_i x).$$

This is a solution of M, including when some squared masses coincide. Continuing once around zero changes it by $-2\pi i\, g(x)$. Start at infinity on this ray, travel to ε, circle zero once, and return to infinity on the continued branch. With the factor e^{-sx} this is a closed rapid-decay twisted cycle: both ends decay at infinity for the stated half-plane, and there is no endpoint at zero. The integral of ω along its small circle is $O(\varepsilon(1 + |\log \varepsilon|))$ and tends to zero. Its two rays therefore give, up to a nonzero constant and orientation, exactly $H_m(s)$. The construction is valid for complex masses: fixed-ray continuations of K_0 have at most exponential growth, which is dominated by e^{-sx} in a sufficiently far right half-plane. Integrals of exact forms on this closed twisted cycle vanish.

Since $\omega, \nabla_s\omega, \dots, \nabla_s^r\omega$ are linearly dependent over K in (VI.A.8), period integration proves

$$\mathrm{ord}_{\mathbb{C}(s)} H_m \leq r. \tag{VI.A.9}$$

This argument accounts for the unilateral endpoint instead of silently dropping its possible rank-one contribution.

VI.A.4 Visibility of the simple Fourier constituent

Let $\mathcal{K} = j_{!*}M$ on the affine x-line. Lemma VI.A.3 and (VI.A.5) imply that $\mathcal{F}\mathcal{K}$ is simple of generic rank r. Its germ at $s = \infty$ has exactly the Jordan blocks of sizes d stated in the theorem.

We show that this constituent occurs in the module generated by the actual integral, not merely in a formal transform. Put $A_x = \mathbb{C}[x]\langle\partial_x\rangle$ and $\mathcal{E} = A_x g$. All its elements are entire functions of exponential type. Its restriction to $\mathbb{G}_m$ is M; therefore its composition factors are $\mathcal{K}$, once, and possibly copies of the point module δ_0. For an entire function f in this module, write $\widehat{f}(s) = \int_0^\infty e^{-sx} f(x)\, dx$ and set

$$\mathcal{L} = \{\widehat{f} : f \in \mathcal{E}\} + \mathbb{C}[s].$$

All integrals converge in a common sufficiently far right half-plane for each finite set of functions involved. Integration by parts gives

$$\widehat{xf} = -\partial_s \widehat{f}, \qquad \widehat{f'} = s\widehat{f} - f(0).$$

Consequently $\mathcal{L}$ is an A_s-module, and

$$\mathcal{L}/\mathbb{C}[s] \cong \mathcal{F}\mathcal{E}. \tag{VI.A.10}$$

The map is injective: $\widehat{f}(s) \to 0$ along the positive real axis, so if it is a polynomial it is zero, and uniqueness of the Laplace transform then gives $f = 0$. Thus (VI.A.10) is an actual isomorphism.

Since Fourier sends δ_0 to the trivial module $\mathbb{C}[s]$, every composition factor of $A_s H_m \subseteq \mathcal{L}$ is either $\mathcal{FK}$ (at most once) or the trivial affine-line connection.

If the former were absent, H_m would lie in an iterated extension of trivial affine-line connections. Every function in such a module is a polynomial: choose successive lifts of horizontal generators; the derivative of a lift is a polynomial combination of earlier lifts, so induction and integration prove the assertion. This contradicts $H_m(s) = s^{-1} + O(s^{-3})$. Hence $\mathcal{FK}$ occurs and $\operatorname{ord} H_m \geq r$. Combining with (VI.A.9) proves equality. Since this constituent already uses the full generic rank, the generic differential module generated by H_m is isomorphic to the irreducible generic module of $\mathcal{FK}$. Its monodromy at infinity is therefore exactly the one obtained by stationary phase.

VI.A.5 Descent and the algebraicity dichotomy

Multiplication by s does not change differential order or irreducibility over $\mathbb{C}(s)$. To descend from $sH_m(s) = F_m(s^{-2})$ to $z = s^{-2}$, use the derivation $\partial_z = -(s^3/2)\partial_s$, which preserves even functions. A relation of order r over $\mathbb{C}(s)$ can be decomposed using

$$\mathbb{C}(s) = \mathbb{C}(s^2) \oplus s\mathbb{C}(s^2).$$

Its even and odd parts are separately zero, because all derivatives of $F_m(s^{-2})$ with respect to z are even. At least one part is a nonzero relation over $\mathbb{C}(s^2) = \mathbb{C}(z)$ of no larger order. Conversely every relation over $\mathbb{C}(z)$ pulls back, so the minimal orders are equal. A nontrivial factorization over $\mathbb{C}(z)$ would pull back to one over $\mathbb{C}(s)$, contrary to the irreducibility already proved. This completes the proof of Theorem VI.A.1.

For $n \geq 2$, the summand $\operatorname{Sym}^n V$ in W supplies a unipotent monodromy block of size $n \geq 2$ at $s = \infty$. This monodromy has infinite order. The minimal differential module of an algebraic function has finite monodromy: after taking a finite normal algebraic extension containing all its conjugates, their constant linear span supplies a finite-branch solution space and a differential annihilator over the base field. Its minimal operator is a factor of this finite-monodromy operator. Thus H_m, and consequently F_m, cannot be algebraic for $n \geq 2$. The explicit formulas when zero or one weights remain prove the converse and Corollary VI.A.2.

The argument proves a statement about a function of the variable z. It does not assert transcendence of every special value at algebraic z, and the rank argument alone does not determine the full differential Galois group. The new group results in Sections VI.2 and VI.3 use the additional arguments given there.

VI.B Supporting approximation and testing estimates

We collect the approximation and testing estimates used in the proof.

Lemma VI.B.1 (Polynomial approximation with two controlled derivatives)**.** *For every 1-Lipschitz $g : [0,1] \to \mathbb{R}$ and every integer $k \geq 1$, there is a real polynomial P_k of degree at most k such that*

$$\|g - P_k\|_\infty \leq C/k, \qquad \|P_k'\|_\infty \leq C, \qquad \|P_k''\|_\infty \leq Ck.$$

All norms in this display are on $[0,1]$.

Proof. Subtract $g(0)$ and reflect g to a 2-periodic 1-Lipschitz function. Convolve with a positive normalized Jackson trigonometric kernel of degree $2s$. The resulting trigonometric polynomial Q_s approximates g within C/s, and $\|Q_s'\|_\infty \leq 1$, because convolution preserves the Lipschitz constant. The trigonometric Bernstein inequality applied to Q_s' gives $\|Q_s''\|_\infty \leq Cs$.

To replace Q_s by an ordinary polynomial, expand its finitely many exponentials $e^{i\pi j(x-1/2)}$, $|j| \leq 2s$, through Taylor degree $100s$. Each Fourier coefficient has magnitude at most the bounded supremum norm of the reflected function. For $|x - 1/2| \leq 1/2$, the exponential-series remainder and its first two derivatives are at most a polynomial in s times e^{-cs}, since $|\pi j(x - 1/2)| \leq \pi s$ while the truncation degree is $100s$. Their sum over $O(s)$ modes is still exponentially small. Take the real part and add back $g(0)$. With $s = \lfloor k/100 \rfloor$, this proves the assertions for large k; bounded k is handled by a constant polynomial and an enlarged absolute constant. □

Lemma VI.B.2 (Degree-elevated Bernstein coefficients)**.** *If P is a polynomial of degree $k \leq m/2$ and*

$$P(x) = \sum_{j=0}^{m} b_j \binom{m}{j} x^j (1-x)^{m-j},$$

then

$$\max_j |b_j| \leq Ce^{2k^2/m} \|P\|_{[0,1]}.$$

Proof. Put $M = \|P\|_{[0,1]}$ and $F(z) = (1+z)^m P(z/(1+z))$; its jth coefficient is $\binom{m}{j} b_j$. The maximum principle in the inverse Joukowski variable gives $|P((2 + w + w^{-1})/4)| \leq M|w|^k$ for $|w| \geq 1$. For $0 < j < m$ take $r = j/(m-j)$, $v = r/(1+r)^2$, and $u = 2\sqrt{v}\,|\sin(\phi/2)|$. Direct inversion of the Joukowski map shows that the larger modulus associated with $re^{i\phi}/(1 + re^{i\phi})$ is $(1+u)/\sqrt{1-u^2}$. Thus

$$\frac{|F(re^{i\phi})|}{M(1+r)^m} \leq (1-u^2)^{(m-k)/2}(1+u)^k \leq \exp(2k^2/m)\exp(-mu^2/8).$$

For the final inequality, use $k \leq m/2$, $\log(1+u) \leq u$, $\log(1-u^2) \leq -u^2$, and complete the square in $ku - mu^2/8$. The limiting point $u = 1$ is interpreted by continuity. The average of $\exp(-mu^2/8)$ over ϕ is at most $C/\sqrt{1+mv}$, by comparison with a Gaussian integral near zero. Stirling's factorial bounds give

$$\binom{m}{j} \left(\frac{j}{m}\right)^j \left(1 - \frac{j}{m}\right)^{m-j} \geq \frac{c}{\sqrt{1+mv}}.$$

Cauchy's coefficient formula cancels these two square-root factors and proves the claim. The endpoint coefficients are $P(0)$ and $P(1)$. □

Lemma VI.B.3 (A finite likelihood-envelope bound)**.** *Let p be a probability density and let $u_1, \ldots, u_M$ be nonnegative envelopes with total masses $U_a \in [1, 1+\eta]$. Suppose that, on an event*

E, a fitted density $\widehat{p}$ is dominated by some u_a and satisfies $\prod_i \widehat{p}(X_i) \geq \prod_i p(X_i)$, where the X_i are independent with density p. Then, for $x \geq 0$,

$$\mathbb{P}\left(E \cap \left\{h(p,\widehat{p}) > \sqrt{2\eta} + \sqrt{\eta + 2(\log M + x)/N}\right\}\right) \leq e^{-x},$$

and $\mathbb{E}[\mathbf{1}_E h(p,\widehat{p})^2] \leq C\eta + C(\log M + 1)/N$.

Proof. Put $r_a = u_a/U_a$. If $u_a \geq \widehat{p}$, then $\|r_a - \widehat{p}\|_1 \leq 2\eta$, so $h(r_a, \widehat{p}) \leq \sqrt{2\eta}$, and the likelihood comparison implies $\prod_i r_a(X_i)/p(X_i) \geq e^{-N\eta}$. The expectation of the square root of this product is the Nth power of the affinity, at most $e^{-Nh(p,r_a)^2/2}$. Markov's inequality and a union bound over the M deterministic envelopes prove the tail inequality. Integrating it proves the second assertion. The event E is only intersected with the bad event; no conditioning or independence assertion on E is used. □

Lemma VI.B.4 (A quantitative moment-matching pair)**.** *For every interval $[a - M, a + M]$ with $M > 0$ and integer $k \geq 1$, there are probability laws F_k^+, F_k^- on that interval which match moments of every degree less than k and satisfy*

$$W_1(F_k^+, F_k^-) \geq \frac{M}{\pi k}.$$

Proof. Push the measures $(1 \pm \frac{1}{2}T_k(x))\,dx/(\pi\sqrt{1-x^2})$ on $[-1,1]$ forward by $x \mapsto a + Mx$. Positivity, normalization and moment matching follow from $|T_k| \leq 1$ and Chebyshev orthogonality. With $x = \cos\theta$, their CDF difference is $\pm\sin(k\theta)/(\pi k)$, with a sign depending only on the orientation. Therefore

$$W_1(F_k^+, F_k^-) = \frac{M}{\pi k}\int_0^\pi |\sin(k\theta)|\sin\theta\,d\theta \geq \frac{M}{\pi k},$$

because $|\sin u| \geq \sin^2 u$ and $\int_0^\pi \sin^2(k\theta)\sin\theta\,d\theta = 1 + (4k^2-1)^{-1}$. □

Part VII
Beyond Generic Parameters: Global Classifications

Manuscript overview. We extend three proof mechanisms from preceding manuscripts to parameter regimes in which generic arguments leave exceptional cases unresolved. First, a finite-cover ramification bound, collision multiplicities, and an elementary reflection argument determine the full orthogonal differential Galois group of every odd-dimensional hypercubic period and every even-propagator banana period, for all nonzero complex weights. Combined with the previously established opposite-parity symplectic theorems, this gives both complete parity classifications. Second, a multiplicity-preserving formal-exponential budget proves the minimum recurrence width of every weighted central binomial power sum: it is $\lceil r/2 \rceil$ at weight 1, $\lfloor r/2 \rfloor$ at weight -1, and r at every other nonzero complex weight, for all $r \geq 2$. Third, for complementary rational beta exponents other than $1/2$, one explicit coefficient functional of a polynomial weight determines whether every algebraic-level quantile is algebraic or every such quantile is transcendental, on every algebraic truncation interval. Finally, a universal lower bound shows why unrestricted heteroscedastic mixture maximum likelihood cannot have expected transport consistency under arbitrary noise profiles, even when consistent estimation remains possible. Supporting results from the preceding work are identified and partly reproduced. These are mathematical claims with proofs and internal checks, not claims of established historical priority or completed external validation.

Integration note. This part completes several parameter classifications and reproduces earlier supporting arguments as identified in its appendices. Part VIII extends its recurrence theorem to arithmetic subsequences and adds different arithmetic applications.

VII.1 What is completed, and what the proof transfers accomplish

The central issue is uniformity. A theorem for generic parameters does not settle a parameter value where singularities collide, nor does an upper recurrence determine the minimum possible width. Likewise, an algebraic primitive test does not by itself exclude algebraic normalized values when the primitive has a logarithmic part. The following results address these specific obstructions.

VII.1.1 Two complete group classifications

For a tuple of nonzero complex weights, let n_j denote the multiplicities of its distinct squared weights, and put

$$Q = \prod_j (n_j + 1), \qquad P(u) = \prod_j (1 + u + \cdots + u^{n_j}), \qquad n = \sum_j n_j.$$

For hypercubic periods the rank is

$$R_H = Q - [u^{\lfloor n/2 \rfloor}]P(u).$$

For banana periods with $n \geq 2$, the rank is

$$R_B = \sum_{d \geq 2} (d-1)a_d, \quad \bigotimes_j \operatorname{Sym}^{n_j} \mathbb{C}^2 = \bigoplus_{d \geq 0} (\operatorname{Sym}^d \mathbb{C}^2)^{\oplus a_d}$$

under the diagonal SL_2. For even n, equivalently, $R_B = Q - [u^{n/2}]P - [u^{n/2-1}]P$. The group conclusions, in their standard representations, are

	n odd	n even
hypercubic	O_{R_H} (this paper)	Sp_{R_H} (prior)
banana	Sp_{R_B} (prior, $n \geq 3$)	O_{R_B} (this paper).

There are no additional restrictions on the collisions or vanishing of signed thresholds. Zero weights can be deleted; the rational zero- and one-factor banana cases are recorded separately.

The new ingredient excludes decomposition over finite algebraic covers. Permuting h components of dimension d requires at least $d(2h-2)$ units of negative-eigenspace multiplicity by Riemann–Hurwitz. An exposed reflection cannot contribute to this budget. Concentration bounds and exact small-dimensional arguments rule out the decomposition; a reflection then forces the full orthogonal group.

This also gives precise functional algebraic information. In the symplectic cases the first R jets are algebraically independent. In the orthogonal cases of rank at least three their relation ideal is generated by a single nondegenerate quadratic polynomial, and their transcendence degree is $R - 1$. These are statements about functions, not arithmetic claims about each individual special value.

VII.1.2 A complete width formula

For

$$B_{r,\lambda}(n) = \lambda^{-n} \sum_{k=0}^{2n} \binom{2n}{k}^r \lambda^k, \quad r \geq 2, \quad \lambda \in \mathbb{C}^*,$$

Theorem VII.4.1 gives

$$\operatorname{minwidth}_{\mathbb{C}(n)} B_{r,\lambda} = \begin{cases} \lceil r/2 \rceil, & \lambda = 1, \\ \lfloor r/2 \rfloor, & \lambda = -1, \\ r, & \lambda \notin \{1, -1\}. \end{cases}$$

The proof counts formal exponential factors with multiplicity after factorial normalization. It therefore retains the information that a count of distinct finite singularities loses at collisions. Katz's classical differential Galois calculation supplies cyclicity; an explicit concomitant handles the adjoint representation at odd r and weight -1. Width means largest shift minus smallest shift. It must not be replaced by the number of nonzero terms in an arbitrarily sparse recurrence.

VII.1.3 An effective arithmetic dichotomy

Fix $r \in \mathbb{Q} \cap (0,1) \setminus \{1/2\}$, algebraic endpoints $0 \le u < v \le 1$, and a real algebraic polynomial $w(x) = \sum_j w_j x^j$ positive on (u, v). Normalize the density $w(x)x^{r-1}(1-x)^{-r}$ on this interval. The single coefficient

$$c_r(w) = \sum_j w_j \frac{(r)_j}{j!}$$

settles every algebraic probability level:

$c_r(w) = 0$	all internal algebraic-level quantiles are algebraic,
$c_r(w) \neq 0$	all internal algebraic-level quantiles are transcendental.

The exclusion argument combines Baker's independence theorem with coefficient conjugation and strict ordering of complementary beta densities. The argument is uniform in both truncation endpoints. Section VII.5 also gives an explicit polynomial equation in the algebraic case and the equivalent linear-independence statement.

VII.1.4 A necessary boundary for likelihood methods

For an exact unrestricted Gaussian mixture NPMLE with known noise standard deviations σ_i, Theorem VII.6.1 proves

$$\mathbb{E}_{\delta_\mu} W_1(\widehat{G}, \delta_\mu) \ge \frac{\max_i \sigma_i}{2\sqrt{2\pi}\, N^2}.$$

With $N - 1$ unit-noise observations and one standard deviation N^3, this expected error diverges, although the minimax error on the specified growing compact-support class tends to zero. This is a failure of an unrestricted extension, not a contradiction of the published bounded-covariance theorem. It illustrates why the likelihood-contact principle must be paired with an information condition before a global optimality conclusion is attempted.

VII.1.5 Dependencies, provenance, and verification status

The new orthogonal group arguments use the prior all-mass rank, irreducibility, pairing, and infinity-monodromy theorems. The hypercubic rank proof, pairing, and opposite-parity group proofs are reproduced in the appendices. The banana rank input is precisely stated and traced to supplied manuscript [032] [49]; its full original development is not reproduced. The arithmetic proof uses classical Baker theorems, and the recurrence proof uses the classical Kloosterman group calculation and existing upper bounds. The proof transfers continue the preceding manuscripts' Bessel and singularity methods, arithmetic separation methods, and likelihood contact methods [65, 66]. Separate internal checks cover the main arguments, exceptional cases, and the invariant-tensor correction. External expert review and full priority assessment remain outstanding.

VII.2 Full orthogonal groups in every odd hypercubic dimension

Let $n \ge 1$ be odd and $m_1, \dots, m_n \in \mathbb{C}^\times$. Define

$$F_m(z) = \sum_{k\ge 0} (2k)! \left(\sum_{a_1+\cdots+a_n=k} \prod_i \frac{m_i^{2a_i}}{a_i!^2} \right) z^k, \qquad H_m(s) = s^{-1} F_m(s^{-2}).$$

Group the masses by equality of their squares. Write the distinct squares as μ_j^2 and their multiplicities as n_j, $1 \le j \le q$. Put

$$Q = \prod_j (n_j + 1), \quad b = [u^{(n-1)/2}] \prod_j (1 + u + \cdots + u^{n_j}), \quad R = Q - b.$$

Theorem VII.2.1 (Every odd dimension, without threshold restrictions)**.** *The full differential Galois groups of H_m over $\mathbb{C}(s)$ and of F_m over $\mathbb{C}(z)$ are $\mathrm{O}_R(\mathbb{C})$ in their standard representations. This holds for every nonzero complex mass tuple, including all collisions and vanishing of signed thresholds. For $n \ge 3$, the minimal order remains R after every finite algebraic extension of the coefficient field. For either function f, the field generated by $f, f', \dots, f^{(R-1)}$ has transcendence degree $R-1$, and its polynomial relation ideal is generated by one nondegenerate quadratic polynomial.*

The new step is the exclusion of every finite-cover decomposition. A reflection and a nontrivial unipotent in an irreducible orthogonal group would not alone exclude such a decomposition. We use the total collision multiplicity together with Riemann–Hurwitz.

VII.2.1 Previously proved differential-module input

We use the all-weight rank theorem (Theorem VII.A.1) and the reflected middle-pairing construction (Lemma VII.B.1) proved in the preceding work. In precise terms, put $B_\mu : (\theta_x^2 - 4\mu^2 x^2) y = 0$ and $M = \bigotimes_j \mathrm{Sym}^{n_j} B_{\mu_j}$. The cyclic module of H_m is the simple positive-rank Fourier transform $\mathcal{F}(j_{!*}M)$, equivalently $H^1_{\mathrm{mid}}(M \otimes e^{-sx})$; its rank is R. It has a nondegenerate horizontal symmetric form when n is odd. The same form descends to F_m: the simultaneous involution $(s, x) \mapsto (-s, -x)$ fixes the cyclic class $s[\eta\, dx]$ and its z-derivatives and preserves the residue pairing.

If

$$\bigotimes_j \mathrm{Sym}^{n_j} \mathbb{C}^2 = \bigoplus_{d\ge 0} (\mathrm{Sym}^d \mathbb{C}^2)^{\oplus a_d}$$

under diagonal SL_2, then infinity monodromy of H_m is unipotent with blocks J_d of multiplicities a_d, $d \ge 1$. In particular its largest block J_n occurs exactly once. These are established all-weight statements, not generic assumptions.

Here are the finite local data, also fixing the convention at zero. The formal germ of M at infinity is

$$\bigoplus_{\lambda\in\Lambda} E^{c_\lambda x} \otimes R_n, \quad \Lambda = \prod_j \{n_j, n_j - 2, \dots, -n_j\}, \quad c_\lambda = 2 \sum_j \lambda_j \mu_j,$$

where R_n is the rank-one regular germ with solution $x^{-n/2}$. Its monodromy is -1. If

$$k_c = \#\{\lambda \in \Lambda : c_\lambda = c\},$$

local Fourier stationary phase gives regular vanishing cycles of dimension k_c with monodromy -1 at $s = c$; it gives zero vanishing cycles elsewhere. This follows from the local Fourier formula at every finite point, including zero [77, Corollary 2.3(1), §3.3], and preservation of regular local

monodromy [41, Remarks 5.2, 5.3(4), 5.7]. Colliding exponential factors are a direct sum already before the transform, so their multiplicities add.

The Fourier module is an intermediate extension. Nearby monodromy therefore has no nontrivial unipotent part: such a part would give an eigenvalue-one vanishing cycle. It is an involution T_c with -1-eigenspace of dimension k_c. The equation is regular singular at every finite point, since the irregular part is unchanged by formal vanishing cycles, and is regular singular at infinity by the rank theorem. In particular

$$T_c^2 = 1, \qquad \operatorname{rank}(T_c - 1) = k_c, \qquad \sum_c k_c = Q. \tag{VII.2.1}$$

Schlesinger's theorem identifies its differential Galois group with the Zariski closure of these local monodromies.

VII.2.2 Two elementary concentration bounds

Lemma VII.2.2 (Maximum threshold multiplicity)**.** *For every mass tuple, $k_c \le b$ for every c.*

Proof. Choose $\xi \in \mathbb{C}^\times$ such that $\operatorname{Re}(\xi\mu_j) \neq 0$ for every j. Change the sign of each μ_j if necessary so these real parts are positive; this permutes Λ. Write $\lambda_j = n_j - 2a_j$ with $0 \le a_j \le n_j$. In the product poset $\prod_j \{0, \dots, n_j\}$ with coordinatewise order, a fiber of $\sum_j a_j\mu_j$ is an antichain: strict coordinatewise increase strictly increases its real projection.

A product of finite chains has a symmetric-chain decomposition, which we recall explicitly. A rectangle $\{0, \dots, a\} \times \{0, \dots, d\}$ with $a \le d$ splits into the chains

$$(i, 0), (i, 1), \dots, (i, d-i), (i+1, d-i), \dots, (a, d-i), \quad 0 \le i \le a.$$

Their endpoint ranks add to $a + d$. Taking products of each existing symmetric chain with a new chain and using this rectangle construction proves the assertion by induction. Every chain in the decomposition passes once through rank $(n-1)/2$. Hence the number of chains is b, and an antichain meets each at most once. This proves the bound. □

Lemma VII.2.3 (Uniform chains compared with coin flips)**.** *For every partition $n = \sum_j n_j$,*

$$\frac{b}{Q} \le 2^{-n} \binom{n}{\lfloor n/2 \rfloor}.$$

Here the lemma allows either parity and defines $b = [u^{\lfloor n/2 \rfloor}] \prod_j (1 + \cdots + u^{n_j})$. More generally, the centered sum of independent uniforms on $\{0, \dots, n_j\}$ has at most the mass of a centered $\mathrm{Bin}(n, 1/2)$ *in every centered interval. Consequently $Q > 4b$ for every odd $n \ge 9$.*

Proof. Let U_m be uniform on $\{0, \dots, m\}$ and let B_m have the binomial distribution with parameters $(m, 1/2)$. Center both at $m/2$. The binomial law places at least as much mass as the uniform law in every centered interval. Indeed its probabilities are symmetric and decrease away from the center, so the average of its central h probabilities is at least the average of all $m + 1$ probabilities, namely $1/(m+1)$.

This comparison is preserved by convolution with a symmetric unimodal law. To see this, for any centered interval I, the function $x \mapsto \mathbb{P}(x + Y \in I)$ is symmetric and decreases with $|x|$ when Y has such a law. Summation by parts expresses its expectation as a nonnegative combination of centered-interval probabilities. The statement applies equally to integer and half-integer centered lattices. Symmetric unimodal laws are closed under convolution: each is a nonnegative mixture of uniform laws on centered lattice intervals, and the convolution of two such uniform laws has probabilities proportional to interval-overlap lengths, which are symmetric and unimodal.

Replace the independent variables U_{n_j} one at a time by B_{n_j}. The probability in any centered interval cannot decrease. Their final sum has the binomial law B_n. For odd n, the two central probabilities of the original sum are both b/Q; compare the centered interval containing precisely these two lattice points. For even n, compare the singleton at the center, whose original probability is b/Q. The asserted inequality follows in both parities. At $n = 9$ its right side is $126/512 < 1/4$. For successive odd indices the ratio is $\binom{2h+3}{h+1}2^{-2h-3}/(\binom{2h+1}{h}2^{-2h-1}) = (2h+3)/(2h+4) < 1$, proving the last assertion. □

VII.2.3 The branch budget of a hypothetical block decomposition

Lemma VII.2.4 (Isotypic blocks and a ramification budget)**.** *Let a regular-singular connection on the punctured projective line have irreducible monodromy with Zariski closure $G \subseteq \mathrm{GL}(V)$. Suppose infinity monodromy is unipotent with a unique largest Jordan block of size $\ell \geq 2$, and every finite monodromy T_c has eigenvalues in $\{1, -1\}$. Write k_c for its -1 generalized eigenspace dimension. Then G^0 is reductive and its isotypic components have multiplicity one. If there are t such components of dimension d, then*

$$\dim V = td, \qquad d \geq \ell, \qquad d(2t-2) \leq \sum_c k_c.$$

If h distinct finite local monodromies are reflections, the last right side can be replaced by $\sum_c k_c - h$. If $t > 1$, also $d \leq \max_c k_c$.

Proof. The unipotent radical of G^0 is characteristic in G^0 and normal in G. Its fixed space is nonzero and G-invariant, so irreducibility and faithfulness make that radical trivial. Thus $V|_{G^0}$ is completely reducible. Its isotypic components are permuted transitively by G; let their common irreducible dimension be d and their common multiplicity be e.

Every unipotent element of G belongs to G^0, since the Zariski closure of its cyclic subgroup is connected. On each isotypic component its Jordan blocks are repeated e times. The unique largest block at infinity forces $e = 1$ and $d \geq \ell$.

The block permutation action defines a connected degree-t cover of the punctured sphere. Complete it to a map from a compact Riemann surface of genus g to $\mathbb{P}^1$. Infinity is unramified because its unipotent monodromy belongs to G^0. For each T_c, its unipotent part belongs to G^0, so the block action is that of its semisimple part. The latter is an involution. If it swaps a_c disjoint pairs of blocks, each swapped pair contributes d eigenvalues -1: its matrix is $\left(\begin{smallmatrix} 0 & A \\ A^{-1} & 0 \end{smallmatrix}\right)$ in suitable bases. Thus $a_c d \leq k_c$. Riemann–Hurwitz, equivalently Euler characteristic for a triangulation with branch values as vertices, gives

$$2t - 2 \leq 2t + 2g - 2 = \sum_c a_c.$$

A reflection has $k_c = 1 < d$, so cannot permute blocks and its contribution can be removed. This proves both budgets. If $t > 1$, some finite monodromy must permute blocks: the finite local loops generate the fundamental group, and the action is transitive. At that point $k_c \geq d$, proving the last claim. □

Assume $n \geq 3$ until the rank-one case in the final proof. Apply the lemma to the equation of H_m, with $\ell = n$. Its Galois group $G \subseteq \mathrm{O}_R$ acts irreducibly by the prior rank theorem. If its identity component has t isotypic blocks of dimension d, then

$$d \geq n, \qquad R = td, \tag{VII.2.2}$$

and (VII.2.1) gives

$$2t - 2 \leq \frac{Q}{d} = \frac{Qt}{Q-b}. \tag{VII.2.3}$$

We must prove $t = 1$.

At least two finite monodromies are individual reflections. Indeed maximize $\mathrm{Re}(\xi c_\lambda)$ as in Lemma VII.2.2. Its unique maximum is attained by the endpoint sign tuple, has positive projection, and has multiplicity one. Its negative also has multiplicity one. Since $d \geq 3$, neither reflection can permute blocks. Removing their contributions from the branch budget improves (VII.2.3) to

$$d(2t - 2) \leq Q - 2. \tag{VII.2.4}$$

Furthermore, if $t > 1$, some finite local monodromy must permute blocks, since these generate the monodromy group and infinity fixes all blocks. Lemma VII.2.2 then forces

$$d \leq b. \tag{VII.2.5}$$

For odd $n \geq 9$, Lemma VII.2.3 gives $Q > 4b$. Inequality (VII.2.3) rearranges to $t \leq 2(Q - b)/(Q - 2b) < 3$. Thus $t > 1$ would mean $t = 2$ and $d = (Q - b)/2 > b$, contradicting (VII.2.5). For $n = 3$, the possible ranks are $3, 4, 5$, and (VII.2.2) already forces $t = 1$.

For completeness, the remaining finite check for $n = 5, 7$ is shown in the table. Its last column lists every $t \geq 2$ satisfying $R = td$, $d \geq n$, $d \leq b$, and (VII.2.4); it is obtained by checking the integer divisors of the displayed R.

(n_j)	Q	b	R	remaining t
(5)	6	1	5	none
$(4,1)$	10	2	8	none
$(3,2)$	12	3	9	none
$(3,1,1)$	16	4	12	none
$(2,2,1)$	18	5	13	none
$(2,1,1,1)$	24	7	17	none
(1^5)	32	10	22	none
(7)	8	1	7	none
$(6,1)$	14	2	12	none
$(5,2)$	18	3	15	none
$(5,1,1)$	24	4	20	none
$(4,3)$	20	4	16	none
$(4,2,1)$	30	6	24	none
$(4,1,1,1)$	40	8	32	none
$(3,3,1)$	32	7	25	none
$(3,2,2)$	36	8	28	none
$(3,2,1,1)$	48	11	37	none
$(3,1,1,1,1)$	64	15	49	none
$(2,2,2,1)$	54	13	41	none
$(2,2,1,1,1)$	72	18	54	none
$(2,1,1,1,1,1)$	96	25	71	none
(1^7)	128	35	93	3

In the last case all seven squared masses are distinct. Choose ξ also to avoid equality between the absolute real projections of any two masses. After changing signs and reordering, write these projections as $0 < a_1 < a_2 < \cdots < a_7$, with oriented masses $\nu_1, \ldots, \nu_7$. Among their subset sums, $0, \nu_1, \nu_2$ are the unique sums with their respective projected values: any other singleton has larger projection than a_2, and every subset with two or more entries has projection at least $a_1 + a_2$. Hence the signed exponents

$$2\sum_j \nu_j, \qquad 2\sum_j \nu_j - 4\nu_1, \qquad 2\sum_j \nu_j - 4\nu_2$$

all have multiplicity one. Their negatives do too, and the six are distinct because $\sum_j a_j > 2a_2$. None of these six reflections can permute blocks. The last possibility $t = 3, d = 31$ would therefore require $31(2 \cdot 3 - 2) \le 128 - 6$, or $124 \le 122$, a contradiction. We have proved that G^0 acts irreducibly in every odd dimension $n \ge 3$.

VII.2.4 A reflection forces the full orthogonal Lie algebra

Lemma VII.2.5. *Let $G \subseteq \mathrm{O}(V, B)$ be an algebraic subgroup such that G^0 acts irreducibly on V, with $\dim V \ge 3$. If G contains an orthogonal reflection, then $G^0 = \mathrm{SO}(V, B)$ and $G = \mathrm{O}(V, B)$.*

Proof. Write the reflection as $r_u(v) = v - 2B(u, v)u/B(u, u)$, where $B(u, u) \neq 0$. Put $\mathfrak{g} = \operatorname{Lie} G^0$ and $W = \mathfrak{g}u \subseteq u^\perp$. For $v, w \in V$, set $L_{v,w}(x) = B(v, x)w - B(w, x)v$. Since G^0 is normal in G, $\mathfrak{g}$ is stable under conjugation by r_u. For $A \in \mathfrak{g}$, direct calculation gives

$$A - r_u A r_u = \frac{2}{B(u, u)} L_{u,Au}.$$

Thus $L_{u,w} \in \mathfrak{g}$ for all $w \in W$. For $v, w \in W$ the identity $[L_{u,w}, L_{u,v}] = B(u, u)L_{w,v}$ gives $L_{w,v} \in \mathfrak{g}$. For $A \in \mathfrak{g}$ and $w \in W$,

$$[A, L_{u,w}] = L_{Au,w} + L_{u,Aw}.$$

The first term on the right is already in $\mathfrak{g}$, so $L_{u,Aw} \in \mathfrak{g}$. Applying it to u shows $B(u, u)Aw - B(Aw, u)u \in W$. Hence $\mathbb{C}u + W$ is $\mathfrak{g}$-invariant, and therefore G^0-invariant since G^0 is connected. Irreducibility yields $\mathbb{C}u + W = V$, so $W = u^\perp$. The operators just obtained span all of $\mathfrak{so}(V, B)$. Therefore $G^0 = \mathrm{SO}(V, B)$. The reflection has determinant -1, giving the other component. □

Completion of Theorem VII.2.1. For $n \ge 3$ we have proved connected irreducibility and exhibited a reflection. The lemma gives the full orthogonal group for H_m. The quadratic base change $z = s^{-2}$ and rational gauge retain its identity component. The descended symmetric pairing puts the group of F_m inside O_R. After the quadratic base change its Galois group is a subgroup of the original group, yet it already equals the full O_R by the result for H_m. Hence the original group of F_m also equals O_R. For $n = 1$, the explicit formula $F_m(z) = (1 - 4m_1^2 z)^{-1/2}$ gives $\mathrm{O}_1 = \{\pm 1\}$ for both variables. Its square is rational; explicitly $H_m(s)^2 = (s^2 - 4m_1^2)^{-1}$. These rational functions are not squares in their respective rational function fields. Thus the one-jet field has transcendence degree zero and its relation ideal is generated by the indicated irreducible quadratic over that base field. No absolute irreducibility is asserted in this rank-one case.

For $n \ge 3$, finite algebraic base extension preserves the connected SO_R, which acts irreducibly. Thus the order cannot fall. The orbit of a nonzero vector under O_R has dimension $R - 1$, whether its quadratic norm is zero or nonzero. The Picard–Vessiot orbit-dimension formula gives the stated transcendence degree. In the cyclic derivative basis, the inverse Gram matrix of the horizontal form gives a quadratic relation $\mathbf{f}^T C^{-1} \mathbf{f} = c$, $c \in \mathbb{C}$. Its quadratic part has rank $R \ge 3$, so this polynomial is absolutely irreducible. The relation ideal is a height-one prime containing this irreducible quadratic, hence is precisely its principal ideal. □

Corollary VII.2.6 (All dimensions and all weights)**.** *Combining Theorem VII.2.1 with the previously proved all-even-dimensional Theorem VII.C.1 gives, for every $n \ge 1$ and every nonzero complex mass tuple,*

$$G(F_m) = G(H_m) = \begin{cases} \mathrm{O}_R(\mathbb{C}), & n \text{ odd}, \\ \mathrm{Sp}_R(\mathbb{C}), & n \text{ even}, \end{cases} \qquad R = \prod_j (n_j + 1) - [u^{\lfloor n/2 \rfloor}] \prod_j (1 + u + \cdots + u^{n_j}).$$

Zero weights may be deleted before applying this classification; if all weights vanish then $F = 1$ and $H = s^{-1}$ have trivial groups. The new contribution of this section is the odd-dimensional case, including every collided-threshold specialization.

VII.3 Full orthogonal groups for every even-propagator banana period

Let $n \geq 2$ be even and let $m_1, \dots, m_n \in \mathbb{C}^\times$. We use the banana period

$$\varpi_m(t) = \sum_{k \geq 0} \frac{c_k(m)}{t^{k+1}}, \qquad c_k(m) = \sum_{a_1+\cdots+a_n=k} \binom{k}{a_1, \dots, a_n}^2 \prod_i m_i^{2a_i}.$$

Group the masses by equality of their squares, and denote the multiplicities by $n_1, \dots, n_q$. Put

$$Q = \prod_j (n_j + 1), \qquad B(u) = \prod_j (1 + u + \cdots + u^{n_j}), \qquad b = [u^{n/2}]B(u), \qquad c = [u^{n/2-1}]B(u),$$

and

$$R = Q - b - c, \qquad \delta = \begin{cases} 1 & \text{if every } n_j \text{ is even,} \\ 0 & \text{otherwise,} \end{cases} \qquad \kappa = (-1)^{n/2}, \qquad S = \frac{Q - \kappa\delta}{2}. \quad \text{(VII.3.1)}$$

Here R is the all-mass minimal order established in the supplied manuscript [49]. Indeed, if $\bigotimes_j \operatorname{Sym}^{n_j} \mathbb{C}^2 = \bigoplus_d (\operatorname{Sym}^d \mathbb{C}^2)^{\oplus a_d}$ under the diagonal SL_2, then $R = \sum_{d \geq 2}(d-1)a_d = Q - 2b + a_0 = Q - b - c$.

Theorem VII.3.1 (No exceptional nonzero mass tuples)**.** *For every even $n \geq 2$ and every $m \in (\mathbb{C}^\times)^n$, the full differential Galois group over $\mathbb{C}(t)$ of the minimal equation of ϖ_m is*

$$\mathrm{O}_R(\mathbb{C}).$$

There is no distinct-threshold or nonzero-threshold hypothesis. For $n \geq 4$, the minimal equation remains irreducible of order R after every finite algebraic extension of $\mathbb{C}(t)$.

The new issue is irreducibility under the connected group. Irreducibility under the full group, a reflection, and a nontrivial unipotent do not suffice: an orthogonal group permuting several isomorphic blocks would satisfy those three conditions. We exclude such a decomposition by counting ramification of its finite permutation cover. Collisions contribute to this count, but the total available multiplicity is independent of the collisions.

VII.3.1 The all-mass local input and its negative-eigenspace budget

We use the previously proved all-mass order and middle-cohomology theorem of [49]: the minimal module is irreducible of rank R, preserves a nondegenerate symmetric form, and has unipotent monodromy at infinity with blocks J_{d-1} for the summands with $d \geq 2$. In particular its largest block is J_{n-1} and occurs exactly once. These facts hold without conditions on signed sums. We now compute the finite semisimple local data needed for the extension of the Galois theorem.

Choose representatives μ_j for the distinct squared masses, and put

$$\Lambda = \prod_j \{n_j, n_j - 2, \dots, -n_j\}, \qquad c_\lambda = \sum_j \lambda_j \mu_j, \qquad z_0 = \#\{\lambda \in \Lambda : c_\lambda = 0\}.$$

The coordinatewise negation involution on Λ has a fixed point precisely when $\delta = 1$.

Lemma VII.3.2 (Finite local multiplicities)**.** *The equation is regular singular everywhere. At every nonzero squared threshold a, its local monodromy T_a is an involution, whose negative eigenspace has dimension*

$$k_a = \frac{1}{2}\#\{\lambda \in \Lambda : c_\lambda^2 = a\}.$$

At zero the semisimple part of T_0 is an involution whose negative eigenspace has dimension

$$k_0 = \frac{z_0 - \kappa\delta}{2}.$$

Consequently

$$k_0 + \sum_{a\neq 0} k_a = S. \tag{VII.3.2}$$

At least one nonzero threshold has $k_a = 1$.

Proof. We recall the two-Fourier-transform construction to account for zero as well as nonzero collisions. Put $y = u^2$ and $g_m(y) = \prod_i I_0(m_i\sqrt{y})$. Its Bessel module $M = \bigotimes_j \operatorname{Sym}^{n_j} B_{\mu_j}$ has, on the formal u-cover at infinity, a direct-sum decomposition into rank-one factors

$$E^{c_\lambda u} \otimes u^{-n/2}, \qquad \lambda \in \Lambda.$$

Invertible formal series have been absorbed by rank-one formal gauges. This is a splitting before equal exponential factors are grouped; a collision therefore introduces no formal extension.

For $c_\lambda \neq 0$, the pair $\lambda, -\lambda$ descends to a rank-two ramified factor at $y = \infty$. The first Fourier transform gives $E^{c_\lambda^2/(4s)}$ times a regular rank-one factor with power $s^{(n-3)/2}$ and monodromy -1. This is the stationary-phase calculation: the measure $2u\,du$, the value of $u^{1-n/2}$ at $u = c_\lambda/(2s)$, and the Gaussian factor $s^{-1/2}$ produce the displayed power. After inversion $s = 1/v$, the rational gauge v^{-1}, and the second Fourier transform, this factor supplies a vanishing-cycle line of monodromy -1 at $t = c_\lambda^2$, with the normalization $t = 4\tau$ for the second Fourier variable. Equal squared thresholds give direct sums of these lines. Intermediate extension identifies the nonunit nearby and vanishing-cycle eigenspaces; the remaining nearby eigenspace is trivial. Thus T_a is an involution with the stated negative multiplicity.

For clarity, at exponent zero the original regular formal part has rank z_0. Each nonfixed pair $\lambda, -\lambda$ contributes one $+1$ and one -1 monodromy eigenvalue upon descent from u to y. If the fixed tuple $\lambda = 0$ exists, its factor is $y^{-n/4}$, of monodromy κ. Hence the negative multiplicity in this original zero-exponent part is $(z_0 - \kappa\delta)/2$.

Local Fourier transform preserves the monodromy of a regular formal germ, up to inversion of the loop convention; this includes the unipotent regular part, in the formal vanishing-cycle convention of [77, Corollary 2.3 and §3.3]. See also [41, 40]. At the first transformed zero, the nonunit -1 part of the regular nearby germ is exactly that of its vanishing cycles, so it has this same dimension. Inversion preserves it, and the second local Fourier transform again preserves it. Finally, at the last transformed zero, passing from vanishing to nearby cycles again leaves the -1 part unchanged. All other eigenvalues in these regular parts are $+1$; intermediate extension may lengthen their unipotent blocks but cannot create another semisimple eigenvalue. This proves the claim about T_0. In particular T_0^2 is unipotent. The same local descriptions show regularity at every finite point; regularity at infinity is part of the prior input.

There are $(Q - z_0)/2$ nonzero pairs in total. Adding their negative multiplicities to k_0 proves (VII.3.2).

Choose $\xi \in \mathbb{C}^\times$ with $\operatorname{Re}(\xi\mu_j) \neq 0$ for every j. The functional $\operatorname{Re}(\xi c_\lambda)$ has the unique maximizer $\lambda_j^* = n_j \operatorname{sign}\operatorname{Re}(\xi\mu_j)$, whose value is strictly positive. Thus $c_* = c_{\lambda^*} \neq 0$. Only λ^* and $-\lambda^*$ can have squared exponent c_*^2, because the only complex square roots of c_*^2 are c_* and $-c_*$. This threshold has $k_{c_*^2} = 1$ and its monodromy is a genuine reflection. □

VII.3.2 A ramification obstruction to a finite permutation of blocks

Lemma VII.3.3 (The permutation-cover bound)**.** *Let $n \geq 4$ be even, and let G be the Galois group of the banana equation. If G^0 acts reducibly, then there are integers $h \geq 2$ and $d \geq n - 1$*

with $R = hd$ such that

$$d(2h-2) \le S-1. \tag{VII.3.3}$$

In particular $S \le R$ is sufficient for G^0 to act irreducibly.

Proof. Apply the general block-budget Lemma VII.2.4 with largest unipotent block size $\ell = n-1$. Lemma VII.3.2 supplies the finite negative-eigenspace budget S and at least one genuine reflection. That general lemma also permits the unipotent part of T_0, since only its semisimple eigenvalues enter the budget. It gives $R = hd$, $d \ge n-1$, and $d(2h-2) \le S-1$. If $S \le R = hd$, this would imply $d(h-2) \le -1$, impossible for $h \ge 2$. □

VII.3.3 The numerical inequality and the two low dimensions

Lemma VII.3.4. *For every multiplicity partition of an even $n \ge 8$, $R > Q/2$ and hence $S \le R$.*

Proof. Let X_j be independent uniforms on $\{0, \dots, n_j\}$, and center their sum by $n/2$. Its probabilities at zero and at one are b/Q and c/Q. The uniform-to-binomial concentration comparison of Lemma VII.2.3 says that its mass in every centered interval is at most the corresponding mass of a centered $\mathrm{Bin}(n, 1/2)$. By symmetry,

$$\begin{aligned}\frac{b+c}{Q} &= \frac{1}{2}\left(\Pr(X_1 + \cdots + X_q = n/2) + \Pr(|X_1 + \cdots + X_q - n/2| \le 1)\right) \\ &\le \frac{\binom{n}{n/2} + \binom{n}{n/2-1}}{2^n} = \frac{\binom{n+1}{n/2}}{2^n}.\end{aligned}$$

At $n = 8$ the last value is $126/256 < 1/2$. For $n = 2k$ its ratio at $k+1$ to its value at k is $(2k+3)/(2k+4) < 1$. Therefore $R/Q = 1 - (b+c)/Q > 1/2$ for every even $n \ge 8$. Since R is an integer and $S \le (Q+1)/2$, this implies $S \le R$. □

For completeness, all smaller multiplicity partitions give the following integer data directly from (VII.3.1).

$n=4$ partition	R	S	$n=6$ partition	R	S
$[4]$	3	2	$[6]$	5	4
$[3,1]$	4	4	$[5,1]$	8	6
$[2,2]$	4	4	$[4,2]$	9	8
$[2,1,1]$	5	6	$[4,1,1]$	12	10
$[1,1,1,1]$	6	8	$[3,3]$	9	8
			$[3,2,1]$	13	12
			$[3,1,1,1]$	17	16
			$[2,2,2]$	14	14
			$[2,2,1,1]$	18	18
			$[2,1,1,1,1]$	23	24
			$[1,1,1,1,1,1]$	29	32

Every row with $S \le R$ is handled by Lemma VII.3.3. The exceptional ranks $5, 23, 29$ are prime. Since a component dimension must divide the rank and be at least $n-1 \ge 3$, these rows also admit no nontrivial permutation of components. Only $n = 4$ with four distinct squared masses remains.

Lemma VII.3.5 (Four distinct masses)**.** *For four nonzero masses with pairwise distinct squares, every nonzero squared threshold has multiplicity $k_a \le 2$.*

Proof. Fix a nonzero signed exponent c_*. Choose a real projection $z \mapsto \mathrm{Re}(\xi z)$ which is nonzero on c_* and on every mass, and gives pairwise distinct absolute values to the masses. Such a choice

excludes only finitely many real lines for ξ. Change individual mass signs and reorder so their projected values satisfy $0 < a_1 < a_2 < a_3 < a_4$. All signed tuples giving c_* yield subsets of these four positive numbers with the same sum v, where $2v \neq a_1 + a_2 + a_3 + a_4$ because the projected signed exponent is nonzero.

At most one singleton and at most one triple can have sum v. Two different equal-sum pairs must be disjoint, because the a_i are distinct; they would be complementary and force $2v = \sum_i a_i$. Thus there is at most one pair. A singleton and a triple of sum v cannot intersect, by positivity, so they too would be complementary and force the forbidden central value. Empty or full subsets cannot share their sum with any other subset. Consequently at most two subsets have sum v. The number of signed tuples giving c_* is therefore at most two. Each pair $\lambda, -\lambda$ over the squared threshold contributes exactly one tuple giving c_*, which proves the claim. □

In the remaining rank-six case, a nontrivial component decomposition would have two components of dimension three. Lemma VII.3.5 shows that no nonzero-threshold involution can exchange them. The infinity loop also fixes them. These loops generate the punctured-sphere group after omitting the loop at zero; hence the entire permutation action is trivial, contradicting transitivity. This finishes connected irreducibility in every even dimension $n \geq 4$.

Proof of Theorem VII.3.1. For $n \geq 4$, the preceding arguments show that G^0 is irreducible. The invariant symmetric form places G inside O_R, and Lemma VII.3.2 supplies a reflection. The elementary reflection-normalizer argument in Lemma VII.2.5 now gives $G^0 = \mathrm{SO}_R$ and $G = \mathrm{O}_R$. Finite algebraic base change preserves the identity component; the standard SO_R representation is irreducible for $R \geq 3$, proving the order assertion.

For $n = 2$, direct summation gives

$$\varpi_m(t) = \frac{1}{\sqrt{(t - (m_1 + m_2)^2)(t - (m_1 - m_2)^2)}}.$$

The two roots are distinct because $m_1 m_2 \neq 0$, so this is a nonrational quadratic algebraic solution. Its rank-one Galois group is $\{\pm 1\} = \mathrm{O}_1$, as required. □

Corollary VII.3.6 (The unique jet relation)**.** *For even $n \geq 4$ and arbitrary nonzero complex masses, the R functions $\varpi_m, \varpi_m', \ldots, \varpi_m^{(R-1)}$ have transcendence degree $R - 1$ over $\mathbb{C}(t)$. Their polynomial-relation ideal is generated by the one nondegenerate irreducible quadratic relation supplied by the invariant form. The same assertion holds after any finite algebraic extension of the coefficient field.*

Proof. The orbit of a nonzero vector in the standard orthogonal representation has dimension $R - 1$, whether its quadratic value is zero or nonzero. The Picard–Vessiot orbit formula therefore gives the stated transcendence degree. The relation ideal is a height-one prime containing the invariant quadratic. For $R \geq 3$ that polynomial is absolutely irreducible, since a product of two affine linear forms has quadratic rank at most two. It thus generates the ideal. After finite base change the connected group is still SO_R and has the same nonzero-vector orbit dimensions. □

Together with the odd-propagator theorem, this classifies the full banana Galois group for every number of nonzero masses. Zero masses may simply be deleted from the defining series. With no nonzero masses or one nonzero mass, the period is, respectively, $1/t$ or $1/(t - m^2)$ and has trivial Galois group. No assertion about a numerical value at a fixed energy, or about the separate leading-coefficient multiplicity conjecture, is implied.

VII.4 Minimum recurrence width for every exponent and complex weight

For an integer $r \geq 2$ and $\lambda \in \mathbb{C}^*$, put

$$a_{r,\lambda}(n) = \sum_{k=0}^{n} \binom{n}{k}^r \lambda^k, \qquad B_{r,\lambda}(n) = \lambda^{-n} a_{r,\lambda}(2n).$$

The width of an eventual nonzero recurrence over $\mathbb{C}(n)$ is the difference between its largest and smallest integer shifts with nonzero coefficients. Arbitrary gaps between shifts are permitted. Geometric rescaling of a sequence preserves its minimum width.

Theorem VII.4.1 (All exponents and all complex weights)**.** *For every integer $r \geq 2$ and every $\lambda \in \mathbb{C}^*$,*

$$\operatorname{minwidth}_{\mathbb{C}(n)} B_{r,\lambda} = \begin{cases} \lceil r/2 \rceil, & \lambda = 1, \\ \lfloor r/2 \rfloor, & \lambda = -1, \\ r, & \lambda \notin \{1, -1\}. \end{cases}$$

In particular no further exceptional algebraic or transcendental weights occur, and collisions between critical values do not lower the width away from the two symmetry weights.

The standard upper recurrences at the symmetry weights come from Stoll [28] and McIntosh [29]; weighted recurrences in low degrees were studied by Matala-Aho [26]. The hyper-Bessel product and binomial-period correspondence is also studied by Gaiur, Rubtsov and van Straten [60]. The new step is a uniform lower-bound mechanism using the differential Galois group of a Kloosterman connection. It works even when several formal exponential factors coincide. This is a width theorem, not a theorem about the minimum number of nonzero terms of an arbitrarily sparse recurrence.

VII.4.1 Hyper-Bessel connections and independence of their scales

Define the entire function

$$\mathcal{F}_r(x) = \sum_{n \geq 0} \frac{x^n}{(n!)^r}.$$

It solves $((x\partial_x)^r - x)y = 0$. Work over $K = \mathbb{C}(t)$, and write M_a for the rank-r differential module whose scalar solutions include

$$f_a(t) = \mathcal{F}_r(at^r), \qquad a \in \mathbb{C}^*.$$

Equivalently its scalar equation is

$$\left\{ \left(\frac{t\partial_t}{r} \right)^r - at^r \right\} y = 0.$$

We use the following precise classical input. The differential Galois group of M_a in its standard representation is

$$G_r = \begin{cases} \mathrm{SL}_r(\mathbb{C}), & r \text{ odd}, \\ \mathrm{Sp}_r(\mathbb{C}), & r \text{ even}. \end{cases} \tag{VII.4.1}$$

This is Katz's Kloosterman-group calculation [59, Corollary 4.4.8]; the connection and its r-fold pullback are identified explicitly in [51, Proposition 2.4 and the proof of Proposition 2.7]. Scaling t by a nonzero constant gives the assertion for every a. In particular these groups are connected and their standard representations are irreducible.

The formal decomposition at $t=\infty$ consists of r rank-one factors with exponential parts

$$\exp(r\alpha\zeta t), \qquad \zeta^r=1, \quad \alpha^r=a. \tag{VII.4.2}$$

Their remaining parts are regular singular. Direct substitution gives formal solutions

$$e^{r\alpha\zeta t}t^{-(r-1)/2}(1+O(t^{-1}));$$

the distinct leading exponential roots recursively determine the formal series. Only the exponential parts, with their multiplicities, will be used below. The duality relation is

$$M_a^* \simeq M_{(-1)^r a}. \tag{VII.4.3}$$

For example, this follows by reversing the cyclic basis for the companion system and alternating its signs, or by formal adjunction with respect to dx/x before the pullback.

Lemma VII.4.2 (Independence of distinct scale classes)**.** *Choose finitely many nonzero scales $a_1,\dots,a_s$. If r is even, assume they are pairwise distinct. If r is odd, assume $a_i\neq\pm a_j$ for $i\neq j$. The connected differential Galois group of $\bigoplus_i M_{a_i}$ is the full product G_r^s.*

Proof. The connected joint group projects onto each individual group: its image has finite index in the connected group G_r. Its Lie algebra is therefore a subdirect product of simple Lie algebras. By the Lie-algebra form of Goursat's lemma, a proper subdirect product identifies two of these simple factors by an automorphism. Automorphisms of $\mathfrak{sp}_r$ are inner; automorphisms of $\mathfrak{sl}_r$ are inner or the standard duality automorphism. Consequently the two standard representations on the connected joint group would be isomorphic, or in the odd case dual.

Passing to the finite algebraic extension corresponding to the identity component makes this an isomorphism of the associated differential modules. But an isomorphism preserves the formal exponential parts. After any common finite ramification at infinity, (VII.4.2) shows that M_a and M_b can be isomorphic only if $a=b$, and that M_a and M_b^* can be isomorphic only if $a=(-1)^r b$. Indeed equality of the multisets of coefficients of the exponential terms, followed by taking their rth powers, gives these equalities. Both possibilities contradict the scale hypotheses. Thus no pair of simple factors is identified and the Lie algebra is the full product. A connected closed subgroup with this Lie algebra is the full product group. □

VII.4.2 A recurrence budget at irregular infinity

For a nonzero differential operator $P=\sum_i b_i(t)\partial_t^i$ with formal Laurent coefficients at infinity, let v be the largest power of t occurring as a leading power of one of the b_i. Define its polynomial principal symbol at infinity by

$$\sigma_\infty(P)(\xi)=\sum_i \left(\lim_{t\to\infty} t^{-v}b_i(t)\right)\xi^i.$$

The limit denotes the constant Laurent coefficient after normalization. The resulting polynomial is nonzero. Since differentiating a Laurent coefficient lowers its leading power of t, one has

$$\sigma_\infty(PQ)=\sigma_\infty(P)\sigma_\infty(Q). \tag{VII.4.4}$$

This statement holds up to the harmless nonzero scalar resulting from a different normalization of the leading power.

Lemma VII.4.3 (Exponential rank is a recurrence-width lower bound)**.** *Suppose*

$$h(t)=\sum_{n\geq 0}\frac{u_n}{((2n)!)^r}t^{2rn}$$

is a solution of a differential module over $\mathbb{C}(t)$. Suppose the differential module generated by this particular scalar solution has, formally at infinity, a decomposition into rank-one factors with exponential parts $e^{\beta t}$ and regular-singular remaining parts. Let q be the total number of factors with $\beta \neq 0$, counted with multiplicity. Then any eventual recurrence for u_n has width at least $q/(2r)$.

Proof. After translating shifts and clearing denominators, write a proposed recurrence as

$$\sum_{j=0}^{L} p_j(n) u_{n-j} = 0, \qquad p_j \in \mathbb{C}[n].$$

Let $D = \max_j \deg p_j$ and $c_j = \mathrm{lc}(p_j)$ when $\deg p_j = D$. Multiplying the coefficient identity by $((2n)_{\underline{2L}})^r/((2n)!)^r$ shows that h satisfies an inhomogeneous differential equation $Ph = Q(t)$ with polynomial right side and

$$P = \sum_{j=0}^{L} t^{2rj} p_j\left(\frac{t\partial_t}{2r} + j\right) \prod_{\ell=0}^{2(L-j)-1} \left(\frac{t\partial_t}{r} - \ell\right)^r.$$

Empty products are one. The polynomial right side accounts only for finitely many initial coefficients. The largest coefficient power of t in this operator is $D + 2rL$, and its principal symbol is

$$\sigma_\infty(P)(\xi) = \sum_{\deg p_j = D} c_j \left(\frac{\xi}{2r}\right)^D \left(\frac{\xi}{r}\right)^{2r(L-j)}. \tag{VII.4.5}$$

The powers contributed by distinct j differ, so this polynomial is nonzero. Its number of nonzero roots, counted with multiplicity, is at most $2rL$. Left multiplication by a polynomial in $t\partial_t$ killing $Q(t)$ makes the equation homogeneous and only multiplies (VII.4.5) by a constant times a power of ξ. Thus the same nonzero-root budget holds for a homogeneous annihilator P_0.

Let R be the monic minimal scalar annihilator of h. The specified formal decomposition implies a formal first-order factorization

$$R = (\partial_t - \beta_d + O(t^{-1})) \cdots (\partial_t - \beta_1 + O(t^{-1})).$$

One way to see this is to take a rank-one formal submodule, use its solution with logarithmic derivative $\beta_i + O(t^{-1})$ to obtain a right first-order factor, and continue on the quotient. Formal Laurent gauge changes affect the logarithmic derivative by $O(t^{-1})$, so the listed exponential coefficients are preserved. Hence $\sigma_\infty(R) = \prod_i (\xi - \beta_i)$. Minimality gives a right division $P_0 = AR$ over $\mathbb{C}(t)$. Equation (VII.4.4), also valid after formal completion, shows that $\sigma_\infty(R)$ divides $\sigma_\infty(P_0)$. Its q nonzero roots counted with multiplicity therefore satisfy $q \leq 2rL$. □

VII.4.3 The cyclic period and all parameter strata

The elementary Cauchy-product identity is

$$\mathcal{F}_r(x)\mathcal{F}_r(\lambda x) = \sum_{n\geq 0} \frac{a_{r,\lambda}(n)}{(n!)^r} x^n. \tag{VII.4.6}$$

Consequently the function to which Lemma VII.4.3 applies is

$$h_\lambda(t) = \tfrac{1}{2}\{f_1(t)f_\lambda(t) + f_{-1}(t)f_{-\lambda}(t)\} = \sum_{n\geq 0} \frac{a_{r,\lambda}(2n)}{((2n)!)^r} t^{2rn}. \tag{VII.4.7}$$

Products of solution spaces are interpreted inside a common Picard–Vessiot field. Multiplication is an equivariant linear map from the tensor product of the two representations. If that tensor

product is irreducible, the map is injective because it is nonzero. Also, a vector with a nonzero component in each of two nonisomorphic irreducible summands generates their entire direct sum. These facts identify the module generated by the specific scalar function (VII.4.7), rather than merely an ambient annihilating system.

Proof of Theorem VII.4.1: lower bounds. First assume $\lambda \notin \{1, -1\}$. For even r, the four scales $1, \lambda, -1, -\lambda$ are distinct; Lemma VII.4.2 gives four independent symplectic factors. The two products in (VII.4.7) belong to nonisomorphic irreducible tensor representations, each of dimension r^2. For odd r, the two scales $1, \lambda$ give independent special linear factors, while the negative scales give their duals by (VII.4.3). The two products then belong to $V \otimes W$ and $V^* \otimes W^*$. These are nonisomorphic irreducibles since $r \geq 3$ and the standard special-linear representation is not self-dual. In both parities the particular sum generates the full direct sum of dimension $2r^2$.

Every formal exponential part of a product is the sum of one exponential coefficient from each of its two factors. A zero sum for $M_a \otimes M_b$ can occur only when $b = (-1)^r a$. This is excluded in the present case, for both products. Hence all $2r^2$ formal factors are nonzero. Lemma VII.4.3 proves $L \geq r$. Coincidences between nonzero sums are harmless, since the budget retains their multiplicities.

For $\lambda = 1$ and even r, the two products are squares for two independent symplectic groups. Their representations are $\operatorname{Sym}^2 V_1$ and $\operatorname{Sym}^2 V_{-1}$, both irreducible and nonisomorphic. In each, the formal exponential coefficients are the unordered sums of two rth-root coefficients. Exactly $r/2$ of these sums vanish (the opposite pairs). Each summand thus has $r(r+1)/2 - r/2 = r^2/2$ nonzero formal factors. The total is r^2, giving $L \geq r/2$. For odd r, the two summands are $\operatorname{Sym}^2 V$ and $\operatorname{Sym}^2 V^*$ for SL_r; they are again nonisomorphic irreducibles. No opposite pair of rth roots exists, so their total nonzero formal rank is $r(r+1)$, giving $L \geq (r+1)/2$.

For $\lambda = -1$, the two products in (VII.4.7) coincide. If r is even, M_1 and M_{-1} are independent symplectic modules. Their tensor product is irreducible of rank r^2 and has no zero exponential part. The resulting lower bound is $L \geq r/2$. If r is odd, the tensor representation is $V \otimes V^* = \mathbf{1} \oplus \mathrm{Ad}$ for SL_r. The multiplication map kills the trivial summand. To check this explicitly, put $\delta = (t/r)\partial_t$. For solutions y of M_1 and z of M_{-1}, the nondegenerate concomitant

$$\mathcal{B}(y, z) = \sum_{j=0}^{r-1} (-1)^j (\delta^j y)(\delta^{r-1-j} z)$$

is constant, since differentiating telescopes and r is odd. Choose bases y_i, z_i dual for this pairing, and let Y, Z be their jet matrices, with rows indexed by δ^j, $0 \leq j < r$. Then $Y^{\mathsf{T}} J Z = I$, where J is the invertible anti-diagonal matrix with anti-diagonal entries $(-1)^j$. The invariant tensor $\sum_i y_i \otimes z_i$ is consequently sent by multiplication to

$$\sum_i y_i z_i = (Y Z^{\mathsf{T}})_{00} = (J^{-\mathsf{T}})_{00} = 0.$$

The product $f_1 f_{-1}$ is nonzero, so multiplication is nonzero on the irreducible adjoint summand and hence injective there. Its cyclic module is therefore the entire adjoint representation. Its nonzero formal factors are the $r(r-1)$ ordered differences of distinct rth-root exponential coefficients. Lemma VII.4.3 gives $L \geq (r-1)/2$. This proves all lower bounds in the theorem. □

Proof of Theorem VII.4.1: upper bounds. For $\lambda \notin \{1, -1\}$ put

$$L_N[p] = \sum_{k=0}^{N} p(k) \binom{N}{k}^r \lambda^k.$$

An index shift gives, for every $h \geq 0$,

$$L_N[k^{r+h} - \lambda (N-k)^r (k+1)^h] = 0.$$

The displayed polynomial has leading coefficient $1-(-1)^r\lambda \neq 0$. Successive elimination expresses every polynomial moment in terms of $L_N[1], \dots, L_N[k^{r-1}]$. Using $\binom{N-1}{k}^r = N^{-r}(N-k)^r\binom{N}{k}^r$ gives a rational r-dimensional backward state relation $M_{N-1} = T(N)M_N$. Its two-step relation expresses $r+1$ consecutive even observations as $r+1$ rows on an r-dimensional state space. Their rational linear dependence proves width at most r; no matrix inverse is needed.

For $\lambda = 1$, the ordinary Franel recurrence of width at most $\lceil r/2 \rceil$ is the known upper bound [28]. Taking every second state of its rational companion system preserves this dimension and gives the same upper bound for the even subsequence. For $\lambda = -1$, McIntosh's construction [29, Theorem 1] directly gives width at most $\lfloor r/2 \rfloor$ for the centered alternating sequence. These upper bounds and the proved lower bounds complete the classification. □

Remark VII.4.4 (Why no collision parameters are omitted). Counting only distinct finite critical values loses information when two such values merge. After factorial normalization, their multiplicities become multiplicities of formal exponential factors of a tensor connection. The connected Galois group makes the chosen period cyclic in the relevant representation, and the principal symbol of every annihilator retains each of those factors. This replaces both a generic-parameter specialization argument and any unproved assertion that individual finite singularities remain visible at exceptional weights.

VII.5 A global algebraic-quantile criterion for complementary beta weights

The arithmetic of a normalized integral can change after truncation, and an algebraic primitive alone does not classify all normalized inverses. For complementary rational beta exponents other than the arcsine exponent, we obtain an exact criterion that is uniform in the truncation interval and in every polynomial weight. One explicit linear functional of the coefficients determines whether all algebraic probability levels have algebraic quantiles or all have transcendental quantiles. There is no intermediate exceptional set.

VII.5.1 The complete criterion

Fix

$$r \in \mathbb{Q} \cap (0,1), \qquad r \neq \tfrac{1}{2}, \qquad b_r(x) = x^{r-1}(1-x)^{-r}.$$

Let u, v be real algebraic numbers with $0 \le u < v \le 1$. Let $w(X) = \sum_{j=0}^{d} w_j X^j$ have real algebraic coefficients and satisfy $w(x) > 0$ for $u < x < v$. No positivity outside this interval is required. Set

$$F_{r,w;u,v}(x) = \frac{\displaystyle\int_u^x w(t) b_r(t)\,dt}{\displaystyle\int_u^v w(t) b_r(t)\,dt}, \qquad u \le x \le v. \tag{VII.5.1}$$

These integrals converge even when $u = 0$ or $v = 1$. The CDF is continuous and strictly increasing. With $(r)_0 = 1$ and $(r)_j = r(r+1)\cdots(r+j-1)$, define the algebraic number

$$c_r(w) = \sum_{j=0}^{d} w_j \frac{(r)_j}{j!}. \tag{VII.5.2}$$

Theorem VII.5.1 (Global quantile dichotomy)**.** *Under the preceding hypotheses, the following five statements are equivalent:*

1. $c_r(w) = 0$.
2. $F_{r,w;u,v}(x)$ *is algebraic for some algebraic* $x \in (u,v)$.
3. $F_{r,w;u,v}(x)$ *is algebraic for every algebraic* $x \in (u,v)$.
4. *The quantile* $F^{-1}_{r,w;u,v}(\lambda)$ *is algebraic for some algebraic* $\lambda \in (0,1)$.
5. *The quantile* $F^{-1}_{r,w;u,v}(\lambda)$ *is algebraic for every algebraic* $\lambda \in (0,1)$.

Thus, if $c_r(w) \neq 0$*, all internal algebraic-cut cumulative probabilities and all internal algebraic-level quantiles are transcendental. If* $r = p/q$ *in lowest terms and* $c_r(w) = 0$*, the quantile at each algebraic level satisfies an explicit nonconstant polynomial equation of degree* $q \deg w$ *with algebraic coefficients.*

The polynomial reduction used below is the same elementary reduction available for general polynomial beta weights. The substantive arithmetic step is the exclusion when $c_r(w) \neq 0$, for every algebraic truncation and every algebraic level. It follows from a coefficient-conjugation consequence of Baker's theorem and strict ordering of two beta densities.

VII.5.2 Fixed logarithms and coefficient conjugation

We use two classical consequences of Baker's theorems [61, 46]; the linear-independence statement is also recorded explicitly as Theorem 1.2 of [62]. For any fixed logarithms of nonzero algebraic numbers:

1. a nonzero linear form with algebraic coefficients in those logarithms is transcendental;
2. logarithms linearly independent over $\mathbb{Q}$ remain linearly independent over $\overline{\mathbb{Q}}$.

These are different inputs: the first separates an algebraic correction from a logarithmic expression; the second controls the zero relations.

Lemma VII.5.2 (Conjugating coefficients in a zero relation)**.** *Let $\ell_1, \dots, \ell_N$ be fixed logarithms of nonzero algebraic numbers, and let $a_1, \dots, a_N \in \overline{\mathbb{Q}}$. Then*

$$\sum_{j=1}^{N} a_j \ell_j = 0 \quad \Longrightarrow \quad \sum_{j=1}^{N} \overline{a_j} \ell_j = 0. \tag{VII.5.3}$$

The logarithms, their arguments, and their branch choices are held fixed. Only the coefficients are conjugated.

Proof. Choose a $\mathbb{Q}$-basis $\eta_1, \dots, \eta_s$ of the $\mathbb{Q}$-span of the ℓ_j, taking the basis from the given list. Write $\ell_j = \sum_k q_{jk} \eta_k$ with $q_{jk} \in \mathbb{Q}$. Baker's linear-independence theorem makes the η_k independent over $\overline{\mathbb{Q}}$. The initial relation therefore implies $\sum_j a_j q_{jk} = 0$ for every k. Conjugating these algebraic equalities, and using that the q_{jk} are rational, proves (VII.5.3). □

For $0 \le y \le 1$ put

$$U_r(y) = \int_0^y b_r(t)\, dt = B_y(r, 1-r).$$

Euler's reflection identity gives

$$U_r(1) = B(r, 1-r) = \frac{\pi}{\sin(\pi r)} = U_{1-r}(1). \tag{VII.5.4}$$

Lemma VII.5.3 (Transfer between complementary beta integrals)**.** *Let $y_1, \dots, y_N$ be algebraic numbers in $[0, 1]$, and let $a_1, \dots, a_N$ be real algebraic numbers. If*

$$\sum_{j=1}^{N} a_j U_r(y_j) = 0, \tag{VII.5.5}$$

then

$$\sum_{j=1}^{N} a_j U_{1-r}(y_j) = 0.$$

Moreover, if the left side of (VII.5.5) is algebraic, it is zero, so the same conclusion holds.

Proof. Write $r = p/q$ in lowest terms, with $1 \le p < q$. For $0 < y < 1$ set

$$s = \left(\frac{y}{1-y} \right)^{1/q} > 0.$$

The substitution $t = z^q/(1+z^q)$ gives

$$U_r(y) = q \int_0^s \frac{z^{p-1}}{1+z^q}\, dz. \tag{VII.5.6}$$

For every ζ with $\zeta^q = -1$, let $\ell_\zeta(s) = \log(1 - s/\zeta)$ be the continuous branch along real $s \geq 0$ with value zero at $s = 0$. The path has no zero: no such ζ is positive real. If $\zeta = -1$, the argument is positive real; otherwise it stays strictly in one open half-plane for $s > 0$. Thus these branches are unambiguous. When y is algebraic, s and every $1 - s/\zeta$ are nonzero algebraic numbers.

The residue of $qz^{p-1}/(1 + z^q)$ at $z = \zeta$ is $\zeta^{p-q} = -\zeta^p$. Partial fractions and the common value zero at $s = 0$ give

$$U_r(y) = -\sum_{\zeta^q=-1} \zeta^p \ell_\zeta(s), \qquad U_{1-r}(y) = -\sum_{\zeta^q=-1} \zeta^{q-p} \ell_\zeta(s). \tag{VII.5.7}$$

Conjugating only the coefficients in the first expression produces

$$-\sum_{\zeta^q=-1} \zeta^{-p} \ell_\zeta(s) = \sum_{\zeta^q=-1} \zeta^{q-p} \ell_\zeta(s) = -U_{1-r}(y),$$

because $\zeta^{-p} = -\zeta^{q-p}$.

The endpoint $y = 0$ contributes zero. At $y = 1$, use the finite logarithmic representation

$$U_r(1) = -\frac{i}{\sin(\pi r)} \log(-1), \qquad \log(-1) = i\pi. \tag{VII.5.8}$$

Here $\sin(\pi r)$ is a nonzero real algebraic number. Conjugating its coefficient yields $-U_{1-r}(1)$, by (VII.5.4). This handles both endpoints without taking limits of the individual logarithms in (VII.5.7).

Represent all terms in (VII.5.5) by these finite logarithmic expressions and apply Lemma VII.5.2. The real coefficients a_j are fixed by conjugation, and every $U_r(y_j)$ changes to $-U_{1-r}(y_j)$. This proves the first assertion. For the last assertion the expression is a linear form with algebraic coefficients in fixed logarithms of algebraic numbers. Baker's nonzero-form transcendence theorem forces it to be zero if it is algebraic. □

VII.5.3 Strict ordering survives every truncation

Lemma VII.5.4 (No common internal truncated probability)**.** *For any real $0 \leq u < x < v \leq 1$ and $r \neq 1/2$,*

$$\frac{U_r(x) - U_r(u)}{U_r(v) - U_r(u)} \neq \frac{U_{1-r}(x) - U_{1-r}(u)}{U_{1-r}(v) - U_{1-r}(u)}. \tag{VII.5.9}$$

For $r > 1/2$ the left side is strictly smaller; for $r < 1/2$ it is strictly larger.

Proof. The ratio of the two positive densities is

$$\frac{b_r(t)}{b_{1-r}(t)} = \left(\frac{t}{1-t}\right)^{2r-1},$$

strictly increasing when $r > 1/2$ and strictly decreasing when $r < 1/2$. For the increasing case, set $A_r = \int_u^x b_r$ and $D_r = \int_x^v b_r$, with analogous notation for $1 - r$. Then

$$A_r D_{1-r} - A_{1-r} D_r = \int_u^x \int_x^v \{b_r(t) b_{1-r}(z) - b_{1-r}(t) b_r(z)\}\, dz\, dt < 0.$$

The integrand is strictly negative for interior $t < z$, and all integrals are finite and positive. This proves $A_r/(A_r + D_r) < A_{1-r}/(A_{1-r} + D_{1-r})$. The decreasing case reverses the inequalities. □

VII.5.4 Polynomial reduction and proof of the criterion

Lemma VII.5.5 (The single logarithmic coefficient)**.** *There is a unique polynomial* $R \in (\overline{\mathbb{Q}} \cap \mathbb{R})[X]$ *such that*

$$w(X) = c_r(w) + X(1-X)R'(X) + (r-X)R(X). \tag{VII.5.10}$$

With

$$H(x) = x^r(1-x)^{1-r}R(x), \qquad H(0) = H(1) = 0, \tag{VII.5.11}$$

one has

$$\int_0^x w(t)b_r(t)\,dt = c_r(w)U_r(x) + H(x). \tag{VII.5.12}$$

If w *is nonconstant of degree* d*, then* $\deg R = d-1$*. If* w *is constant,* $R = 0$*.*

Proof. For $\mathcal{L}_r R = X(1-X)R' + (r-X)R$,

$$\mathcal{L}_r(X^j) = (j+r)X^j - (j+1)X^{j+1}.$$

Descending elimination gives a unique R and constant c with $w = c + \mathcal{L}_r R$, together with the degree assertion. Direct differentiation gives $H' = (\mathcal{L}_r R)b_r$. Since $0 < r < 1$, both endpoint values of H are zero. Integrating over $[0,1]$ gives

$$c = \frac{\int_0^1 w(t)b_r(t)\,dt}{B(r,1-r)} = \sum_j w_j \frac{B(r+j,1-r)}{B(r,1-r)} = \sum_j w_j \frac{(r)_j}{j!} = c_r(w).$$

Integrating to x proves (VII.5.12). The whole-interval integral in this calculation need not be positive when w is positive only on (u,v); accordingly $c_r(w)$ may vanish or have either sign. □

Proof of Theorem VII.5.1. Write $c = c_r(w)$ and use R, H from Lemma VII.5.5. Suppose first that $c \neq 0$ and that an algebraic $x \in (u,v)$ has algebraic cumulative probability $\lambda = F_{r,w;u,v}(x)$. The integral equation becomes

$$c\{U_r(x) - U_r(u) - \lambda[U_r(v) - U_r(u)]\} = -H(x) + H(u) + \lambda[H(v) - H(u)]. \tag{VII.5.13}$$

All terms on the right are algebraic: this includes endpoints zero and one by (VII.5.11). By the last assertion of Lemma VII.5.3, the logarithmic expression in braces must vanish. Its coefficients $1, -\lambda, \lambda - 1$ are real algebraic. The same lemma therefore gives

$$U_r(x) - U_r(u) = \lambda[U_r(v) - U_r(u)],$$
$$U_{1-r}(x) - U_{1-r}(u) = \lambda[U_{1-r}(v) - U_{1-r}(u)].$$

This contradicts Lemma VII.5.4. Thus no algebraic x has an algebraic CDF value when $c \neq 0$; equivalently no algebraic level has an algebraic quantile.

If $c = 0$, (VII.5.12) gives

$$F_{r,w;u,v}(x) = \frac{H(x) - H(u)}{H(v) - H(u)}.$$

The denominator is positive because $H' = wb_r > 0$ on (u,v). All these values are algebraic at algebraic x. For an algebraic level $\lambda \in (0,1)$, set

$$\eta = H(u) + \lambda[H(v) - H(u)] \in \overline{\mathbb{Q}}.$$

The unique quantile solves $H(x) = \eta$. Writing $r = p/q$ in lowest terms, it is therefore a root of

$$X^p(1-X)^{q-p}R(X)^q - \eta^q = 0. \tag{VII.5.14}$$

Here w cannot be a nonzero constant, since its c would be that constant. Thus $d = \deg w \geq 1$, $R \neq 0$, and the polynomial in (VII.5.14) has degree $q + q\deg R = qd$. In particular it is nonconstant, and its coefficients are algebraic. The quantile is algebraic. Raising to the qth power may introduce extra roots, but the original equation $H(x) = \eta$ is strictly monotone on (u,v) and specifies the required root uniquely. All five equivalences now follow. □

VII.5.5 Consequences and sharpness

Corollary VII.5.6 (Exact linear-independence formulation)**.** *Under the hypotheses of Theorem VII.5.1, fix any algebraic $x \in (u, v)$ and put*

$$J_1 = \int_u^x w(t) b_r(t)\, dt, \qquad J_2 = \int_u^v w(t) b_r(t)\, dt.$$

Then

$$\dim_{\overline{\mathbb{Q}}} \operatorname{span}_{\overline{\mathbb{Q}}}\{1, J_1, J_2\} = \begin{cases} 1, & c_r(w) = 0, \\ 3, & c_r(w) \neq 0. \end{cases}$$

In particular, for the unweighted complementary beta density, 1, $B_x(r, 1-r) - B_u(r, 1-r)$, and $B_v(r, 1-r) - B_u(r, 1-r)$ are linearly independent over $\overline{\mathbb{Q}}$ for every algebraic subdivision $0 \le u < x < v \le 1$.

Proof. Let $L_1 = U_r(x) - U_r(u)$ and $L_2 = U_r(v) - U_r(u)$. Any relation $a_0 + a_1 L_1 + a_2 L_2 = 0$ with algebraic coefficients forces $a_0 = 0$ by Baker's nonzero-form theorem. If $a_1 \neq 0$, it gives the algebraic ratio $L_1/L_2 = -a_2/a_1$, contradicting the proof of Theorem VII.5.1 for $w = 1$. If $a_1 = 0$, then $a_2 = 0$ because $L_2 > 0$. Thus $1, L_1, L_2$ are independent. When $c = c_r(w) \neq 0$, the transformation

$$(1, L_1, L_2) \longmapsto (1, cL_1 + H(x) - H(u), cL_2 + H(v) - H(u))$$

is invertible over $\overline{\mathbb{Q}}$, with determinant c^2. This gives dimension three. If $c = 0$, both integrals are algebraic, giving dimension one. □

Corollary VII.5.7 (All positive full-interval polynomial weights)**.** *Let $r \in \mathbb{Q} \cap (0, 1) \setminus \{1/2\}$ and let w be a real-algebraic polynomial positive on $(0, 1)$. Every algebraic-level internal quantile of its normalized beta weighting is transcendental. The same assertion holds after conditioning on any nondegenerate interval with algebraic endpoints in $[0, 1]$.*

Proof. The complete integral in Lemma VII.5.5 is positive, so $c_r(w) > 0$. Apply Theorem VII.5.1 to every interval. □

Corollary VII.5.8 (Every non-half-integral rational integer-sum beta pair)**.** *Let $a, b > 0$ be rational with $a + b \in \mathbb{Z}$ and $a \notin \frac{1}{2}\mathbb{Z}$. Then every algebraic internal probability level of* Beta(a, b) *has a transcendental quantile, including after conditioning on any nondegenerate algebraic interval in $[0, 1]$. The conclusion is unchanged by multiplying the density by any real-algebraic polynomial positive on $(0, 1)$.*

Proof. Write $a = m + r$, $b = n + 1 - r$, where $m, n \ge 0$ are integers and $r \in \mathbb{Q} \cap (0, 1) \setminus \{1/2\}$. The beta density is $x^m(1-x)^n b_r(x)$ up to normalization. The polynomial factor is positive throughout $(0, 1)$, so Corollary VII.5.7 applies, also after multiplying by the additional positive polynomial. □

For example, the conclusion holds for every Beta$(m + 1/3, n + 2/3)$ with integers $m, n \ge 0$, for every algebraic truncation and every algebraic probability level. The prior's two fractions need not have small denominators: the same statement holds for $m + 17/101, n + 84/101$ and, uniformly, for every complementary rational pair other than $1/2, 1/2$. Together with the previously proved closed half-integer classification and the polynomial primitive for integral beta shapes, Corollary VII.5.8 supplies the remaining denominator classes in the full-interval positive rational integer-sum family.

Example VII.5.9 (Both sides of the criterion occur on the same interval)**.** Take $r = 1/3$ and $[u, v] = [1/3, 1]$. For $w(X) = X - 1/3$, one has $c_r(w) = 0$, $R = -1$, and

$$H(x) = -x^{1/3}(1 - x)^{2/3}.$$

Every algebraic-level quantile is algebraic; more precisely the level-λ quantile is the unique $x \in (1/3, 1)$ satisfying

$$x(1 - x)^2 = \frac{4}{27}(1 - \lambda)^3.$$

The left side is strictly decreasing on this interval. For the constant weight $w = 1$ on the very same interval, $c_r(w) = 1$, and every algebraic-level quantile is transcendental.

The restriction $r \neq 1/2$ is essential. The unweighted arcsine CDF is $2 \arcsin \sqrt{x}/\pi$, and every rational level has the algebraic quantile $\sin^2(\pi\lambda/2)$. The coefficient-conjugation transfer then exchanges two identical densities, so the strict-order contradiction correctly disappears. The arcsine exceptional-packet and half-integer pair theorems are separate previously established results, not additional claims of this section.

Scope and relation to the available literature. The polynomial elimination and the algebraic-primitive direction are supporting tools already used for other beta parameter families. The new argument here establishes the complementary-exponent obstruction for every algebraic interval and every algebraic probability level, and hence the exact criterion (VII.5.2) for all polynomial weights positive on the selected interval. It does not use an unproved independence statement about arbitrary beta periods.

Dhillon and Saha study arithmetic properties of unnormalized incomplete beta values [55]; the author's conference presentation [56] explicitly includes rational nonintegral a with $a + b \in \mathbb{N}$ and logarithmic methods. The available slides do not state this all-interval normalized inverse criterion. A comparison with the full published article has not been completed, so no historical priority claim is made. The presentation's separate question about a Schneider-type theorem is not claimed to be resolved here.

VII.6 A single noisy observation can destroy expected transport accuracy

Consider independent observations

$$X_i = \Theta_i + \sigma_i Z_i, \qquad \Theta_i \overset{\text{iid}}{\sim} G, \qquad Z_i \overset{\text{iid}}{\sim} N(0,1), \qquad 1 \le i \le N,$$

with independent latent variables and noises and known positive standard deviations $\sigma_1, \dots, \sigma_N$. Write $\varphi_\sigma(x) = \sigma^{-1}\varphi(x/\sigma)$, where φ is the standard normal density, and $p_{H,\sigma} = H * \varphi_\sigma$. The ordinary joint NPMLE is any measurable maximizer

$$\widehat{G} \in \operatorname*{argmax}_{H \in \mathcal{P}(\mathbb{R})} \sum_{i=1}^{N} \log p_{H,\sigma_i}(X_i).$$

In particular, its optimization domain is not restricted to a known support interval.

Theorem VII.6.1 (A universal noise-outlier lower bound)**.** *For every $N \ge 1$, every positive noise profile, every $\mu \in \mathbb{R}$, and every measurable choice of the exact unrestricted joint NPMLE,*

$$\mathbb{E}_{\delta_\mu} W_1(\widehat{G}, \delta_\mu) \ge \frac{\max_{1 \le i \le N} \sigma_i}{2\sqrt{2\pi}\, N^2}. \tag{VII.6.1}$$

More precisely, for every data set and every i,

$$W_1(\widehat{G}, \delta_\mu) \ge \frac{|X_i - \mu|}{2N} \left(1 - N \exp\left\{-\frac{(X_i - \mu)^2}{2\sigma_i^2}\right\}\right)_+. \tag{VII.6.2}$$

Consequently, for a triangular array of noise profiles, expected W_1 consistency at any point mass requires $\max_i \sigma_{N,i} = o(N^2)$.

The condition just obtained is necessary, not sufficient. The theorem concerns expected transport loss. It does not assert failure of convergence in probability, because rare observations may account for the lower bound.

Soloff, Guntuboyina and Sen establish heteroscedastic NPMLE theory, including the likelihood contact condition and deconvolution guarantees [63]. Their stated deconvolution theorem assumes uniform positive lower and upper bounds on the covariance matrices [64, Theorem 11]. The result here rules out an unrestricted extension to arbitrary unbounded noise profiles in expected W_1 risk. It does not contradict that bounded-covariance theorem.

VII.6.1 Proof of the universal lower bound

For completeness, every maximizer is supported on the sample interval $[\min_i X_i, \max_i X_i]$: projecting a parameter outside this interval to its nearest endpoint strictly increases every Gaussian likelihood contribution. Weak compactness of probability measures on this interval and continuity of the likelihood also prove existence.

Let H be a maximizer. Differentiating its likelihood along $(1-\varepsilon)H + \varepsilon\delta_\theta$ at $\varepsilon = 0+$ gives the standard contact inequality

$$S_H(\theta) := \frac{1}{N} \sum_{j=1}^{N} \frac{\varphi_{\sigma_j}(X_j - \theta)}{p_{H,\sigma_j}(X_j)} \le 1 \qquad (\theta \in \mathbb{R}).$$

Evaluating at $\theta = X_i$ and retaining the ith nonnegative term yields the fitted-density floor

$$p_{H,\sigma_i}(X_i) \ge \frac{\varphi(0)}{N\sigma_i}. \tag{VII.6.3}$$

We need the following elementary Gaussian estimate: for $z > 0$ and $u \in \mathbb{R}$,

$$e^{-(z-u)^2/2} \le e^{-z^2/2} + \frac{2|u|}{z}. \tag{VII.6.4}$$

If $u < 0$, the exponential decreases. If $u \ge z/2$, the term $2|u|/z$ alone is at least one. In the remaining case $0 \le u < z/2$, the mean-value theorem uses an intermediate $t \in [z-u, z] \subset [z/2, z]$, and

$$zte^{-t^2/2} \le 2t^2 e^{-t^2/2} \le 4/e < 2.$$

This proves (VII.6.4); reflection gives the same estimate for $z < 0$ with $|z|$ in the denominator.

Take $z = (X_i - \mu)/\sigma_i$, $u = (\theta - \mu)/\sigma_i$ and integrate (VII.6.4) against H. When $X_i \ne \mu$ it follows that

$$\frac{\sigma_i p_{H,\sigma_i}(X_i)}{\varphi(0)} \le \exp\left\{-\frac{(X_i-\mu)^2}{2\sigma_i^2}\right\} + \frac{2W_1(H,\delta_\mu)}{|X_i - \mu|}.$$

Combining this inequality with (VII.6.3) proves (VII.6.2). If $X_i = \mu$, its right-hand side is zero and the same conclusion is immediate.

Under $G = \delta_\mu$, write $X_i - \mu = \sigma_i Z$ and put $b = \sqrt{2\log N}$, with $b = 0$ when $N = 1$. Direct Gaussian integration gives

$$\mathbb{E}\big[|Z|\mathbf{1}_{\{|Z|>b\}}\big] = \frac{2}{\sqrt{2\pi}N},$$
$$\mathbb{E}\big[|Z|e^{-Z^2/2}\mathbf{1}_{\{|Z|>b\}}\big] = \frac{1}{\sqrt{2\pi}N^2}.$$

Taking expectations in (VII.6.2) therefore yields

$$\mathbb{E}_{\delta_\mu} W_1(H,\delta_\mu) \ge \frac{\sigma_i}{2N}\left\{\frac{2}{\sqrt{2\pi}N} - \frac{1}{\sqrt{2\pi}N}\right\} = \frac{\sigma_i}{2\sqrt{2\pi}N^2}.$$

This holds for each deterministic index i, so taking their maximum proves (VII.6.1). □

VII.6.2 Failure of uniform risk optimality despite many precise observations

The lower bound is not caused by an intrinsically impossible statistical experiment. We give a self-contained benchmark which discards the single imprecise observation. In particular, the comparison requires no assumption about the behavior of another NPMLE.

Lemma VII.6.2 (A sufficient homoscedastic benchmark)**.** *From $n \ge 2$ independent observations with distribution $G * N(0,1)$, where $G \in \mathcal{P}([-A,A])$ and $A > 0$, there is an estimator $\widetilde{G} \in \mathcal{P}([-A,A])$ such that, for every $T > 0$,*

$$\sup_G \mathbb{E}_G W_1(\widetilde{G}, G) \le \frac{C}{T} + CA^{3/2}e^{T^2/2}\sqrt{\frac{T}{n}}, \tag{VII.6.5}$$

where C is absolute. The estimator may use A and T.

Proof. For a finite measure F write $\chi_F(t) = \int e^{itx}F(dx)$, and let $\widehat{\chi}_n(t) = n^{-1}\sum_{j=1}^n e^{itX_j}$. Choose a measurable minimizer, over $H \in \mathcal{P}([-A,A])$, of

$$\int_{-T}^{T} |e^{-t^2/2}\chi_H(t) - \widehat{\chi}_n(t)|^2\, dt.$$

Existence follows from compactness and continuity. Comparing to the true G and applying the triangle inequality in $L^2[-T,T]$ gives

$$\|\chi_{\widetilde{G}} - \chi_G\|_{L^2[-T,T]} \le 2e^{T^2/2}\|\widehat{\chi}_n - \chi_{G*N(0,1)}\|_{L^2[-T,T]}.$$

At each t the empirical characteristic function has variance at most $1/n$. Hence the expected norm on the right is at most $\sqrt{2T/n}$.

Fix a symmetric probability density K whose Fourier transform is supported on $[-1,1]$ and whose first absolute moment is finite. For example, normalize $(\sin(x/4)/(x/4))^4$. Put $K_T(x) = TK(Tx)$. Given a 1-Lipschitz test function f on $[-A,A]$, subtract $f(0)$ and extend it linearly to zero on $[-2A,-A]$ and $[A,2A]$, and then by zero outside $[-2A,2A]$. The resulting g is globally 1-Lipschitz, agrees with $f - f(0)$ on the support, and has $\|g\|_2 \le 2A^{3/2}$. Also $\|g * K_T - g\|_\infty \le C/T$. Fourier inversion, Cauchy–Schwarz and Plancherel yield, uniformly in these tests,

$$\begin{aligned}\left|\int f\, d(\widetilde{G} - G)\right| &\le \frac{C}{T} + \left|\int (g * K_T)\, d(\widetilde{G} - G)\right| \\ &\le \frac{C}{T} + CA^{3/2}\|\chi_{\widetilde{G}} - \chi_G\|_{L^2[-T,T]}.\end{aligned}$$

Here $|\chi_{K_T}| \le 1$ and it vanishes outside $[-T,T]$. The Kantorovich–Rubinstein dual formula and the preceding expectation bound prove (VII.6.5). □

Corollary VII.6.3 (Divergent MLE risk in a consistently estimable experiment)**.** *Let $N \ge 3$,*

$$A_N = \sqrt{\log(eN)}, \qquad \sigma_1 = \cdots = \sigma_{N-1} = 1, \qquad \sigma_N = N^3, \qquad \mathcal{G}_N = \mathcal{P}([-A_N, A_N]).$$

Then every measurable unrestricted joint NPMLE satisfies

$$\sup_{G \in \mathcal{G}_N} \mathbb{E}_G W_1(\widehat{G}, G) \ge \frac{N}{2\sqrt{2\pi}}, \tag{VII.6.6}$$

whereas the minimax expected loss obeys

$$\inf_{\widetilde{G}} \sup_{G \in \mathcal{G}_N} \mathbb{E}_G W_1(\widetilde{G}, G) \le \frac{C}{\sqrt{\log(eN)}}. \tag{VII.6.7}$$

Thus ordinary unrestricted joint likelihood is not uniformly optimal over all known heteroscedastic noise profiles, even when all but one observation have unit noise and the true mixing support grows only as $\sqrt{\log N}$.

Proof. The law δ_0 belongs to $\mathcal{G}_N$, and Theorem VII.6.1 gives (VII.6.6). For the upper bound discard observation N, take $n = N-1$ and $T = \sqrt{\frac{1}{2}\log n}$ in Lemma VII.6.2. Its second term is at most $C\log(eN)n^{-1/4}$, which is bounded by $C/\sqrt{\log(eN)}$ after increasing the absolute constant. This proves (VII.6.7). □

The benchmark uses a known interval, as is permitted in the minimax comparison. The point-mass lower bound itself needs no support information. No conclusion is drawn here about a likelihood estimator constrained to a supplied interval, or about bounded noise profiles. The mechanism is specific: each datum imposes the density floor (VII.6.3), and an unusually large standardized observation at a very large noise scale forces a nonnegligible transport distance from a point mass. A valid all-profile theorem for ordinary unrestricted likelihood must account for this obstruction.

VII.7 The exact boundary of the conclusions

The global classifications proved here concern the explicitly defined period families, the stated central binomial power sums, and the specified complementary beta densities. They do not settle a general classification of all periods or of every rational-parameter beta quantile. The excluded arcsine exponent $r = 1/2$ has different arithmetic and was treated separately in the preceding manuscript.

For the banana family, the all-parameter Galois group conclusion is distinct from the leading-coefficient multiplicity question in [032]. Appendix VII.E provides a general local lower bound on leading-coefficient zeros. It does not identify the exact multiplicity at every banana singularity or prove the separate conjecture.

For heteroscedastic likelihood, the necessary condition $\sigma_{\max} = o(N^2)$ is not asserted to be sufficient. The theorem concerns expected W_1 loss, and no failure of convergence in probability is deduced from the rare-event lower bound. The minimax comparison specifies the support class and constructs its own comparator, avoiding reliance on an inapplicable homoscedastic extension.

The recurrence theorem counts minimum width, not sparse term count. The group theorems concern differential-algebraic relations between functions, not transcendence of all algebraic-input special values. The arithmetic theorem does address individual algebraic cuts and probability levels, but only under its stated rational-exponent and algebraic-coefficient hypotheses.

A correction to the preceding prospectus. The preceding draft described a rational-lemniscate extension too broadly as prospective work. Supplied manuscript [024] already contains the bounded connected rational-lemniscate length result, including its critical-boundary treatment. That statement is not counted as a new result or an unresolved extension here.

VII.A Previously established hypercubic rank and algebraicity theorem

Reproduced supporting proof from the preceding manuscript. This appendix is not counted as a new result.

For $n \geq 1$ and nonzero complex numbers $m_1, \dots, m_n$, define

$$c_k(m) = \sum_{a_1+\cdots+a_n=k} \binom{k}{a_1, \dots, a_n}^2 \prod_i m_i^{2a_i}, \tag{VII.A.1}$$

$$F_m(z) = \sum_{k\geq 0} \binom{2k}{k} c_k(m) z^k. \tag{VII.A.2}$$

The coefficient is the weighted number of length-$2k$ closed walks on the n-dimensional hypercubic lattice when each step in either coordinate direction i has weight m_i. Complex weights are allowed in the analytic statement. The series converges near zero.

Partition the masses by equality of their squares. Write their distinct nonzero squared values as $\mu_1^2, \dots, \mu_q^2$, with multiplicities $n_1, \dots, n_q$. Thus $\sum_j n_j = n$. For the standard representation $V = \mathbb{C}^2$ of SL_2, put

$$W = \bigotimes_{j=1}^{q} \mathrm{Sym}^{n_j} V = \bigoplus_{d\geq 0} (\mathrm{Sym}^d V)^{\oplus a_d}, \qquad Q = \prod_j (n_j + 1), \qquad b = \sum_d a_d. \tag{VII.A.3}$$

The decomposition in the middle uses the diagonal SL_2; the tensor product itself is irreducible for the product group SL_2^q.

Theorem VII.A.1 (All weights and all dimensions)**.** *The minimal order of a nonzero operator in $\mathbb{C}(z)\langle\partial_z\rangle$ annihilating F_m is*

$$r(m) = Q - b = \sum_{d\geq 1} d\, a_d = \prod_j (n_j + 1) - [u^{\lfloor n/2 \rfloor}] \prod_j (1 + u + \cdots + u^{n_j}). \tag{VII.A.4}$$

The minimal operator is irreducible over $\mathbb{C}(z)$. No hypothesis on distinct signed sums, thresholds, or generic choices of the nonzero masses is required.

More precisely, define

$$H_m(s) = s^{-1} F_m(s^{-2}).$$

Its minimal operator over $\mathbb{C}(s)$ is irreducible of the same order $r(m)$. The local monodromy of this equation at $s = \infty$ is unipotent, with one Jordan block of size d for each of the a_d summands with $d \geq 1$ in (VII.A.3)*.*

Corollary VII.A.2 (Algebraicity classification)**.** *Allow zero weights and delete them before counting the dimension. Then $F_m(z)$ is algebraic over $\mathbb{C}(z)$ if and only if at most one weight is nonzero. With no nonzero weights it is 1; with one nonzero weight m, it is $(1 - 4m^2 z)^{-1/2}$. For two or more nonzero weights it is transcendental, even when signed thresholds collide or vanish.*

For equal squared masses, (VII.A.4) is $r = n$. For pairwise distinct squared masses it is

$$r = 2^n - \binom{n}{\lfloor n/2 \rfloor},$$

whose first values are $1, 2, 5, 10, 22, 44, 93$. For example the three-dimensional multiplicity patterns $[3]$, $[2, 1]$, $[1, 1, 1]$ give orders $3, 4, 5$. The first two cases agree with classical isotropic and tetragonal formulae. Generic three-dimensional order five has also been announced in recent work of Elkies, Grimm and Schwartz; it is not presented here as a new generic-three-dimensional discovery. The scope of Theorem VII.A.1 is its uniform formula in all dimensions and all squared-weight multiplicities, including every nonzero complex specialization.

VII.A.1 Provenance and standard differential-module inputs

The Bessel representation and the central-binomial relation to hypercubic walks are classical; see Guttmann [37]. Delves and Joyce's anisotropic three-dimensional order-four equation concerns the tetragonal family with two equal coordinate weights [38]. The recent public announcement [39] describes a generic order-five result and a genus-two period formula in dimension three; its stated exceptional-set restriction and the absence of a linked full paper are kept separate from the theorem proved here. A literature search does not certify historical priority for the all-dimensional statement.

We use the following standard facts about algebraic connections on a curve. They are stated explicitly to fix the extension convention. The Fourier–Laplace transform is the Weyl-algebra equivalence $x \mapsto -\partial_s$, $\partial_x \mapsto s$, and preserves simple holonomic modules. For an irreducible connection M on $\mathbb{G}_m$, its intermediate extension $j_{!*}M$ to $\mathbb{A}^1$ is simple. At a regular singular point with unipotent monodromy T, its vanishing cycles are $\operatorname{im}(T-1)$, with the restricted monodromy. A Jordan block of size $d+1$ therefore contributes a block of size d, and a size-one block contributes nothing. The last assertion follows directly from the nearby/vanishing-cycle maps: absence of a point-supported quotient makes the canonical map surjective, while absence of a point-supported submodule makes the variation map injective, with composite $T-1$.

The stationary-phase theorem gives, if M has only the finite singularity zero and all slopes at infinity are at most one,

$$\operatorname{rank} \mathcal{F}(j_{!*}M) = \dim \operatorname{im}(T_0 - 1). \tag{VII.A.5}$$

The formal germ of the Fourier transform at infinity is regular singular with this vanishing-cycle monodromy. Slope-one exponential factors at the original infinity contribute at finite transformed points, not to this rank formula through a slope-greater-than-one term. These are the formal stationary-phase statements of García López and Sabbah [40, 41]; in particular one must use the intermediate extension, rather than retain the entire nearby cycle space at zero.

We also use the curve de Rham index formula and the local-to-global compact-support sequence: for a connection N on $\mathbb{G}_m$,

$$\chi_{\mathrm{dR}}(\mathbb{G}_m, N) = -\operatorname{irr}_0(N) - \operatorname{irr}_\infty(N),$$

and

$$H^1_c \longrightarrow H^1 \longrightarrow H^1_{\mathrm{loc},0} \oplus H^1_{\mathrm{loc},\infty} \longrightarrow H^2_c \longrightarrow 0, \qquad H^2_c \cong H^0(N^\vee)^\vee.$$

All applications below have zero H^0 for the connection and its dual, and no formal slope-zero part at infinity. These standard index and duality inputs [45], as well as the Bessel differential Galois fact used next, are not asserted as new results.

VII.A.2 The Bessel connection and its complete local partition

Set $\theta_x = x\partial_x$. Let B_μ be the rank-two connection associated with

$$(\theta_x^2 - 4\mu^2 x^2)y = 0. \tag{VII.A.6}$$

It has solutions $I_0(2\mu x), K_0(2\mu x)$. Its differential Galois group over $\mathbb{C}(x)$ is SL_2; this is the classical non-half-integral Bessel case [42, 43]. At zero, I_0 is even and analytic, while

$$K_0(2\mu x) = -(\log x) I_0(2\mu x) + \text{an even analytic germ.}$$

The local monodromy is therefore a nontrivial unipotent block. At infinity the formal exponential factors are $e^{2\mu x}, e^{-2\mu x}$, with a common regular prefactor $x^{-1/2}$.

Lemma VII.A.3. *If the μ_j^2 are pairwise distinct and nonzero, the direct sum of the B_{μ_j} has differential Galois group $\mathrm{SL}_2^{\,q}$. Consequently*

$$M = \bigotimes_j \operatorname{Sym}^{n_j} B_{\mu_j}$$

is irreducible of rank Q, and the function $g(x) = \prod_i I_0(2m_i x)$ generates this connection. At zero its monodromy has blocks of sizes $d+1$, with multiplicities a_d from (VII.A.3); its slopes at infinity are at most one.

Proof. The projections of the Galois group, and of its identity component, onto each SL_2 factor are surjective. The Goursat lemma for products of simple groups [44, Proposition 1.8.2] says that failure of the full product would identify two standard rank-two representations, after a finite algebraic base change and a rank-one twist. At a point over infinity their exponential-factor pairs would be

$$\{2\mu_i x, -2\mu_i x\} = \{2\mu_j x + q(x), -2\mu_j x + q(x)\}.$$

Comparing the sums gives $q(x) = 0$ for their irregular parts, and comparing their differences gives $\mu_i = \pm\mu_j$, a contradiction. The same argument remains valid after ramification at infinity. Thus the full product acts, and its external tensor product of symmetric powers is irreducible.

Evaluation at the product of the I_0 solutions is a nonzero morphism from this simple differential module to a differential field. Its kernel is zero, so its image is precisely the cyclic module generated by g. At zero all factors have the same nonzero logarithmic-monodromy constant. The tensor monodromy is $\exp(\kappa e)$ for the diagonal raising operator e on W. On $\operatorname{Sym}^d V$ this has one block of size $d+1$. At infinity tensor products of the Bessel formal factors have exponentials $\exp(2x\sum_j \lambda_j\mu_j)$, where $\lambda_j = n_j, n_j-2, \dots, -n_j$. Some sums may coincide or vanish; this only produces slope-zero or slope-one components, never slopes greater than one. □

Every summand in (VII.A.3) has the parity of n and contains exactly one weight-zero vector if n is even, or exactly one weight-one vector if n is odd. Thus b is the corresponding central weight multiplicity. The character

$$\prod_j (u^{n_j} + u^{n_j-2} + \cdots + u^{-n_j})$$

proves the coefficient formula in (VII.A.4). In particular $\operatorname{Sym}^n V$ occurs once, so $r \ge n$.

VII.A.3 An upper bound that eliminates the endpoint ambiguity

For $\operatorname{Re} s > 2\sum_i |m_i|$, termwise integration gives

$$H_m(s) = \int_0^\infty e^{-sx} g(x)\,dx = \sum_{k\ge 0} \frac{(2k)!}{s^{2k+1}} \sum_{|a|=k} \prod_i \frac{m_i^{2a_i}}{a_i!^2} = s^{-1} F_m(s^{-2}). \tag{VII.A.7}$$

To prove its order bound one cannot simply identify a unilateral Laplace transform with a Fourier transform: derivatives produce endpoint terms. We instead place this particular integral in a middle de Rham space and construct its closed cycle explicitly.

Work over $K = \mathbb{C}(s)$ and twist M by e^{-sx}; call the resulting connection N. On the cyclic vector η evaluating to g, its x-connection is $\nabla_x^M - s$ and its commuting s-connection is $\nabla_s = \partial_s - x$. The Bessel bases $b_\mu, \theta_x b_\mu$ and their symmetric tensor products give a $K[x]$-lattice N_0 stable under θ_x. Its residue at zero is nilpotent with the same b Jordan blocks as M.

The connection N is irreducible of rank $Q \ge 2$, also after extending the constant field algebraically, so it and its dual have no global horizontal vector. Since s is transcendental

over the mass constants, every formal exponential factor at infinity is nonzero and linear in x after the twist. Thus $\mathrm{irr}_\infty(N) = Q$, while zero is regular singular. The index formula gives $\dim_K H^1_{\mathrm{dR}}(\mathbb{G}_m, N) = Q$. Local cohomology at infinity vanishes, while at zero it is the cokernel of the nilpotent residue and has dimension b. The compact-support sequence therefore gives

$$H^1_{\mathrm{mid}}(N) := \ker\{H^1_{\mathrm{dR}}(N) \longrightarrow H^1_{\mathrm{loc},0}(N)\}, \qquad \dim_K H^1_{\mathrm{mid}}(N) = Q - b = r. \tag{VII.A.8}$$

Here $\omega = [\eta\, dx]$ belongs to this kernel. To check this without conflating dx and dx/x, complete the lattice to $\Lambda = N_0 \otimes K[[x]]$. The operator θ_x acts on $x^k\Lambda/x^{k+1}\Lambda$ as $k + \mathrm{Res}$, which is invertible for $k \geq 1$. It is therefore bijective on $x\Lambda$. There is $v \in x\Lambda$ with $\theta_x v = x\eta$, hence $\nabla_x v = \eta$, so $\eta\, dx$ is formally exact at zero. The commuting connection ∇_s preserves N_0, acts on both global and local cohomology, and hence preserves the kernel in (VII.A.8).

It remains to verify that (VII.A.7) is a period of this space. Choose branches along the positive ray for

$$h(x) = K_0(2m_1 x) \prod_{i>1} I_0(2m_i x).$$

This is a solution of M, including when some squared masses coincide. Continuing once around zero changes it by $-2\pi i\, g(x)$. Start at infinity on this ray, travel to ε, circle zero once, and return to infinity on the continued branch. With the factor e^{-sx} this is a closed rapid-decay twisted cycle: both ends decay at infinity for the stated half-plane, and there is no endpoint at zero. The integral of ω along its small circle is $O(\varepsilon(1 + |\log \varepsilon|))$ and tends to zero. Its two rays therefore give, up to a nonzero constant and orientation, exactly $H_m(s)$. The construction is valid for complex masses: fixed-ray continuations of K_0 have at most exponential growth, which is dominated by e^{-sx} in a sufficiently far right half-plane. Integrals of exact forms on this closed twisted cycle vanish.

Since $\omega, \nabla_s\omega, \dots, \nabla_s^r\omega$ are linearly dependent over K in (VII.A.8), period integration proves

$$\mathrm{ord}_{\mathbb{C}(s)}\, H_m \leq r. \tag{VII.A.9}$$

This argument accounts for the unilateral endpoint instead of silently dropping its possible rank-one contribution.

VII.A.4 Visibility of the simple Fourier constituent

Let $\mathcal{K} = j_{!*}M$ on the affine x-line. Lemma VII.A.3 and (VII.A.5) imply that $\mathcal{F}\mathcal{K}$ is simple of generic rank r. Its germ at $s = \infty$ has exactly the Jordan blocks of sizes d stated in the theorem.

We show that this constituent occurs in the module generated by the actual integral, not merely in a formal transform. Put $A_x = \mathbb{C}[x]\langle\partial_x\rangle$ and $\mathcal{E} = A_x g$. All its elements are entire functions of exponential type. Its restriction to $\mathbb{G}_m$ is M; therefore its composition factors are $\mathcal{K}$, once, and possibly copies of the point module δ_0. For an entire function f in this module, write $\widehat{f}(s) = \int_0^\infty e^{-sx} f(x)\, dx$ and set

$$\mathcal{L} = \{\widehat{f} : f \in \mathcal{E}\} + \mathbb{C}[s].$$

All integrals converge in a common sufficiently far right half-plane for each finite set of functions involved. Integration by parts gives

$$\widehat{xf} = -\partial_s \widehat{f}, \qquad \widehat{f'} = s\widehat{f} - f(0).$$

Consequently $\mathcal{L}$ is an A_s-module, and

$$\mathcal{L}/\mathbb{C}[s] \cong \mathcal{F}\mathcal{E}. \tag{VII.A.10}$$

The map is injective: $\widehat{f}(s) \to 0$ along the positive real axis, so if it is a polynomial it is zero, and uniqueness of the Laplace transform then gives $f = 0$. Thus (VII.A.10) is an actual isomorphism.

Since Fourier sends δ_0 to the trivial module $\mathbb{C}[s]$, every composition factor of $A_s H_m \subseteq \mathcal{L}$ is either $\mathcal{FK}$ (at most once) or the trivial affine-line connection.

If the former were absent, H_m would lie in an iterated extension of trivial affine-line connections. Every function in such a module is a polynomial: choose successive lifts of horizontal generators; the derivative of a lift is a polynomial combination of earlier lifts, so induction and integration prove the assertion. This contradicts $H_m(s) = s^{-1} + O(s^{-3})$. Hence $\mathcal{FK}$ occurs and $\operatorname{ord} H_m \geq r$. Combining with (VII.A.9) proves equality. Since this constituent already uses the full generic rank, the generic differential module generated by H_m is isomorphic to the irreducible generic module of $\mathcal{FK}$. Its monodromy at infinity is therefore exactly the one obtained by stationary phase.

VII.A.5 Descent and the algebraicity dichotomy

Multiplication by s does not change differential order or irreducibility over $\mathbb{C}(s)$. To descend from $sH_m(s) = F_m(s^{-2})$ to $z = s^{-2}$, use the derivation $\partial_z = -(s^3/2)\partial_s$, which preserves even functions. A relation of order r over $\mathbb{C}(s)$ can be decomposed using

$$\mathbb{C}(s) = \mathbb{C}(s^2) \oplus s\mathbb{C}(s^2).$$

Its even and odd parts are separately zero, because all derivatives of $F_m(s^{-2})$ with respect to z are even. At least one part is a nonzero relation over $\mathbb{C}(s^2) = \mathbb{C}(z)$ of no larger order. Conversely every relation over $\mathbb{C}(z)$ pulls back, so the minimal orders are equal. A nontrivial factorization over $\mathbb{C}(z)$ would pull back to one over $\mathbb{C}(s)$, contrary to the irreducibility already proved. This completes the proof of Theorem VII.A.1.

For $n \geq 2$, the summand $\operatorname{Sym}^n V$ in W supplies a unipotent monodromy block of size $n \geq 2$ at $s = \infty$. This monodromy has infinite order. The minimal differential module of an algebraic function has finite monodromy: after taking a finite normal algebraic extension containing all its conjugates, their constant linear span supplies a finite-branch solution space and a differential annihilator over the base field. Its minimal operator is a factor of this finite-monodromy operator. Thus H_m, and consequently F_m, cannot be algebraic for $n \geq 2$. The explicit formulas when zero or one weights remain prove the converse and Corollary VII.A.2.

The argument proves a statement about a function of the variable z. It does not assert transcendence of every special value at algebraic z, and the rank argument alone does not determine the full differential Galois group. The group classifications use the additional arguments in Sections VII.2 and VII.C.

VII.B Previously established reflected Poincaré pairing

Reproduced supporting proof from the preceding manuscript.

The next construction works in every dimension and for every squared-mass multiplicity pattern. We give it directly rather than infer the sign of a form from a Fourier-transform convention.

Lemma VII.B.1 (A form for every choice of weights)**.** *For n nonzero complex weights, the middle connection $\mathcal{P}_m$ has a nondegenerate horizontal bilinear form of parity $(-1)^{n+1}$. Here parity $+1$ means symmetric. Thus its Galois group is contained in an orthogonal group for odd n and a symplectic group for even n.*

Proof. In the Bessel basis $b_\mu, \theta_x b_\mu$, the form $B_\mu(b_\mu, \theta_x b_\mu) = 1$ is alternating, nondegenerate and horizontal. Indeed, $\theta_x^2 b_\mu = 4\mu^2 x^2 b_\mu$ makes its connection matrix infinitesimally symplectic. Symmetric powers and tensor products give M a horizontal nondegenerate form B of parity $\epsilon = (-1)^n$.

Let $\iota : x \mapsto -x$. Its canonical action fixes each Bessel basis vector $b_\mu, \theta_x b_\mu$; the defining matrix is even in x. This identifies $\iota^* M$ with M and preserves B. Consequently

$$\iota^* N_s = M \otimes e^{sx} \simeq N_s^\vee,$$

compatibly both with the x-connection and with the s-connections $\partial_s + x$ on the left and the dual of $\partial_s - x$ on the right. Curve Poincaré duality therefore supplies a perfect pairing of $\mathcal{P}_m$ with itself. The use of the middle image makes this pairing nondegenerate: the annihilator of the image of $H_c^1 \to H^1$ is the kernel of the dual map, so the perfect compact-support pairing descends to the two images. This is the same duality-and-involution construction used for exponential connections in [51, §2, especially (2.12) and Corollary 2.15].

For completeness, its sign can be checked by residues. Represent $a, b \in \mathcal{P}_m$ by N_s-valued one-forms α, β, with formal primitives f_a, f_b at 0 and ∞. The sum of local residues gives

$$P(a,b) = \sum_{v=0,\infty} \operatorname{Res}_v B(f_a, \iota^* \beta). \tag{P}$$

Local exactness at both points is precisely the middle condition; at infinity the twisted connection has no local cohomology. Formula (P) is independent of the choices by the compact-support duality just described. Pullback by ι preserves the sum of residues, and ι is an isometry for B. Hence

$$P(b,a) = \sum_v \operatorname{Res}_v B(\iota^* f_b, \alpha) = \epsilon \sum_v \operatorname{Res}_v B(\alpha, \iota^* f_b).$$

The one-form

$$dB(f_a, \iota^* f_b) = B(\alpha, \iota^* f_b) + B(f_a, \iota^* \beta)$$

has zero residues. It follows that $P(b,a) = -\epsilon P(a,b)$. Residues commute with ∂_s, and the coefficient pairing is horizontal for the two commuting parameter connections, so P is horizontal. This proves the assertion, including nondegeneracy and its sign. □

VII.C Previously established even-dimensional symplectic theorem

Reproduced supporting proof; the new opposite-parity result is Theorem VII.2.1.

Keep the hypercubic series F_m and its Laplace form H_m, now for an arbitrary even number $n \geq 2$ of nonzero complex weights. Let the distinct squared weights have multiplicities $n_1, \dots, n_q$, and set

$$R = \prod_{j=1}^{q} (n_j + 1) - [u^{n/2}] \prod_{j=1}^{q} (1 + u + \cdots + u^{n_j}).$$

The all-weight order theorem identifies R as the rank of the irreducible middle connection (VII.A.8), with no conditions on signed sums.

Theorem VII.C.1 (All even dimensions and all nonzero complex weights)**.** *For every even $n \geq 2$ and every $m \in (\mathbb{C}^\times)^n$, the full differential Galois groups of H_m over $\mathbb{C}(s)$ and F_m over $\mathbb{C}(z)$ are $\mathrm{Sp}_R(\mathbb{C})$ in their standard representations. Thus R is even, and for each function $f = H_m$ or F_m the R functions $f, f', \dots, f^{(R-1)}$ are algebraically independent over its coefficient field. These assertions hold even when signed mass sums collide or vanish.*

Orthogonal and symplectic structures for lattice Green differential operators have long been studied; see, for example, [53]. The statement here determines the full group uniformly over every nonzero complex weight tuple in every even dimension, including collided thresholds. No historical-priority claim is inferred from a literature search.

Here Sp_R denotes the group preserving a nondegenerate alternating form on an R-dimensional space. We first isolate the linear-algebra argument that will identify the group.

Lemma VII.C.2 (One transvection in a connected irreducible group)**.** *Let (V, ω) be a finite-dimensional complex symplectic space. If a connected algebraic subgroup $G \subseteq \mathrm{Sp}(V)$ acts irreducibly and contains a nonidentity transvection, then $G = \mathrm{Sp}(V)$.*

Proof. Write $N_u(v) = \omega(u, v)u$. A symplectic transvection is $1 + aN_u$ for some $u \neq 0$ and $a \neq 0$; this follows directly by writing its rank-one difference as $v \mapsto \ell(v)u$ and using the symplectic identity to conclude $\ell \in \mathbb{C}\omega(u, -)$. Rescaling u absorbs a. Its powers are $1 + kN_u$, so their Zariski closure is $\{1 + tN_u : t \in \mathbb{C}\}$. Consequently $N_u \in \mathfrak{g} = \mathrm{Lie}\, G$, as is N_v for every $v \in \mathcal{O} = G \cdot u$. The span of $\mathcal{O}$ is a nonzero invariant space, hence is V.

Join two orbit vectors when their symplectic pairing is nonzero. The spans W_i of different connected components of this graph are mutually orthogonal. Their sum is direct: if a finite sum $\sum_i w_i = 0$ with $w_i \in W_i$, then w_i is orthogonal to W_i by this equality and to all other W_j by construction; it is orthogonal to V, hence zero. Thus there are finitely many components. The group G permutes their spans. The stabilizer of each span is a closed subgroup of finite index, so connectedness makes every span G-invariant. Irreducibility forces a single component. Choose finitely many orbit vectors spanning V and add the finitely many vertices of paths connecting them. This produces a finite connected spanning set $S \subseteq \mathcal{O}$.

For $v, w \in V$ put

$$L_{v,w}(x) = \omega(v, x)w + \omega(w, x)v.$$

The identities

$$[N_v, N_w] = \omega(v, w)L_{v,w}, \qquad [L_{v,w}, N_y] = \omega(v, y)L_{w,y} + \omega(w, y)L_{v,y} \tag{T}$$

follow by direct multiplication. The first gives $L_{v,w} \in \mathfrak{g}$ for edges of the graph, and $L_{v,v} = 2N_v$ is also in $\mathfrak{g}$. Induct on the length of a path from v to y. If $\omega(v, y) \neq 0$, the first identity already applies. Otherwise, let w be the penultimate vertex. By induction $L_{v,w} \in \mathfrak{g}$, and the second identity gives $L_{v,y} \in \mathfrak{g}$ because $\omega(w, y) \neq 0$. Hence all $L_{v,w}$ with $v, w \in S$ belong to $\mathfrak{g}$. Since S spans V, these span $\mathrm{Sym}^2 V \simeq \mathfrak{sp}(V)$. Thus $\mathfrak{g} = \mathfrak{sp}(V)$, and connectedness proves the lemma. □

VII.C.1 Formal factors and collided thresholds

We state exactly the local Fourier input. If a holonomic module $\mathcal{K}$ has formal germ at infinity

$$\widehat{\mathcal{K}}_{\infty} = \bigoplus_{c\in\mathbb{C}} E^{cx} \otimes R_c, \qquad R_c \text{ regular},$$

then stationary phase at the finite transformed point $s = c$ gives

$$\phi_c(\mathcal{F}\mathcal{K}) \simeq \mathcal{F}^{(\infty,0)}(R_c). \tag{LF}$$

Here ϕ_c is the formal vanishing-cycle germ, including its regular unipotent part. For a regular germ the local transform on the right is regular of the same rank and preserves its monodromy (up to the harmless inversion convention). In particular a trivial rank-one germ becomes the one-dimensional vanishing cycles of δ_0, with identity monodromy. This statement includes $c = 0$. It follows either by applying the global Fourier transform to the trivial connection or from the inverse regular local-transform conventions. Precise references are Arinkin [77, Corollary 2.3(1), §3.3], García López [40, Proposition 2.1], and Sabbah [41, Remarks 5.2, 5.3(4), 5.7].

For an intermediate extension of positive rank, with nearby monodromy T, the canonical map from nearby to vanishing cycles is surjective and the variation map in the opposite direction is injective. Their composites are $T - 1$. Consequently

$$\phi_c \text{ has identity monodromy} \quad\Longrightarrow\quad T \text{ is unipotent and } (T-1)^2 = 0, \qquad \operatorname{rank}(T-1) = \dim \phi_c. \tag{VC}$$

Indeed the vanishing cycles contain every nonidentity generalized eigenspace of T and the image of $T - 1$ on its unipotent part. Thus (VC) does not discard a constant formal factor.

Lemma VII.C.3. *For even n, the minimal equation of H_m is regular singular everywhere. Each finite local monodromy is unipotent with square-zero difference from the identity. At some finite nonzero point this difference has rank one.*

Proof. Let μ_j^2 be the distinct squared weights and put $\Lambda = \prod_j \{n_j, n_j - 2, \dots, -n_j\}$. For $M = \bigotimes_j \operatorname{Sym}^{n_j} B_{\mu_j}$ the Bessel formal solutions at infinity give the direct-sum decomposition

$$\widehat{M}_{\infty} \simeq \bigoplus_{\lambda\in\Lambda} E^{c_\lambda x} \otimes R_n, \qquad c_\lambda = 2\sum_j \lambda_j \mu_j, \tag{B}$$

where R_n is rank one with a formal solution $x^{-n/2}$. The invertible formal power series in the Bessel expansions are absorbed by rank-one formal gauges. Since n is even, R_n is the trivial formal connection. Distinct indices with equal c_λ remain a direct sum of trivial regular factors; collision introduces no extension or logarithm in (B).

The all-weight theorem identifies the equation of H_m with the generic connection of the simple module $\mathcal{F}(j_{!*}M)$. Its positive rank ensures that this is an intermediate extension at every finite point. If $d_c = \#\{\lambda \in \Lambda : c_\lambda = c\}$, (LF) says its formal vanishing cycles at c are regular, of dimension d_c, and have identity monodromy. At other points the vanishing cycles are zero. The purely irregular part of a holonomic germ is unchanged by formal vanishing cycles, so regularity of these vanishing cycles excludes an irregular nearby part. By (VC), all these points are regular singular with the asserted unipotence. The infinity statement of the all-weight theorem already gives regularity and unipotence at infinity.

Choose $\xi \in \mathbb{C}^\times$ such that $\operatorname{Re}(\xi\mu_j) \neq 0$ for every j. Define $\lambda_j^* = n_j \operatorname{sign}\operatorname{Re}(\xi\mu_j)$. This is the unique maximizer of the real functional $\operatorname{Re}(\xi c_\lambda)$ on Λ, and its value is $2\sum_j n_j |\operatorname{Re}(\xi\mu_j)| > 0$. Thus $c_{\lambda^*} \neq 0$ and $d_{c_{\lambda^*}} = 1$. The corresponding local monodromy is a nonidentity transvection. □

Proof of Theorem VII.C.1 for H_m. The middle pairing in Lemma VII.B.1 is alternating, so the Galois group G lies in Sp_R. The all-weight rank theorem gives irreducibility. By Lemma VII.C.3 the equation is regular singular everywhere, so Schlesinger's density theorem identifies G with the Zariski closure of its monodromy group [43]. The fundamental group of the punctured sphere is generated by the local loops. All their matrices are unipotent, including the one at infinity, so all belong to G^0. Thus $G = G^0$. Lemma VII.C.3 supplies one transvection, and Lemma VII.C.2 gives $G = \mathrm{Sp}_R$. □

VII.C.2 Descent to the ordinary generating function

It remains to exclude a finite similitude multiplier on descending from s to $z = s^{-2}$. This can be checked on the explicit pairing. Let $\omega = [\eta\, dx]$ be the cyclic class evaluating to H_m in (VII.A.8), where η is the even tensor product of the Bessel cyclic vectors. The simultaneous involution

$$J : (s, x) \longmapsto (-s, -x)$$

preserves N_s and induces a semilinear involution of $\mathcal{P}_m$ covering $s \mapsto -s$. It is an isometry for (P): applying J to both entries and then changing x to $-x$ leaves the sum of residues unchanged. On the cyclic class it has $J\omega = -\omega$ because dx changes sign. Therefore $\zeta = s\omega$ is fixed. The parameter derivative

$$\nabla_z = -\frac{s^3}{2}\nabla_s$$

commutes with J, so all $\nabla_z^i \zeta$ are fixed. They are a basis by the order theorem and the parity descent for $F_m(s^{-2}) = sH_m(s)$. In this basis the Gram matrix of (P) is fixed by $s \mapsto -s$, and hence belongs to $\mathbb{C}(s^2) = \mathbb{C}(z)$. It is nondegenerate, alternating and z-horizontal. Thus the full Galois group of F_m lies in Sp_R. Its identity component is unchanged by the quadratic extension to $\mathbb{C}(s)$, where the group just proved is Sp_R. The full group over $\mathbb{C}(z)$ must therefore be Sp_R as well.

Finally Sp_R acts transitively on the nonzero vectors of its standard representation. The orbit of the nonzero solution H_m has dimension R, and the same holds for F_m. Picard–Vessiot Galois correspondence gives transcendence degree R for the field generated by the first R jets. Since it has exactly R generators, they are algebraically independent. This completes the proof of Theorem VII.C.1.

VII.D Previously established odd-propagator symplectic theorem

Reproduced supporting proof; the new opposite-parity result is Theorem VII.3.1.

The same transvection argument has a second application to the banana periods in the supplied manuscript [49]. Here the number of Bessel factors is odd, whereas it is even in the hypercubic application. The two integral transforms are different, and so are their parity laws. This section removes the nonzero, distinct-threshold hypothesis from the odd-factor Galois-group theorem in that manuscript. It does not prove the remaining leading-coefficient multiplicity conjecture.

For nonzero complex masses $m_1, \dots, m_n$, define

$$c_k(m) = \sum_{a_1+\cdots+a_n=k} \binom{k}{a_1, \dots, a_n}^2 \prod_i m_i^{2a_i}, \qquad \varpi_m(t) = \sum_{k\geq 0} \frac{c_k(m)}{t^{k+1}}.$$

The series converges near infinity. Group the masses by equality of their squares, with multiplicities $n_1, \dots, n_q$, and write

$$\bigotimes_{j=1}^{q} \operatorname{Sym}^{n_j} \mathbb{C}^2 = \bigoplus_{d\geq 0} (\operatorname{Sym}^d \mathbb{C}^2)^{\oplus a_d}$$

for restriction to the diagonal SL_2.

Precise prior input. The all-mass order theorem of [49] states that the minimal connection of ϖ_m is irreducible, has rank

$$o(m) = \sum_{d\geq 2} (d-1)a_d,$$

and has unipotent monodromy at infinity with blocks J_{d-1}, one for each summand with $d \geq 2$. In particular, the block J_{n-1} occurs once. Its middle-cohomology construction gives an invariant nondegenerate alternating form when n is odd. These statements do not require distinct or nonzero signed thresholds. They are used here as previously proved inputs, rather than counted again as new results. The cited manuscript contains their proofs; the new removal of threshold restrictions is proved below.

Theorem VII.D.1 (Every odd number of propagators, every nonzero mass tuple)**.** *Let $n \geq 3$ be odd and $m_i \in \mathbb{C}^\times$. The differential Galois group over $\mathbb{C}(t)$ of the minimal equation of ϖ_m is*

$$\mathrm{Sp}(o(m), \mathbb{C}).$$

This holds when nonzero thresholds coincide and when signed mass sums vanish. Consequently

$$\varpi_m, \varpi_m', \dots, \varpi_m^{(o(m)-1)}$$

are algebraically independent over $\mathbb{C}(t)$, and remain so over every finite algebraic extension of that differential field.

VII.D.1 Why collisions do not obstruct connectedness

Put $y = u^2$ and $g_m(y) = \prod_i I_0(m_i\sqrt{y})$. Its Bessel connection is

$$M = \bigotimes_j \operatorname{Sym}^{n_j} B_{\mu_j}, \qquad B_\mu : \quad (\theta_y^2 - \mu^2 y/4)f = 0, \quad \theta_y = y\partial_y.$$

On the formal u-cover at infinity, it splits as a direct sum of rank-one factors indexed by

$$\lambda_j \in \{n_j, n_j - 2, \dots, -n_j\}, \qquad c(\lambda) = \sum_j \lambda_j \mu_j.$$

Each factor has the form

$$e^{c(\lambda)u} u^{-n/2}(1 + O(u^{-1})).$$

The splitting comes from the two formal Bessel solutions in each factor before taking symmetric and tensor powers. Equality of two exponential factors therefore gives a direct sum with multiplicity, not a new nontrivial formal extension.

For $c \neq 0$, the pair indexed by $\lambda, -\lambda$ descends to

$$A_c(n) = [u \mapsto y = u^2]_+(E^{cu} \otimes U_n), \qquad U_n : \ u^{-n/2}.$$

The standard local Fourier calculation sends this pair, after the first Fourier transform, to

$$E^{c^2/(4s)} \otimes R_c, \qquad R_c \text{ regular of rank one.}$$

The power in its Laplace asymptotic is $s^{(n-3)/2}$: under $y = u^2$, the measure contributes $2u\,du$, evaluation at the critical point $u = c/(2s)$ contributes $s^{n/2-1}$, and the Gaussian fluctuation contributes $s^{-1/2}$. Equivalently this is the formal stationary-phase formula [41, 40]. Thus the monodromy of R_c is $(-1)^{n-3}$, independent of c. Only its value 1 for odd n is needed here.

The construction of the banana connection first takes the Fourier transform of $j_{!*}M$, then pulls its restriction to $\mathbb{G}_m$ back by $s = 1/v$, and then takes the Fourier transform of the intermediate extension in v. The rational gauge v^{-1} in the scalar integral representation has no effect on monodromy. If τ is the second Fourier variable, the period normalization is $t = 4\tau$; the factor $E^{c^2 v/4}$ is therefore located at $t = c^2$. The factor above produces vanishing cycles at $t = c^2$ with the same monodromy. If several pairs give the same nonzero c^2, their vanishing-cycle spaces are the direct sum of these lines. For odd n their monodromy is the identity.

The final Fourier module is simple of positive rank. Hence it is the intermediate extension at each finite point. If T_a denotes its local monodromy at a nonzero threshold a, the nearby/vanishing-cycle description gives

$$\phi_a \cong \operatorname{im}(T_a - I), \qquad (T_a - I)^2 = 0.$$

In particular T_a is unipotent, even at a collision. The regular part arising from $c = 0$ is still regular through the two transforms; it can affect the monodromy at zero, but creates no further nonzero singularity. All formal local factors of the final module are regular, so its global equation is regular singular.

The monodromy group is generated by the loops around infinity and the nonzero thresholds: in the fundamental group relation one can omit the loop around zero. Every one of these generators is unipotent. The Zariski closure of the cyclic group generated by a unipotent matrix in characteristic zero is connected. Schlesinger's theorem therefore shows that the differential Galois group G is connected. This argument needs no assertion that the monodromy at zero itself is unipotent.

VII.D.2 A simple threshold always survives

Choose $\xi \in \mathbb{C}$ such that $\operatorname{Re}(\xi\mu_j) \neq 0$ for every j, and maximize $\operatorname{Re}(\xi c(\lambda))$ over the finite index set. There is a unique maximizer, given by

$$\lambda_j^* = n_j \operatorname{sign} \operatorname{Re}(\xi\mu_j).$$

Its value has strictly positive real projection, so $c_* := c(\lambda^*) \neq 0$. The only tuple giving c_* is λ^*; the only one giving $-c_*$ is $-\lambda^*$. Over $\mathbb{C}$, equality $c(\lambda)^2 = c_*^2$ means $c(\lambda) = c_*$ or $-c_*$.

Consequently the squared threshold c_*^2 has exactly one contributing pair, even if other thresholds collide or vanish.

Its vanishing-cycle space is one-dimensional with trivial monodromy. Since the final module is an intermediate extension, the corresponding T_* satisfies

$$\operatorname{rank}(T_* - I) = 1, \qquad (T_* - I)^2 = 0.$$

It is a nontrivial symplectic transvection.

Proof of Theorem VII.D.1. The prior input gives irreducibility and an invariant alternating form. The preceding arguments give connectedness and a nontrivial transvection. Lemma VII.C.2 therefore gives $G = \mathrm{Sp}(o(m))$.

Let E be a Picard–Vessiot extension and v the nonzero solution vector corresponding to ϖ_m. The field generated by its first $o(m)$ derivatives is the fixed field of its stabilizer. Its transcendence degree is consequently the dimension of the G-orbit of v. A symplectic group is transitive on nonzero vectors, so this dimension is $o(m)$. There are exactly $o(m)$ generators, and hence they are algebraically independent. Algebraic extension of the base field preserves their transcendence degree; equivalently, the connected Galois group is unchanged by finite algebraic base extension. □

For pairwise distinct squared masses, the rank in this theorem is the established formula

$$o(m) = 2^n - \binom{n+1}{\lfloor (n+1)/2 \rfloor}.$$

For all equal squared masses it is $n-1$. The theorem concerns functional algebraic independence, not arithmetic independence of values at a fixed energy. The separate reflection argument for even n is given in Section VII.3.

VII.E Leading-coefficient multiplicity bounds monodromy variation

The following local statement applies to arbitrary singularities, including irregular ones. It does not infer convergence of formal solutions at an irregular point.

Lemma VII.E.1 (A simple leading zero)**.** *Let*

$$ty^{(N)} + \sum_{j=0}^{N-1} b_j(t)y^{(j)} = 0, \qquad b_j \in \mathbb{C}\{t\}, \quad N \geq 1.$$

Its local monodromy T on the N-dimensional solution space satisfies $\operatorname{rank}(T - I) \leq 1$.

Proof. For $N = 1$ the assertion is immediate. Suppose $N \geq 2$ and substitute $y = \sum_{n\geq 0} c_n t^n$. At coefficient t^k, the coefficient multiplying the new unknown c_{k+N-1} is

$$\frac{(k+N-1)!}{k!}\big(k + b_{N-1}(0)\big).$$

All remaining terms involve earlier coefficients. The multiplier has at most one zero for $k \geq 0$; there the recurrence imposes at most one constraint and introduces one new free coefficient. Thus the formal Taylor solution space has dimension at least $N - 1$. After division by t, each pole order is at most $1 \leq N - j$, so the equation is regular singular. Frobenius convergence gives $N - 1$ single-valued solutions and hence $\dim \ker(T - I) \geq N - 1$. □

Theorem VII.E.2 (Multiplicity bound, with no regularity hypothesis)**.** *Let*

$$L = \sum_{j=0}^{N} a_j(t)\partial_t^j, \qquad a_j \in \mathbb{C}\{t\}, \quad a_N \not\equiv 0,$$

and let $m = \operatorname{ord}_0(a_N)$. If T is the local monodromy of $Ly = 0$ around zero, then

$$\operatorname{rank}(T - I) \leq \min\{m, N\}.$$

Proof. The case $m = 0$ is ordinary. For $m \geq 1$, choose a small closed disk D centered at zero with no other zero of a_N and no zero of a_N' away from zero; this is possible since $a_N = t^m u(t)$ with $u(0) \neq 0$. Replace only the leading coefficient:

$$L_\varepsilon = (a_N(t) - \varepsilon)\partial_t^N + \sum_{j=0}^{N-1} a_j(t)\partial_t^j.$$

For small nonzero ε, Rouché's theorem gives exactly m zeros of $a_N - \varepsilon$ in D, all simple by the derivative condition, and none on ∂D.

At each of these m points, division by a holomorphic unit puts the operator into Lemma VII.E.1. The corresponding local monodromy has variation rank at most one. Monodromy T_ε around ∂D is a product of conjugates of these local monodromies. The identity $AB - I = A(B - I) + (A - I)$ gives $\operatorname{rank}(T_\varepsilon - I) \leq m$.

Normalize the companion fundamental matrix to the identity at a fixed point of the circle. Its coefficient matrix converges uniformly there to that of L, so parameter dependence of solutions gives $T_\varepsilon \to T$. Vanishing of the $(m + 1)$-minors is closed; therefore $\operatorname{rank}(T - I) \leq m$. The bound by N is automatic. □

Corollary VII.E.3 (Several singular points)**.** *If a scalar operator with polynomial coefficients annihilates a local system whose monodromies T_c at distinct finite points c are visible on its solution space, then its leading coefficient has degree at least*

$$\sum_c \operatorname{rank}(T_c - I).$$

Here "visible" may mean restriction to a monodromy-invariant subspace; restriction cannot increase the rank of $T_c - I$.

Proof. Apply Theorem VII.E.2 at every c and sum the required zero multiplicities of the leading coefficient. □

Part VIII

Entropy–Transport, Critical Values, and Laplace Monodromy

Manuscript overview. This expanded revision supplies proofs for several remaining ranges and extends the structural conclusions of the preceding manuscript. For Chebyshev coefficients in the Bernstein basis, we prove that $\log 2$ is the optimal universal exponential constant when an absolute multiplicative prefactor is allowed, and determine the exact exponential profile in proportional and moderate degrees. Incorporating the previously established full estimation curves closes the Poisson support transition and the worst-null binomial testing transition. We classify the minimum recurrence width of every arithmetic subsequence of every nontrivially weighted binomial-power sum, and identify the entire space of algebraic-number-valued Apéry limits. A local Fourier-transform argument replaces an insufficiently specified global-period input and gives an iterated monodromy annihilator bound. Applications determine exact Bessel determinant modules and algebraic relations among initial derivatives; for bounded-height tableau exponential generating functions, the arithmetic conclusions hold at every nonzero algebraic argument. Finally, for positive rational beta parameters with integral sum, arbitrary algebraic cuts, and positive algebraic polynomial tilts, a two-interval Angelesco polynomial has algebraic coefficients exactly in the polynomial weight case or on the diagonal of its multi-index. The earlier results used in the revision are distinguished from the new extensions, and the unresolved statements are listed explicitly.

Integration note. This is the latest expanded revision, incorporating the complete Poisson and binomial estimation curves and the repaired local Fourier argument. Its new classifications and its explicit remaining questions are preserved without altering their hypotheses.

VIII.1 What this revision establishes

The starting point is the thirty-page draft entitled *Entropy–Transport, Critical Values and Laplace Monodromy*. This revision retains its principal conclusions with their precise hypotheses, replaces the global-period dependency in its Laplace argument, and incorporates relevant results proved in the intervening development. Every unpublished supporting argument needed here is reproduced; published structural inputs are identified where they enter. No earlier local file or separate bibliography is needed to compile or read this manuscript.

Topic	Starting point	Conclusion in this revision
Chebyshev coefficients	Constant-one bound $e^{m^2/t}$ and a bulk energy bound	Sharp fixed-prefactor exponent $\log 2$; exact proportional and moderate exponential profiles.
Poisson mixtures	A remaining window between polynomial support scales and $B \geq N$	Full known-support minimax curve, incorporating the earlier all-support proof; crossover $B \asymp N/\log(eN)$.
Binomial testing	Near-linear replication ranges	Full worst-null separation curve for all $N, t \geq 1$, by the complete estimation theorem.
Franel recurrences	Ordinary and alternating minimum widths	All complex weights $\lambda \neq 0$ and all progressions $qn + b$, including every parity exception.
Apéry limits	Rational-valued eventual uniqueness	A complete K-linear classification for every $\mathbb{Q} \subseteq K \subseteq \overline{\mathbb{Q}}$ and every progression.
Laplace monodromy	A rapid-decay period input needing clarification	A local Fourier proof and the iterated bound $(T_r - I)^r p(T_r) = 0$.
Bessel determinants	Three ordinary-generating-function algebraicity thresholds	Exact modules and jet relations; tableau specialization at every nonzero algebraic point.
Angelesco polynomials	Arithmetic obstruction for incomplete beta integrals	An exact algebraicity criterion for every two-component multi-index, with polynomial tilts.

Here "global" refers to the explicitly quantified parameter family in each theorem. It does not remove hypotheses concerning known support, algebraic parameters, positivity, or regular singularities. In particular, the arithmetic uniqueness statements concern eventual solutions with algebraic-number values; they do not assert uniqueness in the complex solution space. The new Angelesco theorem requires integral parameter sum, and the sharp coefficient theorem allows a fixed prefactor.

Three mechanisms connect the results. Integrating a coefficient contour retains the Gaussian localization lost by a supremum bound. Likelihood scores produce an exact posterior coupling, allowing different polynomial scales to use different statistical norms. In the differential arguments, formal exponential multiplicities and local Jordan blocks retain information that distinct critical values alone can lose. Arithmetic specialization then uses classical transcendence theorems with their exceptional-point hypotheses checked explicitly.

VIII.2 The constant-one Chebyshev–Bernstein theorem

This section retains the attached draft's constant-one theorem and energy refinement. To make its finite verification fully reproducible, we use the earlier conversation manuscript's rational certificate, printed in full below, in place of an unprinted interval-arithmetic check. The next section gives a new sharp exponent theorem with a fixed universal prefactor.

Write $B_j^t(x) = \binom{t}{j}x^j(1-x)^{t-j}$ and define $C(t,m,j)$ by

$$T_m(2x-1) = \sum_{j=0}^{t} C(t,m,j)B_j^t(x), \qquad 0 \le m \le t.$$

Theorem VIII.2.1 (The coefficient conjecture)**.** *For all integers $t \ge 1$, $0 \le m \le t$, and $0 \le j \le t$,*

$$|C(t,m,j)| \le \exp(m^2/t). \tag{VIII.2.1}$$

The inequality is strict for $m > 0$. The proof below is a finite-certificate proof: its only computational part uses exact rational arithmetic for $t < 512$. The complete verifier is included in the text.

This is the precise constant-one assertion posed in [70, Supplement, Remark 4.1, equation (23)], rather than a bound with an unspecified multiplicative constant. The distinction was retained in the subsequent discussion [71, Appendix C.2] and in our preceding manuscript. The universal exponent with an unspecified fixed prefactor is determined in Theorem VIII.3.1; the prefactor-one optimum remains separate.

Lemma VIII.2.2 (Three complementary coefficient bounds)**.** *For $1 \le m \le t$,*

$$|C(t,m,j)| \le 2^m, \tag{VIII.2.2}$$

$$|C(t,m,j)| \le (t+1)\exp\{tH(m/t)\} \le (t+1)\exp\{(\log 2)m^2/t\}, \tag{VIII.2.3}$$

where $H(a) = \frac{1}{2}[(1+a)\log(1+a) + (1-a)\log(1-a)]$ and $0\log 0 = 0$. For $m < t$ there is also the bound

$$|C(t,m,j)| \le D(t,m) := \prod_{l=1}^{m-1} \frac{t+l}{t-l}. \tag{VIII.2.4}$$

In particular, $D(t,m) \le e^{m^2/t}$ whenever

$$(m+6)(m-1)^2 \le 6t^2. \tag{VIII.2.5}$$

Proof. At degree m the coefficients are $(-1)^{m-i}\binom{2m}{2i}/\binom{m}{i}$. The successive absolute-value ratio is $(2m-2i-1)/(2i+1)$, so their maximum occurs at a middle index. Vandermonde's identity bounds the numerator at that index by $2^m \max_i \binom{m}{i}$, proving (VIII.2.2) at degree m. Degree elevation replaces coefficients by convex combinations and proves the bound at every $t \ge m$.

For (VIII.2.3), put

$$F(z) = (1+z)^t T_m\left(\frac{z-1}{z+1}\right) = \sum_{j=0}^{t} \binom{t}{j} C(t,m,j)z^j.$$

Endpoint coefficients have absolute value one. For $0 < j < t$, set $r = j/(t-j)$ and, on $z = re^{i\theta}$, set $u = 2\sqrt{r}\,|\sin(\theta/2)|/(1+r)$. The larger inverse Joukowski modulus of $(z-1)/(z+1)$ is $\sqrt{(1+u)/(1-u)}$. Consequently

$$\frac{|F(re^{i\theta})|}{(1+r)^t} \le (1-u^2)^{t/2}\left(\frac{1+u}{1-u}\right)^{m/2} \le \exp\{tH(m/t)\}.$$

The last maximum is attained at $u = m/t$; limiting values handle $u = 1$. Cauchy's formula and $\binom{t}{j} r^j/(1+r)^t \ge 1/(t+1)$ give the first inequality in (VIII.2.3). This probability is a mode of a Binomial$(t, j/t)$ law, which proves the stated lower bound. Finally

$$H(a) = \sum_{k\ge 1} \frac{a^{2k}}{2k(2k-1)} \le a^2 \sum_{k\ge 1} \frac{1}{2k(2k-1)} = a^2 \log 2.$$

To apply the older bound (VIII.2.4), combine the hypergeometric formula with the identity

$$x^r = \sum_j (j)_{\underline{r}} B_j^t(x)/(t)_{\underline{r}}.$$

This gives

$$C(t,m,j) = (-1)^m \, {}_3F_2\left(\begin{matrix} -m, m, -j \\ 1/2, -t \end{matrix}; 1\right) = (-1)^m Q_m(j; -1/2, -1/2, t).$$

Dette's Hahn bound [73, Corollary 3.6(b)] is exactly (VIII.2.4). Its stipulated range $m < t$ is respected here. Thus this inequality itself is prior art. For the final assertion, using the positive series for arctanh gives

$$\begin{aligned} \log D(t,m) &= 2 \sum_{l=1}^{m-1} \operatorname{arctanh}(l/t) \\ &\le \frac{m(m-1)}{t} + \frac{m^2(m-1)^2}{6t^3\{1-(m-1)^2/t^2\}}. \end{aligned}$$

The second term is at most m/t precisely under (VIII.2.5). □

Proof of Theorem VIII.2.1. The case $m = 0$ is immediate. Suppose first $t \ge 512$. If $m^2/t \ge \frac{10}{3}\log(t+1)$, the contour bound proves the assertion because $\log 2 < 7/10$. Otherwise,

$$m^2 < \frac{10}{3} t \log(t+1) \le \frac{25}{9} t^{4/3}, \qquad m < \frac{5}{3} t^{2/3}.$$

Here $\log(t+1) \le (5/6)t^{1/3}$ for $t \ge 512$: at 512 it follows from $513 < 729$, $\log 3 < 11/10$, and $33/5 < 20/3$; the difference $(5/6)t^{1/3} - \log(t+1)$ has positive derivative thereafter. In particular $m < t$, and

$$(m+6)(m-1)^2 \le m^3 + 6m^2 \le \left(\frac{125}{27} + \frac{25}{96}\right) t^2 < 6t^2.$$

The last estimate uses $t^{2/3} \ge 64$. The Hahn bound therefore applies. The elementary logarithmic inequalities just used can, for example, be verified by positive Taylor lower bounds for $e^{7/10}$ and $e^{11/10}$.

For $1 \le t < 512$, the following verifier covers every pair (t, m). Its first branch uses (VIII.2.5); its second uses $2^m \le e^{7m/10} \le e^{m^2/t}$; its third uses the contour bound with $\log 2 < 7/10$. Every other pair is checked with the exact identity

$$C(t,m,j) = \frac{1}{\binom{t}{j}} \sum_{i=\max(0,j-t+m)}^{\min(m,j)} (-1)^{m-i} \binom{2m}{2i} \binom{t-m}{j-i}. \tag{VIII.2.6}$$

The function `exp_lower` returns a rational number no larger than e^x for $x \ge 0$, so each successful assertion is a rigorous inequality in the required direction. There are no floating-point comparisons. The complete output is

$$(34784, 39518, 56329, 185), \qquad 12734 \text{ direct coefficient comparisons}, \qquad \max t_{\mathrm{direct}} = 145.$$

The four counts sum to $511 \cdot 512/2$, as required. □

```
from fractions import Fraction
from math import comb

def exp_lower(x, degree=32):
    term = total = Fraction(1)
    for k in range(1, degree + 1):
        term *= x / k
        total += term
    return total

counts = [0, 0, 0, 0]
comparisons = largest_direct = 0
for t in range(1, 512):
    for m in range(1, t + 1):
        if m < t and (m+6)*(m-1)**2 <= 6*t*t:
            counts[0] += 1
            continue
        if 10*m >= 7*t:
            counts[1] += 1
            continue
        if exp_lower(Fraction(3*m*m, 10*t)) >= t+1:
            counts[2] += 1
            continue
        lower = exp_lower(Fraction(m*m, t))
        for j in range(t + 1):
            numerator = sum(
                (-1)**(m-i)*comb(2*m, 2*i)*comb(t-m, j-i)
                for i in range(max(0,j-t+m), min(m,j)+1)
            )
            assert Fraction(abs(numerator), comb(t,j)) <= lower
            comparisons += 1
        counts[3] += 1
        largest_direct = max(largest_direct, t)
print(counts, comparisons, largest_direct)
assert sum(counts) == 511*512//2
```

VIII.2.1 Krawtchouk identities and the energy refinement

The following identities also justify the weighted norm used in the next section. Define

$$K_k(x;N)=\sum_{h}(-1)^h\binom{x}{h}\binom{N-x}{k-h},\qquad \sum_{k=0}^{N}K_k(x;N)u^k=(1-u)^x(1+u)^{N-x}.$$

Lemma VIII.2.3. *For $0\le m,j\le t$,*

$$\binom{t}{j}C(t,m,j)=(-1)^{m+j}K_{2j}(t+m;2t).$$

Moreover

$$\sum_{x=0}^{N}\binom{N}{x}K_k(x;N)K_\ell(x;N)=2^N\binom{N}{k}\mathbf{1}_{k=\ell}.$$

For fixed t,j, write $y=2j/t-1$ and $C_m=C(t,m,j)$. Then

$$C_0=1,\quad C_1=y,\qquad (t-m)C_{m+1}=2tyC_m-(t+m)C_{m-1}\quad(1\le m\le t-1).$$

Proof. In the coefficient generating polynomial $F(z)$ above, put $z=w^2$. The even part of $(1+iw)^{t+m}(1-iw)^{t-m}$ is

$$\frac{1}{2}\{(1+iw)^{t+m}(1-iw)^{t-m}+(1-iw)^{t+m}(1+iw)^{t-m}\}.$$

Multiplication by $(-1)^m$ gives $F(z)$. Comparing its even powers with the Krawtchouk generating function, and using $K_k(N-x;N) = (-1)^k K_k(x;N)$, gives the first identity. Multiplying two generating functions and summing with weight $\binom{N}{x}$ gives $(2+2uv)^N$; comparison of coefficients gives orthogonality. Finally, the same generating function verifies

$$(N-x)K_k(x+1;N) + xK_k(x-1;N) = (N-2k)K_k(x;N).$$

Take $N = 2t$, $k = 2j$, $x = t+m$ and insert the signs in the first identity to obtain the recurrence. □

The one-term consequence of this orthogonality identity is the exact squared bound used in the attached draft,

$$C(t,m,j)^2 \le \frac{4^t\binom{2t}{2j}}{\binom{2t}{t+m}\binom{t}{j}^2}.$$

It is not needed by the rational-certificate proof above.

Proposition VIII.2.4 (The energy bound retained from the draft)**.** *For fixed $t \ge 2, j$, set $y = 2j/t - 1$, $C_m = C(t,m,j)$, and*

$$\alpha_0 = 1, \qquad \alpha_{m+1} = \alpha_m \frac{(t-m)^2}{(t+m)(t-m+1)}.$$

For $1 \le m \le t$ define

$$Q_m = \alpha_m C_m^2 + \alpha_{m-1} C_{m-1}^2 - \frac{2t\alpha_m}{t-m+1} y C_m C_{m-1}, \qquad \kappa_m = \frac{t^2}{(t+m-1)(t-m+2)}.$$

Then, for $1 \le m < t$,

$$Q_{m+1} - Q_m = -\frac{2(m-1)\alpha_{m-1}}{(t+m-1)(t-m+2)} C_{m-1}^2 \le 0, \qquad Q_1 = 1 - \frac{t}{t+1} y^2.$$

If $\kappa_m y^2 < 1$, this yields

$$C(t,m,j)^2 \le D(t,m) \frac{t+1}{t-m+1} \frac{1 - ty^2/(t+1)}{1 - \kappa_m y^2},$$

where $D(t,m) = \prod_{i=1}^{m-1}(t+i)/(t-i)$ also at $m = t$. In particular, for each fixed $0 < \delta < 1$, uniformly for $|y| \le 1-\delta$ and $m \le \delta t/2$,

$$|C(t,m,j)| \le K_\delta \exp\{m(m-1)/(2t) + O(m^4/t^3)\}.$$

Proof. Substitute the three-term recurrence into Q_{m+1} and cancel its $y^2 C_m^2$ terms. The result is

$$Q_{m+1} = \alpha_m C_m^2 - \frac{2t\alpha_m}{t-m+1} y C_m C_{m-1} + \alpha_m \frac{t+m}{t-m+1} C_{m-1}^2.$$

Subtracting Q_m and using the ratio of consecutive α's gives the displayed difference. Completing the square gives

$$Q_m = \alpha_{m-1}\left(C_{m-1} - \frac{t\alpha_m y}{(t-m+1)\alpha_{m-1}} C_m\right)^2 + \alpha_m C_m^2 (1 - \kappa_m y^2).$$

Telescoping the defining product gives $\alpha_m^{-1} = D(t,m)(t+1)/(t-m+1)$. These identities and $Q_m \le Q_1$ prove the bound. In the stated bulk range, $(t+m-1)(t-m+2) \ge t^2 - m^2$, so $\kappa_m y^2 \le (1-\delta)^2/(1-\delta^2/4) < 1$ uniformly. The remaining rational factors are bounded, and the positive series for artanh gives $\log D(t,m) = m(m-1)/t + O(m^4/t^3)$ for $m \le t/2$. □

VIII.3 The optimal exponent with a universal prefactor

Let $M_{t,m} = \max_{0\le j\le t} |C(t,m,j)|$ and put

$$H(a) = \tfrac{1}{2}\{(1+a)\log(1+a) + (1-a)\log(1-a)\}, \qquad 0 \le a \le 1,$$

where $0\log 0 = 0$. The following strengthens the coefficient results of the attached draft. Its distinction between a fixed multiplicative constant and multiplicative constant one is essential.

Theorem VIII.3.1 (A sharp exponential constant)**.** *There is an absolute constant K such that, for all $t \ge 1$, $0 \le m \le t$ and $0 \le j \le t$,*

$$|C(t,m,j)| \le K(1+m^2/t)^{1/4}\exp\{tH(m/t)\}, \tag{VIII.3.1}$$

$$|C(t,m,j)| \le K\,2^{m^2/t}. \tag{VIII.3.2}$$

The exponent $\log 2$ in the second inequality is optimal among inequalities $|C(t,m,j)| \le K_c\exp(cm^2/t)$ with a finite constant K_c independent of t,m,j. This does not prove the same inequality with prefactor exactly one.

The improvement comes from integrating the coefficient contour, rather than replacing its integrand by its supremum. The resulting prefactor depends on m^2/t, instead of on t alone.

Lemma VIII.3.2 (A Gaussian integral against the arcsine law)**.** *For every $A, B \ge 0$,*

$$(1+B)\int_0^{\pi/2} \exp\{-\tfrac{1}{2}(B\sin v - A)^2\}\,dv \le K\sqrt{1+A}.$$

Proof. The case $B \le 1$ is immediate. Suppose $B > 1$. The substitution $y = B\sin v$ gives the measure $dy/\sqrt{B^2-y^2}$ on $[0,B]$. If $B \le 2(A+1)$, use $\sqrt{B^2-y^2} \ge \sqrt{B(B-y)}$ to bound the integral by

$$B^{-1/2}\int_0^B e^{-(y-A)^2/2}(B-y)^{-1/2}\,dy \le B^{-1/2}(2+\sqrt{2\pi}).$$

For the last inequality put $u = B-y$, split at $u=1$, and use $u^{-1/2} \le 1$ on the remaining interval. Multiplication by $1+B \le 2B$ proves the claim in this case.

If $B > 2(A+1)$, the part $0 \le y \le 3B/4$ is at most K/B, because the denominator is at least $\sqrt{7}B/4$ and the Gaussian integral is bounded. On $3B/4 \le y \le B$ we have $y - A > B/4$; this part is at most $(\pi/2)e^{-B^2/32}$. Multiplying by $1+B$ gives a uniform constant, which is at most $K\sqrt{1+A}$. □

Proof of (VIII.3.1). Endpoint coefficients have absolute value one. For $0 < j < t$, put $p = j/t$, $r = p/(1-p)$, $v = p(1-p)$, $a = m/t$, and

$$F(z) = (1+z)^t T_m\left(\frac{z-1}{z+1}\right) = \sum_{j=0}^{t}\binom{t}{j}C(t,m,j)z^j.$$

For $z = re^{i\theta}$ set $u = 2\sqrt{v}\,|\sin(\theta/2)|$. The inverse Joukowski formula and $|T_m(w)| \le |\zeta|^m$ when $w = (\zeta+\zeta^{-1})/2$, $|\zeta| \ge 1$, give

$$\frac{|F(re^{i\theta})|}{(1+r)^t} \le \exp\{tf_a(u)\}, \qquad f_a(u) = \tfrac{1}{2}\log(1-u^2) + a\,\operatorname{artanh} u.$$

Explicitly, one may take $\zeta = (\sqrt{z}+i)/(\sqrt{z}-i)$ or its reciprocal; choosing the larger modulus gives $|\zeta| = \sqrt{(1+u)/(1-u)}$. Limits cover the removable endpoint configurations. Since

$f_a'(u) = (a-u)/(1-u^2)$ and $f_a(a) = H(a)$, integration of this derivative on the interval between u and a shows

$$f_a(u) \le H(a) - \tfrac{1}{2}(u-a)^2. \tag{*}$$

This uses $1/(1-w^2) \ge 1$ and also holds at $a = 1$ by a limit.

The binomial mass $\pi_j = \binom{t}{j} p^j (1-p)^{t-j}$ is a mode of $\mathrm{Bin}(t,p)$. Chebyshev's probability inequality on the interval of radius $2\sqrt{tv}$ around its mean j implies

$$\pi_j \ge \frac{3/4}{4\sqrt{tv}+1} \ge \frac{c}{1+\sqrt{tv}}.$$

Cauchy's coefficient formula, (*) and Lemma VIII.3.2, with $A = m/\sqrt{t}$ and $B = 2\sqrt{tv}$, therefore yield

$$|C(t,m,j)| \le K e^{tH(a)} \sqrt{1+m/\sqrt{t}} \le K(1+m^2/t)^{1/4} e^{tH(a)}.$$

The final inequality follows from $(1+u)^2 \le 2(1+u^2)$. □

Proof of (VIII.3.2) *and optimality.* Write $x = m^2/t$ and $\Delta(a) = (\log 2)a^2 - H(a)$. The positive power series

$$H(a) = \sum_{h \ge 1} \frac{a^{2h}}{2h(2h-1)}$$

shows that $H(a)/a^2$ increases strictly on $(0,1)$ and tends to $\log 2$ at one. Hence for $0 \le a \le 99/100$ there is an absolute $c_0 > 0$ such that $t\Delta(a) \ge c_0 x$. The factor $(1+x)^{1/4} e^{-c_0 x}$ is bounded, proving the assertion in this range.

Suppose $99/100 \le a < 1$ and write $\delta = 1-a$, $k = t-m$. Then k is an integer with $1 \le k \le t/100$. Since $H'(a) = \operatorname{artanh} a$,

$$\begin{aligned} \Delta(1-\delta) &\ge -2(\log 2)\delta + \tfrac{1}{2}\int_0^{\delta} \log(1/u)\,du \\ &= \delta\{\tfrac{1}{2}\log(1/\delta) + \tfrac{1}{2} - 2\log 2\} \ge \tfrac{1}{4}\delta \log(1/\delta). \end{aligned}$$

The last inequality follows from $\delta \le 1/100$; for example $\log 100 > 4$ and $\log 2 < 7/10$ suffice. The function $k \mapsto k\log(t/k)$ increases for $1 \le k \le t/100$, so

$$t\Delta(a) \ge \tfrac{1}{4} k \log(t/k) \ge \tfrac{1}{4}\log t.$$

Since $x \le t$, this absorbs the prefactor in (VIII.3.1). At $m = t$, the elementary degree bound $|C(t,t,j)| \le 2^t$ proves the result directly.

Finally, for even t the exact central coefficient is

$$|C(t,t,t/2)| = \frac{\binom{2t}{t}}{\binom{t}{t/2}} = 2^{t-1/2}(1+o(1)).$$

Thus no fixed prefactor permits an exponent $c < \log 2$. □

VIII.3.1 Exact weighted energy and matching exponential profiles

The next identity is the classical Hahn orthogonality in this normalization. We prove it directly from the Krawtchouk identities; its sharp-profile consequences are the additional conclusions needed here. Put

$$w_{t,j} = 4^{-t}\binom{2j}{j}\binom{2t-2j}{t-j}, \qquad A_{t,m} = \frac{(t+m)!(t-m)!}{(t!)^2}.$$

Proposition VIII.3.3 (Exact weighted inverse norm)**.** *The weights $w_{t,j}$ sum to one. If $P(x)=\sum_{m=0}^{t} a_m T_m(2x-1)=\sum_{j=0}^{t} b_j B_j^t(x)$, with real coefficients, then*

$$\sum_{j=0}^{t} w_{t,j} b_j^2 = a_0^2 + \frac{1}{2}\sum_{m=1}^{t} A_{t,m} a_m^2.$$

In particular, for $m \geq 1$,

$$M_{t,m} \geq \sqrt{A_{t,m}/2}. \tag{VIII.3.3}$$

Proof. The generating series of $4^{-j}\binom{2j}{j}$ is $(1-z)^{-1/2}$, so convolution proves the assertion about the weights. Write $K_k(x;N)$ for the Krawtchouk polynomials defined in the preceding section. Their combinatorial formula gives the duality

$$\frac{K_k(x;N)}{\binom{N}{k}} = \frac{K_x(k;N)}{\binom{N}{x}}.$$

Indeed, multiplying either side by $\binom{N}{x}\binom{N}{k}$ counts ordered subsets of sizes x, k, with the sign of their intersection. Together with the orthogonality identity from that section this gives

$$\sum_{k=0}^{N} \frac{K_k(x;N)K_k(y;N)}{\binom{N}{k}} = \frac{2^N}{\binom{N}{x}}\mathbf{1}_{x=y}.$$

Using $K_k(N-y;N)=(-1)^k K_k(y;N)$, its even-$k$ part is half the sum of the right sides for y and $N-y$.

Take $N=2t$, $x=t+m$, $y=t+\ell$, and use $\binom{t}{j}C(t,m,j)=(-1)^{m+j}K_{2j}(t+m;2t)$ and

$$\frac{w_{t,j}}{\binom{t}{j}^2} = \frac{(2t)!}{4^t (t!)^2}\frac{1}{\binom{2t}{2j}}.$$

For $0 \leq m, \ell \leq t$, the resulting weighted inner product is zero unless $m=\ell$. Its value is one for $m=\ell=0$ and $A_{t,m}/2$ otherwise. Expanding $b_j=\sum_m a_m C(t,m,j)$ proves the identity. Averaging $C(t,m,j)^2 \leq M_{t,m}^2$ proves the lower bound. □

Corollary VIII.3.4 (Sharp proportional-degree and moderate-degree scales)**.** *If $t\to\infty$ and $m/t \to a \in [0,1]$, then*

$$\frac{1}{t}\log M_{t,m} \longrightarrow H(a).$$

If instead $m=o(t)$ and $m^2/t \to \infty$, then

$$\log M_{t,m} = (\tfrac{1}{2}+o(1))m^2/t.$$

The latter maximum is over every coefficient, without a restriction keeping j/t away from zero and one.

Proof. The elementary factorial estimate $\log(n!)=n\log n - n + O(\log(n+1))$, valid also at $n=0$ with $0\log 0=0$, gives $\log A_{t,m}=2tH(m/t)+O(\log(t+1))$ uniformly. Combine (VIII.3.3) with (VIII.3.1); the case $m=0$ is immediate. For $m=o(t)$, the product

$$A_{t,m}=\prod_{i=1}^{m}\frac{t+i}{t-i+1}$$

and $\log(1+u)=u+O(u^2)$ give $\log A_{t,m}=m^2/t+O(m^3/t^2)$. Also $tH(m/t)=m^2/(2t)+O(m^4/t^3)$. The logarithm of the prefactor in (VIII.3.1) is $O(1+\log(1+m^2/t))=o(m^2/t)$, proving the second assertion. □

Remark VIII.3.5 (What remains open in this argument). The conjecturally stronger statement $M_{t,m} \le 2^{m^2/t}$ with prefactor one is not proved here. Conversely, a global exponent $1/2$ with either a fixed or polynomial prefactor is impossible: the degree-t central coefficient grows as $2^{t-1/2}$. The exponent $1/2$ is the sharp moderate-degree limit, whereas $\log 2$ is the sharp universal exponent when a fixed prefactor is allowed.

VIII.4 The entropy–transport transfer retained from the source

Let $\mathcal{X}$ be a countable metric space, let K_θ be component probability kernels, and write $p_H = \int K_\theta \, dH(\theta)$. For an empirical law q, assume the exact mixing NPMLE H exists and has positive probability on the observed support. Zero empirical cells are omitted in score sums.

Theorem VIII.4.1 (Component concentration transfers through likelihood contact)**.** *Suppose that $\psi : [0,\infty) \to [0,\infty)$ is nondecreasing and concave and that*

$$W_1(R, K_\theta) \le \psi(D(R\|K_\theta))$$

for every component and every probability law with finite entropy. Then

$$W_1(q, p_H) \le \psi(D(q\|p_H)) \le \psi(D(q\|p_G))$$

for every admissible comparator mixing law G for which the final entropy is finite.

Proof. The directional derivative toward a point mass gives $S(\theta) = \sum_x q(x)K_\theta(x)/p_H(x) \le 1$. Its H-integral equals one, so $S = 1$ almost surely under H. Thus $R_\theta(x) = q(x)K_\theta(x)/p_H(x)$ is a probability kernel H-almost surely, and $\int R_\theta \, dH = q$. The entropy chain rule gives

$$\int D(R_\theta\|K_\theta) \, dH(\theta) = D(q\|p_H).$$

Mixing couplings, applying the component bound, and then Jensen's inequality prove the first conclusion. Likelihood optimality gives $D(q\|p_H) \le D(q\|p_G)$, proving the second. All applications below have finite right sides; truncation of transport costs gives the same conclusion before passing to an unbounded cost. □

This is the source manuscript's transfer principle. Its application to binomial kernels has $\psi(u) = \sqrt{u/(2t)}$ for normalized count cost. For Poisson mixtures, a parameter-free occupancy channel will import that same inequality while preserving the original likelihood fit.

The argument does not normalize every stratum of an arbitrary heterogeneous joint likelihood: only the aggregate score is normalized. Consequently an arbitrary-profile theorem does not follow verbatim. The preceding work's separate two-depth result uses an affine posterior repair; the present full-plane results concern common known trial counts or common known exposures.

VIII.5 Closure of the Poisson critical window

Consider the experiment

$$p_G(j) = \int_0^B e^{-\lambda} \frac{\lambda^j}{j!}\, dG(\lambda), \qquad X_1, \dots, X_N \overset{\text{iid}}{\sim} p_G,$$

where the support radius $B > 0$ is known. Let $\widehat{G}$ be any measurable maximizer of the original Poisson mixture likelihood over all probability laws on $[0, B]$. There is no restriction on the number of mixing atoms. Set $L = \log(eN)$ and, for $N \geq 1$, define

$$r(N, B) = \begin{cases} B \min\{1, L^{-1} \log(e + L/B)\}, & 0 < B \leq L, \\ \max\{\sqrt{B/L}, B/\sqrt{N}\}, & B > L. \end{cases} \tag{VIII.5.1}$$

The two expressions at the breakpoint agree up to absolute constants.

Theorem VIII.5.1 (Global Poisson population recovery)**.** *There exist absolute constants $0 < c < C < \infty$ such that, for every $N \geq 1$ and every $B > 0$,*

$$c\, r(N, B) \leq \inf_{\widetilde{G}} \sup_{G \in \mathcal{P}([0,B])} \mathbb{E}_G W_1(\widetilde{G}, G) \leq \sup_{G \in \mathcal{P}([0,B])} \mathbb{E}_G W_1(\widehat{G}, G) \leq C\, r(N, B).$$

In particular, the ordinary NPMLE is minimax up to absolute factors throughout the entire support/sample-size plane. For observations $X_i \mid \Theta_i \sim \mathrm{Poi}(t\Theta_i)$ with $G \in \mathcal{P}([0, M])$, the corresponding rate is $r(N, tM)/t$.

The exact baseline in the supplied draft is its Theorem B:

$$\sup_G \mathbb{E}_G W_1(\widehat{G}, G) \leq C \left\{ 1 + \sqrt{\frac{B}{\log(1 + N/B)}} \right\}.$$

Together with the lower bounds cited there and its Theorem C, this proves optimality for $c_0 \log N \leq B \leq N^{1-\varepsilon}$ and for $B \geq N$, but leaves the transition window recorded in its Section 14. Theorem VIII.5.1 closes that window for the same ordinary likelihood estimator. Its growing-support crossover is $B \asymp N/\log(eN)$: below that scale the deconvolution term $\sqrt{B/L}$ dominates, and above it the latent-sampling term $B/\sqrt{N}$ dominates. There is no residual logarithmic loss.

Miao et al. [78, Theorem 6.1(a)] established the small-support rate for the NPMLE, and their Theorem 6.1(b) gives $\sqrt{B/\log N}$ under $B \leq C_0 N^{1/3-\varepsilon_0}$. The present revision incorporates the all-support argument established in the preceding development, rather than counting the same theorem as a second discovery. It also includes the diameter clipping needed for arbitrarily small B; an unclipped lower-rate formula cannot be extrapolated beyond the maximum possible loss B. All ingredients used below are proved here, including the two-scale binomial inverse inequality in Proposition VIII.5.6. The estimator is never replaced by a transformed-data MLE.

For $N \in \{1, 2\}$ the rate in (VIII.5.1) is comparable to B. The diameter bound and the two-point latent-sampling lower bound given below therefore suffice. In the proof of the nontrivial regimes we may assume $N \geq 3$.

VIII.5.1 A parameter-free occupancy channel

For an integer $m \geq 1$, define a Markov kernel A_m as follows. Given a count x, independently assign x balls uniformly to m boxes, and output the number of occupied boxes. Poisson splitting proves

$$A_m \,\mathrm{Poi}(\lambda) = \mathrm{Bin}(m, \phi_m(\lambda)), \qquad \phi_m(\lambda) = 1 - e^{-\lambda/m}. \tag{VIII.5.2}$$

Indeed the box counts are independent Poisson random variables of mean λ/m, so their nonempty indicators are independent Bernoulli variables. In particular, if $F = \phi_m \# G$, then $A_m p_G$ is exactly the binomial mixture with mixing law F. On $[0, \phi_m(B)]$ the inverse map has derivative at most $me^{B/m}$, and consequently

$$W_1(G, H) \le me^{B/m} W_1(\phi_m \# G, \phi_m \# H). \tag{VIII.5.3}$$

This channel is a proof device. We do not transform the data before fitting, or assert that a transformed MLE remains an MLE.

We use $W_{1,m}$ for transportation on $\{0, \dots, m\}$ with cost $|i-j|/m$, and $h(p,q)^2 = \sum_j (\sqrt{p_j} - \sqrt{q_j})^2$.

Lemma VIII.5.2 (Empirical entropy on the infinite alphabet)**.** *Let q be the empirical law of N observations from p_G. Uniformly over $G \in \mathcal{P}([0, B])$,*

$$\mathbb{E} D(q \| p_G) \le C(B + L)/N. \tag{VIII.5.4}$$

Proof. For every coordinate with probability $p_j > 0$, convexity and $\log u \le u - 1$ give

$$0 \le \mathbb{E}\{q_j \log(q_j/p_j) - q_j + p_j\} \le \min\{(1 - p_j)/N,\ p_j \log(1/p_j)\}.$$

The second bound also uses $q_j \log q_j \le 0$. The expression inside braces is nonnegative, so its infinite sum can be averaged by Tonelli's theorem. Coordinates with $p_j = 0$ contribute zero almost surely. For $j \ge B$, the Poisson mass is increasing in its mean throughout $[0, B]$, and hence

$$p_G(j) \le e^{-B} B^j / j! \le (eB/j)^j \le e^{-j} \quad (j \ge e^2 B).$$

Take $J = \lceil e^2 B + 3L \rceil$. The first $J+1$ coordinates contribute at most $(J+1)/N$. Since $u \log(1/u)$ is increasing on $(0, e^{-1}]$, the remaining contribution is bounded by $\sum_{j>J} j e^{-j} \le C(J+1)e^{-J} \le C/N$. This proves (VIII.5.4). Notice that the expected empirical chi-square divergence on the full infinite alphabet would not give this conclusion. □

Lemma VIII.5.3 (Root-N prediction after occupancy)**.** *For $m = \lceil \max(B, L) \rceil$ and every Poisson NPMLE $\widehat{G}$,*

$$\mathbb{E} W_{1,m}(A_m p_{\widehat{G}}, A_m p_G) \le C/\sqrt{N}. \tag{VIII.5.5}$$

Proof. Existence follows from compactness of the mixing parameter space and upper semicontinuity of the likelihood; the admissible fit δ_B is strictly positive at every observed count. Write $H = \widehat{G}$. The likelihood directional derivative gives

$$S(\lambda) = \sum_{j : q_j > 0} \frac{q_j K_\lambda(j)}{p_H(j)} \le 1 \quad (0 \le \lambda \le B).$$

Since $\int S \, dH = 1$, equality holds H-almost everywhere. Thus $R_\lambda(j) = q_j K_\lambda(j)/p_H(j)$ is a probability kernel H-almost everywhere and $\int R_\lambda \, dH = q$. Its entropy chain rule and the likelihood comparison yield

$$\int D(R_\lambda \| K_\lambda) \, dH(\lambda) = D(q \| p_H) \le D(q \| p_G). \tag{VIII.5.6}$$

The binomial transport-entropy inequality is

$$W_{1,m}(R, \mathrm{Bin}(m, p))^2 \le D(R \| \mathrm{Bin}(m, p))/(2m).$$

For completeness, a normalized-count 1-Lipschitz function changes by at most $1/m$ on changing one of the m Bernoulli bits. The conditional Hoeffding bound gives centered log-mgf at most

$u^2/(8m)$; entropy duality and Wasserstein duality give the displayed inequality, including degenerate endpoints whenever the entropy is finite. Apply it to $A_m R_\lambda$, use entropy data processing, and integrate (VIII.5.6). Convexity of transportation under mixtures and Cauchy–Schwarz give

$$W_{1,m}(A_m q, A_m p_H) \le \sqrt{D(q\|p_G)/(2m)}.$$

For each count threshold, the CDF of $A_m q$ is an average of independent $[0,1]$-valued functions of the X_i. Its expected absolute error is at most $1/(2\sqrt{N})$. The CDF formula for $W_{1,m}$ therefore gives $\mathbb{E}W_{1,m}(A_m q, A_m p_G) \le 1/(2\sqrt{N})$. Now use Lemma VIII.5.2 and $(B+L)/m \le 2$. □

VIII.5.2 A growing-support Hellinger bound

Lemma VIII.5.4 (Poisson mixture entropy and likelihood control)**.** *For every fixed absolute $b_0 > 0$, uniformly over $b_0 L \le B \le N$,*

$$\mathbb{E}h(p_{\widehat{G}}, p_G) \le C_{b_0} B^{1/4} L^{3/4}/\sqrt{N}. \tag{VIII.5.7}$$

Proof. For complex z the vector $K_z = (e^{-z} z^j/j!)_{j\ge 0}$ belongs to ℓ_1 and satisfies $\|K_z\|_1 = e^{|z|-\operatorname{Re} z}$. On the ellipse $z = B(1+\cos(\theta+iu))/2$, direct calculation gives

$$|z| - \operatorname{Re} z = \tfrac{B}{2}(\cosh u - 1)(1-\cos\theta) \le B(\cosh u - 1).$$

The kth vector Chebyshev coefficient v_k of $x \mapsto K_{Bx}$ therefore satisfies, by contour shifting,

$$\|v_k\|_1 \le 2\exp\{B(\cosh u - 1) - ku\} \le 2\exp\{-c\min(k^2/B, k)\}.$$

For the second inequality take $u = k/(2B)$ if $k \le 2B$, using $\cosh u - 1 \le u^2$ for $0 \le u \le 1$, and take $u = 1$ otherwise. It follows by summing the Gaussian and exponential tails that truncating the expansion at

$$d = \left\lceil C\left\{\sqrt{B\log(C(B+1)/\beta)} + \log(C(B+1)/\beta)\right\}\right\rceil$$

has uniform ℓ_1 error at most $\beta/8$. Every Chebyshev moment of a probability law on $[0,B]$ belongs to $[-1,1]$, and $\|v_k\|_1 \le 2$ also follows directly from its real integral formula. Grid the first d moments with mesh $\beta/(16(d+1))$. Choose one actual mixture in each occupied grid box. This gives an ℓ_1 cover of radius β, with centers in the mixture class and cardinality M satisfying

$$\log M \le Cd\log(C(d+1)/\beta). \tag{VIII.5.8}$$

Take $J = \lceil e^2 B + 6L\rceil$ and $\beta = N^{-3}/(J+1)$. In the indicated range, $d \le C_{b_0}\sqrt{BL}$ and $\log M \le C_{b_0}\sqrt{B}L^{3/2}$. For every cover center p_a, define the pointwise envelope

$$u_a(j) = \begin{cases} p_a(j) + \beta, & j \le J, \\ e^{-j}, & j > J. \end{cases}$$

Every mixture is bounded coordinatewise by its associated envelope, whose total mass U_a lies in $[1, 1+\eta]$ for $\eta = CN^{-3}$, whenever that envelope is used for at least one mixture. Unused envelopes can be discarded. Normalize $r_a = u_a/U_a$. An associated mixture p satisfies $h(p, r_a) \le \sqrt{2\eta}$.

Here is the elementary likelihood argument, to specify the treatment of the infinite alphabet. If $\widehat{p}$ is a likelihood maximizer and $u_a \ge \widehat{p}$, then

$$\prod_{i=1}^{N} \frac{r_a(X_i)}{p_G(X_i)} \ge e^{-N\eta}.$$

Under p_G, the expectation of the square root of this product is the Nth power of the affinity, at most $\exp\{-Nh(p_G, r_a)^2/2\}$. Markov's inequality followed by a union bound shows, with probability at least $1 - e^{-x}$,

$$h(p_G, \widehat{p}) \le \sqrt{2\eta} + \sqrt{\eta + 2(\log M + x)/N}.$$

Integration gives $\mathbb{E}h(p_G, \widehat{p})^2 \le C\eta + C(\log M + 1)/N$, which proves (VIII.5.7). □

VIII.5.3 The two-scale argument closes every growing-support regime

The binomial inverse inequality proved below states that if F, H are mixing laws on $[0,1]$, b_F, b_H their m-trial binomial mixtures, and $1 \le S \le K \le m/2$, then

$$W_1(F,H) \le C/K + Ce^{4S^2/m} W_{1,m}(b_F, b_H) + Ce^{2K^2/m} h(b_F, b_H)/S. \tag{VIII.5.9}$$

Its small high-frequency amplitude $1/S$ is indispensable. The proof approximates a Lipschitz test function at two polynomial degrees, preserves a bounded derivative at the lower degree, and uses the degree-elevated Bernstein coefficient bounds. This is Proposition VIII.5.6.

Suppose $b_0 L \le B \le N$, with $b_0 > 0$ an absolute constant to be fixed below. Set $m = \lceil \max(B, L) \rceil$, so $m \asymp_{b_0} B$. For sufficiently large absolute N, take

$$S = \lfloor \sqrt{m}/16 \rfloor, \qquad K = \left\lfloor \min\{m/2, \sqrt{amL}\} \right\rfloor, \qquad a = 1/64.$$

These are admissible, $S \asymp \sqrt{m}$, $e^{4S^2/m} \le C$, and $m/K \le C\sqrt{m/L}$ because $m \ge L$. Apply (VIII.5.9) to the pushforwards under ϕ_m, then use (VIII.5.3), Lemmas VIII.5.3 and VIII.5.4, and Hellinger data processing. This gives

$$\mathbb{E}W_1(\widehat{G}, G) \le C_{b_0} \left\{ \sqrt{B/L} + \frac{B}{\sqrt{N}} + \frac{e^{2aL} B^{3/4} L^{3/4}}{\sqrt{N}} \right\}.$$

The ratio of the last term to $\sqrt{B/L}$ is at most $Ce^{2aL} N^{-1/4} L^{5/4}$, which is uniformly bounded since $B \le N$ and $2a < 1/4$. We have proved

$$\mathbb{E}W_1(\widehat{G}, G) \le C_{b_0} \{ \sqrt{B/L} + B/\sqrt{N} \}. \tag{VIII.5.10}$$

Bounded N is absorbed in the absolute constant by the diameter bound; this is uniform in B because $B \ge b_0 L$ in this step.

If $B > N$, no Hellinger estimate or approximation is necessary. For any binomial mixing law, coupling its parameter p with its normalized count gives expected cost at most $1/(2\sqrt{m})$. Thus

$$W_1(F,H) \le m^{-1/2} + W_{1,m}(b_F, b_H).$$

Together with (VIII.5.3) and Lemma VIII.5.3, this gives $\mathbb{E}W_1(\widehat{G}, G) \le C(\sqrt{B} + B/\sqrt{N}) \le CB/\sqrt{N}$. Every $B \ge L$ is now covered.

VIII.5.4 The small-support regime, including arbitrarily small radii

We include a direct proof to ensure uniformity as $B \downarrow 0$. Let f be 1-Lipschitz on $[0, B]$ and normalize $f(0) = 0$. Jackson approximation gives a degree-k polynomial $P(\lambda) = \sum_{l=0}^{k} a_l \lambda^l$ such that

$$\|f - P\|_{[0,B]} \le CB/k, \qquad \|P\|_{[0,B]} \le CB.$$

Its monomial coefficients satisfy

$$|a_l| \le BC_1^k B^{-l}. \tag{VIII.5.11}$$

Indeed expand $P(Bx)$ in shifted Chebyshev polynomials: each Chebyshev coefficient has magnitude at most $2\|P\|$, and the sum of absolute monomial coefficients of $T_j(2x-1)$ is at most 7^j by its three-term recurrence. Summing the finite geometric series proves (VIII.5.11) after enlarging C_1.

Take $k \ge 2eB$ and put $Q_k(\lambda) = \sum_{r=0}^{k} \lambda^r/r!$. The exponential remainder implies

$$\|e^{-\lambda}P(\lambda)Q_k(\lambda) - f(\lambda)\|_{[0,B]} \le CB/k.$$

For example, $0 \le 1 - e^{-\lambda}Q_k(\lambda) \le C(eB/(k+1))^{k+1} \le C2^{-k}$, which suffices. Write this weighted polynomial as $\sum_{j=0}^{2k} b_j K_\lambda(j)$. Since

$$b_j = \sum_{\substack{0\le l\le k\\ 0\le j-l\le k}} a_l \frac{j!}{(j-l)!},$$

equation (VIII.5.11) gives

$$\max_{0\le j\le 2k} |b_j| \le B(C_2 k/B)^k. \tag{VIII.5.12}$$

The factors $k+1$ and C_1^k have been absorbed into C_2^k; $2k/B \ge 1$ by hypothesis. Wasserstein duality consequently gives the deterministic inequality

$$W_1(F,H) \le CB/k + B(C_2k/B)^k \|p_F - p_H\|_1. \tag{VIII.5.13}$$

For the MLE, likelihood comparison, $h^2 \le D$, the Hellinger triangle inequality and $\|p-q\|_1 \le 2h(p,q)$ imply

$$\mathbb{E}\|p_{\widehat{G}} - p_G\|_1 \le 4\sqrt{\mathbb{E}D(q\|p_G)} \le C\sqrt{L/N} \qquad (B \le L).$$

Put $d_0 = \log(e + L/B)$ and $k = \lfloor \gamma L/d_0 \rfloor$, where $\gamma > 0$ is a sufficiently small absolute constant. If $\gamma L/d_0 < 2$, the diameter bound $W_1 \le B$ is already at most a constant times $B\min(1, d_0/L)$. Otherwise $k \ge \gamma L/(2d_0)$. Choose a fixed $b_0 > 0$ sufficiently small that $B \le b_0 L$ implies $\gamma(L/B)/(2\log(e+L/B)) \ge 2e$; then $k \ge 2eB$. Furthermore

$$k\log(C_2k/B) \le C_3\gamma L \le L/8$$

after decreasing γ once, since $\log(C_2\gamma(L/B)/d_0) \le C_3 d_0$. The stochastic term in (VIII.5.13) is therefore at most $CBe^{L/8}\sqrt{L/N} \le CB/k$: here $k \le \gamma L$ and $e^{-3L/8}L^{3/2}$ is bounded. This proves the required rate for $B \le b_0 L$. On $b_0 L \le B \le L$, equation (VIII.5.10) is bounded by an absolute constant, whereas $B\min(1, d_0/L)$ is bounded below by a positive absolute constant. This proves the upper bound in Theorem VIII.5.1 for every B.

VIII.5.5 Matching lower bounds without support-range restrictions

We first record a convenient moment-matching pair. On an interval $[a-M, a+M]$, push forward the probability measures

$$(1 \pm \tfrac{1}{2}T_k(x))\frac{dx}{\pi\sqrt{1-x^2}}, \qquad -1 < x < 1,$$

under $x \mapsto a + Mx$. Denote them by F_k^+, F_k^-. They match all moments of degree less than k. Their CDF difference is, up to orientation, $\sin(k\theta)/(\pi k)$ for $x = \cos\theta$. Consequently

$$W_1(F_k^+, F_k^-) = \frac{M}{\pi k}\int_0^\pi |\sin(k\theta)| \sin\theta \, d\theta \ge \frac{M}{\pi k}. \tag{VIII.5.14}$$

The last inequality follows from $|\sin u| \ge \sin^2 u$ and $\int_0^\pi \sin^2(k\theta)\sin\theta\,d\theta = 1 + (4k^2-1)^{-1}$.

For $B \le L$, use this pair on $[0, B]$. Expanding $e^{-\lambda}$ and using the matched moments gives the elementary bound

$$\|p_{F_k^+} - p_{F_k^-}\|_1 \le 2 \sum_{r \ge k} \frac{(2B)^r}{r!}. \tag{VIII.5.15}$$

Indeed the term with observed count j and exponential-expansion index l vanishes if $j + l < k$; summing the absolute values of all remaining terms gives (VIII.5.15). Set $d_0 = \log(e + L/B)$ and $k = \lceil 100L/d_0 \rceil$. Writing $t = L/B \ge 1$, we have $t = e^{d_0} - e \ge e^{d_0}/(e+1)$ and $e^{d_0/2}/d_0 \ge e/2$. These inequalities show

$$\log \frac{k}{2eB} \ge \frac{d_0}{2}, \qquad k \log \frac{k}{2eB} \ge 50L.$$

They also ensure $k \ge 4B$. Thus the tail in (VIII.5.15) is at most $4(2eB/k)^k \le 4e^{-50L}$. The two N-sample laws have TV at most $2Ne^{-50L}$, whereas (VIII.5.14) is at least $cB \min\{1, d_0/L\}$. Le Cam's two-point inequality proves the small-support lower bound, including arbitrarily tiny B.

For $B \ge L$, set $a = B/2$, $M = \sqrt{BL}/8$, and $k = \lceil 20L \rceil$. The moment-matching interval lies inside $[0, B]$. The exact Poisson identity

$$\sum_{x=0}^{\infty} \frac{K_\lambda(x) K_\mu(x)}{K_a(x)} = \exp\{(\lambda - a)(\mu - a)/a\}$$

shows, upon integrating against the signed difference $F_k^+ - F_k^-$, that

$$\sum_x \frac{(p_{F_k^+}(x) - p_{F_k^-}(x))^2}{K_a(x)} \le 4 \sum_{r \ge k} \frac{(M^2/a)^r}{r!} = 4 \sum_{r \ge k} \frac{(L/32)^r}{r!}.$$

All exchanges are justified by boundedness of λ, μ on the integration interval and the absolutely convergent exponential series. The last tail is at most Ce^{-40L}, for instance by $r! \ge (r/e)^r$ and a geometric-tail bound. Cauchy–Schwarz bounds the predictive TV by half the square root of this quantity. Le Cam's inequality and (VIII.5.14) therefore give the lower rate $c\sqrt{B/L}$.

Finally consider mixing laws on $\{0, B\}$ with respective weights $1/2 \pm c_0/\sqrt{N}$ at B, for a sufficiently small absolute $c_0 > 0$. Their latent N-sample laws have TV bounded away from one, by the Bernoulli KL formula and Pinsker's inequality. The Poisson sample is a Markov image of that latent sample, so its TV is no larger. The mixing-law distance is $2c_0 B/\sqrt{N}$. Le Cam's inequality yields the remaining lower rate $cB/\sqrt{N}$. Taking the maximum completes the proof of Theorem VIII.5.1.

The proof uses a common known exposure and a known compact support. It makes no assertion for heterogeneous exposures, unknown support radii, or an MLE constrained to a prescribed number of components. The gain over a direct one-scale inverse estimate is structural: the occupancy channel imports a binomial concentration inequality, likelihood contact supplies posterior entropy, and the two-scale inverse places predictive transportation and high-frequency Hellinger error in different terms.

VIII.5.6 Self-contained binomial inversion tools

These approximation tools are reproduced to make the closure proof independent of the earlier supplement. In this subsection write $B_j^t(x) = \binom{t}{j} x^j (1-x)^{t-j}$ and $p_F(j) = \int B_j^t \, dF$. The integer t is a binomial trial count, $W_{1,t}$ uses cost $|i-j|/t$, and $h(p,q)^2 = \sum_j (\sqrt{p_j} - \sqrt{q_j})^2$.

Lemma VIII.5.5 (Stable approximation and bounded-polynomial inversion)**.** *There is an absolute C with the following properties. For every 1-Lipschitz $f : [0,1] \to \mathbb{R}$ and integer $k \ge 1$, there is a polynomial P_k of degree at most k such that*

$$\|P_k - f\|_\infty \le C/k, \qquad \|P_k'\|_\infty \le 2.$$

If $P=\sum_{j=0}^{t} b_j B_j^t$ has degree $k\le t/2$, then

$$\max_j |b_j| \le Ce^{2k^2/t}\|P\|_\infty. \tag{VIII.5.16}$$

For the derivative-stable P_k, its coefficient vector also satisfies

$$\max_{j<t} t|b_{j+1}-b_j| \le Ce^{4k^2/t}. \tag{VIII.5.17}$$

Proof. Subtract $f(0)$, reflect f on $[0,1]$ to a period-two 1-Lipschitz function, and convolve with the normalized positive Jackson kernel $c_s[\sin(s\pi u/2)/\sin(\pi u/2)]^4$. Its first absolute moment is $O(1/s)$, so the resulting trigonometric polynomial Q_s has error $O(1/s)$ and derivative norm at most one. Its frequencies have magnitude at most $2s$, and its Fourier coefficients have modulus at most one. Replacing each exponential by its Taylor polynomial at $1/2$ of degree $100s$ changes the function and derivative by at most e^{-100s}. Indeed $|\pi l(x-1/2)|\le \pi s$ and the factorial inequality $M!\ge (M/e)^M$ bounds both remainders, including the $4s+1$ terms and the extra derivative factor. Take $s=\lfloor k/100\rfloor$ for $k\ge 100$; use a constant polynomial and enlarge C for smaller k.

For (VIII.5.16), put $M=\|P\|_\infty$ and $F(z)=(1+z)^tP(z/(1+z))$. The maximum principle in the inverse Joukowski variable gives $|P((2+w+w^{-1})/4)|\le M|w|^k$ for $|w|\ge 1$. For $0<j<t$, set $r=j/(t-j)$, $v=r/(1+r)^2$, and $u=2\sqrt{v}|\sin(\phi/2)|$. The larger inverse Joukowski modulus for $re^{i\phi}/(1+re^{i\phi})$ is $(1+u)/\sqrt{1-u^2}$. Hence

$$\frac{|F(re^{i\phi})|}{M(1+r)^t} \le (1-u^2)^{(t-k)/2}(1+u)^k \le e^{2k^2/t}e^{-tu^2/8}.$$

The last inequality follows from $k\le t/2$ and completing the square. The average of the final Gaussian factor is at most $C/\sqrt{1+tv}$. Cauchy's coefficient formula, together with $\binom{t}{j}(j/t)^j(1-j/t)^{t-j}\ge c/\sqrt{1+tv}$, proves (VIII.5.16). The last lower bound follows directly from Stirling's factorial bounds; endpoint coefficients are $P(0)$ and $P(1)$. Finally differentiate the Bernstein expansion: $P_k'=t\sum_{j<t}(b_{j+1}-b_j)B_j^{t-1}$. Apply (VIII.5.16) at degree $k-1\le (t-1)/2$ and use $2(k-1)^2/(t-1)\le 4k^2/t$. □

Proposition VIII.5.6 (Two-scale deterministic inversion)**.** *For integers $1\le s\le k\le t/2$ and any two mixing laws G,H,*

$$W_1(G,H)\le \frac{C}{k}+Ce^{4s^2/t}W_{1,t}(p_G,p_H)+\frac{Ce^{2k^2/t}}{s}h(p_G,p_H). \tag{VIII.5.18}$$

Proof. For any 1-Lipschitz test f, take the polynomials P_k,P_s from Lemma VIII.5.5. The approximation error in integrating P_k instead of f against $G-H$ is at most C/k. The Bernstein coefficients of P_s have normalized-grid Lipschitz constant at most $Ce^{4s^2/t}$. On the other hand,

$$\|P_k-P_s\|_\infty \le C/k+C/s\le C/s.$$

Thus the Bernstein coefficients of P_k-P_s have supremum norm at most $Ce^{2k^2/t}/s$. Test P_s by count transportation and P_k-P_s by $\|p_G-p_H\|_1\le 2h(p_G,p_H)$. All constants are uniform over f, so taking the Wasserstein dual supremum proves the result. □

VIII.6 Closing the entire binomial random-effects testing transition

The attached manuscript left open the testing transition $t \asymp N/\log N$. Its sharp deterministic inverse modulus also showed that global count-Wasserstein error alone cannot close that transition. The previously proved complete estimation theorem uses two different statistical norms on two different parts of a test polynomial. The low-degree part is controlled by likelihood transportation. The remaining polynomial has small amplitude, and is controlled by a predictive Hellinger bound. This yields the full replication curve, including its transition and its parametric plateau.

For a probability measure G on $[0,1]$, put

$$K_x(j) = B_j^t(x) = \binom{t}{j} x^j (1-x)^{t-j}, \qquad p_G(j) = \int B_j^t(x)\, dG(x), \quad 0 \le j \le t.$$

We observe N independent counts from p_G. Their empirical law is q. The unrestricted mixing-measure MLE is any maximizer $\widehat{G}$ of $\ell(H) = \sum_j q_j \log p_H(j)$. Write $\widehat{p} = p_{\widehat{G}}$, $L_N = \log(eN)$, and

$$\mathcal{R}_{N,t} = \inf_{\widetilde{G}} \sup_{G} \mathbb{E}_G W_1(\widetilde{G}, G), \qquad r_{N,t} = \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{tL_N}} + \frac{1}{\sqrt{N}}\right\}.$$

The infimum allows randomized estimators. Wasserstein distance on the count space uses the normalized cost $|j-l|/t$ and is denoted $W_{1,t}$. Our Hellinger convention is $h(p,v)^2 = \sum_j (\sqrt{p_j} - \sqrt{v_j})^2$, also for nonnormalized nonnegative vectors v.

Theorem VIII.6.1 (All-replication optimality of the unrestricted MLE)**.** *There are absolute constants $0 < c < C < \infty$ such that, for every pair of integers $N, t \ge 1$,*

$$c r_{N,t} \le \mathcal{R}_{N,t} \le \sup_G \mathbb{E}_G W_1(\widehat{G}, G) \le C r_{N,t}. \tag{VIII.6.1}$$

The upper bound holds for every measurable choice of an unrestricted MLE. In particular, at $t \asymp N/\log(eN)$ its risk is $O(N^{-1/2})$, and the same order holds for every larger t.

The statistical question extends the replication-range conjectures of Vinayak, Kong, Valiant and Kakade [70] and the discussion in Section 6.2 of Musco, Musco, Rosenblatt and Singh [71]. The preceding near-linear result already answered the fixed-polynomial-range question in that discussion. The estimation theorem is reproduced from the preceding compilation; its consequence for the full testing curve is derived below. It is not counted as a new estimation result here.

VIII.6.1 An elementary finite-envelope bound for predictive likelihood

The following entropy argument is stated with its details because it must apply even when the true distribution has zero cells. Neither an interior mixing density nor a regular parametric likelihood is assumed.

Lemma VIII.6.2 (Small covers of the binomial moment body)**.** *For $0 < \beta < 1$ the set $\{p_G : G \in \mathcal{P}([0,1])\}$ has a finite ℓ_1 cover of radius β, with centers in that same set, whose cardinality M satisfies*

$$\log M \le m \log \frac{C(m+1)}{\beta}, \qquad m = \min\left\{t, \left\lceil \sqrt{2t \log(16t/\beta)} \right\rceil\right\}. \tag{VIII.6.2}$$

Proof. View $K_x = (B_0^t(x), \dots, B_t^t(x))$ as a vector-valued polynomial. Its shifted Chebyshev expansion is

$$K_x = \sum_{k=0}^{t} v_k T_k(2x-1).$$

We claim $\|v_k\|_1 \le 2\exp\{-k^2/(2t)\}$ for $k \ge 1$. Indeed, the even periodic function $F(z) = K_{(1+\cos z)/2}$ has Fourier coefficient $v_k/2$. For $z = \theta + iu$, the ellipse identity gives

$$\left|\frac{1+\cos z}{2}\right| + \left|\frac{1-\cos z}{2}\right| = \cosh u.$$

The binomial formula therefore yields $\|F(\theta+iu)\|_1 = (\cosh u)^t \le e^{tu^2/2}$. Shift the Fourier contour in the appropriate direction and take $u = k/t$. This proves the claim. In particular, truncating after degree $m < t$ has vector ℓ_1 error at most $2te^{-(m+1)^2/(2t)} \le \beta/8$, uniformly in x. For $m = t$ the error is zero.

Each mixing law contributes its first m shifted Chebyshev moments in $[-1,1]^m$. Partition that cube into cells of coordinate diameter at most $\beta/[8(m+1)]$, and choose one mixing-law representative from every nonempty cell. If G, H lie in one cell, their truncated transforms differ in ℓ_1 norm by at most $2m\beta/[8(m+1)] \le \beta/4$. The two tails contribute at most $\beta/4$. Thus their predictive laws have distance at most β. The number of cells is at most $[C(m+1)/\beta]^m$, as asserted. □

Lemma VIII.6.3 (Likelihood envelopes, including zero cells)**.** *Let $\varepsilon \ge 0$ and $\eta > 0$. Let p be the true law on a finite alphabet with d cells, and suppose a model admits an ℓ_1 cover of radius η/d, with M probability centers in the model. Let $\widetilde{p}$ be any fitted model law satisfying*

$$\sum_{i=1}^{N} \log \widetilde{p}(X_i) \ge \sum_{i=1}^{N} \log p(X_i) - N\varepsilon.$$

For every $x \ge 0$, simultaneously for all such fitted laws, except on an event of probability at most e^{-x},

$$h(p, \widetilde{p}) \le \sqrt{\eta} + \sqrt{\varepsilon + \eta + 2(\log M + x)/N}. \tag{VIII.6.3}$$

Consequently

$$\mathbb{E}h(p, \widetilde{p})^2 \le C\left\{\varepsilon + \eta + \frac{\log M + 1}{N}\right\}. \tag{VIII.6.4}$$

Proof. For each center p_a, form the nonnegative vector $u_a(j) = p_a(j) + \eta/d$. Every model law is bounded coordinatewise by at least one u_a, and $\sum_j u_a(j) = 1 + \eta$. If $\widetilde{p} \le u_a$, then

$$h(u_a, \widetilde{p})^2 \le \sum_j (u_a(j) - \widetilde{p}(j)) = \eta.$$

Thus $h(p, \widetilde{p}) > \sqrt{\eta} + r$ implies $h(p, u_a) > r$. Moreover the likelihood comparison implies $\prod_i u_a(X_i)/p(X_i) \ge e^{-N\varepsilon}$. The ratios need only be defined on the support of p, where all sampled observations lie almost surely. Markov's inequality with exponent $1/2$ gives

$$\begin{aligned}
\Pr\left\{\prod_i \frac{u_a(X_i)}{p(X_i)} \ge e^{-N\varepsilon}\right\} &\le e^{N\varepsilon/2}\left(\sum_j \sqrt{p(j)u_a(j)}\right)^N \\
&= e^{N\varepsilon/2}\left(1 + \frac{\eta}{2} - \frac{h(p,u_a)^2}{2}\right)^N \\
&\le \exp\{N(\varepsilon + \eta - r^2)/2\}
\end{aligned}$$

when $h(p, u_a) > r$. Taking the union over the M envelopes proves (VIII.6.3). Integrating that tail, and using $(a+b)^2 \le 2a^2 + 2b^2$, proves (VIII.6.4). □

Corollary VIII.6.4 (A sufficient predictive Hellinger rate)**.** *For $1 \le t \le N$, every exact MLE satisfies*

$$\mathbb{E}_G h(p_G, \widehat{p}) \le \frac{C t^{1/4} L_N^{3/4}}{\sqrt{N}}. \tag{VIII.6.5}$$

More generally, if $\ell(\widetilde{G}) \ge \ell^* - \varepsilon$, the right side can be increased by $C\sqrt{\varepsilon}$.

Proof. Set $\eta = (eN)^{-2}$ and $\beta = \eta/(t+1)$ in the preceding lemmas. For $t \le N$, (VIII.6.2) gives $m \le C\sqrt{tL_N}$ and $\log M \le C\sqrt{t}\, L_N^{3/2}$. The MLE likelihood is at least that of the true law. The approximate likelihood condition loses at most $N\varepsilon$. Apply (VIII.6.4) and Jensen's inequality. □

The logarithmic factors in (VIII.6.5) need not be sharp. The argument below only needs the strict improvement from effective dimension t to a quantity of order $\sqrt{t}$ up to logarithms.

Lemma VIII.6.5 (Likelihood transport, recalled)**.** *For every binomial-mixture MLE,*

$$W_{1,t}(q, \widehat{p})^2 \le \frac{D(q\|\widehat{p})}{2t} \le \frac{D(q\|p_G)}{2t}. \tag{VIII.6.6}$$

Consequently

$$\mathbb{E}_G W_{1,t}(\widehat{p}, p_G) \le C/\sqrt{N}. \tag{VIII.6.7}$$

Proof. In score sums, terms with $q_j = 0$ are set to zero; every observed cell has positive fitted probability. The likelihood score condition gives $S(x) = \sum_j q_j K_x(j)/\widehat{p}(j) \le 1$, and its integral against $\widehat{G}$ is one. Thus $S = 1$ almost surely under $\widehat{G}$. The posterior vectors $R_x(j) = q_j K_x(j)/\widehat{p}(j)$ are therefore probability laws, with mixture q, and

$$\int D(R_x\|K_x)\, d\widehat{G}(x) = D(q\|\widehat{p}).$$

Every normalized-count binomial kernel satisfies $W_{1,t}(R, K_x)^2 \le D(R\|K_x)/(2t)$: a 1-Lipschitz test of the normalized sum of t Bernoulli variables has centered logarithmic moment generating function at most $\lambda^2/(8t)$ by bounded differences; entropy duality and optimization over λ give the inequality. Mix the couplings, use Cauchy–Schwarz, and then use likelihood optimality to prove (VIII.6.6). Also $\mathbb{E}D(q\|p_G) \le t/N$, by $D(q\|p_G) \le \sum_{j:p_G(j)>0}(q_j - p_G(j))^2/p_G(j)$. Finally $\mathbb{E}W_{1,t}(q, p_G) \le 1/(2\sqrt{N})$ by integrating the standard deviation of the empirical distribution function. The triangle inequality proves (VIII.6.7). □

Upper bound in Theorem VIII.6.1. It suffices first to treat $t \le N$. Bounded N or bounded t are handled by the loss bound one. Fix a sufficiently small absolute $a > 0$, for example $a = 1/64$, and take

$$k = \left\lfloor \min\{t/2, \sqrt{atL_N}\} \right\rfloor, \qquad s = \lfloor \sqrt{t}/4 \rfloor.$$

For sufficiently large N and t, these satisfy $1 \le s \le k \le t/2$, $s \asymp \sqrt{t}$, and

$$1/k \le C\{1/t + (tL_N)^{-1/2}\}, \qquad e^{4s^2/t} \le C.$$

Use Proposition VIII.5.6, Lemma VIII.6.5, and Corollary VIII.6.4. This gives

$$\mathbb{E}_G W_1(\widehat{G}, G) \le \frac{C}{k} + \frac{C}{\sqrt{N}} + \frac{C e^{2k^2/t} L_N^{3/4}}{\sqrt{N}\, t^{1/4}}. \tag{VIII.6.8}$$

The final term, multiplied by $\sqrt{tL_N}$, is at most

$$C(eN)^{2a} N^{-1/2} t^{1/4} L_N^{5/4} \le C(eN)^{2a} N^{-1/4} L_N^{5/4} \le C.$$

The last bound is uniform because $2a < 1/4$. This proves the desired upper bound when $t \le N$.

For $t \ge N$, couple every mixing parameter x to its normalized binomial count. The expected absolute difference is at most $\sqrt{x(1-x)/t} \le 1/(2\sqrt{t})$. Hence, after identifying count laws with measures on $[0,1]$,

$$W_1(\widehat{G}, G) \le 1/\sqrt{t} + W_{1,t}(\widehat{p}, p_G).$$

Take expectations and apply (VIII.6.7). This is $O(N^{-1/2})$ for $t \ge N$, and completes the proof of the upper bound. □

VIII.6.2 Matching lower bounds

Lower bound in Theorem VIII.6.1. The $N^{-1/2}$ term follows from mixing laws supported on $\{0,1\}$ with weights $1/2 \pm c/\sqrt{N}$. Each count then reveals exactly one endpoint label, regardless of t. Their N-sample total variation stays bounded away from one for sufficiently small absolute c.

For the other terms let $\pi(dx) = dx/[\pi\sqrt{x(1-x)}]$ and $h_k(x) = T_k(2x-1)$. Define $G_k^{\pm} = (1 \pm h_k/2)\pi$. Integration of their signed cumulative function gives

$$W_1(G_k^+, G_k^-) = \frac{1}{2\pi k}\int_0^\pi |\sin(k\theta)| \sin\theta \, d\theta \asymp 1/k.$$

If $k = t+1$, the count laws are identical, proving the $1/t$ lower bound. For $k \le t$, draw $X \sim \pi$, $J \mid X \sim \mathrm{Binomial}(t, X)$ and put $Qh(j) = \mathbb{E}[h(X) \mid J = j]$. The posterior is $\mathrm{Beta}(j+1/2, t-j+1/2)$. The self-adjoint operator Q^*Q preserves polynomial degrees and has leading degree-k coefficient

$$\lambda_{t,k} = \prod_{i=0}^{k-1} \frac{t-i}{t+1+i}.$$

Orthogonality of the Chebyshev polynomials therefore gives $Q^*Qh_k = \lambda_{t,k}h_k$ and $\mathbb{E}[Qh_k(J)^2] = \lambda_{t,k}/2$. This is the classical beta–binomial diagonalization [72, Section 4.1]. The count laws are $p_\pi(1 \pm Qh_k/2)$, so

$$\chi^2(p_{G_k^+} \| p_{G_k^-}) \le 2\mathbb{E}[Qh_k(J)^2] = \lambda_{t,k} \le e^{-k^2/(2t)}.$$

Take $k = \lceil\sqrt{8t\log(8N)}\rceil$ if this is at most t. The chi-squared divergence of the N-fold products is $(1+\chi^2)^N - 1 \le e^{N(8N)^{-4}} - 1$, so their total variation is bounded by an absolute constant smaller than one. The two-point testing bound yields risk at least $c/k \ge c'/\sqrt{tL_N}$. If this choice has $k > t$, then $t \le CL_N$ apart from bounded t, and $1/\sqrt{tL_N} \le C'/t$; the exact moment-matching lower bound already suffices. The maximum of the three lower bounds is comparable to their sum clipped at one, proving (VIII.6.1). □

VIII.6.3 The full worst-null testing curve

Fix $\alpha, \beta \in (0,1)$ with $\alpha + \beta < 1$. A null-dependent test ψ_{G_0} has level α if $\mathbb{E}_{G_0}\psi_{G_0} \le \alpha$. Define $\varepsilon^*_{N,t}(\alpha,\beta)$ as the infimum of the separations ε for which such tests exist for every G_0, with $\mathbb{E}_G(1-\psi_{G_0}) \le \beta$ whenever $W_1(G, G_0) \ge \varepsilon$. Thus the rate is uniform in the null law; it need not describe the difficulty at each particular fixed null.

Theorem VIII.6.6 (All trial counts, including the critical window)**.** *For all integers $N, t \ge 1$,*

$$\varepsilon^*_{N,t}(\alpha,\beta) \asymp_{\alpha,\beta} \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{t\log(eN)}} + \frac{1}{\sqrt{N}}\right\}.$$

An upper bound is attained by the plug-in test based on any unrestricted NPMLE. In particular the parametric plateau begins at $t \asymp N/\log(eN)$, and the window left in the attached manuscript has no remaining logarithmic gap.

Proof. Let $r = r_{N,t}$ and let C be the upper-bound constant in Theorem VIII.6.1. Reject when $W_1(\widehat{G}, G_0) > Cr/\alpha$. Markov's inequality gives level α. If $W_1(G, G_0) > Cr(\alpha^{-1} + \beta^{-1})$, the triangle inequality bounds acceptance probability by $\Pr_G\{W_1(\widehat{G}, G) > Cr/\beta\} \le \beta$. Using a slightly larger constant handles non-strict separation.

For the lower bound use the same three pairs constructed in the lower proof of Theorem VIII.6.1, taking one member as the null and the other as the alternative. The degree-$t+1$ pair gives identical sample laws and separation at least c/t. For the middle term choose $k = \lceil \sqrt{C_{\alpha,\beta} t \log(eN)} \rceil$ when $k \le t$; the displayed chi-square estimate makes the product-law total variation smaller than $1 - \alpha - \beta$ by taking $C_{\alpha,\beta}$ large. If $k > t$, the $1/t$ pair already gives the required order. Finally choose the endpoint-weight perturbation of size $c_{\alpha,\beta}/\sqrt{N}$ with $c_{\alpha,\beta}$ small, giving the $N^{-1/2}$ obstruction. For any two sample laws P_0, P_1, the sum of type I and type II errors is at least $1 - \|P_0 - P_1\|_{\mathrm{TV}}$. These pairs therefore prove the maximum of the three lower bounds. Their maximum is comparable to their sum clipped at one, as required. □

Kania, Wasserman and Balakrishnan [69, Section 3, Theorem 3] give the corresponding worst-null lower curve and discuss the gap in the intermediate replication range. The attached manuscript's Corollary D used only the near-linear estimation bound. Theorem VIII.6.6 closes its remaining range by inserting the complete estimation theorem; the testing transfer itself is the same elementary Markov argument.

VIII.7 Global Franel recurrences and arithmetic Apéry limits

The positive-moment argument in the attached draft explains why definite critical values force recurrence complexity. We retain that criterion, but remove it from the logical dependencies of the Franel classification. The replacement uses multiplicities of formal exponential factors and therefore covers complex weights and colliding critical values. We then extend the classification to every arithmetic subsequence and identify the entire space of Apéry limits over the algebraic numbers.

VIII.7.1 The critical-value criterion and its scope

The width of an eventual nonzero recurrence $\sum_{j\in J} p_j(n)u_{n+j} = 0$, with $p_j \in \mathbb{C}(n)$, is $\max J - \min J$ after zero coefficients are removed. Here J is a finite set of integers; gaps are allowed. The zero sequence has minimum width zero. A sequence with positive minimum width is understood modulo finitely many initial terms. Width is not the number of its nonzero terms and is not the order of an ordinary differential equation.

Proposition VIII.7.1 (Visibility of same-side definite critical values)**.** *Let Φ be real analytic on a compact real analytic manifold of positive dimension, equipped with a positive analytic density $d\mu$. Set $u_n = \int \Phi^n \, d\mu$. Suppose that a nonzero critical value v has finitely many critical points above it, all nondegenerate local maxima, or all nondegenerate local minima. Regular points in the same fiber are allowed. Then some analytic continuation of $\sum_{n\geq 0} u_n z^n$ is singular at $z = 1/v$. Consequently, s distinct such nonzero critical values force minimum recurrence width at least s.*

Proof. The analytic Morse lemma and radial integration give, near v, a pushforward density with singular part $\mathbf{1}_{\{\epsilon u>0\}}(\epsilon u)^{d/2-1}h(\epsilon u)$, where $u = x - v$, d is the manifold dimension, ϵ is the common side and h is analytic with $h(0) > 0$. The regular contribution is smooth. Positivity and the common side prevent cancellation of the first nonsmooth term. This remains true when $d/2-1$ is an integer: the one-sided polynomial then has a derivative jump. The Cauchy transform of this measure cannot be holomorphic through v, because its boundary imaginary part is a nonzero constant multiple of this nonsmooth density. Inverting the Cauchy variable gives the asserted singularity of the moment generating function.

For the recurrence assertion, translate shifts to $0, \dots, L$ and clear denominators. With $\theta = z\partial_z$, the coefficient recurrence gives an inhomogeneous differential equation whose operator has the form $\sum_{j=0}^{L} z^{L-j} p_j(\theta - j)$. Its highest derivative coefficient is $z^D H(z)$ with $\deg H \leq L$. An operator polynomial in θ removes the polynomial inhomogeneous term without creating new nonzero finite zeros in that leading coefficient. Every branch is regular away from those zeros and zero. Thus at most L distinct nonzero finite singularities are possible. □

This proposition requires the stated real positivity and Hessian conditions. It does not assert that every complex critical value is visible. None of those conditions is needed in the next theorem.

VIII.7.2 Every weight and every arithmetic subsequence

For $r \geq 2$ and $\lambda \in \mathbb{C}^*$ write

$$a_{r,\lambda}(N) = \sum_{k=0}^{N} \binom{N}{k}^r \lambda^k, \qquad u_{r,\lambda;q,b}(n) = a_{r,\lambda}(qn + b), \quad q \geq 1, \quad 0 \leq b < q.$$

Theorem VIII.7.2 (Complete subsequence classification)**.** *For every integer $r \geq 2$, every $\lambda \in \mathbb{C}^*$, and all q, b as above, the minimum eventual recurrence width of $u_{r,\lambda;q,b}$ over $\mathbb{C}(n)$ is*

parameter stratum	*minimum width*
$\lambda \notin \{1, -1\}$	r
$\lambda = 1$	$\lceil r/2 \rceil$
$\lambda = -1,\ q$ *odd*	$2\lfloor r/2 \rfloor$
$\lambda = -1,\ q$ *even*, b *even*	$\lfloor r/2 \rfloor$
$\lambda = -1,\ q$ *even*, b *odd*	0.

In the last case the sequence is identically zero. The same minimum holds over every field $K(n)$ with $\mathbb{Q}(\lambda) \subseteq K \subseteq \mathbb{C}$.

The $q = 2, b = 0$ case contains the all-weight centered classification. For arbitrary q, b, the conclusion also rules out a loss of width under sampling, except for the stated alternating parity symmetry. Stoll's and McIntosh's constructions provide the symmetry-weight upper bounds [28, 29]. Weighted low-degree recurrences and the higher-Bessel product interpretation have established precedents [26, 60]. The theorem above is proved here by a uniform lower bound, including multiplicity at exceptional weights; it is not presented as a previously named conjecture about subsequences.

VIII.7.3 The differential Galois input

Define the entire function

$$\mathcal{F}_r(x) = \sum_{n \geq 0} \frac{x^n}{(n!)^r}.$$

It solves $((x\partial_x)^r - x)y = 0$. Work over $K = \mathbb{C}(t)$, and write M_a for the rank-r differential module whose scalar solutions include

$$f_a(t) = \mathcal{F}_r(at^r), \qquad a \in \mathbb{C}^*.$$

Equivalently its scalar equation is

$$\left\{ \left(\frac{t\partial_t}{r} \right)^r - at^r \right\} y = 0.$$

We use the following precise classical input. The differential Galois group of M_a in its standard representation is

$$G_r = \begin{cases} \mathrm{SL}_r(\mathbb{C}), & r \text{ odd}, \\ \mathrm{Sp}_r(\mathbb{C}), & r \text{ even}. \end{cases} \tag{VIII.7.1}$$

This is Katz's Kloosterman-group calculation [59, Corollary 4.4.8]; the connection and its r-fold pullback are identified explicitly in [51, Proposition 2.4 and the proof of Proposition 2.7]. Scaling t by a nonzero constant gives the assertion for every a. In particular these groups are connected and their standard representations are irreducible.

The formal decomposition at $t = \infty$ consists of r rank-one factors with exponential parts

$$\exp(r\alpha\zeta t), \qquad \zeta^r = 1, \quad \alpha^r = a. \tag{VIII.7.2}$$

Their remaining parts are regular singular. Direct substitution gives formal solutions

$$e^{r\alpha\zeta t} t^{-(r-1)/2} (1 + O(t^{-1}));$$

the distinct leading exponential roots recursively determine the formal series. Only the exponential parts, with their multiplicities, will be used below. The duality relation is

$$M_a^* \simeq M_{(-1)^r a}. \tag{VIII.7.3}$$

For example, this follows by reversing the cyclic basis for the companion system and alternating its signs, or by formal adjunction with respect to dx/x before the pullback.

Lemma VIII.7.3 (Independence of distinct scale classes)**.** *Choose finitely many nonzero scales $a_1, \dots, a_s$. If r is even, assume they are pairwise distinct. If r is odd, assume $a_i \neq \pm a_j$ for $i \neq j$. The connected differential Galois group of $\bigoplus_i M_{a_i}$ is the full product G_r^s.*

Proof. The connected joint group projects onto each individual group: its image has finite index in the connected group G_r. Its Lie algebra is therefore a subdirect product of simple Lie algebras. By the Lie-algebra form of Goursat's lemma, a proper subdirect product identifies two of these simple factors by an automorphism. Automorphisms of $\mathfrak{sp}_r$ are inner; automorphisms of $\mathfrak{sl}_r$ are inner or the standard duality automorphism. Consequently the two standard representations on the connected joint group would be isomorphic, or in the odd case dual.

Passing to the finite algebraic extension corresponding to the identity component makes this an isomorphism of the associated differential modules. But an isomorphism preserves the formal exponential parts. After any common finite ramification at infinity, (VIII.7.2) shows that M_a and M_b can be isomorphic only if $a = b$, and that M_a and M_b^* can be isomorphic only if $a = (-1)^r b$. Indeed equality of the multisets of coefficients of the exponential terms, followed by taking their rth powers, gives these equalities. Both possibilities contradict the scale hypotheses. Thus no pair of simple factors is identified and the Lie algebra is the full product. A connected closed subgroup with this Lie algebra is the full product group. □

VIII.7.4 A factorial-normalization budget with an arbitrary sampling interval

For a nonzero differential operator $P = \sum_i b_i(t)\partial_t^i$ with formal Laurent coefficients at infinity, let v be the largest leading power of t among its coefficients and put

$$\sigma_\infty(P)(\xi) = \sum_i \left(\lim_{t\to\infty} t^{-v} b_i(t) \right) \xi^i.$$

The limit means the constant coefficient after Laurent normalization. Differentiating a Laurent coefficient lowers its degree, so

$$\sigma_\infty(PQ) = \sigma_\infty(P)\sigma_\infty(Q) \tag{VIII.7.4}$$

(up to a harmless nonzero constant if another normalization is used).

Lemma VIII.7.4 (Nonzero exponential multiplicities bound width)**.** *Let*

$$H(t) = \sum_{n\geq 0} \frac{u_n}{((qn+b)!)^r} t^{r(qn+b)}.$$

Suppose the differential module generated by this scalar function has at infinity a decomposition into rank-one factors with exponential parts $e^{\beta t}$ and regular-singular remaining parts. If ν of these factors have $\beta \neq 0$, counted with multiplicity, then every eventual recurrence of u_n has width at least $\nu/(qr)$.

Proof. Write a proposed width-L recurrence as $\sum_{j=0}^{L} p_j(n)u_{n-j} = 0$, with $p_j \in \mathbb{C}[n]$. Let $D = \max \deg p_j$ and let c_j be the leading coefficient when $\deg p_j = D$. Multiplication by $((qn+b)_{\underline{qL}})^r/((qn+b)!)^r$ gives $PH = Q(t)$, with Q polynomial and

$$P = \sum_{j=0}^{L} t^{qrj} p_j \left(\frac{t\partial_t}{qr} + j - \frac{b}{q}\right)^{q(L-j)-1} \prod_{\ell=0} \left(\frac{t\partial_t}{r} - \ell\right)^r.$$

Indeed the coefficient contributed by the jth term at exponent $r(qn+b)$ is $p_j(n)u_{n-j}/((qn+b-qL)!)^r$ for sufficiently large n. The finitely many other coefficients account for Q. The largest coefficient degree in t is $D + qrL$, and

$$\sigma_\infty(P)(\xi) = \sum_{\deg p_j = D} c_j \left(\frac{\xi}{qr}\right)^D \left(\frac{\xi}{r}\right)^{qr(L-j)}. \tag{VIII.7.5}$$

Distinct j give distinct powers, so this is nonzero and has at most qrL nonzero roots with multiplicity. Left multiplication by a polynomial in $t\partial_t$ killing Q makes the equation homogeneous and adds only zero roots to the principal symbol.

Let R be the monic minimal scalar annihilator of H. Its formal rank-one decomposition gives a first-order factorization with factors $\partial_t - \beta_i + O(t^{-1})$: take a formal rank-one solution, right-divide by its logarithmic-derivative factor, and repeat on the quotient. Laurent gauge changes add only $O(t^{-1})$. Thus $\sigma_\infty(R) = \prod_i(\xi - \beta_i)$. The homogeneous annihilator is a left multiple of R. Equation (VIII.7.4) then forces its principal symbol to contain all ν nonzero factors of R, with multiplicity. Consequently $\nu \le qrL$. □

VIII.7.5 Cyclicity after root-of-unity filtering

The Cauchy-product identity gives

$$\mathcal{F}_r(x)\mathcal{F}_r(\lambda x) = \sum_{N\ge 0} \frac{a_{r,\lambda}(N)}{(N!)^r} x^N.$$

Writing $\mu_q = \{\epsilon : \epsilon^q = 1\}$, its root-of-unity filter is

$$H_{q,b}(t) = \frac{1}{q} \sum_{\epsilon\in\mu_q} \epsilon^{-b} f_\epsilon(t) f_{\lambda\epsilon}(t) = \sum_{n\ge 0} \frac{a_{r,\lambda}(qn+b)}{((qn+b)!)^r} t^{r(qn+b)}. \tag{VIII.7.6}$$

All products lie in one Picard–Vessiot field. Multiplication of scalar solutions is an equivariant linear map from their tensor product. On an irreducible representation a nonzero such map is injective. Distinct nonisomorphic irreducible images have direct sum, and a vector with a nonzero component in each generates the whole direct sum. These facts establish cyclicity for the specific filtered function, not just for an ambient system that annihilates it.

Lower bounds in Theorem VIII.7.2. First suppose $\lambda \ne \pm 1$. Each product in (VIII.7.6) belongs to an irreducible rank-r^2 tensor representation. For even r the two scales are distinct independent symplectic factors. For odd r they are distinct modulo sign, and hence independent special-linear factors. Use Lemma VIII.7.3 for the finite set of all scale classes occurring in the filter.

The q product representations are pairwise nonisomorphic even when their scales overlap. For even r, their two-factor supports agree only if the unordered pairs $\{\epsilon, \lambda\epsilon\}$ agree. For odd r, a representation also records whether it is standard or dual at each signed scale class: these are inequivalent for odd $r \ge 3$. Hence isomorphism again requires equality of the unordered pairs of actual scales. For two distinct indices ϵ, η, such equality would require $\eta = \lambda\epsilon$ and $\lambda\eta = \epsilon$, forcing $\lambda^2 = 1$, which is excluded. Since every coefficient ϵ^{-b}/q is nonzero, the filter is cyclic

in all q irreducible summands. A formal exponential coefficient in $M_a \otimes M_c$ can vanish only if $c = (-1)^r a$. This too is excluded. Therefore $\nu = qr^2$ and the budget gives $L \geq r$.

For $\lambda = 1$, the summands are $\operatorname{Sym}^2 M_\epsilon$. Their representations are irreducible and pairwise nonisomorphic. For even r this follows from distinct symplectic scale factors. For odd r, opposite scales if present give $\operatorname{Sym}^2 V$ and $\operatorname{Sym}^2 V^*$, which are nonisomorphic for $r \geq 3$. In each summand the formal exponents are the unordered sums of two rth-root exponents. There are $r/2$ zero sums when r is even, and none when r is odd. The nonzero rank per summand is thus $r^2/2$ for even r, and $r(r+1)/2$ for odd r. The q nonzero filter components give respectively $L \geq r/2$ and $L \geq (r+1)/2$.

Now let $\lambda = -1$. The products indexed by ϵ and $-\epsilon$ coincide. If q is even, their filter coefficients sum to zero for odd b, and to $2\epsilon^{-b}/q$ for even b. This proves the asserted vanishing in the first case. In the second case there are $q/2$ surviving products. If q is odd, none of these pairs repeats and there are q surviving products.

For even r, each surviving product is an irreducible tensor of two independent symplectic modules. Distinct surviving products have disjoint scale pairs, so they are nonisomorphic. Each has r^2 nonzero formal factors, since zero would require the scales to be equal rather than opposite.

For odd r, each product lies in $V \otimes V^* = \mathbf{1} \oplus \mathrm{Ad}$ for the special-linear factor indexed by the class $\{\epsilon, -\epsilon\}$. The multiplication map kills the trivial summand. To check this without assuming that invariant scalar functions are constant, put $\delta = (t/r)\partial_t$. For solutions y, z of $\delta^r y = \epsilon t^r y$, $\delta^r z = -\epsilon t^r z$, the nondegenerate concomitant

$$\mathcal{B}(y,z) = \sum_{j=0}^{r-1} (-1)^j (\delta^j y)(\delta^{r-1-j} z)$$

is constant. Indeed its derivative telescopes to zero. Choose dual solution bases y_i, z_i for this pairing and jet matrices Y, Z whose rows are δ^j, $0 \leq j < r$. Then $Y^{\mathsf{T}} J Z = I$, with J invertible and anti-diagonal, whence

$$\sum_i y_i z_i = (Y Z^{\mathsf{T}})_{00} = (J^{-\mathsf{T}})_{00} = 0 \qquad (r \geq 2).$$

The selected product is nonzero, so the multiplication map is injective on the irreducible adjoint summand. Different surviving products use different independent scale classes. Each therefore contributes the $r(r-1)$ nonzero ordered differences of distinct rth-root exponents. There are q such products for odd q, or $q/2$ for even q and even b. Dividing their nonzero ranks by qr gives respectively $2\lfloor r/2 \rfloor$ and $\lfloor r/2 \rfloor$, in both parities of r. □

Upper bounds and coefficient fields. For $\lambda \neq \pm 1$ set $\mathcal{L}_N[p] = \sum_{k=0}^{N} p(k) \binom{N}{k}^r \lambda^k$. An index shift gives

$$\mathcal{L}_N[k^{r+h} - \lambda (N-k)^r (k+1)^h] = 0 \qquad (h \geq 0).$$

The polynomial displayed here has leading coefficient $1 - (-1)^r \lambda \neq 0$. Successive elimination reduces every moment to $\mathcal{L}_N[1], \ldots, \mathcal{L}_N[k^{r-1}]$. The identity $\binom{N-1}{k}^r = N^{-r}(N-k)^r \binom{N}{k}^r$ therefore gives a backward r-dimensional state relation over $\mathbb{Q}(\lambda)(N)$. Apply it q times between observations: $r+1$ consecutive observations on the progression $qn + b$ are rows on the same r-dimensional state, so they satisfy a recurrence of width at most r. This argument needs no matrix inverse.

For $\lambda = 1$, Stoll's ordinary Franel recurrence has width at most $m = \lceil r/2 \rceil$ over $\mathbb{Q}(n)$ [28]. A rational companion system has dimension m; its q-step system gives the same upper bound on every progression. For $\lambda = -1$, symmetry gives $a_{r,-1}(N) = 0$ for odd N. McIntosh's recurrence [29, Theorem 1] gives width at most $h = \lfloor r/2 \rfloor$ for $A_r(n) = (-1)^n a_{r,-1}(2n)$. Geometric rescaling preserves width, so even progressions have the claimed upper bound h. Interlacing this recurrence

with the zero odd sequence gives a width-$2h$ recurrence for $a_{r,-1}(N)$: a recurrence on its even entries with shifts j becomes one with shifts $2j$, valid also on the zero odd entries. Taking a q-step companion system gives the upper bound $2h$ for odd q. All upper constructions have coefficients in $\mathbb{Q}(\lambda)(n)$. Since the lower bounds hold over $\mathbb{C}(n)$, extending the coefficient field anywhere between these fields cannot change the minimum. □

Corollary VIII.7.5 (The Franel and McIntosh statements of the draft)**.** *For every $r \geq 2$,*

$$\operatorname{minwidth}_{\mathbb{C}(n)} \sum_{k=0}^{n} \binom{n}{k}^r = \lceil r/2 \rceil, \qquad \operatorname{minwidth}_{\mathbb{C}(n)} \sum_{k=-n}^{n} (-1)^k \binom{2n}{n+k}^r = \lfloor r/2 \rfloor.$$

More generally, the centered weighted sequence $\lambda^{-n} a_{r,\lambda}(2n)$ has width $\lceil r/2 \rceil$ at $\lambda = 1$, $\lfloor r/2 \rfloor$ at $\lambda = -1$, and r elsewhere.

Proof. Use Theorem VIII.7.2 with $(q, b) = (1, 0)$ or $(2, 0)$. The centered alternating sum equals $(-1)^n a_{r,-1}(2n)$, and geometric rescaling preserves width. □

The alternating conclusion is Theorem E of the attached draft, now independent of a critical-value visibility argument. The unweighted conclusion removes the restriction to telescoping recurrences in [74, Theorem 1.1]. Critical-value collisions do not create exceptions in Theorem VIII.7.2: the budget counts formal exponential factors with multiplicity, not distinct critical values.

VIII.7.6 The whole arithmetic space of Apéry limits

Put $A_r(N) = a_{r,1}(N)$ and $m = \lceil r/2 \rceil$. In this subsection the notation A_r denotes the ordinary, not the alternating, Franel sequence. Define rational sequences $A_{r,j}(N)$ by

$$\sum_{j \geq 0} A_{r,j}(N) z^{2j} = \sum_{k=0}^{N} \binom{N}{k}^r \left\{ \prod_{\ell=1}^{k} \left(1 - \frac{z}{\ell}\right) \prod_{\ell=1}^{N-k} \left(1 + \frac{z}{\ell}\right) \right\}^{-r}. \tag{VIII.7.7}$$

The right side is even by replacing k with $N - k$, and $A_{r,0} = A_r$. Let

$$\phi_{r,j} = [z^{2j}] \left(\frac{z}{\sin z}\right)^r \in \mathbb{Q}_{>0}.$$

The input from Straub–Zudilin [74, Theorem 1.3] is that $A_{r,j}$, $0 \leq j < m$, satisfy every telescoping recurrence for $\binom{N}{k}^r$ on a sufficiently late tail, and have limits

$$\lim_{N \to \infty} \frac{A_{r,j}(N)}{A_r(N)} = \phi_{r,j} \pi^{2j}. \tag{VIII.7.8}$$

Van Hoeij's invariant-submodule calculation [76, Remark 6] gives a telescoper of order m for this summand. These are published inputs; the transfer to the unrestricted minimal recurrence is supplied by the preceding theorem.

For $q \geq 1$ and $0 \leq b < q$, let $L_{r;q,b}$ denote the monic minimal eventual recurrence of $A_r(qn+b)$, with shifts normalized to $0, \ldots, m$. If K is a field between $\mathbb{Q}$ and $\overline{\mathbb{Q}}$, write $\mathcal{S}_{r;q,b}(K)$ for the K-valued solutions on a sufficiently late regular tail, identifying sequences that agree eventually.

Theorem VIII.7.6 (Complete arithmetic Apéry-limit classification)**.** *For every $r \geq 2$, $q \geq 1$, $0 \leq b < q$ and $\mathbb{Q} \subseteq K \subseteq \overline{\mathbb{Q}}$, the following statements hold.*

1. *$L_{r;q,b}$ has order m and coefficients in $\mathbb{Q}(n)$. For $(q, b) = (1, 0)$ it is also the monic minimal telescoper for the summand $\binom{n}{k}^r$.*

2. *The sequences $A_{r,j}(qn+b)$, $0 \le j < m$, are a basis of $\mathcal{S}_{r;q,b}(K)$. Every complex tail solution also has a unique expansion in this same basis, with complex constants.*

3. *Every $X \in \mathcal{S}_{r;q,b}(K)$ has an Apéry limit, and*

$$X \longmapsto \lim_{n\to\infty} \frac{X(n)}{A_r(qn+b)} \quad \text{is a } K\text{-linear isomorphism onto} \quad \bigoplus_{j=0}^{m-1} K\pi^{2j}. \tag{VIII.7.9}$$

In particular two algebraic-number-valued tail solutions with the same limit are equal eventually.

Proof. For $q = 1, b = 0$, the minimal recurrence has order m by Corollary VIII.7.5. Stoll provides an annihilator of this order over $\mathbb{Q}(n)$, so its monic normalization is the unique minimal operator even over $\mathbb{C}(n)$. The minimal telescoper supplied by van Hoeij has the same order and annihilates the sequence. Right division by the monic minimal recurrence leaves a lower-order annihilator as remainder, necessarily zero. Equality of orders thus makes the two monic operators equal. Equation (VIII.7.8) therefore applies to solutions of this minimal recurrence.

Pass to a tail beyond all poles and integer zeros of its leading and trailing coefficients, and beyond the finitely many exceptional indices for the $A_{r,j}$. A companion system has dimension m. Its q-step system yields one common rational-coefficient recurrence of order at most m for every sampled solution. By Theorem VIII.7.2, the sampled sequence $A_r(qn+b)$ has minimum order exactly m. Hence this common operator has order m and, after normalization, is $L_{r;q,b}$.

The m rational sequences $A_{r,j}(qn+b)$ are linearly independent over $\mathbb{C}$. Indeed a complex dependence among rational columns of an infinite matrix implies a rational dependence: the row space is finite-dimensional and can be spanned by finitely many rational rows, whose nullspace is defined over $\mathbb{Q}$. A rational dependence on any tail would, after division by $A_r(qn+b)$ and passage to the limit, contradict the transcendence of π in (VIII.7.8). The solution space of a regular order-m tail recurrence has dimension m, so these columns are a basis over $\mathbb{C}$.

Evaluation at m consecutive sufficiently late indices is an isomorphism from this solution space to $\mathbb{C}^m$. The resulting basis matrix has rational entries and nonzero determinant, hence rational inverse. A solution taking values in K consequently has all its basis coefficients in K, proving the basis assertion over K. Its limit is $\sum_{j=0}^{m-1} c_j \phi_{r,j} \pi^{2j}$. The numbers $1, \pi^2, \dots, \pi^{2m-2}$ are linearly independent over $\overline{\mathbb{Q}}$, and every $\phi_{r,j}$ is nonzero rational. This proves injectivity and the stated image. □

An algebraic-number-valued solution need not be assumed in advance to lie in one number field: its m initial tail values generate a finite extension of $\mathbb{Q}$, and the rational recurrence then keeps its entire tail in that field. Thus the theorem covers all such solutions, including those initially described only as $\overline{\mathbb{Q}}$-valued.

Corollary VIII.7.7 (The normalized limits in Theorem F, over algebraic numbers)**.** *For $1 \le j < m$ put*

$$D_{r,j}(N) = \frac{A_{r,j}(N)}{2^{\binom{r+2j-1}{2j}}}.$$

Then $D_{r,j}(0) = 0$, $D_{r,j}(1) = 1$, and it is a solution of the minimal Franel recurrence on a tail, with

$$\lim_{N\to\infty} \frac{D_{r,j}(N)}{A_r(N)} = \frac{\phi_{r,j}\pi^{2j}}{2^{\binom{r+2j-1}{2j}}}.$$

It is the unique algebraic-number-valued tail solution with that limit. The same uniqueness holds for its sampled sequence $D_{r,j}(qn+b)$ relative to $L_{r;q,b}$ and $A_r(qn+b)$. In particular, for $r \ge 3$

and $r \geq 5$, respectively,

$$\lim \frac{D_{r,1}(N)}{A_r(N)} = \frac{\zeta(2)}{r+1}, \qquad \lim \frac{D_{r,2}(N)}{A_r(N)} = \frac{3(5r+2)\zeta(4)}{(r+1)(r+2)(r+3)}.$$

Proof. At $N = 0$ the deformation is 1, while at $N = 1$ it is $(1-z)^{-r}+(1+z)^{-r}$. The stated initial values follow by coefficient extraction. The limits and uniqueness follow from Theorem VIII.7.6. Finally $\phi_{r,1} = r/6$ and $\phi_{r,2} = r(5r+2)/360$ give the two special formulas, already computed in [74, Examples 1.5 and 1.7]. □

VIII.7.7 Exact scope of the conjectural statements

Chamberland–Straub [75, Conjectures 9 and 11] state the normalized $\zeta(2)$ and $\zeta(4)$ limits, respectively for $r \geq 3$ and $r \geq 5$, and include uniqueness. Their wording does not specify a coefficient field for the secondary solution. The numerical values of the limits and the deformed rational sequences are already in Straub–Zudilin. What the argument here supplies is their membership in the unrestricted minimal recurrence, complete uniqueness over $\overline{\mathbb{Q}}$, and the same conclusions for every arithmetic subsequence. It does not establish unrestricted complex uniqueness.

In fact, on the complex tail solution space the limit map has a kernel of dimension $m-1$. For $r \geq 7$ this obstruction persists even with the two displayed initial normalizations in the eventual sense. To make it explicit, set $c_j = A_{r,j}(1)$ and $\ell_j = \phi_{r,j}\pi^{2j}$, $j = 1, 2, 3$, and put

$$Z = (c_2\ell_3 - c_3\ell_2)A_{r,1} + (c_3\ell_1 - c_1\ell_3)A_{r,2} + (c_1\ell_2 - c_2\ell_1)A_{r,3}.$$

Then $Z(0) = Z(1) = 0$, $Z/A_r \to 0$, and Z is not the zero tail solution: its coefficient of $A_{r,3}$ is nonzero by the transcendence of π, and the basis is independent. Thus $D_{r,1} + cZ$ has the same two initial values and limit for every $c \in \mathbb{C}$, and satisfies the recurrence eventually. An assertion requiring the recurrence to hold at every small nonnegative index needs a separate regularity argument; no such assertion is inferred here from eventual annihilation. The coefficient-degree prediction in [74, Remark 4.1] is likewise not settled by a width bound.

VIII.8 A local Fourier proof and sharpening of the Laplace–Jordan principle

Theorem J in the attached draft can be proved without identifying global Fourier solutions with rapid-decay periods or rotating their integration chains. The replacement uses formal stationary phase and the elementary nearby-cycle/vanishing-cycle description at one point. It also identifies which eigenvalue can acquire an additional Jordan step.

Lemma VIII.8.1 (Nearby and vanishing monodromy)**.** *Let V, W be finite-dimensional complex vector spaces, with maps $u : V \to W$ and $v : W \to V$, and suppose*

$$T = I_V + vu, \qquad S = I_W + uv$$

are invertible. For every polynomial $p \in \mathbb{C}[z]$,

$$(T - I_V)p(T) = v\,p(S)u. \tag{VIII.8.1}$$

In particular, $p(S) = 0$ implies $(T - I_V)p(T) = 0$. For each $\lambda \neq 1$, the restrictions of T and S to their λ-generalized eigenspaces are isomorphic. Only the 1-eigenspace can acquire a further Jordan step in passing from S to T.

Proof. The relation $uT = Su$ implies $up(T) = p(S)u$. Multiplication by v proves (VIII.8.1). On a λ-generalized eigenspace with $\lambda \neq 1$, $T - I_V$ and $S - I_W$ are invertible. Thus u and v give mutually inverse isomorphisms after composing one of them with the appropriate inverse of $T - I_V$ or $S - I_W$. The identity then proves the assertion at $\lambda = 1$ as well. □

We use local monodromy at infinity in the positive local coordinate $1/s$. With this convention, regular local Fourier transform preserves its monodromy conjugacy class. A change of Fourier sign is a homothety of the target and does not alter that class.

Proposition VIII.8.2 (The Fourier monodromy bound)**.** *Let $\mathcal{M}$ be a holonomic module on the complex affine s-line, whose generic connection is regular singular at infinity. Let M be the monodromy of that regular germ. Write T for the monodromy at $x = 0$ of the generic connection underlying its Fourier transform. Then the transformed connection has no singularity at a finite $x \neq 0$, is regular singular or ordinary at 0, and*

$$p(M) = 0 \quad \Longrightarrow \quad (T - I)p(T) = 0.$$

The same annihilating polynomial applies to the monodromy of every differential subquotient of the transformed generic connection. No middle-extension hypothesis on $\mathcal{M}$ is required.

Proof. Formal stationary phase gives

$$\Phi_x(\mathcal{F}\mathcal{M}) \simeq \mathcal{F}^{(\infty,x)}\big((\Psi_\infty \mathcal{M})^{\leq 1,(x)}\big)$$

for every finite x; this is [77, Corollary 2.3(1)]. Because $\Psi_\infty\mathcal{M}$ is regular, its exponential-type summand is zero unless $x = 0$. At $x = 0$ its local Fourier transform is again regular and has monodromy M: [41, Remark 5.2 and Remark 5.3(4)]. These statements concern the extended formal local Fourier functor, which includes trivial regular summands. For example, a trivial rank-one infinity germ contributes the trivial rank-one vanishing cycles of a delta module; it is not discarded.

For a holonomic module on a formal disc, its irregular part is the same in nearby and vanishing cycles. Thus the regularity of the vanishing cycles proves regularity of the nearby germ, and their vanishing at $x \neq 0$ proves smoothness there. On the regular part, nearby and vanishing cycles

have the description (V, W, u, v) with monodromies $I + vu$ and $I + uv$, respectively; see [77, §3.3, formula (3.3)]. The vanishing monodromy S is conjugate to M. Lemma VIII.8.1 therefore proves the asserted identity and bound. Restricting an annihilating polynomial to an invariant subspace or its quotient preserves the identity. The argument also works for the dual local system of solutions, since local Fourier transform is compatible with duality and both source and target monodromies are dualized. □

Theorem VIII.8.3 (Eigenvalue-resolved iterated Borel bound)**.** *Let $A_0(t) = \sum_{n\ge 0} a_n t^n$ be a nonzero D-finite germ with positive radius of convergence. Suppose its minimal equation is regular singular or ordinary at 0, and denote its monodromy by M. For $r \ge 0$, put*

$$A_r(x) = \sum_{n\ge 0} \frac{a_n x^n}{(n!)^r}.$$

Each minimal equation is regular singular or ordinary at 0. If $p(M) = 0$, then its monodromy T_r satisfies

$$(T_r - I)^r p(T_r) = 0. \tag{VIII.8.2}$$

In particular, nonunit monodromy eigenvalues acquire no larger Jordan blocks under any number of Borel transforms. The largest Jordan-block size at eigenvalue 1 increases by at most r. If A_0 is algebraic, every nonunit block of T_r is semisimple, and every block at eigenvalue 1 has size at most $r + 1$.

Proof. It suffices to prove the assertion for one transform and iterate. Put $f(s) = s^{-1}A_0(s^{-1})$ and $\mathcal{M} = \mathbb{C}[s]\langle\partial_s\rangle \cdot f$. Its generic connection is the rational gauge and variable transform of the minimal equation of A_0. It is regular at infinity, and its local monodromy in the coordinate $1/s$ is M.

Here is the exact relation with the series transform. On $\mathbb{C}[s] \oplus s^{-1}\mathbb{C}[[s^{-1}]]$, define

$$\mathcal{B}(s^j) = 0 \quad (j \ge 0), \qquad \mathcal{B}(s^{-n-1}) = x^n/n! \quad (n \ge 0).$$

Direct checking on monomials gives $\mathcal{B}(sf) = \partial_x\mathcal{B}(f)$ and $\mathcal{B}(\partial_s f) = -x\mathcal{B}(f)$. Consequently, with Fourier convention $s \mapsto \partial_x$, $\partial_s \mapsto -x$, there is a surjection of holonomic modules

$$\mathcal{F}(\mathcal{M}) \longrightarrow \mathbb{C}[x]\langle\partial_x\rangle \cdot A_1.$$

The solution local system of the minimal equation of A_1 is therefore a subquotient of the Fourier solution local system. Proposition VIII.8.2 gives $(T_1 - I)p(T_1) = 0$ and regularity. Applying the same argument successively with the polynomial $(z-1)^j p(z)$ proves (VIII.8.2).

For algebraic A_0, the local monodromy of its minimal equation has finite order: its solutions are spanned by finitely many algebraic conjugates. Its minimal polynomial is a product of distinct roots-of-unity factors. Multiplication by $(z-1)^r$ therefore gives exactly the stated eigenvalue-specific bounds. □

Theorem J of the attached draft follows by taking $p(z) = (z^N - 1)^b$ and $r = 1$, since $(z^N - 1)^{b+1}$ is divisible by $(z-1)(z^N-1)^b$. Its algebraicity consequences use $r = 1$ and $r = 2$. More generally, a block of size L at eigenvalue 1 in A_r forces a block of size at least $L - r$ at eigenvalue 1 in A_0, whenever the regularity hypothesis on A_0 holds. In particular, a block of size $r + 2$ excludes algebraicity of A_0. This replaces the rapid-decay-period input for the attached algebraicity classification; it does not supply any assertion about numerical values of the ordinary generating functions.

VIII.9 Bessel determinants and the three algebraicity classifications

This section supplies the combinatorial definitions, Bessel reductions, and scalar ordinary-point result used in the differential and arithmetic theorems of Section VIII.10. The Laplace input is Theorem VIII.8.3. In particular, no unpublished tableau operator theorem is needed for the arithmetic specialization.

VIII.9.1 Definitions and classical determinant identities

Let $y_k(n)$ count standard Young tableaux of size n with at most k rows, and let $u_k(n)$ count permutations of $\{1,\dots,n\}$ whose longest increasing subsequence has length at most k. Let $f_k(n)$ count perfect matchings of $\{1,\dots,2n\}$ with no k mutually crossing arcs. Thus a forbidden crossing consists of arcs (a_i,b_i) with $a_1<\cdots<a_k<b_1<\cdots<b_k$. All three empty objects have count 1. Define the ordinary generating functions

$$\widetilde{Y}_k(t)=\sum_{n\ge0}y_k(n)t^n,\qquad \widetilde{U}_k(t)=\sum_{n\ge0}u_k(n)t^n,\qquad \widetilde{F}_k(t)=\sum_{n\ge0}f_k(n)t^n,$$

and the normalized functions

$$Y_k(x)=\sum_{n\ge0}y_k(n)\frac{x^n}{n!},\quad U_k(s)=\sum_{n\ge0}u_k(n)\frac{s^n}{(n!)^2},\quad F_k(x)=\sum_{n\ge0}f_k(n)\frac{x^{2n}}{(2n)!}.$$

Put $b_j(x)=I_j(2x)$, with $b_{-j}=b_j$. The classical Gordon–Gessel and Weyl-chamber determinant identities are

$$Y_{2r}(x)=\det[b_{i-j}(x)+b_{i+j-1}(x)]_{i,j=1}^{r}, \tag{VIII.9.1}$$
$$Y_{2r+1}(x)=e^x\det[b_{i-j}(x)-b_{i+j}(x)]_{i,j=1}^{r}, \tag{VIII.9.2}$$
$$U_k(s)=\det[I_{i-j}(2\sqrt{s})]_{i,j=1}^{k}, \tag{VIII.9.3}$$
$$F_k(x)=\det[b_{i-j}(x)-b_{i+j}(x)]_{i,j=1}^{k-1}. \tag{VIII.9.4}$$

Empty determinants equal 1. The tableau formulas are stated in [80, §1.2, equations (6)–(7)], with their attribution to Gordon and Gessel; the permutation formula is Gessel's [79]. Formula (VIII.9.4) is the Grabiner–Magyar formula [81], also displayed with exactly this normalization in [82, equation (1.4)]. These enumerative identities are the imported combinatorial inputs.

VIII.9.2 Reduction to two Bessel coordinates

Write

$$y(x)=I_0(2x)=\sum_{m\ge0}\frac{x^{2m}}{(m!)^2},\qquad y_s(s)=I_0(2\sqrt{s})=\sum_{m\ge0}\frac{s^m}{(m!)^2}.$$

Their scalar equations and logarithmic companion solutions are

$$y''=4y-x^{-1}y',\qquad \kappa(x):=K_0(2x)=-(\log x)y(x)+h(x), \tag{VIII.9.5}$$
$$y_s''=s^{-1}(y_s-y_s'),\qquad \kappa_s(s):=K_0(2\sqrt{s})=-\tfrac12(\log s)y_s(s)+h_s(s), \tag{VIII.9.6}$$

where h,h_s are holomorphic at zero. The integer-order Bessel recurrences and these local expansions may be found in [57, §§10.29, 10.31].

Lemma VIII.9.1 (Bessel group and independence)**.** *The two Bessel equations have differential Galois group* SL_2 *over their rational function fields. The pairs* (y, y') *and* (y_s, y_s') *are algebraically independent over* $\mathbb{C}(x)$ *and* $\mathbb{C}(s)$*, respectively.*

Proof. Kolchin's Bessel-group theorem gives SL_2 for the ordinary Bessel equation of order zero [42]; the precise non-half-integer criterion is also stated in [83, Example 1.3.32]. A nonzero linear change of variable gives the modified equation (VIII.9.5). Equation (VIII.9.6) pulls back to it under $s = x^2$. Its Galois group contains that of this finite algebraic pullback and is contained in SL_2 because its Wronskian is a constant times $1/s$. It is therefore SL_2 as well.

For either equation, let v be the analytic Bessel solution and w an independent solution in a Picard–Vessiot extension. If $R(v, v') = 0$ with rational-function coefficients, the group SL_2, transitive on nonzero solution vectors, gives $R(c_1 v + c_2 w, c_1 v' + c_2 w') = 0$ for all constants c_1, c_2. The fundamental matrix is invertible. A polynomial over the Picard–Vessiot field which vanishes for all $(c_1, c_2) \in \mathbb{C}^2$ is zero. Hence $R = 0$. □

Lemma VIII.9.2 (Determinants are homogeneous forms)**.** *Each determinant in* (VIII.9.1)–(VIII.9.4) *is a nonzero homogeneous polynomial in the corresponding Bessel pair, with rational Laurent-polynomial coefficients. The degrees are* $\lfloor k/2 \rfloor$ *for* $e^{-\varepsilon x} Y_k(x)$*, where* $k = 2\lfloor k/2 \rfloor + \varepsilon$*;* k *for* $U_k(s)$*; and* $k - 1$ *for* $F_k(x)$*.*

Proof. The recurrences

$$b_0 = y, \qquad b_1 = y'/2, \qquad b_{j+1} = b_{j-1} - (j/x) b_j$$

express every b_j as a rational Laurent-polynomial linear form in y, y'. Thus a determinant of size d is homogeneous of degree d.

For the s-variable put $J_j(s) = s^{-j/2} I_j(2\sqrt{s})$ for $j \geq 0$. Then $J_0 = y_s$, $J_1 = y_s'$ and $s J_{j+1} = J_{j-1} - j J_j$. Multiplying row i of (VIII.9.3) by $s^{-i/2}$ and column j by $s^{j/2}$ does not change its determinant, and changes its entry to $s^{\max(0, j-i)} J_{|i-j|}(s)$. All coefficients are therefore in $\mathbb{Q}[s, s^{-1}]$. Each resulting form is nonzero because the normalized counting series has constant coefficient 1. □

Lemma VIII.9.3 (The logarithmic degree is exact)**.** *Let* v *be either analytic Bessel solution above, in variable* z*, and let* $\Phi \in \mathbb{C}(z)[X_0, X_1]$ *be nonzero homogeneous of degree* $d \geq 1$*. The function* $\Phi(v, v')$ *has minimal differential order* $d + 1$*, and its minimal equation at zero has one unipotent Jordan block of size* $d + 1$*. Multiplication by* $e^{\varepsilon z}$ *does not change these two assertions.*

Proof. Extend differentiation to the polynomial ring in X_0, X_1 using the relevant Bessel equation. If a rational differential operator L annihilates $\Phi(v, v')$, then the polynomial $L\Phi$ vanishes at (v, v'). Lemma VIII.9.1 implies $L\Phi = 0$ as a polynomial. Thus L also annihilates $\Phi(w, w')$ for the logarithmic companion w.

By (VIII.9.5) or (VIII.9.6), this latter solution has the convergent expansion

$$\Phi(w, w') = \sum_{j=0}^{d} (\log z)^j g_j(z), \qquad g_d = c^d \Phi(v, v') \neq 0, \quad c = -1 \text{ or } -\tfrac{1}{2},$$

where the g_j are meromorphic at zero. Therefore $(T - I)^d \Phi(w, w') = d!(2\pi i)^d g_d \neq 0$. On the other hand, all derivatives of $\Phi(v, v')$ belong to the $(d + 1)$-dimensional rational span of $v^d, v^{d-1} v', \ldots, (v')^d$. Its minimal order is at most $d + 1$, so the block just found occupies the entire solution space and the order is exactly $d + 1$. The exponential factor gives an invertible rational differential gauge $\partial_z \mapsto \partial_z - \varepsilon$ and is single-valued at zero. □

VIII.9.3 Ordinary generating functions

Theorem VIII.9.4 (The three complete algebraicity classifications)**.** *For positive integer parameters:*

1. *$\widetilde{Y}_k$ is algebraic if and only if $k \le 3$;*
2. *the ordinary generating function for permutations avoiding an increasing subsequence of length ℓ is algebraic if and only if $\ell \le 3$;*
3. *$\widetilde{F}_k$ is algebraic if and only if $k \le 2$.*

Proof. The small tableau cases are

$$\widetilde{Y}_1(t) = \frac{1}{1-t}, \qquad \widetilde{Y}_2(t) = \frac{\sqrt{(1+2t)/(1-2t)}-1}{2t}, \qquad \widetilde{Y}_3(t) = \frac{1-t-\sqrt{1-2t-3t^2}}{2t^2}.$$

The apparent denominators at zero are removable in the chosen formal branches. The latter two formulas follow respectively from $y_2(n) = \binom{n}{\lfloor n/2 \rfloor}$ and the Motzkin enumeration [80, equation (3)]. For $k \ge 4$, Lemmas VIII.9.2–VIII.9.3 give a block of size $\lfloor k/2 \rfloor + 1 \ge 3$ in Y_k. If $\widetilde{Y}_k$ were algebraic, the onefold Borel bound of Theorem VIII.8.3 would make every block at eigenvalue 1 have size at most two, a contradiction.

For permutations, avoidance of length ℓ is counted by $u_{\ell-1}(n)$ when $\ell \ge 2$. The cases $\ell = 1, 2, 3$ give respectively 1, $(1-t)^{-1}$, and the Catalan series $(1-\sqrt{1-4t})/(2t)$. For $\ell \ge 4$, put $k = \ell - 1 \ge 3$. The series U_k has a block of size $k+1 \ge 4$, whereas algebraicity of $\widetilde{U}_k$ and the twofold Borel bound permit size at most three.

For matchings, $k = 1$ allows only the empty matching and $k = 2$ gives the Catalan series. If $k \ge 3$ and $\widetilde{F}_k$ were algebraic, then $\widetilde{F}_k(t^2)$ would be algebraic. Its exponential generating function, obtained by padding the sequence with zeros at odd indices, is $F_k(x)$. The onefold Borel bound gives block size at most two, contrary to its size $k \ge 3$ from Lemma VIII.9.3.

In each contradiction the assumed algebraic power series is convergent near zero and its minimal equation is regular singular there, as its solutions are spanned by algebraic conjugates with convergent Puiseux expansions. Thus all hypotheses of Theorem VIII.8.3 are satisfied. □

VIII.9.4 A published Jacobi system and a scalar tableau equation

The arithmetic specialization below requires every nonzero point to be ordinary for the minimal tableau scalar equation. We now prove that property directly, using an explicit scalar elimination of a published Jacobi system.

For $N \ge 1$, $\alpha > -1$, $b > 0$, $\beta > 0$, and $0 \le p \le N$, let

$$H_p(z) = \binom{N}{p}^{-1} \int_{[0,1]^N} e^{-z\sum u_i} \prod_i u_i^{\alpha}(1-u_i)^{b-1} \prod_{i<j} |u_i - u_j|^{\beta} \cdot e_p(1-u_1, \dots, 1-u_N)\, du,$$

where e_p is the elementary symmetric polynomial and $e_0 = 1$. The Forrester–Kumar system [84, Corollary 2.5, equations (2.19)–(2.20)] states that

$$(N-p)zH_{p+1} = ((N-p)z + B_p)H_p + zH_p' - D_pH_{p-1}, \tag{VIII.9.7}$$

where

$$B_p = p\left(\alpha + b + 1 + \frac{\beta}{2}(2N-p-1)\right), \qquad D_p = p\left(b + \frac{\beta}{2}(N-p)\right).$$

Terms outside the range $0 \le p \le N$ are omitted. This is the only published differential-system input to the construction; the elimination and its specialization are given explicitly next.

Lemma VIII.9.5 (Scalar elimination)**.** *There is a scalar differential operator* $\mathcal{A} = \sum_{m=0}^{N+1} a_m(z)\partial_z^m$ *annihilating* H_0 *such that*

$$a_m \in \mathbb{R}[z], \qquad a_{N+1} = z^N, \qquad \deg a_m \le N, \qquad z^{m-1} \mid a_m \quad (m \ge 1).$$

The coefficients are rational when the parameters are rational.

Proof. Write $D = \partial_z$ and construct rational operators

$$P_0 = 1, \qquad P_1 = 1 + D/N,$$

and, for $1 \le p < N$,

$$(N-p)P_{p+1} = \left(D + N - p + \frac{B_p}{z}\right) P_p - \frac{D_p}{z} P_{p-1}. \tag{VIII.9.8}$$

Equation (VIII.9.7) gives $H_p = P_p H_0$. Write $P_p = \sum_{m=0}^{p} q_{p,m} D^m$ for $p \ge 1$. Induction using $D \circ (qD^m) = qD^{m+1} + q'D^m$ gives

$$q_{p,p} = \frac{(N-p)!}{N!}, \qquad q_{p,m} \in \operatorname{span}\{z^{-(p-1)}, \dots, 1\}, \qquad \operatorname{ord}_0 q_{p,m} \ge m - p.$$

For the induction, differentiation or division by z lowers the least Laurent exponent by at most one, and no positive exponent is introduced. The highest derivative comes only from $D \circ P_p$. The coefficient of D^m in that product is $q'_{p,m} + q_{p,m-1}$, whose order is at least $m - p - 1$; the other terms in (VIII.9.8) have the same bound. The case $p = 1$ involving P_0 is immediate.

The $p = N$ equation in (VIII.9.7) now shows that

$$\mathcal{A} = N! z^{N-1}\bigl(zD \circ P_N + B_N P_N - D_N P_{N-1}\bigr) \tag{VIII.9.9}$$

annihilates H_0. The bracket has Laurent exponents between $-(N-1)$ and 1, so multiplication by z^{N-1} makes its coefficients polynomials of degree at most N. Its top coefficient is $z/N!$, giving $a_{N+1} = z^N$. The coefficient of D^m in the bracket has order at least $m - N$, proving the stated divisibility after multiplication. The formulas also prove rationality. □

Proposition VIII.9.6 (No nonzero tableau scalar singularities)**.** *Let* $k = 2r + \varepsilon \ge 2$*, with* $\varepsilon \in \{0, 1\}$*. There is an order-*$r+1$ *operator over* $\mathbb{Q}[x]$ *annihilating* Y_k *whose leading coefficient is* x^r*. The minimal monic operator has order* $r + 1$ *and is ordinary at every* $x \neq 0$*.*

Proof. First derive the required Jacobi integral from the determinant. The Bessel integral

$$b_j(x) = \frac{1}{\pi} \int_0^{\pi} e^{2x\cos\theta} \cos(j\theta)\, d\theta$$

and the identities

$$\cos((i-j)\theta) + \cos((i+j-1)\theta) = 2\cos((i - \tfrac{1}{2})\theta)\cos((j - \tfrac{1}{2})\theta),$$

$$\cos((i-j)\theta) - \cos((i+j)\theta) = 2\sin(i\theta)\sin(j\theta)$$

convert the two tableau matrices into Gram matrices. The quotients $\cos((i - \frac{1}{2})\theta)/\cos(\theta/2)$ and $\sin(i\theta)/\sin\theta$ are monic polynomials of degree $i - 1$ in $u = 2\cos\theta$. Putting $a = \varepsilon - \frac{1}{2}$ and $w(u) = (2-u)^a(2+u)^{1/2}$, both Gram matrices have entries

$$\frac{1}{2\pi} \int_{-2}^{2} e^{xu} Q_i(u) Q_j(u) w(u)\, du.$$

Expanding the determinant and applying Fubini, or equivalently using Andreief's identity, gives

$$Y_k(x) = \frac{e^{\varepsilon x}}{r!(2\pi)^r} \int_{[-2,2]^r} e^{x\sum u_i} \Delta(u)^2 \prod_i w(u_i)\, du. \tag{VIII.9.10}$$

Here the determinants of the monic polynomial bases are the Vandermonde Δ.

Substitute $u_i = 4v_i - 2$ in (VIII.9.10) and take

$$N = r, \qquad \alpha = \tfrac{1}{2}, \qquad b = \varepsilon + \tfrac{1}{2}, \qquad \beta = 2$$

in the preceding Jacobi system. The precise identity is

$$Y_k(x) = \frac{4^{r(r+\varepsilon)}}{r!(2\pi)^r} e^{(\varepsilon-2r)x} H_0(-4x).$$

Indeed the squared Vandermonde, weights, and Jacobian contribute $4^{r(r-1)}$, $4^{\varepsilon r}$, and 4^r. In the operator of Lemma VIII.9.5, substitute

$$z = -4x, \qquad \partial_z = -\tfrac{1}{4}(\partial_x - (\varepsilon - 2r)).$$

After multiplication by -4, its leading coefficient is x^r. All coefficients are rational polynomials and its order is $r+1$. Lemma VIII.9.3 already proves that this is the minimal order. Dividing this operator by x^r therefore gives the minimal monic operator. Its coefficients are holomorphic at every nonzero point, proving the assertion. □

Only the ordinary-point conclusion of Proposition VIII.9.6 is needed in Theorem VIII.10.4. Possible cancellation of a common power of x in a primitive polynomial normalization is irrelevant to that conclusion. This explicit construction supplies the required statement without invoking a separate internal minimal-operator manuscript.

VIII.10 Exact Bessel modules, algebraic base changes, and tableau values

The algebraicity classifications of Theorem VIII.9.4 admit stronger differential and arithmetic conclusions. The general homogeneous-form theorem below determines the normalized equations, their complete local partition, and all algebraic relations between their initial jets. The ordinary-point result needed for arithmetic specialization was proved explicitly in Proposition VIII.9.6.

VIII.10.1 A homogeneous-form theorem

Let $K = \mathbb{C}(z)$ and let y denote either

$$y(z) = I_0(2z), \qquad y'' = 4y - z^{-1}y', \quad \text{or} \quad y(z) = I_0(2\sqrt{z}), \qquad y'' = z^{-1}(y - y').$$

Both rank-two connections have differential Galois group SL_2. Their local monodromy at zero is a nontrivial unipotent block of size two. These Bessel inputs, including their primary references, are established in Lemma VIII.9.1. For $d \geq 1$, let $\Phi \in K[X_0, X_1]$ be a nonzero homogeneous form of degree d, and put

$$F(z) = e^{\varepsilon z}\Phi(y(z), y'(z)), \qquad \varepsilon \in \{0, 1\}.$$

Theorem VIII.10.1 (Exact module and stability)**.** *The minimal differential operator of F over K is irreducible of order $d+1$. Its differential Galois group, in its solution representation, is*

$$\begin{cases} \mathrm{Sym}^d(\mathrm{SL}_2), & \varepsilon = 0, \\ \{c\,\mathrm{Sym}^d(g) : c \in \mathbb{C}^*,\ g \in \mathrm{SL}_2\}, & \varepsilon = 1. \end{cases}$$

In the first case this is isomorphic to SL_2 for odd d and to PSL_2 for even d. At zero the complete local monodromy partition consists of one unipotent block $J_{d+1}(1)$.

All these orders and irreducibility assertions remain valid over every finite algebraic extension of K. After a ramified algebraic change of variable, the local block at a point above zero still has size $d+1$.

Proof. Let V be the two-dimensional Bessel solution space in a Picard–Vessiot extension. Polarizing $u \mapsto \Phi(u, u')$ gives a SL_2-equivariant map from $\mathrm{Sym}^d V$ to the differential field. It is nonzero because y, y' are algebraically independent over K, by Lemma VIII.9.1. The representation $\mathrm{Sym}^d V$ is irreducible, so this map is injective. The Galois orbit of any nonzero image spans the entire image. The standard Galois-orbit characterization of a minimal scalar operator therefore gives the claimed order, irreducibility, and solution representation.

For $\varepsilon = 1$, the compositum of the Bessel Picard–Vessiot extension and that of $u' = u$ has group $\mathrm{SL}_2 \times \mathbb{G}_m$. Indeed, a proper subdirect product would give a nontrivial common quotient of SL_2 and $\mathbb{G}_m$, which is impossible. The preceding argument applies to the irreducible representation $\mathrm{Sym}^d V$ tensored with the standard character of $\mathbb{G}_m$.

The symmetric power of a nontrivial $J_2(1)$ is a single $J_{d+1}(1)$: in the usual degree-d polynomial representation, its logarithm acts by a nonzero multiple of $X_0\partial_{X_1}$, with one chain of length $d+1$. The exponential twist is single-valued and nonvanishing at zero and does not alter this partition.

Both displayed Galois groups are connected. Finite algebraic extension of the base leaves the identity component unchanged, and hence leaves the full connected group and its irreducible representation unchanged. A ramification of index e replaces the local monodromy by its eth power; a unipotent Jordan block in characteristic zero retains its size under this operation. □

Corollary VIII.10.2 (The three determinant families)**.** *With the notation of Section VIII.9, the following orders are exact:*

function	d	*minimal order*	*monodromy at zero*
$Y_k(x),\ k \geq 2$	$\lfloor k/2 \rfloor$	$\lfloor k/2 \rfloor + 1$	$J_{\lfloor k/2 \rfloor + 1}(1)$
$U_k(s),\ k \geq 1$	k	$k+1$	$J_{k+1}(1)$
$F_k(x),\ k \geq 2$	$k-1$	k	$J_k(1)$.

The permutation and matching groups are respectively $\mathrm{Sym}^k(\mathrm{SL}_2)$ *and* $\mathrm{Sym}^{k-1}(\mathrm{SL}_2)$*. For tableaux, write* $k = 2d+\varepsilon$*; the group is the corresponding group in Theorem VIII.10.1. Every listed minimal order persists after finite algebraic extension of the coefficient field.*

Proof. The determinant reductions in Lemma VIII.9.2 give nonzero homogeneous forms of exactly the displayed degrees. Their nonvanishing also follows from the constant term 1 of each generating function. Apply Theorem VIII.10.1. □

VIII.10.2 The complete algebraic relations among initial jets

Write $\mathbf{J}_F = (F, F', \dots, F^{(d)})^{\mathsf{T}}$ and

$$\mathbf{m}_d = (y^d, y^{d-1}y', \dots, (y')^d)^{\mathsf{T}}.$$

Differentiating and reducing y'' by the Bessel equation produces a rational matrix $P(z)$ with

$$\mathbf{J}_F = e^{\varepsilon z} P(z) \mathbf{m}_d. \tag{VIII.10.1}$$

Theorem VIII.10.3 (Jet cone)**.** *The matrix* P *belongs to* $\mathrm{GL}_{d+1}(K)$*. After the invertible rational change of jet coordinates* $\mathbf{Z} = P^{-1}\mathbf{J}_F$*, the ideal of all polynomial relations over* K *is generated by*

$$Z_i Z_{j+1} - Z_{i+1} Z_j, \qquad 0 \leq i < j \leq d-1. \tag{VIII.10.2}$$

For $d = 1$ *the ideal is zero. Consequently*

$$\mathrm{trdeg}_K\, K(F, F', \dots, F^{(d)}) = 2.$$

Every two distinct entries of this jet vector are algebraically independent over K*, and remain so over every finite algebraic extension of* K*. When* $\varepsilon = 0$*, one has the further exact field identity*

$$K(F, F', \dots, F^{(d)}) = K(y^d, y^{d-1}y', \dots, (y')^d) = K(y, y')^{\mu_d},$$

where μ_d *acts by simultaneous scalar multiplication on* y, y'*. Thus its index in* $K(y, y')$ *is exactly* d*.*

Proof. If the rows of P were dependent, (VIII.10.1) would give a nonzero rational differential equation for F of order at most d, contradicting Theorem VIII.10.1. Therefore P is invertible.

The kernel of the substitution $Z_j \mapsto X_0^{d-j} X_1^j$ is the homogeneous ideal (VIII.10.2) of the affine cone over the degree-d rational normal curve. For completeness, these minors identify all products of Z_j having the same number of factors and the same sum of indices; their images are the same monomial in X_0, X_1. Grouping monomials by those two integers proves that they generate the entire kernel. Algebraic independence of y, y' shows that this is exactly the relation ideal for $\varepsilon = 0$. For $\varepsilon = 1$, the three functions y, y', e^z are algebraically independent: the group $\mathrm{SL}_2 \times \mathbb{G}_m$ acts transitively on the nonzero Bessel solution vector and independently on the exponential coordinate. Multiplying every degree-d monomial by the same independent scalar preserves the same homogeneous kernel. This proves the ideal assertion and its dimension.

Two different rows of P are linearly independent, so their degree-d forms A, B are not proportional. Two nonproportional homogeneous forms of the same positive degree in two variables are algebraically independent: their ratio is a nonconstant function of X_1/X_0, while simultaneous scaling changes their common magnitude. Equivalently, their Jacobian is nonzero. The same conclusion holds after multiplying both by the independent scalar e^z. This proves pairwise independence; an algebraic extension of the coefficient field cannot decrease transcendence degree. Finally the displayed monomial field is the fixed field of the faithful scalar action of μ_d on the rational field $K(y, y')$, proving the index statement. □

VIII.10.3 Arithmetic specialization without exceptional nonzero points

For tableaux Proposition VIII.9.6 supplies an additional essential input: an order-$d+1$ annihilator of Y_k has leading coefficient x^d, where $d = \lfloor k/2 \rfloor$, and $d+1$ is the exact minimal order. Thus every $\alpha \neq 0$ is an ordinary point of the minimal scalar equation. This property is stronger than having a connection which is merely gauge-equivalent to one regular at α; it is what rules out exceptional zeros of the scalar determinant form.

Theorem VIII.10.4 (All nonzero algebraic tableau arguments)**.** *Let $k \geq 2$, $d = \lfloor k/2 \rfloor$, and $\alpha \in \overline{\mathbb{Q}}^*$. Then*

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(Y_k(\alpha), Y_k'(\alpha), \dots, Y_k^{(d)}(\alpha)) = 2.$$

Every two distinct numbers in this displayed tuple are algebraically independent over $\overline{\mathbb{Q}}$. All their algebraic relations are obtained from (VIII.10.2) by the invertible algebraic change of coordinates $\mathbf{Z} = P(\alpha)^{-1}\mathbf{J}_{Y_k}(\alpha)$. In particular $Y_k(\alpha)$ is transcendental and nonzero. For $k = 1$, $Y_1(\alpha) = e^\alpha$ is transcendental.

Proof. Put $y = I_0(2x)$. The functions y, y', e^x are E-functions over $\mathbb{Q}$; their first-order system is

$$\frac{d}{dx}\begin{pmatrix} y \\ y' \\ e^x \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 4 & -1/x & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} y \\ y' \\ e^x \end{pmatrix}.$$

They are algebraically independent over $\mathbb{C}(x)$ by the group argument in Theorem VIII.10.3. The Siegel–Shidlovskii theorem in [68, Theorem 1.1], with common denominator x, therefore says that $y(\alpha), y'(\alpha), e^\alpha$ are algebraically independent over $\overline{\mathbb{Q}}$ for every nonzero algebraic α.

The determinant reductions use coefficients in $\mathbb{Q}[x, x^{-1}]$, so P in (VIII.10.1) has no pole at α. Moreover $P(\alpha)$ is invertible. Indeed the symmetric-power Bessel system is ordinary there, and so is the minimal scalar equation of Y_k by the preceding input. Evaluation of its first $d+1$ jets is an isomorphism from its solution space to $\mathbb{C}^{d+1}$. The injective polarized solution map of Theorem VIII.10.1 is an isomorphism between solution spaces of that same dimension. Their evaluation matrices are related by $P(\alpha)$, forcing its determinant to be nonzero.

We may therefore specialize the monomial-coordinate argument of Theorem VIII.10.3. The independence of the three basic values gives exactly the Veronese-cone relation ideal and transcendence degree two. Different rows of the invertible $P(\alpha)$ are nonproportional homogeneous forms, giving pairwise algebraic independence by the same Jacobian argument. The case $k = 1$ is the classical transcendence of e^α. □

Remark VIII.10.5 (Scope of the arithmetic statement). No corresponding absence of exceptional nonzero scalar points is assumed here for the permutation and matching determinants. Their functional order, Galois, and jet-cone conclusions hold for all parameters as stated. The arithmetic theorem above uses Proposition VIII.9.6 explicitly; functional Galois information alone would not justify its all-points specialization.

VIII.11 An arithmetic classification for split beta orthogonality

The critical-value argument uses positivity to prevent cancellation of local contributions. A different use of positivity gives an arithmetic classification for multiple orthogonal polynomials: once a logarithmic moment identity forces orthogonality on both sides of a cut, the required sign changes can exceed the degree. The classification below is uniform in both orthogonality indices, in the algebraic cut, and in every positive algebraic polynomial modification of the weight.

VIII.11.1 The complete parameter and index classification

Let $a, b > 0$ be rational with $a + b \in \mathbb{Z}$, let $\xi \in \overline{\mathbb{Q}} \cap (0, 1)$, and let $w \in (\overline{\mathbb{Q}} \cap \mathbb{R})[X]$ satisfy $w(x) > 0$ for $0 < x < 1$. Define the positive integrable weight

$$\rho(x) = w(x)x^{a-1}(1 - x)^{b-1}.$$

For $m, n \geq 0$, the monic type-II Angelesco polynomial $P_{m,n}$ has degree $m + n$ and satisfies

$$\int_0^\xi P_{m,n}(x)x^j\rho(x)\,dx = 0 \qquad (0 \leq j < m), \tag{VIII.11.1}$$

$$\int_\xi^1 P_{m,n}(x)x^j\rho(x)\,dx = 0 \qquad (0 \leq j < n). \tag{VIII.11.2}$$

Empty lists of conditions are allowed, and $P_{0,0} = 1$. Existence and uniqueness will be proved below.

Theorem VIII.11.1 (Exact algebraic-coefficient classification)**.** *For the preceding family, the following are equivalent:*

1. *Every coefficient of $P_{m,n}$ is algebraic.*
2. *Every zero of $P_{m,n}$ is algebraic.*
3. *Either $a, b \in \mathbb{Z}$, or $m = n$.*

In the algebraic cases the coefficients belong to

$$K = \mathbb{Q}(\xi, w_0, \ldots, w_d), \qquad w(X) = \sum_{j=0}^{d} w_j X^j.$$

Consequently, when a, b are nonintegers and $m \neq n$, $P_{m,n}$ has at least one transcendental coefficient and at least one transcendental zero. All its zeros are nevertheless real and simple, with precisely m in $(0, \xi)$ and n in $(\xi, 1)$.

The rationality of a, b and the integer-sum condition are essential to the stated arithmetic reduction. The theorem includes the arcsine pair $a = b = 1/2$; no exclusion of that pair is needed. We assert that at least one zero is transcendental off the diagonal, not that every zero is.

VIII.11.2 Positivity and normality

Lemma VIII.11.2 (Sign changes and existence)**.** *For a positive integrable weight on $(0, 1)$, each system (VIII.11.1)–(VIII.11.2) has a unique monic solution of degree $m + n$. That solution has exactly the zero distribution stated in Theorem VIII.11.1.*

Proof. A nonzero real polynomial Q orthogonal to $1, x, \dots, x^{k-1}$ on an interval with a positive weight has at least k sign changes in its interior. Indeed, if the sign-change points were $t_1, \dots, t_s$ with $s < k$, the polynomial $q(x) = \prod_{i=1}^{s}(x - t_i)$ would make Qq have one constant nonzero sign except at finitely many points. Its weighted integral could not vanish, contradicting orthogonality to q.

A nonzero polynomial satisfying both lists therefore has at least $m + n$ distinct real zeros in the two disjoint interiors. In particular, there is no such polynomial of degree at most $m + n - 1$. The linear map from that $(m+n)$-dimensional coefficient space to its $m+n$ moments is injective, hence bijective. It follows that the lower coefficients of a monic degree-$m + n$ solution exist uniquely. The same sign-change count exhausts its degree, proving simplicity and the precise locations of all zeros. The case $m + n = 0$ is immediate. □

VIII.11.3 A logarithmic moment forces orthogonality on both sides

The transcendence input is the classical consequence of Baker's theorem: a nonzero linear form, with algebraic coefficients, in fixed logarithms of nonzero algebraic numbers is transcendental [46]. We reproduce the needed integral reduction, so the argument does not require any unproved period-independence assertion.

For $r \in \mathbb{Q} \cap (0, 1)$ write

$$b_r(x) = x^{r-1}(1 - x)^{-r}, \qquad U_r(y) = \int_0^y b_r(x)\, dx.$$

Lemma VIII.11.3 (Two transcendental base integrals)**.** *If $\xi \in \overline{\mathbb{Q}} \cap (0, 1)$, then both $U_r(\xi)$ and $\int_\xi^1 b_r(x)\, dx$ are transcendental. This includes $r = 1/2$.*

Proof. Write $r = p/q$ with $1 \le p < q$ and put $s = (\xi/(1 - \xi))^{1/q} > 0$. The substitution $x = z^q/(1 + z^q)$ and partial fractions give

$$U_r(\xi) = q \int_0^s \frac{z^{p-1}}{1 + z^q}\, dz = -\sum_{\zeta^q = -1} \zeta^p \log(1 - s/\zeta).$$

For each root ζ, take the continuous logarithm along positive real s, starting at zero when $s = 0$. Its argument never vanishes because no root of $\zeta^q = -1$ is positive real. The displayed identity follows by differentiation and the common value at $s = 0$. All logarithm arguments at the specified s are nonzero algebraic numbers, and all coefficients are algebraic. The integral is strictly positive, so Baker's theorem proves its transcendence. Finally,

$$\int_\xi^1 b_r(x)\, dx = U_{1-r}(1 - \xi),$$

and the same argument applies. □

Lemma VIII.11.4 (Arithmetic synchronization of moments)**.** *Let $r \in \mathbb{Q} \cap (0, 1)$ and $S \in (\overline{\mathbb{Q}} \cap \mathbb{R})[X]$. Then*

$$\int_0^\xi S(x) b_r(x)\, dx = 0 \quad \Longleftrightarrow \quad \int_\xi^1 S(x) b_r(x)\, dx = 0.$$

More precisely, define the unique constant $c(S)$ and polynomial R_S by

$$S(X) = c(S) + X(1 - X)R_S'(X) + (r - X)R_S(X). \tag{VIII.11.3}$$

Either vanishing condition is equivalent to the two algebraic linear conditions

$$c(S) = 0, \qquad R_S(\xi) = 0. \tag{VIII.11.4}$$

If S has coefficients in a field $K \supset \mathbb{Q}$ containing ξ, then $c(S)$ and all coefficients of R_S belong to K.

Proof. The operator in (VIII.11.3) sends X^j to $(j+r)X^j - (j+1)X^{j+1}$. Descending elimination therefore gives a unique R_S and constant $c(S)$, over the coefficient field. Put $H_S(x) = x^r(1-x)^{1-r}R_S(x)$. Direct differentiation and $H_S(0) = H_S(1) = 0$ give

$$\int_0^\xi Sb_r = c(S)U_r(\xi) + H_S(\xi),$$
$$\int_\xi^1 Sb_r = c(S)\{U_r(1) - U_r(\xi)\} - H_S(\xi).$$

Here $H_S(\xi)$ is algebraic. Either integral can vanish only if $c(S) = 0$, by Lemma VIII.11.3. It then forces $H_S(\xi) = 0$, equivalently $R_S(\xi) = 0$ since $0 < \xi < 1$. Conversely, these two conditions make both integrals zero. All operations in descending elimination take place over K. □

VIII.11.4 Proof of the classification

Proof of Theorem VIII.11.1. Suppose first that a, b are integers. Since they are positive, ρ is a polynomial with coefficients in K. Each moment on either interval belongs to K, by termwise integration and $\xi \in K$. The invertible linear system in Lemma VIII.11.2 consequently has its monic solution in $K[X]$.

Otherwise neither shape is an integer, since their sum is an integer. Write

$$a = A + r, \qquad b = B + 1 - r, \qquad A, B \in \mathbb{Z}_{\geq 0}, \quad r \in \mathbb{Q} \cap (0,1).$$

Then $\rho = gb_r$, where $g(X) = w(X)X^A(1-X)^B \in K[X]$ is positive on $(0,1)$.

Assume that $P_{m,n}$ has algebraic coefficients and set $h = \max(m,n)$. For every $0 \leq j < h$, at least one of (VIII.11.1) and (VIII.11.2) holds. Apply Lemma VIII.11.4 to the algebraic polynomial $S = X^j P_{m,n} g$. Both integrals then vanish. Thus $P_{m,n}$ satisfies h orthogonality conditions on each interval. Positivity forces at least $2h$ distinct zeros, so

$$m + n = \deg P_{m,n} \geq 2\max(m,n).$$

This is possible only when $m = n$.

Conversely let $m = n = k$. The $2k$ equations

$$c(X^j P g) = 0, \qquad R_{X^j P g}(\xi) = 0, \qquad 0 \leq j < k, \tag{VIII.11.5}$$

are homogeneous linear equations over K in the $2k+1$ coefficients of P of degree at most $2k$. They have a nonzero solution in $K[X]$. Lemma VIII.11.4 shows that this solution satisfies both orthogonality lists. By the sign-change argument it has degree exactly $2k$. Divide by its nonzero leading coefficient. Uniqueness identifies the resulting monic polynomial with $P_{k,k}$, proving $P_{k,k} \in K[X]$.

Finally, a monic polynomial has all roots algebraic if and only if all its coefficients are algebraic: one direction uses elementary symmetric functions, and the other uses the algebraic closedness of $\overline{\mathbb{Q}}$. The zero distribution follows from Lemma VIII.11.2. □

VIII.11.5 Quadrature, an explicit example, and the literature boundary

Corollary VIII.11.5 (Algebraic cuts destroy all-algebraic Gaussian nodes)**.** *If $a, b > 0$ are rational nonintegers with integer sum, then for every algebraic $\xi \in (0,1)$ and every $N \geq 1$, the N-node Gaussian quadrature rule for ρ restricted to either $[0,\xi]$ or $[\xi,1]$ has at least one transcendental node. In contrast, all zeros of every balanced $P_{N,N}$ are algebraic. These assertions remain true under every positive algebraic polynomial factor w.*

Proof. The Gaussian nodes are the zeros of the degree-N monic orthogonal polynomial, namely $P_{N,0}$ or $P_{0,N}$. Apply the theorem. The balanced assertion is the case $(m, n) = (N, N)$. □

For $w = 1$ and $(a, b) = (r, 1 - r)$, the classical diagonal Rodrigues formula specializes to

$$P_{k,k}(x) = \frac{(-1)^k}{(2k)_k\, b_r(x)} \frac{d^k}{dx^k} \left[x^{k+r-1}(1-x)^{k-r}(x-\xi)^k \right], \qquad k \geq 1. \tag{VIII.11.6}$$

Leibniz' rule makes the right side a polynomial of degree at most $2k$. Integration by parts k times on each interval proves the orthogonality: derivatives of order at most $k - 1$ of the expression in brackets vanish at all relevant endpoints, and $\deg(x^j) < k$. Its leading coefficient before normalization is $(-1)^k(2k)_k$, as can be read from the highest-degree terms. This establishes (VIII.11.6) and its monic normalization. For example,

$$P_{1,1}(x) = x^2 - \frac{1 + r + \xi}{2} x + \frac{r\xi}{2}.$$

At $r = 1/3$ and $\xi = 1/2$ its two zeros are

$$\frac{11 - \sqrt{73}}{24}, \qquad \frac{11 + \sqrt{73}}{24},$$

one on each side of the cut. Every off-diagonal polynomial for the same measures has a transcendental coefficient and a transcendental zero, including the degree-one polynomial for either single interval.

The diagonal formula is classical. After an affine change of variable, this is the Jacobi–Angelesco system with zero exponent at the common endpoint; see Van Assche and Coussement [67, §3.5, Eq. (3.45)]. Their equations (3.48)–(3.50) also express near-diagonal polynomials through incomplete moment ratios. We do not count the diagonal Rodrigues formula, Angelesco normality, polynomial beta reduction, or Baker's theorem as new. The addition proved here is the exact arithmetic exclusion of every unbalanced index throughout the specified parameter family, including positive polynomial modifications. We make no historical-priority claim. Nor does this result assert algebraic independence of different zeros or settle an unrestricted problem for arbitrary rational beta shapes with noninteger sum.

VIII.12 Exact scope and remaining questions

The gaps in the source draft are not all of the same kind. The Poisson and binomial transition gaps are removed by incorporating the complete estimation results and proving the testing consequence. The local Fourier argument supplies a precise replacement for the rapid-decay period assertion. The Franel theorem goes beyond the original central sums to all arithmetic subsequences. The Angelesco and tableau-value theorems are additional arithmetic applications rather than assertions about unnamed classical conjectures.

The following distinctions remain part of the mathematical conclusions.

1. The inequality $|C(t, m, j)| \leq 2^{m^2/t}$ with prefactor exactly one is not proved. The exponent $\log 2$ is optimal when an absolute prefactor is allowed. An exponent $1/2$ is valid in the moderate-degree asymptotic regime but impossible globally with a polynomial prefactor.

2. The mixture theorems use a known support interval and an ordinary unrestricted maximum-likelihood maximizer. They do not establish an unknown-support adaptation theorem. The testing rate is worst-null, not a claim that every fixed null is equally difficult. Arbitrary heterogeneous observation kernels still require a separate normalization or comparison argument; the homogeneous posterior identity cannot simply be applied within each stratum.

3. Recurrence width and differential order are different invariants. The Franel results classify width over rational-function coefficient fields. They do not determine the least possible coefficient degrees of every minimum-width recurrence.

4. The Apéry results concern regular eventual tails. Initial singular indices of an arbitrary chosen polynomial recurrence are not silently filled in. Complex linear combinations can share a limit; injectivity is asserted over subfields of the algebraic numbers.

5. The iterated Fourier statement assumes the specified regularity at the source point. It is not a bound for arbitrary irregular source connections. For Bessel determinant values, absence of exceptional nonzero points is proved for tableaux; it is not asserted for every rational homogeneous form or every scalar determinant realization.

6. For Angelesco polynomials, a noninteger parameter sum, an arbitrary nonpolynomial tilt, or more than two intervals lies outside the theorem. In the nonalgebraic cases the proof gives at least one transcendental coefficient and at least one transcendental zero, not transcendence of every zero.

The finite verification in Section VIII.2 is a proof certificate, not numerical evidence: every comparison uses rational arithmetic and a one-sided Taylor bound. It checks all 130816 pairs with $1 \leq m \leq t < 512$ through four exhaustive branches, including 12734 direct coefficient comparisons. Symbolic checks of representative Angelesco formulas supplement, but do not replace, the all-parameter proofs. The structural and statistical arguments underwent separate internal reviews. These checks improve the manuscript's auditability; they do not constitute external peer review or establish historical priority.

Part IX

Exact Cone Gluing and Collision Tests for Weighted Trace Laws

Manuscript overview. We develop the endpoint analysis of earlier Bessel–Jacobi work into an exact local contour identity. The discrepancy between the two physical germs across a cone vertex is an outward-cone period with a fixed phase. The proof works at resonance without dividing by exponent differences, and without first constructing a differential equation. A polyhedral analyticity argument applies this identity to positive weighted traces with heterogeneous endpoint powers, nonnegative pairwise repulsion exponents, and a positive analytic tilt. Coincident vertex heights are handled by adding their full germs. Further applications give the exact logarithmic and differentiability classification at every ordinary Jacobi trace knot; a global endpoint expansion for sums of independent beta variables; and a finite polynomial criterion for all subset-sum collisions in integer-parameter Jacobi transforms. In the last setting the minimal differential order equals the number of genuine density knots. A three-variable example shows that interaction restores a frequency whose leading corner amplitudes cancel. Classical beta convolutions, contour continuation, and spline theory are explicitly acknowledged; these results are not advertised as the resolution of a newly identified historical conjecture.

IX.1 What is extended, and what is not claimed

The supplied manuscript [032] analyzes Bessel and tableau integrals by their corner expansions and positive Laguerre–Selberg amplitudes. Part I of the subsequent collected volume develops complete endpoint Fourier expansions for Jacobi transforms. Parts II and IV use a published Forrester–Kumar differential system to identify physical connection coefficients, first off resonance and then at every real integrable parameter. The present draft changes the logical order: an exact contour calculation proves the local connection first; Fourier asymptotics then describe its singularity and detect cancellations.

The following distinctions delimit the conclusions.

1. The graphical interaction theorem gives exact local discrepancies at *every* real knot, with no generic-position assumption. For an arbitrary analytic tilt it does not assert global complex continuation, holonomicity, or a full connection matrix for arbitrary bases.

2. The independent-beta formula is global along the real support and uses explicitly specified upper boundary values. Special beta convolution and spline formulas are already classical. The examples illustrate the general normalization and its cancellations, rather than establish priority for those distributions.

3. The finite collision theorem gives an exact differential order for every positive weight vector when the density weight is polynomial. Its proof is a finite endpoint calculation; its relationship with classical box splines is retained rather than suppressed.

4. The ordinary Jacobi regularity theorem is a consequence of the explicit nonzero connection coefficient. It adds a complete local singularity classification to the earlier connection formula.

The known Forrester–Kumar system and integer-parameter connection problem are described in [85]. The earlier local identity can be recovered from the general cone theorem, but none of the present all-parameter contour arguments requires that system.

For Jacobi weights we use exponents $a_i, b_i > -1$ in $x_i^{a_i}(1-x_i)^{b_i}$. In the independent-beta section, the letters $a_i, b_i > 0$ instead denote *shape parameters*, so the corresponding endpoint exponents are $a_i - 1, b_i - 1$. This change is stated again there. An upper continuation passes a real singular point through the upper half-plane with no additional winding. Thus $(-s)^\lambda$ continued from negative to positive s becomes $e^{-\pi i\lambda}s^\lambda$.

IX.2 An exact local gluing identity for analytic cone amplitudes

The argument below isolates the local step needed in a weighted-trace connection formula. It uses elementary contour deformation, rather than a differential equation or a nonresonance assumption. It does not, by itself, prove global continuation of a weighted-trace density. The beta integral and its analytic continuation are classical; see, for example, the defining integral and hypergeometric representation in `https://dlmf.nist.gov/8.17.E1` and `https://dlmf.nist.gov/8.17.E7`. No priority claim is made for the local identity.

IX.2.1 The radial statement

Fix real numbers $\alpha, \eta > 0$ and put $D = \alpha + \eta$. Let $a(r, q)$ be holomorphic on the bidisc $|r|, |q| < R$. Choose $0 < \delta < R/4$ and define, for real $|s| < \delta$,

$$I(s) = \int_{\substack{r,q\geq 0\\ r+q\leq\delta}} \delta_0(s - r + q) r^{\alpha-1} q^{\eta-1} a(r, q)\, dr\, dq. \tag{IX.2.1}$$

Here δ_0 is the Dirac delta. Equivalently, with $Q(s) = (\delta - s)/2$,

$$I_+(s) = \int_0^{Q(s)} (q+s)^{\alpha-1} q^{\eta-1} a(q+s, q)\, dq, \qquad 0 < s < \delta, \tag{IX.2.2}$$

$$I_-(s) = \int_{-s}^{Q(s)} (q+s)^{\alpha-1} q^{\eta-1} a(q+s, q)\, dq, \qquad -\delta < s < 0. \tag{IX.2.3}$$

These are real-analytic on their respective open intervals. Let $I_-^{\uparrow}$ denote continuation of the negative-side germ to positive s through the upper half-plane, with no additional winding around zero. Thus the continuation of $(-s)^\lambda$ is $e^{-\pi i\lambda} s^\lambda$.

Theorem IX.2.1 (Exact cone gluing)**.** *The continuation just described exists for sufficiently small $|s| > 0$. For positive s sufficiently close to zero,*

$$\boxed{I_+(s) - I_-^{\uparrow}(s) = e^{-\pi i\eta} F(s), \qquad F(s) = s^{D-1} \int_0^1 (1-v)^{\alpha-1} v^{\eta-1} a\big(s(1-v), -sv\big)\, dv.} \tag{IX.2.4}$$

There is no restriction on whether D is an integer. The analytic factor $s^{1-D}F(s)$ is holomorphic at zero. The same identity holds for complex α, η with positive real parts, using the continued positive-axis branches.

Proof. For $s \neq 0$, the two branch points of the integrand in the q-plane are $-s$ and 0. Start with negative real s, with the integration path in (IX.2.3) and positive real values of its powers. As s moves through the upper half-plane, its endpoint $-s$ moves through the lower half-plane. Transport the integration contour continuously without crossing either moving branch point. The upper endpoint $Q(s)$ remains near $\delta/2$ and causes no obstruction. All contours can be kept in a fixed small bidisc in the variables $(q+s, q)$. This defines holomorphic continuation locally along the path: after straightening a small segment at a moving endpoint, its integrand is a holomorphic bounded factor times a fixed integrable power. Compact subarcs away from the endpoints present only ordinary holomorphic parameter integrals.

For positive real s, the transported contour can be deformed to the segment from $-s$ to 0 approached below, followed by the segment from 0 to $Q(s)$. In making this deformation, small arcs about $-s$ and 0 give vanishing contributions, respectively $O(\varepsilon^{\operatorname{Re}\alpha})$ and $O(\varepsilon^{\operatorname{Re}\eta})$, for each fixed $s \neq 0$. On the first segment $q + s$ is positive, while the continued argument of q is $-\pi$. Consequently

$$I_-^{\uparrow}(s) = I_+(s) + e^{-\pi i(\eta-1)} \int_{-s}^{0} (q+s)^{\alpha-1} (-q)^{\eta-1} a(q+s, q)\, dq.$$

The substitution $q = -sv$ turns the final integral into $F(s)$. Since $e^{-\pi i(\eta-1)} = -e^{-\pi i\eta}$, this proves (IX.2.4). Uniform convergence of the integral defining its analytic factor proves holomorphy at zero. The argument only requires positive real parts of the two exponents, so also proves the final assertion. In particular, no division by $\sin(\pi D)$ and no limit through nonresonant parameters is required. □

Remark IX.2.2 (Empty clusters)*.* If the negative cluster is absent, replace its radial measure by a unit point mass at zero. The physical left germ is zero and the jump is simply the positive-cone density $F(s)$; the phase is one. If the positive cluster is absent, the physical right germ is zero and the same calculation gives $-I_-^{\uparrow} = e^{-\pi i\eta} F$. These cases correspond formally to $\eta = 0$ or $\alpha = 0$, but are not obtained by inserting zero into the beta integral. The case in which both clusters are absent is a point mass, not a density, and is excluded.

IX.2.2 Homogeneous cone measures and the simplex normalization

Let K_+, K_- be cones with positive linear radial functions $r(u)$ and $q(v)$, whose sections $r = 1$ and $q = 1$ are compact. Assume their measures have the polar decompositions

$$d\mu_+(u) = r^{\alpha-1} dr\, d\nu_+(\omega), \quad u = r\omega, \qquad d\mu_-(v) = q^{\eta-1} dq\, d\nu_-(\zeta), \quad v = q\zeta,$$

where $\nu_\pm$ are finite measures. Let $A(u, v)$ be analytic under complex radial scaling near the origin, uniformly on these compact angular sections. More precisely, assume $A(r\omega, q\zeta)$ is holomorphic for $|r|, |q| < R$ and is uniformly bounded on each smaller bidisc. Then

$$a(r, q) = \int A(r\omega, q\zeta)\, d\nu_+(\omega) d\nu_-(\zeta)$$

is holomorphic by dominated integration. Applying Theorem IX.2.1 to this function proves the identity for the pushforward of $A(u, v)d\mu_+(u)d\mu_-(v)$ by $r(u) - q(v)$, truncated by $r(u) + q(v) \le \delta$.

Its canonical outward-cone germ is

$$F(s) = s^{D-1} \int_{r(u)+q(v)=1} A(su, -sv)\, d\sigma_\mu(u, v). \tag{IX.2.5}$$

The measure in this formula is explicitly defined by

$$\int B\, d\sigma_\mu = \int_0^1 \int B((1 - v)\omega, v\zeta)(1 - v)^{\alpha-1} v^{\eta-1}\, d\nu_+(\omega) d\nu_-(\zeta)\, dv.$$

Thus there is no Euclidean surface-measure ambiguity. If $A(0, 0) = 1$, the positive normalization constant is

$$Z = \nu_+(K_+ \cap \{r = 1\})\nu_-(K_- \cap \{q = 1\}) \frac{\Gamma(\alpha)\Gamma(\eta)}{\Gamma(D)}, \qquad F(s) = Zs^{D-1}U(s), \quad U(0) = 1. \tag{IX.2.6}$$

Consequently the coefficient of the normalized corner germ $s^{D-1}U(s)$ is $e^{-\pi i\eta}Z$.

IX.2.3 Every Taylor coefficient and the complete Fourier symbol

Write the convergent expansion $a(r, q) = \sum_{k,l\ge0} a_{kl} r^k q^l$. Termwise integration gives the convergent series

$$F(s) = s^{D-1} \sum_{j\ge0} f_j s^j, \qquad f_j = \sum_{k+l=j} (-1)^l a_{kl} \frac{\Gamma(\alpha + k)\Gamma(\eta + l)}{\Gamma(D + j)}. \tag{IX.2.7}$$

This is the all-coefficient content of the jump formula. In particular, it is the replacement $v \mapsto -v$ in the analytic amplitude that produces the sign $(-1)^l$; using $A(su, sv)$ would be incorrect.

For a smooth cutoff χ equal to one near zero and supported in a sufficiently small interval, the local Fourier transform has expansion

$$\int e^{iRs}\chi(s)I(s)\, ds \sim \sum_{j\ge0} \Gamma(D + j)e^{\pi i(\alpha-\eta+j)/2} f_j R^{-D-j}, \qquad R \to +\infty. \tag{IX.2.8}$$

Indeed, the monomial $r^k q^l$ contributes

$$a_{kl}\Gamma(\alpha + k)\Gamma(\eta + l)e^{\pi i(\alpha+k-\eta-l)/2} R^{-D-k-l}.$$

Summing over $k + l = j$ gives (IX.2.8). Taylor remainders give a complete expansion by the one-dimensional endpoint argument in each radial variable. The truncation boundary $r + q = \delta$ causes no other local contribution: its phase $r - q$ has nonzero tangential derivative, and the cutoff removes its endpoints $s = \pm\delta$. Repeated integration by parts there gives arbitrary decay.

IX.2.4 Collisions, resonance, and the exact scope of gluing

Corollary IX.2.3 (Several corners at one knot)**.** *Suppose a density near a knot τ is the sum of finitely many local cone densities $I_h(t-\tau)$ of the preceding form and a single function $B(t)$ holomorphic through τ. Different corners may have different α_h, η_h, and their values $D_h = \alpha_h + \eta_h$ may coincide or differ by integers. Then*

$$\rho_+(\tau+s) - \rho_-^{\uparrow}(\tau+s) = \sum_h e^{-\pi i \eta_h} F_h(s). \tag{IX.2.9}$$

No distinctness or nonresonance condition is required.

Proof. Continue each of the finitely many identities (IX.2.4); the continuation of B has no discrepancy. □

The condition on B is essential for this exact conclusion. When the D_h are nonintegral and a prior argument gives decompositions into holomorphic backgrounds and convergent noninteger-power series, a complete Fourier expansion containing only the noninteger scales forces the two backgrounds to agree: the first unmatched Taylor coefficient would create an integer Fourier power. If several D_h agree modulo integers, Fourier matching identifies only their combined coefficients. Their separately normalized contributions are still supplied by the individual cone integrals in (IX.2.9). When a D_h is integral, integer Fourier powers cannot separate a background mismatch from that corner's contribution. The direct contour argument remains valid regardless.

One cannot infer exact analytic gluing from full Fourier asymptotics alone. For example, let χ be a compactly supported smooth cutoff equal to one near zero and set

$$h(s) = \begin{cases} \chi(s) e^{-1/s^2}, & s > 0, \\ 0, & s \le 0. \end{cases}$$

This is smooth, and its Fourier transform decays faster than every inverse power. Adding it changes no coefficient in any complete local Fourier power expansion. Yet its right germ is nonzero, whereas its left germ and every continuation of that zero germ are zero. Thus a smooth localization proves a singular expansion only modulo a smooth remainder, unless analyticity or an appropriate finite-dimensional differential system is established separately.

Changing the radial truncation from δ to another nearby positive value changes I by a function holomorphic for small s: the difference of (IX.2.2) is an integral over a segment bounded away from both moving branch points, with holomorphic endpoints. Hence the corner jump is independent of this particular analytic truncation.

IX.2.5 Application data for a weighted Jacobi corner

For $c_i > 0$, $a_i, b_i > -1$, and $\beta \ge 0$, consider the weight

$$|\Delta(x)|^{\beta} \prod_{i=1}^{N} x_i^{a_i} (1-x_i)^{b_i}, \qquad t = \sum_{i=1}^{N} c_i x_i.$$

At the corner indexed by $S \subseteq \{1, \dots, N\}$, put $x_i = u_i$ for $i \notin S$ and $x_j = 1 - v_j$ for $j \in S$. The knot and radial functions are

$$\tau_S = \sum_{j \in S} c_j, \quad r = \sum_{i \notin S} c_i u_i, \quad q = \sum_{j \in S} c_j v_j,$$

and the two homogeneity parameters are

$$\alpha_S = \sum_{i\notin S}(a_i+1) + \beta\binom{N-|S|}{2},$$
$$\eta_S = \sum_{j\in S}(b_j+1) + \beta\binom{|S|}{2}.$$

The homogeneous measures are

$$|\Delta(u)|^\beta \prod_{i\notin S} u_i^{a_i}\,du, \qquad |\Delta(v)|^\beta \prod_{j\in S} v_j^{b_j}\,dv.$$

Their angular measures are finite: the positive c_i give compact simplexes, endpoint powers are integrable, and $\beta \geq 0$ causes no collision blow-up. The analytic amplitude is

$$A_S(u,v) = \prod_{i\notin S}(1-u_i)^{b_i} \prod_{j\in S}(1-v_j)^{a_j} \prod_{\substack{i\notin S\\ j\in S}}(1-u_i-v_j)^\beta.$$

Every power uses its branch equal to one at the corner. Thus the local outward germ is obtained by replacing this amplitude by

$$A_S(su,-sv) = \prod_{i\notin S}(1-su_i)^{b_i} \prod_{j\in S}(1+sv_j)^{a_j} \prod_{\substack{i\notin S\\ j\in S}}(1+s(v_j-u_i))^\beta.$$

The exact local phase is $e^{-\pi i\eta_S}$. Its normalization is the weighted-simplex integral of the two homogeneous measures; there need not be a Selberg product for unequal c_i, a_i, b_i. If several subset sums τ_S coincide, their contributions add as in Corollary IX.2.3.

The next argument proves the required analyticity of the complement. It also allows nonuniform interaction exponents and an analytic tilt. Global continuation along arbitrary complex paths is a separate issue and is not asserted.

IX.2.6 Analyticity of regular polytope slices

Lemma IX.2.4 (Analytic slices)**.** *Let $P \subset \mathbb{R}^N$ be a full-dimensional compact polytope and let $L \neq 0$ be a linear functional. Suppose t_0 is not the value of L at any vertex of P. Let $\ell_1, \dots, \ell_M$ be affine functions nonnegative on P, let e_j be real, and let H be holomorphic near P. If*

$$\int_{P\cap\{L=t_0\}} \prod_j \ell_j(x)^{e_j}\,\frac{dS(x)}{|\nabla L|} < \infty,$$

then the density

$$J(t) = \int_{P\cap\{L=t\}} H(x)\prod_j \ell_j(x)^{e_j},\frac{dS(x)}{|\nabla L|}$$

has a holomorphic germ at t_0. Factors identically zero on P are excluded; zero exponents may be omitted.

Proof. The assertion is immediate when the slice is empty. Otherwise, in a sufficiently small real interval about t_0, the edges of P crossed by $L = t$ are fixed. Each slice vertex is the intersection of one such edge with $L = t$, so is an affine function of t. The face incidences of the slice are fixed on this interval. Fix an ordering of its vertices and use the resulting pulling triangulation for all these slices. Its nondegenerate simplices vary affinely in their vertices and remain nondegenerate

after the interval is shortened. Their intersections have measure zero and hence do not affect the integral.

Choose fixed linear coordinates on $\ker L$. On each simplex the map from a fixed standard simplex has a Jacobian determinant holomorphic in t, nonzero at t_0, and of fixed real sign near t_0. The density factor from the coarea formula is a fixed constant. Write its vertices as $v_0(t), \dots, v_{N-1}(t)$. For barycentric coordinates $\lambda_k \ge 0$, $\sum_k \lambda_k = 1$,

$$\ell_j\Big(\sum_k \lambda_k v_k(t)\Big) = \sum_k \lambda_k a_{jk}(t), \qquad a_{jk}(t) = \ell_j(v_k(t)).$$

Every a_{jk} is affine in t. It is either identically zero or strictly positive at t_0: the vertex $v_k(t_0)$ lies in the relative interior of an original edge of P, and a nonnegative affine function can vanish there only if it vanishes on that whole edge. Thus, outside the fixed zero faces, the ratio of the preceding linear form to its value at t_0 is a convex combination of the ratios $a_{jk}(t)/a_{jk}(t_0)$ over the nonzero coefficients. For complex t in a small disc, all these ratios, uniformly in the barycentric coordinates, lie in a fixed disc about one disjoint from zero. Their powers e_j therefore have a common holomorphic branch and a uniform bound. The integrand is dominated by a fixed constant times its integrable absolute weight at t_0. Holomorphic dominated integration on each simplex proves the claim. For $N = 1$, the slice is a point on an interval; the same argument is simply substitution with the constant coarea factor. □

IX.2.7 All-parameter local connections with graphical interactions

Fix $N \ge 1$, numbers $c_i > 0$, $a_i, b_i > -1$, and $\beta_{ij} \ge 0$ for $1 \le i < j \le N$. Let H be holomorphic on a complex neighborhood of $[0,1]^N$, and real and strictly positive on that cube. Put

$$\begin{aligned} w(x) &= H(x) \prod_i x_i^{a_i}(1-x_i)^{b_i} \prod_{i<j} |x_i - x_j|^{\beta_{ij}}, \\ Z &= \int_{[0,1]^N} w(x)\,dx > 0, \qquad L(x) = \sum_i c_i x_i, \\ \rho(t) &= Z^{-1} \int_{[0,1]^N} \delta_0(t - L(x)) w(x)\,dx. \end{aligned} \tag{IX.2.10}$$

For a subset S, let v_S be the cube vertex whose coordinates in S are one, and write $T = S^c$ and $\tau_S = L(v_S)$. Define

$$\begin{aligned} \alpha_S &= \sum_{i \in T} (a_i + 1) + \sum_{\substack{i<j \\ i,j \in T}} \beta_{ij}, \\ \eta_S &= \sum_{i \in S} (b_i + 1) + \sum_{\substack{i<j \\ i,j \in S}} \beta_{ij}, \qquad D_S = \alpha_S + \eta_S. \end{aligned}$$

An empty sum is zero. The two weighted Laplace integrals are

$$\begin{aligned} W_T^- &= \int_{[0,\infty)^T} e^{-\sum_{i \in T} c_i u_i} \prod_{i \in T} u_i^{a_i} \prod_{\substack{i<j \\ i,j \in T}} |u_i - u_j|^{\beta_{ij}}\,du, \\ W_S^+ &= \int_{[0,\infty)^S} e^{-\sum_{j \in S} c_j v_j} \prod_{j \in S} v_j^{b_j} \prod_{\substack{i<j \\ i,j \in S}} |v_i - v_j|^{\beta_{ij}}\,dv. \end{aligned}$$

Empty integrals are one. These integrals are finite and positive: endpoint powers are integrable, collision factors are locally bounded, and the exponential controls polynomial growth at infinity. Set

$$Z_S = \frac{W_T^- W_S^+}{\Gamma(D_S)}.$$

On the simplex $\sum_{i\in T} c_i u_i + \sum_{j\in S} c_j v_j = 1$, let $d\sigma_S$ be the delta measure multiplied by

$$\prod_{i\in T} u_i^{a_i} \prod_{j\in S} v_j^{b_j} \prod_{\substack{i<j\\ i,j\in T}} |u_i-u_j|^{\beta_{ij}} \prod_{\substack{i<j\\ i,j\in S}} |v_i-v_j|^{\beta_{ij}}.$$

Its total mass is Z_S by the Laplace radial decomposition. This identity also covers an empty cluster without introducing $\Gamma(0)$. For $N=1$ the delta measure is a point mass of size $1/c_1$ multiplied by the displayed homogeneous weight. Define $x_S(s;u,v)$ by $x_i = su_i$ for $i\in T$ and $x_j = 1+sv_j$ for $j\in S$, and put

$$U_S(s) = \frac{1}{Z_S}\int \frac{H(x_S(s;u,v))}{H(v_S)} \prod_{i\in T}(1-su_i)^{b_i} \prod_{j\in S}(1+sv_j)^{a_j} \cdot \prod_{\substack{i\in T\\ j\in S}} (1+s(v_j-u_i))^{\beta_{\min(i,j),\max(i,j)}}\, d\sigma_S(u,v), \tag{IX.2.11}$$

$$F_S(\tau_S+s) = s^{D_S-1}U_S(s), \qquad s>0. \tag{IX.2.12}$$

All analytic factors use their branches equal to one at $s=0$. The simplex is compact, its weight is integrable, and H extends past the corner. Therefore U_S is holomorphic near zero and $U_S(0)=1$.

Theorem IX.2.5 (Weighted graphical Jacobi local connections)**.** *The density* (IX.2.10) *is real-analytic away from the finite set of distinct subset sums* $\{\tau_S\}$*. At every such knot* τ*, its left germ continues to its right through a sufficiently small upper semicircle. On the right of the knot,*

$$\boxed{\rho_+(\tau+s) - \rho_-^{\uparrow}(\tau+s) = \sum_{S:\,\tau_S=\tau} e^{-\pi i\eta_S} \frac{H(v_S)W_T^- W_S^+}{Z\Gamma(D_S)} F_S(\tau+s), \qquad 0<s\ll 1.} \tag{IX.2.13}$$

The exterior physical density is zero at the two extreme endpoints. The formula includes every resonance and every collision among the subset sums. It determines physical local discrepancies in the specified normalization; it does not assert a full connection matrix between arbitrary bases of solutions of a differential system.

Proof. Divide the cube into its $N!$ closed order chambers. Each is a simplex whose vertices are a nested chain of cube vertices. On each chamber every absolute difference is a nonnegative affine function after choosing its sign. The only vertex heights are subset sums. At a height avoiding them, Lemma IX.2.4 therefore proves analyticity once the slice weight is known to be integrable.

Here is a direct integrability check, also used below. A compact slice avoiding all cube vertices can be covered by finitely many regions on each of which one coordinate x_i belongs to $[\varepsilon, 1-\varepsilon]$ for a fixed $\varepsilon>0$. Eliminate this coordinate in the equation $L(x)=t$, with delta Jacobian $1/c_i$. Its endpoint factors are bounded. Every interaction factor is bounded on the cube because its exponent is nonnegative. The remaining weight is bounded by a constant times $\prod_{j\neq i} x_j^{a_j}(1-x_j)^{b_j}$, whose integral on the remaining cube is finite. This proves slice integrability.

Fix a knot τ. For every cube vertex v_S with $\tau_S=\tau$, remove the open corner neighborhood

$$P_S(\delta) = \{x\in[0,1]^N : \sum_i c_i|x_i-(v_S)_i| < \delta\}.$$

Choose $\delta>0$ small enough that their closures are disjoint and each corner neighborhood meets only order chambers containing that corner. The latter is possible because there are finitely many closed chambers, and a vertex has positive distance from every chamber not containing it. We may also require $\delta < \min_i c_i$. It may also be chosen so that no new vertex of the complementary chamber polytopes has height τ. To see this explicitly, consider an edge of an order-chamber

simplex incident to v_S. Its other endpoint is nested with v_S, so all changing coordinates move either from zero towards one, or from one towards zero. Consequently, on that edge,

$$|L(x) - \tau_S| = \sum_i c_i |x_i - (v_S)_i|.$$

Its intersection with the new truncation face has height $\tau_S + \delta$ or $\tau_S - \delta$. A cut sufficiently close to a simplex vertex creates only these edge-intersection vertices. Disjointness of the removed corner closures prevents intersections between their truncation faces inside a chamber. The other vertices are original cube vertices with heights different from τ. Thus all complementary chamber polytopes have no vertex at height τ.

Their slice weights are integrable by the preceding check, since the truncated complement contains no cube vertex of height τ. Lemma IX.2.4 shows that their sum is a single holomorphic background through τ. Boundaries between chambers and between truncated pieces have zero slice measure for sufficiently small nonzero $s = t - \tau$ and do not change the decomposition.

In $P_S(\delta)$ use $x_i = u_i$ for $i \in T$ and $x_j = 1 - v_j$ for $j \in S$. Then $t - \tau_S = r - q$ and its truncation is exactly $r + q < \delta$. The two homogeneous measures have parameters α_S, η_S. The analytic amplitude is $H(v_S)$ times the one whose outward value appears in (IX.2.11). Theorem IX.2.1, the normalization (IX.2.6), and the separate empty-cluster calculation therefore give its discrepancy as

$$e^{-\pi i \eta_S} H(v_S) Z_S F_S.$$

Sum over the corners at τ, cancel the common holomorphic background, and divide by the global normalization Z. This gives (IX.2.13). The same argument supplies the asserted local upper continuation, including at the extreme endpoints. In dimension one it reduces directly to the one-sided beta-weight calculation, with delta Jacobian $1/c_1$; no chamber singularities or interactions are present. □

The theorem is local at every knot. For an arbitrary analytic tilt H, its possible singularities away from the original cube depend on the chosen complex extension of H. Consequently this proof does not assert continuation of every F_S along every path in the complement of the subset sums. Such a global assertion requires a separate argument and potentially a restriction on H.

IX.3 Weighted sums of beta variables: exact corner germs and collided knots

Let $m \geq 1$ and let X_i be independent $\mathrm{Beta}(a_i, b_i)$ variables, $a_i, b_i > 0$, and let $T = \sum_{i=1}^m c_i X_i$, where $c_i > 0$. Write $B_i = B(a_i, b_i)$. All assertions below concern real shape parameters; no integrality or nonresonance assumption is imposed. The exact density of two beta variables and their linear combinations is classical (Pham-Gia–Turkkan, 1998). Consequently the density formula below is not presented as a first-discovery claim. Its explicit connection normalization and collision criterion are the points to compare with existing general-sum formulas.

For a subset $S \subseteq \{1, \dots, m\}$ put

$$v_S = \sum_{i \in S} c_i, \quad d_{S,i} = \begin{cases} b_i & i \in S, \\ a_i & i \notin S, \end{cases} \quad e_{S,i} = a_i + b_i - d_{S,i}, \quad \epsilon_{S,i} = \begin{cases} -1 & i \in S, \\ 1 & i \notin S, \end{cases}$$

$$D_S = \sum_i d_{S,i}, \qquad \eta_S = \sum_{i \in S} b_i, \qquad k_S = \frac{\prod_i c_i^{-d_{S,i}} \Gamma(d_{S,i})/B_i}{\Gamma(D_S)}.$$

On the standard simplex $\Delta_{m-1}=\{u_i\ge 0,\ \sum_i u_i=1\}$, use the measure obtained by eliminating the last coordinate. Define

$$U_S(s)=\frac{\Gamma(D_S)}{\prod_i\Gamma(d_{S,i})}\int_{\Delta_{m-1}}\prod_i u_i^{d_{S,i}-1}\,(1-\epsilon_{S,i}su_i/c_i)^{e_{S,i}-1}\,du. \tag{IX.3.1}$$

This is analytic for $|s|<\min_i c_i$, is real on a real neighborhood of zero, and satisfies $U_S(0)=1$. Set $F_S(s)=s^{D_S-1}U_S(s)$, with the positive determination for small $s>0$. The superscript $+$ below means continuation above the positive real singular points.

Theorem IX.3.1 (Global corner representation)**.** *Away from the finite set of subset sums $\{v_S\}$, the density of T is*

$$f_T(t)=\sum_{S:\ v_S<t} e^{-\pi i\eta_S}k_S F_S^{+}(t-v_S). \tag{IX.3.2}$$

The formula holds for every $a_i,b_i,c_i>0$, including coincident subset sums and integral D_S. It also gives zero outside the support $[0,\sum_i c_i]$. No limiting substitution in a formula with sine denominators is used.

The normalized germ has the explicit Lauricella representation

$$U_S(s)=F_B^{(m)}((d_{S,i})_i,(1-e_{S,i})_i;D_S;(\epsilon_{S,i}s/c_i)_i)\,,$$

where the convention for F_B is

$$F_B^{(m)}(\mathbf{p},\mathbf{q};D;\mathbf{z})=\sum_{\mathbf{n}\ge 0}\frac{\prod_i(p_i)_{n_i}(q_i)_{n_i}}{(D)_{|\mathbf{n}|}}\prod_i\frac{z_i^{n_i}}{n_i!}.$$

Proof. The density of c_iX_i on $(0,c_i)$ is $c_i^{-a_i}x^{a_i-1}(1-x/c_i)^{b_i-1}/B_i$. On the positive half-line define its lower-endpoint extension

$$g_{i,0}(x)=\frac{c_i^{-a_i}}{B_i}x^{a_i-1}(1-(x+i0)/c_i)^{b_i-1}$$

and the upper-endpoint function

$$g_{i,1}(s)=\frac{c_i^{-b_i}}{B_i}s^{b_i-1}(1+s/c_i)^{a_i-1}\quad(s>0).$$

These are locally integrable because the shapes are positive. For $x>c_i$, the upper continuation of the first expression equals $e^{-\pi i(b_i-1)}g_{i,1}(x-c_i)$, and hence

$$f_{c_iX_i}(x)=\mathbf{1}_{x>0}g_{i,0}(x)+e^{-\pi ib_i}\mathbf{1}_{x>c_i}g_{i,1}(x-c_i).$$

This is an exact identity of locally integrable functions. Convolve these m identities and expand the finite product. For the term indexed by S, put $x_i=(t-v_S)u_i$. The power and Jacobian give $(t-v_S)^{D_S-1}$ and the remaining simplex integral is exactly $k_SU_S(t-v_S)$.

For $\operatorname{Im}s>0$ the simplex factors have no zeros in its interior, so the same integral is a holomorphic continuation of its germ. Its boundary values give the specified upper continuations; local integrability holds away from subset sums. This proves (IX.3.2). Its support and reality also follow immediately from the original convolution identity. Finally expand each factor in (IX.3.1) in its convergent binomial series and use the Dirichlet integral to obtain the stated F_B series. □

Theorem IX.3.2 (Exact connection at every collided knot)**.** *Fix a subset sum v. The physical density to its right minus the upper continuation of the physical density to its left is, for small $s = t - v > 0$, exactly*

$$J_v(s) = \sum_{S : v_S = v} e^{-\pi i \eta_S} k_S s^{D_S - 1} U_S(s). \tag{IX.3.3}$$

In particular contributions must be summed before deciding whether the knot is a genuine singularity.

Proof. In (IX.3.2), the subsets with $v_S < v$ represent the density on the preceding interval. Continue each of these terms along the same upper path. Immediately to the right of v, precisely the subsets with $v_S = v$ have been added. Their difference is (IX.3.3). This argument does not divide by differences of subset sums or by trigonometric expressions in the shapes. □

IX.3.1 A coefficient criterion for cancellation and regularity

Lemma IX.3.3 (Moderate growth from an elementary radial integral)**.** *The local complex branches of a beta-sum density have moderate growth at every subset sum. Its local singular part is a finite sum of radial corner integrals*

$$I_+(s) = \int_0^{(\delta - s)/2} (q + s)^{\alpha - 1} q^{\eta - 1} A(q + s, q)\, dq, \qquad \alpha, \eta > 0,$$

and their upper continuations, with A holomorphic near the origin. An empty positive or negative cluster instead gives a one-sided power times a holomorphic function.

Proof. Near a corner, set $x_i = u_i$ at a lower endpoint and $x_i = c_i - v_i$ at an upper endpoint. Group their positive distances into $r = \sum u_i$ and $q = \sum v_i$. Dirichlet integration within the two clusters gives the radial powers with $\alpha = \sum_{i \notin S} a_i$ and $\eta = \sum_{i \in S} b_i$; the remaining factors are holomorphic under radial scaling, uniformly on the compact angular simplexes. Truncating by $r + q < \delta$ gives the displayed integral after imposing $s = r - q$.

For completeness the remaining part has a holomorphic density germ. Subdivide each coordinate interval into two short endpoint intervals and a middle interval. Choose their interior division points so that no artificial sum of division endpoints equals the knot under study; there are only finitely many forbidden equalities. In each resulting box, the moving hyperplane has constant incidence with all faces near that knot unless it meets one of the original corner vertices already isolated above. Triangulate its fiber with vertices affine in s. Pullback to fixed simplexes leaves fixed integrable endpoint powers times holomorphic factors bounded uniformly on a complex neighborhood. Dominated integration proves holomorphy. Replacing a small corner box by the radial truncation changes only such a transverse boundary contribution.

It remains to bound the radial integral. Replace its upper endpoint by a fixed small $\delta_0 > 0$; the intervening integral is holomorphic and bounded near zero. Put $\rho = |s|$ and split the real integration segment at 2ρ. On $q \geq 2\rho$, $|q + s| \geq q/2$, so the absolute integral is bounded by

$$C \int_{2\rho}^{\delta_0} q^{\alpha + \eta - 2}\, dq \leq C' \{1 + \rho^{\alpha + \eta - 1} + |\log \rho|\}.$$

On $0 \leq q \leq 2\rho$, scale $q = \rho u$, $s = \rho e^{i\theta}$. The remaining integral

$$\int_0^2 u^{\eta - 1} |u + e^{i\theta}|^{\alpha - 1}\, du$$

is bounded uniformly for $0 \leq \theta \leq \pi$: its possible singularities at $u = 0$ and $u = 1$ are separate and integrable, because $\alpha, \eta > 0$. Thus this part is $O(\rho^{\alpha + \eta - 1})$. Contour continuation of the

negative-side branch differs by an integral from $-s$ to zero; scaling that segment gives the same power bound. Empty-cluster terms satisfy the assertion directly. These estimates establish moderate growth without a differential equation or a nonresonance assumption. □

The following elementary local reconstruction is useful. Suppose a real density, analytic on both punctured sides of a point, has an upper connection jump

$$J(s) = \sum_{\nu} j_\nu s^\nu, \qquad \nu > -1,$$

where the exponents lie in a finite union of unit-spaced progressions and the grouped series converge. Assume its complex branches have moderate growth at the point. Modulo a function analytic through zero, its real local form is the convergent sum of

$$(\operatorname{Re} j_\nu + \cot(\pi\nu)\operatorname{Im} j_\nu)\, s_+^\nu + \frac{\operatorname{Im} j_\nu}{\sin(\pi\nu)}(-s)_+^\nu, \qquad \nu \notin \mathbb{Z}, \tag{IX.3.4}$$

and

$$\operatorname{Re} j_q\, s_+^q + \frac{\operatorname{Im} j_q}{\pi} s^q \log|s|, \qquad q \in \mathbb{Z}_{\geq 0}. \tag{IX.3.5}$$

Here s_+^ν is zero for $s < 0$ and s^ν for $s > 0$. For (IX.3.4), upper continuation takes $(-s)^\nu$ to $e^{-\pi i\nu}s^\nu$; for (IX.3.5), it takes $\log(-s)$ to $\log s - \pi i$. These observations verify the asserted jump term by term. The sine denominators are bounded away from zero within each noninteger progression, so convergence is preserved. Subtract this explicit local form. The remaining upper branch has real boundary values on both sides and zero discrepancy. Schwarz reflection extends it to a holomorphic function on a punctured disk. Moderate growth makes the isolated singularity a pole or a removable singularity. Local integrability on the real line excludes every pole, proving the claim.

Lemma IX.3.3 verifies this hypothesis for the beta convolutions, including the endpoint and one-variable cases.

Expand each $U_S(s) = \sum_{n\geq 0} u_{S,n} s^n$ using Theorem IX.3.1, and group all equal powers in (IX.3.3):

$$j_{v,\nu} = \sum_{\substack{v_S = v,\ n\geq 0 \\ D_S - 1 + n = \nu}} e^{-\pi i \eta_S} k_S u_{S,n}. \tag{IX.3.6}$$

Theorem IX.3.4 (Exact cancellation and regularity)**.** *A real knot is analytic exactly when all coefficients $j_{v,\nu}$ vanish. Otherwise let ν_0 be the first exponent with nonzero coefficient. If $\nu_0 > 0$, the density admits a local C^k representative at that knot for exactly the nonnegative integers $k < \nu_0$. It is not $C^{\lceil \nu_0 \rceil}$. If $\nu_0 \leq 0$, it is not continuous. A nonzero imaginary coefficient at an integer exponent means a logarithmic singularity, whereas a real coefficient there means a one-sided polynomial discrepancy. Moreover its exact local Sobolev threshold is*

$$f_T \in H^s_{\mathrm{loc}} \text{ at } v \quad \Longleftrightarrow \quad s < \nu_0 + \tfrac{1}{2}.$$

If all jump coefficients vanish, it belongs locally to every H^s. This is a coefficient criterion, not a claim of a finite uniform zero-testing algorithm.

Proof. The convergent reconstruction (IX.3.4)–(IX.3.5) and Lemma IX.3.3 give the analytic and C^k assertions. Only nonnegative integers k occur in the C^k assertion; a negative first exponent gives an unbounded one-sided power, while exponent zero gives a jump or a logarithm.

Take a smooth cutoff χ equal to one near v. The complete localized Fourier expansion, in the convention $e^{iR(t-v)}$, is

$$\int e^{iR(t-v)}\chi(t) f_T(t)\, dt \sim \sum_{\nu} \Gamma(\nu+1) e^{i\pi(\nu+1)/2} j_{v,\nu} R^{-\nu-1}, \qquad R \to +\infty.$$

One can verify this termwise in the displayed local reconstruction. Alternatively a monomial $r^k q^l$ in the radial amplitude contributes $\Gamma(\alpha+k)\Gamma(\eta+l)e^{i\pi(\alpha+k-\eta-l)/2}R^{-\alpha-\eta-k-l}$. Its beta-integral jump coefficient has the factor $e^{-\pi i\eta}(-1)^l\Gamma(\alpha+k)\Gamma(\eta+l)/\Gamma(\alpha+\eta+k+l)$, which gives exactly the universal coefficient above. Taylor remainders and integration by parts away from the corner justify the complete expansion. Reality determines the negative-frequency expansion by conjugation. The first nonzero coefficient therefore gives decay of exact magnitude $R^{-\nu_0-1}$ in both directions. The defining weighted Fourier integral for H^s converges exactly when $2s-2\nu_0-2<-1$, proving the threshold, including failure at equality. □

IX.3.2 Exact examples

A completely cancelled subset-sum knot. For three independent uniforms with $c=(1,1,2)$,

$$f_T(t)=\frac{1}{4}\{t_+^2-2(t-1)_+^2+2(t-3)_+^2-(t-4)_+^2\}.$$

The two contributions at $v=2$ are $S=\{3\}$ and $S=\{1,2\}$. They have opposite signs and identical constant normalized germs, so $J_2=0$: the subset sum 2 is not a singularity at all. This uniform special case is the classical spline phenomenon.

Partial cancellation raises differentiability. Let $X_1\sim\mathrm{Beta}(1,1)$ and $X_2\sim\mathrm{Beta}(2,2)$ with $c_1=c_2=1$. Direct convolution gives

$$f_T(t)=\begin{cases}3t^2-2t^3, & 0<t<1,\\ 1-3(t-1)^2+2(t-1)^3, & 1<t<2,\\ 0, & t\notin[0,2].\end{cases}$$

Both corners at 1 have $D_S=3$, but their order-two coefficients cancel. The exact discrepancy is $J_1(s)=4s^3$. Consequently the density is C^2 but not C^3 there; using only the smallest uncombined corner exponent would miss this improvement.

A resonant logarithmic example. For $X_1\sim\mathrm{Beta}(1/2,1/2)$ and $X_2\sim\mathrm{Beta}(3/2,3/2)$, again with unit weights, both corners at 1 have $D_S=2$ and $k_S=4/\pi$. Their phases are $-i$ and i. The two normalized germs satisfy $U_{\{2\}}(s)=U_{\{1\}}(-s)$ and $U'_{\{1\}}(0)=-1/2$. Hence

$$J_1(s)=\frac{4i}{\pi}s^2+O(s^4).$$

The local density therefore has the term $4s^2\log|s|/\pi^2$ and is C^1 but not C^2 at the center. The value at the center is $f_T(1)=8/\pi^2$, directly from convolution.

Scope and provenance. The exact formulas and examples above are proved here, but no claim of historical first discovery is made. In particular, the two-variable density is covered by T. Pham-Gia and N. Turkkan, *Distribution of the linear combination of two general beta variables and applications*, Communications in Statistics–Theory and Methods 27 (1998), 1851–1869, doi:10.1080/03610929808832194. A recent weighted-sum expansion by Lee–Park, AIMS Mathematics 11 (2026), 28348–28382, doi:10.3934/math.20261129, Theorem 3.5, already gives a finite truncated-polynomial density formula for arbitrary positive weights when all variables have shape $\mathrm{Beta}(1,L-1)$ for a common integer $L\geq 2$; that formula permits subset-sum collisions. Their Lauricella F_A expression concerns a subsequent gamma mixture, not arbitrary positive real beta shapes. The all-shape connection normalization and exact collision criterion are therefore the more specific comparison targets; the short literature check does not establish their historical priority.

IX.4 All collided graphical knots: exact singularity criteria

The reconstruction argument in the preceding section depends only on an analytic cone amplitude, integrable homogeneous angular measures, and a holomorphic transverse contribution. It therefore applies to the full graphical family of Theorem IX.2.5, including its analytic tilt. This gives an exact criterion even when the finite polynomial test of the last section is unavailable.

Theorem IX.4.1 (Grouped germs determine local regularity)**.** *Use the notation of Theorem IX.2.5. At a fixed subset-sum knot τ, expand its complete discrepancy as*

$$J_\tau(s) = \sum_{S:\tau_S=\tau} C_S s^{D_S-1} U_S(s) = \sum_\nu j_{\tau,\nu} s^\nu, \qquad C_S = e^{-\pi i \eta_S} \frac{H(v_S) W_{S^c}^- W_S^+}{Z\Gamma(D_S)}.$$

Terms with the same exponent are combined. This is a convergent series whose exponents belong to finitely many translates of the nonnegative integers, and all exponents exceed -1. The density is analytic across τ if and only if every grouped coefficient $j_{\tau,\nu}$ is zero. Otherwise let ν_τ be the least exponent with a nonzero coefficient. Then, for integers $k \geq 0$,

$$\rho \text{ admits a } C^k \text{ representative near } \tau \quad \Longleftrightarrow \quad k < \nu_\tau.$$

For a smooth cutoff χ equal to one near τ and supported sufficiently close to that knot, and any real σ,

$$\chi\rho \in H^\sigma(\mathbb{R}) \quad \Longleftrightarrow \quad \sigma < \nu_\tau + \tfrac{1}{2}.$$

When ν_τ is a nonnegative integer, its nonzero imaginary coefficient is precisely a logarithmic term with coefficient $\operatorname{Im} j_{\tau,\nu_\tau}/\pi$; its real part is the one-sided power discrepancy. At a knot belonging to exactly one corner, $\nu_\tau = D_S - 1$ and that knot is never removable.

Proof. The proof of Lemma IX.3.3 is unchanged for the two homogeneous cone measures in Theorem IX.2.5: their finite angular integrals give a holomorphic bounded radial amplitude, and the radial exponents are $\alpha_S, \eta_S > 0$ for nonempty clusters. Lemma IX.2.4 supplies the common holomorphic remainder. Empty clusters give one-sided powers directly. Thus the moderate-growth hypothesis and the local reconstruction formulas (IX.3.4)–(IX.3.5) apply. They prove the analyticity and classical differentiability assertions.

Alternatively, the exact Fourier relation makes the sharp Sobolev threshold explicit. By (IX.2.8), a coefficient $j_{\tau,\nu}$ of the connection germ contributes

$$e^{i\tau R}\Gamma(\nu+1)e^{\pi i(\nu+1)/2} j_{\tau,\nu} R^{-\nu-1}$$

to the localized positive-frequency Fourier expansion. Indeed $j_{\tau,\nu} = e^{-\pi i \eta_S} f_{S,n}$ and $\nu+1 = D_S+n$ on each corner, so the factor is independent of the corner after equal exponents are combined. The first surviving term is nonzero; reality gives the conjugate behavior at negative frequency. Integrating its squared modulus with weight $(1+R^2)^\sigma$ proves the sharp threshold, including failure at equality. For a single corner $C_S \neq 0$ and $U_S(0) = 1$, so its leading exponent survives. □

The global real-line thresholds follow by a finite partition of unity: take the minimum of ν_τ over the nonremovable knots. There are at least the two support endpoints. The criterion is exact, but for an arbitrary analytic tilt it is a test of a whole convergent germ, not a finite algorithm. The polynomial theorem below supplies a finite test in its specified integer-parameter regime.

IX.5 Resonance and sharp regularity of Jacobi traces

This section refines the all-parameter physical connection theorem. Its additional conclusions are the exact logarithmic term at resonance, its absence criterion, and sharp regularity. These are consequences of the connection theorem and its complete local Fourier expansion. No independent historical-priority claim is made for the regularity of power singularities.

IX.5.1 Notation and inputs

Fix $N \ge 1$, $a, b > -1$, and $\beta \ge 0$. Let

$$\rho(t) = S_N^{-1} \int_{[0,1]^N} \delta\Big(t - \sum_i x_i\Big) |\Delta(x)|^\beta \prod_i x_i^a (1-x_i)^b \, dx,$$
$$S_N = \int_{[0,1]^N} |\Delta(x)|^\beta \prod_i x_i^a (1-x_i)^b \, dx.$$

Extend the density by zero outside $[0, N]$. For a knot $p \in \{0, \dots, N\}$, put $m = N - p$ and

$$\alpha = m(a+1) + \tfrac{\beta}{2} m(m-1), \quad \eta = p(b+1) + \tfrac{\beta}{2} p(p-1), \quad D = \alpha + \eta > 0.$$

Set

$$W_r(q) = \prod_{j=0}^{r-1} \frac{\Gamma(q+1+j\beta/2)\Gamma(1+(j+1)\beta/2)}{\Gamma(1+\beta/2)}, \qquad W_0(q) = 1.$$

$$K = \binom{N}{p} \frac{W_m(a) W_p(b)}{S_N} > 0.$$

These symbols always refer to the fixed knot p.

For completeness, the canonical analytic factor is the following simplex integral. On $\sum_i u_i + \sum_j v_j = 1$, with m lower and p upper coordinates, define

$$w(u,v) = |\Delta(u)|^\beta |\Delta(v)|^\beta \prod_i u_i^a \prod_j v_j^b, \qquad Z = W_m(a) W_p(b) / \Gamma(D).$$

With Dirac-delta simplex measure $d\sigma$, let

$$U(s) = Z^{-1} \int w(u,v) \prod_i (1 - s u_i)^b \prod_j (1 + s v_j)^a \prod_{i,j} (1 + s(v_j - u_i))^\beta \, d\sigma.$$

The branches equal one at zero. The function U is holomorphic for $|s| < 1$, real on the real axis, and $U(0) = 1$. Empty products are one. Write $F(p+s) = s^{D-1} U(s)$, initially for $s > 0$.

Theorem IX.2.5, specialized to equal weights, common endpoint parameters, a complete interaction graph, and analytic factor one, gives the physical upper continuation jump

$$\rho_{\text{right}} - \rho_{\text{left}}^{\uparrow} = CF, \qquad C = K e^{-\pi i \eta} / \Gamma(D). \tag{IX.5.1}$$

The $\binom{N}{p}$ corners at the same knot have identical normalized functions, so their coefficients add to this nonzero value. Equation (IX.2.8) gives, for a smooth cutoff χ supported close to p and equal to one there, the complete expansion

$$\int_{\mathbb{R}} e^{iRt} \chi(t) \rho(t) \, dt \sim e^{ipR} R^{-D} \sum_{\ell \ge 0} A_\ell R^{-\ell}, \quad A_0 = K e^{\pi i (\alpha - \eta)/2} \ne 0, \quad R \longrightarrow +\infty. \tag{IX.5.2}$$

No $\log R$ terms occur. Both formulas include endpoints and $\beta = 0$.

The moderate-growth proof of Lemma IX.3.3 applies unchanged to homogeneous cone measures with integrable angular weights and an analytic amplitude: radial scaling replaces the Dirichlet angular integrals by the two cluster integrals, leaving exactly the same one-dimensional estimates. Here those weights include the within-cluster Vandermonde powers, which are integrable for the stated parameters. The analytic part away from the corners is supplied by Theorem IX.2.5. Thus the local reconstruction following Lemma IX.3.3 applies to the Jacobi density without any differential-system input.

IX.5.2 Exact resonance

Theorem IX.5.1 (Complete local singularity classification)**.** *There is a real analytic germ A at zero with the following properties. If $D \notin \mathbb{Z}$, then, for sufficiently small real $s \neq 0$,*

$$\rho(p+s) = A(s) + \begin{cases} c_+ s^{D-1} U(s), & s > 0, \\ c_- (-s)^{D-1} U(s), & s < 0, \end{cases} \qquad c_+ = \frac{K \sin(\pi\alpha)}{\Gamma(D)\sin(\pi D)}, \qquad c_- = \frac{K \sin(\pi\eta)}{\Gamma(D)\sin(\pi D)}. \tag{IX.5.3}$$

At least one of c_+, c_- is nonzero.

If $D = d \in \mathbb{Z}_{\geq 1}$, put $k = d-1$. Then

$$\boxed{\rho(p+s) = A(s) + L\, s^k U(s) \log|s| + J\, \mathbf{1}_{\{s>0\}} s^k U(s)} \tag{IX.5.4}$$

where

$$L = -\frac{K \sin(\pi\eta)}{\pi\Gamma(d)} = \frac{(-1)^d K \sin(\pi\alpha)}{\pi\Gamma(d)}, \qquad J = \frac{K\cos(\pi\eta)}{\Gamma(d)}. \tag{IX.5.5}$$

The entire logarithmic coefficient is $L s^k U(s)$.

At resonance there is no logarithmic term if and only if α, η are both integers. In that case $J = (-1)^\eta K/\Gamma(d) \neq 0$. No knot is a point of real analyticity.

Proof. Consider first noninteger D. The proposed piecewise fractional-power term in (IX.5.3) has upper continuation discrepancy

$$(c_+ - c_- e^{-\pi i(D-1)}) s^{D-1} U(s) = C s^{D-1} U(s),$$

as follows by taking real and imaginary parts and using $D = \alpha + \eta$. Its coefficients are real and at least one is nonzero: their simultaneous vanishing would force both α and η to be integers.

At resonance $D = d$, the function $s^k U(s)$ is holomorphic and real on the real axis. Upper continuation takes $\log(-s)$ to $\log s - i\pi$. Therefore the proposed real function

$$Q(s) = L s^k U(s) \log|s| + J \mathbf{1}_{\{s>0\}} s^k U(s)$$

has upper discrepancy $(J + i\pi L) s^k U(s)$. Taking $J = \operatorname{Re} C$ and $L = \operatorname{Im} C/\pi$ gives exactly (IX.5.5) and matches (IX.5.1).

In either case subtract the proposed singular term from the physical density. The remaining upper branch has zero discrepancy and real boundary values on both punctured sides. Schwarz reflection gives a holomorphic function on a punctured disk. The radial moderate-growth bound makes its singularity a pole or a removable singularity. The physical density and the subtracted terms are locally integrable because $D > 0$, so no pole is possible. This proves that the remainder is a common real analytic germ A, exactly as in the local reconstruction following Lemma IX.3.3.

At integer D, $\sin(\pi\eta) = 0$ is equivalent to integrality of both η and α. If $L = 0$, then $J \neq 0$. A nonzero fractional power, logarithm, or analytic jump rules out analyticity across the knot. □

Here an analytic jump means a difference between two analytic germs after removing the common logarithmic term. It need not be a polynomial: integer α, η do not force U to be polynomial.

Corollary IX.5.2 (Renormalized derivative jump)**.** *At resonance, subtract $Ls^k U(s)\log|s|$. The resulting function has matching derivatives below order k; its derivative of order k has right-minus-left jump*

$$k!J = K\cos(\pi\eta).$$

The leading logarithm in the unrenormalized k-th derivative is

$$k!L\log|s| = \frac{(-1)^d K\sin(\pi\alpha)}{\pi}\log|s|.$$

The renormalized function is analytic across the knot precisely when $\cos(\pi\eta) = 0$. For $k = 0$, the derivative jump means the function jump.

IX.5.3 Sharp regularity

Theorem IX.5.3 (Local and global thresholds)**.** *For every integer $j \geq 0$, the density has a C^j representative near p if and only if*

$$j < D_p - 1.$$

For noninteger $D_p > 1$, put $j = \lfloor D_p - 1\rfloor$ and $\theta = D_p - 1 - j$. It is $C^{j,\theta}$, but not $C^{j,\theta'}$ for any $\theta < \theta' \leq 1$. If $D_p = k+1$, $k \geq 1$, it is $C^{k-1,1}$ when $L = 0$. When $L \neq 0$, it is $C^{k-1,\theta}$ for every $0 < \theta < 1$ but not $C^{k-1,1}$.

For every real σ, with a cutoff as above,

$$\chi\rho \in H^\sigma(\mathbb{R}) \quad\Longleftrightarrow\quad \sigma < D_p - \tfrac{1}{2}. \tag{IX.5.6}$$

Here H^σ is the L^2 Bessel-potential space. For the entire zero-extended density,

$$\rho \in H^\sigma(\mathbb{R}) \quad\Longleftrightarrow\quad \sigma < \min_{0\leq p\leq N} D_p - \tfrac{1}{2}. \tag{IX.5.7}$$

Its global C^j threshold is $j < \min_p D_p - 1$. Equality at every stated threshold is excluded.

Proof. Differentiate the local formulas, using $U(0) = 1$. A noninteger power has the stated exact Hölder regularity. The $(k-1)$-st derivative of $s^k\log|s|$ has a nonzero $s\log|s|$ term, which is Hölder of every order below one but not Lipschitz. If $L = 0$, the nonzero J yields a Lipschitz $(k-1)$-st derivative and a jump in derivative k. For $D_p < 1$ the density is unbounded on at least one side. For $D_p = 1$ there is either a logarithmic divergence or a nonzero jump.

By (IX.5.2), the localized Fourier transform has modulus asymptotic to KR^{-D_p} at positive infinity. Reality gives the same modulus at negative infinity. Therefore the Sobolev condition is exactly convergence of $\int_1^\infty R^{2\sigma-2D_p}\,dR$. At equality this integral diverges logarithmically. A smooth partition of unity about the finitely many knots and their complement proves the global sufficient condition. Necessity follows by localizing at a minimizing knot, because multiplication by a smooth cutoff is bounded on H^σ for every real σ. Away from the knots the density is analytic. The classical global threshold follows in the same way. □

The minimizing exponent is explicit:

$$D_p = \beta(p - N/2)^2 + (b-a)(p - N/2) + \frac{N(a+b+2)}{2} + \frac{\beta N(N-2)}{4}.$$

For $\beta > 0$, choose the integer $p \in [0, N]$ nearest to $N/2 + (a-b)/(2\beta)$, clipping to the endpoints. For $\beta = 0$, $\min_p D_p = N\min(a+1, b+1)$. Ties do not alter the threshold.

IX.5.4 Symmetric midpoint

Corollary IX.5.4 (Midpoint parity dichotomy)**.** *Suppose $a = b$, N is even, and $p = N/2$. Then U is even, $\alpha = \eta = D/2$, and the density is even about p. If $D \notin \mathbb{Z}$, its nonanalytic part is*

$$\frac{K}{2\Gamma(D)\cos(\pi D/2)}|s|^{D-1}U(s).$$

If $D = d$ is odd, its nonanalytic part is $Ls^{d-1}U(s)\log|s|$, with $L \neq 0$ and $J = 0$. If $D = d$ is even, its nonanalytic part modulo an even analytic germ is

$$\frac{(-1)^{d/2}K}{2\Gamma(d)}|s|^{d-1}U(s).$$

Thus symmetry never removes an integer midpoint singularity. For odd N, the midpoint is not a knot and the density is analytic there.

Proof. Interchanging the two simplex clusters takes s to $-s$; hence U is even. Reflection $x_i \mapsto 1 - x_i$ proves symmetry of the density. Substitute $\alpha = \eta = D/2$ in the formulas above. For even d, the integer $k = d - 1$ is odd and $\mathbf{1}_{\{s>0\}}s^k = (s^k + |s|^k)/2$. Absorb the analytic odd term into the background; symmetry makes the remaining background even. □

IX.6 Exact collision tests for polynomial Jacobi weights

This section gives a finite exact test at every subset-sum collision for integer endpoint powers and even Dyson index. It also identifies the test with removal of a knot in the trace density. Polynomial densities on slices of cubes belong to the classical spline framework; see [89, 90]. Finite density formulas for positive weighted independent beta variables are also available; for example, [91, Theorem 3.5] treats independent $\mathrm{Beta}(1, \ell - 1)$ variables for integer $\ell \geq 2$ without requiring distinct subset sums. The assertion here is the explicit specialization to Jacobi weights, together with the exact minimal rational differential order and a collision example in which a vanished leading corner term does not remove its frequency. No claim that the underlying polynomial integration or spline representation is new is made.

Let $N \geq 1$, $c_i > 0$, $a_i, b_i, m \in \mathbb{Z}_{\geq 0}$, and put

$$P(u) = \prod_{i<j}(u_j - u_i)^{2m}\prod_{i=1}^{N}u_i^{a_i}(1-u_i)^{b_i}, \qquad D = \sum_i(a_i + b_i) + mN(N-1), \qquad L = N + D.$$

Define the unnormalized transform

$$Z(t) = \int_{[0,1]^N} e^{t\sum_i c_i u_i}P(u)\,du.$$

It is entire and $Z(0) > 0$. Constant normalization by $Z(0)$ has no effect on the differential orders below. For $S \subseteq [N]$, let v_S be its zero-one indicator vector and $\lambda_S = \sum_{i\in S} c_i$. Write $\Lambda = \{\lambda_S : S \subseteq [N]\}$, with repetitions removed. Multi-indices have nonnegative integer components.

Theorem IX.6.1 (Complete finite collision criterion)**.** *For each $\lambda \in \Lambda$ define the polynomial*

$$Q_\lambda(t) = \sum_{\substack{S\subseteq[N]\\ \lambda_S=\lambda}}(-1)^{N-|S|}\sum_{|\alpha|\leq D}\frac{(-1)^{|\alpha|}(\partial^\alpha P)(v_S)}{\prod_i c_i^{\alpha_i+1}}\,t^{D-|\alpha|}. \tag{IX.6.1}$$

Then the following statements hold.

1. *The exact identity*
$$t^L Z(t) = \sum_{\lambda \in \Lambda} e^{\lambda t} Q_\lambda(t) \tag{IX.6.2}$$
holds for every complex t.

2. *The minimal order over* $\mathbb{C}(t)$ *of a nonzero homogeneous linear differential operator annihilating* Z *is*
$$M = \#\{\lambda \in \Lambda : Q_\lambda \not\equiv 0\}. \tag{IX.6.3}$$
In particular, a frequency is removed if and only if, for every $0 \le k \le D$,
$$\sum_{\substack{S \subseteq [N] \\ \lambda_S = \lambda}} (-1)^{N-|S|+k} \sum_{|\alpha|=k} \frac{(\partial^\alpha P)(v_S)}{\prod_i c_i^{\alpha_i+1}} = 0. \tag{IX.6.4}$$

3. *These tests give a finite semialgebraic partition of the positive weight space* $(0,\infty)^N$ *on which the exact order is constant. For algebraic positive weights, they are exact algebraic decision tests.*

Proof. For a polynomial p of degree at most D, repeated integration by parts gives, for $t \neq 0$,
$$\int_0^1 e^{ctu} p(u)\, du = \sum_{k=0}^{D} \frac{(-1)^k}{(ct)^{k+1}} \bigl(e^{ct} p^{(k)}(1) - p^{(k)}(0)\bigr).$$
Apply this identity successively in every coordinate. Derivatives of total order greater than D vanish. Choosing the upper endpoint on S and the lower endpoint on its complement gives exactly the sign and denominator in (IX.6.1). Grouping equal values of λ_S proves (IX.6.2) away from zero. Both sides are entire, so the identity extends to zero.

For each surviving λ, set $y_\lambda = e^{\lambda t} Q_\lambda(t)/t^L$. Its logarithmic derivative is rational. Thus the $\mathbb{C}(t)$-span of the M functions y_λ is closed under differentiation and contains Z; a linear dependence among $Z, Z', \dots, Z^{(M)}$ gives an annihilator of order at most M.

Distinct real exponentials are linearly independent over $\mathbb{C}(t)$. Indeed, after clearing denominators, a purported dependence is a polynomial-exponential sum. Along the positive real axis, divide by its largest exponential with a nonzero polynomial coefficient. A nonzero polynomial cannot equal an exponentially decaying sum, a contradiction. Now suppose a rational operator A annihilates Z. Every Ay_λ is $e^{\lambda t}$ times a rational function. The same independence gives $Ay_\lambda = 0$ for every surviving λ. The functions y_λ are also linearly independent over the constants. On a disk avoiding the poles of the coefficients and the zeros of the leading coefficient, an operator of order r has a solution space of dimension r. Consequently $r \ge M$, proving (IX.6.3).

Coefficient comparison gives (IX.6.4). There are finitely many possible equality patterns among the linear forms λ_S. Fix such a pattern. Each coefficient in (IX.6.1) is a rational function of c with rational coefficients and denominators consisting only of powers of the positive c_i. Multiplication, for example, by $\prod_i c_i^{D+1}$ clears them. Equality or inequality to zero of these finitely many polynomials, together with the subset-sum equality pattern, gives the claimed finite semialgebraic partition. □

Lemma IX.6.2 (An individual corner never disappears)**.** *Let* Q_S *denote the single-*S *summand in* (IX.6.1)*, before colliding frequencies are grouped. Then* $Q_S \neq 0$*. Consequently pairwise distinct subset sums give order* 2^N*.*

Proof. Use local coordinates $u_i = x_i$ on S^c and $u_i = 1 - x_i$ on S. The lowest homogeneous term of P is
$$H(x) = \Delta(x_{S^c})^{2m} \prod_{i \notin S} x_i^{a_i} \Delta(x_S)^{2m} \prod_{i \in S} x_i^{b_i}.$$

It has separate degrees $h_- = \sum_{i\notin S} a_i + m|S^c|(|S^c|-1)$ and $h_+ = \sum_{i\in S} b_i + m|S|(|S|-1)$ in the two clusters. The coefficient of the highest power $t^{D-h_- -h_+}$ in Q_S is therefore

$$(-1)^{N-|S|+h_-}\int_{(0,\infty)^N} e^{-\sum_i c_i x_i} H(x)\,dx.$$

Indeed, each monomial in the local coordinates contributes its factorial moment, and the endpoint changes of sign leave precisely the displayed common sign. The integral is finite and strictly positive. The highest coefficient is consequently nonzero. □

Remark IX.6.3 (Constructing a minimal operator). The theorem is constructive even when the weights are not rational. For the surviving frequencies form the Wronskian of the y_λ. It is nonzero: after factoring out each column's y_λ, its leading determinant at infinity is the Vandermonde in the distinct λ. The monic Wronskian operator with these M solutions has rational coefficients, because exponential factors cancel from its coefficient ratios. It is a minimal operator for Z. Individual y_λ can have poles at zero; the complete sum Z has a removable singularity there. No conclusion about absence of apparent singularities of the scalar operator is needed.

IX.6.1 The same test removes precisely the density knots

Let ρ be the compactly supported density defined by

$$\int_{\mathbb{R}} \varphi(x)\rho(x)\,dx = \int_{[0,1]^N} \varphi\Big(\sum_i c_i u_i\Big) P(u)\,du.$$

Such an integrable density exists by integrating first in u_1, since $c_1 > 0$. It is not normalized; its integral is $Z(0)$.

Theorem IX.6.4 (Differential order equals the number of genuine knots)**.** *The density ρ is a polynomial of degree at most $L-1$ on each component of $\mathbb{R}\setminus\Lambda$. With $\mathcal{D} = d/dx$, its distributional derivative satisfies*

$$\mathcal{D}^L\rho = (-1)^L \sum_{\lambda\in\Lambda} Q_\lambda(-\mathcal{D})\delta_\lambda. \tag{IX.6.5}$$

A candidate knot λ is removable, meaning that one polynomial agrees almost everywhere with ρ on a neighborhood of λ, if and only if $Q_\lambda = 0$. Hence the minimal order in (IX.6.3) *equals the number of genuine knots, including the two support endpoints.*

Proof. For a compactly supported distribution, the transform used here sends $\mathcal{D}$ to multiplication by $-t$ and sends δ_λ to $e^{\lambda t}$. Thus the transforms of both sides of (IX.6.5) agree by (IX.6.2). Injectivity of the Fourier transform on compactly supported distributions proves the identity. On an interval disjoint from Λ, it gives $\mathcal{D}^L\rho = 0$, so ρ is a polynomial distribution of degree less than L.

In a neighborhood containing no other candidate knot, the right side is zero exactly when $Q_\lambda = 0$, since the derivatives of a Dirac mass are linearly independent. Vanishing is equivalent to one polynomial on that neighborhood. Finally, the support is exactly $[0, \sum_i c_i]$: P is positive on the interior away from the diagonals and coordinate endpoints, and every sufficiently small interval inside the image has positive mass. The density is zero outside its support. A polynomial agreeing across either support endpoint would therefore be identically zero near that endpoint, a contradiction. Both endpoints are genuine knots. □

IX.6.2 A leading cancellation that does not remove a frequency

Proposition IX.6.5 (Uniform and interacting three-variable laws)**.** *Let $N = 3$, $a_i = b_i = 0$, and $c = (p, q, p+q)$ with $p, q > 0$. For $\beta = 0$ the exact differential order is six if $p \neq q$ and four if $p = q$. For $\beta = 2$ it is seven if $p \neq q$ and five if $p = q$.*

Proof. For $\beta = 0$, independence gives

$$Z(t) = \frac{(e^{pt}-1)(e^{qt}-1)(e^{(p+q)t}-1)}{pq(p+q)t^3}.$$

The terms at frequency $p+q$ cancel. If $p \neq q$, the remaining frequencies are $0, p, q, 2p+q, p+2q, 2p+2q$, all distinct with nonzero coefficients. If $p = q$, the numerator reduces to $e^{4pt} - 2e^{3pt} + 2e^{pt} - 1$, giving four frequencies.

For $\beta = 2$, $D = 6$ and $L = 9$. The two corners at the central frequency are $(0, 0, 1)$ and $(1, 1, 0)$. Substitution in (IX.6.1) gives

$$Q_{p+q}(t) = \frac{24(p^4+q^4)}{p^4q^4(p+q)^4} t((p+q)^2t^2 + 6), \tag{IX.6.6}$$

which is nonzero. To verify this finite calculation, at the first corner substitute $(u_1, u_2, u_3) = (x, y, 1-z)$ into P:

$$P = (y-x)^2(1-z-x)^2(1-z-y)^2.$$

Expand this polynomial and use (IX.6.1); the coefficients of t^4, t^2, t^0 in the sum of the two central corners are zero, while those of t^3 and t are respectively

$$\frac{24(p^4+q^4)}{p^4q^4(p+q)^2}, \qquad \frac{144(p^4+q^4)}{p^4q^4(p+q)^4}.$$

All other frequencies have one corner if $p \neq q$, and its polynomial is nonzero by Lemma IX.6.2. When $p = q$, the only further collisions pair corners related by interchanging the first two coordinates. Their complete corner polynomials are equal, and hence add rather than cancel. The frequency counts are consequently seven and five. □

For example, the independent uniform laws have

$$Z_{(1,1,2),0}(t) = \frac{e^{4t} - 2e^{3t} + 2e^t - 1}{2t^3}, \qquad Z_{(1,2,3),0}(t) = \frac{e^{6t} - e^{5t} - e^{4t} + e^{2t} + e^t - 1}{6t^3}.$$

Their orders are four and six. At Dyson index two, the central polynomials are respectively $6t(2t^2+3)$ and $17t(3t^2+2)/18$, and the orders increase to five and seven.

The structural reason that a leading cancellation can be misleading is particularly transparent here. Whenever $P(1-u) = P(u)$ and $C = \sum_i c_i$, the substitution $u \mapsto 1-u$ yields

$$Z(t) = e^{Ct}Z(-t), \qquad Q_{C-\lambda}(t) = (-1)^L Q_\lambda(-t).$$

For symmetric integer endpoint powers, D is even. With odd N, the central polynomial must therefore be odd. For independent uniform variables it is constant and hence vanishes; for Δ^2 it can be a nonzero odd polynomial, as (IX.6.6) shows.

Remark IX.6.6 (Scope). The finite criterion covers arbitrary positive weights, all their subset-sum degeneracies, and nonhomogeneous nonnegative integer endpoint powers. It is an exact algorithmic classification, not a short combinatorial list of every possible weight relation. It does not resolve colliding frequencies for arbitrary noninteger endpoint powers or for non-even Dyson index. The elementary polynomial regime is also not evidence for irreducibility or a full differential Galois group.

IX.7 A concrete comparison and the remaining global question

The three-variable family in Proposition IX.6.5 gives a small exact diagnostic of the method. For weights $(1, 2, 3)$ the candidate heights are $0, 1, 2, 3, 4, 5, 6$. Under the independent uniform law, the central height 3 is removable and the transform has minimal order six. Under the normalized Jacobi weight $360 \prod_{i<j}(x_i - x_j)^2$, the central height is genuine and the minimal order is seven. The leading corner contributions at that height still cancel. The complete calculation detects the surviving term.

For the latter law, its central polynomial is

$$Q_3(z) = \frac{17}{18} z(3z^2 + 2), \qquad L = 9, \qquad Z(0) = \frac{1}{360}.$$

Equation (IX.6.5), or direct inverse Laplace transformation, gives the normalized local polynomial discrepancy

$$\rho(3 + s)_{\text{right}} - \rho(3 + s)^{\uparrow}_{\text{left}} = \frac{17}{2} s^5 + \frac{17}{126} s^7.$$

Consequently the density is C^4 but not C^5 at 3; its fifth derivative has right-minus-left jump 1020. Symmetry about 3 allows the entire nonanalytic part to be written

$$\frac{17}{4}|s|^5 + \frac{17}{252}|s|^7$$

modulo an even analytic germ. This contrasts with complete removal under the independent law. The coefficient signs use the convention $Z(z) = \int e^{zt} \rho(t)\, dt$, so a term z^{-2r} corresponds to $t_+^{2r-1}/(2r - 1)!$.

The main question left by this manuscript is a genuinely global one: already for $H = 1$ and arbitrary noninteger repulsion exponents, find an explicit minimal continuation system and an effective finite criterion for complete cancellation at every collided height. Existence of holonomic systems for hypergeometric integrals is a classical input, not a new open claim here. The local formula determines the whole germ, but does not by itself decide whether an arbitrary analytic germ vanishes from finitely many coefficients. In particular, a scalar leading amplitude cannot supply that criterion, as the preceding example proves. A separate task is to determine which of these normalizations and classifications already occur in the extensive literature on Selberg integrals, hypergeometric connections, and splines. Neither task is replaced by the exact calculations here.

Part X

Global Transfers from Recurrences, Periods and Statistical Recovery

Manuscript overview. This note develops six transfers of methods from a supplied collection of 34 manuscripts. It gives exact recurrence widths for multinomial-power sums and all arithmetic subsequences; a classification of algebraic relations among polynomial averages of every regular bounded Duffing orbit at algebraic parameters; spectral pruning and a complete algebraic/transcendental heat-weight classification on complete split graphs; exact two-level Laplacian spectra for arbitrary matching defects; global corner connections for entire interactions; and exact observation thresholds under unknown, time-varying hidden refresh schedules. The proofs and their parameter restrictions are included. Classical Mellin, differential-Galois and elliptic-transcendence inputs are cited. Two explicitly marked additional applications depend on earlier, unpublished results from the supplied collection. The note records new derivations relative to that collection; it does not certify historical priority or the solution of a previously named open conjecture.

Status. Developed with AI assistance and checked by independent internal derivations, adversarial readings and selected exact or numerical checks. This is a research draft, not an externally peer-reviewed publication. Numerical agreement is never substituted for a proof.

Guide to the results and dependencies

The main statements are global within the explicitly specified families. The word "global" does not assert a classification of all graphs, all Hamiltonians, all interactions or all hidden Markov models.

Family	Result proved here	Main input or restriction
Multinomial sums	Exact minimum rational-coefficient recurrence width, including every arithmetic subsequence	Katz's groups and the classical Mellin rank formula
Duffing orbits	Algebraic independence of frequency and the second moment; complete polynomial-average and action classification	Classical Chudnovsky–Philibert independence; regular bounded orbits
Split graphs	Universal inverse-power pruning; exact heat transition; algebraic/transcendental classification of optimal heat weights	Convexity, explicit spectra and Lindemann–Weierstrass
Matching defects	Exact realization and strictness of the full two-level spectral slice for all matching sizes	Explicit Parseval frames; $n \geq 2r + 1$
Corner connections	Complete density connection for entire interactions, including resonances and coincident knots	Endpoint cancellation; exact order has stronger hypotheses
Hidden refresh	Exact thresholds and an all-order inverse with unknown positive, unequal retentions	Known injective emission kernel; population-law identification

Earlier unpublished inputs. The optional banana *coefficient-degree* corollary uses the all-mass differential-order theorem in manuscript [032]; the minimum recurrence-width proof does not use that theorem. The unrestricted Weibull/stable application uses the two-view injectivity theorem in [030]. Those two prior theorems are identified as inputs rather than silently re-proved. The general refresh threshold theorem and its explicit circle example are proved here. The perfect-matching spectral case, arbitrary Dyson exponent, arbitrary hidden transitions, and the universal graph-design leading constant remain outside the conclusions.

X.1 Exact recurrence widths from Mellin rank

This section supplies two extensions not contained in the preceding manuscripts: the removal of every threshold restriction from the banana coefficient-width formula, and an exact all-subsequence classification for powers of multinomial coefficients with arbitrarily many categories. The Mellin rank theorem and the differential Galois groups used below are published inputs, not new results of this work. Historical priority for the applications is not asserted.

X.1.1 A coefficient form of the Mellin rank theorem

Write $\theta = x\partial_x$ and $\mathcal{D} = \mathbb{C}[x, x^{-1}]\langle\theta\rangle$. For a sequence $v = (v_n)$, its minimum width is the least L for which a nonzero relation $\sum_{j=0}^{L} p_j(n) v_{n+j} = 0$, $p_j \in \mathbb{C}(n)$, holds eventually, with both extreme coefficients nonzero. Multiplication by a nonzero sequence h_n with $h_{n+1}/h_n \in \mathbb{C}(n)^*$ preserves this invariant.

We use the following precise classical rank statement. For a holonomic $\mathcal{D}$-module M which is a connection everywhere on $\mathbb{G}_m$, the algebraic Mellin transform has generic rank

$$\dim_{\mathbb{C}(\theta)}\bigl(\mathbb{C}(\theta) \otimes_{\mathbb{C}[\theta]} M\bigr) = \operatorname{irr}_0(M) + \operatorname{irr}_\infty(M). \tag{X.1.1}$$

This is the case with no finite nonzero singularities of the rank formula in the proof of García López's Theorem 4, p. 504 [93]. It also follows from the Loeser–Sabbah rank theorem, with the shifted de Rham convention in [94, Theorem 35, equations (3.5)–(3.7), Appendix B], and the irregular de Rham index formula. In that convention $\chi = \dim H^1_{\mathrm{dR}} - \dim H^0_{\mathrm{dR}}$ on a curve; no unshifted Euler-characteristic sign is being suppressed.

Lemma X.1.1 (Coefficient width and Mellin rank)**.** *Let $g(x) = \sum_{n\ge 0} v_n x^n$ be entire. Suppose $M = \mathcal{D}g$ is a semisimple algebraic connection on all of $\mathbb{G}_m$, and no simple summand is the trivial connection. Then the minimum eventual width of (v_n) is the quantity in* (X.1.1)*.*

Proof. Every element of M is represented by a convergent Laurent series with only finitely many negative powers. Map it to the tail of its coefficient sequence. The action of θ becomes multiplication by n, while x becomes the backward shift $v_n \mapsto v_{n-1}$. The kernel consists of Laurent polynomials belonging to M. If a nonzero Laurent polynomial belongs to M, a polynomial in θ isolates one of its monomials, and multiplication by a power of x gives 1. Thus $\mathbb{C}[x, x^{-1}]$ is a trivial $\mathcal{D}$-submodule of M, contradicting the hypothesis. The coefficient map is therefore injective.

Tail sequences form a vector space over $\mathbb{C}(n)$: poles and zeros of any nonzero rational function affect only finitely many indices. Consequently localization of the coefficient map identifies $\mathbb{C}(\theta) \otimes M$ with the rational span of all integer shifts of (v_n). To see injectivity explicitly, a fraction represented by $h/q(\theta)$ maps to zero only if the coefficients of h are zero eventually, and hence only if $h = 0$ by the preceding paragraph. Surjectivity onto the shift span follows because $\mathcal{D}$ is generated by x, x^{-1}, θ.

For a cyclic shift module of finite dimension R, the minimum recurrence width is R. Indeed, $R+1$ consecutive shifts are dependent. Conversely, a relation of width L with nonzero extremes expresses both the next and the preceding shift through L consecutive shifts. Semilinearity then shows that all integer shifts lie in their span, so $R \le L$. Apply (X.1.1). □

X.1.2 Weighted higher-Bessel products

For $r \ge 2$ put

$$\mathcal{F}_r(x) = \sum_{n\ge 0} \frac{x^n}{(n!)^r}, \qquad (\theta^r - x)\mathcal{F}_r = 0.$$

Choose distinct nonzero $a_1, \dots, a_s \in \mathbb{C}$ and positive integers $h_1, \dots, h_s$. When r is odd impose also $a_i \neq -a_j$ for $i \neq j$. Let $H = \sum h_j$ and define

$$G(x) = \prod_{j=1}^{s} \mathcal{F}_r(a_j x)^{h_j} = \sum_{n \geq 0} \frac{A(n)}{(n!)^r} x^n.$$

Thus $A(n)$ is the sum of rth powers of H-category multinomial coefficients, with each scale a_j repeated h_j times. Choose $\alpha_j^r = a_j$ and a primitive rth root ζ. Set

$$D = \prod_{j=1}^{s} \binom{h_j + r - 1}{r - 1},$$

and let Z be the number of arrays of nonnegative integers $(k_{j,\ell})_{1 \leq j \leq s,\, 0 \leq \ell < r}$ satisfying

$$\sum_{\ell=0}^{r-1} k_{j,\ell} = h_j \quad (1 \leq j \leq s), \qquad \sum_{j=1}^{s} \alpha_j \sum_{\ell=0}^{r-1} k_{j,\ell} \zeta^{\ell} = 0. \tag{X.1.2}$$

The count is independent of the chosen roots α_j.

Theorem X.1.2 (Exact weighted multinomial width)**.** *Under these hypotheses, the minimum eventual recurrence width of $A(n)$ is*

$$\boxed{\frac{D - Z}{r}}.$$

No hypothesis of distinct nonzero formal exponential coefficients is required. Equal nonzero coefficients are counted with their multiplicities; only zero coefficients are subtracted. The same minimum holds over every coefficient field $K(n)$ with $\mathbb{Q}(a_1, \dots, a_s) \subseteq K \subseteq \mathbb{C}$.

Proof. Katz's calculation [92, Corollary 4.4.8] gives differential Galois group SL_r when r is odd and Sp_r when r is even for $\theta^r - ax$. These groups are connected. After $x = t^r$, formal solutions at infinity have the form

$$e^{r\alpha\zeta^{\ell} t}\, t^{-(r-1)/2}(1 + O(t^{-1})), \qquad 0 \leq \ell < r.$$

Substitution in the equation gives the power and the recursively determined formal series; distinct leading exponential roots separate the factors. The dual module at scale a is the module at scale $(-1)^r a$.

The connected joint group of the scale modules is their full product. For completeness, its Lie algebra is a subdirect product of the simple Lie algebras of the individual groups. A proper subdirect product would, by Goursat's lemma, identify two factors. The automorphisms of $\mathfrak{sp}_r$ are inner; those of $\mathfrak{sl}_r$ are inner or standard duality. Initially the identification may be projective: two standard modules can differ by a rank-one character, possibly after duality. Both determinant modules are trivial, so the rth power of that character is trivial. The character therefore has finite image and is killed by a finite algebraic extension. Passing also to the extension corresponding to the identity component, two scale modules would consequently be isomorphic or dual. An isomorphism preserves their formal exponential multisets, even after further ramification. Taking rth powers of their coefficients forces respectively $a_i = a_j$ or $a_i = (-1)^r a_j$, excluded by the hypotheses. Hence the joint connected group is the product.

Each Sym^{h_j} of the standard representation is irreducible for the indicated group, and their external tensor product is irreducible of dimension D. Let $\mathcal{E}$ be the tensor product of these symmetric-power cyclic connections, using the convention in which a scalar solution is a $\mathcal{D}$-linear evaluation map into analytic functions. Evaluating each cyclic vector at $\mathcal{F}_r(a_j x)$ and multiplying defines a nonzero $\mathcal{D}$-linear morphism from $\mathcal{E}$ into functions on the universal cover of $\mathbb{G}_m$; the

tensor cyclic section evaluates to G, and $G(0) = 1$. If the dual convention for differential modules is used, replace $\mathcal{E}$ by its dual throughout; irreducibility, rank, and irregularities are unchanged.

The everywhere nonsingular connection $\mathcal{E}$ is simple as a $\mathcal{D}$-module: an irreducible connection has no positive-rank proper connection subobject, and a nonsingular connection has no submodule or quotient supported at a point. Thus the nonzero evaluation map is injective, and the nonzero tensor cyclic section generates all of $\mathcal{E}$. It follows that $\mathcal{D}G$ is exactly this irreducible connection, globally on $\mathbb{G}_m$, with no punctual submodule or quotient. In particular its generic rank is D and its solution space is the injective multiplication image of the tensor solution representation. This last point matters: using an arbitrary scalar presentation with apparent singularities would not justify (X.1.1) with an empty finite singular set.

The connection is regular singular at zero. On the t-cover its formal factors are indexed exactly by the arrays in (X.1.2), with exponential coefficients $r\sum_j \alpha_j \sum_\ell k_{j,\ell}\zeta^\ell$. Every nonzero factor has slope one in t, hence slope $1/r$ in x. A zero coefficient leaves a regular rank-one factor. The formal splitting exists before equal coefficients are grouped, so collisions introduce no extra extension. Therefore

$$\operatorname{irr}_0 = 0, \qquad \operatorname{irr}_\infty = (D - Z)/r.$$

It is positive: choose a real linear functional nonzero on every finite root set and maximize it separately in each group; the resulting total maximum has positive projection. Lemma X.1.1, followed by factorial rescaling, proves the width formula.

For descent, a recurrence supplied over $\mathbb{C}$ has some finite coefficient-degree bound. For that fixed degree and width, its coefficient identities form a linear system over $\mathbb{Q}(a_1, \dots, a_s)$. The row space has a finite basis over that field, so a nontrivial complex kernel implies a nontrivial kernel over the field. The complex lower bound prevents the extreme terms of a descended minimum-width relation from vanishing. This proves the field assertion. □

X.1.3 All complex banana masses, including zero thresholds

Let $m_1, \dots, m_N \in \mathbb{C}^*$, group their distinct squared masses as $\mu_1^2, \dots, \mu_s^2$ with multiplicities h_j, and put

$$D = \prod_j (h_j + 1), \qquad Z_0 = \#\left\{ (\ell_j) : \ell_j \in \{h_j, h_j - 2, \dots, -h_j\}, \quad \sum_j \ell_j \mu_j = 0 \right\}.$$

Corollary X.1.3 (The complete banana coefficient-width formula)**.** *For*

$$c_n(m) = \sum_{k_1 + \cdots + k_N = n} \binom{n}{k_1, \dots, k_N}^2 \prod_i m_i^{2k_i},$$

the minimum eventual width is $(D - Z_0)/2$, over $\mathbb{C}(n)$ and over $\mathbb{Q}(m_1^2, \dots, m_N^2)(n)$.

Proof. Apply Theorem X.1.2 with $r = 2$, $a_j = \mu_j^2$. The two counts $(k_{j,0}, k_{j,1})$ correspond to $\ell_j = k_{j,0} - k_{j,1}$. There is no extra sign exclusion for even r. □

For masses $(1, 2, 3, 4)$, $D = 16$ and $Z_0 = 2$, so the minimum width is seven. Nevertheless the distinct nonzero squared thresholds are only $4, 16, 36, 64, 100$. Counting distinct singular locations gives merely five; the Mellin calculation proves that the collision multiplicities still require seven shifts. For $(1, 2, 3)$ the width is three, while for $(1, 1, 2)$ it is two. These are genuine zero-threshold specializations, not instances of a noncollision hypothesis.

Corollary X.1.4 (The coefficient-degree optimum also has no threshold restriction)**.** *Assume $N \geq 2$ and use the all-mass irreducibility and order theorem of manuscript [032]. Write*

$\bigotimes_j \operatorname{Sym}^{h_j} \mathbb{C}^2 = \bigoplus_d (\operatorname{Sym}^d \mathbb{C}^2)^{\oplus a_d}$. *The minimum coefficient degree, minimized separately from width, is*

$$o(m) = \sum_{d \geq 2} (d-1) a_d.$$

Proof. A polynomial recurrence of coefficient degree e yields an inhomogeneous differential equation $PF = Q(x)$ of order at most e for $F(x) = \sum c_n(m) x^n$, with Q polynomial. The differential module of F over $\mathbb{C}(x)$ is irreducible by [032]. If $Q \neq 0$, it contains the trivial rank-one module, hence is itself trivial. For $N \geq 3$ its order is at least $N - 1 \geq 2$, excluding this possibility. For $N = 2$, $F = (1 - 2(m_1^2 + m_2^2)x + (m_1^2 - m_2^2)^2 x^2)^{-1/2}$ is nonrational because $m_1 m_2 \neq 0$, again excluding a trivial module. Thus $Q = 0$ and $e \geq o(m)$. Conversely clearing denominators in the order-$o(m)$ homogeneous equation and comparing coefficients gives a recurrence of coefficient degree at most $o(m)$. □

X.1.4 All multinomial exponents and every arithmetic subsequence

For integers $r \geq 2$ and $H \geq 1$ define

$$A_{r,H}(n) = \sum_{k_1 + \cdots + k_H = n} \binom{n}{k_1, \ldots, k_H}^r,$$

and

$$Z_{r,H} = \#\left\{(k_0, \ldots, k_{r-1}) \in \mathbb{Z}_{\geq 0}^r : \sum k_j = H, \quad \sum k_j \zeta^j = 0\right\}, \qquad R_{r,H} = \frac{\binom{H+r-1}{r-1} - Z_{r,H}}{r}.$$

Theorem X.1.5 (All-subsequence multinomial classification)**.** *For every $q \geq 1$ and $0 \leq b < q$, the minimum eventual recurrence width of $A_{r,H}(qn + b)$ is exactly $R_{r,H}$. The minimum holds over $\mathbb{Q}(n)$ as well as $\mathbb{C}(n)$.*

Proof. For $q = 1$, this is Theorem X.1.2 with one scale and multiplicity H. An order-$R_{r,H}$ rational companion system yields, by taking its q-step evolution, a common recurrence of width at most $R_{r,H}$ for every sampled sequence.

For the converse put

$$g_{q,b}(x) = \frac{1}{q} \sum_{\epsilon^q = 1} \epsilon^{-b} \mathcal{F}_r(\epsilon x)^H.$$

Its coefficients are $A_{r,H}(n)/(n!)^r$ on $n \equiv b \pmod q$ and zero elsewhere. The q summands have pairwise nonisomorphic irreducible solution modules. For even r, distinct scales give independent symplectic factors by the Goursat argument above. For odd $r \geq 3$, opposite scales, when present, give $\operatorname{Sym}^H V$ and $\operatorname{Sym}^H V^*$ for the same special-linear factor; these are nonisomorphic since their highest weights are $H\omega_1$ and $H\omega_{r-1}$. Other signed scale classes are independent. Thus the images of the q summands have direct sum, and every nonzero filtered component makes $g_{q,b}$ cyclic in that whole sum. It is a semisimple connection without a trivial summand.

Each summand has irregularity $R_{r,H}$, so Lemma X.1.1 says that the full, zero-interlaced coefficient sequence has minimum width $qR_{r,H}$. A width-L recurrence for the sampled normalized coefficients induces a width-qL recurrence for the full sequence, with shifts divisible by q: replace the sample index by $(n - b)/q$. On the other residue classes every term is zero. The induced relation holds eventually and forces $qL \geq qR_{r,H}$. Factorial normalization preserves sampled width, since $((q(n+1) + b)!/(qn + b)!)^r$ is polynomial in n. Together with the upper bound this proves the assertion. □

Corollary X.1.6 (Closed prime-exponent formula)**.** *For every prime p, every number $H \geq 1$ of categories, and every arithmetic subsequence,*

$$\boxed{\operatorname{minwidth} A_{p,H}(qn+b) = \frac{1}{p}\left(\binom{H+p-1}{p-1} - \mathbf{1}_{p\mid H}\right).}$$

Proof. A polynomial $\sum_{j=0}^{p-1} k_j X^j$ vanishes at a primitive pth root if and only if it is a scalar multiple of $\Phi_p(X) = 1 + X + \cdots + X^{p-1}$. Thus $Z_{p,H}$ is one if $p \mid H$ and zero otherwise. Apply Theorem X.1.5. □

For example, for exponent three the widths for $H = 1, 2, 3, 4, 5, 6$ are $1, 2, 3, 5, 7, 9$. For exponent five and $H = 3$ the width is seven. These assertions concern every progression, not just the full sequence. For composite r, the same theorem remains exact with the finite cyclotomic count $Z_{r,H}$: equivalently count nonnegative-coefficient polynomials of degree below r, coefficient sum H, divisible by Φ_r. For $r = 4$ this gives $Z_{4,H} = 0$ for odd H, and $Z_{4,H} = H/2 + 1$ for even H.

X.1.5 Independent exact certificates and scope

For masses $(1, 1, 2)$, writing $u_n = c_n(1, 1, 2)$, the width-two relation is

$$(n+2)^2 u_{n+2} - 2(10n^2 + 30n + 23)u_{n+1} + 4(4n+3)(4n+5)u_n = 0.$$

For masses $(1, 2, 3)$, a width-three relation has forward coefficients

$$\begin{aligned} p_0 &= -144(4n+3)(4n+5)(4n+9), \\ p_1 &= 196(2n+3)^2(4n+9), \\ p_2 &= -14(16n^3 + 100n^2 + 202n + 129), \\ p_3 &= (n+3)^2(4n+5). \end{aligned}$$

For the three-category cube sum $A_{3,3}(n)$, forward coefficients are

$$\begin{aligned} p_0 &= -729(n+1)^4(21n^2 + 91n + 99), \\ p_1 &= 3(3n+5)(63n^5 + 546n^4 + 1921n^3 + 3435n^2 + 3116n + 1143), \\ p_2 &= -3(189n^6 + 2331n^5 + 11811n^4 + 31362n^3 \\ &\qquad + 45855n^2 + 34891n + 10793), \\ p_3 &= (n+3)^4(21n^2 + 49n + 29). \end{aligned}$$

These identities were independently certified in exact rational arithmetic, not merely fitted: in each case the operator

$$\sum_{j=0}^{L} x^{L-j} p_j(\theta - j)\big(\theta(\theta-1)\cdots(\theta-j+1)\big)^r$$

reduces to the zero vector in the tensor/symmetric-power connection $\theta^r \mathcal{F}_r(ax) = ax\mathcal{F}_r(ax)$. Direct integer evaluation through $n = 100 - L$ provided a second check. For masses $(1, 2, 3, 4)$, finite-field diagnostics found width seven, coefficient degree eighteen, independently modulo 1000003 and 1000033; the latter was checked on 279 coefficients beyond fitting. That diagnostic supports, but is not used in, the general proof.

The results are extensions relative to the supplied manuscripts. Higher-Bessel product formulas, Kloosterman groups and Mellin rank are established theory [92, 93, 94, 95]. The

contribution here is their precise use for unrestricted coefficient recurrences, including all zero-threshold banana degenerations and all arithmetic subsequences of multinomial powers. This does not prove irrationality of a special numerical value, nor the separate leading-coefficient multiplicity conjecture for minimal-order banana Picard–Fuchs operators. The general weighted theorem for odd r retains the explicit exclusion of opposite scales; that additional representation stratum is not claimed to be classified here.

X.2 An arithmetic classification of Duffing actions and polynomial time averages

The following result strengthens the arithmetic-clock conclusions for polynomial Hamiltonians to algebraic independence of dynamical observables. The transcendence input is the classical Chudnovsky–Philibert theorem; the contribution here is an explicit elliptic reduction, an exact classification of all polynomial time averages, and the resulting action–frequency trichotomy. No literature-wide priority claim is made. All coefficients and coordinates below are dimensionless; algebraic rescalings preserve the assertions.

Write $\overline{\mathbb{Q}}_{\mathbb{R}} = \overline{\mathbb{Q}} \cap \mathbb{R}$. Consider

$$H(x,p) = \frac{p^2}{2} + \frac{ax^2}{2} + \frac{bx^4}{4}, \qquad a,b,E \in \overline{\mathbb{Q}}_{\mathbb{R}}. \tag{X.2.1}$$

For a nonconstant periodic orbit of energy E, with least period T, define its angular frequency, normalized action, and time average by

$$\Omega = \frac{2\pi}{T}, \qquad J = \frac{1}{2\pi}\oint p\,dx, \qquad \langle P\rangle = \frac{1}{T}\int_0^T P(x(t),p(t))\,dt, \qquad \xi = \langle x^2\rangle. \tag{X.2.2}$$

The action is oriented by the Hamiltonian flow, so it is positive.

Proposition X.2.1 (All nonconstant bounded periodic regimes)**.** *For $b \neq 0$, the nonconstant bounded periodic orbits are precisely the following cases:*

1. *$b > 0$, $E > 0$, with arbitrary a: one symmetric oval;*
2. *$b > 0$, $a < 0$, $-a^2/(4b) < E < 0$: two ovals, one in each half-plane $x > 0$ and $x < 0$;*
3. *$b < 0$, $a > 0$, $0 < E < -a^2/(4b)$: one bounded symmetric oval.*

Other real components at the same energy may be unbounded. In every listed case $E \neq 0$ and $a^2 + 4bE > 0$, so the complex energy curve is smooth. For $b = 0$, such an orbit exists exactly when $a > 0$ and $E > 0$.

Proof. The potential is $V(x) = ax^2/2 + bx^4/4$ and $V'(x) = x(a + bx^2)$. For $b > 0$, its minima are at zero if $a \geq 0$, and at $x = \pm\sqrt{-a/b}$ with value $-a^2/(4b)$ if $a < 0$. For $b < 0$, a local well exists only when $a > 0$; its two surrounding maxima have value $-a^2/(4b)$. These elementary extrema give the stated compact components. At a minimum the orbit is constant, and at a barrier energy it is a separatrix, not a nonconstant periodic orbit. The roots of $2E - ax^2 - bx^4/2$ are simple exactly when $E(a^2 + 4bE) \neq 0$. The quadratic case is immediate. □

Lemma X.2.2 (Classical elliptic independence input)**.** *Let $g_2, g_3 \in \overline{\mathbb{Q}}$ with $g_2^3 - 27g_3^2 \neq 0$. For the Weierstrass functions with these invariants, let $\omega \neq 0$ be a period and $\eta = \zeta(z+\omega) - \zeta(z)$ its associated quasi-period. Then π/ω and η/ω are algebraically independent over $\overline{\mathbb{Q}}$.*

Attribution and period convention. This is Chudnovsky's theorem in the form proved quantitatively by Philibert [96, Sections 1–2]. Philibert states it for a primitive period. Any nonzero period is $n\omega_0$ for a primitive period ω_0 and a nonzero integer n, and its quasi-period is $n\eta_0$. Thus the first coordinate is multiplied by $1/n$ and the second is unchanged. Algebraic independence over $\mathbb{Q}$ and over its algebraic closure is equivalent here. □

Theorem X.2.3 (Frequency and mean-square independence)**.** *For every orbit in Proposition X.2.1 with $b \neq 0$, the two real numbers Ω and ξ are algebraically independent over $\overline{\mathbb{Q}}$.*

Proof. On the energy curve put $u = x^2$ and $v = xp$. Then

$$v^2 = -\frac{b}{2}u^3 - au^2 + 2Eu, \qquad \frac{du}{v} = 2\frac{dx}{p}.$$

The quotient is a smooth elliptic curve. Set

$$X = -\frac{b}{8}u - \frac{a}{12}, \qquad Y = -\frac{b}{8}v.$$

Direct substitution gives

$$Y^2 = 4X^3 - g_2X - g_3, \qquad g_2 = \frac{a^2 + 3bE}{12}, \quad g_3 = \frac{2a^3 + 9abE}{432}, \tag{X.2.3}$$

and

$$g_2^3 - 27g_3^2 = \frac{E^2b^2(a^2 + 4bE)}{256} \neq 0, \qquad \frac{dX}{Y} = \frac{du}{v} = 2\frac{dx}{p}.$$

The finite algebraic quotient extends to the smooth projective models. Let γ be the oriented physical oval and use its image cycle on the Weierstrass curve. The associated period is

$$\omega = \oint_{q_*\gamma} \frac{dX}{Y} = 2T > 0.$$

This formula uses the image cycle with its multiplicity and is valid whether or not that cycle is primitive. With $X = \wp(z)$, $Y = \wp'(z)$, the associated second-kind integral is

$$\oint_{q_*\gamma} X\frac{dX}{Y} = -\eta.$$

The image of the real oval avoids infinity, and the differential has zero residue there, so this is the usual quasi-period convention. Since $u = -8X/b - 2a/(3b)$, it follows that

$$\Omega = \frac{4\pi}{\omega}, \qquad \xi = \frac{8}{b}\frac{\eta}{\omega} - \frac{2a}{3b}. \tag{X.2.4}$$

This is an invertible affine change over $\overline{\mathbb{Q}}$ of the two coordinates in Lemma X.2.2. □

Theorem X.2.4 (All polynomial phase-space averages)**.** *For each $P \in \overline{\mathbb{Q}}[x, p]$ there are explicitly computable $A_P, B_P, C_P \in \overline{\mathbb{Q}}$ such that*

$$\langle P \rangle = A_P + B_P\xi + C_P\Omega. \tag{X.2.5}$$

They are unique for the fixed orbit, and

$$\langle P \rangle \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad B_P = C_P = 0.$$

In the symmetric regimes $E > 0$, one always has $C_P = 0$, and the field generated by all polynomial time averages is $\overline{\mathbb{Q}}(\xi)$, of transcendence degree one. In the two-well regime $E < 0$, that field is $\overline{\mathbb{Q}}(\xi, \Omega)$, of transcendence degree two. In every regime, adjoining Ω to all polynomial averages gives exactly the latter field.

Proof. Write $m_n = \langle x^{2n} \rangle$, so $m_0 = 1$ and $m_1 = \xi$. Integrating the exact differential $d(x^{2n+1}p)$ around the orbit, and using $p^2 = 2E - ax^2 - bx^4/2$, gives

$$m_{n+2} = \frac{4E(2n+1)}{b(2n+3)}m_n - \frac{4a(n+1)}{b(2n+3)}m_{n+1}, \qquad n \geq 0. \tag{X.2.6}$$

Thus $m_n = A_n + B_n\xi$, where $A_0 = 1, B_0 = 0, A_1 = 0, B_1 = 1$ and both coefficient sequences satisfy the same recurrence.

For a monomial with an odd power of momentum,

$$\int_0^T x^r p^{2s+1}\,dt = \oint x^r (2E - ax^2 - bx^4/2)^s\,dx = 0,$$

because the integrand is the differential of a polynomial. Even powers of p reduce by the energy identity to polynomials in x. On a symmetric orbit $x(t + T/2) = -x(t)$, so odd powers of x average to zero. This proves the assertion in the regimes $E > 0$.

For $E < 0$, write

$$u_\pm = \frac{-a \pm \sqrt{a^2 + 4bE}}{b} > 0, \quad c = -\frac{a}{b}, \quad h = \frac{\sqrt{a^2 + 4bE}}{b}.$$

Here $b > 0$, and the selected oval has a fixed sign $\varepsilon = \operatorname{sgn}(x) \in \{1, -1\}$. Let

$$R_j = \sum_{k=0}^{\lfloor j/2 \rfloor} \binom{j}{2k} c^{j-2k} h^{2k} \frac{\binom{2k}{k}}{4^k} = \frac{1}{\pi}\int_0^\pi (c + h\cos\theta)^j\,d\theta. \tag{X.2.7}$$

These are algebraic, and $R_0 = 1$. Changing variable $u = x^2$ on each half of the orbit yields

$$\begin{aligned}\oint x^{2j+1}\,dt &= \varepsilon\sqrt{\frac{2}{b}}\int_{u_-}^{u_+} \frac{u^j\,du}{\sqrt{(u_+ - u)(u - u_-)}} \\ &= \varepsilon\sqrt{\frac{2}{b}}\,\pi R_j.\end{aligned}$$

Consequently

$$\langle x^{2j+1}\rangle = \frac{\varepsilon R_j}{\sqrt{2b}}\,\Omega. \tag{X.2.8}$$

The recurrence and these reductions compute all three coefficients in (X.2.5). Algebraic independence of ξ, Ω proves uniqueness and the claimed algebraicity criterion. The field equalities follow from $\langle x^2\rangle = \xi$ and, in the intrawell case, $\langle x\rangle = \varepsilon\Omega/\sqrt{2b}$. □

Corollary X.2.5 (Complete polynomial relation ideal)**.** *Let $P_1, \dots, P_s$ be any finite family of polynomial observables, and write their averages as $A_i + B_i\xi + C_i\Omega$. All polynomial relations over $\overline{\mathbb{Q}}$ among the averages are generated by the affine linear relations*

$$\sum_{i=1}^{s} r_i(Z_i - A_i), \qquad \sum_i r_i B_i = \sum_i r_i C_i = 0.$$

The transcendence degree of their field is the rank, over $\overline{\mathbb{Q}}$, of the matrix with rows (B_i, C_i).

Proof. Evaluate the affine linear map $Z_i \mapsto A_i + B_i U + C_i V$ into $\overline{\mathbb{Q}}[U, V]$. A linear change of variables identifies its kernel with the displayed affine linear ideal. The subsequent evaluation $U = \xi, V = \Omega$ is injective by Theorem X.2.3. □

Theorem X.2.6 (Global action–frequency trichotomy)**.** *For every nonconstant bounded periodic orbit of (X.2.1) at algebraic parameters and energy:*

1. *If $b = 0$, then $a > 0$ and $\Omega = \sqrt{a}$, $J = E/\sqrt{a}$ are algebraic.*
2. *If $b \neq 0$ and $a = 0$, necessarily $b > 0, E > 0$. Both J and Ω are transcendental, $J\Omega = 4E/3$, and $\operatorname{trdeg}_{\overline{\mathbb{Q}}}\,\overline{\mathbb{Q}}(J, \Omega) = 1$.*

3. If $ab \neq 0$, then J and Ω are algebraically independent.

Proof. The harmonic formulas are elementary. For $b \neq 0$, the $n = 0$ case of (X.2.6) gives

$$\langle x^4 \rangle = \frac{4E}{3b} - \frac{4a}{3b}\xi.$$

Moreover $\oint p\,dx = \int_0^T p^2\,dt$. Therefore

$$J\Omega = \langle p^2 \rangle = 2E - a\xi - \frac{b}{2}\langle x^4 \rangle = \frac{4E - a\xi}{3}. \tag{X.2.9}$$

If $a = 0$, the nonzero algebraic product and transcendence of Ω prove part 2. If $a \neq 0$, the identities

$$J = \frac{4E - a\xi}{3\Omega}, \qquad \xi = \frac{4E - 3J\Omega}{a}$$

give equality of the two fields $\overline{\mathbb{Q}}(J, \Omega) = \overline{\mathbb{Q}}(\xi, \Omega)$, proving part 3. □

Corollary X.2.7 (Algebraic action levels force transcendental energy)**.** *Fix $a, b \in \overline{\mathbb{Q}}_{\mathbb{R}}$ with $b \neq 0$ and a nonzero algebraic number j. If a nonconstant bounded regular periodic orbit of (X.2.1), at a real energy E, satisfies $J(E) = j$, then E is transcendental. This assertion applies separately on every existing central or intrawell branch; it does not assert that a prescribed action level is attained on each branch.*

Proof. If E were algebraic, Theorem X.2.6 would apply. In the pure-quartic case it makes J transcendental; in the mixed case it makes J algebraically independent of Ω, and therefore transcendental as well. Both contradict $J(E) = j \in \overline{\mathbb{Q}}$. This is a statement about classical action levels, including any chosen algebraic semiclassical action prescription. No conclusion about exact quantum eigenvalues is inferred. □

Remark X.2.8 (Useful pure-quartic check). For $a = 0, b > 0, E > 0$, the recurrence yields

$$\langle x^{4k} \rangle = \left(\frac{4E}{b}\right)^k \frac{(1/4)_k}{(3/4)_k}, \qquad \langle x^{4k+2} \rangle = \left(\frac{4E}{b}\right)^k \frac{(3/4)_k}{(5/4)_k}\,\xi.$$

Thus infinitely many nonconstant observables have algebraic averages; the classification does not confuse transcendence of the period with transcendence of every normalized average.

Remark X.2.9 (Boundary and novelty limitations). The equilibrium and separatrix energies are excluded because they do not have a nonconstant finite-period orbit. The theorem concerns classical dynamics and polynomial time averages, not quantum eigenvalues or irrationality of Euler's constant. The elliptic independence theorem is classical. Targeted searches did not locate this exact full action–frequency and polynomial-average classification, but such searches do not establish publication priority or solve a named prior open problem.

X.3 Topology deletion and transcendental heat-time thresholds

The allocation of conductances to minimize total effective resistance is a classical convex design problem [97]. The present result uses spectral convexity and an explicit three-space decomposition to solve an entire family with arbitrary internal topology. It also identifies an exact support transition for heat-trace optimization whose transition time is transcendental. These are deductions proved here; no claim of historical priority or resolution of a previously named conjecture is made.

Let A, B be disjoint labelled vertex sets with $|A| = m$, $|B| = s$, and $n = m + s$. Let H be any simple graph on B, and let

$$G = I_m \vee H$$

be the graph with no edges inside A, every edge between A and B, and the prescribed edges of H inside B. Conductances are nonnegative and satisfy the normalization

$$\operatorname{tr} L = 2 \sum_{e \in E(G)} w_e = T > 0. \tag{X.3.1}$$

Put $V = \mathbf{1}^{\perp}$. When the positive support is connected, define

$$J_p(L) = \operatorname{tr}_V (L|_V)^{-p}, \qquad p > 0, \qquad \det{}'(L) = \det(L|_V).$$

Disconnected supports have $J_p = +\infty$ and $\det' = 0$.

Theorem X.3.1 (Deletion of an arbitrary internal graph)**.** *Suppose $m > s \geq 1$. For every real $p > 0$, the unique minimizer of J_p under (X.3.1) is the same Laplacian L_*, given by*

$$w_e = w_* := \frac{T}{2ms} \quad (e \text{ between } A \text{ and } B), \qquad w_e = 0 \quad (e \in E(H)). \tag{X.3.2}$$

It is also the unique maximizer of $\det'(L)$. Its positive spectrum is

$$\{(sw_*)^{(m-1)}, (mw_*)^{(s-1)}, nw_*\}. \tag{X.3.3}$$

In particular,

$$\min J_p = \left(\frac{2ms}{T}\right)^p \left((m-1)s^{-p} + (s-1)m^{-p} + n^{-p}\right), \tag{X.3.4}$$

$$\max \det{}'(L) = \left(\frac{T}{2ms}\right)^{n-1} s^{m-1} m^{s-1} n. \tag{X.3.5}$$

If H has an edge, the threshold is exact: for $m \leq s$, the weighting (X.3.2) is not optimal for any $p > 0$, nor for the determinant.

Proof. For a strictly convex twice differentiable f, the spectral function $\operatorname{tr} f(Q)$ is strictly convex on positive definite matrices. For completeness, in an eigenbasis of Q its second derivative in a nonzero symmetric direction D is

$$\sum_i f''(\lambda_i) D_{ii}^2 + 2 \sum_{i<j} \frac{f'(\lambda_i) - f'(\lambda_j)}{\lambda_i - \lambda_j} D_{ij}^2,$$

with continuous interpretation at repeated eigenvalues. Every coefficient is positive for $f(x) = x^{-p}$ and for $f(x) = -\log x$. The edge weights determine the Laplacian injectively, so the resulting objectives on weights are strictly convex as well.

The invariant orthogonal subspaces of L_* are the zero-sum vectors on A, the zero-sum vectors on B, the contrast of the two constant blocks, and the global constant vector. They give (X.3.3). For $q > 0$ and $b_e = e_i - e_j$, put

$$S_q(e) = b_e^{\mathsf{T}} (L_*|_V)^{-q} b_e.$$

The squared projections of a cross-edge vector onto the three nonconstant spaces are $1 - 1/m$, $1 - 1/s$, and $1/m + 1/s$. An internal B-edge vector belongs wholly to the B zero-sum space. Thus their respective sensitivities are

$$\begin{aligned} C_q &= (1 - 1/m)(sw_*)^{-q} + (1 - 1/s)(mw_*)^{-q} + \frac{n}{ms}(nw_*)^{-q}, \\ I_q &= 2(mw_*)^{-q}. \end{aligned}$$

Since $\partial J_p / \partial w_e = -pS_{p+1}(e)$, the optimality condition under the total conductance constraint is $C_{p+1} \geq I_{p+1}$. Direct simplification gives

$$\Delta_p := C_{p+1} - I_{p+1} = \frac{w_*^{-p-1}}{ms} \left((m-1)s^{-p} - (s+1)m^{-p} + n^{-p} \right). \tag{X.3.6}$$

This is strictly positive when $m > s$. Indeed, put $a = s/m$, $b = s/n$. After multiplication by s^p, the bracket is

$$F(p) = m - 1 - (s+1)a^p + b^p.$$

Here $0 < b < a < 1$, $F(0) = m - s - 1 \geq 0$, and

$$F'(p) = (s+1)a^p \log(m/s) - b^p \log(n/s) > 0.$$

The last inequality follows from $(s+1)\log(m/s) > \log(1 + m/s)$. To see this, for $u \geq 1 + 1/s$ the inequality $u^{s+1} > 1 + u$ holds: at the left endpoint it follows from the binomial theorem, and the difference is increasing for $u \geq 1$. Apply this with $u = m/s$.

For the determinant criterion the derivative of $-\log\det'$ is $-S_1(e)$, and

$$C_1 - I_1 = \frac{m - s - 1}{msw_*} \geq 0. \tag{X.3.7}$$

For any feasible L, conservation of the total conductance and the convex tangent inequality now give

$$J_p(L) - J_p(L_*) \geq p\Delta_p \sum_{e \in E(H)} w_e \geq 0. \tag{X.3.8}$$

The same argument with (X.3.7) proves determinant optimality. Strict convexity gives uniqueness, including $m = s+1$, where the first-order determinant margin vanishes. Equations (X.3.4)–(X.3.5) follow from the spectrum.

If $m \leq s$, the bracket in (X.3.6) is at most $(m-s-2)s^{-p} + n^{-p} < 0$. Increase the conductance of any edge of H from zero and decrease all cross-edge conductances equally by the same total amount. The derivative of J_p along this feasible connected perturbation is $p\Delta_p < 0$. For $-\log\det'$ the corresponding derivative is negative by (X.3.7). This proves the final assertion. □

Corollary X.3.2 (A quantitative certificate)**.** *Under the hypotheses of Theorem X.3.1, every feasible connected Laplacian satisfies*

$$J_p(L) - J_p(L_*) \geq p\Delta_p \sum_{e \in E(H)} w_e + \frac{p(p+1)}{2T^{p+2}} \|L - L_*\|_{\mathrm{F}}^2.$$

The constant is independent of the topology of H.

Proof. Every positive eigenvalue along the segment from L_* to L is at most T. In the Hessian formula, all divided differences are bounded below by $\min_{0<x\le T} p(p+1)x^{-p-2} = p(p+1)T^{-p-2}$. Taylor's integral remainder adds the stated quadratic term to (X.3.8). Restriction to V preserves the Frobenius norm because $L - L_*$ annihilates $\mathbf{1}$. □

Theorem X.3.3 (A broader class of spectral criteria)**.** *If $m \ge s+2$, the same L_* uniquely minimizes $\operatorname{tr}_V f(L|_V)$ for every $f \in C^3((0,\infty))$ with $f'' > 0$ and $f''' \le 0$, with optimization over connected supports. If f has a finite continuous extension to zero, the same conclusion holds over all nonnegative weights by continuity. This includes inverse powers, negative logarithm, heat trace, and squared spectral dispersion at fixed trace. If H has an edge, the threshold cannot be decreased to $m = s+1$ for this whole class: the criterion $f(x) = x^2$ fails there.*

Proof. Set $g = -f'$, which is decreasing and convex, and abbreviate

$$A_0 = \frac{m-1}{m}, \qquad B_0 = \frac{s+1}{s}, \qquad C_0 = \frac{n}{ms}.$$

Then $A_0 + C_0 = B_0$. The cross-minus-internal sensitivity is

$$A_0 g(sw_*) + C_0 g(nw_*) - B_0 g(mw_*).$$

Since $B_0 m - A_0 s - C_0 n = m - s - 2 \ge 0$, Jensen's inequality and monotonicity of g show that this expression is nonnegative. The same convex tangent argument and strict convexity finish the proof. For $f(x) = x^2$ the sensitivity difference is $2w_*(m-s-2)$, which is negative at $m = s+1$, giving a strictly improving internal-edge perturbation. □

For heat trace it is convenient to allow disconnected networks as well:

$$\mathcal{H}_t(L) = \operatorname{tr}(e^{-tL}) - 1 = \operatorname{tr}_V(e^{-tL|_V}), \qquad t > 0.$$

This is strictly convex on the full compact weight simplex and has a unique minimizer. Counting the zero eigenvalues in this definition is essential; the fixed constant eigenvalue alone is subtracted.

Theorem X.3.4 (Exact heat-time transition and its transcendence)**.** *Assume $s \ge 2$, $m = s+1$, and that H has at least one edge. Let $z_s > 1$ be the unique root in $(1,\infty)$ of*

$$s^2 z^{s+1} - (s+1)^2 z^s + (2s+1) = 0. \tag{X.3.9}$$

Put

$$t_s = \frac{2s(s+1)}{T} \log z_s. \tag{X.3.10}$$

Then L_* is the unique minimizer of $\mathcal{H}_t$ precisely when $t \ge t_s$. For $0 < t < t_s$, the unique minimizer gives positive weight to at least one edge of H. The threshold is independent of the topology of H. If T is a positive algebraic number, t_s is transcendental.

For comparison, if $m \ge s+2$, L_* minimizes $\mathcal{H}_t$ for every $t > 0$. If $m \le s$ and H has an edge, it does so for no $t > 0$.

Proof. With $g(x) = e^{-tx}$, the omitted positive derivative factor t does not affect the optimality inequality. Thus the necessary and sufficient condition is

$$A_0 e^{-tsw_*} + C_0 e^{-tnw_*} - B_0 e^{-tmw_*} \ge 0.$$

For $m = s+1$, divide by e^{-tmw_*} and put $y = tw_*$. The left side has the sign of

$$D(y) = \frac{s}{s+1} e^{y} + \frac{2s+1}{s(s+1)} e^{-sy} - \frac{s+1}{s}.$$

Here $D(0) = 0$, $D'(0) = -1$, $D''(y) > 0$, and $D(y) \to \infty$ as $y \to \infty$. Hence it has exactly one strictly positive zero y_s, is negative on $(0, y_s)$, and positive on (y_s, ∞). Writing $z = e^y$ and multiplying by $s(s+1)z^s$ gives (X.3.9), with $z_s = e^{y_s}$. This proves existence and uniqueness of the specified algebraic root and the criterion $t \geq y_s/w_* = t_s$, including equality by strict convexity.

Below the threshold an internal-edge perturbation strictly improves L_*. Every weighting with no internal edge has heat trace at least that of L_*: average it over all permutations within A and within B and use strict convexity. Consequently a global minimizer below the threshold must use some internal edge. This averaging statement concerns the complete-bipartite face of the weight simplex and does not require H itself to have any symmetry.

If T is positive algebraic, then w_* and z_s are algebraic and $z_s > 1$. If t_s were algebraic, the nonzero number $w_* t_s$ would be algebraic, while $e^{w_* t_s} = z_s$ would also be algebraic, contrary to the Hermite–Lindemann theorem. Thus t_s is transcendental.

The case $m \geq s + 2$ follows from Theorem X.3.3. For $m \leq s$, after division by e^{-tmw_*} the sensitivity difference is

$$A_0 e^{(m-s)tw_*} + C_0 e^{-stw_*} - B_0 < 0,$$

because it is zero at $t = 0$ and strictly decreases for $t > 0$. An internal-edge perturbation again proves failure of optimality. □

Example X.3.5. For $m = 3$, $s = 2$, $T = 2$, and $H = K_2$, all inverse-power optima and the determinant optimum delete the sole internal edge. Each of the six cross edges has weight $1/6$, and the positive eigenvalues are $\{1/3, 1/3, 1/2, 5/6\}$. In particular,

$$\min J_p = 2 \cdot 3^p + 2^p + (6/5)^p, \qquad \min J_1 = \frac{46}{5}, \qquad \max \det{}' L = \frac{5}{108}.$$

The heat-time threshold is the explicit transcendental number

$$\boxed{t_2 = 6 \log\left(\frac{5 + \sqrt{105}}{8}\right).}$$

Indeed (X.3.9) factors as $(z-1)(4z^2 - 5z - 5) = 0$. The support used by the optimizer changes at this exact time, although the allowed graph and the total conductance remain fixed.

Theorem X.3.6 (Algebraic versus transcendental optimal conductances)**.** *Let $m, s \geq 2$, let $G = I_m \vee K_s$, and assume the trace budget $T > 0$ and heat time $t > 0$ are algebraic numbers. The unique heat-trace optimizer has a common cross conductance a and a common internal conductance b. Its arithmetic type is completely classified as follows.*

1. *If $m \geq s + 2$, then $a = T/(2ms)$ and $b = 0$, both algebraic.*
2. *If $m = s + 1$, then a and b are both transcendental for $t < t_s$, and are $T/(2ms)$ and 0 for $t > t_s$, respectively. Equality is impossible because t_s is transcendental.*
3. *If $m \leq s$, then a and b are both positive and transcendental for every positive algebraic t.*

Proof. Strict convexity gives a unique minimizer on the compact weight simplex. It is invariant under the full permutation groups on A and B, giving the two conductances a, b. Their budget and three positive-eigenvalue expressions are

$$2msa + s(s-1)b = T, \qquad D = sa, \quad V = ma + sb, \quad W = na,$$

with multiplicities $m - 1, s - 1, 1$. Eliminate b to obtain

$$V = \frac{T}{s-1} - \frac{m(s+1)}{s-1} a, \qquad 0 \leq a \leq \frac{T}{2ms}.$$

The heat-trace objective on this interval is

$$F(a) = (m-1)e^{-tsa} + (s-1)e^{-tV} + e^{-tna}.$$

At $a = 0$,

$$F'(0) = -tm(s+1)\left(1 - e^{-tT/(s-1)}\right) < 0.$$

Thus an optimizer always has $a > 0$. Theorem X.3.4 determines exactly when the other endpoint $b = 0$ is optimal. In every remaining case, $a, b > 0$, and stationarity gives

$$s(m-1)e^{-tsa} + ne^{-tna} = m(s+1)e^{-tV}. \tag{X.3.11}$$

If a were algebraic, the budget would make b algebraic, hence D, V, W would be algebraic. Since $a > 0$, $D \neq W$. If V is distinct from both, (X.3.11) contradicts the Lindemann–Weierstrass linear independence theorem for exponentials of distinct algebraic numbers. If $V = D$ or $V = W$, the equality reduces to a nonzero multiple of $e^{-tD} - e^{-tW} = 0$, also impossible. Therefore a is transcendental. The budget expresses b as an affine function of a with nonzero algebraic slope, so b is transcendental as well. In the endpoint cases, the formulas show that both conductances are algebraic. □

In particular, on $K_4 - e = I_2 \vee K_2$, every positive algebraic heat time and positive algebraic budget produce two transcendental optimal conductances. On $K_{3,1,1} = I_3 \vee K_2$, the same statement holds below $6\log((5+\sqrt{105})/8)$ when $T = 2$, and above that time the optimum instead has the rational conductances $1/6$ and 0. The transcendence conclusion concerns the exact optimum, not numerical approximations to it.

The smallest case also has an exact one-root description. On $K_4 - e$, for arbitrary $T, t > 0$, let r be the unique positive root of

$$r^4(1+2r) = 3e^{-tT}.$$

Then the heat-optimal conductances are

$$a = -\frac{\log r}{2t}, \qquad b = \frac{T}{2} - 4a.$$

The root satisfies $e^{-tT/4} < r < 1$, as follows by evaluating the strictly increasing left side at the two endpoints, so both conductances are positive. Substitution into (X.3.11) verifies stationarity and strict convexity gives the global optimum. This root formula is exact; the transcendence assertion for algebraic T, t follows from Theorem X.3.6, not from the root formula alone.

X.4 Two-level spectra with arbitrary matching defects

X.4.1 Statement and normalization

Let $M_r = \{\{1,2\},\{3,4\},\dots,\{2r-1,2r\}\}$. A strict weighted Laplacian on $K_n - M_r$ has zero row sums, zero off-diagonal entries on M_r, and strictly negative off-diagonal entries on every allowed edge. A relaxed realization replaces the last strict inequalities by weak ones. Write $s = n - 2r \geq 1$ and $H_n = I_n - \mathbf{1}\mathbf{1}^{\mathsf{T}}/n$.

Theorem X.4.1 (All matching sizes, complete two-level slice)**.** *Let $r \geq 1$, $n \geq 2r+1$, and $0 < a < b$. The multiset*

$$\Lambda = \{0, a^{(n-r-1)}, b^{(r)}\}$$

has a relaxed weighted-Laplacian realization on $K_n - M_r$ if and only if

$$\frac{b}{a} \geq \frac{n}{n-2}. \tag{X.4.1}$$

It has a strict realization if and only if (X.4.1) holds and it is not the exceptional case

$$n = 2r+2, \qquad \frac{b}{a} = \frac{n}{n-2}.$$

At this exceptional point every relaxed realization has the additional zero entry between the two unmatched vertices.

Put $g = b/a - 1 > 0$ and $c = 1/(ng)$. Every real symmetric matrix with the stated spectrum and kernel $\mathbb{R}\mathbf{1}$ is of the form

$$L = aH_n + agP, \qquad P^2 = P = P^{\mathsf{T}}, \quad P\mathbf{1} = 0, \quad \operatorname{rank} P = r. \tag{X.4.2}$$

Conversely each such projection gives the required spectrum. For $i \neq j$,

$$L_{ij} = ag(P_{ij} - c).$$

Consequently the problem is precisely to arrange

$$P_{2i-1,2i} = c \quad (1 \leq i \leq r), \qquad P_{uv} < c \quad (\{u,v\} \notin M_r, u \neq v), \tag{X.4.3}$$

or the corresponding weak inequalities. Our constructions give a matrix $Z \in \mathbb{R}^{n\times r}$ with $Z^{\mathsf{T}}Z = I_r$ and $Z^{\mathsf{T}}\mathbf{1} = 0$; then $P = ZZ^{\mathsf{T}}$.

X.4.2 Necessity and the exact boundary

Lemma X.4.2 (Compressed matching spectrum)**.** *Let F be the adjacency matrix of M_r. The spectrum of H_nFH_n on $\mathbf{1}^{\perp}$ is*

$$\{-1^{(r)},\ 0^{(s-1)},\ (s/n)^{(1)},\ 1^{(r-1)}\}.$$

The positive spectral subspace consists of vectors constant on each matched pair and constant on the set of unmatched vertices, with total coordinate sum zero.

Proof. Each pair-difference vector has eigenvalue -1. The $r-1$ dimensional space of pair-constant vectors supported on matched vertices and with sum zero has eigenvalue 1. Vectors supported on the unmatched vertices and with sum zero give $s-1$ zero eigenvalues. The remaining vector is constant on all matched vertices, constant on the unmatched vertices, and orthogonal to $\mathbf{1}$; direct substitution gives eigenvalue s/n. These spaces are mutually orthogonal and their dimensions sum to $n-1$. □

If P satisfies the contacts, then

$$\operatorname{tr}(PH_nFH_n) = \operatorname{tr}(PF) = 2rc.$$

The rank-r variational principle and Lemma X.4.2 give

$$2rc \le (r-1) + s/n = \frac{r(n-2)}{n}, \qquad c \le c_{\max} := \frac{n-2}{2n}. \tag{X.4.4}$$

This is exactly (X.4.1).

At equality the top r eigenvalues are separated from the rest because $s/n > 0$. Equality in the variational principle forces P to be the unique projection onto the positive spectral subspace. It is explicitly

$$P_* = \operatorname{diag}\left(frac12J_2, \dots, frac12J_2, \tfrac{1}{s}J_s\right) - \tfrac{1}{n}J_n. \tag{X.4.5}$$

Across different blocks its entries are $-1/n < c_{\max}$. Inside the unmatched block the off-diagonal value is $1/s - 1/n$. If $s \ge 3$, this is strictly less than $1/2 - 1/n = c_{\max}$. For $s = 1$ there is no such off-diagonal entry. For $s = 2$ it equals $c_{\max}$, forcing precisely the extra missing edge stated in the theorem. In every case (X.4.5) gives a relaxed realization.

X.4.3 An explicit frame for all small positive contacts

Set

$$c_0 = \frac{2r-1}{2(2r+1)}.$$

We first construct every $0 < c \le c_0$ on $2r+1$ vertices and add zero rows when $n > 2r+1$.

For $0 < t \le 1$, set $B = 1 + t^2$, $D = (1+t)^2$ and

$$U = (BI_r + DJ_r)^{-1/2}.$$

Let $u_1, \dots, u_r$ be the rows of U. Take the rows of Z to be

$$u_i, tu_i \quad (1 \le i \le r), \qquad v = -(1+t)\sum_{i=1}^{r} u_i, \tag{X.4.6}$$

followed, if necessary, by $s-1$ zero rows. They sum to zero, and

$$Z^{\mathsf{T}}Z = U^{\mathsf{T}}(BI_r + DJ_r)U = I_r.$$

The Gram matrix of the u_i is

$$S = UU^{\mathsf{T}} = \frac{1}{B}I_r - \frac{D}{B(B+rD)}J_r.$$

Its off-diagonal entries are strictly negative. Moreover, $u_i \cdot v = -(1+t)/(B+rD) < 0$. Thus the only positive off-diagonal entries of $P = ZZ^{\mathsf{T}}$ are exactly the prescribed matching contacts, all equal to

$$c(t) = \frac{t\{B + (r-1)D\}}{B(B+rD)}. \tag{X.4.7}$$

All other off-diagonal entries are negative or zero, so strict support is automatic for $c(t) > 0$. The function is continuous, $c(t) \to 0$ as $t \downarrow 0$, and $c(1) = c_0$. This already proves that it covers the required interval.

For an exact parameter formula, put $u = t + t^{-1} \ge 2$. Then

$$c(t) = \frac{ru + 2(r-1)}{u\{(r+1)u + 2r\}}.$$

This expression is strictly decreasing in $u \ge 2$, since the numerator of its derivative is

$$-r(r+1)u^2 - 4(r^2-1)u - 4r(r-1) < 0.$$

For a prescribed $c \in (0, c_0]$, choose the positive solution $u \ge 2$ of

$$c(r+1)u^2 + (2rc - r)u - 2(r-1) = 0, \qquad t = \frac{2}{u + \sqrt{u^2 - 4}}.$$

When $r = 1$ the quadratic also has a zero root, which is discarded.

X.4.4 The remaining contacts and every strictness exception

Suppose $s \ge 2$. Choose $q_1, \dots, q_s \ge 0$ with $\sum_j q_j = 1$, and write $k = \sum_j q_j^2$. Define

$$U = (2I_r + 4kJ_r)^{-1/2}$$

and again denote its rows by u_i. Take the rows of Z to be

$$u_i, u_i \quad (1 \le i \le r), \qquad q_j v \quad (1 \le j \le s), \qquad v = -2\sum_i u_i. \tag{X.4.8}$$

The same calculation shows $Z^{\mathsf{T}}\mathbf{1} = 0$ and $Z^{\mathsf{T}}Z = I_r$. Here

$$UU^{\mathsf{T}} = \frac{1}{2}I_r - \frac{k}{1+2rk}J_r,$$

so the matching contacts are

$$c(k) = \frac{1 + 2(r-1)k}{2(1+2rk)}. \tag{X.4.9}$$

Entries between different matched pairs are negative, and entries between matched and unmatched vertices are $-q_j/(1+2rk) \le 0$. Between unmatched vertices $i \ne j$ they are $2rq_iq_j/(1+2rk)$. The exact margin from the contact is

$$\begin{aligned} c(k) - P_{2r+i,2r+j} &= \frac{1 + 2(r-1)k - 4rq_iq_j}{2(1+2rk)} \\ &= \frac{1 - (q_i+q_j)^2 + (2r-1)(q_i-q_j)^2 + 2(r-1)\sum_{h\ne i,j} q_h^2}{2(1+2rk)}. \end{aligned} \tag{X.4.10}$$

Every term in the last numerator is nonnegative. It vanishes precisely when $q_i = q_j = 1/2$ and all the other q_h are zero. This conclusion also holds when $r = 1$, using $\sum_h q_h = 1$.

Take specifically

$$q_1 = x, \qquad q_2 = \cdots = q_s = \frac{1-x}{s-1}, \qquad 1/s \le x \le 1.$$

Then $k = x^2 + (1-x)^2/(s-1)$ runs continuously and monotonically from $1/s$ to 1. Meanwhile $c(k)$ is strictly decreasing, with

$$c(1/s) = \frac{n-2}{2n} = c_{\max}, \qquad c(1) = c_0.$$

For $s \ge 3$ these shares never consist of precisely two halves, so (X.4.10) is strictly positive throughout. For $s = 2$ it is positive except at $x = 1/2$, which corresponds exactly to $c = c_{\max}$. Thus (X.4.8) covers every $c_0 \le c \le c_{\max}$ with exactly the previously identified strictness exception. Together with (X.4.6), this proves Theorem X.4.1.

The construction is fully explicit. In the large-contact interval,

$$k = \frac{1-2c}{4rc-2(r-1)}, \qquad x = \frac{1+\sqrt{(s-1)(sk-1)}}{s}.$$

For either frame, the positive inverse square root has the elementary formula

$$(BI_r + DJ_r)^{-1/2} = B^{-1/2}I_r + \frac{(B+rD)^{-1/2} - B^{-1/2}}{r}J_r.$$

No nonlinear optimization or uncertified root search is required.

X.4.5 Consequences, independent checks, and remaining targets

Corollary X.4.3 (The previously missing five-vertex case)**.** *On $K_5 - M_2$, the spectrum $\{0, a^{(2)}, b^{(2)}\}$ has a strict realization if and only if $b/a \geq 5/3$.*

Corollary X.4.4 (A growing-rank exceptional boundary)**.** *For every $r \geq 1$, on $2r+2$ vertices the boundary spectrum $\{0, a^{(r+1)}, ((r+1)a/r)^{(r)}\}$ forces the support $K_{2r+2} - M_{r+1}$. Its relaxed realization for the fixed original matching is unique. For every larger ratio, exact strict $K_{2r+2} - M_r$ support is possible.*

Proof. At equality (X.4.5) is unique, and its blocks are all pairs. Its entries outside those pairs are $-1/n$, hence the corresponding off-diagonal entries of L are strictly negative. □

The accompanying independent numerical script constructs P by the displayed scalar formulas and verifies orthonormality, centering, the full requested spectrum, and every allowed and forbidden edge. These checks support implementation correctness; the proof above is exact and does not depend on finite tests. Historical priority beyond the supplied manuscript collection has not been established by this note.

The missing perfect-matching case $n = 2r$ is genuinely outside the proof: the positive compressed eigenspace then has dimension $r-1$, and the extra core row in (X.4.6) is unavailable. Likewise, arbitrary ordered spectra require more than this repeated-eigenvalue slice. The natural reference point is Fallat–Gupta–Lin, *Inverse Eigenvalue Problem for Laplacian Matrices of a Graph*, https://doi.org/10.1137/24M1714472; no assertion that our particular all-r classification appeared there is made.

Attempted all-graph design comparison. The second target considered was the gap between $1/(4\sqrt{2})$ and $1/(2\sqrt{2})$ in the leading constant of the worst D-optimal/A-optimal inverse-trace ratio over all connected n-vertex graphs. This note does not close that gap. To state a precise sufficient remaining inequality, let $d = n-1$, A be the determinant optimal Laplacian on $\mathbf{1}^\perp$, $t = \operatorname{tr} A^{-1}$, and τ the optimal inverse trace, with total conductance one. The currently available matrix dual bounds estimate edge norms by the sum of a bulk term and a bottleneck term. A uniform additive bound of the form

$$\sqrt{\tau} \;\geq\; \frac{d}{\sqrt{2}} + \frac{2t}{d\sqrt{n}} - o(n) \tag{X.4.11}$$

in the only critical regime $t \asymp n^{5/2}$ would imply the sharp upper leading constant $1/(4\sqrt{2})$, matching the known barbells. Indeed, on writing $t = d^2\sqrt{n}\,u$, the proposed bound would give

$$\frac{t}{\tau\sqrt{n}} \leq \frac{u}{(1/\sqrt{2}+2u)^2} + o(1) \leq \frac{1}{4\sqrt{2}} + o(1).$$

Outside that regime the existing bounds $t/\tau \leq \min\{2t/d^2, nd^2/(4t)\}$ already give a negligible normalized ratio. A proof of (X.4.11), or a counterexample to it, remains missing: the elementary shifted-matrix duality argument only supplies the weaker quadrature combination. Thus (X.4.11) is a concrete sufficient target, not a theorem or a claimed equivalent reformulation.

X.5 Global corner connections under entire interactions

The physical-density connection problem for Jacobi traces concerns the coefficients selecting the density from local branches of a differential system. Forrester and Kumar solve the bare-weight problem under integer restrictions on two parameters [98]; the preceding collection extends that formula to all integrable real Jacobi parameters. Here a different proof gives a further extension: arbitrary positive trace weights, nonidentical endpoint exponents, and entire interactions. Its Jacobi specialization includes every even Dyson index and nonlinear external potentials. The proof does not assert a full connection matrix between arbitrary bases, and no historical-priority claim or resolution of a named published conjecture is made.

X.5.1 The model and its canonical cone functions

Fix $N \ge 1$, positive numbers $c_1, \dots, c_N$, and real exponents $a_i, b_i > -1$. Let $Q : \mathbb{C}^N \to \mathbb{C}$ be entire, real valued and nonnegative on $[0,1]^N$, and not identically zero there. Define

$$Z = \int_{[0,1]^N} Q(x) \prod_{i=1}^{N} x_i^{a_i} (1 - x_i)^{b_i} \, dx > 0. \tag{X.5.1}$$

Let X have the probability density given by the integrand divided by Z, and let ρ be the density of $T = \sum_i c_i X_i$, extended by zero outside $[0, \sum_i c_i]$. For $S \subseteq [N]$, set

$$v_{S,i} = \mathbf{1}_{\{i \in S\}}, \qquad \lambda_S = \sum_{i \in S} c_i,$$

$$A_S = \sum_{i \notin S} (a_i + 1) + \sum_{i \in S} (b_i + 1), \qquad d_S = \prod_{i \notin S} c_i^{-a_i - 1} \prod_{i \in S} c_i^{-b_i - 1}, \tag{X.5.2}$$

$$\epsilon_S = (-1)^{|S|} \exp\left(-\pi i \sum_{i \in S} b_i \right). \tag{X.5.3}$$

Write $\mathcal{D} = \{u \in [0,\infty)^N : \sum_i u_i = 1\}$ and use the Dirac-delta simplex measure $d\sigma$, obtained by eliminating one coordinate with Jacobian one. For $N = 1$ this is unit mass.

For $\operatorname{Im} s > 0$, define

$$J_S(s) = d_S s^{A_S - 1} \int_{\mathcal{D}} Q\left(v_S + \left(\frac{s u_i}{c_i} \right)_{i=1}^{N} \right) \prod_{i \notin S} u_i^{a_i} \left(1 - \frac{s u_i}{c_i} \right)^{b_i} \cdot \prod_{i \in S} u_i^{b_i} \left(1 + \frac{s u_i}{c_i} \right)^{a_i} d\sigma. \tag{X.5.4}$$

The powers are continued from sufficiently small positive s. Explicitly, $\arg s \in (0, \pi)$, while $1 - s u_i / c_i$ has argument in $(-\pi, 0)$ and $1 + s u_i / c_i$ has argument in $(0, \pi)$ whenever $u_i > 0$. For $0 < s < \min_i c_i$, (X.5.4) is a convergent ordinary simplex integral with the positive real choices of its nonsingular factors. Let $J_S^+(r)$ denote its upper boundary value at $r > 0$.

Theorem X.5.1 (Global entire-interaction connection formula)**.** *Under the preceding assumptions, each J_S is holomorphic in the upper half-plane and is the analytic continuation through that half-plane of its small positive cone germ. For every real t outside the finite knot set $\mathcal{K} = \{\lambda_S : S \subseteq [N]\}$,*

$$\boxed{\rho(t) = \frac{1}{Z} \sum_{S : \lambda_S < t} \epsilon_S J_S^+(t - \lambda_S).} \tag{X.5.5}$$

The sum is empty for $t < 0$. For $t > \sum_i c_i$, the complete sum in (X.5.5) vanishes identically. The identity also holds as an equality of locally integrable distributions across the knots.

The density is real analytic between consecutive distinct knots. At a knot λ, its physical right germ minus the upper continuation of its physical left germ is

$$\frac{1}{Z}\sum_{S:\lambda_S=\lambda}\epsilon_S J_S(t-\lambda). \tag{X.5.6}$$

Thus the theorem includes coinciding subset sums and every integral resonance of the corner exponents.

Proof. For $x>0$, the elementary boundary-value identity is

$$x^a(1-x)^b\mathbf{1}_{\{x<1\}} = x^a(1-x-i0)^b - e^{-\pi i b}x^a(x-1)^b\mathbf{1}_{\{x>1\}}. \tag{X.5.7}$$

For $x<1$ the second term is zero. For $x>1$ the first term equals $e^{-\pi i b}x^a(x-1)^b$ and is cancelled by the second. All factors are locally integrable because $a,b>-1$.

Multiply (X.5.7) for the N coordinates, multiply by $Q(x)$, and expand the product on the positive orthant. Integrate against $\delta(t-\sum_i c_i x_i)$. In the term indexed by S, put $x_i=1+y_i$ for $i\in S$ and $x_i=y_i$ otherwise. The coefficient is precisely ϵ_S, while $y_i\ge 0$ and $\sum_i c_i y_i = t-\lambda_S$. This cone is empty if $t<\lambda_S$. If $r=t-\lambda_S>0$, substitute $y_i=ru_i/c_i$. The coarea scaling is $r^{N-1}/\prod_i c_i$, and the endpoint factors give the remaining powers in $d_S r^{A_S-1}$. The resulting integral is the boundary value in (X.5.4), proving (X.5.5) whenever the integrals exist.

We supply the analytic details. On a compact subset of $\operatorname{Im} s>0$, none of the factors $1\pm su_i/c_i$ vanishes. The image of the compact simplex under the argument of Q is compact, so Q and its derivatives are bounded there. The factors $u_i^{a_i}$ or $u_i^{b_i}$ have exponents greater than -1. Dominated differentiation therefore proves holomorphy and identifies the function with its initially convergent cone germ.

For a positive real r, the additional singular hyperplanes of the boundary integral are $u_i=c_i/r$ for $i\notin S$. Together with the coordinate faces $u_j=0$, they intersect the simplex transversely except when r is a subset sum of the c_i with $i\notin S$. Indeed, a failure to eliminate a nonsingular coordinate would require every coordinate to be either zero or one of the specified c_i/r, forcing precisely such a subset-sum identity. Away from these exceptional values, eliminate a nonsingular coordinate on each of finitely many patches. The remaining singular factors are products of independent one-dimensional powers with exponents greater than -1. For a negative exponent,

$$|c_i-(r+i\eta)u_i|^{b_i}\le |c_i-ru_i|^{b_i};$$

positive powers are uniformly bounded. This gives dominated convergence as $\eta\downarrow 0$.

To verify continuity in r before using reflection, fix $r_0>0$ away from these exceptional values. In a sufficiently small closed interval about r_0, choose a finite partition of unity on the simplex, locally also in r, such that on each patch one coordinate stays uniformly away from all its singular faces. Eliminate that coordinate. At a coordinate face retain the coordinate u_j itself; at an internal singular face use the moving coordinate $w_i=u_i-c_i/r$. These changes have Jacobian one, and the eliminated coordinate and all nonsingular factors remain uniformly separated from zero on the selected patch. The partition functions may be extended by zero in a common compact coordinate box, with the inequalities $u_j\ge 0$ retained at coordinate faces. The singular factors are then bounded by a fixed integrable product of powers $u_j^{a_j}$ or $u_j^{b_j}$ and $|w_i|^{b_i}$, with every exponent greater than -1, times a uniformly bounded factor. In particular, $c_i-ru_i=-rw_i$, and the preceding negative-power bound is uniform in r on this interval. Dominated convergence in these moving coordinates proves the joint boundary continuity as $r\to r_0$ and $\eta\downarrow 0$. Thus the boundary value is continuous and equals the real cone integral used in the product expansion.

For t in a compact interval, all expanded terms are locally integrable by Fubini: the corresponding part of the positive orthant is bounded, Q is bounded there, and all coordinate

endpoint powers are integrable. This proves the distributional identity, including at the knots. The original density is supported in $[0, \sum_i c_i]$, which also proves the exterior cancellation.

On an open interval between successive knots, fix its active set of subsets. The corresponding sum of the $J_S(t - \lambda_S)$ is holomorphic above the interval and has continuous real boundary values, equal to $Z\rho$. Schwarz reflection extends it holomorphically across the interval. Its continuation past the next knot through the upper half-plane is the same sum. On the right, the physical formula adds exactly the terms whose subset sum equals that knot. Their difference is (X.5.6). No sine denominator or resonant exponent difference has been introduced. □

X.5.2 Jacobi interactions and normalized coefficients

Let $k \geq 0$ be an integer and write $\Delta(x) = \prod_{i<j}(x_j - x_i)$. Take

$$Q(x) = \Delta(x)^{2k} W(x), \tag{X.5.8}$$

where W is entire, real valued, and strictly positive on the cube. This includes $W = e^{-V}$ for every real polynomial V in all N variables; neither separability nor permutation symmetry is required. For S of size p, put $m = N - p$ and

$$h_S = k\{m(m-1) + p(p-1)\}, \qquad \gamma_S = A_S + h_S - 1.$$

For an empty or singleton tuple, its Vandermonde is one.

Corollary X.5.2 (Explicit canonical coefficients at even Dyson index)**.** *For the model* (X.5.8)*,*

$$J_S(s) = K_S s^{\gamma_S}(1 + O(s)), \qquad K_S > 0, \tag{X.5.9}$$

where the convergent positive simplex integral is

$$K_S = d_S W(v_S) \int_{\mathcal{D}} \prod_{i \notin S} u_i^{a_i} \prod_{i \in S} u_i^{b_i} \Delta\left(\left(\frac{u_i}{c_i}\right)_{i \notin S}\right)^{2k} \Delta\left(\left(\frac{u_i}{c_i}\right)_{i \in S}\right)^{2k} d\sigma. \tag{X.5.10}$$

Consequently the explicitly defined cone germ J_S has the scalar normalization $F_S = J_S/K_S$, for which $F_S(s) = s^{\gamma_S}(1 + O(s))$, and its physical-density coefficient is

$$C_S = \epsilon_S K_S / Z. \tag{X.5.11}$$

These formulas include all resonant values of γ_S. The leading-term condition alone is not a uniqueness assertion among all local solutions; the integral (X.5.4) *selects the germ.*

Proof. Every within-cluster Vandermonde contributes its homogeneous power of s, giving the total h_S. Every cross-cluster difference has leading value 1 or -1, whose even power is one. The remaining leading interaction is $W(v_S) > 0$. This proves (X.5.9) and (X.5.10). The angular integrand is nonnegative, integrable, and positive on a set of positive simplex measure; for $N = 1$ the integral is evaluation at its single point. Thus $K_S > 0$. Apply Theorem X.5.1. □

When all $c_i = 1$, all $a_i = a$, all $b_i = b$, and $W = 1$, symmetry groups subsets by cardinality. The summed coefficient agrees with the canonical bare-Jacobi coefficient at $\beta = 2k$. In particular, the additional bare-Jacobi phase $e^{-\pi i k p(p-1)}$ equals one since $p(p-1)$ is even. With unequal trace weights, the angular constants (X.5.10) are generally anisotropic simplex integrals rather than Selberg products.

X.5.3 Exact differential order under polynomial insertions

The connection formula permits general entire W. Exact finite linear differential order requires additional hypotheses.

Theorem X.5.3 (Exact order with polynomial interactions and linear sources)**.** *Suppose that*

$$W(x) = P(x)e^{\sum_i h_i x_i},$$

where $h_i \in \mathbb{R}$ and P is a real polynomial, nonnegative on the cube and strictly positive at every cube vertex. Suppose that all 2^N subset sums λ_S are distinct. Then the Fourier–Laplace transform

$$M(z) = \frac{1}{Z}\int_{[0,1]^N} \Delta(x)^{2k} P(x) \prod_i x_i^{a_i}(1-x_i)^{b_i} e^{\sum_i (h_i + c_i z)x_i}\, dx \tag{X.5.12}$$

has minimal differential order exactly 2^N over $\mathbb{C}(z)$.

Proof. For the upper bound, put

$$f_i(w) = \int_0^1 x^{a_i}(1-x)^{b_i} e^{wx}\, dx.$$

Integration by parts after multiplication by $x(1-x)$ is justified because $a_i, b_i > -1$, and gives the Kummer equation

$$w f_i'' + (a_i + b_i + 2 - w) f_i' - (a_i + 1) f_i = 0. \tag{X.5.13}$$

Thus every derivative of f_i belongs to the $\mathbb{C}(w)$-span of f_i, f_i'. Expand $\Delta^{2k}P$ into finitely many monomials. Every resulting integral in (X.5.12) is a product of derivatives $f_i^{(r_i)}(h_i + c_i z)$. Hence M and all its derivatives belong to the differential-stable $\mathbb{C}(z)$-span of the 2^N products obtained by selecting either f_i or f_i' in every factor. Linear algebra gives differential order at most 2^N. Apparent poles in this rational reduction do not alter the rational-function-field argument.

For the lower bound, at the corner v_S use inward coordinates $x_i = y_i$ for $i \notin S$ and $x_i = 1 - y_i$ for $i \in S$. Set

$$\begin{aligned} D_{S,-} &= \sum_{i \notin S}(a_i + 1) + km(m-1), \\ D_{S,+} &= \sum_{i \in S}(b_i + 1) + kp(p-1), \qquad D_S = D_{S,-} + D_{S,+}. \end{aligned}$$

An empty cluster has degree zero. Define the positive Laplace cone integrals, with empty integral one,

$$\begin{aligned} B_{S,-} &= \int_{[0,\infty)^{[N]\setminus S}} e^{-\sum_{i\notin S} c_i y_i} \prod_{i \notin S} y_i^{a_i} \Delta(y)^{2k}\, dy, \\ B_{S,+} &= \int_{[0,\infty)^S} e^{-\sum_{i\in S} c_i y_i} \prod_{i \in S} y_i^{b_i} \Delta(y)^{2k}\, dy. \end{aligned}$$

Their finiteness and positivity follow from positive c_i and the integrable endpoint powers. Radial integration in each cluster and Taylor expansion of the remaining analytic factors give, to every prescribed finite order,

$$M(iR) \sim \sum_{S \subseteq [N]} e^{i\lambda_S R} R^{-D_S} \sum_{\ell \geq 0} A_{S,\ell} R^{-\ell}, \tag{X.5.14}$$

with nonzero leading coefficients

$$A_{S,0} = \frac{P(v_S) e^{\sum_{i\in S} h_i}}{Z} B_{S,-} B_{S,+} e^{\pi i (D_{S,-} - D_{S,+})/2} \neq 0. \tag{X.5.15}$$

Indeed, the radial endpoint integral of degree $d > 0$ contributes $\Gamma(d)e^{\pm\pi id/2}R^{-d}$, and its angular factor divides out the same $\Gamma(d)$. Outside corner neighborhoods, at least one coordinate is strictly inside $(0, 1)$; repeated integration by parts in that coordinate gives arbitrarily rapid decay. The Vandermonde power and the inserted polynomial have no collision singularities. This also justifies complete remainders. Applying ∂_z^j inserts $(\sum_i c_i x_i)^j$, so its leading amplitude at the same corner is $\lambda_S^j A_{S,0}$.

Suppose $\sum_{j=0}^{r} p_j(z)\partial_z^j$ annihilates M, where the p_j are polynomials after clearing denominators. Let

$$d = \max_j \deg p_j, \qquad q(u) = \sum_{j=0}^{r} [z^d]p_j(z)u^j.$$

Then q is nonzero and has degree at most r. The leading amplitude at frequency λ_S in the resulting oscillatory expansion is $i^d A_{S,0}q(\lambda_S)$, with power R^{d-D_S}.

For clarity, finite complete expansions with distinct real frequencies have unique coefficients, even if their initial real powers differ. If a coefficient were nonzero, select the largest exponent among the first nonzero coefficients at the finitely many frequencies. Divide by that power. The remaining leading sum is a finite trigonometric polynomial $\sum_j b_j e^{i\lambda_j R}$ plus a uniform $o(1)$ remainder, with at least one $b_j \neq 0$. Its mean square satisfies

$$\frac{1}{T}\int_T^{2T} \left|\sum_j b_j e^{i\lambda_j R}\right|^2 dR \longrightarrow \sum_j |b_j|^2 > 0,$$

because each cross integral for distinct frequencies is $O(T^{-1})$. This contradicts the identically zero function. Equivalently, one may successively remove the highest powers: above any fixed power only finitely many terms of the finitely many expansions occur. Therefore $q(\lambda_S) = 0$ for every S. There are 2^N distinct roots, forcing $r \geq 2^N$. This matches the upper bound. □

X.5.4 Branch checks and exact examples

Example X.5.4 (One coordinate and the continuation sign)**.** For $N = 1$, $0 < t < c$, the single active term is

$$\rho(t) = \frac{1}{Zc}Q(t/c)(t/c)^a(1 - t/c)^b.$$

For $t > c$, its upper continuation has factor $e^{-\pi ib}$. The new cone term has coefficient $-e^{-\pi ib}$ and the same remaining real amplitude, so the exterior density cancels exactly. This holds for asymmetric, noninteger $a, b > -1$ and any admissible Q.

Example X.5.5 (Independent uniforms and colliding knots)**.** For $Q = 1$, $a_i = b_i = 0$, and arbitrary positive c_i, one has $Z = 1$ and

$$J_S(s) = \frac{s^{N-1}}{(N-1)!\prod_i c_i}.$$

The theorem gives the classical weighted-uniform identity

$$\rho(t) = \frac{1}{(N-1)!\prod_i c_i}\sum_{S\subseteq[N]} (-1)^{|S|}(t - \lambda_S)_+^{N-1},$$

with the natural step-function convention when $N = 1$. For $N = 2$, $c_1 = c_2 = 1$, the two singleton terms share the knot 1 and give $t_+ - 2(t-1)_+ + (t-2)_+$. Thus colliding knots require summation, not selection of one corner coefficient.

Example X.5.6 (A correlated resonant example with exact order four)**.** Take $N = 2$, $c = (1, 2)$, $a_i = b_i = 0$, $k = 1$, and $W = 1$. Then $Z = \int_0^1 \int_0^1 (x - y)^2 \, dx \, dy = 1/6$. Direct cone integration gives

$$J_\varnothing(s) = s^3/8, \qquad J_{\{1\}}(s) = s/2 + s^2/4 + s^3/8,$$
$$J_{\{2\}}(s) = s/2 - s^2/4 + s^3/8, \qquad J_{\{1,2\}}(s) = s^3/8.$$

Consequently, writing $s_+ = \max(s, 0)$,

$$\rho(t) = 6\Big\{ \frac{t_+^3}{8} - \frac{(t-1)_+}{2} - \frac{(t-1)_+^2}{4} - \frac{(t-1)_+^3}{8} - \frac{(t-2)_+}{2} + \frac{(t-2)_+^2}{4} - \frac{(t-2)_+^3}{8} + \frac{(t-3)_+^3}{8} \Big\}.$$

The polynomial sum vanishes identically for $t > 3$, and integration on the three intervals gives total mass one. Its four subset sums are distinct, so Theorem X.5.3 proves exact transform order four. Every exponent in this example is resonant.

A separate high-precision numerical diagnostic used $c = (1, \sqrt{2})$, $a = (0.2, -0.3)$, $b = (0.4, -0.2)$, $k = 1$, and $W(x) = \exp(0.3x_1^2 + 0.2x_1x_2)$. Direct physical-slice quadrature and the active cone sum agreed to at least 39 decimal places at $t = 0.4, 1.2, 1.7$. At $t = 2.6$, outside the support, the absolute residual was below $9 \cdot 10^{-41}$. Each quadrature was split at internal branch crossings. These calculations test signs and continuation; they are not used as a proof. The nonlinear potential satisfies the connection theorem but is outside the exact-order theorem.

X.5.5 Limits of the transfer

The key identity is coordinatewise endpoint cancellation with an *entire* common interaction. For a non-even Dyson index, $|\Delta(x)|^\beta$ does not supply such an interaction: collision hyperplanes carry branch information that the coordinatewise identity does not control. Thus no extension to arbitrary β and arbitrary weighted traces is inferred. Likewise, arbitrary entire external potentials need not have a finite differential equation of the order in Theorem X.5.3; that theorem explicitly uses a polynomial insertion and linear sources. Subset-sum collisions are admitted by the connection formula, but can lower differential order and are excluded from the exact-order assertion.

The independent polynomial-density specialization belongs to an already studied family of generalized Irwin–Hall formulas. Accordingly, the result is presented as a proved extension of the supplied collection, with a different proof and explicit parameter scope, rather than a claim that endpoint inclusion–exclusion or every special case is new.

X.6 Recovery under arbitrary unknown refresh schedules

X.6.1 Model and the earlier inputs

Let Θ and $\mathcal{X}$ be standard Borel spaces and let $\theta \mapsto P_\theta$ be a known probability kernel with an injective component map. For a finite signed measure ν on Θ, define

$$A_m\nu = \int_\Theta P_\theta^{\otimes m}\, d\nu(\theta), \qquad m \geq 1.$$

Let $q \in \mathbb{N} \cup \{\infty\}$ be the least m for which A_m is injective on all Borel probability measures, with $q = \infty$ if there is none. Injectivity on probabilities is equivalent to injectivity on finite signed measures: a signed kernel element has mass zero, and its normalized Jordan parts give a probability counterexample. If $q < \infty$, component injectivity follows automatically from injectivity of A_q.

Given a population G, start with $\theta_1 \sim G$ and use the transitions

$$M_i(\theta, d\eta) = \rho_i\delta_\theta(d\eta) + (1-\rho_i)G(d\eta), \qquad 0 < \rho_i \leq 1. \tag{X.6.1}$$

Conditionally on the latent path, the observations X_i are independent, with laws P_{θ_i}. Every latent marginal is G, even when the transition kernels differ. Write T_I for the joint observed law on a consecutive interval I, and $T_n = T_{[1,n]}$. All identification statements refer to these population laws, rather than to exact recovery from one realized finite record. The norm $\|\cdot\|_{\mathrm{var}}$ is full total variation; two probability measures have distance at most two.

The compact, constant-retention, $q = 2$ three-visit argument occurs in [101, Part I, theorem "Blind three-visit recovery"]. The present argument handles all q, standard Borel spaces, and a different unknown retention at every edge. The unrestricted two-view theorem used later is an explicit input from [100, Part V, theorem "Two observations without support or moment restrictions"]; it is not proved again by the temporal transfer.

X.6.2 All-order identification and inversion

Lemma X.6.1 (The zero-covariance exception)**.** *Put $\mu = A_1G$ and $C_G = A_2G - \mu^{\otimes 2}$. Then $C_G = 0$ if and only if P_θ is constant G-almost surely. Under component injectivity this is equivalent to G being a point mass. At a point population every retention is unidentifiable at every observation length.*

Proof. Choose a countable determining algebra $\mathcal{B}$ for the standard Borel space $\mathcal{X}$. For $B \in \mathcal{B}$,

$$C_G(B \times B) = \mathrm{Var}_G(P_\theta(B)).$$

If $C_G = 0$, intersect the countably many full-measure sets on which these coordinates are constant. The component laws agree there on $\mathcal{B}$, hence as measures. Component injectivity makes this full-measure set a singleton. The converse is immediate. For a point population the entire observed process is iid with one component law, whatever the retentions. □

Lemma X.6.2 (Recovery of every retention)**.** *Suppose G is not a point mass. Any n-visit law, $n \geq 3$, identifies the entire vector $(\rho_1, \ldots, \rho_{n-1})$.*

Proof. Let Q_{ij} be the law of (X_i, X_j), $i < j$. The refresh construction gives

$$Q_{ij} = \mu^{\otimes 2} + \left(\prod_{h=i}^{j-1} \rho_h\right) C_G. \tag{X.6.2}$$

If no refresh occurs, both observations share a latent variable; after at least one refresh the final latent variable is independent of the initial one. This statement remains true if a refreshed draw happens to equal the initial state.

Set $D_{ij} = Q_{ij} - \mu^{\otimes 2}$. By Lemma X.6.1, $C_G \neq 0$, and all the retentions are positive. For every consecutive triple,

$$\rho_i = \frac{\|D_{i,i+2}\|_{\mathrm{var}}}{\|D_{i+1,i+2}\|_{\mathrm{var}}}, \qquad \rho_{i+1} = \frac{\|D_{i,i+2}\|_{\mathrm{var}}}{\|D_{i,i+1}\|_{\mathrm{var}}}. \tag{X.6.3}$$

Overlapping triples recover all edges. All measures in these formulas are marginals of the observed n-visit law. □

For $S \subseteq \{1, \dots, n-1\}$ let $\pi(S)$ be the consecutive partition obtained by cutting after the indices in S. Tensor products over its blocks retain the chronological coordinate order.

Theorem X.6.3 (Explicit interval-partition inverse)**.** *For known positive retentions,*

$$T_n = \sum_{S \subseteq [n-1]} \left(\prod_{i \notin S} \rho_i \right) \left(\prod_{i \in S} (1 - \rho_i) \right) \bigotimes_{B \in \pi(S)} A_{|B|} G, \tag{X.6.4}$$

$$A_n G = \left(\prod_{i=1}^{n-1} \rho_i \right)^{-1} \sum_{S \subseteq [n-1]} (-1)^{|S|} \left(\prod_{i \in S} (1 - \rho_i) \right) \bigotimes_{B \in \pi(S)} T_B. \tag{X.6.5}$$

Here $[n-1] = \{1, \dots, n-1\}$, and empty products equal one.

Proof. Condition on the independent Bernoulli refresh indicators. A run between successive cuts has law $A_{|B|}G$, and distinct runs use independent draws from G. This proves (X.6.4).

Expand each T_B on the right of (X.6.5) using (X.6.4). Fix the final refined cut set R. After division by $\prod_i \rho_i$, its coefficient is

$$\prod_{i \in R} \frac{1 - \rho_i}{\rho_i} \sum_{S \subseteq R} (-1)^{|S|}.$$

This vanishes unless $R = \varnothing$, in which case it is one. Only $A_n G$ survives. No convergence issue occurs, since every sum is finite. □

Theorem X.6.4 (Exact threshold transfer)**.** *Consider all Borel populations and arbitrary unknown positive retentions.*

1. *If $q < \infty$, $\max(q, 3)$ visits identify G; for non-point G they also identify every retention in the record.*

2. *If $3 \leq q < \infty$, the exact universal visit threshold is q, both for population recovery and for joint recovery on the non-point class.*

3. *If $q = 1$ and the family has at least two components, one visit identifies G, and exactly two identify (G, ρ_1) on the non-point class.*

4. *If $q = 2$, the universal threshold for population and non-point joint recovery lies between two and three. Both values are attained by emission families described below.*

5. *If $q = \infty$, no finite visit count identifies all populations, even if the retentions are known.*

Proof. For a non-point population, Lemma X.6.2 recovers the retentions. Formula (X.6.5), followed by the A_q inverse, then recovers G. If the population is a point mass, the observable equality $D_{12} = 0$ and positive retention force every competing population to be a point mass by Lemma X.6.1. The component law identifies that point. This proves the upper bound.

For $k < q$, take $G \neq H$ with $A_k G = A_k H$. All A_j with $j \leq k$ also agree by marginalization, so (X.6.4) gives identical k-visit laws for any common known retention vector. When $q \geq 3$, use $k = q - 1 \geq 2$. Such witnesses cannot be point populations: equality $A_2 H = P_\theta^{\otimes 2}$ would force H to be the same point population. Their shorter marginals also give the required failures. If $q = \infty$, the argument applies at each finite k; if needed, averaging both witnesses with one fixed non-point population makes them non-point without changing equality or their nonzero difference.

If $q = 1$, the marginal identifies G, hence C_G. For non-point G, two visits identify $\rho_1 = \|D_{12}\|_{\mathrm{var}} / \|C_G\|_{\mathrm{var}}$. A single visit contains no retention parameter, giving necessity. At $q = 2$, the one-view lower bound and the three-view upper bound give the stated alternatives. □

X.6.3 Sharp known-retention amplification

Theorem X.6.5 (A product conditioning law)**.** *Fix positive retentions and $n \geq 2$. If two populations have observed n-visit laws T, U and $\epsilon = \|T - U\|_{\mathrm{var}}$, then*

$$\|A_n G - A_n H\|_{\mathrm{var}} \leq L_n(\rho)\epsilon, \tag{X.6.6}$$

where

$$L_n(\rho) = \prod_{i=1}^{n-1} \frac{2 - \rho_i}{\rho_i} \left(1 + \sum_{i=1}^{n-1} \frac{1 - \rho_i}{2 - \rho_i}\right).$$

If $A_{n-1} G = A_{n-1} H$, there is also the exact identity

$$T - U = \left(\prod_{i=1}^{n-1} \rho_i\right) (A_n G - A_n H). \tag{X.6.7}$$

Consequently, whenever the replication threshold is n, the best uniform inverse amplification has order $\prod_i \rho_i^{-1}$, up to constants depending only on n.

Proof. Each marginal difference has variation at most ϵ. A telescoping product of k probability measures therefore has difference at most $k\epsilon$. Apply (X.6.5) and set $c_i = 1 - \rho_i$. The sum of its absolute coefficients, weighted by the block count, is

$$\sum_{S \subseteq [n-1]} (|S| + 1) \prod_{i \in S} c_i = \prod_i (1 + c_i) \left(1 + \sum_i \frac{c_i}{1 + c_i}\right).$$

This proves (X.6.6).

If $A_{n-1} G = A_{n-1} H$, every term of (X.6.4) with at least one cut is the same for the two populations. Only the no-cut term remains, proving (X.6.7). At threshold n there are distinct such populations, and their A_n laws differ. Their exact norm ratio is $\prod_i \rho_i^{-1}$. The displayed upper constant is at most $n 2^{n-1} \prod_i \rho_i^{-1}$, completing the order comparison. □

Remark X.6.6. For constant retention, $L_n(\rho) = \rho^{1-n}(2 - \rho)^{n-2}[n - (n-1)\rho]$. These bounds assume known retentions. Formulas (X.6.3) establish exact unknown-retention identification, but do not by themselves give a uniform noisy inverse without a covariance lower bound. No such general bound is claimed.

X.6.4 An exact three-visit theorem for Weibull and stable populations

We use the following precise preceding input. For Weibull kernels

$$P_{\alpha,B}(W > x) = e^{-Bx^{\alpha}}, \quad x > 0, \qquad \alpha, B > 0,$$

the map A_2 is injective on all finite complex measures on $(0,\infty)^2$. Equivalently, the transform

$$\nu \longmapsto \left\{ \int e^{-B(u^{\alpha}+v^{\alpha})}, d\nu(\alpha, B) : u, v > 0 \right\}$$

is injective. This is [100, Part V, theorem "Two observations without support or moment restrictions"], proved there by a weighted integer-dilation inversion with Gamma-function endpoint control. The same input gives A_2 injectivity for symmetric stable components with characteristic function $e^{-B|u|^{\alpha}}$, on $(0,2] \times (0,\infty)$.

Theorem X.6.7 (Unrestricted three-visit threshold)**.** *For either of these emission families, three visits with unknown positive, possibly different, retentions identify every Borel population G. If G is not a point mass, both retentions are also identified. Two visits fail to identify all populations even with a single retention in $(0,1)$. Thus the exact universal threshold is three. There are no support, moment, finite-component, or latent-independence restrictions on G.*

Proof. The positive assertion is Theorem X.6.4 and the preceding A_2 input. For sharpness choose $0 < a < b$, with $b \le 2$ in the stable case. A positive stable variable V of index a/b has

$$\mathbb{E}e^{-sV} = e^{-s^{a/b}}, \qquad s > 0.$$

This is a classical distribution; see [102]. Its Laplace exponent follows from the Lévy measure $(a/b)x^{-1-a/b}\,dx/\Gamma(1-a/b)$, and it is strictly positive because its Laplace transform tends to zero at infinity.

Put $\theta_0 = (a, 1)$ and let ν be the law of (b, V). Substitution of $s = x^b$ in the Weibull survival transform, or $s = |u|^b$ in the stable characteristic transform, gives

$$A_1\nu = P_{\theta_0} =: \mu.$$

The measures ν and δ_{θ_0} are distinct. Hence $C_\nu = A_2\nu - \mu^{\otimes 2} \neq 0$ by the supplied injectivity of A_2. For $0 < s \le 1$ set

$$G_s = (1-s)\delta_{\theta_0} + s\nu.$$

Every G_s has marginal μ, and

$$A_2G_s = \mu^{\otimes 2} + sC_\nu, \qquad T_2(G_s, \rho) = \mu^{\otimes 2} + \rho s C_\nu. \tag{X.6.8}$$

Thus $(G_1, \rho = 1/4)$ and $(G_{1/2}, \rho = 1/2)$ give exactly the same entire two-visit probability law. Both populations are non-point, and both retentions lie strictly between zero and one. This proves failure for populations themselves, not only for retention recovery. For comparison their lag-two contrasts in the constant-retention three-visit experiment are $C_\nu/16$ and $C_\nu/8$, respectively. □

Remark X.6.8. Under $R = 1/\alpha$, $Q = -\log B/\alpha$, a Weibull observation has logarithm $Q + R\log E$ with E unit exponential. The same theorem therefore applies to unrestricted reflected-Gumbel location–scale populations, and by reflection to the usual Gumbel family. This is a coordinate-equivalent consequence, not a separate identification theorem.

X.6.5 Why two visits sometimes suffice despite a one-view defect

Lemma X.6.9 (Spherical calibration)**.** *Suppose there are bounded real measurable functions $\phi_1, \dots, \phi_d$ on $\mathcal{X}$ such that*

$$v(\theta) = \left(\int \phi_1 \, dP_\theta, \dots, \int \phi_d \, dP_\theta \right)$$

is injective and has constant squared norm c. For every non-point G, two refresh visits identify the retention by

$$\rho = \frac{\mathbb{E}\langle \phi(X_1), \phi(X_2) \rangle - \|\mathbb{E}\phi(X_1)\|^2}{c - \|\mathbb{E}\phi(X_1)\|^2}. \tag{X.6.9}$$

If A_2 is injective, these two visits then identify G as well.

Proof. Put $\bar{v} = \int v \, dG$. The denominator is $\int \|v - \bar{v}\|^2, dG$, which is positive unless G is a point mass, by injectivity of v. Under no refresh the conditional expectation of the inner product is $\|v(\theta)\|^2 = c$; under a refresh it is $\|\bar{v}\|^2$. This proves (X.6.9). Equation (X.6.2) then recovers $A_2 G$ from the observed pair law. □

Proposition X.6.10 (An entire two-visit example)**.** *There is a strictly positive jointly entire periodic emission density with replication threshold $q = 2$ for which two visits identify every population, and the unknown retention for every non-point population.*

Proof. Take $\Theta = \mathcal{X} = \mathbb{R}/(2\pi\mathbb{Z})$ with Haar probability measure m and set

$$k(x \mid \theta) = 1 + \sum_{n=2}^{\infty} e^{-n^2} \cos(n(x - \theta)).$$

Gaussian coefficient decay proves locally uniform convergence with all derivatives on $\mathbb{C}^2$, hence joint entirety and periodicity. The integral in x is one, and $k \geq 1 - \sum_{n \geq 2} e^{-n^2} > 0$.

The observed Fourier frequencies are $S = \mathbb{Z} \setminus \{-1, 1\}$. Put $a_0 = 1$, $a_n = e^{-n^2}/2$ for $|n| \geq 2$, and $a_{\pm 1} = 0$. The Fourier coefficient of $A_2 G$ at (j, k) is $a_j a_k \widehat{G}(j + k)$. Since $S + S = \mathbb{Z}$, all Fourier coefficients of G are recovered from $A_2 G$. This proves its injectivity. The one-view laws of $(1 + \epsilon \cos\theta)m$ and $(1 - \epsilon \cos\theta)m$ coincide for $0 < \epsilon < 1$, so $q = 2$.

Use the bounded vector of observation functions

$$\phi(x) = \left(\frac{\cos 2x}{a_2}, \frac{\sin 2x}{a_2}, \frac{\cos 3x}{a_3}, \frac{\sin 3x}{a_3} \right).$$

Its conditional mean is $v(\theta) = (\cos 2\theta, \sin 2\theta, \cos 3\theta, \sin 3\theta)$, whose squared norm is two. Equality of two such vectors makes both $2(\theta - \eta)$ and $3(\theta - \eta)$ multiples of 2π, hence $\theta = \eta$ on the circle. Apply Lemma X.6.9. Point populations are handled by Lemma X.6.1. □

X.6.6 Scope and comparison

Positive entire kernels with every replication threshold $q \in \mathbb{N} \cup \{\infty\}$ are constructed in [99, Part V]. Theorem X.6.4 transfers their exact thresholds $q \geq 3$ to unknown nonconstant refresh schedules without increasing the visit count. The exceptional $q = 2$ behavior is real: Proposition X.6.10 and Theorem X.6.7 realize its two alternatives.

Classical grouped-sample work [103] gives sharp bounds for finite mixtures of arbitrary components. Finite-state nonparametric HMM identification under full-rank transitions and linearly independent emissions is treated in [104]. General-state translation HMMs with unknown noise and tail hypotheses are treated in [105]. These are distinct experiments and assumptions. The present statements use known emission families and the restricted refresh transitions (X.6.1).

The interval-partition inversion is elementary Möbius inversion, not a newly claimed general principle. A targeted search of primary sources did not establish historical priority for the combined all-order schedule theorem.

No conclusion is asserted here for arbitrary unknown transitions, state-dependent retention, mixtures of unknown retention schedules, or finite-sample minimax rates in the unrestricted models. The known-retention bound in Theorem X.6.5 is not an unknown-retention stability theorem. An inverse modulus on unbounded Weibull parameters does not follow merely from injectivity.

X.7 What has and has not been closed

The proved conclusions are exact theorems about the parameter families stated above. Several mechanisms transfer to a new setting: local irregularity determines an exact global recurrence width; elliptic period independence classifies averages in classical mechanics; spectral convexity combined with exponential independence determines the arithmetic nature of an optimizer; and interval-partition inversion turns arbitrary positive refresh schedules into shared-latent data. The strongest statements concern an entire family, rather than a fitted recurrence, a numerical sample or one exceptional parameter value.

This note does not establish a famous conjecture, the irrationality of Euler's constant, an exact quantum spectrum, or an unrestricted inverse spectral classification. In particular the all-graph design-comparison leading constant remains unproved, and the condition displayed at the end of the matching section is a proposed sufficient condition, not a theorem. Historical novelty of the combined statements requires a wider specialist literature review. Assessment at the level of a particular journal or award is not a mathematical consequence of the proofs.

Verification record. The analytic arguments were cross-read independently. Symbolic checks covered the Duffing elliptic transform and moment recurrence, exact recurrence certificates, graph boundary spectra and the split-graph stationarity equation. Frame computations covered 960 matching cases; selected high-precision quadratures checked the Duffing and corner formulas. These checks support error detection; the proofs, not the checks, carry the universal statements.

Part XI
Diagonal Grade and Homogeneous Algorithms

Manuscript overview. We address two existing, explicitly stated mathematical questions. First, the squared-multinomial generating function on n nonzero complex weights has minimum rational-diagonal variable count exactly n, including colliding and nonreal weights. This proves the all-order banana diagonal-grade expectation of Harder and Kramer–Miller, Example 3.10. The lower bound follows from a size-$(n-1)$ unipotent monodromy block, obtained through two Fourier transforms of Bessel connections, and their published monodromy bound for rational diagonals. Second, a closed bounded balanced star-shaped subset of ℓ_1 admits exact recovery of a continuous linear scalar functional with two adaptive norm-one linear measurements, while every globally positively homogeneous algorithm using any finite number of the admissible measurements has worst-case error at least one. This gives a negative answer to the general comparison in Krieg–Kritzer Open Problem 1, already under the weakening of convexity alone. The separation persists for the linear span of the measurement pool; an approximate version holds on absorbing sets with nonempty interior. The two arguments are presented in full with their external inputs.

Status. An AI-assisted research draft, internally cross-checked and not externally peer reviewed. The cited source statements were checked directly; no exhaustive historical priority claim or journal-level assessment is made. These are two distinct published targets: an all-order expectation and an open comparison question. They are not a proof of Christol's full conjecture or of every homogeneous-information variant.

Exact targets, proof dependencies and scope

Harder–Kramer-Miller's published Example 3.10 verifies banana diagonal grade in the first five loop orders and leaves the equality in higher orders as an explicit expectation. Section 1 proves all orders, as well as the weighted extension. The proof does not assume the unpublished all-mass minimal-order theorem from the supplied manuscript [032]. It adapts its method but supplies the Bessel irreducibility, two-transform block argument, boundary-term comparison and specialization needed here. The external inputs are standard Bessel differential Galois theory, formal stationary phase for holonomic differential modules, and the published diagonal-grade lower bound.

Krieg–Kritzer's Open Problem 1 asks for universal constants comparing globally positively homogeneous algorithms with unrestricted adaptive algorithms when one or more assumptions of a linear problem are weakened. Section 2 disproves that universal assertion by weakening only convexity. It follows the original convention that the algorithm is homogeneous on the ambient vector space. It leaves variants retaining convexity and the paper's separate randomized question open. The proof is elementary and independent of any unpublished result in the supplied collection.

The derivations were checked independently for the following potential failure points: the vanishing-cycle rather than nearby-cycle convention in Fourier transform; survival of the simple factor after both analytic Laplace transforms; specialization preserving nonzero weights and every coefficient identity; the domain of the homogeneity requirement; and realizability of the finite adversarial transcript.

XI.1 The banana diagonal-grade expectation in every dimension

XI.1.1 The existing statement and the precise conclusion

Harder and Kramer–Miller [106, Example 3.10] consider

$$H_\ell(z) = \sum_{k\ge 0} \left(\sum_{k_0+\cdots+k_\ell=k} \binom{k}{k_0,\dots,k_\ell}^2 \right) z^k.$$

They establish $\mathrm{dg}(H_\ell) = \ell$ for $\ell \le 5$ and explicitly expect the same equality for every $\ell \ge 6$. Their convention is that diagonal grade is the minimum number of rational-function variables minus one. This is an explicitly stated expectation in an existing paper, not a new conjecture introduced here. We prove that expectation, as well as the following uniform version.

Theorem XI.1.1. *Let $n \ge 2$ and $a_1,\dots,a_n \in \mathbb{C}^\times$. Put*

$$c_k(a) = \sum_{k_1+\cdots+k_n=k} \binom{k}{k_1,\dots,k_n}^2 \prod_{i=1}^n a_i^{k_i}, \qquad F_a(z) = \sum_{k\ge 0} c_k(a) z^k.$$

The smallest number of variables in a rational function, regular at the origin and with complex coefficients, whose complete diagonal is F_a, is exactly n. In particular, for every $\ell \ge 1$,

$$\mathrm{dg}(H_\ell) = \ell.$$

No distinctness, positivity, or noncollision assumption on the weights is required. If the weights lie in a number field K, the same minimum holds when the representing rational function is required to be over K.

The proof below does not use the full minimal-order theorem for banana periods. It proves only the monodromy lower bound needed here, by adapting the two-Fourier-transform argument of the supplied manuscript [032]. The external geometric input for diagonal grade is the published theorem of Harder–Kramer-Miller; the Fourier input is stated explicitly below.

XI.1.2 An explicit upper bound

Set

$$P_a(x_1,\dots,x_{n-1}) = (1 + x_1 + \cdots + x_{n-1}) \left(a_n + \frac{a_1}{x_1} + \cdots + \frac{a_{n-1}}{x_{n-1}} \right).$$

Then $\mathrm{CT}(P_a^k) = c_k(a)$ by the multinomial theorem. Thus

$$F_a = \mathrm{Diag} \frac{1}{1 - x_0 x_1 \cdots x_{n-1} P_a(x_1,\dots,x_{n-1})}. \tag{XI.1.1}$$

The denominator is a polynomial with constant term one: the exponent of each x_i in P_a is at least -1, and its multiplier contains x_0. In the geometric expansion, the exponent of x_0 selects k; the remaining equal-exponent conditions select exactly the constant term of P_a^k. This proves the upper bound n over the field generated by the weights.

XI.1.3 The published lower-bound theorem

For a regular singular differential module E, let $\nu_0(E)$ be the largest Jordan-block size of the unipotent part of its local monodromy at zero. The following consequence of [106, Theorem 2.5 and Corollary 2.6] is the only diagonal-grade theorem we need:

> If a nonrational series over a number field is the diagonal of a rational function in $N \geq 2$ variables, the unipotent part of the local monodromy of its minimal differential module at zero has Jordan-block size at most $N-1$.

We use the theorem only in this range; a rational function is already a diagonal in one variable. In particular, a block of size $n-1 \geq 2$ gives the required lower bound $N \geq n$. Notice that a block occurring in any composition factor suffices: passing to subobjects and quotients cannot increase the nilpotence index of a unipotent matrix.

XI.1.4 A monodromy block without assuming minimal order

Write $a_i = m_i^2$ with $m_i \neq 0$ and set

$$G(y) = \prod_{i=1}^{n} I_0(2m_i\sqrt{y}) = \sum_{k\geq 0} \frac{c_k(a)}{(k!)^2} y^k, \qquad H(v) = \sum_{k\geq 0} \frac{c_k(a)}{k!} v^k.$$

These definitions are independent of the signs of the square roots. Let $\mathcal{D}_y = \mathbb{C}[y]\langle\partial_y\rangle$ and $j : \mathbb{G}_m \hookrightarrow \mathbb{A}^1$.

Lemma XI.1.2. *The connection $M = (\mathcal{D}_y G)|_{\mathbb{G}_m}$ is irreducible. At zero its monodromy is unipotent and has a Jordan block of size $n+1$. At infinity all its slopes are at most $1/2$.*

Proof. For each distinct weight b_j occurring with multiplicity h_j, let B_j be the connection of $\theta_y^2 - b_j y$. The Bessel differential Galois group is SL_2; it is also the $r = 2$ case of [107, Corollary 4.4.8]. For distinct b_j, the simultaneous group is $\prod_j \mathrm{SL}_2$. Indeed a proper connected subdirect product would identify two standard representations up to duality and a rank-one character. Trivial determinants make that character finite, so it disappears over a finite algebraic extension. But the formal exponential sets $\{2\sqrt{b_i}\sqrt{y}, -2\sqrt{b_i}\sqrt{y}\}$ and $\{2\sqrt{b_j}\sqrt{y}, -2\sqrt{b_j}\sqrt{y}\}$ can then agree only if $b_i = b_j$. Duality does not change either set. This contradiction proves the product assertion by Goursat's lemma for the simple Lie algebra $\mathfrak{sl}_2$.

Consequently $\bigotimes_j \mathrm{Sym}^{h_j} B_j$ is an irreducible connection. Evaluation on the indicated Bessel solutions defines a nonzero morphism of this connection into analytic functions, with the tensor cyclic vector evaluating to G. An everywhere nonsingular irreducible connection on $\mathbb{G}_m$ is simple as a $\mathcal{D}$-module; it has no punctual submodule or quotient. Thus the evaluation map is injective and its image is exactly M.

At zero each B_j has one unipotent block of size two. With V_h denoting the hth symmetric power of the standard $\mathfrak{sl}_2$ representation, decompose

$$W = \bigotimes_j V_{h_j} = \bigoplus_{d\geq 0} V_d^{\oplus e_d}.$$

The diagonal raising operator gives a monodromy block of size $d+1$ for each V_d. The highest summand V_n occurs once because $\sum h_j = n$, so a block of size $n+1$ occurs. At infinity the formal solutions are products of $e^{\pm 2\sqrt{b_j}\sqrt{y}} y^{-1/4}$ times formal series; their slopes are $1/2$, or zero when an exponential sum vanishes. Collisions of nonzero sums do not introduce higher slopes. □

We record precisely the Fourier facts used next. With the convention $y \mapsto -\partial_s$, $\partial_y \mapsto s$, Fourier transform is an exact equivalence of holonomic Weyl modules and interchanges $\mathcal{O}$ and δ_0. For a minimal extension of a regular singular germ with unipotent monodromy T, its vanishing cycles are $\mathrm{im}(T-1)$, with the induced monodromy. Thus a block of size b becomes a block of size $b-1$, disappearing for $b = 1$. If the only finite singularity is zero and the slopes at infinity are at most one, formal stationary phase identifies the germ at infinity of its Fourier transform with precisely these vanishing cycles. In particular its rank is their dimension. For input slopes less

than one, the Fourier transform has no nonzero finite singularity; a slope $q < 1$ at the original infinity becomes $q/(1-q)$ at zero. These are the formal stationary-phase theorem of [108, §1], with the regular holonomic extension convention in [109, Remark 5.7, p. 326]. In particular, [109, equations (5.16)–(5.18), p. 329] explicitly identifies the minimal-extension regular contribution with $\phi = \mathrm{im}(T-1)$ and separates the slope-> 1 contribution, which is absent here. The slope correspondence follows from [109, Theorem 5.1 and Remarks 5.3].

Let

$$K_1 = j_{!*}M, \qquad N = \iota^*\big(\mathcal{F}K_1|_{\mathbb{G}_m}\big), \quad \iota(v) = 1/v, \qquad S = \mathcal{F}(j_{!*}N).$$

Fourier transform and intermediate extension preserve simplicity. The first transform has positive rank $\sum_d d e_d$, is nonsingular on $\mathbb{G}_m$, and has at infinity the blocks of sizes d for $d \geq 1$. At zero its slopes are at most one. After inversion, N therefore has those blocks at zero and slopes at most one at infinity. A second application gives

$$\operatorname{rank} S = \sum_{d\geq 2}(d-1)e_d > 0, \qquad S_\infty : \quad \text{blocks of sizes } d-1 \ (d \geq 2). \tag{XI.1.2}$$

In particular S_∞ is regular singular and contains a block of size $n-1$. The simple module S is not $\mathcal{O}$: otherwise its inverse Fourier transform $j_{!*}N$ would be punctual, although N has positive rank.

Lemma XI.1.3. *The simple module S is a composition factor of the Weyl module generated by*

$$\varpi(t) = t^{-1}F_a(t^{-1}).$$

Consequently the minimal differential module of F_a at zero has unipotent nilpotence index at least $n-1$.

Proof. Put $R = \sum_i |m_i|$. We have $|c_k(a)| \leq R^{2k}$, so G has growth $O(e^{2R\sqrt{|y|}})$ and H has exponential type at most R^2. Termwise integration, justified by these bounds, gives

$$\widehat{G}(s) = \int_0^\infty e^{-sy}G(y)\,dy = s^{-1}H(s^{-1}), \qquad \varpi(t) = \int_0^\infty e^{-tv}H(v)\,dv. \tag{XI.1.3}$$

The first identity holds for $\Re s > 0$, the second for $\Re t > R^2$.

Here is the boundary-term bookkeeping. If $\mathcal{K}$ is the Weyl module generated by finitely many entire functions of exponential type, then their Laplace transforms together with $\mathbb{C}[s]$ form a Weyl module $\mathcal{L}$, and

$$0 \longrightarrow \mathbb{C}[s] \longrightarrow \mathcal{L} \longrightarrow \mathcal{F}\mathcal{K} \longrightarrow 0.$$

Integration by parts gives $\widehat{yh} = -\partial_s\widehat{h}$ and $\widehat{h'} = s\widehat{h} - h(0)$, proving stability and the quotient description. The map into that quotient is injective because a Laplace transform tends to zero on the positive real axis; if it is polynomial, it is zero, and uniqueness of the Laplace transform gives $h = 0$. Derivatives and polynomial multiples retain exponential-type bounds, by Cauchy's estimates.

Since $\mathcal{D}_y G$ restricts to M, its composition factors are K_1 and possibly copies of δ_0. The first Laplace transform therefore puts $\mathcal{D}_s\widehat{G}$ in a module whose factors are $\mathcal{F}K_1$ once and copies of $\mathcal{O}$. Inverting s and multiplying by the unit v^{-1}, then extending the entire function H across zero, shows that the factors of $\mathcal{D}_v H$ are among $j_{!*}N$ once, $\mathcal{O}$, and δ_0. Applying the second Laplace transform in (XI.1.3), the factors of $\mathcal{D}_t\varpi$ are among S once, $\mathcal{O}$, and δ_0.

Suppose S did not occur. On $\mathbb{G}_m$, the module $E = \mathbb{C}[t,t^{-1}]\mathcal{D}_t\varpi$ would then be an iterated extension of trivial connections, realized by functions single-valued near infinity. Choose successive lifts u_i of horizontal generators of its trivial quotients. They can be chosen to form an $\mathbb{C}[t,t^{-1}]$-basis, because each quotient is free, and $\partial_t u_i$ is a Laurent-polynomial combination of earlier u_j.

Induction shows that every u_i is a Laurent polynomial: an antiderivative of a Laurent polynomial is a Laurent polynomial plus a multiple of $\log t$, and single-valuedness eliminates the logarithm. Thus ϖ would be a Laurent polynomial. This would make G a polynomial, by its coefficient definition. But every nonzero-scale $I_0(2m_i\sqrt{y})$ is an entire function with infinitely many zeros; their product has infinitely many zeros and equals one at zero. This contradiction proves that S occurs.

Regular singular local monodromy is exact on subquotients. Equation (XI.1.2) therefore gives a block of size at least $n-1$ in the local monodromy of $\mathcal{D}_t\varpi$ at infinity. Changing variables $t = 1/z$ and multiplying by z^{-1} preserves the Jordan-block sizes, yielding the asserted bound for F_a. □

XI.1.5 Proof of the diagonal-grade theorem and descent

First suppose the weights are algebraic and $n \geq 3$. Lemma XI.1.3 gives a block of size $n-1 \geq 2$, so F_a is nonrational. The Harder–Kramer-Miller bound implies that any number-field rational diagonal representation requires at least n variables. Formula (XI.1.1) attains that bound. For $n = 2$ the elementary identity

$$F_{a,b}(z) = \bigl(1 - 2(a+b)z + (a-b)^2 z^2\bigr)^{-1/2}$$

shows nonrationality when $ab \neq 0$: if $a \neq b$, the quadratic has discriminant $16ab \neq 0$, while if $a = b$ it is a nonconstant linear polynomial. Thus one variable is impossible and two suffice.

Finally suppose complex weights admitted a rational representation P/Q in fewer than n variables. Normalize $Q(0) = 1$ and let A be the finitely generated $\mathbb{Q}$-subalgebra of $\mathbb{C}$ generated by the coefficients of P, Q, the weights a_i, and their inverses. Every Taylor coefficient of P/Q belongs to A, and every diagonal coefficient identity belongs to A. Choose a maximal ideal of A. By Zariski's lemma its residue field is a finite extension of $\mathbb{Q}$. Specialization preserves $Q(0) = 1$, all infinitely many coefficient identities, and the nonvanishing of every a_i (their inverses lie in A). It would therefore produce a forbidden number-field representation in fewer than n variables. This proves the assertion even for complex coefficients and arbitrary nonzero complex weights. The explicit upper bound proves the assertion over the original number field as well. □

Corollary XI.1.4. *If a finite tuple of complex weights contains exactly h nonzero entries, its squared-multinomial generating function has minimum rational-diagonal variable count* $\max(1, h)$.

Proof. Zero entries can be deleted in the coefficient formula. The theorem handles $h \geq 2$. For $h = 1$ the series is $(1 - az)^{-1}$, and for $h = 0$ it is 1; both are rational and use one variable. □

XI.1.6 What remains open in the Lairez–Vanhove direction

Lairez–Vanhove [110, Conjecture 4.0.1, equations (4.6)–(4.8)] predict, generically,

$$q_o(t) = t^{\alpha(n)} \prod_{\mu} (t - \mu)\widetilde{q}(t), \qquad \mu = \left(\sum_i \varepsilon_i m_i\right)^2,$$

where the product keeps distinct thresholds and $\widetilde{q}$ accounts for apparent singularities. Their conjecture does not specify a formula for $\alpha(n)$; it only says $\alpha(n) \in \mathbb{N}$. Neither the above diagonal theorem nor the exact Mellin recurrence-width formula proves simplicity of threshold factors in the minimal-order operator. A concrete local obstruction is that

$$L_3 = 2x\partial_x^3 + 3\partial_x^2$$

has solution space $\langle 1, x, x^{1/2} \rangle$ and a simple leading zero, whereas its order-two right factor with solution space $\langle x, x^{1/2} \rangle$ is

$$L_2 = 2x^2\partial_x^2 - x\partial_x + 1, \qquad L_3 = (x^{-1}\partial_x)L_2.$$

Thus passing from a simple-threshold higher-order equation to a minimal factor can increase the leading multiplicity. The finite-jet condition identified in [032, Proposition `bs:prop:jetdefect`] is still needed in that approach. We claim no closure of the remaining Lairez–Vanhove leading-coefficient assertion.

XI.2 Failure of homogeneous optimality on star-shaped sets

XI.2.1 The externally posed question and its conventions

Krieg and Kritzer [111, Section 3, Open Problem 1] ask whether positively homogeneous algorithms remain optimal when one or more of the assumptions defining a linear problem are weakened. The original assumptions are: a convex balanced input set F in a vector space X, a linear solution operator $S : X \to Y$, and a pool Λ of linear measurements. Their weakening $(1')$ permits any input set satisfying

$$x \in F, \quad 0 \le t \le 1 \quad \Longrightarrow \quad tx \in F.$$

Other allowed weakenings replace linearity of S or of the measurements by positive homogeneity. The question asks for universal constants $C > 0$ and $b \in \mathbb{N}$ such that

$$\inf_{\substack{A \in \mathcal{A}_{bn} \\ A \text{ positively homogeneous}}} \operatorname{err}(A; S, F) \;\le\; C \inf_{A \in \mathcal{A}_n} \operatorname{err}(A; S, F) \tag{XI.2.1}$$

for every such problem and every $n \in \mathbb{N}$.

Here $\mathcal{A}_n$ consists of deterministic algorithms using at most n measurements from Λ, chosen adaptively from the preceding exact real answers; their selection rules and recovery maps are unrestricted. The error is

$$\operatorname{err}(A; S, F) = \sup_{x \in F} |S(x) - A(x)|.$$

An algorithm is a map $A : X \to Y$. As explicitly stipulated in [111, Section 1.2], positive homogeneity means

$$A(tx) = tA(x) \qquad (x \in X,\ t \ge 0). \tag{XI.2.2}$$

Thus the condition is imposed on the ambient space, not merely on pairs x, tx belonging to F. We use precisely this convention.

XI.2.2 The example

Work over the real scalars. Let $X = \ell_1(\mathbb{N})$ with its usual norm, let e_j denote its standard unit vectors, and set

$$F = \bigcup_{j \ge 1} \{te_j : -1 \le t \le 1\}, \qquad S(x) = \sum_{j=1}^{\infty} x_j. \tag{XI.2.3}$$

Define continuous linear functionals

$$L_0(x) = \sum_{j=1}^{\infty} \frac{x_j}{j}, \qquad L_k(x) = \sum_{j=1}^{k} x_j \quad (k \ge 1), \qquad \Lambda = \{L_0, L_1, L_2, \ldots\}. \tag{XI.2.4}$$

All these functionals, and S, have operator norm one. The set F is nonempty, bounded, balanced, and star-shaped. It is closed: if a norm-convergent sequence from F has nonzero limit x, some coordinate of x is nonzero; that coordinate is then nonzero in all sufficiently late terms, forcing all those terms onto the same coordinate segment. The limit belongs to that closed segment. The set is not convex, since $(e_1 + e_2)/2 \notin F$.

Theorem XI.2.1 (Exact separation)**.** *For the problem* (XI.2.3)–(XI.2.4)*,*

$$\inf_{A \in \mathcal{A}_2} \operatorname{err}(A; S, F) = 0, \qquad \inf_{\substack{A \in \mathcal{A}_n \\ A \text{ positively homogeneous}}} \operatorname{err}(A; S, F) = 1 \quad \text{for every finite } n \ge 1. \tag{XI.2.5}$$

The zero in the first equality is attained by a specified two-query algorithm, rather than being merely an unattained infimum.

Proof. First read $y = L_0(x)$. If $y = 0$, output zero. Otherwise set

$$k(y) = \left\lceil \frac{1}{|y|} \right\rceil,$$

read $L_{k(y)}(x)$, and output that answer. This defines an algorithm on all of X; in particular $k(y)$ is a positive finite integer for every nonzero finite y. For $x = te_j \in F$ with $t \neq 0$, we have $y = t/j$ and therefore

$$\frac{1}{|y|} = \frac{j}{|t|} \geq j.$$

Consequently $k(y) \geq j$, and the second answer is $L_{k(y)}(te_j) = t = S(te_j)$. On F, the first answer vanishes only when $x = 0$. The algorithm is exact on F.

Now fix any algorithm $A \in \mathcal{A}_n$ satisfying (XI.2.2), where n is finite. Put $v_j = je_j \in X$. We claim that all sufficiently late v_j produce one and the same complete transcript for A.

To see this, construct the transcript successively. Whenever the selected query is L_0, prescribe the answer 1. Whenever it is L_k for $k \geq 1$, prescribe the answer 0. The transcript fixes each successive query, even if selection is discontinuous. At most n prefix queries occur, so their indices have a finite maximum M (put $M = 0$ if none occurs). For every $j > M$ we have

$$L_0(v_j) = 1, \qquad L_k(v_j) = 0$$

for every prefix query on this transcript. Induction on the stages shows that each such v_j really produces the prescribed transcript, including the stopping decision if fewer than n queries are used. Hence there is a single real number c such that

$$A(v_j) = c \qquad (j > M).$$

By positive homogeneity on X,

$$A(e_j) = \frac{A(v_j)}{j} = \frac{c}{j} \qquad (j > M).$$

Since $e_j \in F$ and $S(e_j) = 1$, it follows that

$$\operatorname{err}(A; S, F) \geq \sup_{j>M} \left| 1 - \frac{c}{j} \right| \geq 1.$$

Finally the zero algorithm is positively homogeneous and has error $\sup_{x \in F} |S(x)| = 1$, proving equality. □

Corollary XI.2.2 (Negative answer to Open Problem 1)**.** *There are no universal $C > 0$ and finite integer $b \geq 1$ satisfying (XI.2.1) for the class of problems specified in Krieg–Kritzer Open Problem 1. This fails already when only condition (1) is weakened to (1′), while F remains closed, bounded and balanced, the ambient space is Banach, and the solution and all admissible measurements are continuous linear functionals of norm one.*

Proof. Apply (XI.2.1) to this one problem at $n = 2$. Its right side is zero for every C, while its left side is one for every finite b by Theorem XI.2.1. □

XI.2.3 Robustness of the obstruction and exact scope

Proposition XI.2.3 (Allowing finite linear combinations)**.** *The separation in Theorem XI.2.1 remains valid if the measurement pool is enlarged to the full real linear span of $\{L_0, L_1, L_2, \ldots\}$, or to its intersection with the closed unit ball of X^*.*

Proof. Every functional in the enlarged span has the form

$$L(x) = aL_0(x) + \sum_{r=1}^{q} b_r L_{k_r}(x)$$

with finitely many indices. Its coefficient a is intrinsic, because the coefficient sequence of L_0 is not finitely supported. For all $j > \max_r k_r$, one has $L(v_j) = a$. In the lower-bound proof, prescribe this answer a to each selected query. Only finitely many queries are selected, so one can choose a single finite M beyond all their prefix supports. The same common transcript and homogeneity argument follows. The original two-query algorithm and the zero algorithm remain admissible. □

Proposition XI.2.4 (Input sets with nonempty interior)**.** *For $0 < \delta < 1$, replace F by*

$$F_\delta = F \cup \delta B_{\ell_1}.$$

This set is closed, bounded, balanced and star-shaped, and contains a neighborhood of zero. With the same solution and measurement pool,

$$\inf_{A \in \mathcal{A}_2} \operatorname{err}(A; S, F_\delta) \le \delta, \qquad \inf_{\substack{A \in \mathcal{A}_n \\ A \text{ positively homogeneous}}} \operatorname{err}(A; S, F_\delta) = 1$$

for every finite n. Therefore the universal comparison also fails if the allowed star-shaped input sets are required to be absorbing and to have nonempty interior.

Proof. The stated properties of F_δ follow directly from those of its two constituents. The two-query algorithm is exact on F. For $x \in \delta B_{\ell_1}$, if its first answer vanishes its error is $|S(x)| \le \delta$; otherwise its error is

$$|S(x) - L_{k(y)}(x)| = \left| \sum_{j > k(y)} x_j \right| \le \|x\|_1 \le \delta.$$

The homogeneous lower bound follows from $F \subset F_\delta$, and the zero algorithm still has error one because $F_\delta \subset B_{\ell_1}$. For any proposed universal C, b, choose $0 < \delta < \min\{1, 1/C\}$ and set $n = 2$. □

Remark XI.2.5 (Why the two-query algorithm is not globally homogeneous)*.* Although the displayed two-query algorithm agrees with the linear map S on F, its extension to X is not homogeneous. For $j > 1$, the input $v_j = je_j$ gives first answer 1, so the second query is L_1 and the output is zero, whereas the input e_j gives output one. No alternative finite-query globally homogeneous extension can avoid the lower bound. If homogeneity were required only within F, the exact two-query algorithm would satisfy that different condition.

Remark XI.2.6 (Information cost and other open questions)*.* The model counts queried functionals, not the arithmetic cost of evaluating a particular prefix functional L_k. The second index can become arbitrarily large when $|y|$ is small; this is permitted by the adaptive exact-information model in the cited question. Every selected functional is continuous and norm one. The query selection is Borel measurable, so no nonmeasurable choice is used. This example is nonconvex and does not contradict the positive theorem for convex balanced sets in [111]. It does not settle the variants that retain convexity but weaken the solution or measurement assumption, the randomized Open Problem 2 of that paper, or the unrestricted two-continuous-query problem in $\mathbb{R}^3$ posed in [112].

Corollary XI.2.7 (The same separation with standard information)**.** *The exact separation can also be realized on a Banach space of functions using the complete class of point evaluations as information.*

Proof. Let $D = \{0, 1, 2, \ldots\}$ and let X consist of the functions $f : D \to \mathbb{R}$ for which $(f(k))_{k\geq 1}$ converges and

$$f(0) = \sum_{k\geq 1} \frac{f(k)}{k(k+1)}.$$

With the supremum norm this is a Banach space isometric to the space of convergent sequences. Define $S(f) = \lim_{k\to\infty} f(k)$ and take all evaluations $f \mapsto f(k)$, $k \in D$, as measurements. The solution and every measurement have norm one. Let $h_j(k) = \mathbf{1}_{\{k\geq j\}}$ for $k \geq 1$ and $h_j(0) = 1/j$. The telescoping sum $\sum_{k\geq j} 1/[k(k+1)] = 1/j$ shows $h_j \in X$. Put $F = \bigcup_{j\geq 1}[-h_j, h_j]$. This set is closed, bounded, balanced and star-shaped: convergence of points on different segments can have a limit only at zero, since their nonzero steps remain separated in supremum norm.

For $f = th_j \in F$, the first observation $y = f(0) = t/j$ is followed, when nonzero, by evaluation at $k = \lceil 1/|y| \rceil$. Then $k \geq j$ and the returned value is $t = S(f)$. For the homogeneous lower bound use $v_j = jh_j$. Its evaluation at zero is one, while its evaluation at every fixed positive integer is eventually zero as $j \to \infty$. The finite-transcript argument of Theorem XI.2.1 applies verbatim and gives $A(h_j) = c/j$ for all sufficiently large j, whereas $S(h_j) = 1$. The error is therefore at least one, and the zero algorithm attains one. The nonempty-interior variant also follows by adjoining δB_X as above. □

Source and status check. The primary arXiv record for [111], checked on 19 October 2026, identifies version 2 of 3 March 2024 as the latest version. Its Section 3 still states Open Problem 1, and Section 1.2 explicitly imposes ambient-space homogeneity. The construction and proof above are self-contained. A targeted literature search found no earlier resolution, but it is not an exhaustive priority review.

XI.3 Other attempted published targets

The following targets were investigated in parallel. None is counted as a further complete conjecture resolution in this note.

1. The remaining Lairez–Vanhove leading-coefficient assertion: the obstruction is discussed at the end of Section 1. Simple local monodromy does not by itself force simple zeros in the scalar minimal operator's leading coefficient.

2. Saito's cubic Pisot question: the assertion that $\lfloor \beta^{3^k} \rfloor$ is composite infinitely often for every cubic Pisot number was not proved. Trace-factor and congruence arguments did not eliminate the exceptional real-conjugate cases.

3. Erchenko–Jakobson–Tsypin's suspension question: the required general spectral-volume monotonicity was not proved. A reduction to a union-of-events inequality does not control the overlaps needed for the desired lower bound.

4. The fifth Bessel-moment modular-value conjecture: common differential and moment recurrences do not fix the missing Bessel-to-modular period normalization. Pairing and determinant identities alone leave a free parameter, so they do not prove the individual-period formula.

5. The Ismail–Miller Bessel-ratio conjecture was excluded as an unsolved target because a published-source preprint already claims and proves it. A separate squared-Bessel McKay strictness question yielded finite-range cumulant obstructions in this investigation, but no proof in its whole parameter range.

The two main conclusions have exact mathematical scopes. The first determines the minimum variable count of a complete infinite family; the second supplies a counterexample to an explicitly universal comparison. Neither statement determines a journal placement or award level, and independent specialist review remains the next verification step before asserting a formally established literature contribution.

Part XII

Floquet Rigidity, Logistic Mixtures, and Factorial Reduction

Manuscript overview. We give complete proofs of two statements explicitly left open in the sources cited below. First, a complex periodic potential on the integer lattice with every rectangular period at most three is Floquet isospectral to zero only if it vanishes. Together with the known existence theorem for larger periods, this gives the full rectangular period classification. The argument uses dimension reduction by coordinate averaging, followed by quadratic and cubic perturbation identities and separation of their linear poles. Second, finite mixtures of the Malik–Abraham logistic family are identifiable with arbitrary component-specific positive diagonal scales. A generic coordinate slice reduces the problem to univariate singularity separation; the proof also permits component-specific positive shapes. Finally, an elementary common-denominator theorem proves the explicit factorial-reduction family of Ben David et al. on its nondegenerate domain. Undefined continued fractions allowed by the literal printed quantifiers are identified explicitly.

XII.1 Scope, provenance, and proof status

The principal targets and conclusions are as follows.

1. **Rectangular Floquet rigidity.** Cobb–Faust–Kretschmer [113], Section 4.4, leave the all-period-three case in dimension at least three unresolved. Faust et al. [114], Problem 1, include it in the period-classification problem. Section 2 proves the all-period-three case in every dimension and hence the full rectangular classification. The existence direction for a period exceeding three is an existing result of [113]. Arbitrary nonrectangular lattices are outside the conclusion.

2. **Logistic mixture identifiability.** Shi–Zheng–Han [115], at the end of Section 2.2, explicitly ask for finite-mixture identifiability when the positive scale vector may vary between components. Section 3 proves precisely this statement, then gives a positive-shape extension. The cited source is a public preprint, not a journal publication verified here.

3. **Factorial reduction.** Section 4 proves the explicit rational-root family of Conjecture 1.3 in [116] on the nonterminating domain, and more generally whenever the canonical continuant pair is nonzero infinitely often. It also gives the corresponding infinite rational family for fixed split even-degree b. The statement is qualified because the printed parameters include examples with every later convergent equal to $0/0$. The classification in Conjecture 1.4 remains outside the result.

These are full proof arguments, not computational evidence or proposals to assume a missing lemma. The main arguments were independently checked during this research session. They have not undergone external peer review. The cited versions were inspected directly; the literature search is not an exhaustive priority certification. No journal-level or prize-level assessment is asserted.

XII.2 Rigidity for every rectangular period at most three

Theorem XII.2.1. *Let $d \geq 1$ and let $V : \mathbb{Z}^d \to \mathbb{C}$ be $(3\mathbb{Z})^d$-periodic. If $\Delta + V$ and Δ are Floquet isospectral, then $V = 0$.*

Here $\Delta f(n) = \sum_{j=1}^{d} (f(n + e_j) + f(n - e_j))$ and Floquet isospectrality means equality of the characteristic polynomials at every Floquet parameter on the unit torus. Equality there is an identity of Laurent polynomials, so we may use arbitrary nonzero complex parameters.

Put $G_d = (\mathbb{Z}/3\mathbb{Z})^d$, $N = 3^d$, and let ω be a primitive cube root of unity. For $h \in G_d$, write

$$b_h = \frac{1}{N} \sum_{n \in G_d} V(n) \omega^{-h \cdot n}.$$

After writing the Floquet multipliers as $z_j = t_j^3$, applying the diagonal gauge $f(n) = t^n u(n)$, and applying the finite Fourier transform, the Floquet matrix becomes

$$M_V(t) = D(t) + B, \qquad D(t)_{kk} = \sum_{j=1}^{d} (t_j \omega^{k_j} + t_j^{-1} \omega^{-k_j}), \qquad B_{k\ell} = b_{k-\ell}. \tag{XII.2.1}$$

These transformations are similarities for every $t \in (\mathbb{C}^*)^d$.

Lemma XII.2.2 (Averaging preserves zero isospectrality)**.** *If V is Floquet isospectral to zero, then, for every coordinate j, the $(d-1)$-dimensional potential*

$$\overline{V}_j(n_1, \ldots, \widehat{n}_j, \ldots, n_d) = \frac{1}{3} \sum_{r=0}^{2} V(n_1, \ldots, r, \ldots, n_d)$$

is Floquet isospectral to zero.

Proof. It suffices to take $j = d$. Regard the remaining $t_1, \ldots, t_{d-1}$ as indeterminates and work over their rational function field. Partition the Fourier basis by $k_d = r \in \mathbb{Z}/3\mathbb{Z}$. With $s = t_d$,

$$M_V(t) = sS + C + s^{-1} S^{-1}, \qquad S = \operatorname{diag}(I_m, \omega I_m, \omega^2 I_m), \quad m = 3^{d-1}.$$

The diagonal block C_{rr} is precisely the Fourier matrix of $\Delta_{d-1} + \overline{V}_d$ at the remaining parameters: the within-block potential entries are $b_{(k'-\ell',0)}$. Fix r and substitute $\lambda = s\omega^r + \mu$ in the characteristic polynomial. Dividing by s^{2m} and taking its leading coefficient as $s \to \infty$ gives

$$\left(\prod_{q \neq r} (\omega^r - \omega^q)^m \right) \det(\mu I_m - C_{rr}).$$

Indeed, the two complementary diagonal blocks have nonzero scalar leading terms of order s, and all couplings are of order one; the same formula also follows from their Schur complement. The corresponding limit for $V = 0$ has the identical nonzero scalar factor and the free $(d-1)$-dimensional characteristic polynomial. The assumed polynomial identity therefore proves the claim. □

Proof of Theorem XII.2.1. We induct on d, using the trivial zero-dimensional statement as the base case. In dimension zero the only Floquet matrix is the scalar potential itself, so isospectrality to zero forces that scalar to be zero. The averaging lemma and induction imply that every coordinate average $\overline{V}_j$ vanishes. Consequently

$$b_h = 0 \quad \text{whenever some } h_j = 0. \tag{XII.2.2}$$

In particular, $b_0 = 0$, and only $H = \{1,2\}^d \subset G_d$ can support b.

Take algebraically independent $a_1, \dots, a_d$ and set $t_j = sa_j$. Dividing (XII.2.1) by s gives, with $\varepsilon = s^{-1}$,

$$A(\varepsilon) = D + \varepsilon B + \varepsilon^2 E, \quad D_{kk} = \alpha_k = \sum_j a_j \omega^{k_j}, \quad E_{kk} = \eta_k = \sum_j a_j^{-1} \omega^{-k_j}.$$

The α_k are distinct in $K = \mathbb{C}(a_1, \dots, a_d)$. Thus the characteristic roots of $A(\varepsilon)$ have unique formal power series expansions in $K[[\varepsilon]]$, indexed by their constant terms α_k. Isospectrality gives exactly $\lambda_k(\varepsilon) = \alpha_k + \varepsilon^2 \eta_k$.

For completeness, normalize an eigenvector for the root with constant term α_0 by setting its zeroth component to one, and expand the remaining components as $\varepsilon u_h + \varepsilon^2 v_h + \cdots$. Define

$$L_h = \alpha_0 - \alpha_h = \sum_{j=1}^{d} (1 - \omega^{h_j}) a_j.$$

The first two off-diagonal coefficient equations, using $B_{00} = b_0 = 0$, give

$$u_h = \frac{B_{h0}}{L_h}, \qquad v_h = \sum_{\ell \neq 0} \frac{B_{h\ell} B_{\ell 0}}{L_h L_\ell} \quad (h \neq 0).$$

The ε^2 and ε^3 coefficients in the zeroth component equation therefore yield

$$0 = \sum_{h \neq 0} \frac{B_{0h} B_{h0}}{L_h} = \sum_{h \in H} \frac{b_{-h} b_h}{L_h}, \tag{XII.2.3}$$

$$0 = \sum_{h,\ell \neq 0} \frac{B_{0h} B_{h\ell} B_{\ell 0}}{L_h L_\ell} = \sum_{h,\ell \neq 0} \frac{b_{-h} b_{h-\ell} b_\ell}{L_h L_\ell}. \tag{XII.2.4}$$

The diagonal matrix E makes no contribution at order three: its only potential off-diagonal terms vanish, and its zeroth diagonal entry already accounts for $\varepsilon^2 \eta_0$. The displayed eigenvector calculation verifies this directly, without appealing to a perturbation formula with implicit hypotheses.

In a nonzero summand of (XII.2.4), $h, \ell, h - \ell$ all belong to H. In each coordinate, two nonzero elements of $\mathbb{F}_3$ have nonzero difference if and only if they are opposites. Hence $\ell = -h$, and $h - \ell = -h$. Thus

$$0 = \sum_{h \in H} \frac{b_{-h}^3}{L_h L_{-h}}. \tag{XII.2.5}$$

We now separate these rational functions by their linear poles. Set $u = 1 - \omega$ and $v = 1 - \omega^2$. Each coefficient vector of L_h has entries in $\{u, v\}$. If h contains both values 1 and 2, then L_g can be proportional to L_h, for $g \in H$, only when $g = h$. To verify this, a proportionality constant must send both u and v into $\{u, v\}$. If it does not fix them, it must interchange them, which would require $(u/v)^2 = 1$; but $u/v = -\omega \notin \{1, -1\}$. If h is constant, the only proportional forms are the two constant vectors $h = (1, \dots, 1)$ and $-h = (2, \dots, 2)$.

For every mixed h, the pole at $L_h = 0$ in (XII.2.3) therefore occurs in just one summand. Multiplying by L_h and restricting to its generic point gives $b_h b_{-h} = 0$. For the constant pair, put $S = a_1 + \cdots + a_d$; its contribution is

$$\frac{b_h b_{-h}}{S} \left(\frac{1}{u} + \frac{1}{v} \right) = \frac{b_h b_{-h}}{S},$$

because $u + v = uv = 3$. Its coefficient is also zero. We have proved

$$b_h b_{-h} = 0 \qquad (h \in H). \tag{XII.2.6}$$

Grouping (XII.2.5) into unordered opposite pairs gives

$$0 = \sum_{\{h,-h\}\subset H} \frac{b_h^3 + b_{-h}^3}{L_h L_{-h}}.$$

For a mixed pair, either of its two linear factors occurs in no other denominator, so its numerator must be zero. The constant pair is the only term with its pole at $S = 0$, and has numerator $(b_h^3 + b_{-h}^3)/3$ over S^2, so its numerator is zero as well. Therefore $b_h^3 + b_{-h}^3 = 0$ for every pair. Together with (XII.2.6), this forces $b_h = b_{-h} = 0$. All Fourier coefficients vanish, and hence $V = 0$. □

Corollary XII.2.3 (Complete rectangular classification)**.** *Let $\Lambda = q_1\mathbb{Z} \oplus \cdots \oplus q_d\mathbb{Z}$, with positive integer periods. There exists a nonzero complex Λ-periodic potential Floquet isospectral to zero if and only if $\max_j q_j \geq 4$.*

Proof. Existence for $\max q_j \geq 4$ is Theorem C of Cobb–Faust–Kretschmer. For completeness, the reduction of all smaller periods to the theorem above can be verified as follows. Coordinates of period one add scalar terms $z_j + z_j^{-1}$ to both free and perturbed matrices and may be discarded. In every coordinate of period two set its Floquet multiplier to -1. Both connecting entries $1 + z_j$ and $1 + z_j^{-1}$ then vanish. If there are r period-two coordinates and m period-three coordinates, the resulting matrix is the direct sum of 2^r period-three matrices, with potentials given by the corresponding slices of V. Therefore

$$\prod_{\nu=1}^{2^r} P_{V_\nu}(z, \lambda) = P_0(z, \lambda)^{2^r},$$

where every factor is monic of degree 3^m in λ.

The free period-three polynomial P_0 is irreducible over $F = \mathbb{C}(z_1, \ldots, z_m)$. Indeed, in $K = \mathbb{C}(t_1, \ldots, t_m)$ with $z_j = t_j^3$, the extension K/F is Galois with group $(\mathbb{Z}/3\mathbb{Z})^m$, acting by $t_j \mapsto \omega^{k_j} t_j$. The conjugates of $\sum_j (t_j + t_j^{-1})$ are exactly the 3^m distinct free eigenvalues $\sum_j (\omega^{k_j} t_j + \omega^{-k_j} t_j^{-1})$. Its minimal polynomial thus has degree 3^m and is P_0. For $m = 0$ the same assertion reads $P_0 = \lambda$. Unique factorization in $F[\lambda]$ and equal monic degrees force $P_{V_\nu} = P_0$ for every slice. Theorem XII.2.1 (or the scalar case $m = 0$) makes every slice zero. □

Relation to the published open problem. Cobb, Faust and Kretschmer, *Inverse Eigenvalue Problems, Floquet Isospectrality and the Hilbert–Chow Morphism*, arXiv:2601.09604v1, Theorem C and Section 4.4, prove existence when some rectangular period exceeds three and rigidity when at most two periods equal three. They explicitly identify $(3\mathbb{Z})^d$, $d \geq 3$, as the remaining rectangular case. Faust, Friedman, O'Malley, Ramos and Sharma, *Algebraic Properties of the Ideal of Spectral Invariants for the Discrete Laplacian*, arXiv:2602.08304v1, Problem 1, ask for a classification of arbitrary full-rank period lattices and identify the same rectangular barrier. The proof above addresses the complete rectangular subproblem; it does not classify arbitrary nonrectangular lattices. This is an internally checked proof proposal, not a claim of peer review or priority.

XII.3 The full multivariate logistic mixture problem

XII.3.1 The external target and its scope

Shi, Zheng and Han, *Identifiability of multivariate logistic mixture models*, arXiv:1208.3546v2 (1 July 2014), end of Section 2.2, printed page 3, explicitly ask whether all finite mixtures from

$$L_{\mu,\sigma}(x) = \left(1 + \sum_{k=1}^{p} e^{-(x_k-\mu_k)/\sigma_k}\right)^{-1}, \qquad \mu \in \mathbb{R}^p, \quad \sigma \in (0,\infty)^p, \tag{XII.3.1}$$

are identifiable. Their Theorem 1 treats $p = 1$, and their Theorem 2 treats $p > 1$ with one scale vector shared by all components. The open problem allows each component to have its own positive scale vector. The source is a public preprint, not a journal publication verified in this investigation. Its exact problem is answered below. A targeted literature search did not locate a later solution, but this does not establish historical priority.

The proof combines a real analytic choice of a coordinate slice with separation by the nearest complex singularities. It is independent of the proof of the source's univariate lemma.

Theorem XII.3.1 (Full finite-mixture identifiability)**.** *For every positive integer p, the functions in (XII.3.1) are finitely linearly independent over $\mathbb{C}$: if (μ_i, σ_i), $1 \le i \le m$, are pairwise distinct and*

$$\sum_{i=1}^{m} c_i L_{\mu_i,\sigma_i}(x) = 0 \qquad \text{for every } x \in \mathbb{R}^p,$$

then every c_i is zero. Consequently, equality of two finite mixture distributions implies equality of their mixing measures, including the number of distinct components, their weights, their locations, and their positive coordinate scales, up to permutation.

XII.3.2 A self-contained proof

Lemma XII.3.2 (Univariate separation by the nearest poles)**.** *The functions*

$$\ell_{a,s}(x) = \frac{1}{1 + e^{-(x-a)/s}}, \qquad a \in \mathbb{R}, \quad s > 0,$$

are finitely linearly independent.

Proof. Suppose a finite linear relation holds for distinct pairs (a_i, s_i). Each function extends meromorphically to $\mathbb{C}$, with simple poles

$$a_i + (2n+1)\pi i s_i, \qquad n \in \mathbb{Z},$$

and residue s_i at each pole. The real-axis relation therefore holds as a meromorphic identity, by the identity theorem. Let s_* be the smallest remaining scale. For every index with $s_i = s_*$, the pole $a_i + \pi i s_*$ belongs to that summand alone: a larger scale has no pole at that height, and the other summands of the same scale have different real parts of their poles. Taking residues gives $c_i s_* = 0$. Remove these terms and repeat over the finitely many scales. □

Lemma XII.3.3 (One slice separates a finite parameter list)**.** *Let $p \ge 2$, and let (μ_i, σ_i) be finitely many pairwise distinct parameters. There is a point $y \in \mathbb{R}^{p-1}$ such that the pairs*

$$(\mu_{i1} - \sigma_{i1} \log C_i(y),\ \sigma_{i1}), \qquad C_i(y) = 1 + \sum_{k=2}^{p} e^{(\mu_{ik}-y_k)/\sigma_{ik}},$$

are pairwise distinct.

Proof. Pairs whose first-coordinate scales differ are already distinct. For two indices with a common first-coordinate scale s, coincidence of the effective locations is equivalent to

$$C_i(y) = e^{(\mu_{i1}-\mu_{j1})/s} C_j(y). \tag{XII.3.2}$$

The difference of the two sides is real analytic. It is not identically zero. Indeed, if (XII.3.2) held for every y, letting all its coordinates tend to $+\infty$ would give $\mu_{i1} = \mu_{j1}$. Fixing any $k \geq 2$ and letting all the other coordinates tend to $+\infty$ would then give

$$e^{(\mu_{ik}-y_k)/\sigma_{ik}} = e^{(\mu_{jk}-y_k)/\sigma_{jk}} \qquad (y_k \in \mathbb{R}).$$

Equality of these affine logarithms forces $\sigma_{ik} = \sigma_{jk}$ and $\mu_{ik} = \mu_{jk}$. This would make the original parameters identical, a contradiction.

The zero set of a nonzero real analytic function on the connected space $\mathbb{R}^{p-1}$ has empty interior and is closed. A finite union of these sets cannot cover $\mathbb{R}^{p-1}$. Choose y outside their union. □

Proof of Theorem XII.3.1. For $p = 1$, use Lemma XII.3.2. For $p \geq 2$, choose y as in Lemma XII.3.3. On the line $x = (t, y)$, each component is exactly

$$L_{\mu_i,\sigma_i}(t, y) = C_i(y)^{-1} \ell_{\mu_{i1}-\sigma_{i1}\log C_i(y),\,\sigma_{i1}}(t). \tag{XII.3.3}$$

The univariate parameter pairs are distinct, so the restricted relation and Lemma XII.3.2 give $c_i/C_i(y) = 0$ for every i. Since $C_i(y) > 0$, every $c_i = 0$.

For the mixture claim, subtract the two finite mixtures and merge identical component parameters. Linear independence makes every resulting signed weight zero, which is precisely equality of the mixing measures. □

The slice is chosen for a fixed finite union of component lists. This does not assert that one predetermined slice identifies every possible mixture. The statement is also restricted to finitely supported mixing measures; no conclusion for arbitrary infinite mixtures is used.

XII.3.3 A shape-parameter strengthening

The same proof works with an additional, component-dependent positive shape parameter. This strengthening is proved here as a class theorem; it is not counted as a second externally posed open problem.

Theorem XII.3.4 (Positive shapes as well as arbitrary diagonal scales)**.** *For fixed $p \geq 1$, set*

$$L_{\alpha,\mu,\sigma}(x) = \left(1 + \sum_{k=1}^{p} e^{-(x_k-\mu_k)/\sigma_k}\right)^{-\alpha}, \qquad \alpha > 0, \quad \mu \in \mathbb{R}^p, \quad \sigma \in (0,\infty)^p.$$

These are cumulative distribution functions. The family is finitely linearly independent, so its finite mixtures identify all three component parameters (α, μ, σ) and their weights.

Proof. The mixed derivative of order p is

$$(\alpha)_p \left(\prod_{k=1}^{p} \frac{e^{-(x_k-\mu_k)/\sigma_k}}{\sigma_k}\right) \left(1 + \sum_{k=1}^{p} e^{-(x_k-\mu_k)/\sigma_k}\right)^{-\alpha-p},$$

which is nonnegative. The required limits at the coordinate boundaries and at $(+\infty, \ldots, +\infty)$ are zero and one. Repeated integration therefore shows that this density has the displayed CDF and total mass one. Here $(\alpha)_p = \alpha(\alpha+1)\cdots(\alpha+p-1)$.

For the univariate assertion, consider a finite relation among $[1+e^{-(z-a)/s}]^{-\alpha}$ with distinct triples (α,a,s). Put $s_*=\min s$. On the simply connected strip $|\operatorname{Im} z|<\pi s_*$ every denominator is nonzero, so it has a holomorphic logarithm agreeing with its real logarithm on the real axis. Use this logarithm to define the powers. The relation holds throughout the strip by the identity theorem.

Choose a location a occurring with scale s_*. Among terms at this location and scale, let α_* be the largest shape. Approach $z_*=a+\pi i s_*$ along $z=z_*-i\varepsilon$, where $\varepsilon\downarrow 0$. For the selected denominator,

$$1+e^{-(z-a)/s_*}=1-e^{i\varepsilon/s_*}=-i\varepsilon/s_*+O(\varepsilon^2).$$

Multiplication of the relation by ε^{α_*} kills the terms at this pole with lower shape, as well as every term whose denominator is nonzero at z_*. The selected term tends to its coefficient times a nonzero complex constant. Its coefficient is therefore zero. Repeating removes every term with the smallest scale, then all remaining scales. This proves univariate linear independence without requiring integer shapes.

For $p\ge 2$, effective triples on a coordinate slice are

$$(\alpha_i,\ \mu_{i1}-\sigma_{i1}\log C_i(y),\ \sigma_{i1}).$$

Only pairs with equal α_i and equal σ_{i1} can collide. For those pairs the proof of Lemma XII.3.3 applies unchanged. Choose a slice avoiding their finitely many proper analytic collision sets. The factorization

$$L_{\alpha_i,\mu_i,\sigma_i}(t,y)=C_i(y)^{-\alpha_i}\left[1+\exp\left(-\frac{t-\mu_{i1}+\sigma_{i1}\log C_i(y)}{\sigma_{i1}}\right)\right]^{-\alpha_i}$$

reduces the desired relation to the proved univariate result. Its nonzero factors $C_i(y)^{-\alpha_i}$ give the claim. □

Verified source. Z.Q. Shi, T.R. Zheng and J.Q. Han, *Identifiability of multivariate logistic mixture models*, arXiv:1208.3546v2, 1 July 2014, equation (3), Section 2.2, Theorems 1–2, and the final unnumbered Open Problem on printed page 3. https://arxiv.org/html/1208.3546v2.

XII.4 Factorial reduction from split first-order factors

XII.4.1 Statement and the published target

For $a, b \in \mathbb{Z}[n]$, the canonical continuants of the polynomial continued fraction PCF$[a, b]$ satisfy

$$u_n = a(n)u_{n-1} + b(n)u_{n-2}, \qquad n \geq 1, \qquad \begin{matrix} p_{-1} = 1, & p_0 = a(0), \\ q_{-1} = 0, & q_0 = 1. \end{matrix} \tag{XII.4.1}$$

In the balanced case $\deg a = d$, $\deg b = 2d$, factorial reduction means

$$\limsup_{n\to\infty} \frac{1}{n} \log \frac{\gcd(p_n, q_n)}{(n!)^d} > -\infty. \tag{XII.4.2}$$

These are the initialization and definition in [116].

Conjecture 1.3 of that article proposes factorial reduction for

$$b(n) = B \prod_{i=1}^{2d} (n - x_i), \tag{XII.4.3}$$

$$a(n) = \frac{B}{m} \prod_{i=1}^{d} (n + 1 - x_i) - m \prod_{i=d+1}^{2d} (n - x_i), \tag{XII.4.4}$$

where the parameters are rational, $B \neq 0$, and $m \neq 0, \pm\sqrt{|B|}$. Rational coefficients are interpreted through constant inflation, as specified in that article. The restriction $B \neq 0$ is implicit in the prescribed degree of b.

Here is the stronger arithmetic statement that supplies the proof.

Theorem XII.4.1. *Let $f, g \in \mathbb{Q}[n]$ both have degree $d \geq 1$ and split completely over $\mathbb{Q}$. Set*

$$a(n) = f(n + 1) - g(n), \qquad b(n) = f(n)g(n).$$

Suppose that $a, b \in \mathbb{Z}[n]$. For any fixed finite family of integral solutions of (XII.4.1), there exist a constant $C \geq 1$ and integers $D_n \geq 1$ such that

$$D_n \leq C^{n+1}, \qquad D_n \frac{u_n}{(n!)^d} \in \mathbb{Z} \tag{XII.4.5}$$

for every solution in the family and every $n \geq 0$. Consequently, for canonical p, q, at every index where $(p_n, q_n) \neq (0, 0)$,

$$\gcd(p_n, q_n) \geq \frac{(n!)^d}{C^{n+1}}. \tag{XII.4.6}$$

If $b(n) \neq 0$ for every positive integer n, the pair is never zero and the limsup in (XII.4.2) may be replaced by liminf.

There is no assertion here concerning irrationality of a continued fraction's limit, its exact factorial-reduction constant, or the full classification in Conjecture 1.4 of [116]. The product–sum mechanism is classical Euler continued-fraction algebra; see, for example, [117, Theorem 5]. The arithmetic point requiring proof here is the uniform joint denominator estimate for all rational parameters and for both canonical solutions.

XII.4.2 A joint denominator estimate

Write $(A)_k = A(A+1)\cdots(A+k-1)$ and $(A)_0 = 1$.

Lemma XII.4.2. *Fix positive rational numbers $A_1, \dots, A_d, B_1, \dots, B_d$ and $c \in \mathbb{Q} \setminus \{0\}$. Put*

$$h_k = c^k \frac{\prod_{i=1}^{d} (A_i)_k}{\prod_{i=1}^{d} (B_i)_k}.$$

There is a constant K such that, for every $M \geq 0$, a positive integer $E_M \leq \exp(K(M+1))$ clears the denominators of all $h_0, \dots, h_M$ simultaneously.

Proof. Choose a positive integer D with $DA_i = a_i \in \mathbb{Z}_{>0}$ and $DB_i = b_i \in \mathbb{Z}_{>0}$ for all i. Declare bad the finitely many primes dividing D or the numerator or denominator of c. All other primes are good. For $0 \leq k \leq M$, cancellation of D^{dk} gives

$$h_k = c^k \frac{\prod_i \prod_{j=0}^{k-1} (a_i + Dj)}{\prod_i \prod_{j=0}^{k-1} (b_i + Dj)}.$$

All integer factors in these products are at most $C_0(M+1)$, for a constant $C_0 \geq 1$ independent of k, M.

Let p be good. For each $\ell \geq 1$ and each arithmetic progression in the display, the number of its first k entries divisible by p^ℓ is either $\lfloor k/p^\ell \rfloor$ or $\lceil k/p^\ell \rceil$. This follows because D is invertible modulo p^ℓ. There are exactly d progressions in the numerator and d in the denominator. Their count difference is therefore at least $-d$ at each prime power. No prime power greater than $C_0(M+1)$ divides any entry. Thus

$$v_p(h_k) \geq -d \left\lfloor \frac{\log(C_0(M+1))}{\log p} \right\rfloor, \qquad 0 \leq k \leq M. \tag{XII.4.7}$$

Writing $T = \lceil C_0(M+1) \rceil$, the good-prime part of every denominator divides $\operatorname{lcm}(1, \dots, T)^d$. The elementary Chebyshev estimate $\log \operatorname{lcm}(1, \dots, T) = O(T)$ makes this an exponential bound in M.

For completeness the finitely many bad primes also give only exponential factors. Fix such a prime p and a positive rational A. If $v_p(A) < 0$, then $v_p(A+j) = v_p(A)$ for every integer $j \geq 0$. If $v_p(A) \geq 0$, write $A = a/b$ in lowest terms, so $p \nmid b$. The number of $j < k$ for which $p^\ell \mid a + bj$ is at most $k/p^\ell + 1$, and only $O(\log(k+1))$ powers occur. In both cases,

$$|v_p((A)_k)| \leq C_{A,p}(k+1).$$

Applying this to the fixed parameters, and including $kv_p(c)$, gives $v_p(h_k) \geq -K_p(M+1)$ uniformly for $k \leq M$. The integer

$$E_M = \operatorname{lcm}(1, \dots, T)^d \prod_{p \text{ bad}} p^{\lceil K_p(M+1) \rceil}$$

has all the required properties. In particular, this proves a common denominator bound; it does not infer one from separate bounds for individual denominators. □

XII.4.3 Factoring the recurrence

Proof of Theorem XII.4.1. Write

$$f(n) = c \prod_{i=1}^{d} (n - \alpha_i), \qquad g(n) = e \prod_{i=1}^{d} (n - \beta_i),$$

where c, e are nonzero rationals and all roots are rational. Choose an integer $N \geq 1$ greater than every α_i, β_i. For a solution u, set

$$w_n = u_n - f(n+1)u_{n-1}.$$

Substitution into the recurrence yields the exact identity

$$\begin{aligned} w_n &= -g(n)u_{n-1} + f(n)g(n)u_{n-2} \\ &= -g(n)w_{n-1}. \end{aligned}$$

Define

$$F_N = 1, \qquad F_{N+r} = \prod_{j=N+1}^{N+r} f(j+1).$$

These numbers are nonzero. Since $u_n/F_n - u_{n-1}/F_{n-1} = w_n/F_n$, summing gives

$$u_{N+r} = F_{N+r}\left(u_N + w_N \sum_{k=1}^{r} R_k\right), \qquad R_k = \prod_{j=N+1}^{N+k} \frac{-g(j)}{f(j+1)}. \tag{XII.4.8}$$

Both factors in the normalized formula have balanced Pochhammer forms:

$$\frac{F_{N+r}}{((N+r)!)^d} = \frac{c^r}{(N!)^d} \frac{\prod_{i=1}^{d}(N+2-\alpha_i)_r}{(N+1)_r^d}, \tag{XII.4.9}$$

$$R_k = \left(-\frac{e}{c}\right)^k \frac{\prod_{i=1}^{d}(N+1-\beta_i)_k}{\prod_{i=1}^{d}(N+2-\alpha_i)_k}. \tag{XII.4.10}$$

All parameters in these products are positive rational numbers. Lemma XII.4.2 supplies a common denominator at most $\exp(O(M+1))$ for the expressions (XII.4.9) with $r \leq M$, and another such denominator for the R_k with $k \leq M$. The latter also clears every partial sum in (XII.4.8). The fixed numbers u_N, w_N are rational. Multiplying these two denominators and one fixed denominator, common to the chosen finite family of solutions, proves (XII.4.5) for $n \geq N$. Increasing the constant deals with the finitely many earlier indices.

For the canonical pair, put $H_n = (n!)^d$. The integer conditions give $H_n \mid D_n p_n$ and $H_n \mid D_n q_n$. Hence $H_n \mid D_n \gcd(p_n, q_n)$ whenever the pair is nonzero, proving (XII.4.6). Finally the continuant determinant satisfies

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n+1} \prod_{j=1}^{n} b(j).$$

If every $b(j)$ is nonzero, neither row of this two-by-two matrix can vanish. Taking logarithms in (XII.4.6) now proves the stated liminf bound. □

XII.4.4 Rational coefficients and the conjectured families

Corollary XII.4.3. *The family (XII.4.3)–(XII.4.4) has factorial reduction whenever its canonical continuants are nonzero infinitely often. In particular this proves Conjecture 1.3 of [116] for every nonterminating input $b(n) \neq 0$ for $n \geq 1$, in every degree and for all the stated rational parameters.*

Proof. Take

$$f(n) = \frac{B}{m} \prod_{i=1}^{d}(n - x_i), \qquad g(n) = m \prod_{i=d+1}^{2d}(n - x_i).$$

Choose a positive integer L such that $Lf, Lg \in \mathbb{Z}[n]$. Then

$$a'(n) = La(n) = (Lf)(n+1) - (Lg)(n), \qquad b'(n) = L^2 b(n) = (Lf)(n)(Lg)(n)$$

satisfy Theorem XII.4.1. The canonical continuants obey $p'_n = L^{n+1}p_n$ and $q'_n = L^n q_n$ by their initial conditions and recurrence. This is exactly constant inflation.

The property is independent of the chosen integral scale. Indeed, for an already integral pair and an additional integer scale L,

$$L^n \gcd(p_n, q_n) \le \gcd(p'_n, q'_n) \le L^{n+1} \gcd(p_n, q_n).$$

Two different initial scales can be compared through a common multiple. Thus the lower bound proves factorial reduction in the rational-coefficient convention of the source. A nonzero subsequence suffices for its limsup definition. □

Corollary XII.4.4. *Fix a rational polynomial b of even degree $2d$, with nonzero leading coefficient, all roots rational, and $b(n) \neq 0$ for $n \ge 1$. There are infinitely many distinct rational polynomials a of degree d for which* PCF$[a, b]$ *has factorial reduction, in the rational-coefficient convention above.*

Proof. Partition the $2d$ roots into two lists of length d, and vary the allowed nonzero rational parameter m in (XII.4.4). The leading coefficient $B/m - m$ takes infinitely many values: each fixed value has at most two preimages. Every resulting polynomial has the asserted property by Corollary XII.4.3. □

This is the infinite-family assertion of Conjecture 1.2 on the balanced, nonterminating domain. It uses the source's rational-parameter convention; it does not assert an infinite integer-coefficient family for one fixed integer b without inflation.

XII.4.5 Degenerate parameters and limits of the conclusion

The printed quantifiers of Conjecture 1.3 permit positive integer roots. They can produce a formally zero continuant pair and an undefined continued fraction. For example,

$$d = 1, \quad B = 2, \quad m = 1, \quad x_1 = 2, \quad x_2 = 1$$

give

$$a(n) = n - 1, \qquad b(n) = 2(n-2)(n-1).$$

Here $(p_0, q_0) = (-1, 1)$ but $p_n = q_n = 0$ for every $n \ge 1$. Thus an unqualified assertion about numerical gcds is false on this undefined input if $\gcd(0, 0) = 0$. This is why Corollary XII.4.3 states the domain explicitly. The stronger denominator theorem remains valid even on such inputs.

The proof does not establish Conjecture 1.4, does not calculate the exact value of the exponential gcd constant, and does not supply a new irrationality result for any individual constant. The cited 2024 source explicitly presents the families as conjectures; a targeted literature search did not locate a subsequent proof. That search is not an exhaustive priority certification.

Part XIII

Further Open Problems: Proofs and Counterexamples

Manuscript overview. We develop four routes from established methods to explicitly stated open questions: local intersection multiplicity for periodic spectra, resonant Fourier coefficients for heat-flow entropies, adaptive information on Banach spaces, and heat-content tests for Bessel expansions. The main results prove an antisymmetric isospectral existence conjecture, disprove its proposed nonzero singularity pattern, classify all positive finite entropy orders with universal alternating derivatives, and refute a measurement-only extension of homogeneous algorithm optimality. Two Bessel conjectures are treated by a two-term kernel obstruction and a one-sign heat-content summability obstruction; positive heat-kernel mixtures also prove the underlying unrestricted realization conclusion unconditionally. An appendix proves the entire sufficient direction of a quadratic factorial-reduction conjecture, while explicitly leaving its necessary direction open.

Verification and priority. These are complete mathematical arguments in a research draft, checked against the cited problem statements and subjected to separate internal audits. They have not received external peer review. A targeted literature search through 19 October 2026 did not locate the new conclusions below; this does not certify priority. Existing results and methods are credited where used. In particular, the original Shannon counterexample, the first global negation of the Tsallis conjecture, and earlier unrestricted isospectral existence results are not counted as new results here.

Status register. The entries distinguish whole conjectures, a specified branch of an open problem, and a complete parameter classification. Multiple clauses of a single conjecture are not counted as independent conjectures.

Published target	Conclusion established here
Faust et al., Conjecture 1	Proved for every period $n > 3$, with exact residual length $2^m(m! - 1)$.
Faust et al., Conjecture 3	Disproved at $n = 15$ and 20; its origin multiplicity formula is proved.
Wu–Yu–Guo, all-orders parameter question	Tsallis: precisely $\alpha = 2$. Rényi: no $\alpha > 0$. Fixed derivative-order ranges remain outside the claim.
Krieg–Kritzer, Open Problem 1, measurement-only weakening	Refuted on a convex Banach unit ball with a linear target and globally Lipschitz homogeneous measurements.
Acosta Babb, Conjecture 1	Refuted already by the two-term kernel at $\alpha = 1/2$.
Acosta Babb, Conjecture 2	Refuted for $a_m = 1/\log(m + 1)$, for either parity.
Acosta Babb, conditional Corollary 5.2	Proved unconditionally, with a nonnegative realizing function.

Partial result, not a closed conjecture: Ben David et al., Conjecture 1.4: both sufficient families are proved; the necessity of those families under the published limsup-gcd definition is not proved.

XIII.1 Antisymmetric isospectral potentials: existence and singularities

The target in this section is Conjectures 1 and 3 of M. Faust, L. Friedman, G. O'Malley, R. Ramos and A. Sharma, *Algebraic Properties of the Ideal of Spectral Invariants for the Discrete Laplacian*, arXiv:2602.08304v1 (2026), Sections 4–5. Conjecture 1 asserts existence of nonzero reflection-antisymmetric potentials Floquet isospectral to zero for every period greater than three. Conjecture 3 predicts regularity of all nonzero points for odd periods, one nonzero singularity orbit for even periods greater than six, and a specific multiplicity at zero. We prove the existence assertion and the multiplicity formula, and disprove both proposed nonzero-singularity patterns. The full nonrectangular-lattice classification remains open.

Let $n \geq 3$, $m = \lfloor n/2 \rfloor$, and let C_n denote the adjacency matrix of the cycle. Put

$$E_n = \{v \in \mathbb{C}^n : v_{n-1-j} = -v_j \ \ (0 \leq j < n)\}.$$

This is an m-dimensional vector space, parametrized in the even and odd cases, respectively, by $(u_1, \ldots, u_m, -u_m, \ldots, -u_1)$ and $(u_1, \ldots, u_m, 0, -u_m, \ldots, -u_1)$. Define

$$F_n(v; \lambda) = \det(\lambda I - C_n - \operatorname{diag} v) - \det(\lambda I - C_n).$$

Reflection and determinant expansion imply that on E_n this polynomial has the form

$$F_n(v(u); \lambda) = \sum_{k=1}^{m} f_k(u) \lambda^{n-2k}.$$

Indeed reflection takes v to $-v$ without changing the characteristic polynomial. Every potential-dependent determinant term uses some diagonal entries and disjoint edges, so its parity in v agrees with the parity of n minus its degree in λ. The full-cycle permutation terms are independent of v and cancel. Thus the other coefficients vanish. Let $Z_n \subset \mathbb{A}^m$ be the finite scheme defined by $(f_1, \ldots, f_m)$. Signs of individual generators are immaterial. For a one-dimensional cycle, the only dependence of the characteristic polynomial on the Floquet multiplier z is the potential-independent term $-(z + z^{-1})$. Therefore Z_n is precisely the specialized scheme of Floquet-isospectral potentials used in the cited conjectures.

Theorem XIII.1.1. *For every $n \geq 3$, the multiplicity of the origin in Z_n is $2^{\lfloor n/2 \rfloor}$. For every $n > 3$, there exists a nonzero $n\mathbb{Z}$-periodic reflection-antisymmetric potential Floquet isospectral to zero. More precisely, the total scheme-theoretic length away from the origin is*

$$2^m(m! - 1), \qquad m = \lfloor n/2 \rfloor.$$

Here "$n\mathbb{Z}$-periodic" does not require n to be the minimal period.

Proof. Write Q_k for the homogeneous quadratic part of f_k. We first prove that the only common zero of $Q_1, \ldots, Q_m$ is the origin. Let

$$A_r(v) = \sum_{j=0}^{n-1} v_j v_{j+r}, \qquad r \in \mathbb{Z}/n\mathbb{Z},$$

with cyclic indices, and let U_a be the Chebyshev polynomial of the second kind, so the characteristic polynomial of the path on a vertices is $U_a(\lambda/2)$ and $U_0 = 1$. Deleting two vertices at cyclic distance r leaves two paths on $r - 1$ and $n - r - 1$ vertices. Expansion with respect to the diagonal potential therefore gives the exact quadratic term

$$\sum_{k=1}^{m} Q_k(u) \lambda^{n-2k} = \sum_{r=1}^{m} c_r A_r(v(u)) P_r(\lambda), \qquad P_r = U_{r-1}(\lambda/2) U_{n-r-1}(\lambda/2), \tag{XIII.1.1}$$

where $c_r = 1$, except $c_m = 1/2$ when $n = 2m$. The exceptional factor corrects the double counting of diametrically opposite pairs.

The polynomials $P_1, \dots, P_m$ are linearly independent. In fact, $P_1 = U_{n-2}(\lambda/2)$ and the elementary Chebyshev product identity gives

$$P_r - P_{r-1} = U_{n-2r}(\lambda/2), \qquad 2 \le r \le m.$$

The polynomials on the right have distinct degrees $n-4, n-6, \dots, n-2m$, all below the degree $n-2$ of P_1. Thus $Q_k(u) = 0$ for every k implies $A_r(v) = 0$ for $1 \le r \le m$. Since $A_{n-r} = A_r$, all A_r with $r \neq 0$ vanish. Antisymmetry gives $\sum_j v_j = 0$, and hence

$$\sum_{r=0}^{n-1} A_r = (\sum_j v_j)^2 = 0$$

also gives $A_0 = 0$.

Let $\omega = e^{2\pi i/n}$ and $b_h = n^{-1} \sum_j v_j \omega^{-hj}$. Fourier inversion of the autocorrelations yields

$$A_r = n \sum_h b_h b_{-h} \omega^{-hr}, \qquad b_h b_{-h} = 0 \quad \text{for all } h.$$

On the other hand, $v_{n-1-j} = -v_j$ implies $b_{-h} = -\omega^{-h} b_h$. Consequently every $b_h^2 = 0$, so $v = 0$ and $u = 0$.

The m homogeneous quadrics Q_k in m variables therefore form a regular sequence. The usual filtered regular-sequence argument shows that a local complete intersection whose lowest homogeneous terms form a regular sequence has the same associated graded ring as that homogeneous complete intersection. In the present case,

$$\operatorname{gr}(\mathbb{C}[[u_1, \dots, u_m]]/(f_1, \dots, f_m)) \cong \mathbb{C}[u_1, \dots, u_m]/(Q_1, \dots, Q_m).$$

One way to verify the filtered statement is to cancel a lowest-degree relation among the Q_k by Koszul relations, and repeat to increasing degree; regularity says that all relations are generated by these Koszul relations. Completeness justifies the iteration. The Hilbert series of the resulting homogeneous quotient is

$$\frac{(1-t^2)^m}{(1-t)^m} = (1+t)^m.$$

Its length, and therefore the multiplicity at the origin, is 2^m.

Finally, the highest homogeneous part of f_k is $(-1)^k e_k(u_1^2, \dots, u_m^2)$ and $\deg f_k = 2k$. These highest forms have no common projective zero: if all elementary symmetric functions of the u_j^2 vanish, then every $u_j^2 = 0$. Homogenization and Bezout's theorem therefore give a zero-dimensional complete intersection of total affine length $\prod_{k=1}^m 2k = 2^m m!$. This is also Theorem 4.6 of the cited paper. Subtracting the local length at zero proves the displayed formula. For $n > 3$, $m \ge 2$ and that residual length is positive, proving existence. □

XIII.1.1 Period repetition creates singular nonzero points

Lemma XIII.1.2. *Let $q \ge 3$ divide n, with $n > q$, and let $v \in E_q$ be Floquet isospectral to zero. Repeat its entries to form $w \in E_n$. Then $w \in Z_n$, and the Jacobian of the defining equations of Z_n at w has rank at most $\lfloor q/2 \rfloor$. In particular every nonzero such w is a singular point of Z_n.*

Proof. Antisymmetry is preserved because $n - 1 - j \equiv q - 1 - j \pmod q$. Floquet isospectrality is preserved under period enlargement. Explicitly, write $n = rq$ and let $T_q(\lambda)$ be the 2×2 monodromy matrix of the equation $y_{j+1} + y_{j-1} + v_j y_j = \lambda y_j$ over one q-period. Its determinant

is one, and its trace equals the free trace by Floquet isospectrality. Over the repeated period the monodromy is $T_q(\lambda)^r$, whose trace is determined polynomially by the trace of $T_q(\lambda)$ and its determinant. Thus it too has the free trace.

For the rank assertion use the full characteristic-coefficient map $\Phi_n : \mathbb{C}^n \to \mathbb{C}^n$, before restricting it to E_n. It is invariant under cyclic coordinate translations, by permutation similarity of C_n. If T is translation by q, then $Tw = w$ and

$$d\Phi_n(w)\, T = d\Phi_n(w).$$

Consequently $d\Phi_n(w) = d\Phi_n(w)P$, where $P = r^{-1}\sum_{a=0}^{r-1} T^a$ averages over the repetitions. Reflection conjugates T to T^{-1}, so P preserves E_n. Its image there is exactly the repeated copy of E_q, of dimension $\lfloor q/2 \rfloor$. Restriction and deletion of identically zero coefficients now give the claimed rank bound for $d(f_1, \dots, f_m)_w$. As $\lfloor n/2 \rfloor > \lfloor q/2 \rfloor$ and Z_n is zero-dimensional, its Zariski tangent space at w is nonzero. Its local ring is therefore not a field; equivalently, this point is singular and has multiplicity greater than one. □

Theorem XIII.1.3. *The odd-period regularity assertion in Conjecture 3 is false at $n = 15$. The even-period uniqueness assertion is false at $n = 20$. Both failures occur in infinite families of periods.*

Proof. Choose $a \in \mathbb{C}$ with $a^2 = 1 - 2i$, and put

$$v^{(5)} = (a, ia, 0, -ia, -a).$$

Direct determinant expansion gives

$$F_5(v^{(5)}; \lambda) = -a^2\lambda(a^2 - 1 + 2i) = 0.$$

Hence its threefold repetition is a nonzero point of Z_{15}. Lemma XIII.1.2 bounds the rank of the seven-variable Jacobian by two, so the point is singular. The same example works for every odd multiple of five larger than five.

For the even-period assertion, choose b with $b^2 = -2i$, and put

$$v^{(4)} = (b, ib, -ib, -b).$$

Here direct expansion gives $F_4(v^{(4)}; \lambda) = -b^2(b^2 + 2i) = 0$. At $n = 20$, repeat $v^{(4)}$ five times and $v^{(5)}$ four times. Both are nonzero singular points by the lemma. They cannot belong to the same symmetry orbit: their minimal periods are four and five, respectively, and cyclic translations, reversal, negation and complex conjugation all preserve minimal period. These are the symmetries listed in the source paper. Thus there are at least two distinct nonzero singularity orbits. The same construction works at every positive multiple of twenty. □

Exact status. This proves Conjecture 1 in its stated full range. For Conjecture 3, the origin-multiplicity clause holds in full, whereas each of its nonzero-singularity assertions is false. Conjecture 2, which imposes conjugate-reflection symmetry instead of antisymmetry, and Problem 1 for arbitrary nonrectangular period lattices are not resolved here.

XIII.2 Exact parameter obstructions for entropy complete monotonicity

Literature status. Wu, Yu, and Guo, *On the Complete Monotonicity of Rényi Entropy*, IEEE Transactions on Information Theory 71 (2025), 4895–4914, doi:10.1109/TIT.2025.3561384, arXiv:2312.01819v4, formulate in Conjecture 1 the alternating-derivative property for Tsallis entropy for all $1 < \alpha < 2$. Their Question 1 asks about the possible parameter range for the analogous Rényi property at all derivative orders; Question 2 asks the different fixed-derivative-order question, which we do not resolve. The all-parameter Tsallis conjecture is already false as a consequence of the Shannon counterexample of Gu–Sellke, arXiv:2605.11656, by continuity in α near 1. We do not claim its first disproof. The result below instead supplies a counterexample for *every fixed* parameter, and the exact universal parameter classification. The two-frequency circle construction and Gaussian-window transfer are methodological precedents of Zou, Fan, Gao, and Wang, *An Explicit Family of Log-Concave Counterexamples to the Gaussian Completely Monotone Conjecture*, arXiv:2608.30275 (2026). Their four-page paper treats Shannon entropy; the present calculation extends that mechanism to the full Tsallis and Rényi parameter families and uses a qualitative finite-difference transfer.

Let $P_t = e^{t\partial_x^2}$ on $\mathbb{R}$. For a smooth positive probability density f, write

$$F_\alpha(f) = \int_{\mathbb{R}} f(x)^\alpha \, dx, \qquad T_\alpha(f) = \frac{1 - F_\alpha(f)}{\alpha - 1}, \qquad R_\alpha(f) = \frac{-\log F_\alpha(f)}{\alpha - 1}.$$

At $\alpha = 1$ both entropies mean $-\int f \log f$. The choice of generator changes the original paper's time variable by a positive factor only.

Theorem XIII.2.1 (Universal parameter classification)**.** *Among parameters $\alpha > 0$, the inequalities*

$$(-1)^{m-1} \frac{d^m}{dt^m} T_\alpha(P_t f) \geq 0 \qquad (m \geq 1,\ t > 0)$$

hold for every smooth strictly positive probability density f with Gaussian tails if and only if $\alpha = 2$. For Rényi entropy, there is no *parameter $\alpha > 0$ for which the corresponding inequalities hold for all such f. All counterexamples can be chosen from the explicit family*

$$f_{R,a,b}(x) = \frac{(2\pi R^2)^{-1/2} e^{-x^2/(2R^2)} (1 + a\cos x + b\cos 2x)}{1 + a e^{-R^2/2} + b e^{-2R^2}}, \qquad |a| + |b| < 1.$$

The theorem asserts existence of finite R, derivative order m, and time t for each fixed excluded parameter; it does not specify a uniform derivative order or a uniform window size.

Lemma XIII.2.2 (A negative exponential coefficient)**.** *Suppose a real function has an exponential expansion $H(t) = \sum_{k \geq 0} c_k e^{-kt}$ absolutely convergent for $t \geq t_0$, and differentiable termwise for $t > t_0$. If $c_j < 0$ for some integer $j \geq 1$, then $(-1)^m H^{(m)}(t) < 0$ for some $m \geq 1$ and $t > 0$. Indeed, this holds at $t = t_0 + m/j$ for all sufficiently large m.*

Proof. At the stated time, divide the signed derivative by $(j/e)^m$. The resulting series is

$$\sum_{k \geq 1} c_k e^{-kt_0} \left(\frac{k}{j} e^{1 - k/j} \right)^m.$$

The parenthesized factor belongs to $(0, 1]$, and equals 1 only at $k = j$. Dominated convergence gives the strictly negative limit $c_j e^{-jt_0}$. □

Use normalized Lebesgue measure on the circle and put

$$u_t(x) = 1 + a e^{-t} \cos x + b e^{-4t} \cos 2x, \qquad Q_\alpha(t) = \frac{1}{2\pi} \int_0^{2\pi} u_t(x)^\alpha \, dx.$$

Define the circle negative-entropy curves

$$H_\alpha^T(t)=\frac{Q_\alpha(t)-1}{\alpha-1},\qquad H_\alpha^R(t)=\frac{\log Q_\alpha(t)}{\alpha-1}.$$

At $\alpha=1$ their common value is $H_1(t)=(2\pi)^{-1}\int u_t\log u_t$. These functions are analytic functions of $q=e^{-t}$ near $q=0$, so Lemma XIII.2.2 applies to their power-series coefficients in q.

Lemma XIII.2.3 (The resonant coefficient)**.** *For $\alpha\neq 1$, the coefficient of q^6 in H_α^T is*

$$\frac{\alpha(\alpha-2)}{8}a^2b+\frac{5}{16}\frac{\binom{\alpha}{6}}{\alpha-1}a^6.$$

The coefficient of q^6 in H_α^R is

$$\frac{\alpha(\alpha-2)}{8}a^2b+\frac{a^6}{\alpha-1}\left\{\frac{5}{16}\binom{\alpha}{6}-\frac{3}{16}\binom{\alpha}{2}\binom{\alpha}{4}+\frac{1}{24}\binom{\alpha}{2}^3\right\}.$$

At $\alpha=1$, the coefficient of q^6 in H_1 is $-a^2b/8+a^6/96$. At $\alpha=2$, the coefficient of q^4 in H_2^R is $-a^4/8$.

Proof. Only the monomials $(aq\cos x)^6$ and $3(aq\cos x)^2(bq^4\cos 2x)$ contribute to $[q^6]Q_\alpha$. Their circle averages are respectively $5a^6/16$ and $3a^2b/4$. Moreover,

$$[q^2]Q_\alpha=\tfrac{1}{2}\binom{\alpha}{2}a^2,\qquad [q^4]Q_\alpha=\tfrac{3}{8}\binom{\alpha}{4}a^4,$$

and there are no odd powers before q^6. Expanding the logarithm gives $[q^6]\log Q=c_6-c_2c_4+c_2^3/3$, proving the formula. For $\alpha=1$, expand $(1+v)\log(1+v)=v+\sum_{r\ge2}(-1)^r v^r/[r(r-1)]$. Finally, $Q_2=1+a^2q^2/2+b^2q^8/2$, which proves the last assertion. □

For any fixed $\alpha>0$ different from $1,2$, take $a=\varepsilon>0$ and $b=-\operatorname{sgn}(\alpha-2)\varepsilon$. The q^6 coefficient of either negative entropy is then $-\alpha|\alpha-2|\varepsilon^3/8+O_\alpha(\varepsilon^6)$, which is negative for sufficiently small positive ε. For $\alpha=1$, take $a=b=\varepsilon$ and use the same lemma; for Rényi order 2, take $a\neq0$, $b=0$. Thus every excluded parameter has a circle counterexample.

For the conjectured interval $1<\alpha<2$, the Tsallis part admits the fixed circle amplitudes $a=b=1/4$: indeed

$$\frac{\binom{\alpha}{6}}{\binom{\alpha}{3}}=\frac{(\alpha-3)(\alpha-4)(\alpha-5)}{120}\in(-1/5,0),$$

so the negative cubic contribution dominates the sixth-degree term.

Lemma XIII.2.4 (Transfer from the circle to the line)**.** *Fix $\alpha>0$ and $|a|+|b|<1$. If all densities $f_{R,a,b}$, $R>0$, satisfy the entropy alternating-derivative inequalities on the line, then the corresponding circle curve has $(-1)^mH^{(m)}(t)\ge0$ for every $m\ge1$ and $t>0$.*

Proof. Let $S=R^2+2t$ and $Z_R=1+ae^{-R^2/2}+be^{-2R^2}$. A Gaussian convolution calculation gives exactly

$$P_tf_{R,a,b}(x)=\frac{g_S(x)}{Z_R}\left[1+ae^{-R^2t/S}\cos\frac{R^2x}{S}+be^{-4R^2t/S}\cos\frac{2R^2x}{S}\right],$$

where $g_S(x)=(2\pi S)^{-1/2}e^{-x^2/(2S)}$. Writing $F_{\alpha,R}(t)=\int(P_tf_{R,a,b})^\alpha$, and averaging a rapidly oscillating bounded smooth periodic function against a Gaussian, yields, for each fixed $t\ge0$,

$$C_RF_{\alpha,R}(t)\longrightarrow Q_\alpha(t),\qquad C_R=\sqrt{\alpha}\,(2\pi R^2)^{(\alpha-1)/2}.$$

For completeness, substitute $x = \sqrt{S/\alpha}\, y$. The periodic factor then has frequency $R^2/\sqrt{\alpha S} \to \infty$; its amplitudes tend to ae^{-t} and be^{-4t}. Approximate it uniformly by trigonometric polynomials and use the Gaussian Fourier transform on every nonconstant mode. The prefactor is $Z_R^{-\alpha}(R^2/S)^{(\alpha-1)/2} \to 1$.

For $\alpha \neq 1$, apply this limit to

$$K_R^T(t) = \frac{C_R F_{\alpha,R}(t) - 1}{\alpha - 1}, \qquad K_R^R(t) = \frac{\log(C_R F_{\alpha,R}(t))}{\alpha - 1}.$$

Each is a positive constant multiple of a line negative-entropy curve plus a constant, or a line negative-entropy curve plus a constant, respectively. They converge pointwise to H_α^T and H_α^R.

If a smooth function K has $(-1)^m K^{(m)} \geq 0$, then $(-1)^m \Delta_h^m K(t) \geq 0$ for all $h > 0$, by repeated integration over $[0,h]^m$. These finite-difference inequalities pass to a pointwise limit. Dividing by h^m and taking $h \downarrow 0$ recovers the derivative inequalities for the smooth circle limit.

At $\alpha = 1$, write $U_{R,t}$ for the bracket in the exact convolution formula. Then

$$\int P_t f_{R,a,b} \log(P_t f_{R,a,b}) = -\tfrac{1}{2}\log(2\pi S) - \log Z_R - \frac{\mathbb{E}[X_t^2]}{2S} + Z_R^{-1} \int g_S U_{R,t} \log U_{R,t}.$$

Gaussian Fourier differentiation shows $\mathbb{E}[X_0^2]/R^2 \to 1$, hence $\mathbb{E}[X_t^2]/S \to 1$ for fixed t. Adding $\frac{1}{2}\log(2\pi R^2) + \frac{1}{2}$ therefore gives the pointwise limit $H_1(t)$. The same finite-difference argument applies. □

Proof of Theorem XIII.2.1. If all Gaussian-window densities satisfied the relevant property, Lemma XIII.2.4 would imply the circle derivative inequalities. Lemmas XIII.2.2 and XIII.2.3 contradict these inequalities for each excluded parameter. Consequently at least one finite R fails. The densities are normalized, analytic, strictly positive, and bounded above and below by positive multiples of a Gaussian, so their entropies and all time derivatives at positive time are well defined. For the surviving Tsallis parameter $\alpha = 2$, Plancherel gives

$$F_{2,R}(t) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-2t\xi^2} |\widehat{f}(\xi)|^2 \, d\xi,$$

whose derivatives alternate in sign. Since $T_2 = 1 - F_2$, this is exactly the required property. □

Scope and priority. The result concerns the ordinary linear heat flow on $\mathbb{R}$, not nonlinear diffusion, quantum entropy, or monotonicity in the entropy parameter. It supplies no low-order optimal counterexample. The original conjecture's first global negation belongs to the existing Shannon counterexample and its immediate continuity consequence. Our claimed mathematical increment is the complete fixed-parameter classification above. Searches of the original paper, the 2026 Shannon counterexample, and subsequent indexed literature, including the two-frequency Shannon construction of Zou–Fan–Gao–Wang, did not locate this classification; this is not a certificate of priority.

XIII.3 Convex input sets do not rescue homogeneous algorithms

The published problem and the new case. Krieg and Kritzer, *Homogeneous algorithms and solvable problems on cones*, Journal of Complexity **83** (2024), 101840, doi:10.1016/j.jco.2024.101840, pose in Section 3, Open Problem 1, the following question. In the deterministic adaptive information model, are positively homogeneous algorithms optimal up to universal constants in information cost and error if one or more of the assumptions of a linear problem are relaxed? The three assumptions are a convex balanced input set, a linear solution operator, and linear measurements; the corresponding relaxations are a star-shaped input set, a positively homogeneous solution operator, and positively homogeneous measurements. Algorithms are defined on the full ambient vector space, and homogeneity is required there.

An earlier counterexample treated the first relaxation, with a nonconvex input set and linear measurements. The following result treats a different case: the input set remains a closed Banach unit ball and the solution operator remains bounded linear. Only the measurement assumption is relaxed. All measurements in the example can be globally Lipschitz with Lipschitz constant at most one. In fact they are homogeneous for all real scalars, not merely positively homogeneous.

Theorem XIII.3.1 (Convex unit-ball separation)**.** *There exist a real Banach space X, its closed unit ball F, a bounded linear functional $S : X \to \mathbb{R}$ of norm one, and a countable pool Λ of globally 1-Lipschitz homogeneous measurements such that*

$$\inf_{A\in\mathcal{A}_2}\sup_{x\in F}|S(x)-A(x)|=0, \qquad \inf_{\substack{A\in\mathcal{A}_n\\ A \text{ positively homogeneous}}}\sup_{x\in F}|S(x)-A(x)|=1 \quad (n\ge 1).$$

The first infimum is attained by an explicit two-query adaptive algorithm. Here $\mathcal{A}_n$ permits arbitrary adaptive selection of at most n measurements from Λ and arbitrary recovery from their exact values.

Proof. Take $X=\ell_1(\mathbb{N})$, let e_j be its standard unit vectors, and set

$$S(x)=\sum_{j\ge 1}x_j, \qquad Tx=(x_j/j)_{j\ge 1}.$$

The operator $T : X \to X$ is bounded, injective, and has norm one. Define $\phi : [0,\infty) \to [0,1]$ by

$$\phi(t)=\begin{cases}0, & 0\le t\le \tfrac12,\\ 2t-1, & \tfrac12\le t\le 1,\\ 1, & t\ge 1.\end{cases}$$

For $x\neq 0$ put

$$Q(x)=S(x)\frac{\|Tx\|_1}{\|x\|_1}, \qquad R_k(x)=S(x)\phi\Big(\frac{k|Q(x)|}{\|x\|_1}\Big) \quad (k\ge 1),$$

and set $Q(0)=R_k(0)=0$. The admissible measurements are

$$L_0=Q/3, \qquad L_k=R_k/(1+8k) \quad (k\ge 1).$$

Every displayed function is homogeneous over the real scalars.

We first check global Lipschitz continuity, including the origin. If $\|x\|_1\ge\|y\|_1>0$, then

$$\Big|\frac{\|Tx\|_1}{\|x\|_1}-\frac{\|Ty\|_1}{\|y\|_1}\Big|\le\frac{2\|x-y\|_1}{\|x\|_1}.$$

Since $|S(y)| \le \|y\|_1$ and $\|Tx\|_1/\|x\|_1 \le 1$, this gives $|Q(x) - Q(y)| \le 3\|x - y\|_1$. The same bound holds if one point is zero. Also $|Q(x)| \le \|x\|_1$, so for the same ordering of the norms,

$$\left| \frac{|Q(x)|}{\|x\|_1} - \frac{|Q(y)|}{\|y\|_1} \right| \le \frac{4\|x - y\|_1}{\|x\|_1}.$$

Using $0 \le \phi \le 1$ and $\mathrm{Lip}(\phi) = 2$ now yields

$$|R_k(x) - R_k(y)| \le (1 + 8k)\|x - y\|_1.$$

Again the bound at zero is immediate. Therefore every L_k is globally 1-Lipschitz.

For the upper bound, query $y = L_0(x)$. If $y = 0$, return zero. Otherwise choose the finite positive integer

$$k = \left\lceil \frac{1}{3|y|} \right\rceil,$$

query $L_k(x)$, and return $(1 + 8k)L_k(x) = R_k(x)$. If $x \in F$ and $y = 0$, injectivity of T implies $S(x) = 0$. If $y \ne 0$, then $Q(x) = 3y$ and

$$\frac{k|Q(x)|}{\|x\|_1} \ge 1.$$

Thus $\phi = 1$ and the output is exactly $S(x)$. The algorithm is defined on the whole of X, though exactness is asserted only on F.

For the lower bound, fix any finite-query algorithm A that is positively homogeneous on X. Put $v_j = je_j$. Then

$$L_0(v_j) = \tfrac{1}{3}, \qquad L_k(v_j) = 0 \quad \text{whenever } j \ge 2k.$$

Follow the virtual transcript that returns $1/3$ whenever L_0 is queried and zero whenever L_k, $k \ge 1$, is queried. Every finite prefix is realized by all sufficiently large v_j, so only attainable information histories occur. Along this transcript there are only finitely many queries, so their positive indices have a finite maximum M (put $M = 0$ if there are none). All v_j with $j \ge 2M$ follow this exact transcript, including any stopping decision. Their outputs consequently equal one fixed real number c. Positive homogeneity implies

$$A(e_j) = A(v_j)/j = c/j \quad (j \ge \max\{1, 2M\}).$$

Since $e_j \in F$ and $S(e_j) = 1$, the worst-case error is at least $\lim_{j\to\infty} |1 - c/j| = 1$. The zero algorithm attains error one. □

Corollary XIII.3.2 (The measurement-only weakening in Open Problem 1)**.** *No universal constants $C > 0$ and finite $b \in \mathbb{N}$ can satisfy*

$$\inf_{\substack{A \in \mathcal{A}_{bn} \\ A \text{ positively homogeneous}}} \mathrm{err}(A) \le C \inf_{A \in \mathcal{A}_n} \mathrm{err}(A)$$

for all problems with a convex balanced input set, a bounded linear solution operator, and homogeneous Lipschitz measurements. This failure already occurs for the closed unit ball of ℓ_1 and scalar-valued solutions.

Proof. Use Theorem XIII.3.1 at $n = 2$: the right-hand side is zero, while the left-hand side is one for every finite b. □

Corollary XIII.3.3 (Continuous functions on a compact interval)**.** *The same separation is realized on $X = C([0, 1])$ with its supremum norm, with F its closed unit ball and $S(f) = f(0)$. The measurements may again be taken to be countably many globally 1-Lipschitz homogeneous maps.*

Proof. Use the same formulas with the injective norm-one Volterra operator

$$(Tf)(t) = \int_0^t f(s)\,ds.$$

The Lipschitz estimates and the exact two-query algorithm use only $\|S\| = \|T\| = 1$ and injectivity of T, so they are unchanged. Put

$$f_j(t) = \max\{1 - jt, 0\}, \qquad v_j = 2jf_j.$$

Then $\|f_j\|_\infty = S(f_j) = 1$ and $\|Tf_j\|_\infty = 1/(2j)$. Consequently $Q(v_j) = 1$ and $R_k(v_j) = 0$ whenever $j \geq k$. The common-transcript proof gives $A(f_j) = c/(2j)$ eventually and hence worst-case error at least one. □

Proposition XIII.3.4 (Randomization does not remove this obstruction)**.** *For the same problem, every randomized algorithm using at most n of the admissible measurements whose realizations are almost surely positively homogeneous satisfies*

$$\sup_{x \in F} \mathbb{E}|S(x) - A(x)| \geq 1.$$

Equality is attained by the zero algorithm, whereas the deterministic two-query algorithm still has zero error.

Proof. For almost every realization ω, the preceding transcript argument provides a finite $M(\omega)$ and a finite $c(\omega)$ such that $A_\omega(e_j) = c(\omega)/j$ for every sufficiently large j. Hence $A_\omega(e_j) \to 0$ almost surely. Fatou's lemma gives

$$1 = \mathbb{E} \liminf_{j\to\infty} |1 - A_\omega(e_j)| \leq \liminf_{j\to\infty} \mathbb{E}|1 - A_\omega(e_j)| \leq \sup_{x \in F} \mathbb{E}|S(x) - A(x)|.$$

□

Scope and conventions. The counterexample does not solve Krieg–Kritzer Open Problem 2, which requires linear measurements in its randomized question. Nor does it settle the remaining deterministic case in which the input set is convex balanced and all measurements are linear, but only the solution operator is allowed to be homogeneous. The displayed measurements are nonlinear; their uniform Lipschitz bound is not a substitute for linearity. The information model charges one unit per queried functional, not the arithmetic cost of evaluating it. Query selection is Borel measurable: the only nonconstant selection rule is a ceiling of a continuous function away from zero. The lower bound needs no continuity of an algorithm's selection or recovery maps. Global homogeneity is essential: an algorithm that is exact on F automatically agrees there with the homogeneous map S, but need not admit a homogeneous finite-query realization on the ambient space.

Finally, the same exact separation persists if the pool is enlarged to its finite real linear span. For a query $L = \sum_{k=0}^{K} a_k L_k$, the answer at v_j is eventually $a_0/3$. This coefficient is intrinsic, since an identically zero finite combination has zero eventual value. Prescribing these eventual answers produces the same common-transcript argument. The original two-query upper bound remains available.

XIII.4 Two negative answers for Bessel–Fourier kernels and coefficients

Published target and scope. Acosta Babb, *On the rate of growth of Bessel–Fourier coefficients for integrable functions*, J. Math. Anal. Appl. **559** (2026), 130440, Section 5, Conjecture 1 (printed p. 11), proposes positivity of modified Fejér kernels. Its Conjecture 2 (printed p. 13) asks for arbitrary nonnegative sequences $a_m \to 0$ to be realized by an integrable radial function with nonnegative Bessel–Fourier coefficients of size at least $\sqrt{j_m}a_m$, supported on just one parity. We prove that this assertion is false. Both conjectures are refuted below, by different arguments. We also prove the unrestricted coefficient-realization conclusion of Corollary 5.2 unconditionally, using a positive heat-kernel mixture rather than either conjecture.

XIII.4.1 An exact two-term counterexample to kernel positivity

The kernel in the source is

$$K_{0,M}^{\alpha}(x) = \sum_{m=1}^{M} \left(1 - \frac{m}{M+1}\right) j_{0,m}^{\alpha} J_0(j_{0,m}x).$$

Conjecture 1 asserts its nonnegativity on $[0,1]$ for $0 \le \alpha \le 1/2$. We use $\alpha = 1/2$ and $M = 2$, entirely within the stated order-zero setting.

Proposition XIII.4.1 (A rational-point counterexample)**.** *One has*

$$K_{0,2}^{1/2}(9/10) < -\frac{1}{100}.$$

Moreover, $(K_{0,2}^{1/2})'(1) > 1/6$*, so the kernel is negative on an interval immediately to the left of* 1*.*

Proof. Put $a = j_{0,1}$ and $b = j_{0,2}$. The following are rational enclosures; every terminating decimal in this proof denotes its exact rational value:

$$2.4048 < a < 2.4049, \qquad 5.5200 < b < 5.5201, \qquad \sqrt{a} < 1.5508, \quad \sqrt{b} > 2.3494.$$

Here is a reproducible finite certificate for the Bessel evaluations used below:

quantity	enclosure
$J_0(2.4048)$	$(0.000013, 0.000014)$
$J_0(2.4049)$	$(-0.000039, -0.000038)$
$J_0(5.5200)$	$(-0.000027, -0.000026)$
$J_0(5.5201)$	$(0.000007, 0.000008)$
$J_1([2.4048, 2.4049])$	$(0.519, 0.520)$
$J_1([5.5200, 5.5201])$	$(-0.343, -0.338)$
$J_0(0.9[2.4048, 2.4049])$	$(0.13017, 0.13035)$
$J_0(0.9[5.5200, 5.5201])$	$(-0.18907, -0.18694)$

To obtain these enclosures using only rational arithmetic, truncate

$$J_\nu(x) = \sum_{k=0}^{30} \frac{(-1)^k (x/2)^{2k+\nu}}{k!(k+\nu)!} + R_{30,\nu}(x), \qquad \nu \in \{0,1\}.$$

For $0 \le x \le 6$, the alternating-series bound gives

$$|R_{30,\nu}(x)| \le \frac{(x/2)^{62+\nu}}{31!(31+\nu)!} < 10^{-35}.$$

For interval arguments $[l,u]$, evaluate each positive monomial at l,u and each negative monomial at u,l, respectively, and then enlarge both ends by 10^{-35}. This finite rule proves the table.

For completeness the sign changes in the first four rows identify the first two zeros, not merely some zeros. The alternating series gives $J_0(x)>0$ on $[0,2]$. On $x\ge 2$, the function $y(x)=\sqrt{x}J_0(x)$ satisfies $y''+(1+1/(4x^2))y=0$. For any consecutive zeros $s<t$ there, integration by parts and the Dirichlet Poincaré inequality give

$$\frac{\pi^2}{(t-s)^2}\int_s^t y^2 \le \int_s^t (y')^2 = \int_s^t \left(1+\frac{1}{4x^2}\right)y^2 \le \frac{17}{16}\int_s^t y^2.$$

Thus $t-s\ge 4\pi/\sqrt{17}>3$. The zero between 2.4048 and 2.4049 must be the first, and the zero between 5.5200 and 5.5201 must be the second: two further gaps would exceed 6. The square-root bounds follow by squaring the displayed rational numbers.

The formula for the kernel and the table now give

$$\begin{aligned} K_{0,2}^{1/2}(9/10) &= \frac{2\sqrt{a}J_0(0.9a)+\sqrt{b}J_0(0.9b)}{3} \\ &< \frac{2(1.5508)(0.13035)-(2.3494)(0.18694)}{3} = -\frac{8725819}{750000000} < -\frac{1}{100}. \end{aligned}$$

Also $K_{0,2}^{1/2}(1)=0$ and $J_0'=-J_1$, so

$$\begin{aligned} (K_{0,2}^{1/2})'(1) &= \frac{b^{3/2}|J_1(b)|-2a^{3/2}J_1(a)}{3} \\ &> \frac{(5.52)(2.3494)(0.338)-2(2.4049)(1.5508)(0.520)}{3} = \frac{157724021}{937500000} > \frac{1}{6}. \end{aligned}$$

This proves both assertions. □

XIII.4.2 The independent one-parity obstruction

Write $j_m=j_{0,m}$ for the positive zeros of J_0, in increasing order, and put $d\mu(r)=r\,dr$ on $(0,1)$. The coefficient convention in the source is

$$c_m(f)=\frac{2}{J_1(j_m)^2}\int_0^1 f(r)J_0(j_m r)\,r\,dr. \tag{XIII.4.1}$$

These integrals exist for every $f\in L^1(\mu)$, since $|J_0(x)|\le 1$ for real x.

Lemma XIII.4.2 (Bounded heat-content tests)**.** *For $t>0$ define*

$$h_t(r)=\sum_{m=1}^{\infty} e^{-tj_m^2}\frac{2J_0(j_m r)}{j_m J_1(j_m)}. \tag{XIII.4.2}$$

This series converges absolutely and uniformly on $[0,1]$. It satisfies $0\le h_t(r)\le 1$, and $h_t(r)\to 1$ as $t\downarrow 0$ at every $0\le r<1$. For $f\in L^1(\mu)$,

$$\int_0^1 f(r)h_t(r)\,r\,dr=\sum_{m=1}^{\infty} e^{-tj_m^2}c_m(f)\frac{J_1(j_m)}{j_m}. \tag{XIII.4.3}$$

In particular, the absolute value of the right-hand side is at most $\|f\|_{L^1(\mu)}$, and its limit as $t\downarrow 0$ is $\int_0^1 f\,d\mu$.

Proof. Let T_t be the Dirichlet heat semigroup on the unit disk. The function $T_t 1$ is radial; write it as $h_t(r)$. Positivity and the parabolic maximum principle give $0 \le T_t 1 \le 1$. Local convergence of the heat solution to its initial data gives $h_t(r) \to 1$ for $r < 1$. The radial Dirichlet eigenfunctions are $J_0(j_m r)$ with eigenvalues j_m^2, and

$$\int_0^1 J_0(j_m r)^2 r\,dr = \frac{1}{2} J_1(j_m)^2, \qquad \int_0^1 J_0(j_m r) r\,dr = \frac{J_1(j_m)}{j_m}.$$

Expanding the initial function 1 in the radial $L^2(\mu)$ basis therefore gives (XIII.4.2). The standard asymptotics

$$j_m \sim \pi m, \qquad J_1(j_m) = (-1)^{m+1} \sqrt{\frac{2}{\pi j_m}} (1 + O(m^{-1})) \tag{XIII.4.4}$$

show that its coefficient before the heat factor is $O(m^{-1/2})$. Together with $|J_0| \le 1$, this proves absolute uniform convergence for every $t > 0$. Thus integration against an arbitrary $L^1(\mu)$ function is permitted term by term. Substitution of (XIII.4.1) gives (XIII.4.3). The bound follows from $0 \le h_t \le 1$, and dominated convergence gives the stated limit. □

Theorem XIII.4.3 (One-sign heat coefficients must be summable)**.** *Let $f \in L^1((0,1), r\,dr)$ have real Bessel–Fourier coefficients. Suppose that all the numbers*

$$b_m = c_m(f) \frac{J_1(j_m)}{j_m}$$

have the same sign. Then

$$\sum_{m=1}^{\infty} |c_m(f)| \frac{|J_1(j_m)|}{j_m} = \left| \int_0^1 f(r) r\,dr \right| \le \int_0^1 |f(r)| r\,dr. \tag{XIII.4.5}$$

If the sign condition holds outside a finite set, the series on the left is still finite. Consequently, if $c_m(f)$ are nonnegative and vanish on one parity outside a finite set, then

$$\sum_{m=1}^{\infty} \frac{|c_m(f)|}{m^{3/2}} < \infty. \tag{XIII.4.6}$$

Proof. If $b_m \ge 0$, then $e^{-t j_m^2} b_m$ increases to b_m as $t \downarrow 0$. Monotone convergence for the sum, followed by Lemma XIII.4.2, gives

$$\sum_m b_m = \lim_{t \downarrow 0} \sum_m e^{-t j_m^2} b_m = \int_0^1 f\,d\mu.$$

The same argument with signs reversed treats $b_m \le 0$. A finite exceptional set can be subtracted without affecting summability. The zeros of J_0 are simple and its signs alternate, so $J_1(j_m) = -J_0'(j_m)$ has sign $(-1)^{m+1}$. Thus nonnegative coefficients on a single parity satisfy the one-sign condition. Finally, (XIII.4.4) gives $|J_1(j_m)|/j_m \asymp m^{-3/2}$, proving (XIII.4.6). □

Corollary XIII.4.4 (Negative answer to Acosta Babb's Conjecture 2)**.** *Set*

$$a_m = \frac{1}{\log(m+1)}, \qquad m \ge 1.$$

There is no $f \in L^1((0,1), r\,dr)$ satisfying either of the following two alternatives from the published conjecture:

$$\begin{aligned} c_{2m}(f) &= 0, & c_{2m+1}(f) &\ge \sqrt{j_m}\, a_m & (m \ge 1),\\ c_{2m+1}(f) &= 0, & c_{2m}(f) &\ge \sqrt{j_{2m}}\, a_m & (m \ge 1). \end{aligned}$$

This remains impossible even without the further requirement that the corresponding normalized coefficients tend to zero.

Proof. The sequence a_m is nonnegative and tends to zero. In the first alternative, the term with index 1 is immaterial. If f is complex, replace it by its real part; this preserves every prescribed tail coefficient. Then Theorem XIII.4.3 and $j_m \asymp m$ would imply

$$\infty > \sum_{m\ge 1} \frac{c_{2m+1}(f)}{(2m+1)^{3/2}} \ge C \sum_{m\ge 1} \frac{a_m}{m}.$$

The last series diverges. The even alternative gives the same contradiction. More generally, either realization necessarily requires $\sum_m a_m/m < \infty$. □

XIII.4.3 Positive realization survives without either conjecture

Theorem XIII.4.5 (Unrestricted positive coefficient realization)**.** *Let $a_m \ge 0$ satisfy $a_m \to 0$. There exists a nonnegative $f \in L^1((0,1), r\,dr)$ such that*

$$0 \le a_m \le \frac{c_m(f)}{\sqrt{j_m}} \longrightarrow 0.$$

This proves the conclusion of Acosta Babb's Corollary 5.2 without either Conjecture 1 or 2.

Proof. Let $G_t(r)$ be the radial Dirichlet heat kernel with source at the center of the unit disk, normalized relative to $r\,dr$. Equivalently, it is 2π times the ordinary planar Dirichlet heat kernel with one argument at the center. Positivity and loss of mass at the boundary give

$$G_t \ge 0, \qquad \int_0^1 G_t(r) r\,dr \le 1.$$

Its radial eigenfunction expansion is

$$G_t(r) = \sum_{m\ge 1} \frac{2e^{-tj_m^2}}{J_1(j_m)^2} J_0(j_m r).$$

As in Lemma XIII.4.2, this series is absolutely uniform for each $t > 0$. Put

$$d_m = \frac{2}{j_m J_1(j_m)^2}, \qquad d_- := \inf_m d_m > 0, \quad d_+ := \sup_m d_m < \infty.$$

The bounds follow from $d_m \to \pi$ and the nonvanishing of $J_1(j_m)$.

For $n \ge 1$, let

$$A_n = \max\{a_m : 2^n \le j_m < 2^{n+1}\}, \qquad w_n = \frac{e^4}{d_-} 2^{-n/2} A_n,$$

where an empty maximum is zero. The A_n are bounded and tend to zero, so $\sum_n w_n < \infty$. Define

$$f(r) = \sum_{n\ge 1} w_n G_{4^{-n}}(r).$$

Tonelli's theorem gives $f \ge 0$ and $\int f\,d\mu \le \sum_n w_n$. Since $|J_0| \le 1$, the same mass bound justifies termwise evaluation of (XIII.4.1). Therefore

$$\frac{c_m(f)}{\sqrt{j_m}} = d_m \sqrt{j_m} \sum_{n\ge 1} w_n e^{-4^{-n} j_m^2}. \tag{XIII.4.7}$$

If $2^k \le j_m < 2^{k+1}$, the summand $n = k$ alone is at least A_k, and hence at least a_m.

For the limit, the same block inequalities yield

$$\frac{c_m(f)}{\sqrt{j_m}} \le \frac{\sqrt{2} d_+ e^4}{d_-} \sum_{n\ge 1} A_n\, 2^{(k-n)/2} e^{-4^{k-n}}.$$

The sequence $q_l = 2^{l/2} e^{-4^l}$ belongs to $\ell^1(\mathbb{Z})$: its negative tail is geometric and its positive tail decreases superexponentially. Extend A_n by zero for $n \le 0$. Then the last sum is $\sum_{l\in\mathbb{Z}} q_l A_{k-l} \to 0$ by dominated convergence, since A_n is bounded and tends to zero. This proves the theorem. □

Why parity matters. The bounded tests $h_t = T_t 1$ already detect the obstruction before any unregularized partial sum is considered. The integrals of the radial eigenfunctions alternate in sign. Removing one parity while retaining nonnegative coefficients removes exactly the cancellation needed to keep these heat-content tests bounded. The positive realization theorem retains both parities, so the heat-content terms can cancel even though the Bessel–Fourier coefficients themselves are nonnegative. The source's proposed divergent example uses a_m of order $1/\log m$, for which that cancellation is indispensable.

Status. The published Conjectures 1 and 2 are explicitly contradicted above. This does not prove that every L^1 radial eigenfunction expansion has a norm-convergent subsequence; it only excludes the one-parity positive coefficient construction proposed in the source. A targeted search by title, DOI, author, and conjecture number on 19 October 2026 did not locate a subsequent resolution. This is not a certificate of priority, and the argument is a short application of classical heat semigroup positivity rather than a new semigroup theorem.

XIII.A Quadratic factorial reduction: the complete sufficient direction

Scope. This appendix is a partial resolution of a different open conjecture. It is not included among the fully settled targets in the status register. It also gives an exact classification of a stronger joint-denominator property, which should not be conflated with the original limsup question.

XIII.A.1 The published question and the exact conclusion

Let

$$a(n) = An + k, \qquad b(n) = B(n - x_1)(n - x_2), \qquad A, B, k, x_1, x_2 \in \mathbb{Q}, \quad AB \neq 0.$$

As usual, rational coefficients may be made integral by constant inflation $a \mapsto La$, $b \mapsto L^2 b$. Canonical continuants have initial values

$$p_{-1} = 1, \quad p_0 = k, \qquad q_{-1} = 0, \quad q_0 = 1,$$

and satisfy

$$u_n = (An + k)u_{n-1} + B(n - x_1)(n - x_2)u_{n-2}. \tag{XIII.A.1}$$

Conjecture 1.4 of [124] proposes that factorial reduction occurs precisely in the two families

$$A = B/m - m, \quad k \text{ arbitrary}, \tag{XIII.A.2}$$

$$k = \tfrac{A}{2}(1 - x_1 - x_2), \tag{XIII.A.3}$$

with the source's nonzero/excluded-parameter conventions. We prove the entire sufficient direction, including all resonances, and also certain parameter values omitted by those conventions.

Theorem XIII.A.1. *Put $s = x_1 + x_2$. Suppose either*

$$A^2 + 4B \text{ is a nonzero square in } \mathbb{Q}, \qquad \text{or} \qquad k = \frac{A(1 - s)}{2}. \tag{XIII.A.4}$$

After constant inflation to integral coefficients, there is $C \geq 1$ such that, at every index where $(p_n, q_n) \neq (0, 0)$,

$$\gcd(p_n, q_n) \geq \frac{n!}{C^{n+1}}.$$

In particular, if $b(n) \neq 0$ for all positive integers n, factorial reduction holds with a uniform lower bound, stronger than the published limsup definition. Both families (XIII.A.2)–(XIII.A.3) *are covered.*

The nontermination condition is only needed to ensure the numerical gcd never vanishes. The denominator statement below holds for every rational initial condition, whether or not a continued fraction is defined.

Theorem XIII.A.2. *The following are equivalent:*

1. *Condition* (XIII.A.4) *holds.*
2. *For any fixed finite family of rational solutions of* (XIII.A.1)*, there exist integers $D_N \geq 1$ and $C \geq 1$ such that*

$$D_N \leq C^{N+1}, \qquad D_N \frac{u_n}{n!} \in \mathbb{Z} \quad (0 \leq n \leq N)$$

   for every solution in the family.
3. *The two solutions with $(u_0, u_1) = (1, 0)$ and $(0, 1)$ have factorial-normalized generating functions that are G-functions.*

Here the recurrence for freely specified (u_0, u_1) is imposed for $n \geq 2$.

XIII.A.2 The generating equation and its exponents

For a solution put $v_n = u_n/n!$ and $V(z) = \sum_{n\ge 0} v_n z^n$. For $n \ge 2$, equation (XIII.A.1) becomes

$$n(n-1)v_n = (An+k)(n-1)v_{n-1} + B(n-x_1)(n-x_2)v_{n-2}.$$

With $\theta = z\,d/dz$, summation gives

$$[\theta(\theta-1) - z\theta(A\theta + A + k) - Bz^2(\theta+2-x_1)(\theta+2-x_2)]V = 0.$$

After dividing by z^2 this is the ordinary-at-zero equation

$$(1 - Az - Bz^2)V'' - [2A + k + B(5-s)z]V' - B(2-x_1)(2-x_2)V = 0. \tag{XIII.A.5}$$

In particular every solution with rational initial data has rational Taylor coefficients and a positive radius of convergence.

Suppose first that $\Delta^2 = A^2 + 4B \neq 0$ and choose one square root Δ. The finite singularities are

$$\rho_\pm = \frac{-A \pm \Delta}{2B}.$$

The exponents at infinity are $2 - x_1, 2 - x_2$. At $\rho_\pm$ the exponents are 0 and

$$r_\pm = \frac{s-3}{2} \pm \frac{A(1-s)/2 - k}{\Delta}. \tag{XIII.A.6}$$

Indeed, if $P = 1 - Az - Bz^2$ and $Q = -[2A + k + B(5-s)z]$, the nonzero indicial root at a simple zero ρ of P is $1 - Q(\rho)/P'(\rho)$; substitution yields (XIII.A.6). All three singularities are regular singular. Since the numerator $A(1-s)/2 - k$ is rational, both finite exponents are rational if and only if (XIII.A.4) holds.

XIII.A.3 Arithmetic input, including resonant parameters

We use the following established facts about G-operators, in the form given by Fresán–Jossen [125, Theorems 1.8 and 3.3]:

1. A balanced hypergeometric operator $Q(\theta) - tP(\theta)$, where P, Q are nonzero polynomials of equal degree with rational roots, is a G-operator.

2. Pullback by a nonconstant rational function with algebraic coefficients preserves G-operators. This includes translations by algebraic points, not just substitutions fixing zero.

3. At any ordinary algebraic point, a G-operator has a basis of holomorphic G-functions with algebraic Taylor coefficients. Conversely, an operator with such a basis is a G-operator.

4. Every G-operator is regular singular, with rational exponents.

The first fact has no nonresonance or irreducibility hypothesis. The third is equivalently their Definition 1.6, Lemma 1.5, and the pullback statement in Theorem 1.8. These facts are used as existing arithmetic theorems, not claimed as new results here.

For clarity, a rational-coefficient G-function $\sum c_n z^n$ has coefficients of at most exponential size and positive integers $D_N \le C^{N+1}$ simultaneously clearing $c_0, \dots, c_N$. An analytic solution with rational initial values of an ordinary G-operator is a G-function: choose a G-basis, solve the invertible algebraic initial-value matrix, and take that algebraic linear combination.

XIII.A.4 Proof of both theorems

Proof of Theorem XIII.A.2. Assume (XIII.A.4) and first let $\Delta \neq 0$. Apply the affine substitution

$$t = \frac{z - \rho_-}{\rho_+ - \rho_-}.$$

Equation (XIII.A.5) becomes a Gauss equation with parameters

$$a = 2 - x_1, \qquad b = 2 - x_2, \qquad c = 1 - r_-.$$

All three are rational. One can check the equation directly; equivalently these parameters give exponents $0, r_-$ at $t = 0$, $0, r_+$ at $t = 1$, and a, b at infinity, since $1 - r_- - a - b = r_+$ by (XIII.A.6). The Gauss operator, multiplied by t, is

$$\theta_t(\theta_t + c - 1) - t(\theta_t + a)(\theta_t + b),$$

which is balanced hypergeometric with rational roots. The arithmetic input therefore makes it a G-operator, and its algebraic affine pullback (XIII.A.5) is a G-operator as well. Zero is an ordinary point. Consequently the two rationally normalized solutions are G-functions, and so are all rational linear combinations. This proves (1)⇒(3)⇒(2).

If $\Delta = 0$, condition (XIII.A.4) means $k = A(1-s)/2$. Put $w = 1 - Az/2$. Equation (XIII.A.5) reduces to

$$w^2 V_{ww} + (5 - s) w V_w + (2 - x_1)(2 - x_2) V = 0.$$

Its solutions are $w^{x_1 - 2}, w^{x_2 - 2}$ when $x_1 \neq x_2$, and $w^{x_1 - 2}, w^{x_1 - 2} \log w$ otherwise. Around $z = 0$ these are algebraic functions or products of an algebraic function and a logarithm of an algebraic function; they are G-functions. Thus the same implications hold in this case too.

Conversely, (2) gives exponential joint denominators for the two normalized solutions. Their positive analytic radii give the other coefficient-growth condition, so (3) holds. If (3) holds, the two functions form a basis of (XIII.A.5); hence that operator is a G-operator. When $\Delta \neq 0$, its local exponents (XIII.A.6) must be rational. If their common rational numerator $A(1 - s)/2 - k$ is nonzero, this forces $\Delta \in \mathbb{Q}$; otherwise the second condition in (XIII.A.4) holds.

Finally, let $\Delta = 0$. Its double finite root is $\rho = 2/A$. A scalar second-order equation with leading polynomial having a double root is regular singular there only if the coefficient of V' also vanishes there. In the present case that value is

$$Q(2/A) = A(1 - s)/2 - k.$$

Regular singularity of a G-operator forces it to vanish, giving the second condition again. This proves (3)⇒(1). □

Proof of Theorem XIII.A.1. After choosing a constant integral inflation, apply Theorem XIII.A.2 to the canonical pair. At $N = n$ it gives $n! \mid D_n p_n$ and $n! \mid D_n q_n$. Thus, whenever the pair is nonzero,

$$n! \leq D_n \gcd(p_n, q_n) \leq C^{n+1} \gcd(p_n, q_n).$$

The continuant determinant

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n+1} \prod_{j=1}^{n} b(j)$$

shows that a nonterminating pair is never zero. Constant inflation preserves both conditions in (XIII.A.4); it changes factorial reduction bounds only by an exponential factor.

For the first published family, $A^2 + 4B = (B/m + m)^2$. The excluded parameters in the source ensure the square is nonzero. Conversely, if $\Delta \in \mathbb{Q} \setminus \{0\}$, then $m = (-A + \Delta)/2$ is rational and nonzero and satisfies $A = B/m - m$. The second published family is exactly the second condition in (XIII.A.4). □

XIII.A.5 What this does not prove

The source defines factorial reduction by the scalar condition

$$\limsup_{n\to\infty} \frac{1}{n} \log \frac{\gcd(p_n, q_n)}{n!} > -\infty.$$

Theorem XIII.A.2 instead classifies a uniform exponential *joint-denominator* property. A large numerical gcd alone does not immediately give this property: prime divisibility must be compared with $n!$, and the source asks only for a limsup. We have not supplied that implication. Therefore the necessary direction of Conjecture 1.4 is not claimed solved here. In particular, this note proves all of its proposed examples, not the assertion that no other examples exist.

Part XIV

Open Problems Across Further Domains

Manuscript overview. This collection addresses explicitly stated questions in nonlinear differential equations, quantum information, distributional skewness, finite groups, factorial arithmetic, cumulative games, and enumerative geometry. The arguments use analytic dependence at infinity, positivity of a Jordan product, median-reflection identities, equivariant orbit refinement, prime valuations, exact backward induction, and a classical real-rootedness preservation theorem. Each section identifies the precise published target and separates its conclusion from results already available in the literature. The manuscript contains complete proofs or finite integer certificates, rather than numerical evidence for the claimed conclusions.

Verification and priority. These are research arguments with separate internal mathematical checks, not externally refereed results. Targeted searches and the latest accessible source versions were checked through 19 October 2026. Not locating a prior solution does not certify priority. In particular, higher-dimensional quantum nonconvexity was already proved; formal existence of the ODE series was already proved; and the enumerative result below is a short application of Wagner's 1992 theorem. No claim of an Annals-level or Fields-level breakthrough is made.

Precise status register. The following is a list of targets and their scope, not a count of independent major discoveries. An explicit branch of a larger question is labelled as such.

Published target	Conclusion in this manuscript
Habib–Hildebrand, Conjecture 7.3	Exponential coefficient bounds and actual convergence for every allowed complex polynomial.
Afham–Ferrie, Problem 2, fixed-base convexity	Necessary and sufficient condition for convexity in one state; strictly positive qubit counterexample. Other convexity and DPI questions are not classified.
Hunter, Section 5, convolution conjecture	Refuted by an independent uniform and exponential pair, both log-concave and right van Zwet.
Hunter, Section 5, weak-limit assertion	Refuted for the literal atom-inclusive definition; no assertion about continuous-limit or left-limit variants.
Aivazidis–Loukaki–Müller, Problem 5.2	Normal-enlargement inequality proved for all finite groups, with exact equality cases.
Same authors, Problem 5.6	Distinct-integer factorial-product uniqueness refuted by an exact rational identity.
Same authors, Problem 5.3, coprime branch	Product-group formula proved for coprime orders; full equality classification remains open here.
Bhagat et al., Conjecture 4 and Problem 6	Exact finite counterexamples to friendly dominance and to the proposed restriction on unilateral gains.
Same authors, Conjecture 17	Uniform discrepancy bound $\min S - 1$, valid for every heap and every pair of deterministic tie conventions.
Tielker, Conjecture 4.7	The stated real-rootedness follows from the published polytope identities and Wagner's theorem; treated as an existing-theorem application.

XIV.1 Convergence of the Habib–Hildebrand exponential–polynomial expansions

Published target and scope. Habib and Hildebrand, *Necessary conditions for the existence of exponential-polynomial expansions for solutions of certain differential equations*, arXiv:2606.21145v1, Conjecture 7.3, ask whether the formal expansion for

$$u''u - (u')^2 = e^{-x}P(u) - 1 \tag{XIV.1.1}$$

has exponentially bounded polynomial coefficients and consequently converges. Their conditions are $P(0) = 0$ and $P'(0) = P''(0)/2$. Hildebrand and Habib, arXiv:2606.24182v1, Theorem 2.5, subsequently proved the existence of the formal polynomial expansion, resolving Conjecture 7.1. Section 5 of that follow-up explicitly leaves convergence open. We use their formal-existence theorem and their regularization from Lemma 3.1 to prove Conjecture 7.3. We make no assertion about the separate coefficient-positivity Conjecture 7.2.

Theorem XIV.1.1. *Let $P \in \mathbb{C}[t]$ be nonzero, of degree d, and satisfy $P(0) = 0$ and $P'(0) = P''(0)/2$. Let*

$$p_0(z) = z, \qquad p_k(z) = \sum_{j=0}^{(d-2)k+1} c_{k,j} z^j \quad (k \geq 1)$$

be the unique polynomials of the formal expansion in arXiv:2606.24182v1, Theorem 2.5. There is a constant $C \geq 1$, depending on P only, such that

$$|c_{k,j}| \leq C^k \qquad (k \geq 1,\ j \geq 0). \tag{XIV.1.2}$$

Consequently, for every $c \in \mathbb{C}$, the series

$$u_c(x) = \sum_{k=0}^{\infty} p_k(x + c)e^{-kx} \tag{XIV.1.3}$$

converges absolutely and locally uniformly in

$$\mathcal{D}_c = \left\{ x \in \mathbb{C} : Ce^{-\operatorname{Re} x} \max\{1, |x + c|^{d-2}\} < 1 \right\}. \tag{XIV.1.4}$$

It is holomorphic there, satisfies (XIV.1.1), and has $u_c(x) - (x + c) \longrightarrow 0$ along the positive real axis. For $P = 0$ the corresponding assertion is the trivial solution $u_c = x + c$.

XIV.1.1 An analytic family selected at infinity

Write

$$P(t) = \pi t(t + 1) + t^3 Q(t), \qquad \pi = P'(0), \qquad Q \in \mathbb{C}[t].$$

Introduce a parameter $a \in \mathbb{C}$ and the nonsingular system

$$\begin{aligned} u' &= 1 - a\pi e^{-x}u + u^2 v, \\ v' &= ae^{-x}Q(u) + a\pi e^{-x}v - uv^2. \end{aligned} \tag{XIV.1.5}$$

Direct differentiation gives

$$u''u - (u')^2 = ae^{-x}P(u) - 1. \tag{XIV.1.6}$$

This identity does not require $u \neq 0$; this is the useful feature of the regularization.

We construct solutions of (XIV.1.5) on the real ray $x \ge 1$ with $u = x + w$ and exponentially decreasing w, v. Let X_w, X_v be the complex Banach spaces of continuous functions on $[1, \infty)$ with respective norms

$$\|w\|_w = \sup_{x \ge 1} \frac{e^{x/2}|w(x)|}{(1+x)^2}, \qquad \|v\|_v = \sup_{x \ge 1} e^{x/2}|v(x)|.$$

Put $X = X_w \times X_v$ and define $T = (T_w, T_v)$ by

$$T_w(a, w, v)(x) = a\pi \int_x^\infty e^{-s}(s + w(s))\, ds - \int_x^\infty (s + w(s))^2 v(s)\, ds, \tag{XIV.1.7}$$

$$T_v(a, w, v)(x) = -a \int_x^\infty e^{-s} Q(s + w(s))\, ds - a\pi \int_x^\infty e^{-s} v(s)\, ds + \int_x^\infty (s + w(s)) v(s)^2\, ds. \tag{XIV.1.8}$$

Lemma XIV.1.2. *For some $r > 0$ there is a holomorphic map $a \mapsto (w_a, v_a) \in X$, $|a| < r$, with $(w_0, v_0) = (0, 0)$, such that $(w_a, v_a) = T(a, w_a, v_a)$. The functions $u_a = x + w_a, v_a$ solve (XIV.1.5) and satisfy $w_a(x) \to 0$, $v_a(x) \to 0$ as $x \to \infty$.*

Proof. All integrals in (XIV.1.7)–(XIV.1.8) define bounded multilinear maps after expansion into monomials in a, w, v. Here are details of the estimates that ensure this assertion. The only term without an extra exponential factor beyond that in v is the bounded operator

$$Bv(x) = \int_x^\infty s^2 v(s)\, ds, \qquad \|Bv\|_w \le 26 \|v\|_v,$$

as follows from

$$\int_x^\infty s^2 e^{-s/2}\, ds = 2e^{-x/2}(x^2 + 4x + 8).$$

The remaining terms in T_w contain either e^{-s} or at least two factors from w, v. Their integrands are bounded by a fixed polynomial in s times e^{-s}, multiplied by the appropriate product of input norms. Every term in T_v has the same property: in $e^{-s}Q(s + w)$ the additional factor e^{-s} suffices, while $(s + w)v^2$ has at least two exponentially decreasing factors. For every fixed integer $N \ge 0$,

$$\sup_{x \ge 1} e^{x/2} \int_x^\infty (1 + s)^N e^{-s}\, ds < \infty.$$

These estimates also cover higher powers of w, v, since their additional exponential factors only improve the bound. The degree of Q is fixed, so only finitely many such constants are required. Thus $T : \mathbb{C} \times X \to X$ is a polynomial, in particular holomorphic, map.

At $(a, w, v) = (0, 0, 0)$ its derivative in X is

$$D_X T(0, 0, 0)(h, k) = (-Bk, 0).$$

Hence the derivative in X of $F(a, w, v) = (w, v) - T(a, w, v)$ is

$$(h, k) \longmapsto (h + Bk, k),$$

with bounded inverse $(h, k) \mapsto (h - Bk, k)$. Since $F(0, 0, 0) = 0$, the complex analytic implicit-function theorem on Banach spaces supplies the asserted holomorphic fixed point. Differentiating its absolutely convergent integrals gives (XIV.1.5). Its decay follows from membership in $X_w \times X_v$. □

XIV.1.2 Uniform continuation through the apparent singularity

Lemma XIV.1.3. *After decreasing r, the family (u_a, v_a) from Lemma XIV.1.2 has a jointly holomorphic continuation in (a, z) to a neighborhood of $\{|a| < r,\ |z| \le 1\}$. For some $\rho > 0$ and $M < \infty$,*

$$|u_a(z)| \le M \qquad (|a| \le \rho,\ |z| \le 1). \tag{XIV.1.9}$$

Proof. The initial values $(w_a(1), v_a(1))$ are holomorphic in a and vanish at $a = 0$. Unlike (XIV.1.6), system (XIV.1.5) has no singularity at $u = 0$. For completeness, its continuation can be seen directly on the complex disk $|z - 1| < 3$. In the variables $Y = (w, v)$ it is

$$Y' = A(z)Y + R(z, a, Y), \qquad A(z) = \begin{pmatrix} 0 & z^2 \\ 0 & 0 \end{pmatrix},$$

where R is entire in its arguments and, uniformly for $|z - 1| \le 3$ and small Y, a,

$$R(z, a, Y) = O(|a|) + O(|a|\,|Y|) + O(|Y|^2).$$

The fundamental matrix normalized at $z = 1$ is

$$\Phi(z) = \begin{pmatrix} 1 & (z^3 - 1)/3 \\ 0 & 1 \end{pmatrix}.$$

Its inverse is also bounded on this disk. The variation-of-constants equation

$$Y(z) = \Phi(z)\left[Y(1) + \int_1^z \Phi(\zeta)^{-1} R(\zeta, a, Y(\zeta))\, d\zeta\right]$$

is a contraction on a sufficiently small sup-norm ball of holomorphic functions continuous on the closed disk, first choosing the ball small and then $|a|$ small. Indeed, straight integration paths have length at most 3, and the Lipschitz constant of R in this ball is $O(|a|) + O(\|Y\|_\infty)$. Also $Y(1) = O(a)$. Its iterates, and hence its fixed point, are holomorphic in a. Local uniqueness identifies this solution with the ray solution near $z = 1$. The unit disk lies strictly inside $|z - 1| < 3$; compactness after a further decrease in the parameter radius gives (XIV.1.9). □

XIV.1.3 Identification of coefficients and the exponential bound

Proof of Theorem XIV.1.1. Expand the analytic family from Lemma XIV.1.2:

$$u_a(x) = x + \sum_{k \ge 1} a^k U_k(x).$$

Since this is a Banach-valued Taylor expansion, $U_k \in X_w$. In particular $U_k(x) \to 0$ as $x \to \infty$ for every $k \ge 1$. The same coefficients are holomorphic near the closed unit disk by Lemma XIV.1.3.

We claim that

$$U_k(x) = p_k(x) e^{-kx}. \tag{XIV.1.10}$$

The formal-existence theorem cited above says that $x + \sum_{k \ge 1} a^k p_k(x) e^{-kx}$ formally solves (XIV.1.6). This follows equivalently by matching each power of ae^{-x} in that equation. Suppose by induction that the identification holds below order k. At order a^k, the only terms containing the unknown coefficient in $u''u - (u')^2$ are

$$xU_k'' - 2U_k'.$$

All terms on the right-hand side, and all other terms on the left, involve only lower coefficients. Thus the difference $H = U_k - p_k e^{-kx}$ satisfies

$$xH'' - 2H' = 0 \qquad (x \ge 1).$$

Its general solution is $H = A + Bx^3$. Both U_k and $p_k e^{-kx}$ tend to zero at infinity, so $A = B = 0$. This proves the induction, starting at $k = 1$. By the identity theorem, the same equality holds on the holomorphic continuation containing the unit disk.

Cauchy's estimate in the parameter a, using (XIV.1.9), now gives

$$\sup_{|z|=1} |U_k(z)| \le M\rho^{-k}.$$

Since $p_k(z) = e^{kz}U_k(z)$, a second Cauchy estimate, this time for the polynomial variable z, yields

$$|c_{k,j}| \le \sup_{|z|=1} |p_k(z)| \le M(e/\rho)^k.$$

Absorbing the fixed factor M into a constant to the power k proves (XIV.1.2).

The degree bound from the formal-existence theorem implies

$$|p_k(z)| \le ((d-2)k+2)\, C^k \max\{1, |z|^{(d-2)k+1}\}.$$

Putting $z = x + c$ proves absolute and locally uniform convergence in (XIV.1.4), by comparison with a geometric series times a linear factor in k. Holomorphic derivatives converge locally uniformly as well. Polynomial products may therefore be formed term by term, and the formal coefficient identities prove (XIV.1.1) for the convergent sum. On the positive real axis the geometric ratio tends to zero exponentially and its polynomial prefactor cannot prevent $u_c(x) - (x + c) \to 0$. □

What is proved and what is not. Estimate (XIV.1.2) is exactly the hypothesis of Lemma 6.1 in arXiv:2606.21145v1, with $\alpha = \log C$. Thus this resolves its Conjecture 7.3 for every allowed complex polynomial P, including the transformed Painlevé cases $P(t) = t^3, t^4, t^6$. The formal-existence result and the regularization are credited to Hildebrand and Habib; the new step here is the analytic selection at infinity followed by two Cauchy estimates. This argument does not establish positivity of the coefficients, a maximal convergence domain, or any global single-valued continuation outside (XIV.1.4).

XIV.2 Quantum information: the exact convexity region and a qubit obstruction

XIV.2.1 Published target and current scope

Afham and Ferrie, *Riemannian-geometric generalizations of quantum fidelities and Bures–Wasserstein distance*, J. Math. Phys. **66** (2025), 082201, DOI 10.1063/5.0252803, arXiv:2410.04937, Section 10, Problem 2, ask about convexity and joint convexity of their base-dependent distances. A subsequent paper by Reza Rajaei, *Generalized Fidelity, the Data Processing Inequality, and Convexity*, arXiv:2609.09753v2 (2 October 2026), Theorem 3, already disproves convexity in dimensions $d \ge 3$. We do not claim that general nonconvexity is new. The results below give an exact criterion for fixed-base one-variable convexity in every dimension, and an explicit strictly positive two-dimensional counterexample, beyond the dimension range of that theorem. The data-processing question for general qubit inputs is not settled here.

For positive definite Hermitian matrices P, Q, R, put

$$F_R(P,Q) = \operatorname{Tr}\Big[(R^{1/2}PR^{1/2})^{1/2}R^{-1}(R^{1/2}QR^{1/2})^{1/2}\Big],$$
$$B_R(P,Q) = \operatorname{Tr}(P+Q) - 2\operatorname{Re} F_R(P,Q).$$

Here B_R is the squared distance; the distance itself is $\sqrt{B_R}$. Write $\mathcal{P}_d$ for the positive definite cone and $\mathcal{D}_d^\circ = \{P \in \mathcal{P}_d : \operatorname{Tr} P = 1\}$. Multiplying R by a positive scalar changes neither functional.

Theorem XIV.2.1 (Exact fixed-R, Q criterion)**.** *Let $d \geq 2$, and fix $R, Q \in \mathcal{P}_d$. Define*

$$S = (R^{1/2} Q R^{1/2})^{1/2}, \qquad C_{R,Q} = \frac{R^{-1} S + S R^{-1}}{2}.$$

The following are equivalent:

1. *$C_{R,Q}$ is positive semidefinite.*
2. *$P \mapsto B_R(P, Q)$ is convex on $\mathcal{P}_d$.*
3. *$P \mapsto B_R(P, Q)$ is convex on $\mathcal{D}_d^\circ$.*

Proof. Set $A = R^{1/2} P R^{1/2}$. Cyclicity of trace and taking the Hermitian part give

$$\operatorname{Re} F_R(P, Q) = \operatorname{Tr}(C_{R,Q} \sqrt{A}). \tag{Q1}$$

If $C_{R,Q} \succeq 0$, operator concavity of the positive square root implies concavity of (Q1) as a function of P. Thus (1) implies (2), and restriction gives (3).

For completeness, operator concavity follows from

$$\sqrt{A} = \frac{1}{\pi} \int_0^\infty t^{-1/2} A (A + tI)^{-1} \, dt.$$

For $t > 0$, the map $A \mapsto (A+tI)^{-1}$ is operator convex: on every affine positive definite line $D(s)$, its second derivative is $2D(s)^{-1} H D(s)^{-1} H D(s)^{-1} \succeq 0$. Consequently $A(A+tI)^{-1} = I - t(A+tI)^{-1}$ is operator concave; integration preserves the inequality.

To prove (3) implies (1), suppose that $C = C_{R,Q}$ has a negative eigenvalue. In an orthonormal eigenbasis $e_1, \ldots, e_d$, write $C e_j = c_j e_j$, with $c_1 < 0$, and let $\kappa_j = \langle e_j, R^{-1} e_j \rangle > 0$. If $d > 2$, choose $\varepsilon > 0$ so that $b = 1 - \varepsilon \sum_{j=3}^d \kappa_j > 0$; for $d = 2$, take $b = 1$. For sufficiently small $t > 0$, define, in this basis,

$$A(t) = \operatorname{diag}\left(t, \frac{b - \kappa_1 t}{\kappa_2}, \varepsilon, \ldots, \varepsilon\right), \qquad P(t) = R^{-1/2} A(t) R^{-1/2}.$$

The last $d - 2$ entries are absent if $d = 2$. This is an affine path of positive definite matrices satisfying $\operatorname{Tr} P(t) = 1$. Let $a(t) = (b - \kappa_1 t)/\kappa_2$. Then

$$\frac{d^2}{dt^2} \operatorname{Re} F_R(P(t), Q) = -\frac{c_1}{4t^{3/2}} - \frac{c_2 (\kappa_1/\kappa_2)^2}{4 a(t)^{3/2}}.$$

The first summand tends to $+\infty$, and the second remains bounded as $t \downarrow 0$. Hence $B_R(P(t), Q)'' < 0$ for all sufficiently small positive t, contradicting (3). □

Corollary XIV.2.2 (Which bases work for every second argument?)**.** *For $d \geq 2$, a fixed $R \in \mathcal{P}_d$ makes $P \mapsto B_R(P, Q)$ convex for every $Q \in \mathcal{D}_d^\circ$ if and only if R is a scalar multiple of the identity. The assertion holds whether the P-domain is $\mathcal{P}_d$ or $\mathcal{D}_d^\circ$.*

Proof. If $R = rI$, then $C_{R,Q} = r^{-1} S \succeq 0$. Conversely, suppose R^{-1} has two distinct eigenvalues $\alpha, \beta > 0$, with corresponding orthonormal eigenvectors e_1, e_2. Set $u = e_1 + e_2$ and $S_\varepsilon = uu^* + \varepsilon I$. At $\varepsilon = 0$, the compression of $(R^{-1} S_\varepsilon + S_\varepsilon R^{-1})/2$ to their span is

$$\begin{pmatrix} \alpha & (\alpha+\beta)/2 \\ (\alpha+\beta)/2 & \beta \end{pmatrix}, \qquad \det = -\frac{(\alpha - \beta)^2}{4} < 0.$$

For sufficiently small $\varepsilon > 0$, the same compression still has negative determinant, whereas $S_\varepsilon \succ 0$. Put $Q_0 = R^{-1/2} S_\varepsilon^2 R^{-1/2}$ and $Q = Q_0 / \operatorname{Tr} Q_0$. The resulting $C_{R,Q}$ is a positive scalar multiple of the displayed Jordan product, so it is not positive semidefinite. Theorem XIV.2.1 applies. □

XIV.2.2 An explicit two-dimensional counterexample

Theorem XIV.2.3 (Nonconvexity for strictly positive qubits)**.** *Let*

$$R=\begin{pmatrix}1/10 & 0\\ 0 & 9/10\end{pmatrix},\qquad Q=\begin{pmatrix}1/2 & 12/25\\ 12/25 & 1/2\end{pmatrix},\qquad P(t)=\begin{pmatrix}9/10 & 3/10-t/3\\ 3/10-t/3 & 1/10\end{pmatrix}.$$

For $1/100\le t\le 1/25$*, these are strictly positive density matrices, and* $t\mapsto B_R(P(t),Q)$ *is strictly concave. In particular, with* $P_1=P(1/100)$ *and* $P_2=P(1/25)$*,*

$$B_R\left(\frac{P_1+P_2}{2},Q\right)>\frac{B_R(P_1,Q)+B_R(P_2,Q)}{2}. \tag{Q2}$$

The same strict inequality holds after replacing every occurrence of B_R *in (Q2) by* $\sqrt{B_R}$*.*

Proof. The eigenvalues of Q are $49/50$ and $1/50$, and $\det P(t)=t(9-5t)/45>0$ throughout the interval.

Base scaling lets us replace R by $R_0=\mathrm{diag}(1,9)$. For this base,

$$A(t)=R_0^{1/2}P(t)R_0^{1/2}=\begin{pmatrix}9/10 & 9/10-t\\ 9/10-t & 9/10\end{pmatrix}.$$

Its eigenvalues are $9/5-t$ and t, with fixed unit eigenvectors $2^{-1/2}(1,1)^T$ and $2^{-1/2}(1,-1)^T$. The two-by-two square-root formula $\sqrt{M}=(M+\sqrt{\det M}\,I)/\sqrt{\mathrm{Tr}\,M+2\sqrt{\det M}}$ gives

$$S=\frac{1}{\sqrt{146/25}}\begin{pmatrix}23/25 & 36/25\\ 36/25 & 123/25\end{pmatrix},\qquad C=\frac{1}{\sqrt{146/25}}\begin{pmatrix}23/25 & 4/5\\ 4/5 & 41/75\end{pmatrix}.$$

The diagonal entries of C in the indicated eigenbasis of $A(t)$ are respectively $23/(15\sqrt{146/25})$ and $-1/(15\sqrt{146/25})$. Consequently

$$F_R(P(t),Q)=\frac{23\sqrt{9/5-t}-\sqrt{t}}{3\sqrt{146}},\qquad B_R(P(t),Q)=2-\frac{2(23\sqrt{9/5-t}-\sqrt{t})}{3\sqrt{146}}. \tag{Q3}$$

Here F_R is real. Differentiating twice yields

$$\frac{d^2}{dt^2}B_R(P(t),Q)=\frac{23(9/5-t)^{-3/2}-t^{-3/2}}{6\sqrt{146}}<0,$$

since $(9/5-t)/t\ge 44$ and $44^{3/2}>23$.

The values in (Q3) are positive; indeed $23\sqrt{9/5-t}-\sqrt{t}<23\sqrt{9/5}<3\sqrt{146}$, where the last inequality follows by squaring. The strict midpoint inequality for $\sqrt{B_R}$ now follows from strict increase and concavity of the scalar square root:

$$\sqrt{B_R(P(1/40),Q)}>\sqrt{\frac{B_R(P_1,Q)+B_R(P_2,Q)}{2}}\ge\frac{\sqrt{B_R(P_1,Q)}+\sqrt{B_R(P_2,Q)}}{2}.$$

□

Interpretation. The obstruction is the failure of a Jordan product of positive matrices to remain positive. The classification concerns convexity in P for fixed R,Q, not joint convexity of the unsquared distance. The explicit counterexample rules out the latter in dimension two as well. Nothing above establishes data processing or a classification of convexity in the base R.

XIV.3 A uniform–exponential counterexample to van Zwet convolution closure

Published target. Tim Hunter, *A note on the median of skewed distributions*, Statistical Papers **67** (2026), article 64, doi:10.1007/s00362-026-01831-0, Definition 1, calls a distribution with CDF F right van Zwet if one of its medians m satisfies

$$F(m-x)+F(m+x)\le 1 \qquad (x>0).$$

Left van Zwet means the reverse inequalities. The last paragraph of Section 5 conjectures closure under summation for log-concave van Zwet distributions. We refute this even when the independent summands are both right van Zwet and have continuous CDFs.

Theorem XIV.3.1 (Failure of convolution closure)**.** *Let $X\sim\mathrm{Exp}(2)$ and $Y\sim\mathrm{Uniform}[0,1]$ be independent. Both distributions are log-concave and right van Zwet. Their sum $Z=X+Y$ is neither right nor left van Zwet.*

Proof. The densities $2e^{-2x}\mathbf{1}_{[0,\infty)}(x)$ and $\mathbf{1}_{[0,1]}(y)$ are log-concave. Symmetry makes Y both right and left van Zwet, with median $1/2$. At the exponential median $m_X=(\log 2)/2$, its CDF F satisfies

$$F(m_X-x)+F(m_X+x)-1=\begin{cases}1-\cosh(2x), & 0<x<m_X,\\ -e^{-2(m_X+x)}, & x\ge m_X.\end{cases}$$

Both expressions are negative, so X is right van Zwet.

Let h and H denote the density and CDF of Z. Direct convolution and integration give, for $0\le s\le 1$,

$$h(s)=1-e^{-2s},\qquad H(s)=s-\frac{1-e^{-2s}}{2}.$$

Since $H(0)=0$ and $H(1)=(1+e^{-2})/2>1/2$, the unique median M of Z lies in $(0,1)$. Its defining equation is

$$2M-2+e^{-2M}=0.$$

Consequently, for every $0<x<\min\{M,1-M\}$,

$$\begin{aligned}H(M-x)+H(M+x)-1&=2M-2+e^{-2M}\cosh(2x)\\&=e^{-2M}\bigl(\cosh(2x)-1\bigr)>0.\end{aligned}$$

Thus Z is not right van Zwet. For every $x>M$, on the other hand,

$$H(M-x)+H(M+x)-1=H(M+x)-1<0,$$

since Z is nonnegative and has an unbounded right tail. Thus Z is not left van Zwet either. □

Scope. The sum itself is still log-concave: its density is $1-e^{-2s}$ on $(0,1)$ and $(e^2-1)e^{-2s}$ on $[1,\infty)$; the logarithm is concave on both intervals and its derivative jumps downward at 1. What fails is the median-centered skewness condition. The conjecture assumes ordinary log-concavity, not strict log-concavity. The definition and the conjecture impose no requirement that densities be positive on all of $\mathbb{R}$ or that CDFs be globally strictly increasing. The source explicitly treats Uniform$(0,1)$ as right van Zwet in Section 4. Both medians here, and the median of the sum, are unique, so there is no choice-of-median ambiguity.

Status and priority. This is a complete counterexample to the published convolution-closure conjecture, not a partial result. It does not contradict the source's median-superadditivity theorem. Searches for the source title and for subsequent van Zwet convolution results did not locate an earlier resolution as of 19 October 2026; this does not certify priority. The result is an elementary correction to a specialist conjecture, not evidence of an Annals- or Fields-level breakthrough.

XIV.3.1 The weak-limit closure assertion also fails as stated

The first paragraph of Hunter's Section 5 also posits weak-limit closure of the van Zwet class. Definition 1 explicitly permits atoms and uses the ordinary right-continuous CDF at both reflected points. With that exact convention, weak closure fails, even with a common unique median along the entire sequence.

Proposition XIV.3.2 (Failure of weak closure)**.** *For $n \geq 3$, define*

$$\mu_n = \frac{1}{4}\delta_{-1+1/n} + \frac{1}{2}\delta_0 + \frac{3}{20}\delta_{1+1/n} + \frac{1}{10}\delta_2.$$

Every μ_n is right van Zwet and has unique median zero, but the weak limit

$$\mu = \frac{1}{4}\delta_{-1} + \frac{1}{2}\delta_0 + \frac{3}{20}\delta_1 + \frac{1}{10}\delta_2$$

is neither right nor left van Zwet.

Proof. The mass strictly below and strictly above zero is $1/4$ in every case, while the mass at zero is $1/2$, so the median is uniquely zero. Write $D_n(x) = F_n(-x) + F_n(x) - 1$. Exact evaluation gives

$$D_n(x) = \begin{cases} 0, & 0 < x \leq 1 - 1/n, \\ -1/4, & 1 - 1/n < x < 1 + 1/n, \\ -1/10, & 1 + 1/n \leq x < 2, \\ 0, & x \geq 2. \end{cases}$$

Hence every μ_n is right van Zwet, and clearly μ_n converges weakly to μ. For the limiting CDF F, however,

$$F(-1) + F(1) - 1 = \frac{1}{4} + \frac{9}{10} - 1 = \frac{3}{20} > 0,$$

whereas $F(-x) + F(x) - 1 = -1/10$ whenever $1 < x < 2$. Thus neither orientation is possible. □

Continuous approximating distributions. The failure is not caused by requiring the approximating distributions to have atoms. For $0 < \varepsilon < 1/4$, replace the four atoms above by uniform distributions on, respectively,

$$[-1 + \varepsilon, -1 + 2\varepsilon], \quad [-\varepsilon, \varepsilon], \quad [1 + \varepsilon, 1 + 2\varepsilon], \quad [2, 2 + \varepsilon],$$

with the same weights $1/4, 1/2, 3/20, 1/10$. These distributions have continuous CDFs and unique median zero. Their central uniform component contributes zero to the reflected CDF defect. The total noncentral mass on each side is $1/4$, and every positive observation is farther from zero than every negative observation. Thus their right tail beyond any radius is at least their left tail beyond that radius, proving the right van Zwet condition. They converge weakly to the same μ as $\varepsilon \downarrow 0$.

Precise limitation. This second counterexample uses atoms in the limiting law and the source's literal CDF convention. It does not disprove weak closure when the limiting CDF is required to be continuous, or when the condition is imposed only almost everywhere in the radius. A modified condition $F(m - x-) + F(m + x) \leq 1$, using a left limit at the left reflection, is a different definition and is not addressed by this counterexample.

Related claims not counted as new. The sentence after Proposition 2 in Hunter's paper describes Gamma, Lévy and certain Beta cases as lacking formal proofs. However, K. M. Abadir, *The mean–median–mode inequality: counterexamples*, Econometric Theory **21** (2005), 477–482, doi:10.1017/S0266466605050267, explicitly states the same CDF condition and already identifies Gamma and Beta distributions as satisfying it. Consequently we do not count their verification as new resolutions. The primary source text is available in the author's uploaded full text at https://www.researchgate.net/publication/23564935_THE_MEAN-MEDIAN-MODE_INEQUALITY_COUNTEREXAMPLES. The present two counterexamples belong to one probability problem family, rather than two unrelated breakthroughs.

XIV.4 Finite groups and factorial arithmetic

XIV.4.1 Definitions and the orbit product formula

For a subgroup H of a finite group G, let $\mathrm{P}_G(H)$ be the probability that a uniformly chosen left transversal for H is also a right transversal. Equivalently,

$$\mathrm{P}_G(H) = \frac{|\mathrm{DT}_G(H)|}{|H|^{[G:H]}}.$$

Set $f(n) = n!/n^n$ for positive integers n, so $f(1) = 1$ and

$$\frac{f(n+1)}{f(n)} = \left(\frac{n}{n+1}\right)^n < 1. \tag{XIV.4.1}$$

Let H act by left multiplication on the left coset space G/H. If its orbit sizes are $t_1, \ldots, t_s$, then

$$\mathrm{P}_G(H) = \prod_{i=1}^{s} f(t_i). \tag{XIV.4.2}$$

This is Proposition 3.2 in [136], expressed as an orbit formula. For completeness, an H-orbit of size t on G/H corresponds to a double coset HgH, containing exactly t left cosets and t right cosets of H. Each left coset intersects each right coset in exactly $w = |H|/t$ elements. A left transversal inside that double coset has $|H|^t$ choices; those which also represent each right coset correspond to a bijection between its left and right cosets, followed by an element in each chosen intersection, giving $t!w^t$ choices. The ratio is $f(t)$, and choices in distinct double cosets are independent. In particular,

$$\mathrm{P}_G(H) = 1 \quad \Longleftrightarrow \quad H \trianglelefteq G. \tag{XIV.4.3}$$

Indeed, all orbit sizes equal one exactly when H fixes every coset, equivalently $g^{-1}Hg \subseteq H$ for every $g \in G$, which in a finite group is equivalent to normality.

XIV.4.2 Full answer to Problem 5.2

Problem 5.2 of [136] asks whether $\mathrm{P}_G(H) \leq \mathrm{P}_G(HN)$ for every $H \leq G$ and $N \trianglelefteq G$, and asks for the equality cases.

Theorem XIV.4.1. *Let G be a finite group, $H \leq G$, and $N \trianglelefteq G$. Then*

$$\mathrm{P}_G(H) \leq \mathrm{P}_G(HN).$$

Equality holds if and only if $N \leq H$ or $H \trianglelefteq G$.

Proof. Put $L = HN$ and $m = [L : H]$. The natural map

$$\pi : G/H \longrightarrow G/L, \qquad gH \longmapsto gL,$$

is H-equivariant and every fiber has cardinality m. Since $N \trianglelefteq G$ and $N \leq L$, every $n \in N$ fixes every coset gL: indeed $ngL = g(g^{-1}ng)L = gL$. Thus the H-orbits on G/L are exactly the L-orbits.

Fix one such orbit O of size t. Decompose its preimage into H-orbits,

$$\pi^{-1}(O) = O_1 \sqcup \cdots \sqcup O_r.$$

Each equivariant map $O_j \to O$ is surjective, and all its fibers have the same positive integer size u_j. Hence

$$|O_j| = tu_j, \qquad u_j \geq 1, \qquad \sum_{j=1}^{r} u_j = m.$$

By (XIV.4.1), each $f(tu_j) \leq f(t) \leq 1$, and therefore

$$\prod_{j=1}^{r} f(tu_j) \leq f(t). \tag{XIV.4.4}$$

Multiplying (XIV.4.4) over the L-orbits in G/L and using (XIV.4.2) proves the asserted inequality.

If $m = 1$, then $L = H$, giving equality; this is exactly the case $N \leq H$. Assume now $m > 1$. If some base orbit has $t > 1$, its inequality (XIV.4.4) is strict: for $r = 1$ one has $u_1 = m > 1$ and $f(tm) < f(t)$, while for $r \geq 2$ one has

$$\prod_{j=1}^{r} f(tu_j) \leq f(t)^r < f(t).$$

Thus equality for $m > 1$ requires every base orbit to have size one, or $L \trianglelefteq G$. In this case $\mathrm{P}_G(L) = 1$, and equality is equivalent, by (XIV.4.3), to $H \trianglelefteq G$. Conversely, if $H \trianglelefteq G$, then $HN \trianglelefteq G$ and both probabilities are one. □

Corollary XIV.4.2. *For any homomorphism $\varphi : G \to K$ from a finite group and any $H \leq G$,*

$$\mathrm{P}_G(H) \leq \mathrm{P}_{\varphi(G)}(\varphi(H)).$$

Equality holds if and only if $\ker \varphi \leq H$ *or* $H \trianglelefteq G$.

Proof. Take $N = \ker \varphi$. Since $N \leq HN$, the coset actions of HN on G/HN and HN/N on $(G/N)/(HN/N)$ have identical orbits. Formula (XIV.4.2) gives

$$\mathrm{P}_G(HN) = \mathrm{P}_{G/N}(HN/N) = \mathrm{P}_{\varphi(G)}(\varphi(H)),$$

and Theorem XIV.4.1 applies. □

XIV.4.3 Counterexample to Problem 5.6

Problem 5.6 of [136] asks whether equality of products of $f(n) = \Gamma(n+1)/n^n$ at distinct integers greater than one forces the lists of integers to coincide up to rearrangement. No primality, relative-primality, fixed-sum, or equal-cardinality condition is imposed in the problem. The preceding Theorem 4.6 in that article concerns a product at distinct primes on one side; the problem proposes to replace these by distinct integers.

Theorem XIV.4.3. *Let*

$$X = \{3, 5, 6, 7, 8, 14, 16, 20, 36\}, \qquad Y = \{4, 10, 12, 15, 18, 21, 32\}.$$

Then X and Y are disjoint finite sets of integers greater than one, of cardinalities nine and seven respectively, and

$$\prod_{x\in X} \frac{x!}{x^x} = \prod_{y\in Y} \frac{y!}{y^y} = \frac{11^6\,13^5\,17^3\,19^2\,23\,29\,31}{2^{123}\,3^{37}\,5^4\,7^8}. \tag{XIV.4.5}$$

Consequently the uniqueness assertion in Problem 5.6 is false, even if distinctness is required across both lists together.

Proof. For a prime p, Legendre's formula gives

$$v_p(f(n)) = \sum_{j\geq 1} \left\lfloor \frac{n}{p^j} \right\rfloor - n v_p(n).$$

Applying this formula to the two displayed finite sets gives the following complete prime-valuation certificate:

p	2	3	5	7	11	13	17	19	23	29	31
$\sum_{x\in X} v_p(f(x))$	−123	−37	−4	−8	6	5	3	2	1	1	1
$\sum_{y\in Y} v_p(f(y))$	−123	−37	−4	−8	6	5	3	2	1	1	1

Every valuation at a prime $p > 36$ is zero. Unique factorisation of positive rational numbers now proves (XIV.4.5) exactly. □

XIV.4.4 The coprime-order conjecture in Problem 5.3

Write $\mathrm{tp}(G) = \min_{H\leq G} \mathrm{P}_G(H)$. Problem 5.3 asks for a general characterisation of equality in

$$\mathrm{tp}(G \times K) \leq \min\{\mathrm{tp}(G)^{|K|}, \mathrm{tp}(K)^{|G|}\},$$

and explicitly conjectures equality whenever $\gcd(|G|, |K|) = 1$. We prove this entire stated coprime-order branch; the general equality-classification request is not settled here.

Lemma XIV.4.4. *For positive integers a, b, one has*

$$(ab)! \geq (a!)^b (b!)^a, \qquad f(ab) \geq f(a)^b f(b)^a.$$

Proof. On an a-by-b grid, let R be the group of permutations acting independently inside each of its b columns, and C the group acting independently inside each of its a rows. Then $|R| = (a!)^b$, $|C| = (b!)^a$, and $R \cap C = \{1\}$, since a permutation preserving every row and every column fixes each cell. Consequently the multiplication map $R \times C \to S_{ab}$ is injective, giving the factorial inequality. Division by $(ab)^{ab} = a^{ab}b^{ab}$ gives the other inequality. □

Theorem XIV.4.5. *For finite groups G, K of coprime orders,*

$$\mathrm{tp}(G \times K) = \min\{\mathrm{tp}(G)^{|K|}, \mathrm{tp}(K)^{|G|}\}.$$

Proof. Every subgroup of $G \times K$ is of the form $H \times J$ with $H \leq G$ and $J \leq K$. Indeed, for each (g,k) in a subgroup, coprimality of the orders of g and k supplies powers of (g,k) equal to $(g,1)$ and $(1,k)$, by the Chinese remainder theorem.

Let a_i be the H-orbit sizes on G/H, and b_j the J-orbit sizes on K/J. The $(H \times J)$-orbit sizes on $(G \times K)/(H \times J)$ are $a_i b_j$. Put

$$A = -\log \mathrm{P}_G(H), \quad B = -\log \mathrm{P}_K(J), \quad \alpha = -\log \mathrm{tp}(G), \quad \beta = -\log \mathrm{tp}(K).$$

All four numbers are nonnegative and $A \leq \alpha$, $B \leq \beta$. By Lemma XIV.4.4 and (XIV.4.2),

$$\begin{aligned} -\log \mathrm{P}_{G\times K}(H \times J) &\leq \sum_{i,j} [b_j(-\log f(a_i)) + a_i(-\log f(b_j))] \\ &= [K : J]A + [G : H]B. \end{aligned}$$

If $|H|, |J| \geq 2$, and $M = \max\{|K|\alpha, |G|\beta\}$, this is at most

$$\frac{|K|\alpha}{|J|} + \frac{|G|\beta}{|H|} \leq M\left(\frac{1}{|J|} + \frac{1}{|H|}\right) \leq M.$$

If $J = 1$, the orbit formula instead gives exactly

$$\mathrm{P}_{G\times K}(H \times 1) = \mathrm{P}_G(H)^{|K|} \geq e^{-M},$$

and the case $H = 1$ is symmetric. Thus every subgroup has probability at least e^{-M}. Conversely, choosing a minimizing subgroup in each factor and taking its product with the trivial subgroup of the other factor gives both upper bounds $\mathrm{tp}(G)^{|K|}$ and $\mathrm{tp}(K)^{|G|}$. Their minimum is e^{-M}, proving the theorem. □

XIV.4.5 Scope and source status

Theorem XIV.4.1 resolves the full inequality and equality request of Problem 5.2. Theorem XIV.4.3 is a full negative answer to the particular uniqueness assertion of Problem 5.6. It does not contradict Theorem 4.6, whose prime hypothesis is essential. Theorem XIV.4.5 resolves the explicit coprime-order conjecture in Problem 5.3, but not its unrestricted equality classification. No claim is made here concerning Problems 5.1, 5.4–5.5, 5.7, or 5.8.

The statements were checked against the 2024 published version, pages 1280–1281, and arXiv:2209.00986v2 (31 May 2023; latest version displayed on arXiv on 19 October 2026). Exact-title, author, problem-number, and counterexample searches did not locate a subsequent solution. A related 2024 article, Aivazidis–Loukaki–Sambale, *Common transversals and complements in abelian groups*, J. Algebraic Combin. 59, 581–595, addresses common transversals of three cyclic subgroups in finite abelian groups; its stated scope does not resolve either question here. Absence of a located follow-up is a limitation of this literature check, not a proof of novelty.

XIV.5 Three questions on cumulative subtraction games

Primary source and scope. Our targets are Conjectures 4 and 17 and Problem 6 of A. Bhagat, T. Kulkarni, U. Larsson, and D. Murali, *Tie-breaking in self interest cumulative subtraction games*, arXiv:2510.24280v2, 20 January 2026. The first conjecture is false, the second is true, and the question in Problem 6 has a negative answer. The proofs below use the paper's exact deterministic, uniform, history-independent tie-breaking conventions. Its two-action main theorem is unaffected. The source lists its related discrepancy paper as "in preparation"; a search for that exact title and subsequent resolutions did not reveal a published resolution. These are proof results for the cited statements, not a claim that publication priority has been exhaustively established.

XIV.5.1 Definitions and integer recurrence

For a nonempty subtraction set $S \subseteq \mathbb{N}$, players alternately remove an element of S from a heap, provided it does not exceed the remaining heap. Each player's payoff is the total amount they remove. The game ends when the remaining heap is less than $m = \min S$. At a decision node, a player maximizes their own eventual payoff. Among equally good continuations, convention F maximizes the other player's payoff, whereas convention A minimizes it.

Write $O_X(h) = (O_X^1(h), O_X^2(h))$ for the first and second player's payoffs, where $X \in \{FF, FA, AF, AA\}$ names their conventions. Let $\bar{X}$ reverse the two letters. The exact recurrence is

$$O_X(h) = \operatorname*{argmax}_{\text{lex},\,\varepsilon_X} \left\{ \left(s + O_{\bar{X}}^2(h-s), O_{\bar{X}}^1(h-s) \right) : s \in S,\ s \le h \right\}, \tag{XIV.5.1}$$

with $O_X(h) = (0,0)$ for $h < m$. Here the selector first maximizes the first coordinate, and then maximizes ε_X times the second coordinate, with $\varepsilon_X = 1$ for $X \in \{FF, FA\}$ and -1 otherwise. This is exactly Definition 2 of the source. Induction on h proves that it gives the unique payoff vector of the specified pure subgame-perfect equilibrium, regardless of any residual ties between actions producing the same pair.

XIV.5.2 A five-action counterexample to Conjecture 4

Theorem XIV.5.1. *For $S = \{5, 11, 15, 22, 24\}$ and $h = 89$,*

$$O_{FF}(89) = (50, 39), \qquad O_{AA}(89) = (49, 40).$$

Consequently the second player benefits when both conventions change from friendly to antagonistic, disproving Conjecture 4.

Proof. Evaluate the integer recurrence (XIV.5.1) through $h = 89$. The complete, reproducible calculation is given below. For clarity, the final choices, including every legal first move, are

s	candidate under FF	candidate under AA
5	$(40, 48)$	$(40, 48)$
11	$(41, 48)$	$(41, 48)$
15	$(41, 48)$	$(41, 48)$
22	$(49, 40)$	$(49, 40)$
24	$(50, 39)$	$(48, 39)$

Thus the unique optimal first move is 24 under FF and 22 under AA, yielding the claimed values. All operations in the certificate are addition, integer comparison, and finite iteration; no numerical approximation is involved. □

XIV.5.3 A four-action negative answer to Problem 6

Theorem XIV.5.2. *For $S = \{3, 5, 17, 25\}$ and $h = 166$,*

$$O_{AF}(166) = (85, 80), \qquad O_{AA}(166) = (85, 81).$$

Thus, with the first player's antagonistic convention fixed, the second player strictly benefits from changing from friendly to antagonistic. This answers Problem 6 negatively, even when only the deviating player's gain is considered.

Proof. The same recurrence gives these values; the complete certificate below checks all four conventions up to heap 166. □

Lemma XIV.5.3 (Dilation)**.** *For every positive integer q, $0 \le r < q$, and every convention X,*

$$O_{X,qS}(qh + r) = qO_{X,S}(h).$$

Proof. Every legal move is qs with $s \in S$. The condition $qs \le qh + r$ is equivalent to $s \le h$, and the successor is $q(h - s) + r$. Induction on h, using that positive scaling preserves both lexicographic orders in (XIV.5.1), proves the formula. □

The two counterexamples therefore give infinite families, with payoff improvements q at heaps $89q + r$ and $166q + r$, respectively. This is unbounded improvement across different rulesets, consistent with the fixed-ruleset bound proved next.

XIV.5.4 Conjecture 17: a uniform bound for all tie-breaking rules

Lemma XIV.5.4 (Payoff diameter in almost constant-sum games)**.** *Suppose a two-player strategic-form game has $L \le u_1(s) + u_2(s) \le U$ for every strategy profile s. If σ and τ are Nash equilibria, then*

$$|u_i(\sigma) - u_i(\tau)| \le U - L, \qquad i = 1, 2.$$

Proof. Apply equilibrium optimality to the cross profile $c = (\tau_1, \sigma_2)$:

$$u_1(\sigma) \ge u_1(c), \qquad u_2(\tau) \ge u_2(c).$$

Consequently $u_1(\sigma) + u_2(\tau) \ge L$. Subtracting this inequality from $u_1(\tau) + u_2(\tau) \le U$ gives $u_1(\tau) - u_1(\sigma) \le U - L$. Subtracting it instead from $u_1(\sigma) + u_2(\sigma) \le U$ gives $u_2(\sigma) - u_2(\tau) \le U - L$. Interchanging the equilibria proves both reverse bounds. □

Theorem XIV.5.5. *For every nonempty subtraction set $S \subseteq \mathbb{N}$, every heap h, every pair of deterministic tie-breaking conventions X, Y as in the source, and both players i,*

$$|O^i_X(h) - O^i_Y(h)| \le \min S - 1. \tag{XIV.5.2}$$

In particular, Conjecture 17 holds, including for infinite subtraction sets. For finite S this also gives the upper bound $\max S - 1$ *requested in Problem 1, without asserting its sharpness.*

Proof. Here O_X extends the earlier four-letter notation to the payoff under any deterministic convention that resolves ties only after maximizing the mover's own payoff. At every terminal history the unclaimed remainder r satisfies $0 \le r \le m - 1$, where $m = \min S$. Therefore the total payoff lies in $[h - m + 1, h]$ for every strategy profile. For each fixed h the game tree is finite, even when S is infinite, since only subtractions at most h are legal and each move reduces the heap. Every tie-breaking convention under consideration chooses own-payoff-maximizing continuations and hence induces a pure subgame-perfect equilibrium of the original own-payoff game. Such a profile is a Nash equilibrium. Lemma XIV.5.4 with $U = h$ and $L = h - m + 1$ proves (XIV.5.2). For $h < m$ all payoffs are zero, so the same conclusion holds directly. □

XIV.5.5 Complete short certificate

This self-contained Python 3 code is the complete finite computation needed for Theorems XIV.5.1 and XIV.5.2. Every intermediate entry for all four conventions was also checked against an independent recursive implementation retaining fixed Alice/Bob identities. The code below contains everything needed to reproduce the claimed counterexamples.

```
def outcomes(S, H):
    kinds = ("FF", "FA", "AF", "AA")
    out = {t: [(0, 0)] * (H + 1) for t in kinds}
    for h in range(min(S), H + 1):
        for t in kinds:
            choices = [
                (s + out[t[::-1]][h-s][1],
                 out[t[::-1]][h-s][0])
                for s in S if s <= h
            ]
            sign = 1 if t[0] == "F" else -1
            out[t][h] = max(
                choices, key=lambda p: (p[0], sign*p[1]))
    return out

C = outcomes((5, 11, 15, 22, 24), 89)
assert C["FF"][89] == (50, 39)
assert C["AA"][89] == (49, 40)
D = outcomes((3, 5, 17, 25), 166)
assert D["AF"][166] == (85, 80)
assert D["AA"][166] == (85, 81)
print("Both exact counterexamples verified.")
```

The recurrence proves the code's correctness by induction on the heap: every referenced entry has strictly smaller heap, every legal move is evaluated, and the two-stage maximization is precisely the published tie-breaking rule. The certificates require no oracle, random sample, floating-point computation, or optimization software.

XIV.6 Tielker's generalized Eulerian real-rootedness conjecture

The precise target is Conjecture 4.7 of E. Tielker, *Weighted Ehrhart Series and a Type-B Analogue of a Formula of MacMahon*, Discrete Comput. Geom. **75** (2026), 205–240, first published online April 4, 2025, https://doi.org/10.1007/s00454-025-00731-8. We use the descent statistics in that paper, Definitions 2.1 and 2.3–2.4. In particular, repeated negative letters contribute descents according to its standardization convention. Other conventions for signed multiset permutations define different polynomials.

The argument below proves the full stated conjecture, by applying an existing general theorem. It is a short corollary of Wagner's theorem, not a new preservation theorem or a claim of a major new technique. Priority of this explicit application has not been established.

The established preservation theorem

For a nonzero real polynomial p of degree d, define $W(p)$ by

$$\sum_{k\geq 0} p(k)t^k = \frac{W(p)(t)}{(1-t)^{d+1}}.$$

Wagner's Theorem 0.2 states that if $W(p)$ and $W(q)$ have only nonpositive real zeros, then $W(pq)$ has only nonpositive real zeros. See D. G. Wagner, *Total positivity of Hadamard products*, J. Math. Anal. Appl. **163** (1992), 459–483, https://doi.org/10.1016/0022-247X(92)90261-B. This precise formulation also appears as Theorem 1.1 in P. Brändén, L. Ferroni and K. Jochemko,

Preservation of inequalities under Hadamard products, arXiv:2408.12386v2 (2025). Constant nonzero polynomials satisfy the real-zero condition vacuously. Iterating the theorem is legitimate, because a product of the input polynomials is again a polynomial and the degree is additive.

Theorem XIV.6.1. *For every composition $\eta = (\eta_1, \dots, \eta_r)$ of a positive integer, both $d_{S_\eta}(t)$ and $d_{B_\eta}(t)$, with Tielker's definitions, have only negative real zeros (or are constant). Consequently Conjecture 4.7 holds in its entirety.*

Proof. Let

$$\Delta_d = \{x \in \mathbb{R}^d_{\geq 0} : x_1 + \cdots + x_d \leq 1\}, \qquad C_d = \{x \in \mathbb{R}^d : |x_1| + \cdots + |x_d| \leq 1\}.$$

Their Ehrhart polynomials, denoted respectively by a_d and b_d, satisfy, for integers $k \geq 0$,

$$a_d(k) = \binom{k+d}{d}, \qquad b_d(k) = \sum_{s=0}^{d} 2^s \binom{d}{s}\binom{k}{s}.$$

For the second identity, choose the s nonzero coordinates and their signs; the number of positive integer s-tuples of sum at most k is $\binom{k}{s}$. Both displayed expressions are polynomials in k of degree exactly d. The usual binomial generating-function identity gives

$$\sum_{k\geq 0} a_d(k)t^k = \frac{1}{(1-t)^{d+1}},$$

and

$$\begin{aligned}\sum_{k\geq 0} b_d(k)t^k &= \sum_{s=0}^{d} 2^s \binom{d}{s} \frac{t^s}{(1-t)^{s+1}} \\ &= \frac{1}{1-t}\left(1 + \frac{2t}{1-t}\right)^d = \frac{(1+t)^d}{(1-t)^{d+1}}.\end{aligned}$$

Therefore $W(a_d) = 1$ and $W(b_d) = (1+t)^d$, so every factor has only nonpositive real zeros.

Put $n = \eta_1 + \cdots + \eta_r$. Lattice-point counts in Cartesian products multiply; thus the Ehrhart polynomials of $\prod_i \Delta_{\eta_i}$ and $\prod_i C_{\eta_i}$ are $\prod_i a_{\eta_i}$ and $\prod_i b_{\eta_i}$, each of degree n. The $q = 1$ specializations of Tielker's Theorems 3.1 and 3.2 identify their numerator polynomials exactly as

$$d_{S_\eta} = W\left(\prod_{i=1}^{r} a_{\eta_i}\right), \qquad d_{B_\eta} = W\left(\prod_{i=1}^{r} b_{\eta_i}\right).$$

Wagner's theorem, applied $r - 1$ times, proves the required nonpositive real-rootedness. Their constant terms are both 1, so zero is not a root. This proves the assertion. □

Scope. This proves the univariate descent-polynomial assertion at $q = 1$, which is exactly the numbered conjecture. It does not assert stability or real-rootedness for the two-variable major-index refinements at arbitrary q. The type-A part is also consistent with classical real-rootedness results for multiset Eulerian polynomials; it should not be counted as an independent new discovery. The contribution recorded here is the explicit deduction of the stated A/B conjecture from the product descriptions and Wagner's established theorem.

Part XV
Structural Results on Published Open Problems

Manuscript overview. We develop complete arguments for several explicitly posed mathematical questions. For generalized qubit Bures–Wasserstein distance, we determine the sharp global expansion coefficient of every depolarizing channel, thereby giving positive definite counterexamples to data processing. A parameter-uniform arithmetic denominator theorem produces irrational continued-fraction limits for every nonterminating split quadratic numerator and all sufficiently large slopes in a fixed lattice. For arbitrary finite preorders, transfer maps and a multivariate binomial identity yield polar Ehrhart duality, ordinal-sum factorization, matrix Ehrhart–Zeta duality, an explicit gamma expansion, and full weighted zeta reciprocity. We also classify all common-transversal minimizers in odd-order finite groups and exhibit a periodic lattice term that invalidates a numerical monotonicity observation for the Poisson friendship paradox. Each claim is separated from the larger questions it does not resolve.

Research and verification status. This is a collection of proposed proofs, checked through separate in-session derivations and exact arithmetic where appropriate. It has not undergone external peer review. Original statements, cited arithmetic inputs, and available follow-up papers were checked through 19 October 2026; targeted searches do not certify priority. Several numbered conjectures in the preorder paper follow from a common mechanism and are not counted as independent major discoveries. No claim about journal tier or prize-level significance is made.

What is stronger here. The quantum result is an optimal constant on an entire channel family, not only an isolated counterexample. The arithmetic constant is uniform in the varying slope, which is essential to the irrationality conclusion. The preorder results cover arbitrary equivalence classes and quotient posets, and the gamma coefficients have a direct counting interpretation. The finite-group result covers every odd order and every minimizing subgroup. The probability argument isolates an effect invisible in the source's finite numerical plot.

Precise scope register

External target	Conclusion and boundary
Afham–Ferrie Problem 1; the qubit case left open by Rajaei	Negative answer, with a strictly positive rational certificate. For depolarizing parameter $-1/3 \le \lambda < 1$, the optimal squared-distance coefficient is $\lambda^2/(1-\lambda^2)$. All qubit channels are not classified.
Ben David et al., Section 2.8, Observation after Theorem 3	Every nonterminating rational-root quadratic numerator admits a whole cofinal lattice family of irrationality-proving linear denominators. The necessary half of Conjecture 1.4 and the higher-degree assertion remain open here.
Chapoton–Athanasiadis Conjectures 6.2 and 6.3	Polar reversal duality and ordinal-sum factorization for every preorder, with an explicit lattice transfer, a shear, and a finite polar h^* formula.
Same source, Conjectures 4.2, 7.2, and 4.4	Full multichain-matrix duality, double reciprocity, and the dependent descent interpretation, using cloning and lattice translation.
Same source, Conjectures 5.2 and 4.9	Full gamma positivity and full q-zeta reciprocity from one multivariate binomial identity. Conjecture 5.1(a,b) follows; real-rootedness and simplicial-polytope realization do not.
Aivazidis–Loukaki–Müller Problem 5.8	The entire odd-order case, strengthened to all minimizers. The unrestricted even-order case remains open here.
Hazra–den Hollander–Parvaneh, numerical observation after Theorem 4.3	The Poisson probability has eventual increasing and decreasing pairs. Conjecture 2.6 is proved only for sufficiently large Poisson means, not for every mean or all the other distributions.

Existing work not claimed as new. Preorder support-enumerator duality, source Conjecture 5.4, was already proved by Dai et al. The special octopus and lopsided-octopus gamma cases were already treated by Menon. The arithmetic proof uses the established Galochkin condition for rational hypergeometric operators. The quantum proof uses Rajaei's qubit fidelity formula. These inputs are credited in the relevant arguments and references.

XV.1 Quantum information: the sharp depolarizing coefficient

Published target. Afham and Ferrie, *Riemannian-geometric generalizations of quantum fidelities and Bures–Wasserstein distance*, J. Math. Phys. **66** (2025), 082201, Section 10, Problem 1, ask about data processing for their generalized distance. Rajaei, arXiv:2609.09753v2 (2 October 2026), disproves the general inequality in dimensions at least three and explicitly leaves the general qubit case unresolved. We give a negative answer in dimension two and, more strongly, compute the optimal coefficient for the entire family of qubit depolarizing channels. This does not classify all qubit channels.

For trace-one positive semidefinite P, Q and positive definite R, define

$$F_R(P,Q) = \operatorname{Tr}\Big[(R^{1/2}PR^{1/2})^{1/2}R^{-1}(R^{1/2}QR^{1/2})^{1/2}\Big],$$
$$B_R(P,Q) = 2 - 2\operatorname{Re} F_R(P,Q).$$

Let $\mathcal{D}_2^\circ$ be the strictly positive qubit density matrices. The depolarizing channel is

$$\Delta_\lambda(X) = \lambda X + (1-\lambda)\operatorname{Tr}(X)\frac{I_2}{2}, \qquad -\frac{1}{3} \le \lambda \le 1.$$

These are exactly its completely positive parameters: its Pauli-channel weights are $(1+3\lambda)/4$ for the identity and $(1-\lambda)/4$ for each of the three Pauli conjugations.

Theorem XV.1.1 (Optimal global expansion coefficient)**.** *For every $-1/3 \le \lambda < 1$,*

$$\boxed{\sup_{\substack{P,Q,R\in\mathcal{D}_2^\circ \\ P\neq Q}} \frac{B_{\Delta_\lambda(R)}(\Delta_\lambda(P),\Delta_\lambda(Q))}{B_R(P,Q)} = \frac{\lambda^2}{1-\lambda^2}.} \tag{Q1}$$

At $\lambda = 1$, the supremum is 1. Consequently universal data processing holds for this family exactly when

$$-\frac{1}{3} \le \lambda \le \frac{1}{\sqrt{2}} \quad \text{or} \quad \lambda = 1. \tag{Q2}$$

The optimal coefficient for the unsquared distance $\sqrt{B_R}$ is $|\lambda|/\sqrt{1-\lambda^2}$ for $\lambda < 1$.

XV.1.1 A hemisphere representation

Write a qubit state as $\rho(p) = (I_2 + p\cdot\sigma)/2$, with $p \in \mathbb{R}^3$, $|p| \le 1$. Define

$$X(p) = (p, \sqrt{1-|p|^2}) \in \mathbb{R}^4, \qquad N(v) = v/|v|.$$

Here and below Euclidean norms are used for real vectors. For $P = \rho(p), Q = \rho(q), R = \rho(r)$, Rajaei's qubit formula gives

$$\begin{aligned}\operatorname{Re} F_R(P,Q) &= \frac{1 + X(p)\cdot X(q) + X(p)\cdot X(r) + X(q)\cdot X(r)}{2\sqrt{(1+X(p)\cdot X(r))(1+X(q)\cdot X(r))}} \\ &= N(X(p)+X(r))\cdot N(X(q)+X(r)).\end{aligned} \tag{XV.1.1}$$

The formula also follows directly from the two-by-two square-root identity

$$\sqrt{A} = \frac{A + \sqrt{\det A}\, I_2}{\sqrt{\operatorname{Tr} A + 2\sqrt{\det A}}}.$$

Strict positivity of R ensures that the denominators do not vanish. Thus

$$B_R(P,Q) = |N(X(p)+X(r)) - N(X(q)+X(r))|^2. \tag{Q3}$$

Let $F_U(P,Q)=\operatorname{Tr}\sqrt{\sqrt{P}Q\sqrt{P}}$ and $B_U(P,Q)=2-2F_U(P,Q)$. We shall use

$$\operatorname{Re}F_R(P,Q)\le F_U(P,Q),\qquad B_U(P,Q)\le B_R(P,Q). \tag{Q4}$$

To see this, put $Y_P=R^{-1/2}(R^{1/2}PR^{1/2})^{1/2}$, so that $Y_PY_P^*=P$ and $F_R(P,Q)=\operatorname{Tr}(Y_P^*Y_Q)$. Writing $Y_P=\sqrt{P}U_P$ and $Y_Q=\sqrt{Q}U_Q$, with unitaries extended on any kernels, the trace-norm inequality proves (Q4). The familiar qubit fidelity formula further yields

$$|X(p)-X(q)|^2=4(1-F_U(P,Q)^2)=2(1+F_U(P,Q))B_U(P,Q)\le 4B_U(P,Q). \tag{Q5}$$

XV.1.2 Global upper bound

For any nonzero Euclidean vectors u,v,

$$|N(u)-N(v)|^2=\frac{|u-v|^2-(|u|-|v|)^2}{|u||v|}\le\frac{|u-v|^2}{|u||v|}. \tag{Q6}$$

Put $a=|\lambda|<1$. Since the last coordinate of $X(\lambda p)$ is at least $\sqrt{1-a^2}$, we have

$$|X(\lambda p)+X(\lambda r)|\ge 2\sqrt{1-a^2}. \tag{Q7}$$

Also,

$$|X(\lambda p)-X(\lambda q)|\le a\,|X(p)-X(q)|. \tag{Q8}$$

Indeed, the first three coordinates scale by λ, and the last coordinate is obtained from $s=\sqrt{1-|p|^2}$ by the function $g_a(s)=\sqrt{1-a^2+a^2s^2}$, whose derivative has absolute value at most a on $[0,1]$. Applying (Q6)–(Q8), then (Q5) and (Q4), gives

$$\begin{aligned}B_{\Delta_\lambda(R)}(\Delta_\lambda(P),\Delta_\lambda(Q))&\le\frac{|X(\lambda p)-X(\lambda q)|^2}{4(1-a^2)}\\&\le\frac{a^2}{4(1-a^2)}|X(p)-X(q)|^2\\&\le\frac{a^2}{1-a^2}B_U(P,Q)\le\frac{a^2}{1-a^2}B_R(P,Q).\end{aligned}$$

This proves the required upper bound on the full state space.

XV.1.3 Sharpness and strictly positive counterexamples

Fix $0<\theta<\pi$, and let

$$p=(\sin\theta,0,\cos\theta),\quad q=(0,0,1),\quad r_\varepsilon=(\sin\varepsilon,0,-\cos\varepsilon),\qquad 0<\varepsilon<\pi-\theta.$$

These are pure-state Bloch vectors. Their corresponding real state vectors may be chosen as

$$|\psi_p\rangle=(\cos(\theta/2),\sin(\theta/2))^T,\quad |\psi_q\rangle=(1,0)^T,\quad |\psi_r\rangle=(\sin(\varepsilon/2),\cos(\varepsilon/2))^T.$$

All relevant overlaps are positive. Although the rank-one base is outside the original inverse-based definition, its limit from strictly positive bases exists here and equals

$$\operatorname{Re}F_{\rho(r_\varepsilon)}(\rho(p),\rho(q))=\cos(\theta/2). \tag{Q9}$$

For example, this follows by taking the rank-one limit in

$$F_R(P,Q)=\frac{\operatorname{Tr}(RPQ)}{\sqrt{\operatorname{Tr}(RP)\operatorname{Tr}(RQ)}},\qquad P,Q\text{ pure}.$$

The right side is defined for the rank-one base in question, since neither denominator factor vanishes.

For fixed θ, ε, replace all three Bloch vectors by $\mu p, \mu q, \mu r_\varepsilon$, $0 < \mu < 1$. These are strictly positive inputs. Letting $\mu \uparrow 1$, then $\varepsilon \downarrow 0$, is legitimate by continuity of (XV.1.1); the input value in (Q9) remains constant during the second limit. Since $|\lambda| < 1$, all output states remain strictly positive even at these limits. Writing $s = \sqrt{1-\lambda^2}$, the limiting output fidelity is

$$\frac{s}{\sqrt{s^2 + \lambda^2 \sin^2(\theta/2)}}. \tag{Q10}$$

This follows immediately from (Q3): the two vectors to normalize are

$$(\lambda \sin\theta, 0, \lambda(\cos\theta - 1), 2s) \quad \text{and} \quad (0, 0, 0, 2s).$$

Therefore the supremum in (Q1) is at least

$$\frac{1 - s/\sqrt{s^2 + \lambda^2 \sin^2(\theta/2)}}{1 - \cos(\theta/2)}.$$

Letting $\theta \downarrow 0$ gives $\lambda^2/(1-\lambda^2)$. This proves sharpness using only limits of triples of strictly positive states. The case $\lambda = 0$ is immediate.

If $1/\sqrt{2} < \lambda < 1$, choose $\theta > 0$ so small that

$$0 < \sin^2(\theta/2) < 2 - \lambda^{-2}.$$

Then (Q10) is strictly smaller than (Q9), as one checks by squaring the two positive quantities. Taking ε sufficiently small and then μ sufficiently close to one preserves the strict inequality. This constructs genuine positive definite counterexamples for every such channel, rather than just singular limiting violations. At $\lambda = 1$ the channel is the identity. This completes the proof of Theorem XV.1.1.

XV.1.4 An exact rational positive definite certificate

The limiting argument above can be supplemented by a concrete rational example. Put $\mu = 999/1000$, $\lambda = 9/10$, and

$$p = \mu(3/5, 0, 4/5), \qquad q = \mu(0, 0, 1), \qquad r = \mu(20/101, 0, -99/101).$$

Each of the three density matrices has eigenvalues $1/2000$ and $1999/2000$, so all inputs are strictly positive. The hemisphere formula reduces their real generalized fidelity, when their common squared Bloch radius is t, to

$$f(t) = \frac{2020 - 1942t}{\sqrt{(2020 - 1682t)(2020 - 2000t)}}.$$

The input has $t = 998001/10^6$, and the output has $t' = 81t/100$. Clearing the common denominator in the three factors gives

$$(n, a, b) = (81882058, 341362318, 23998000)$$

at the input, and

$$(n', a', b') = (45012446698, 66030347758, 40323838000)$$

at the output. The following exact integer comparisons establish the strict violation without floating-point arithmetic:

$$100n^2 - 81ab = 6914096735052400 > 0,$$
$$49a'b' - 64(n')^2 = 795552362406874390944 > 0.$$

Thus $f(t) > 9/10$ and $f(t') < 7/8$. In particular,

$$B_{\Delta_\lambda(R)}(\Delta_\lambda(P), \Delta_\lambda(Q)) - B_R(P, Q) = 2(f(t) - f(t')) > \frac{1}{20}.$$

Consequence and limitation. The general qubit data-processing question has a negative answer. Moreover, channels arbitrarily close to the identity have arbitrarily large optimal expansion factors when the base is allowed to approach the boundary: the squared-distance coefficient is asymptotic to $1/[2(1-\lambda)]$ as $\lambda \uparrow 1$. This is a supremum over changing input triples; it is not divergence for a fixed strictly positive triple. The exact classification (Q2) concerns depolarizing channels only. It does not characterize all quantum channels, nor does it contradict ordinary Bures data processing.

XV.2 Uniform arithmetic and irrational continued-fraction limits

XV.2.1 The external question and the result

Write

$$b(n) = B(n - x_1)(n - x_2), \qquad s = x_1 + x_2, \qquad B \in \mathbb{Q} \setminus \{0\}, \quad x_1, x_2 \in \mathbb{Q}.$$

For a rational parameter A, set

$$a_A(n) = A\left(n + \frac{1-s}{2}\right). \tag{XV.2.1}$$

Our continued-fraction convention is

$$L_A = a_A(0) + \cfrac{b(1)}{a_A(1) + \cfrac{b(2)}{a_A(2) + \cdots}}.$$

Its canonical continuants satisfy

$$p_{-1} = 1, \quad p_0 = a_A(0), \qquad p_n = a_A(n)p_{n-1} + b(n)p_{n-2},$$
$$q_{-1} = 0, \quad q_0 = 1, \qquad q_n = a_A(n)q_{n-1} + b(n)q_{n-2}.$$

Ben David et al. [138, Section 2.8, Observation after Theorem 3] expect an infinite collection of irrationality-proving linear denominators for every split quadratic b. Their argument assumes a suitable bound on the exponential part of factorial reduction as the parameter varies. A bound separately for each A does not justify that step.

Theorem XV.2.1. *Assume $b(n) \neq 0$ for every positive integer n. There exist an integer $H \geq 1$ and a real $A_0 > 0$, depending only on b, such that for every $A \in H\mathbb{Z}$ with $A > A_0$, the continued fraction L_A converges to a finite irrational number. The proof uses its own canonical convergents and a factorial-reduction estimate uniform in A.*

The nonvanishing hypothesis excludes terminating continued fractions, whose limits are rational. A constant inflation $a \mapsto Da$, $b \mapsto D^2 b$, with D fixed independently of A, makes the polynomial coefficients integral; it multiplies the limit by D and does not change irrationality. It is therefore enough to work with $b \in \mathbb{Z}[n]$, $B \in \mathbb{Z} \setminus \{0\}$, and A in an integral lattice ensuring $a_A \in \mathbb{Z}[n]$.

XV.2.2 The parameter-uniform denominator theorem

Theorem XV.2.2. *Fix $B \in \mathbb{Z} \setminus \{0\}$ and $x_1, x_2 \in \mathbb{Q}$, and put $s = x_1 + x_2$. There exist positive integers D_N and a constant $C \geq 1$, independent of $A \in \mathbb{Z}$, with $D_N \leq C^{N+1}$, having the following property. If $A^2 + 4B \neq 0$ and $u_0, u_1 \in \mathbb{Z}$, then the solution of*

$$u_n = A\left(n + \frac{1-s}{2}\right) u_{n-1} + B(n - x_1)(n - x_2)u_{n-2}, \qquad n \geq 2, \tag{XV.2.2}$$

satisfies

$$D_N \frac{u_n}{n!} \in \mathbb{Z} \qquad (0 \leq n \leq N).$$

The same D_N works for every choice of the two integral initial values.

Proof. Let $V(z) = \sum_{n \geq 0} u_n z^n / n!$. Direct summation gives

$$P(z)V'' + \frac{5-s}{2}P'(z)V' - B(2 - x_1)(2 - x_2)V = 0, \qquad P(z) = 1 - Az - Bz^2. \tag{XV.2.3}$$

Set

$$\Delta^2 = A^2 + 4B, \qquad t = \frac{2Bz + A + \Delta}{2\Delta}, \qquad t_0 = \frac{A + \Delta}{2\Delta}.$$

Then $dt/dz = B/\Delta$, $t_0(1 - t_0) = B/\Delta^2$, and (XV.2.3) becomes the fixed Gauss equation

$$t(1-t)f'' + [c - (a + b + 1)t]f' - abf = 0, \quad a = 2 - x_1,\ b = 2 - x_2,\ c = \frac{5-s}{2}. \tag{XV.2.4}$$

In particular $a + b + 1 = 2c$. The three parameters are rational and do not depend on A.

For $Y = (f, tf')^{\mathsf{T}}$, this equation is $Y' = G(t)Y$, with

$$T(t) = t(1 - t), \qquad G(t) = \frac{N(t)}{T(t)}, \qquad N(t) = \begin{pmatrix} 0 & 1 - t \\ ab\,t & 1 - c + (a + b)t \end{pmatrix}.$$

Define $G_0 = I$ and $G_{m+1} = G_m' + G_m G$, so that $Y^{(m)} = G_m Y$, and put

$$U_m(t) = \frac{T(t)^m G_m(t)}{m!}.$$

These are polynomial matrices of degree at most m. Indeed $U_0 = I$, and

$$U_{m+1} = \frac{TU_m' - mT'U_m + U_m N}{m + 1}$$

preserves that degree bound.

We now invoke the established arithmetic input. Every Gauss operator with rational parameters is a G-operator, without an irreducibility or nonresonance requirement [139, Theorem 3.3]. Equivalence with the Galochkin condition is explained in [139, Section 1.7]; its matrix form and invariance under change of system are given in [140, Definition 6 and Section 2]. Consequently there are positive integers $d_N \leq C_0^{N+1}$ such that

$$d_N U_m(t) \in M_2(\mathbb{Z}[t]) \qquad (0 \leq m \leq N).$$

All objects in this sentence concern the one fixed system (XV.2.4); in particular C_0 and d_N are independent of A.

Taking the first component of the derivative formula gives

$$\frac{V^{(m)}(0)}{m!} = e_1^{\mathsf{T}} \Delta^m U_m(t_0) \begin{pmatrix} u_0 \\ \dfrac{A + \Delta}{2B} u_1 \end{pmatrix}. \tag{XV.2.5}$$

Here the cancellation is exact:

$$\left(\frac{B}{\Delta}\right)^m T(t_0)^{-m} = \Delta^m.$$

Since Δ is an algebraic integer and $\deg U_m \le m$, the entries of

$$2^m d_N \Delta^m U_m\left(\frac{A+\Delta}{2\Delta}\right)$$

are algebraic integers. Thus $2^{N+1}|B|d_N V^{(m)}(0)/m!$ is an algebraic integer for every $m \le N$. It is rational by (XV.2.2), and hence is an integer. Take $D_N = 2^{N+1}|B|d_N$ and enlarge the constant C if necessary. □

Corollary XV.2.3. *Suppose $a_A, b \in \mathbb{Z}[n]$, $A^2 + 4B \neq 0$, and $b(n) \neq 0$ at all positive integers. With a constant C independent of the integral slope A, the canonical continuants satisfy*

$$g_n := \gcd(p_n, q_n) \ge \frac{n!}{C^{n+1}}.$$

Proof. The canonical u_0, u_1 are integral. Theorem XV.2.2 gives $n! \mid D_n p_n$ and $n! \mid D_n q_n$. The continuant determinant

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n+1} \prod_{j=1}^{n} b(j)$$

is nonzero, so the pair is not zero and $n! \le D_n g_n$. □

XV.2.3 Large slope turns the bound into irrationality

We give elementary estimates; no asymptotic theorem for recurrences or existence of an exact exponential gcd order is required.

First assume, after translating the index by a fixed integer if necessary, that

$$x_1, x_2 \le 0, \qquad c_* := \frac{1-s}{2} \ge 0.$$

Choose constants $K \ge 1$ and $J \ge 1$, depending only on b, such that

$$|b(n)| \le Kn^2, \qquad n \le n + c_* \le (1 + c_*)n \quad (n \ge 1), \qquad J = 1 + c_* + 4K.$$

For $A \ge 1$ with $A^2 \ge 8K$, induction gives

$$\frac{A}{2} n \le \frac{q_n}{q_{n-1}} \le JAn, \qquad \left(\frac{A}{2}\right)^n n! \le q_n \le (JA)^n n!. \tag{XV.2.6}$$

The base case is $q_0 = 1$, $q_1 = A(1 + c_*)$. For $n \ge 2$, the inductive lower bound implies

$$\frac{|b(n)|}{q_{n-1}/q_{n-2}} \le \frac{2Kn^2}{A(n-1)} \le \frac{4K}{A} n.$$

Substitution in $q_n/q_{n-1} = A(n + c_*) + b(n)/(q_{n-1}/q_{n-2})$ proves both bounds and also positivity.

Set $r = 4K/A^2 < 1$. The determinant identity and (XV.2.6) show that

$$\left|\frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}}\right| \le \frac{A}{2} n r^n.$$

Thus the convergents have a finite limit L_A and

$$\left|L_A - \frac{p_n}{q_n}\right| \le \frac{A(n+1)r^{n+1}}{2(1-r)^2}. \tag{XV.2.7}$$

Writing the reduced convergent as P_n/Q_n, Corollary XV.2.3 gives

$$Q_n \le C^{n+1}(JA)^n.$$

Consequently, for each fixed $A > 4KCJ$,

$$Q_n \left| L_A - \frac{P_n}{Q_n} \right| \le \frac{ACr}{2(1-r)^2}(n+1)\left(\frac{4KCJ}{A}\right)^n \longrightarrow 0. \tag{XV.2.8}$$

If $L_A = u/v$ were rational, every unequal reduced convergent would satisfy $Q_n|L_A - P_n/Q_n| \ge 1/|v|$. Hence (XV.2.8) would force all sufficiently late convergents to equal L_A, contradicting the nonzero adjacent determinant. This proves irrationality.

For arbitrary rational roots, choose a fixed integer N so large that $x_i - N \le 0$. The tail has

$$b_N(n) = B(n-(x_1-N))(n-(x_2-N)), \quad a_{A,N}(n) = A\left(n + \frac{1-(s-2N)}{2}\right),$$

so it is exactly another family of the same type. Its denominator constant C remains independent of A. The preceding proof establishes irrationality of this tail for every sufficiently large allowed A. Restoring the finitely many initial levels acts by a fractional linear transformation with rational coefficients and nonzero determinant, because all the original $b(j)$ are nonzero. Such a transformation takes a finite irrational number to a finite irrational number. Choosing the integral lattice and the fixed inflation described above completes the proof of Theorem XV.2.1.

XV.2.4 Scope and priority status

The new step relative to the earlier individual factorial-reduction argument is uniformity in the slope: one C works for the whole symmetric family. The result settles the nonterminating quadratic assertion in the Observation after Theorem 3 of [138]; it does not settle that paper's higher-degree expectation, nor the necessary half of Conjecture 1.4. We have checked the original publication and searched for subsequent work; a priority claim requires a broader specialist review.

This argument does not name a previously unresolved classical constant as irrational. It proves the externally requested existence theorem for every split quadratic numerator, with a whole cofinal parameter family and direct irrationality certificates.

XV.3 Preorder polytopes: transfer maps, matrix duality, and reciprocity

Source and scope. Athanasiadis and Chapoton, *Polytopes and posets associated to preorders*, arXiv:2605.26916, formulate the statements addressed below as Conjectures 4.2, 4.4, 6.2, 6.3, and 7.2. The arguments apply to every finite preorder, including nontrivial equivalence classes, and every indicated parameter. The five conjecture numbers represent three mechanisms: a transfer construction, a cloning construction using Postnikov's duality, and lattice translation with Ehrhart reciprocity. Conjecture 4.4 is a consequence of 7.2, not an independent mechanism. The support-enumerator duality of their Conjecture 5.4 was already established by Dai–Hou–Liu–Thawinrak–Wang, arXiv:2608.16037, and is not claimed here.

Let τ be a preorder on E, with $|E| = n \ge 1$. Write $\mathcal{V}$ for its equivalence classes, ordered by the induced partial order, and $v(e)$ for the class containing e. An ideal always means a nonempty order ideal unless otherwise stated. Put

$$Q_\tau(r,s) = \left\{ x \in \mathbb{R}^E_{\ge 0} : \sum_{e\in I} x_e \le r|I| + s \quad \text{for every ideal } I \right\}, \qquad Q_\tau = Q_\tau(1,0).$$

For nonnegative integers r, s these are lattice polytopes, and $Q_\tau(r,s) = rQ_\tau + s\Delta_E$, where $\Delta_E = \operatorname{conv}(0, e_e : e \in E)$; these facts are proved in the source paper. Let $P_\tau = Q_\tau \cap \mathbb{Z}^E$ with coordinatewise order.

XV.3.1 A transfer code for the polar polytopes

Set

$$O_\tau = \operatorname{conv}\{\mathbf{1}_I : I \text{ an order ideal, including } \varnothing\}, \qquad K_\tau = \operatorname{conv}(O_\tau \cup (-\Delta_E)).$$

Here O_τ is equivalently the set of vectors constant on each class, order-decreasing on the quotient poset, and lying in $[0,1]^E$. The polar polytope $\mathcal{R}_\tau^\vee$ of the source paper is precisely K_τ: its listed vertices are $-e_e$ and the nonempty ideal indicators, and adding 0 does not change their convex hull. The origin is interior to K_τ. Indeed $\mathbf{1}_E$ is in O_τ, and the simplex with vertices $\mathbf{1}_E, -e_e$ is full dimensional with 0 strictly inside.

Let $g_\tau(x) = \inf\{t \ge 0 : x \in tK_\tau\}$ be its gauge. For $x \in \mathbb{R}^E$ define

$$V_v(x) = \max(\{0\} \cup \{x_e : v(e) \ge v\}), \qquad M(x) = \max_{e \in E}(0, x_e).$$

Lemma XV.3.1 (Exact gauge)**.** *For every $x \in \mathbb{R}^E$,*

$$g_\tau(x) = M(x) + \sum_{v \in \mathcal{V}} \sum_{e \in v} (V_v(x) - x_e). \tag{XV.3.1}$$

In particular, g_τ is integer valued on $\mathbb{Z}^E$.

Proof. Membership in tK_τ is equivalent to writing $x = a - b$ with $a \in sO_\tau$, $b \ge 0$, $\sum_e b_e \le t - s$, and $0 \le s \le t$. Thus a must be nonnegative, order-decreasing, constant on classes, and satisfy $a_e \ge x_e$. Its coordinatewise smallest possible value is $a_e = V_{v(e)}(x)$. Every feasible representation therefore has $s \ge M(x)$ and

$$t \ge s + \sum_e (a_e - x_e) \ge M(x) + \sum_e (V_{v(e)}(x) - x_e).$$

Taking $a_e = V_{v(e)}(x)$, $s = M(x)$, and $b = a - x$ attains this bound. □

For each v, define

$$U_v = \max(\{0\} \cup \{V_w : w > v\}), \qquad d_v = V_v - U_v, \qquad \delta_e = V_{v(e)} - x_e.$$

All these numbers are nonnegative. They satisfy the local condition

$$d_v > 0 \quad \Longrightarrow \quad \min_{e \in v} \delta_e = 0. \tag{XV.3.2}$$

Conversely, take any nonnegative arrays (d_v) and (δ_e) satisfying (XV.3.2). Reconstruct V from maximal classes downward by

$$V_v = d_v + \max(\{0\} \cup \{V_w : w > v\}), \qquad x_e = V_{v(e)} - \delta_e.$$

These constructions are inverse. To check this assertion, proceed downward through the quotient poset. If $d_v > 0$, the minimum-zero condition makes $\max_{e \in v} x_e = V_v$; if $d_v = 0$, the upper envelope already equals V_v and every $x_e \le V_v$ is allowed. Thus the reconstructed V is exactly the upper envelope of the reconstructed x. This also proves a bijection between integer x and integer codes.

The max-plus recurrence for V gives

$$g_\tau(x) = \max_{C \text{ a chain in } \mathcal{V}} \sum_{v \in C} d_v + \sum_{e \in E} \delta_e. \tag{XV.3.3}$$

The empty chain, of weight zero, may be included. A chain in the quotient poset remains a chain on reversing its order.

Theorem XV.3.2 (Polar reversal duality; source Conjecture 6.2)**.** *For every finite preorder τ and integer $m \geq 0$,*

$$\#(mK_\tau \cap \mathbb{Z}^E) = \#(mK_{\tau^*} \cap \mathbb{Z}^E).$$

Consequently their Ehrhart polynomials and h^*-polynomials agree. More generally, these enumerators depend only on the equivalence classes and the family of chains of the quotient poset.

Proof. Encode x by (d, δ) and reconstruct it with the reversed quotient order. The set of permissible codes does not change, and (XV.3.3) is unchanged because its chain family is unchanged. The resulting map is a lattice bijection preserving the gauge. It restricts to a bijection on every dilate. The same proof applies to any two quotient orders on the same classes having the same chain subsets. □

Theorem XV.3.3 (Ordinal sums and a geometric explanation; source Conjecture 6.3)**.** *If $\tau = \tau_1 \oplus \tau_2$ is an ordinal sum, then*

$$h^*(K_\tau, z) = h^*(K_{\tau_1}, z)h^*(K_{\tau_2}, z).$$

In fact K_τ is integrally piecewise-linearly equivalent to the free sum of K_{τ_1} and K_{τ_2} by a gauge-preserving map.

Proof. Write $x = (x^{(1)}, x^{(2)})$ and let $M_2 = \max(\{0\} \cup \{x_e^{(2)} : e \in E_2\})$. The explicit map is

$$\Phi(x^{(1)}, x^{(2)}) = (x^{(1)} - M_2 \mathbf{1}_{E_1}, x^{(2)}).$$

It is a piecewise integral linear shear with inverse obtained by adding $M_2\mathbf{1}_{E_1}$. For a class v in the first summand its upper envelope equals M_2 plus the upper envelope, computed in τ_1, of $x^{(1)} - M_2\mathbf{1}$. The envelopes in the second summand are unchanged. Formula (XV.3.1) therefore yields

$$g_\tau(x) = g_{\tau_1}(x^{(1)} - M_2\mathbf{1}) + g_{\tau_2}(x^{(2)}).$$

The sum of the two gauges is exactly the gauge of the free sum. This establishes the asserted geometric equivalence, including the lattice bijection on every dilate.

For completeness, the Ehrhart-series calculation requires no theorem about free sums. Set $S_\sigma(z) = \sum_{x \in \mathbb{Z}^{E_\sigma}} z^{g_\sigma(x)}$. Its coefficients are finite because the unit gauge ball is bounded. The displayed identity and the lattice bijection give $S_\tau = S_{\tau_1} S_{\tau_2}$. Since the gauge is integer valued,

$$\sum_{m \geq 0} \#(mK_\sigma \cap \mathbb{Z}^{E_\sigma}) z^m = \frac{S_\sigma(z)}{1 - z}, \qquad h^*(K_\sigma, z) = (1 - z)^{|E_\sigma|} S_\sigma(z).$$

The stated product follows. □

Corollary XV.3.4 (A finite formula for the polar h^*-polynomial)**.** *Let $m_v = |v|$. For $S \subseteq \mathcal{V}$, let $W_S(z)$ be the descent polynomial of linear extensions of the naturally labeled induced poset $\mathcal{V}|_S$, and put $W_\varnothing(z) = 1$. Then*

$$h^*(K_\tau, z) = \sum_{S \subseteq \mathcal{V}} z^{|S|} W_S(z^{-1}) \prod_{v \in S} (1 + z + \cdots + z^{m_v - 1}). \tag{XV.3.4}$$

Proof. For a fixed code d, summing over the deficiencies δ contributes $(1 - z)^{-m_v}$ at a class with $d_v = 0$. At a class with $d_v > 0$, the minimum-zero condition contributes $(1 - z^{m_v})(1 - z)^{-m_v}$, by subtracting the vectors all of whose coordinates are positive. Using the shell-series identity from the preceding proof therefore gives

$$h^*(K_\tau, z) = \sum_{S \subseteq \mathcal{V}} \left(\prod_{v \in S} (1 - z^{m_v}) \right) \sum_{d \in \mathbb{Z}^S_{>0}} z^{\max_C \sum_{v \in C} d_v}.$$

Here the chains are those of the induced poset on S; vertices outside S have weight zero. For $r = |S| > 0$, the inner series equals

$$\frac{z^r W_S(z^{-1})}{(1-z)^r}.$$

To see this, let C_S be Stanley's chain polytope. Positive integer d of chain height at most j are precisely the interior lattice points of $(j+1)C_S$. Stanley's order–chain transfer theorem gives the Ehrhart numerator W_S for C_S; Ehrhart reciprocity then gives the displayed exact-height series after multiplication by $1 - z$. Cancelling $(1 - z)^r$ proves (XV.3.4); the empty subset contributes one. This derivation uses Stanley's classical order–chain-polytope theorem, not a new claim about that theorem. □

XV.3.2 Cloning and the full multichain matrix

Let $\nabla_\tau(k, \ell)$ count k-multichains in the coordinatewise poset of lattice points in ℓQ_τ. A zero-length multichain is counted once.

Theorem XV.3.5 (Matrix Ehrhart–Zeta duality; source Conjecture 4.2)**.** *For every finite preorder and all integers $k, \ell \geq 0$,*

$$\nabla_\tau(k, \ell) = \nabla_{\tau^*}(\ell, k).$$

Proof. The cases $k = 0$ or $\ell = 0$ both have value one. Suppose both are positive. Taking successive differences, with initial vector zero, identifies the multichains with arrays $a_{e,i} \geq 0$, $e \in E$, $1 \leq i \leq k$, such that

$$\sum_{e \in I} \sum_{i=1}^{k} a_{e,i} \leq \ell |I| \quad \text{for every order ideal } I. \tag{XV.3.5}$$

Every intermediate partial sum is feasible because these inequalities have nonnegative coefficients.

Form the bipartite graph B with left vertices $E \times [k]$, right vertices $E \times [\ell]$, and an edge from (e, i) to (f, j) exactly when $f \leq_\tau e$. For any bipartite graph define

$$D(B) = \{a \in \mathbb{Z}_{\geq 0}^{L(B)} : \sum_{p \in A} a_p \leq |N_B(A)| \text{ for all } A \subseteq L(B)\}.$$

If A projects onto $S \subseteq E$, its neighborhood consists of all ℓ clones of the ideal $\downarrow S$. Thus (XV.3.5) implies every inequality defining $D(B)$. Conversely, take $A = I \times [k]$ for an ideal I; reflexivity and downward closure give $N_B(A) = I \times [\ell]$. Hence the two constraint systems are equivalent, and $|D(B)| = \nabla_\tau(k, \ell)$.

Postnikov's duality implies $|D(B)| = |D(B^*)|$. To connect the conventions precisely, add one vertex to each part of B, join each new vertex to all vertices in the opposite part, and call the resulting graph G. The G-draconian sequences on the left are exactly the elements $a \in D(B)$ with an extra coordinate $a_0 = |R(B)| - \sum_p a_p$. For subsets not containing the new vertex, the draconian bound $|N_G(A)| - 1$ is $|N_B(A)|$; for subsets containing it, the bound follows from the fixed total. The new coordinate is nonnegative by the constraint for all original left vertices. Postnikov's Lemma 11.7 and Corollary 11.8 identify the numbers of these sequences on the two sides. Finally B^* is the same cloned construction for τ^* with k and ℓ interchanged. This proves the theorem. □

XV.3.3 Double reciprocity and the word formula

Let $\mathbb{E}_\tau(r, s) = \#(Q_\tau(r, s) \cap \mathbb{Z}^E)$ on nonnegative integers, extended as its polynomial of total degree n.

Theorem XV.3.6 (Double reciprocity; source Conjecture 7.2)**.** *As a polynomial identity,*

$$\mathbb{E}_\tau(-u,-v) = (-1)^n \mathbb{E}_\tau(u-1, v-1).$$

Proof. Take integers $r, s \geq 1$. The full-dimensional polytope $Q_\tau(r,s)$ has integer interior points described by

$$x_e \geq 1, \qquad \sum_{e \in I} x_e \leq r|I| + s - 1.$$

The translation $y = x - \mathbf{1}$ identifies them with all lattice points of $Q_\tau(r-1, s-1)$. Ordinary Ehrhart reciprocity applied to the fixed lattice polytope $Q_\tau(r,s)$, whose dilation polynomial is $t \mapsto \mathbb{E}_\tau(tr, ts)$, consequently gives

$$\mathbb{E}_\tau(-r,-s) = (-1)^n \mathbb{E}_\tau(r-1, s-1).$$

A polynomial in two variables vanishing on every pair of positive integers is identically zero, which proves the result. □

For $a \in \mathbb{N}^E$ write $|a| = \sum_e a_e$. The maximal elements of P_{τ^*} are precisely those of total n, as follows from its integral polymatroid description, or from the purity result in the source paper. The explicit formula there is

$$\mathbb{E}_\tau(r,s) = \sum_{a \in P_{\tau^*}} \binom{s+n-|a|}{n-|a|} \prod_{e \in E} \binom{r+a_e-1}{a_e}.$$

Double reciprocity with $(u,v) = (-m, 0)$, followed by this formula at $(-m-1, -1)$, annihilates all terms with $|a| < n$ and gives

$$\mathrm{Ehr}(Q_\tau, m) = \sum_{a \in \mathrm{Max}(P_{\tau^*})} \prod_{e \in E} \binom{m+1}{a_e}. \tag{XV.3.6}$$

Indeed $\binom{n-|a|-1}{n-|a|} = 0$ when $|a| < n$, while $\binom{-m+a_e-2}{a_e} = (-1)^{a_e}\binom{m+1}{a_e}$.

Corollary XV.3.7 (Descent interpretation; source Conjecture 4.4)**.** *Fix any total order on E. Let $\mathcal{W}_\tau$ consist of the words of length n in E in which the letters of every ideal I appear at least $|I|$ times. Then*

$$h^*(Q_\tau, z) = \sum_{w \in \mathcal{W}_\tau} z^{n-1-\mathrm{des}(w)}.$$

Proof. The content vectors of these words are exactly the maximal elements of P_{τ^*}: use $|a| = n$ and complement ideals to filters. For such a content vector, the classical multiset Eulerian identity is

$$\sum_{m \geq 0} \left(\prod_e \binom{m+1}{a_e} \right) z^m = \frac{\sum_{w \text{ of content } a} z^{n-1-\mathrm{des}(w)}}{(1-z)^{n+1}}.$$

This is the identity used in Remark 7.3 of the source paper. Summing it over (XV.3.6) proves the claim. □

A dependent specialization, not a new mechanism. The unweighted part of source Conjecture 4.9 also follows: $\mathcal{Z}(P_\tau, -1) = (-1)^n \# \mathrm{Max}(P_\tau)$. Indeed Ehrhart–Zeta duality gives $\mathcal{Z}(P_\tau, -1) = \mathrm{Ehr}(Q_{\tau^*}, -2)$; evaluating (XV.3.6) at $m = -2$ makes every summand $\prod_e \binom{-1}{a_e} = (-1)^n$. A multivariate strengthening and a linear-functional argument give the full q-refinement in the accompanying q-zeta section, which also proves gamma positivity of the support enumerator. The real-rootedness, magic positivity, and lattice-existence conjectures are not resolved here.

Primary references. The source statements and notation are in https://arxiv.org/abs/2605.26916, Sections 4, 6, and 7. Postnikov's original duality is in *Permutohedra, associahedra, and beyond*, *International Mathematics Research Notices* 2009, 1026–1106, Lemma 11.7 and Corollary 11.8, DOI https://doi.org/10.1093/imrn/rnn153. The later support-duality result, deliberately excluded from the claims, is https://arxiv.org/abs/2608.16037. Stanley's order–chain transfer theorem is in *Two poset polytopes*, *Discrete & Computational Geometry* 1 (1986), 9–23, https://math.mit.edu/~rstan/pubs/pubfiles/66.pdf.

XV.4 A multivariate identity, gamma positivity, and full q-zeta reciprocity

The preceding reciprocity mechanism admits a multivariate extension. It proves Athanasiadis–Chapoton's Conjecture 5.2 and the full q-refinement of their Conjecture 4.9 for every finite preorder. These are consequences of one common identity, rather than independent proof mechanisms. Write

$$B_a(r) = \binom{r+a-1}{a}, \qquad D_a(r) = \binom{r+1}{a}, \qquad a \geq 0,$$

and let $\mathrm{Max}(P_\tau)$ denote the maximal elements of the integer point poset. All of these have coordinate sum $n = |E|$. Indeed, suppose $|a| < n$ but no coordinate of a can be increased. For each coordinate there would then be an ideal containing it whose inequality is tight. Tight ideals are closed under union: modularity of coordinate sums and cardinalities, together with feasibility for the union and intersection, forces both inequalities to be equalities. The union of the chosen ideals is E, contradicting $|a| < n$. Conversely, the constraint for E makes every point of total n maximal.

Theorem XV.4.1 (Multivariate binomial identity)**.** *For every finite preorder τ, the following identity holds in the independent indeterminates $(r_e)_{e\in E}$:*

$$\sum_{a\in P_\tau} \prod_{e\in E} B_{a_e}(r_e) = \sum_{a\in \mathrm{Max}(P_\tau)} \prod_{e\in E} D_{a_e}(r_e). \tag{XV.4.1}$$

Proof. We first work with the dual indexing convention of the previous section. For a preorder σ, put $U_e = \{f : e \leq_\sigma f\}$ and $\Delta_A = \mathrm{conv}(0, e_f : f \in A)$. For nonnegative integer parameters $\mathbf{r} = (r_e)$ and s, define

$$Q_\sigma(\mathbf{r}, s) = \sum_{e\in E} r_e \Delta_{U_e} + s\Delta_E.$$

Its inequalities are

$$x_e \geq 0, \qquad x(I) \leq \sum_{e\in I} r_e + s \quad \text{for every nonempty ideal } I. \tag{XV.4.2}$$

For completeness, necessity follows because U_e meets an ideal I exactly when $e \in I$. For sufficiency, every subset $A \subseteq E$ has down-closure $I = {\downarrow} A$, and the displayed inequalities imply $x(A) \leq \sum_{e\in I} r_e + s$. These are exactly the capacity inequalities for distributing the nonnegative vector x among sources of capacities r_e with allowed targets U_e, and one source of capacity s with all targets allowed. The max-flow/min-cut theorem gives such a distribution, which is the asserted Minkowski-sum membership.

Postnikov's lattice-point formula, with the coefficients r_e kept independent rather than identified, gives the polynomial

$$E_\sigma(\mathbf{r}, s) = \sum_{a\in P_{\sigma^*}} \binom{s+n-|a|}{n-|a|} \prod_{e\in E} B_{a_e}(r_e). \tag{XV.4.3}$$

This is the same augmented bipartite graph and the same draconian-sequence set as in the source paper's Proposition 7.1; only its simplex coefficients are left as separate variables. The extra coordinate is $a_0 = n - |a|$, and its factor is $\binom{s+a_0}{a_0}$.

If all $r_e, s \geq 1$, this is a full-dimensional lattice polytope. By (XV.4.2), translating each interior integer point by $-\mathbf{1}$ gives exactly the integer points of $Q_\sigma(\mathbf{r}-\mathbf{1}, s-1)$. Ehrhart reciprocity for this fixed polytope, followed by polynomial continuation in all parameters, proves

$$E_\sigma(-\mathbf{r}, -s) = (-1)^n E_\sigma(\mathbf{r} - \mathbf{1}, s - 1).$$

In particular, $E_\sigma(\mathbf{r}, 0) = (-1)^n E_\sigma(-\mathbf{r} - \mathbf{1}, -1)$. In (XV.4.3), the latter specialization annihilates every term with $|a| < n$, and

$$B_{a_e}(-r_e - 1) = (-1)^{a_e} D_{a_e}(r_e).$$

For the surviving terms $|a| = n$, the signs cancel. This proves (XV.4.1) for $\tau = \sigma^*$, and hence for every preorder. □

XV.4.1 An explicit gamma expansion for all preorders

Recall that $h(\tau, z) = \sum_{a \in P_\tau} z^{|\operatorname{supp}(a)|}$ is the support-enumerator polynomial of the source paper; it is distinct from $h^*(Q_\tau, z)$.

Theorem XV.4.2 (Gamma positivity; source Conjecture 5.2)**.** *For every finite preorder on n elements,*

$$h(\tau, z) = \sum_{j=0}^{\lfloor n/2 \rfloor} \gamma_j(\tau) z^j (1+z)^{n-2j},$$

where the coefficients have the explicit nonnegative integer interpretation

$$\gamma_j(\tau) = \#\{a \in \operatorname{Max}(P_\tau) \cap \{0,1,2\}^E : \#\{e : a_e = 2\} = j\}.$$

Consequently $h(\tau, z)$ is palindromic and unimodal.

Proof. Since $B_a(r)$ has degree a and leading coefficient $1/a!$, these polynomials form a basis of $\mathbb{Q}[r]$. Define a linear functional $T_z : \mathbb{Q}[r] \to \mathbb{Q}[z]$ by

$$T_z(B_0) = 1, \qquad T_z(B_a) = z \quad (a \geq 1).$$

The formal identities

$$\sum_{a \geq 0} B_a(r) u^a = (1-u)^{-r}, \qquad \sum_{a \geq 0} D_a(r) v^a = (1+v)^{r+1}$$

with $u = v/(1+v)$ give

$$\sum_{a \geq 0} T_z(D_a) v^a = (1+v)\left(1 + \frac{zu}{1-u}\right) = 1 + (1+z)v + zv^2.$$

Apply T_z separately in every variable of (XV.4.1). The left side becomes $h(\tau, z)$. Only maximal vectors with coordinates at most two remain on the right, each contributing $z^{\#2}(1+z)^{\#1}$. Their coordinate sum is n, so $\#1 = n - 2\#2$, proving the asserted formula. Each basis polynomial $z^j(1+z)^{n-2j}$ is palindromic about $n/2$ and unimodal; their nonnegative sum has the same properties. □

XV.4.2 The full weighted reciprocity identity

For a finite graded poset P, the source defines its q-zeta polynomial by

$$\mathcal{Z}_q(P,[m]_q)=\sum_{p_1\le\cdots\le p_{m-1}} q^{\rho(p_1)+\cdots+\rho(p_{m-1})},\qquad [m]_q=\frac{q^m-1}{q-1},\quad m\ge 2.$$

For $P=P_\tau$, the rank is $\rho(a)=|a|$. All manipulations below take place over $\mathbb{Q}(q)$, so q is an indeterminate.

Theorem XV.4.3 (Full q-zeta reciprocity; source Conjecture 4.9)**.** *For every preorder τ on an n-element set,*

$$\mathcal{Z}_q(P_\tau,[-1]_q)=(-1)^n\sum_{a\in\mathrm{Max}(P_\tau)} q^{-|\operatorname{supp}(a)|}.$$

The support size is the number of elements of P_τ covered by a, so this is exactly the conjectured formula, including its stated exponent.

Proof. The poset P_τ is an integer downset. Fixing the final element a of a multichain and summing independently in each coordinate therefore yields

$$\mathcal{Z}_q(P_\tau,[m]_q)=\sum_{a\in P_\tau} q^{|a|}\prod_{e\in E}\begin{bmatrix}m+a_e-2\\ a_e\end{bmatrix}_q. \tag{XV.4.4}$$

Here the bracket denotes a Gaussian binomial, interpreted through its product formula. Both sides are polynomials in $[m]_q$ over $\mathbb{Q}(q)$; the identity for positive integers m therefore permits evaluation at $[-1]_q=-q^{-1}$. At $m=-1$, the one-coordinate factor is

$$q^a\begin{bmatrix}a-3\\ a\end{bmatrix}_q=\begin{cases}1, & a=0,\\ -(1+q^{-1}), & a=1,\\ q^{-1}, & a=2,\\ 0, & a\ge 3.\end{cases}$$

Thus, with $t=q^{-1}$, the left side in the theorem is

$$\sum_{a\in P_\tau\cap\{0,1,2\}^E}(-(1+t))^{\#\{e:a_e=1\}}t^{\#\{e:a_e=2\}}. \tag{XV.4.5}$$

Define another linear functional L_t on the same binomial basis by

$$L_t(B_0)=1,\quad L_t(B_1)=-(1+t),\quad L_t(B_2)=t,\quad L_t(B_a)=0\ (a\ge 3).$$

Using again $u=v/(1+v)$ in the two formal generating functions gives

$$\sum_{a\ge 0}L_t(D_a)v^a=(1+v)\left(1-(1+t)\frac{v}{1+v}+t\frac{v^2}{(1+v)^2}\right)=1-\frac{tv}{1+v}.$$

Consequently $L_t(D_0)=1$ and $L_t(D_a)=(-1)^a t$ for $a\ge 1$. Apply L_t separately in every variable of (XV.4.1). Its left side is (XV.4.5); its right side is

$$\sum_{a\in\mathrm{Max}(P_\tau)}(-1)^{|a|}t^{|\operatorname{supp}(a)|}=(-1)^n\sum_{a\in\mathrm{Max}(P_\tau)}t^{|\operatorname{supp}(a)|}.$$

Finally, because P_τ is a downset, the lower covers of a are exactly $a-e_e$ for its positive coordinates. This identifies the exponent with the source's $\operatorname{cov}(a)$ and completes the proof. □

Scope. The gamma formula proves source Conjecture 5.2 and its stated palindromicity and unimodality consequences. It does not prove real-rootedness of $h(\tau, z)$, magic positivity of the Ehrhart polynomial, or the stronger simplicial-polytope realization conjectures. The weighted formula proves all of Conjecture 4.9, not just its specialization $q = 1$. The shared input is Postnikov's lattice-point formula and the multivariate translation reciprocity proved above. The target statements are in `https://arxiv.org/html/2605.26916v1`, Sections 4.4 and 5.

XV.5 Common-transversal minimizers in every odd-order finite group

XV.5.1 The published question and the orbit formula

For a subgroup H of a finite group G, let $\mathrm{P}_G(H)$ be the proportion of left transversals of H that are also right transversals, and put

$$\mathrm{tp}(G) = \min_{H \le G} \mathrm{P}_G(H).$$

Problem 5.8 of [141], posed on published page 1281 and described there as the most important relevant question, asks whether there is a cyclic p-subgroup H attaining this minimum such that

$$[H : \mathrm{core}_G(H)] \le p.$$

We prove this for every nontrivial finite group of odd order. This is a complete answer on that class, not a solution for arbitrary finite groups.

Set

$$f(t) = \frac{t!}{t^t}, \qquad \ell(t) = -\log f(t), \qquad k(t) = \frac{\ell(t)}{t} \quad (t \ge 1).$$

If the orbit sizes of H on G/H are $t_1, \dots, t_m$, the double-coset formula of [141, Proposition 3.2] is

$$\mathrm{P}_G(H) = \prod_{i=1}^{m} f(t_i), \qquad A_G(H) := -\log \mathrm{P}_G(H) = \sum_{i=1}^{m} \ell(t_i). \tag{XV.5.1}$$

Indeed, a double coset with t left cosets has t right cosets, and every left/right intersection has $|H|/t$ elements. It contributes $t!(|H|/t)^t$ double-transversal choices out of $|H|^t$ left-transversal choices. Distinct double cosets contribute independently.

Lemma XV.5.1. *For every integer $t \ge 1$, one has $0 \le k(t) < 1$. For every integer $t \ge 3$, one has $k(t) > 1/2$. Consequently, for every $H \le G$,*

$$A_G(H) < [G : H]. \tag{XV.5.2}$$

Proof. The first assertion follows from

$$\log(t!) \ge \int_1^t \log x \, dx = t \log t - t + 1,$$

which gives $\ell(t) \le t - 1 < t$, together with $t! \le t^t$. For the second assertion, $k(3) = \log(9/2)/3 > 1/2$. For example, the last strict inequality follows from $e < 68/25$ and $(68/25)^3 < 81/4$; the elementary bound on e follows by summing its power series through $1/6!$ and bounding the remaining tail by $1/4410$. Furthermore,

$$\ell(t+1) - \ell(t) = t \log(1 + 1/t) > \frac{t}{t+1} \ge \frac{1}{2}.$$

Induction therefore gives $\ell(t) > t/2$ for all $t \ge 3$. Finally, summing $\ell(t_i) < t_i$ in (XV.5.1), and using $\sum_i t_i = [G : H]$, proves (XV.5.2). □

XV.5.2 Minimal non-normal cyclic subgroups

Lemma XV.5.2. *Let $H \le G$ be non-normal. Then H contains a non-normal cyclic prime-power subgroup C. Such a subgroup can be chosen so that, writing $|C| = p^a$, its subgroup C^p is normal in G. For this choice,*

$$\mathrm{core}_G(C) = C^p, \qquad [C : \mathrm{core}_G(C)] = p.$$

Proof. If every cyclic prime-power subgroup contained in H were normal in G, then every cyclic subgroup of H would be normal: decompose it into its primary subgroups and take their product. As H is generated by its cyclic subgroups, this would make H normal in G, a contradiction. Choose C of smallest order among the non-normal cyclic prime-power subgroups contained in H. Its subgroup C^p is trivial or has smaller prime-power order, and so is normal in G. The core of C contains C^p and is a proper subgroup of C. Since $[C : C^p] = p$, the core must be C^p. □

Lemma XV.5.3. *Suppose that C is a non-normal cyclic subgroup of order p^a and $C^p \trianglelefteq G$. Put $c = |C|$ and $s = [G : N_G(C)]$. Then*

$$A_G(C) = \frac{|G| - |N_G(C)|}{c}\, k(p) = \frac{|G|}{c}\left(1 - \frac{1}{s}\right) k(p), \tag{XV.5.3}$$

and

$$\mathrm{P}_G(C) = \left(\frac{p!}{p^p}\right)^{(|G|-|N_G(C)|)/(cp)}. \tag{XV.5.4}$$

Proof. Normality of C^p gives $C^p \le C \cap C^g$ for every $g \in G$. Hence $C \cap C^g$ is either C or C^p, the first case occurring exactly when $g \in N_G(C)$. Thus every orbit of C on G/C has size 1 or p. There are $|N_G(C)|/c$ fixed points and therefore $(|G| - |N_G(C)|)/(cp)$ orbits of size p. Formula (XV.5.1) gives both assertions. □

XV.5.3 The entire odd-order case of Problem 5.8

Theorem XV.5.4. *Let G be a finite group of odd order and let $H \le G$ be non-normal. Either*

1. *H is cyclic of order p^a for an odd prime p, and $\mathrm{core}_G(H) = H^p$; or*
2. *there is a proper subgroup $C < H$, cyclic of order p^a for an odd prime p, with $\mathrm{core}_G(C) = C^p$ and*
$$\mathrm{P}_G(C) < \mathrm{P}_G(H).$$

In particular, if G is not Dedekind, then every subgroup H attaining $\mathrm{tp}(G)$ has the form in the first alternative. Thus Problem 5.8 has an affirmative answer for every nontrivial finite group of odd order.

Proof. Choose $C \le H$ as in Lemma XV.5.2. If $C = H$, the first alternative holds. Suppose $C < H$. Since G has odd order, both

$$r = [H : C] \quad \text{and} \quad s = [G : N_G(C)]$$

are odd integers greater than one, so $r, s \ge 3$. Also $p \ge 3$. Lemmas XV.5.1 and XV.5.3 give

$$A_G(C) = \frac{|G|}{|C|}\left(1 - \frac{1}{s}\right) k(p) > \frac{|G|}{3|C|} \ge \frac{|G|}{|H|} > A_G(H).$$

Exponentiating the negatives proves the strict probability inequality.

If G is not Dedekind, some subgroup has probability less than one, so every minimizing subgroup is non-normal. The second alternative would contradict minimality, proving the claimed form of all minimizers. If G is Dedekind and nontrivial, choose a subgroup of prime order; it is cyclic and normal, has probability one, and its index over its core is one. This supplies the existence assertion also in that case. □

Corollary XV.5.5. *Let G have odd order and not be Dedekind. Let $\mathcal{M}(G)$ be the set of non-normal cyclic prime-power subgroups C whose proper subgroups are all normal in G. If $|C| = p_C^{a_C}$, then*

$$\operatorname{tp}(G) = \min_{C \in \mathcal{M}(G)} \left(\frac{p_C!}{p_C^{p_C}} \right)^{\frac{|G| - |N_G(C)|}{p_C{}^{|C|}}}.$$

Every subgroup attaining the minimum over all subgroups belongs to $\mathcal{M}(G)$.

Proof. For a cyclic p-group C, normality of C^p implies normality of every proper subgroup of C, since each is characteristic in C^p. Apply Theorem XV.5.4 and (XV.5.4). □

Corollary XV.5.6. *Let G be a finite non-Dedekind p-group for an odd prime p. Let c be the smallest order of a non-normal cyclic subgroup of G. Then every minimizer of P_G has order c. Among the non-normal cyclic subgroups of order c, the minimizers are exactly those with smallest normalizer order. Writing that smallest normalizer order as ν,*

$$\operatorname{tp}(G) = \left(\frac{p!}{p^p} \right)^{(|G| - \nu)/(cp)}.$$

Proof. Choose a non-normal cyclic subgroup C of order c. It has $C^p \trianglelefteq G$ and $[G : N_G(C)] \geq p$. Therefore

$$A_G(C) \geq \frac{|G|}{c} \frac{p-1}{p} k(p) > \frac{|G|}{pc},$$

where $(p-1)k(p) > 1$ follows from $p \geq 3$ and $k(p) > 1/2$. Any subgroup H of order greater than c has $|H| \geq pc$, and consequently $A_G(H) < |G|/(pc) < A_G(C)$. A non-normal subgroup of order less than c would contradict Lemma XV.5.2; a non-normal subgroup of order c must itself be cyclic, by that lemma and the definition of c. Thus every minimizer is cyclic of order c. The formula and normalizer characterization follow from (XV.5.4). □

XV.5.4 A restriction on possible even-order counterexamples

Proposition XV.5.7. *Let G be any finite group that is not Dedekind, and let H attain $\operatorname{tp}(G)$. Choose $C \leq H$ as in Lemma XV.5.2, and put*

$$|C| = p^a, \qquad r = [H : C], \qquad s = [G : N_G(C)].$$

Then

$$r \left(1 - \frac{1}{s} \right) k(p) < 1. \tag{XV.5.5}$$

If $C < H$, this implies $r \leq 5$ for $p = 2$, $r \leq 3$ for $p = 3$, and $r \leq 2$ for $p \geq 7$. For $p = 3$ and $s \geq 3$ it implies $r \leq 2$. In particular, in a finite 2-group a minimizing subgroup containing such a proper C can have only $[H : C] = 2$ or 4.

Proof. Since H minimizes probability, $A_G(H) \geq A_G(C)$. Combine (XV.5.2) with (XV.5.3) and cancel $|G|/|C|$ to obtain (XV.5.5). For $p = 2$, $s \geq 2$ and $k(2) = \log 2/2$, giving $r < 4/\log 2 < 6$. For $p = 3$, $k(3) > 1/2$ gives $r < 4$, and $s \geq 3$ improves this to $r < 3$. For $p \geq 7$, the exact bound $7^7/7! > 2^7$ gives $k(7) > \log 2 > 2/3$: indeed, $823543 > 128 \cdot 5040 = 645120$. The increment

estimate in Lemma XV.5.1 gives $\ell(t+1)-\ell(t) > t/(t+1) > 2/3$ for $t \geq 7$. Thus $k(t) > 2/3$ for every $t \geq 7$, and (XV.5.5) gives $r < 3$. The final assertion follows because r is a power of two. □

XV.5.5 Scope and literature check

Theorem XV.5.4 answers the odd-order restriction of the published Problem 5.8, with an additional classification of all minimizers. The general even-order problem is not solved here. The independent commuting-probability conjecture, Problem 5.5, is also not solved here. No computer enumeration is used in these proofs.

The original problem was checked in the published 2024 version, page 1281, and in arXiv:2209.00986v2. Exact-title, author, and problem searches on 19 October 2026 found the original paper and the authors' publication listings, but no subsequent resolution of this odd-order branch. This search does not certify novelty or priority.

XV.6 A lattice correction and eventual oscillations in the friendship paradox

Source and scope. Hazra, den Hollander and Parvaneh, *The friendship paradox for sparse random graphs*, Probability Theory and Related Fields 195 (2026), 109–131, doi:10.1007/s00440-025-01365-w, Conjecture 2.6 asks whether

$$\mathbb{P}\left\{\sum_{i=1}^{X_0} X_i \geq X_0(X_0-1)\right\} \geq \tfrac{1}{2}$$

for independent identically distributed binomial, geometric, or Poisson variables. For the Poisson family the paper's discussion after Theorem 4.3 reports numerical evidence that this probability is strictly decreasing in the mean parameter. We prove that this monotonicity observation does not extend to arbitrarily large parameters. The half-probability conjecture for all parameters is *not* proved or refuted here. In particular, the monotonicity observation is not itself the numbered Conjecture 2.6.

Write Φ for the standard normal distribution function and put

$$S_s(a) = \sum_{j=1}^{\infty} \frac{e^{-2\pi^2 s^2 j^2}}{\pi j} \sin(2\pi j a), \qquad s > 0.$$

This series and all its derivatives converge uniformly. Its periodicity records a lattice effect that ordinary central-limit approximations suppress.

Theorem XV.6.1 (Poisson friendship probability)**.** *Let $X_0, X_1, \dots$ be independent Poisson variables of mean $\lambda > 0$, and define*

$$P(\lambda) = \mathbb{P}\left\{\sum_{i=1}^{X_0} X_i \geq X_0(X_0-1)\right\}.$$

As $\lambda \to \infty$,

$$P(\lambda) = \frac{1}{2} + \frac{7/6 + S_1(\lambda)}{\sqrt{2\pi\lambda}} + o(\lambda^{-1/2}), \tag{XV.6.1}$$

uniformly in the fractional part of λ. Consequently:

1. *$P(\lambda) > 1/2$ for every sufficiently large λ;*

2. for every sufficiently large integer n,

$$P(n+5/4) > P(n+3/4), \qquad P(n+3/4) < P(n+1/4).$$

Thus P has increasing and decreasing pairs at arbitrarily large means.

Lemma XV.6.2 (The central Poisson distribution function)**.** *For $m \to \infty$ through integers and $0 \le a < 1$, uniformly in a,*

$$\mathbb{P}\{\mathrm{Pois}(m+a) \le m\} = \frac{1}{2} + \frac{2/3 - a}{\sqrt{2\pi(m+a)}} + o(m^{-1/2}). \tag{XV.6.2}$$

Also, for each fixed integer r,

$$\sqrt{m+a}\,\mathbb{P}\{\mathrm{Pois}(m+a) = m+r\} \longrightarrow (2\pi)^{-1/2}$$

uniformly in a, and the left side is bounded by a universal constant for all $m + a \ge 1$ and all r for which $m + r \ge 0$.

Proof. The mass estimates follow directly from Stirling's formula, including the uniform bound at the mode. For the distribution function let $N = m + 1$ and let T_N be a sum of N independent exponential variables of mean one. The Poisson–gamma identity gives

$$\mathbb{P}\{\mathrm{Pois}(\lambda) \le m\} = \mathbb{P}\{T_N > \lambda\}.$$

Stirling's formula for the gamma density, followed by Taylor expansion, gives for the density f_N of $(T_N - N)/\sqrt{N}$,

$$f_N(z) = \frac{e^{-z^2/2}}{\sqrt{2\pi}} \left[1 + \frac{z^3 - 3z}{3\sqrt{N}} + O\left(\frac{1+|z|^8}{N}\right)\right]$$

on $|z| \le N^{1/12}$. The constant in the remainder can be chosen independent of N. Exponential Chernoff bounds make the integrals outside this window $o(N^{-1/2})$. Integrating the displayed expansion, using $\int_{-\infty}^{x}(z^3 - 3z)\varphi(z)\,dz = (1 - x^2)\varphi(x)$, and substituting $x = (a-1)/\sqrt{N}$ proves (XV.6.2). The Taylor errors and tail bounds are uniform for $a \in [0, 1]$. □

Proof of the theorem. Put $m = \lfloor \lambda \rfloor$, $a = \lambda - m$, and

$$p_\lambda(k) = \mathbb{P}\{\mathrm{Pois}(\lambda) = k\}, \qquad g_\lambda(k) = \mathbb{P}\{\mathrm{Pois}(k\lambda) \ge k(k-1)\},$$

with $g_\lambda(0) = g_\lambda(1) = 1$. Conditional on $X_0 = k$, the sum of its k independent neighbors has law $\mathrm{Pois}(k\lambda)$. Hence

$$P(\lambda) - \mathbb{P}\{X_0 \le m\} = \sum_{r \in \mathbb{Z}} p_\lambda(m+r)\left[g_\lambda(m+r) - \mathbf{1}_{\{r \le 0\}}\right], \tag{XV.6.3}$$

where terms with $m + r < 0$ are set equal to zero. For fixed r, the normal approximation to $\mathrm{Pois}((m+r)\lambda)$ gives, uniformly in a,

$$g_\lambda(m+r) \longrightarrow \Phi(1 + a - r).$$

Indeed the standardized cutoff is

$$\frac{k(k-1) - k\lambda}{\sqrt{k\lambda}} = \sqrt{k/\lambda}\,(r - a - 1) \longrightarrow r - a - 1.$$

The integer endpoint convention changes the standardized cutoff by at most $1/\sqrt{k\lambda}$ and therefore has no effect on this limit.

We next justify summation of the limits. If $r \ge 3$, the upper Poisson Chernoff inequality yields

$$g_\lambda(k) \le \exp\left[-\frac{k(k-1-\lambda)^2}{2(k-1)}\right] \le e^{-(r-2)^2/2}.$$

If $r \le -2$ and $k \ge 2$, the lower Chernoff bound yields

$$1 - g_\lambda(k) \le \exp\left[-\frac{k(\lambda-k+1)^2}{2\lambda}\right].$$

When $k \ge \lambda/2$, the exponent is at least $r^2/4$. When $2 \le k < \lambda/2$, it is at least $\lambda/4 \ge |r|/4$. For $k = 0, 1$ the difference is zero. Together with the uniform bound on $\sqrt{\lambda}\, p_\lambda(k)$, these estimates give a summable majorant independent of λ and a. Dominated convergence in (XV.6.3), first on a finite range of r and then in its tails, therefore gives uniformly in a,

$$\sqrt{2\pi\lambda}\left(P(\lambda) - \frac{1}{2}\right) = \frac{2}{3} - a + \sum_{r\in\mathbb{Z}} \Big[\Phi(1+a-r) - \mathbf{1}_{\{r\le 0\}}\Big] + o(1).$$

For a standard normal Z, absolute integrability permits the identity

$$\sum_{r\in\mathbb{Z}} \Big[\Phi(1+a-r) - \mathbf{1}_{\{r\le 0\}}\Big] = \mathbb{E}\lfloor 1 + a - Z\rfloor.$$

Thus the main term is $5/3 - \mathbb{E}\{a - Z\}$, where braces denote fractional part. The Fourier series for the sawtooth function, smoothed by the Gaussian, is absolutely convergent and gives

$$\frac{1}{2} - \mathbb{E}\{a - Z\} = \sum_{j\ge 1} \frac{e^{-2\pi^2 j^2}}{\pi j} \sin(2\pi j a) = S_1(a).$$

This proves (XV.6.1).

The bound $|S_1(a)| \le \sum_{j\ge1} e^{-2\pi^2 j^2}/(\pi j) < 1/6$ proves the first consequence. Moreover,

$$S_1(1/4) = \frac{1}{\pi}\sum_{\ell\ge 0} \frac{(-1)^\ell e^{-2\pi^2(2\ell+1)^2}}{2\ell+1} > 0, \qquad S_1(3/4) = -S_1(1/4).$$

Positivity follows from the alternating-series test. Applying the uniform asymptotic expansion at the stated pairs of means shows that their differences, multiplied by $\sqrt{n}$, tend respectively to $2S_1(1/4)/\sqrt{2\pi}$ and $-2S_1(1/4)/\sqrt{2\pi}$. These limits are strictly positive and strictly negative, respectively. □

Why the numerical evidence misses the effect. The first Fourier amplitude is $e^{-2\pi^2}/\pi$, approximately 8.5×10^{-10}. No numerical threshold for the first increasing pair is claimed. The proof establishes eventual increases by a nonzero limiting coefficient, rather than by rounding-sensitive floating-point subtraction.

XV.7 What remains unresolved in this work

The results above should be read with their quantified hypotheses. The general even-order case of the common-transversal question is not settled. The friendship majority conjecture still requires all positive Poisson means and the full binomial and geometric families. For preorder polytopes, gamma positivity does not establish real-rootedness, magic positivity, or the stronger simplicial-polytope realization statements. The arithmetic theorem establishes no new

irrationality result for a named classical constant and does not prove the necessary direction of the source's factorial-reduction classification.

Three further targets were investigated without a complete proof. For Habib–Hildebrand's coefficient-positivity Conjecture 7.2, a positive inverse linear operator does not by itself control the nonlinear residual; finite exact coefficient checks do not resolve that obstruction. The general Wishart principal-minor product inequality of Genest–Ouimet–Richards, Conjecture 1.1, is also left unresolved. Luo's symmetric entropy-curvature questions after Gaussian smoothing require differentiated remainder control that was not obtained here. None of these is counted as a solved target.

Verification record. The quantum upper bound, boundary limit order, and rational certificate were checked separately; the displayed certificate uses exact integer comparisons. The arithmetic argument was rederived from its recurrence, with particular attention to uniformity, reducible and resonant Gauss parameters, and reconstruction of the finite initial segment. Separate readings checked the preorder transfer bijection, weighted-polytope inequalities, multivariate reciprocity, both binomial-basis functionals, and the negative q-integer specialization. The group proof was checked from the orbit product rather than inferred from enumeration. The probability proof was checked for its central-CDF constant, one-unit threshold shift, uniform summable tails, and Fourier sign. These are internal mathematical checks, not formal-machine verification or external acceptance.

Part XVI

Spectral Suprema, Heat-Kernel Selection, and Cellular-Automaton Products

Manuscript overview. Three independent arguments address questions in operator algebras, geometric statistics, and the category of generalized cellular automata. We prove spectral-supremum submajorization for positive measurable operators in arbitrary semifinite von Neumann algebras, including unbounded operators, and give an overlap remainder in the integrable case. We construct positive analytic densities on the circle with two intrinsic means and a single diffusion mean at every positive time, and classify the time-dependent transition in the same density family. We disprove general existence of categorical products for generalized cellular automata by counting morphisms from finite probes. Further sections give an explicit palindromicity classification for bipartite support enumerators using established structure, a short weighted proof of delayed-star unimodality, and a spectral special case of the vector-chromatic energy conjecture. The final section records an unresolved arithmetic obstruction.

Status of the manuscript. These are proposed mathematical proofs with separate in-session derivations and checks. They have not undergone external peer review. The source questions and available follow-up literature were checked through 19 October 2026. Targeted searches do not certify priority. The scope register on the next page distinguishes complete answers, applications of existing results, and partial progress. No assertion about journal tier, prize-level significance, or universal solvability is made.

Different mechanisms, not a single universal method. The operator argument combines projection dimension with a variational formula. The statistical construction uses exact Fourier coefficients and a global polynomial inequality. The categorical obstruction uses the multiplicativity that a universal product would impose on morphism counts. Their common discipline is to preserve the exact quantifiers of the external question and to identify the step that supplies the global conclusion.

Precise scope register

External target	Result and boundary
Bourin–Lee, Conjecture 3.10	Complete proposed proof in every semifinite von Neumann algebra for positive measurable operators, with arbitrary finite families. No type III or infinite-family assertion.
Hundrieser–Eltzner–Huckemann, Section 3.4	Affirmative answer to the proper-subset existence question. A positive analytic circle density has two intrinsic means and one diffusion mean for every positive time. A parameter family has an exact merging threshold; products give 2^d versus one on a flat torus. These are consequences of one construction, not separate open-problem closures.
Castillo-Ramirez–Vazquez-Aceves–Zaldivar-Corichi, end of Section 4	Negative answer to general binary-product existence in $\mathrm{GCA}_{\mathbf{Set}}$. All candidate product objects are excluded for explicit broad families, including $A^{\mathbb{Z}}$ and $B^{\mathbb{Z}}$ for finite alphabets of size at least two. Some other pairs do have products.
Dai et al., Problem 5.3	Complete explicit classification of the palindromic support enumerators: after deleting isolated vertices, a perfect matching identifies the graph with a preorder. This is an application of established Gorenstein structure, with a direct coefficient-defect proof. The prior-round preorder gamma identity is reproduced as an input.
Ibarra et al., Conjecture 4.1	Complete affirmative argument for every delayed star with delays in $\{1, 2\}$, strengthened to nonnegative weights. The proof is an elementary uniform-convolution consequence of the source formula. Delays in $\{1, 2, 3\}$ are not resolved.
Elphick–Tang–Zhang, Conjecture 6.3	Only a spectral class is proved: all negative adjacency eigenvalues have one magnitude. This covers strongly regular graphs and all real $p \geq 2$. The general conjecture remains unresolved here.
Ben David et al., Section 2.8	No new higher-degree closure. An exact family exhibits why the available denominator estimate does not prove irrationality through its own canonical convergents.

Novelty boundaries. Finite matrix Grassmann identities are credited to Aouichaoui–Lee. The bipartite two-coefficient criterion is already covered by Hatasa, and its structural classification also follows from Ohsugi–Hibi. The delayed-star argument is short and is not presented as a major topological advance. A reproduced anti-unimodal general-tree example and a synchronizing-automata candidate were excluded after earlier resolutions were located.

XVI.1 Spectral suprema in arbitrary semifinite von Neumann algebras

External question and scope. Bourin and Lee, *Averages over matrix unitary orbits and spectral order*, arXiv:2606.15624v2 (18 June 2026), Conjecture 3.10, ask whether their finite-dimensional submajorization $A \vee B \prec_w A + B$ extends to all positive τ-measurable operators affiliated with an arbitrary semifinite von Neumann algebra. The theorem below proves this assertion, including unbounded operators and infinite traces, and allows any finite number of summands. The argument uses classical projection comparison and scalar rearrangement formulas. It does not assert that these ingredients are new.

The subsequent paper of Aouichaoui and Lee, *A Grassmann formula for the spectral order of matrices*, arXiv:2609.39659v1 (30 September 2026), treats finite-dimensional trace identities and equality in Schatten norms. Its finite-dimensional results are not counted as new here. A bounded literature search through 19 October 2026 found no treatment of the stated semifinite conjecture; this is not a certification of priority.

XVI.1.1 Notation and the global theorem

Let $(\mathcal{M}, \tau)$ be a von Neumann algebra with a faithful normal semifinite trace. Write $S(\mathcal{M}, \tau)^+$ for its positive τ-measurable affiliated operators. For X in this cone, set

$$P_X(u) = \mathbf{1}_{(u,\infty)}(X), \qquad d_X(u) = \tau(P_X(u)), \qquad \mu_t(X) = \inf\{u \geq 0 : d_X(u) \leq t\}.$$

The integrated singular-value function is

$$K_t(X) = \int_0^t \mu_v(X)\, dv \quad (t > 0).$$

All integrals and traces of positive operators below take values in $[0, \infty]$. We write $X \prec\prec Y$ when $K_t(X) \leq K_t(Y)$ for every $t > 0$. The symbol $\vee_{\mathrm{sp}}$ denotes the join in Olson's spectral order; joins of projections are their usual lattice joins.

Theorem XVI.1.1 (Semifinite spectral-supremum submajorization)**.** *Let $m \geq 1$ and $A_1, \ldots, A_m \in S(\mathcal{M}, \tau)^+$. Their spectral supremum $C = \bigvee_{j=1}^m {}_{\mathrm{sp}} A_j$ exists in $S(\mathcal{M}, \tau)^+$. With $S = \sum_{j=1}^m A_j$, where sums are taken in the algebra of τ-measurable operators, one has, for every $s \geq 0$,*

$$\tau((C - sI)_+) \leq \sum_{j=1}^m \tau((A_j - sI)_+) \leq \tau((S - sI)_+). \tag{XVI.1.1}$$

Consequently,

$$\bigvee_{j=1}^m {}_{\mathrm{sp}} A_j \prec\prec \sum_{j=1}^m A_j. \tag{XVI.1.2}$$

In particular, every fully symmetric operator-space norm satisfies $\|C\|_E \leq \|S\|_E$ whenever $S \in E$. The constant one is sharp, already when all but one summand vanish.

XVI.1.2 A positive-part trace lemma

Lemma XVI.1.2. *For $A_1, \ldots, A_m \in S(\mathcal{M}, \tau)^+$ and $s \geq 0$,*

$$\sum_{j=1}^m \tau((A_j - sI)_+) \leq \tau\left(\left(\sum_{j=1}^m A_j - sI\right)_+\right).$$

Proof. First suppose all A_j are bounded. For a bounded positive operator T the elementary variational formula is

$$\tau((T-sI)_+) = \sup\{\tau((T-sI)e) : e \in \mathcal{M} \text{ a projection, } \tau(e) < \infty\}. \tag{XVI.1.3}$$

Indeed, every displayed compression is at most the positive-part trace. Conversely, finite-trace subprojections of $\mathbf{1}_{(s,\infty)}(T)$ increase to that projection, by semifiniteness; normality of the trace gives the reverse inequality. This uses a net if the algebra is not countably decomposable.

Choose arbitrary finite-trace projections $e_1, \dots, e_m$, and let $r = e_1 \vee \cdots \vee e_m$. Projection comparison gives $\tau(r) \le \sum_j \tau(e_j) < \infty$. Since $r \ge e_j$ and $A_j \ge 0$, $\tau(A_j r) \ge \tau(A_j e_j)$. Therefore

$$\begin{aligned}\tau((S-sI)_+) &\ge \tau((S-sI)r) \\ &= \sum_j \tau(A_j r) - s\tau(r) \\ &\ge \sum_j [\tau(A_j e_j) - s\tau(e_j)].\end{aligned}$$

Taking independent suprema over the e_j proves the bounded case, also if one or more suprema are infinite.

For general A_j, put $A_{j,n} = \min(A_j, nI)$ by functional calculus. Then $0 \le \sum_j A_{j,n} \le S$ in the usual positive order. The standard order monotonicity of generalized singular numbers (Fack–Kosaki, Lemma 2.5(iii), together with their Corollary 2.8) yields

$$\tau\left(\left(\sum_j A_{j,n} - sI\right)_+\right) \le \tau((S-sI)_+).$$

Apply the bounded case and then normal monotone convergence to $(A_{j,n}-sI)_+ \uparrow (A_j - sI)_+$. No subtraction of infinite quantities is used. □

XVI.1.3 Proof of the theorem

Proof of Theorem XVI.1.1. Define a decreasing projection family, for $u \ge 0$, by

$$Q(u) = \bigvee_{j=1}^{m} P_{A_j}(u).$$

It has the appropriate right continuity: if $u_n \downarrow u$, then $P_{A_j}(u_n) \uparrow P_{A_j}(u)$, and the lattice identity

$$\bigvee_n \bigvee_j P_{A_j}(u_n) = \bigvee_j \bigvee_n P_{A_j}(u_n)$$

gives $Q(u_n) \uparrow Q(u)$. Moreover,

$$\tau(Q(u)) \le \sum_{j=1}^{m} d_{A_j}(u) \longrightarrow 0 \quad (u \longrightarrow \infty). \tag{XVI.1.4}$$

Here each tail tends to zero because A_j is τ-measurable. Thus $Q(u)$ is the strict upper spectral family of a positive densely defined affiliated operator C, and (XVI.1.4) makes C τ-measurable. Its spectral projections show immediately that it is the least spectral upper bound of all the A_j.

The layer-cake identity and projection-join trace bound now give

$$\begin{aligned}\tau((C-sI)_+) &= \int_s^\infty \tau(Q(u))\,du \\ &\le \sum_{j=1}^{m} \int_s^\infty d_{A_j}(u)\,du = \sum_{j=1}^{m} \tau((A_j - sI)_+).\end{aligned}$$

Lemma XVI.1.2 proves the second inequality in (XVI.1.1).

Finally, the scalar rearrangement identity

$$K_t(X) = \inf_{s\geq 0}\{ts + \tau((X - sI)_+)\} \tag{XVI.1.5}$$

holds for every positive τ-measurable X. To see it directly, write the trace as $\int_0^\infty (\mu_v(X) - s)_+\, dv$. Every s gives an upper bound for $\int_0^t \mu_v(X)\, dv$; the threshold $s = \mu_t(X)$ gives equality, with the usual extended integral convention. Applying (XVI.1.5) to the first and last terms of (XVI.1.1) proves (XVI.1.2). The norm conclusion is the defining monotonicity property of a fully symmetric space. □

XVI.1.4 An overlap remainder in the integrable case

The same argument retains quantitative information which is lost in the final submajorization inequality.

Proposition XVI.1.3 (Spectral-overlap remainder)**.** *Suppose $A, B \in L^1(\mathcal{M}, \tau)^+$, put $C = A \vee_{\mathrm{sp}} B$ and $S = A + B$, and define*

$$I_s(A, B) = \int_s^\infty \tau(P_A(u) \wedge P_B(u))\, du.$$

Then, for every $s \geq 0$,

$$\tau((C - sI)_+) + I_s(A, B) = \tau((A - sI)_+) + \tau((B - sI)_+) \leq \tau((S - sI)_+). \tag{XVI.1.6}$$

In particular, for every $t > 0$,

$$K_t(S) - K_t(C) \geq I_{\mu_t(S)}(A, B). \tag{XVI.1.7}$$

Proof. For $u > 0$ the projections $P_A(u)$ and $P_B(u)$ have finite trace. The projection dimension formula

$$\tau(P \vee Q) + \tau(P \wedge Q) = \tau(P) + \tau(Q)$$

follows from Murray–von Neumann equivalence of $(P \vee Q) - Q$ and $P - (P \wedge Q)$. Integrating it gives the equality in (XVI.1.6); the lemma gives the inequality. All terms are finite because A, B are integrable. In (XVI.1.5), use $s = \mu_t(S)$ for both C and S. This threshold minimizes the expression for S, so (XVI.1.6) yields (XVI.1.7). □

Existing ingredients and limitations. The background facts on measurable operators, generalized singular numbers, order monotonicity, and their trace formulas are classical; see T. Fack and H. Kosaki, *Generalized s-numbers of τ-measurable operators*, Pacific J. Math. **123** (1986), 269–300, https://doi.org/10.2140/pjm.1986.123.269. The principal conclusion here is the full semifinite extension explicitly asked in Bourin–Lee Conjecture 3.10, not a new proof of their finite matrix theorem. The overlap identity extends the finite-dimensional Grassmann calculation in Aouichaoui–Lee and is credited accordingly. We make no assertion about type III algebras with no semifinite trace, arbitrary infinite families, or two-unitary Loewner domination. The latter is a distinct question and is not a consequence of submajorization.

Primary sources.

- J.-C. Bourin and E.-Y. Lee, arXiv:2606.15624v2, https://arxiv.org/html/2606.15624v2, Conjecture 3.10.

- M. A. Aouichaoui and E.-Y. Lee, arXiv:2609.39659v1, https://arxiv.org/html/2609.39659v1, Theorems 2.1 and 3.1.
- T. Fack and H. Kosaki, publisher's full text: https://msp.org/pjm/1986/123-2/pjm-v123-n2-p03-s.pdf.

XVI.2 Geometric statistics: strict loss of intrinsic means under heat-kernel selection

External question and scope. Hundrieser, Eltzner and Huckemann, *A lower bound for estimating Fréchet means* (2026), Section 3.4, explicitly ask whether the zero-time limit of diffusion means can be a proper subset of the intrinsic mean set. The following construction answers that existence question affirmatively. It gives strictly positive real-analytic densities, and the diffusion mean is unique for *every* positive diffusion time. Products give examples with 2^d intrinsic means and just one diffusion mean on a d-dimensional flat torus. This is a resolution of that stated existence question, not of the different Wasserstein-rate conjecture in Section 3.5 of the same paper.

XVI.2.1 An exact family on the circle

Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, equipped with its circumference-one Riemannian metric and normalized length measure. Its heat kernel for the generator $\frac{1}{2}\Delta$ is

$$K_t(u) = \sum_{n\in\mathbb{Z}} e^{-2\pi^2 n^2 t} e^{2\pi i n u} = \frac{1}{\sqrt{2\pi t}} \sum_{k\in\mathbb{Z}} e^{-(u+k)^2/(2t)}, \qquad t > 0.$$

For a density f, define the intrinsic and diffusion objectives by

$$F_f(y) = \int_{\mathbb{T}} d(x,y)^2 f(x)\,dx, \qquad L_{f,t}(y) = -\int_{\mathbb{T}} \log K_t(x-y) f(x)\,dx.$$

Their sets of minimizers are denoted by $M_I(f)$ and $M_{D,t}(f)$.

Theorem XVI.2.1 (A proper limiting subset for every positive time)**.** *Suppose*

$$a > 0, \qquad b \geq 8a, \qquad b + 10a < 1,$$

and put

$$f_{a,b}(x) = 1 - a\cos(2\pi x) - b\cos(4\pi x) + 9a\cos(6\pi x). \tag{XVI.2.1}$$

Then $f_{a,b}$ is a strictly positive real-analytic probability density, and

$$M_I(f_{a,b}) = \{0, \tfrac{1}{2}\}, \qquad M_{D,t}(f_{a,b}) = \{\tfrac{1}{2}\} \quad \text{for every } t > 0.$$

Consequently the diffusion mean sets converge, in particular in Hausdorff distance, to a proper subset of the intrinsic mean set as $t \downarrow 0$.

Proof. The density integrates to one, and $f_{a,b} \geq 1 - b - 10a > 0$. Let $\theta = 2\pi y$. The absolutely convergent Fourier series

$$d(u,0)^2 = \frac{1}{12} + \frac{1}{\pi^2} \sum_{n\geq 1} \frac{(-1)^n}{n^2} \cos(2\pi n u)$$

gives, by orthogonality,

$$\begin{aligned} F_{f_{a,b}}(y) &= \frac{1}{12} + \frac{1}{2\pi^2}\left(a\cos\theta - \frac{b}{4}\cos(2\theta) - a\cos(3\theta)\right) \\ &= \frac{1}{12} - \frac{b}{8\pi^2} + \frac{b + 8a\cos\theta}{4\pi^2}\sin^2\theta. \end{aligned} \tag{XVI.2.2}$$

The last term is nonnegative. It vanishes precisely when $\theta \in \{0, \pi\}$ modulo 2π: this remains true at $b = 8a$, where the extra factor $1 + \cos\theta$ vanishes only at $\theta = \pi$. This proves the intrinsic-mean assertion.

We now compute the diffusion objective exactly. Set $s = 2\pi^2 t$ and $q = e^{-s} \in (0,1)$. Jacobi's product formula (NIST DLMF, equation 20.5.3) gives

$$K_t(u) = \prod_{k\geq 1} (1-q^{2k})(1+q^{2k-1}e^{2\pi i u})(1+q^{2k-1}e^{-2\pi i u}).$$

Expanding the logarithms, taking real parts, and summing the geometric series yields

$$\log K_t(u) = C_t + \sum_{n\geq 1} \frac{(-1)^{n+1}}{n\sinh(ns)} \cos(2\pi n u), \qquad C_t = \sum_{k\geq 1} \log(1-q^{2k}). \tag{XVI.2.3}$$

All these expansions are absolutely and uniformly convergent in real u for each fixed $t > 0$. Indeed the sum of the absolute logarithmic series is bounded by $2\sum_{k,n\geq 1} q^{n(2k-1)}/n < \infty$. Thus termwise integration against the density is justified. With $c = \cos\theta \in [-1,1]$, it gives

$$L_{f_{a,b},t}(y) = -C_t + \frac{1}{2} G_s(c), \qquad G_s(c) = \frac{ac}{\sinh s} - \frac{b(2c^2-1)}{2\sinh(2s)} - \frac{3a(4c^3-3c)}{\sinh(3s)}. \tag{XVI.2.4}$$

An exact polynomial factorization is

$$G_s(c) - G_s(-1) = (c+1)B_s(c), \qquad B_s(c) = \frac{a}{\sinh s} + \frac{b(1-c)}{\sinh(2s)} - \frac{3a(2c-1)^2}{\sinh(3s)}.$$

The quadratic B_s is concave on $[-1,1]$. We verify positivity at both endpoints. First,

$$B_s(1) = a\left(\frac{1}{\sinh s} - \frac{3}{\sinh(3s)}\right) > 0,$$

since $\sinh(3s) > 3\sinh s$ for $s > 0$. For the other endpoint let $z = \cosh s > 1$. Using $b \geq 8a$ and the double- and triple-angle identities,

$$\begin{aligned} \frac{\sinh s}{a} B_s(-1) &= 1 + \frac{b/a}{z} - \frac{27}{4z^2-1} \\ &\geq 1 + \frac{8}{z} - \frac{27}{4z^2-1} = \frac{4(z-1)(z^2+9z+2)}{z(4z^2-1)} > 0. \end{aligned}$$

Concavity now implies $B_s(c) > 0$ throughout $[-1,1]$. Therefore $G_s(c) > G_s(-1)$ whenever $c > -1$. The unique minimizer on $\mathbb{T}$ is consequently $y = \frac{1}{2}$, for every $t > 0$. □

One rational-coefficient example. Take $a = 1/200$ and $b = 2/25 = 16a$. The density is bounded below by $87/100$, and

$$f(x) = 1 - \frac{1}{200}\cos(2\pi x) - \frac{2}{25}\cos(4\pi x) + \frac{9}{200}\cos(6\pi x),$$

$$F_f(y) = \frac{1}{12} - \frac{1}{100\pi^2} + \frac{1}{100\pi^2}\sin^2(2\pi y)(2+\cos(2\pi y)).$$

The two intrinsic minima are both nondegenerate in this example. The phenomenon therefore does not require atoms, vanishing densities, singular support, or a degenerate intrinsic minimum.

XVI.2.2 Exact time-dependent classification in the same family

The Fourier computation also describes the transition outside the parameter range of Theorem XVI.2.1.

Proposition XVI.2.2 (The merging threshold)**.** *Let $a > 0$, $b \geq 0$ and $b + 10a < 1$. For $s = 2\pi^2 t$ and $z = \cosh s$, define*

$$b_*(s) = \frac{4az(7 - z^2)}{4z^2 - 1}.$$

If $b \geq b_*(s)$, the diffusion mean is the singleton $\{\frac{1}{2}\}$. If $b < b_*(s)$, the diffusion mean set consists of exactly the two points with

$$\cos(2\pi y) = c_-(s), \qquad c_-(s) = \frac{-B - \sqrt{B^2 + 4AC}}{2A} \in (-1, 0),$$

where

$$A = \frac{36a}{\sinh(3s)}, \qquad B = \frac{2b}{\sinh(2s)}, \qquad C = \frac{a}{\sinh s} + \frac{9a}{\sinh(3s)}.$$

For $0 \leq b < 8a$ there is exactly one merging time $t_* > 0$: the mean set has two points for $0 < t < t_*$ and is $\{\frac{1}{2}\}$ for $t \geq t_*$. In particular, $b \geq 8a$ is necessary and sufficient for the singleton assertion to hold at all positive times in this family.

Proof. The factorization in the preceding proof remains valid. Since $B_s(1) > 0$ and B_s is concave, the endpoint $c = -1$ minimizes G_s if and only if $B_s(-1) \geq 0$. When equality holds, concavity and $B_s(1) > 0$ still imply $B_s(c) > 0$ for $c > -1$, so the minimizer is unique. Rearranging $B_s(-1) \geq 0$ gives exactly $b \geq b_*(s)$. If the inequality fails, $G_s'(-1) = B_s(-1) < 0$. Moreover

$$G_s'(c) = C - Bc - Ac^2$$

has one negative and one positive root, since $A, C > 0$. Its negative root is in $(-1, 0)$ and is the only interior local minimum. Finally,

$$G_s(1) - G_s(-1) = 2a\left(\frac{1}{\sinh s} - \frac{3}{\sinh(3s)}\right) > 0.$$

The negative root, whose value is smaller than $G_s(-1)$, is therefore the unique global minimizer in the variable c.

For completeness, the derivative of b_*/a with respect to z is

$$\frac{d}{dz}\frac{4z(7 - z^2)}{4z^2 - 1} = -\frac{4(4z^4 + 25z^2 + 7)}{(4z^2 - 1)^2} < 0.$$

The threshold decreases from $8a$ at $z = 1$ to zero at $z = \sqrt{7}$. Thus the stated merging time is unique; explicitly $t_* = \operatorname{arcosh}(z_*)/(2\pi^2)$, where $z_* \in (1, \sqrt{7}]$ is the unique solution of

$$4az_*^3 + 4bz_*^2 - 28az_* - b = 0.$$

□

Corollary XVI.2.3 (Arbitrarily many discarded intrinsic means)**.** *For every integer $d \geq 1$, the flat torus $\mathbb{T}^d$ supports a strictly positive real-analytic probability density whose intrinsic mean set has exactly 2^d elements, whereas its diffusion mean set is a singleton for every $t > 0$.*

Proof. Use the product density $f_{a,b}^{\otimes d}$. The squared Riemannian distance on the product is the sum of the squared circle distances, and the heat kernel for $\frac{1}{2}\Delta$ is the product of the circle heat kernels. Both expected objectives therefore split as sums of their one-dimensional counterparts. Their mean sets are respectively $\{0, \frac{1}{2}\}^d$ and $\{(\frac{1}{2}, \ldots, \frac{1}{2})\}$. □

An exact indication of how the tie is broken. Theorem XVI.2.1 supplies a global proof rather than just an asymptotic comparison. Nevertheless its endpoint gap has a useful interpretation:

$$L_{f_{a,b},t}(0) - L_{f_{a,b},t}(\tfrac{1}{2}) = a\left(\frac{1}{\sinh s} - \frac{3}{\sinh(3s)}\right) = \frac{4a}{3}s + O(s^3) > 0, \qquad s = 2\pi^2 t \downarrow 0.$$

After multiplication by $2t$, this distinction disappears at order t^2, while the leading objective converges to the intrinsic objective. Convergence of objectives therefore preserves the inclusion of limiting minimizers but need not preserve every minimizer.

Source and verification record. The external question is in S. Hundrieser, B. Eltzner and S. F. Huckemann, *A lower bound for estimating Fréchet means*, Annals of the Institute of Statistical Mathematics (published 22 June 2026), Section 3.4, https://doi.org/10.1007/s10463-026-00989-6. The classical product formula is recorded at https://dlmf.nist.gov/20.5.E3. The two polynomial factorizations used above were independently checked by exact symbolic arithmetic. The proof itself uses no numerical optimization. A search of the available primary literature did not locate an earlier answer to this specific existence question; this is not a certification of priority or external acceptance.

XVI.3 A categorical obstruction for generalized cellular automata

External question and scope. Castillo-Ramirez, Vazquez-Aceves and Zaldivar-Corichi, *A categorical framework for cellular automata*, arXiv:2602.04049v1 (3 February 2026), end of Section 4, ask whether categorical products exist in $\mathrm{GCA}_{\mathbf{Set}}$. Their Theorem 4.7 constructs weak products, but does not establish products. The obstruction below excludes every possible product object for a broad family of pairs, not only their proposed weak product. The morphisms in this category are underlying functions; a group homomorphism witnessing that a map is a generalized cellular automaton is not part of the morphism data.

Write C_p for the cyclic group of prime order p, and 1 for the trivial group. For groups G, K, a generalized cellular automaton $T : D^G \to C^K$ has a homomorphism $\phi : K \to G$, a finite memory set $S \subseteq G$, and a local rule $\mu : D^S \to C$ such that

$$T(x)(k) = \mu\big((x(s\phi(k)))_{s\in S}\big).$$

The direction of the witnessing homomorphism is contravariant in the universe groups.

Theorem XVI.3.1 (Nonexistence of products)**.** *Let A, B be finite sets with $a = |A| \geq 2$ and $b = |B| \geq 2$. Let G, H be groups. Suppose that for some prime p, the sets $\mathrm{Hom}(G, C_p)$ and $\mathrm{Hom}(H, C_p)$ are finite, and at least one contains a nontrivial homomorphism. Then the objects A^G and B^H have no categorical product in $\mathrm{GCA}_{\mathbf{Set}}$. In particular, the category does not have binary products.*

Lemma XVI.3.2 (Finite cyclic probe)**.** *Let D, C be finite sets with $d = |D| \geq 2$ and $c = |C| \geq 2$, and let K be any group. Set*

$$r_K = |\,\mathrm{Hom}(K, C_p)|, \qquad N_p(d) = \frac{d^p + (p-1)d}{p}, \qquad m_p(d) = d^p - N_p(d).$$

If r_K is finite, then

$$|\mathrm{GCA}_{\mathbf{Set}}(D^{C_p}, C^K)| = r_K c^{d^p} - (r_K - 1)c^{N_p(d)} = c^{N_p(d)}\big[r_K c^{m_p(d)} - (r_K - 1)\big].$$

If r_K is infinite, this morphism set is infinite.

Proof. Because the source universe C_p is finite, its whole group may be used as memory. For a fixed homomorphism $\phi : K \to C_p$, all its cellular automata are exactly

$$T_{\phi,f}(x)(k) = f(\sigma_{\phi(k)}x), \qquad f : D^{C_p} \to C,$$

where σ denotes cyclic translation. There are c^{d^p} such maps for this ϕ, since evaluation at the identity of K recovers f.

Suppose $\phi \neq \psi$ and $T_{\phi,f} = T_{\psi,g}$. Evaluation at the identity first gives $f = g$. Choose k with $\phi(k) \neq \psi(k)$. The equality at k says that f is invariant under the nonidentity translation $\phi(k) - \psi(k)$. As p is prime, this translation generates C_p, so f is invariant under every cyclic translation. Conversely, every invariant f yields the same map for every ϕ. Thus all the classes have precisely the same pairwise intersection, namely the invariant local rules.

There are $N_p(d)$ translation orbits on D^{C_p}: the d constant configurations have orbit size one, and all other configurations have orbit size p. Hence there are exactly $c^{N_p(d)}$ invariant local rules. Removing this common intersection from each class proves the finite formula. For infinite r_K, fix one noninvariant f; such a rule exists because $d, c \geq 2$. Then $\phi \mapsto T_{\phi,f}$ is injective by the preceding intersection argument, proving infinitude. □

Proof of Theorem XVI.3.1. Suppose a product exists, and write its configuration object as C^K, allowing C and K to be arbitrary sets and groups at this stage. Let $E = \{*\}^1$. A generalized cellular automaton from E to C^K is precisely a constant configuration with value in C. Thus the product bijection on morphisms out of E forces

$$|C| = |A|\,|B| = ab.$$

In particular C is finite and has at least four elements.

Put $r_G = |\operatorname{Hom}(G, C_p)|$ and define r_H, r_K analogously. For every finite alphabet D, the product property gives

$$|\mathrm{GCA}_{\mathbf{Set}}(D^{C_p}, C^K)| = |\mathrm{GCA}_{\mathbf{Set}}(D^{C_p}, A^G)||\mathrm{GCA}_{\mathbf{Set}}(D^{C_p}, B^H)|.$$

The right side is finite by the hypotheses and Lemma XVI.3.2; that lemma therefore also implies that r_K is finite. Substitute its formula, cancel $(ab)^{N_p(d)}$, and write $m = m_p(d)$. We obtain, for every $d \geq 2$,

$$r_K(ab)^m - (r_K - 1) = \big[r_G a^m - (r_G - 1)\big]\big[r_H b^m - (r_H - 1)\big]. \tag{XVI.3.1}$$

As $d \to \infty$, $m_p(d) \to \infty$. Divide by $(ab)^m$ and take limits to get $r_K = r_G r_H$. Using this identity in (XVI.3.1) gives

$$r_G(r_H - 1)a^m + r_H(r_G - 1)b^m = 2r_G r_H - r_G - r_H.$$

Both r_G, r_H are positive integers because trivial homomorphisms exist. At least one is greater than one. The left side consequently tends to infinity with d, whereas the right side is fixed. This is a contradiction. □

Corollary XVI.3.3 (Two explicit families)**.** *For all finite alphabets A, B of cardinality at least two:*

1. *A^1 and B^{C_p} have no product for any prime p;*
2. *$A^{\mathbb{Z}}$ and $B^{\mathbb{Z}}$ have no product in* $\mathrm{GCA}_{\mathbf{Set}}$.

More generally, Theorem XVI.3.1 applies to any pair of finitely generated groups when at least one has a nontrivial finite abelian quotient.

Proof. For a finitely generated group, there are finitely many homomorphisms to C_p. A nontrivial finite abelian quotient admits a quotient C_p for some prime p. The specific homomorphism counts for the first family are $r_1 = 1$ and $r_{C_p} = p$; for the second family both are p. □

XVI.3.1 Finite nonabelian probes

The same obstruction extends beyond cyclic quotients when the two alphabets have the same cardinality.

Theorem XVI.3.4 (Obstruction from any finite quotient)**.** *Let A, B be finite alphabets of the same cardinality $a \geq 2$. Suppose there is a nontrivial finite group F such that* $\operatorname{Hom}(G, F)$ *and* $\operatorname{Hom}(H, F)$ *are finite, and at least one contains a nontrivial homomorphism. Then A^G and B^H have no product in* $\mathrm{GCA}_{\mathbf{Set}}$. *In particular this applies to any pair of finitely generated groups of which at least one has a nontrivial finite quotient, without an abelian hypothesis on that quotient.*

Proof. Write $q = |F| \geq 2$. For an alphabet D of cardinality $d \geq 2$, put $M = d^q$. For any finite alphabet V of cardinality $v \geq 2$ and group J with $r_J = |\operatorname{Hom}(J, F)| < \infty$, let

$$N_v(J, d) = |\mathrm{GCA}_{\mathbf{Set}}(D^F, V^J)|.$$

For each $\phi : J \to F$, evaluation at the identity identifies the ϕ-cellular automata with all v^M local rules $f : D^F \to V$. The union of these r_J classes therefore has cardinality

$$N_v(J, d) = r_J v^M - E_v(J, d), \qquad E_v(J, d) \geq 0.$$

We need the following two estimates:

$$E_v(J, d) \leq \binom{r_J}{2} v^{M/2 + d^{q/2}}, \tag{XVI.3.2}$$

$$r_J > 1 \quad \Longrightarrow \quad E_v(J, d) \geq v^{M/q}. \tag{XVI.3.3}$$

For the upper estimate, maps belonging to distinct homomorphism classes have the same local rule f, and this rule is invariant under a nonidentity translation of F. If the translating element has order $h \geq 2$, Burnside's orbit-counting formula bounds the number of its orbits on D^F by

$$\frac{M}{h} + \frac{h-1}{h} d^{q/2} \leq \frac{M}{2} + d^{q/2}.$$

Indeed every nonidentity power of that translation fixes at most $d^{q/2}$ configurations. The intersection of two classes has at most $v^{M/2 + d^{q/2}}$ maps. The total overcount in a union is at most the sum of its pairwise intersection sizes, proving (XVI.3.2). For the lower estimate, all rules invariant under the entire group F belong to every homomorphism class. There are at least M/q orbits under F, hence at least $v^{M/q}$ such rules. If $r_J > 1$, these rules alone contribute at least $v^{M/q}$ to the overcount.

Now suppose a product is C^K. The singleton probe again gives $|C| = a^2$. The morphism set from the finite probe $\{0, 1\}^F$ into this product must be finite. Choosing a local rule that records the input at the identity embeds $\operatorname{Hom}(K, F)$ into this morphism set: distinct homomorphisms select distinct independent input coordinates. Thus r_K is finite, and the estimates above apply also to K.

The product property yields

$$N_{a^2}(K, d) = N_a(G, d) N_a(H, d).$$

By (XVI.3.2), dividing by a^{2M} and letting $d \to \infty$ gives $r_K = r_G r_H$. After subtracting the leading terms, the product identity becomes

$$E_{a^2}(K, d) = r_H a^M E_a(G, d) + E_a(H, d) N_a(G, d).$$

Assume without loss of generality $r_G > 1$. The right side is at least $a^M E_a(G, d) \geq a^{(1+1/q)M}$. But (XVI.3.2) bounds the left side by

$$\binom{r_K}{2} a^{M + 2d^{q/2}}.$$

Since $d^q/q - 2d^{q/2} \to \infty$, these two bounds are incompatible. Finally, a finitely generated group has only finitely many homomorphisms into any fixed finite group, proving the stated consequence. □

A certificate using only two small probes. For $A = B = \{0,1\}$, $G = 1$, $H = C_2$, the singleton probe forces $|C| = 4$. The four-point probe $X = \{0,1\}^{C_2}$ has

$$|\mathrm{GCA}(X, A^1)| = 16, \qquad |\mathrm{GCA}(X, B^{C_2})| = 24.$$

It would therefore require 384 morphisms into the product. But Lemma XVI.3.2 gives

$$|\mathrm{GCA}(X, C^K)| = 256 r_K - 64(r_K - 1) = 192 r_K + 64.$$

The equation $384 = 192 r_K + 64$ would force $r_K = 5/3$. This excludes all possible alphabets and all possible universe groups for a product of this concrete pair.

What this does not assert. The theorem does not say that no pair in the category has a product. For example, two objects over the trivial group have the usual product over the trivial group. It also does not contradict the fixed-universe category $\mathrm{CA}_{\mathbf{Set}}(G)$ having products: the obstruction uses generalized morphisms with varying universe homomorphisms, which are additional morphisms in the larger category. The nonabelian theorem also covers finitely generated perfect groups with nontrivial finite quotients when the alphabet sizes agree. No classification is claimed for all possible groups or alphabet sizes.

Source and novelty audit. The exact externally posed question is in the final paragraph of Section 4 of `https://arxiv.org/html/2602.04049v1`. Definition 4.1 and the paragraph after Theorem 4.3 specify the category used above. A targeted search on 19 October 2026 did not locate a published resolution of this product-existence question; this is a search result rather than a certification of priority.

XVI.4 A complete palindromicity classification for bipartite support enumerators

External question. Dai, Hou, Liu, Thawinrak, and Wang, *Counting Lattice Points in Minkowski Sums of Cross Polytopes*, arXiv:2608.16037v2, Problem 5.3, ask which bipartite graphs have support enumerators satisfying palindromicity, unimodality, and real-rootedness properties. We classify the entire palindromicity branch. Every palindromic case also has an explicit nonnegative gamma expansion. This does not classify all unimodal or all real-rooted cases. The gamma expansion for preorders was obtained in the preceding research round; its proof is reproduced here as an input, not counted again as a new conjecture resolution.

For a finite bipartite graph $G = (L \sqcup R, E)$ define

$$D(G) = \{a \in \mathbb{Z}^L_{\geq 0} : a(S) \leq |N_G(S)| \text{ for every } S \subseteq L\}, \qquad F_G(z) = \sum_{a \in D(G)} z^{|\operatorname{supp}(a)|}.$$

We ignore multiple edges, which do not change these definitions. An isolated left vertex forces a zero coordinate and an isolated right vertex changes no neighborhood, so deleting isolated vertices preserves F_G. Palindromicity is taken with respect to the actual polynomial degree. The empty active graph has $F_G = 1$.

Theorem XVI.4.1 (Classification)**.** *Suppose G has at least one edge, delete its isolated vertices, and set $d = \deg F_G$. The following are equivalent.*

1. *$F_G(z)$ is palindromic.*
2. *Both active parts have the same size n, and there is a perfect matching such that, after labeling its pairs (ℓ_i, r_i), the relation*

$$i \preceq j \quad \Longleftrightarrow \quad \ell_i r_j \in E$$

   is transitive. It is automatically reflexive, so it is a preorder.
3. *$F_G(z)$ is gamma-positive of degree d.*

If these conditions hold, $d = n$, and every choice of perfect matching in item 2 gives a transitive relation. Moreover,

$$F_G(z) = \sum_{j=0}^{\lfloor n/2 \rfloor} \gamma_j z^j (1+z)^{n-2j}, \qquad \gamma_j = \#\{a \in D(G) \cap \{0,1,2\}^n : |a| = n,\ \#\{i : a_i = 2\} = j\}.$$

In particular every palindromic support enumerator in this entire graph family is unimodal.

XVI.4.1 The top coefficient forces a balanced perfect matching

An indicator vector $\mathbf{1}_S$ belongs to $D(G)$ precisely when S can be matched injectively into R, by Hall's theorem. If $a \in D(G)$, then $\mathbf{1}_{\operatorname{supp}(a)} \leq a$ is also feasible. Consequently the degree of F_G is the maximum matching size d. Its constant coefficient is one.

Suppose that its top coefficient is also one. There is then a unique matchable subset of L of cardinality d. Every nonisolated left vertex belongs to some maximum matching: start with one incident edge and apply augmenting paths until a maximum matching is reached; augmentation preserves the set of already matched left vertices. Hence that unique subset must be all active left vertices. Thus $d = |L|$ and there is a matching saturating L.

If any active right vertex r were unmatched, choose an incident left vertex ℓ_i. The saturating matching together with $\ell_i r$ realizes the demand vector $\mathbf{1} + e_i$: each demand unit is assigned to a

distinct right vertex adjacent to its left coordinate. Such an assignment implies all inequalities defining $D(G)$. The distinct feasible full-support vectors $\mathbf{1}$ and $\mathbf{1}+e_i$ would make the top coefficient at least two. Therefore the parts are balanced and the matching is perfect. This proves the asserted reduction using only $[z^d]F_G = [z^0]F_G = 1$.

XVI.4.2 The two adjacent coefficients detect every missing transitive edge

Fix any perfect matching and identify both parts with $[n]$, so every diagonal edge is present. Write $i \to j$ for an edge and let $i \rightsquigarrow j$ denote directed reachability. Set $F_G(z) = \sum_{k=0}^{n} h_k z^k$.

Lemma XVI.4.2 (Exact coefficient defect)**.** *For a balanced graph with a perfect matching,*

$$h_1 = n + \#\{(i,j) : i \neq j,\ i \to j\}, \qquad h_{n-1} = n + \#\{(i,j) : i \neq j,\ i \rightsquigarrow j\}.$$

Consequently

$$h_{n-1} - h_1 = \#\{(i,j) : i \neq j,\ i \rightsquigarrow j,\ i \not\to j\}. \tag{XVI.4.1}$$

For $n = 1$, the formula for h_{n-1} is instead $h_0 = 1$ and the same zero-defect conclusion holds directly; the displayed formulas apply when $n \geq 2$.

Proof. A vector supported only at i may have any positive value up to $|N(i)|$, giving the first formula. For $n \geq 2$, a support of size $n-1$ omits a unique coordinate j. Because $|a| \leq n$, the possible vectors are $\mathbf{1} - e_j$ and $\mathbf{1} - e_j + e_i$, with $i \neq j$. The first is feasible. We claim that the second is feasible exactly when $i \rightsquigarrow j$.

For this demand vector the only potentially restrictive Hall inequalities are those for sets S containing i but not j. They require $|N(S)| \geq |S| + 1$. Since diagonal edges give $N(S) \supseteq S$, this is equivalent to the existence of an edge leaving S. If i reaches j, every such S has an edge leaving it. If i does not reach j, choose S to be the set of all vertices reachable from i; it is closed under outgoing edges and excludes j, so its inequality fails. This proves the claim and the second formula. Every edge is a reachable pair, yielding (XVI.4.1). □

If F_G is palindromic, $h_{n-1} = h_1$. Thus every reachable pair is already an edge, which is exactly transitivity. The argument used an arbitrary perfect matching. For $n = 1$ the single active edge already defines the unique reflexive preorder. This proves necessity in Theorem XVI.4.1.

XVI.4.3 A self-contained sufficiency argument and the gamma coefficients

For a preorder τ on $[n]$, put

$$P_\tau = \{a \in \mathbb{Z}^n_{\geq 0} : a(I) \leq |I| \text{ for every ideal } I\}.$$

In the matching convention above $N(S) = \uparrow_\tau S$. The defining inequalities for $D(G)$ are therefore equivalent to those for all upper ideals of τ: necessity uses $S = I$, and sufficiency uses $S \subseteq \uparrow S$ and nonnegativity. Thus $D(G) = P_{\tau^*}$.

We recall the following identity, including its argument so the present classification does not depend on an unavailable companion manuscript:

$$\sum_{a \in P_\sigma} \prod_i \binom{r_i + a_i - 1}{a_i} = \sum_{\substack{a \in P_\sigma \\ |a| = n}} \prod_i \binom{r_i + 1}{a_i}. \tag{XVI.4.2}$$

To prove it, for the dual preorder $\rho = \sigma^*$ consider

$$Q_\rho(\mathbf{r}, s) = \sum_i r_i \Delta_{\uparrow_\rho i} + s\Delta_{[n]}, \qquad \Delta_A = \operatorname{conv}(0, e_i : i \in A).$$

For nonnegative parameters its inequalities are $x_i \geq 0$ and $x(I) \leq \sum_{i \in I} r_i + s$ for nonempty ideals I of ρ. Necessity is immediate; sufficiency follows from the capacitated bipartite matching criterion, since the sources reaching a target set S are exactly $\downarrow S$.

Postnikov's lattice-point formula for sums of simplices gives

$$E_\rho(\mathbf{r}, s) = \sum_{a \in P_\sigma} \binom{s + n - |a|}{n - |a|} \prod_i \binom{r_i + a_i - 1}{a_i}.$$

This is an established formula, with independent simplex coefficients. For integral $r_i, s \geq 1$, interior integer points of $Q_\rho(\mathbf{r}, s)$ translate by $-\mathbf{1}$ exactly to integer points of $Q_\rho(\mathbf{r} - \mathbf{1}, s - 1)$. Ehrhart reciprocity along each ray and polynomial continuation therefore give

$$E_\rho(-\mathbf{r}, -s) = (-1)^n E_\rho(\mathbf{r} - \mathbf{1}, s - 1).$$

Evaluate this polynomial identity to express $E_\rho(\mathbf{r}, 0)$ as $(-1)^n E_\rho(-\mathbf{r} - \mathbf{1}, -1)$. In the latter formula every term with $|a| < n$ vanishes, and the signs in the terms with $|a| = n$ cancel. This proves (XVI.4.2).

Now $B_j(r) = \binom{r+j-1}{j}$ is a polynomial basis. Define a linear functional T_z by $T_z(B_0) = 1$ and $T_z(B_j) = z$ for $j \geq 1$. The generating functions $(1 - u)^{-r}$ and $(1 + v)^{r+1}$, with $u = v/(1 + v)$, show that

$$\sum_{j \geq 0} T_z\left(\binom{r+1}{j}\right) v^j = (1 + v)(1 + zv) = 1 + (1 + z)v + zv^2.$$

Apply this functional in every variable of (XVI.4.2). The left side becomes the support enumerator. On the right only vectors with coordinates in $\{0, 1, 2\}$ survive, each contributing $z^{\#2}(1 + z)^{\#1}$. Since $|a| = n$, we have $\#1 = n - 2\#2$. This gives the stated gamma expansion. Gamma positivity implies palindromicity and unimodality, completing all implications.

Corollary XVI.4.3 (Two coefficients suffice)**.** *For a nonconstant support enumerator $F_G(z) = \sum_{k=0}^{d} h_k z^k$, the conditions $h_d = 1$ and $h_{d-1} = h_1$ are equivalent to palindromicity, and in this case imply the gamma expansion above.*

Proof. The top-coefficient argument forces a balanced perfect matching, and the coefficient-defect formula forces transitivity. The theorem then applies. For $d = 1$ the claim follows directly from $F_G = 1 + h_1 z$. The converse is immediate from palindromicity and $h_0 = 1$. □

This numerical criterion is not claimed as new: Hatasa's *A palindromicity criterion for the h-polynomials of bipartite edge rings*, arXiv:2605.26865, Theorem 1.1, proves the corresponding criterion in the larger class of bipartite edge rings. The source's root-polytope realization makes the criterion above a special case. Our argument identifies its explicit matching-and-preorder meaning for the support enumerators in Problem 5.3.

Scope, novelty, and verification. The assertion presented here is the concrete necessary-and-sufficient classification for all bipartite graphs, together with the coefficient defect (XVI.4.1). Sufficiency uses the previously developed preorder identity, reproduced above; Postnikov's counting formula is an existing input. No classification of all unimodal or all real-rooted support enumerators is claimed. An independent derivation checked the argument, and exact enumeration of all 1×4, 2×4, 3×4, and 4×3 bipartite graphs, totaling 8464, agreed with the criterion. This finite check is a diagnostic, not the proof. Priority is not certified. The classification also has a classical algebraic route: Ohsugi–Hibi's *Special simplices and Gorenstein toric rings*, https://arxiv.org/abs/math/0503666, Theorem 2.1(a'), characterizes Gorenstein bipartite edge rings through perfect matchings and acceptable sets. Applied to the augmented graph in

the source's root-polytope realization, its condition simplifies to the preorder condition above. Its Corollary 3.2(a') also gives the associated unimodality conclusion. Accordingly, we regard this section as an explicit combinatorial answer to the stated graph-classification question using established structure, not as a new general Gorenstein or palindromicity criterion. The primary question is at https://arxiv.org/pdf/2608.16037, Problem 5.3; Postnikov's formula is https://doi.org/10.1093/imrn/rnn153, also recalled as Lemma 2.7 in the source paper.

XVI.5 A short closure of the delayed-star unimodality conjecture

Ibarra, Landry, Montoya-Vega and Przytycki formulate Conjecture 4.1 in *Exploring unimodality of the plucking polynomial with delay function*, https://arxiv.org/pdf/2311.18059, asking whether the delayed plucking polynomial of every star is unimodal when all delays belong to $\{1, 2\}$. The result below answers that statement affirmatively. Its proof is a short convolution argument, not a claim of a new general theorem in topology.

Put $[m]_q = 1 + q + \cdots + q^{m-1}$ for $m \geq 1$ and $[m]_q! = \prod_{j=1}^{m} [j]_q$, with $[0]_q! = 1$. A finite nonnegative sequence is unimodal if it is nondecreasing up to some index and nonincreasing thereafter. We regard the zero sequence as unimodal.

Lemma XVI.5.1 (Uniform convolution)**.** *If a polynomial $A(q)$ has nonnegative unimodal coefficients, then $[m]_q A(q)$ has unimodal coefficients for every integer $m \geq 1$.*

Proof. Extend the coefficient sequence (a_k) of A by zero to all integer indices, and choose a mode r. If b_k denotes the coefficient of the product, then

$$b_k - b_{k-1} = a_k - a_{k-m}.$$

For $k \leq r$ this difference is nonnegative; for $k \geq r + m$ it is nonpositive. For the intermediate indices $r \leq k \leq r + m$, the sequence a_k is nonincreasing while a_{k-m} is nondecreasing. Their difference is therefore nonincreasing on that interval. Thus the first differences can change sign only from nonnegative to nonpositive, as required. □

Theorem XVI.5.2 (Weighted delayed-star unimodality)**.** *For every integer $n \geq 2$ and every polynomial $p(q) = \sum_{i=0}^{n-1} c_i q^i$ with real coefficients $c_i \geq 0$, the polynomial*

$$p(q)[n-1]_q!$$

is unimodal.

Proof. Set $m = n - 1$ and first form $B(q) = p(q)[m]_q$. With the convention $c_i = 0$ outside $0 \leq i \leq m$, its first differences are

$$b_k - b_{k-1} = c_k - c_{k-m}.$$

For $k < m$ they are nonnegative, for $k > m$ they are nonpositive, and the single difference at $k = m$ may have either sign. Thus B is unimodal. Multiplication successively by $[1]_q, \ldots, [m-1]_q$ preserves this property by Lemma XVI.5.1 and gives the stated polynomial. □

Corollary XVI.5.3 (Source Conjecture 4.1)**.** *For every star with n rays and every delay function taking values in $\{1, 2\}$, the delayed plucking polynomial is unimodal.*

Proof. Order the leaves from right to left as $v_0, \ldots, v_{n-1}$. The first plucked leaf must have delay one, and choosing v_i contributes q^i. After that pluck all remaining delays equal one, and the remaining star has ordinary plucking polynomial $[n-1]_q!$. Hence

$$Q(T, f) = \left(\sum_{i : f(v_i) = 1} q^i \right) [n-1]_q!.$$

This is also Proposition 2.5 of the source paper. Apply Theorem XVI.5.2. For $n = 1$ the polynomial is either 1 or 0, and for the tree with no rays it is 1. □

Scope and prior work. The argument proves all ray numbers and requires no symmetry of the delay pattern. It strengthens the binary coefficient statement to arbitrary nonnegative coefficients. It gives weak unimodality, not a classification of strict unimodality or log-concavity. The source's adjacent question about star delays in $\{1, 2, 3\}$ is not resolved here. The general-tree anti-unimodal-delay question, Open Problem 4.2, is *not* counted as a new result: an independently reproduced eight-edge counterexample agrees with the witness already published by Jonas J. Gebendorfer on 14 October 2026, in *Nonunimodal plucking polynomials with anti-unimodal delays*, DOI 10.13140/RG.2.2.27965.83682. The new proof above is an elementary consequence of the source's formula; a search finding no earlier statement of this consequence does not establish priority.

XVI.6 A spectral class of the vector-chromatic p-energy conjecture

External target and scope. Elphick, Tang and Zhang, *A Spectral Lower Bound on Chromatic Numbers using p-Energy*, arXiv:2504.01295v5, Conjecture 6.3, conjecture that every graph G and every real $p \geq 2$ satisfy

$$\chi_v(G) \geq 1 + \max\left\{\frac{E_p^+(G)}{E_p^-(G)}, \frac{E_p^-(G)}{E_p^+(G)}\right\}^{1/(p-1)}. \tag{XVI.6.1}$$

Here $E_p^{\pm}$ are the sums of the pth powers of the magnitudes of the positive and negative adjacency eigenvalues, and χ_v is the vector chromatic number. The theorem below proves this for a spectral class containing every strongly regular graph with an edge. It does *not* settle the conjecture for general graphs. No priority claim is made for the restricted theorem.

Theorem XVI.6.1. *Let G be a finite simple graph with at least one edge. Suppose all its negative adjacency eigenvalues are equal to $-s$, where $s > 0$. Set $d = \lambda_{\max}(A(G))$ and $h = d/s$. Then, for every real $p \geq 2$,*

$$\max\left\{\frac{E_p^+(G)}{E_p^-(G)}, \frac{E_p^-(G)}{E_p^+(G)}\right\} \leq h^{p-1}. \tag{XVI.6.2}$$

Consequently (XVI.6.1) *holds for G.*

Proof. Write the positive eigenvalues as $a_1, \dots, a_r$, and let m be the multiplicity of $-s$. The Perron bound gives $0 < a_i \leq d$ and $h \geq 1$. As the adjacency matrix has zero diagonal and nonnegative entries,

$$\sum_i a_i = ms, \qquad \sum_i a_i^3 - ms^3 = \operatorname{tr}(A^3) \geq 0. \tag{XVI.6.3}$$

For every $p \geq 1$,

$$E_p^+(G) = \sum_i a_i^p \leq d^{p-1} \sum_i a_i = ms^p h^{p-1} = E_p^-(G) h^{p-1}.$$

For the reverse ratio first suppose $2 \leq p \leq 3$. Since $a_i \leq d$ and $p - 3 \leq 0$,

$$E_p^+(G) = \sum_i a_i^3 a_i^{p-3} \geq d^{p-3} \sum_i a_i^3 \geq ms^3 d^{p-3}.$$

It follows that

$$\frac{E_p^-(G)}{E_p^+(G)} \leq h^{3-p} \leq h^{p-1}.$$

For $p \geq 3$, put $w_i = a_i/(ms)$, so that $\sum_i w_i = 1$. The second inequality in (XVI.6.3) says $\sum_i w_i a_i^2 \geq s^2$. The monotonicity of weighted power means therefore gives

$$\sum_i w_i a_i^{p-1} \geq \left(\sum_i w_i a_i^2\right)^{(p-1)/2} \geq s^{p-1}.$$

Thus $E_p^+(G) \geq ms^p = E_p^-(G)$, which proves (XVI.6.2) in the remaining range.

For completeness, the vector version of Hoffman's bound can be proved directly. If u_v are unit vectors giving a vector k-coloring, their Gram matrix C is positive semidefinite, has diagonal 1, and satisfies $C_{vw} \leq -1/(k-1)$ on edges. Choose a nonnegative unit Perron eigenvector x of A, and set $X = \operatorname{diag}(x) C \operatorname{diag}(x)$. Then $X \succeq 0$, $\operatorname{tr} X = 1$, and

$$-s \leq \operatorname{tr}(AX) \leq -\frac{x^{\mathsf{T}} A x}{k-1} = -\frac{d}{k-1}.$$

Hence $k \geq 1 + h$. Taking the infimum over vector colorings and applying (XVI.6.2) proves the claim. □

Corollary XVI.6.2. *Conjecture 6.3 holds for every strongly regular graph with at least one edge, for all real $p \geq 2$.*

Proof. For a strongly regular graph with parameters (v, k, λ, μ),

$$A^2 = (k - \mu)I + (\lambda - \mu)A + \mu J.$$

On the orthogonal complement of the all-ones vector its eigenvalues solve $x^2-(\lambda-\mu)x-(k-\mu) = 0$. Since $k \geq \mu$, at most one root is negative. Together with the eigenvalue $k > 0$, this proves that the negative eigenvalues all have a common value. The theorem applies. The same argument includes the imprimitive cases; a disjoint union of equal complete graphs can also be checked directly. □

Sharpness within the class. For a complete balanced q-partite graph with a vertices in each part, the nonzero adjacency eigenvalues are $(q - 1)a$ once and $-a$ with multiplicity $q - 1$. Thus $h = q - 1$ and $E_p^+/E_p^- = h^{p-1}$. Its vector chromatic number is q, by the clique lower bound and its ordinary q-coloring. Hence the restricted theorem is sharp for every real $p \geq 2$.

Why the proof does not settle the general conjecture. The two scalar moments in (XVI.6.3) compare every positive eigenvalue with the *same* negative scale s. With several negative magnitudes, the cubic moment does not control their pth moment in the needed direction. Replacing this missing comparison by an appeal to interpolation would be an unjustified step. The general conjecture remains open in this work.

Other attempted graph target. Li, Wang and Guo, *The (Signless) Laplacian Spectral Radius with a Fixed Induced Subgraph of a Graph*, Electronic Journal of Combinatorics 33(3) (2026), P3.81, DOI:10.37236/12935, Conjectures 1–2, propose that the maximum signless Laplacian, respectively Laplacian, spectral radius of a graph containing an induced H and having t additional edges is achieved by attaching t leaves to one vertex of H. Neither conjecture was resolved here. Random exploration of extensions obtained by adding one vertex gave no counterexample and is not evidence of a proof.

Primary sources.

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- W.-J. Li, Z. Wang and J.-M. Guo, *The (Signless) Laplacian Spectral Radius with a Fixed Induced Subgraph of a Graph*, https://doi.org/10.37236/12935 (25 September 2026), Conjectures 1–2.

XVI.7 An arithmetic boundary: higher degree is not yet resolved

The higher-degree expectation in Section 2.8 of Ben David et al., *On the Connection Between Irrationality Measures and Polynomial Continued Fractions*, Arnold Math. J. 10 (2024), 529–566, doi:10.1007/s40598-024-00250-z, requires a polynomial continued fraction whose own convergents prove irrationality. No new closure of that expectation is claimed here.

The distinction from proving that its limit is irrational can be made explicit. Fix $d \geq 1$ and an integer $k \geq 2$, and put

$$b(n) = -n^{2d}, \qquad a_k(n) = -k(n+1)^d - k^{-1}n^d.$$

Constant inflation by $-k$ gives the equivalent fraction with

$$a'_k(n) = k^2(n+1)^d + n^d, \qquad b'_k(n) = -k^2 n^{2d}.$$

Writing $K = k^2$ and

$$H_N = \sum_{j=1}^{N} \frac{1}{K^j j^d},$$

the nth canonical convergent of this inflated fraction is exactly H_{n+1}^{-1}. Indeed its continuants are

$$p_n = K^{n+1}((n+1)!)^d, \qquad q_n = p_n H_{n+1}.$$

The first formula satisfies the continuant recurrence directly. For the second, using $H_{n+1} - H_n = K^{-n-1}(n+1)^{-d}$ verifies the same recurrence; the initial data are $p_0 = K$ and $q_0 = 1$. Consequently the limit is $\mathrm{Li}_d(K^{-1})^{-1}$.

Let $D_N = \mathrm{lcm}(1, \ldots, N)$. The standard denominator clearing factor $K^N D_N^d$ for H_N bounds the reduced denominator of H_N^{-1} by a fixed constant times $K^N D_N^d$. Meanwhile,

$$0 < \mathrm{Li}_d(K^{-1}) - H_N \leq \frac{K^{-N-1}}{(N+1)^d(1-K^{-1})}.$$

The reciprocal error satisfies the same bound up to a fixed factor. Multiplying these two estimates leaves a factor of order $D_N^d/(N+1)^d$, whose exponential growth is $\exp(dN + o(N))$. Thus this estimate does not tend to zero as required by the elementary rationality contradiction. Making k large does not improve its exponential sign: the K^N and K^{-N} factors cancel.

This is a limitation of the available estimate, not a nonexistence theorem for better cancellations. Proving irrationality of the limiting hypergeometric value by other methods also does not establish that these particular convergents have the positive approximation exponent demanded by the external question. A new uniform cancellation theorem or a different recurrence is still needed.

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